# Yang–Mills Mass Gap on the Lattice and Its Reduction to One Cube of the Continuum

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(Dated: 2 November 2026)

*We prove that four-dimensional lattice Yang–Mills theory with any compact simple gauge group has a unique infinite-volume state and an explicit transfer-matrix gap in a forest gauge, up to $\beta_W = 0.351$ for $\mathrm{SU}(2)$ and $\beta_W = 0.77$ for $\mathrm{SU}(3)$, and that a Dobrushin–Shlosman estimate in one cube of fixed physical size gives a mass gap in every continuum limit. Towards that estimate we label the blocks of Bałaban's averages, prove that the weight of bad islands factorises through kernels on the good layers between them, with cross-ratios exponentially small in the widths of the layers, and give the explicit coupling past which bad islands are rarer than their entropy at every scale up to $1/8$ of the scale that defects reach. The averaged covariance of the labels and the effective action across scales remain open.*

## I. INTRODUCTION

Wilson's lattice gauge theory [93] assigns an element $Q_e$ of a compact group $G \subset \mathrm{U}(N)$ to every link $e$ of $\mathbb{Z}^4$ and weighs a configuration by $\exp(N\beta \sum_p \mathrm{Re}\,\mathrm{Tr}\, Q_p)$, where $Q_p$ is the ordered product of the four link variables around the plaquette $p$.[1] At small $|\beta|$ the theory is under control: it has a unique infinite-volume state, correlations decay exponentially, and the transfer matrix has a gap [75, 82].[2] The question is how far into the coupling range such statements can be proved with explicit constants, since the problem of the continuum limit [56] begins where strong coupling ends and Bałaban's weak-coupling analysis [6–8] takes over.

Two methods give explicit ranges. The Bakry–Émery method [4, 5] of Shen, Zhu and Zhu [84] bounds the Hessian of the action; in [16] we showed that the sharp bound is $4d = 16$ and that no curvature argument for the ungauged measure goes beyond $|\beta| < 1/32$. The Dobrushin method [34, 35][3] bounds the influence of each link on the conditional law of another; for $\mathrm{SU}(2)$ the single-link conditional law is a von Mises–Fisher law on the three-sphere, the sharp Kantorovich modulus[4] of

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*2020 Mathematics Subject Classification.* Primary 81T25, 82B20; Secondary 81T13, 60K35, 82B26, 22E30. *PACS.* 11.15.Ha, 11.15.Tk, 05.50.+q, 02.50.Cw. *Keywords.* Lattice gauge theory; mass gap; Dobrushin uniqueness; block averages; reflection positivity. An earlier version of this paper is available at `https://hexagonmath.org/2610.00341v6`. New in this version is Sec. XIII: the product law on good windows with each path link determined by its pair (Lemma 174, Proposition 175), the rough background with negative constrained curvature (Proposition 172), the nonconvex bad side and its three-preimage branch family (Propositions 176 and 177), the two sides of the path-link law: the response to the links that enter only the action, the good side to radius $1/32$, and the fold across which the law jumps for every coupling (Lemma 178, Propositions 179 and 180), the bounded response of the 24-link site of a bad pair, which carries its inside links among its variables, to every other link (Theorem 183), the exact factorisation of the weight of two bad islands through one kernel on the good layer between them, on a domain that their labels do not move, with the covariance bound of the chart for the layer law in the coordinates of the labels and a cross-ratio exponentially small in the width of the layer (Theorems 207 and 209, Proposition 208, Corollaries 187 and 210), the evaluation of the prefactor of that cross-ratio (Lemma 211), the bound on the cross-ratio of two islands given the labels of all the others (Theorem 215), the growth by a factor four per step of an error bounded by a multiple of the Euclidean form, which forces the one-step errors of an effective action to be relative and gauge invariant (Proposition 195), the floor $a^2/16$ on the covariance of rare labels (Proposition 189), the explicit island threshold $\beta^*(\rho, s)$ past which the islands of bad blocks at one scale are rarer than their entropy (Theorem 199), its composition to every scale up to $a_{k_*}/8$ at the witness radius $10^{-4}$, sharp for the cut-off at $\kappa_c = 0.08376$, with the abelian field that stops the fine witnesses (Theorems 201 and 205, Propositions 202 and 204), the coupling-cost minimality lemmas (Lemmas 190 and 191), the exact block response $\frac{1}{4}(1 + I_2(\beta_W)/I_1(\beta_W))$ and its crossing at $\beta_W = 0.3666$ (Proposition 192), the weighted Schur locality lemma with its perturbation ledger (Lemma 193, Proposition 194), and the positive-time spectral bounds (Lemmas 196 and 197, Proposition 198); the abstract, the introduction and the remark after Theorem 3 state them.

[1] This is the normalisation of Shen, Zhu and Zhu [84] and of our earlier paper [16]. The conventional inverse coupling is $\beta_W = N^2\beta$, so the weight is $\exp(\frac{\beta_W}{N}\sum_p \mathrm{Re}\,\mathrm{Tr}\, Q_p)$; for $\mathrm{SU}(2)$, identified with the unit quaternions, $\frac{1}{2}\mathrm{Tr}\, Q_p = \mathrm{Re}\, Q_p$ and $\beta_W = 4\beta$. Standard accounts of the lattice theory are the books of Montvay and Münster [71] and of Gattringer and Lang [44]; the Hamiltonian version is due to Kogut and Susskind [61].

[2] Strong-coupling results of this kind go back to Osterwalder and Seiler [75], who proved exponential clustering and a gap by a convergent expansion, and to Seiler's monograph [82]; the general framework for lattice gauge fields and their continuum limits is that of Brydges, Fröhlich and Seiler [25]. More recent work treats the strong-coupling regime by probabilistic methods: the survey of Chatterjee [28], his large-$N$ solution of $\mathrm{SO}(N)$ lattice gauge theory at strong coupling [29] and its planar extension by Basu and Ganguly [11], the Langevin dynamics of Shen, Zhu and Zhu [84], and the area law in an enlarged region by Cao, Nissim and Sheffield [27]. For finite gauge groups Adhikari and Cao proved correlation decay at weak coupling [1]. In two dimensions the continuum Yang–Mills measure has been constructed by Gross, King and Sengupta [47], Driver [37] and Lévy [62]; in four dimensions Bałaban's renormalisation group, as expounded by Dimock [32], gives ultraviolet stability in finite volume, and Magnen, Rivasseau and Sénéor constructed the theory with an infrared cutoff [67]. None of these results gives a mass gap in physical units.

[3] Dobrushin's theorem: if the conditional law of each variable given all others moves, in Kantorovich distance, by at most $C_{xy}$ times the distance by which the variable at $y$ moves, and $\sup_x \sum_y C_{xy} < 1$, then the Gibbs measure is unique and correlations decay at the rate given by the Neumann series of the matrix $C$. Dobrushin and Shlosman [36] replaced single variables by finite boxes, which is the form used in Sec. X.

[4] The Kantorovich distance between two probability measures on a metric

that family is one quarter [16] (Theorem 254 in Appendix I), and every link has eighteen partners, which gives $\beta_W < 2/9$ and nothing beyond, because the coefficients are attained.

Both limits come from the same count. A link of $\mathbb{Z}^4$ lies in six plaquettes and shares each with three other links. Gauge invariance makes part of this redundant: in a box, the variables on a forest of links can be set to the identity without changing any gauge-invariant expectation [30].[5] The forest we use is a field of straight lines with one line through every site, chosen so that every plaquette contains exactly one fixed link. Then every free link has exactly twelve free partners, two in each of its plaquettes (Lemma 9), and both methods improve.

**Theorem 1** (SU(2))**.** *Let $G = \mathrm{SU}(2)$ and $0 \le \beta_W < \frac{1}{3}$. For every gauge-invariant local observable $O$ the expectations $W_\Lambda(O)$ in boxes with free boundary converge, along every sequence of boxes that exhausts $\mathbb{Z}^4$, to a limit $\omega(O)$ that does not depend on the sequence. The state $\omega$ is invariant under translations and the hypercubic group and reflection positive;*[6] *truncated correlations of Lipschitz observables decay at every rate below $-\log(3\beta_W)$; and the transfer matrix has spectral gap*

$$m \ge -\log(3\beta_W) > 0.$$

*For $-\frac{1}{3} < \beta_W < 0$ all statements except reflection positivity and the gap hold.*

The step behind Theorem 1 is short once the line field is in place. The number $\frac{1}{3}$ is also the end of the road for this kind of argument.

**Theorem 2** (Ceiling)**.** *Let $F'$ be any set of links of $\mathbb{Z}^4$ without cycles, and give two distinct links outside $F'$ that share a plaquette the coefficient $|\beta|$, the largest single-link Kantorovich coefficient of* SU(2)*. Then the weighted Dobrushin condition with these coefficients fails for $\beta_W \ge \frac{1}{3}$. For the line field the coefficients are attained, so no single-link Kantorovich argument in that gauge reaches $\beta_W = \frac{1}{3}$.*

The ceiling concerns coefficients between single links. Take instead as a block the three free links of one plaquette. The free links of the line field split into such triples with period two (Lemma 12), the coupling inside the block plaquette is no longer paid for, and each link keeps ten outside partners. The price is the sensitivity of a law on three spheres instead of one, which we bound analytically by integrating out the middle link of the block.

**Theorem 3** (SU(2) with plaquette blocks)**.** *Let $G = \mathrm{SU}(2)$ and $0 \le \beta_W \le 0.351$. The number $\alpha(\beta_W)$ of (4) is increasing, $\alpha(\frac{1}{3}) \le 0.9428$, $\alpha(0.345) \le 0.9805$ and $\alpha(0.351) \le 0.99995$. The conclusions of Theorem 1 hold with decay at every rate below $-\frac{1}{2}\log\alpha(\beta_W)$ and a transfer-matrix gap*

$$m \ge -\tfrac{1}{2}\log\alpha(\beta_W) > 0;$$

*at $\beta_W = \frac{1}{3}$ this gives $m \ge 0.0294$. For $-0.351 \le \beta_W < 0$ all statements except reflection positivity and the gap hold.*

At small coupling the rate of Theorem 1 is the better one, so both bounds stand side by side; Theorem 3 matters past $\beta_W = \frac{1}{3}$, where the gap of Theorem 1 has closed.

The curvature method gains as well, and for every $N$. Let $\mathrm{Hess}\, S_0$ be the Hessian of $\sum_p \mathrm{Re}\,\mathrm{Tr}\, Q_p$ on $G^E$ and $F$ the set of fixed links.

**Theorem 4** (Curvature in the gauge)**.** *For every $N$, every field $Q$ and every tangent vector $v = (X_e Q_e)$ with $X_e = 0$ for $e \in F$,*

$$|\mathrm{Hess}\, S_0(v,v)| \le \tfrac{29}{2}|v|^2,$$

*against the sharp constant* 16 *without gauge fixing. Consequently, for $G = \mathrm{SU}(N)$ and $|\beta| < \frac{1}{29}$, or $G = \mathrm{SO}(N)$, $N \ge 3$, and $|\beta| < \frac{N-2}{58N}$, the conclusions of Theorem 1 hold with the rate and gap of Theorem 5. For* SU(3) *this is $\beta_W < \frac{9}{29} = 0.310$.*

The constant $\frac{29}{2}$ is a statement about $\mathrm{U}(N)$, so it holds on every closed subgroup, and the curvature of the group does the rest. This gives the theorem for every compact simple gauge group, the exceptional groups included, with an explicit gap. Let $\rho : G \to \mathrm{U}(N)$ be a faithful unitary representation of a compact connected Lie group $G$, weigh a configuration by $\exp(N\beta\sum_p \mathrm{Re}\,\mathrm{Tr}\,\rho(Q_p))$, and let $\kappa_G$ be the least Ricci[7] curvature of $G$ for the bi-invariant metric $\langle X, Y\rangle = -\mathrm{Tr}(\rho(X)\rho(Y))$; we identify $G$ with $\rho(G) \subset \mathrm{U}(N)$.

**Theorem 5** (Every compact simple group)**.** *Suppose that the Lie algebra of $G$ is semisimple, so that $\kappa_G > 0$, and that*

$$N|\beta| < \tfrac{2}{29}\,\kappa_G.$$

*Then the conclusions of Theorem 1 hold, for $\beta < 0$ all of them except reflection positivity and the gap, with decay at every rate below*

$$m_G(\beta) = \operatorname{arccosh}\Big(1 + \frac{1}{12}\Big(\frac{\kappa_G}{N|\beta|} - \frac{29}{2}\Big)\Big) > 0$$

*and a transfer-matrix gap $m \ge m_G(\beta)$. For a simple $G$, $\kappa_G = h^\vee/(2\ell_\rho)$, where $h^\vee$ is the dual Coxeter number and $\ell_\rho$ the Dynkin index of $\rho$; Table I lists the values.*

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space is the least expected distance over their couplings, equivalently the largest difference of their integrals over 1-Lipschitz functions [90]. The modulus of a family of laws indexed by a metric space is the best Lipschitz constant of the map from the index to the law in that distance.

[5] Creutz used a maximal tree to remove the gauge freedom completely; the temporal gauge, with all links in one direction fixed, is the special case of a family of straight lines. In a maximal tree almost every link is fixed in some region and none in others, so the number of partners of a free link is not uniform, and the temporal gauge leaves fourteen partners per link. The point here is a pattern that fixes exactly one link in every plaquette.

[6] That is, $\omega(\vartheta(\bar{F})F) \ge 0$ for every $F$ depending on the links on one side of a lattice hyperplane, $\vartheta$ being the reflection in it. It gives the Hilbert space and the transfer matrix of Osterwalder and Schrader [73, 74] and of Osterwalder and Seiler [75], reconstructed in Appendix F.

[7] For a bi-invariant metric on a compact Lie group the Ricci curvature is $-\frac{1}{4}\mathrm{Tr}((\mathrm{ad}X)^2)$ in the direction of $X$. It is positive for every $X \ne 0$ exactly when the Lie algebra is semisimple, and $\kappa_G$ is its minimum over unit vectors.

TABLE I. Least Ricci curvature $\kappa_G = h^\vee/(2\ell_\rho)$ of the compact simple groups in a faithful representation $\rho$ of least dimension $N$, and the range of Theorem 5 for the conventional inverse coupling $\beta_W = N^2\beta$.

| Group | $\rho$ | $h^\vee$ | $\ell_\rho$ | $\kappa_G$ | $\beta_W <$ |
|---|---|---|---|---|---|
| $\mathrm{SU}(N)$, $N \geq 2$ | $\mathbb{C}^N$ | $N$ | 1 | $\frac{N}{2}$ | $\frac{N^2}{29}$ |
| $\mathrm{SO}(N)$, $N \geq 5$ | $\mathbb{R}^N$ | $N-2$ | 2 | $\frac{N-2}{4}$ | $\frac{N(N-2)}{58}$ |
| $\mathrm{SO}(3)$ | $\mathbb{R}^3$ | 2 | 4 | $\frac{1}{4}$ | $\frac{3}{58}$ |
| $\mathrm{Sp}(n)$, $n \geq 2$ | $\mathbb{C}^{2n}$ | $n+1$ | 1 | $\frac{n+1}{2}$ | $\frac{2n(n+1)}{29}$ |
| $G_2$ | 7 | 4 | 2 | 1 | $\frac{14}{29}$ |
| $F_4$ | 26 | 9 | 6 | $\frac{3}{4}$ | $\frac{39}{29}$ |
| $E_6$ | 27 | 12 | 6 | 1 | $\frac{54}{29}$ |
| $E_7$ | 56 | 18 | 12 | $\frac{3}{4}$ | $\frac{84}{29}$ |
| $E_8$ | 248 | 30 | 60 | $\frac{1}{4}$ | $\frac{124}{29}$ |

For $\mathrm{SU}(N)$, $\kappa_G = \frac{N}{2}$, and for $\mathrm{SO}(N)$, $\kappa_G = \frac{N-2}{4}$, so Theorem 5 contains the ranges of Theorem 4 and supplies their rate.[8] For small $|\beta|$ the existence of a gap is classical for every compact group, with no explicit range [75, Thm. 3.6]; what is new is an explicit range for every group, past the range $N|\beta| < \kappa_G/16$ that the Hessian bound 16 gives without gauge fixing, and an explicit gap.[9] As $\beta \to 0$ the gap grows like $\log \frac{\kappa_G}{6N|\beta|}$.

No Bakry–Émery argument for the ungauged measure can pass $|\beta| = \frac{1}{32}$ [16, Thm. C]; the gauge fixing changes the measure, not the metric, which is why it can. For $\mathrm{SU}(2)$ the curvature range $\beta_W < \frac{4}{29}$ lies inside that of Theorem 1, and no curvature argument for the gauge-fixed $\mathrm{SU}(2)$ measure passes $\beta_W = \frac{1}{3}$: when one active link $e$ is $-1$ and all others are 1, the six plaquettes through $e$ have variable $-1$, and along a geodesic through $Q_e$ the Hessian of $-\frac{\beta_W}{2}\sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p$ is $-6\beta_W$ times the metric of the unit sphere $S^3$, whose Ricci curvature is 2, so the Bakry–Émery tensor is negative there once $\beta_W > \frac{1}{3}$. For $\mathrm{SU}(3)$ the Dobrushin route goes much further.

**Theorem 6** ($\mathrm{SU}(3)$)**.** *Let $G = \mathrm{SU}(3)$ and $0 \leq \beta_W \leq \frac{3}{4}$. Then the conclusions of Theorem 1 hold, with decay at every rate below $-\log(1.33319\,\beta_W)$ and a transfer-matrix gap*

$$m \geq -\log(1.33319\,\beta_W) > 0.$$

The proof of Theorem 6 bounds the Kantorovich modulus of the one-link measure $\nu_b(dU) \propto e^{\mathrm{Re}\,\mathrm{Tr}(Ub)}dU$ on $\mathrm{SU}(3)$ uniformly over staple sums $b$. Its analytic part is a lower bound for the spectral gap of $\nu_b$ through Bochner's formula with a cross term (Lemma 29), a corrector chosen box by box among polynomials of degree two in the matrix entries (Lemma 30), the reduction of $b$ to diagonal matrices (Lemma 31), the character expansion of the moments of $\nu_b$ (Lemma 32) and an interpolation inequality on boxes of staple sums (Lemma 33). The remaining step evaluates the resulting bound on 2341 boxes with rigorous error bounds; the theorem rests on that computation, which is described in full in Sec. VII G and Appendix E.

The plaquette blocks of Theorem 3 work for $\mathrm{SU}(3)$ as well. They use the same modulus of the one-link measure, but only through the three links of a block, so they need $\Lambda(2\beta_W)$ small and not $2r\Lambda(r) < 1$: near the end of Theorem 6 they give a much larger gap, and they go past it.

TABLE II. Explicit ranges of uniqueness, decay and a transfer-matrix gap for four-dimensional lattice Yang–Mills theory. The first three rows hold for the ungauged measure; the last three are the results of this paper.

| Method | $\mathrm{SU}(2)$, $\beta_W$ | $\mathrm{SU}(3)$, $\beta_W$ |
|---|---|---|
| Curvature, Shen–Zhu–Zhu [84] | $< \frac{1}{12}$ | $< \frac{3}{16}$ |
| Curvature, sharp [16] | $< \frac{1}{8}$ | $< \frac{9}{32}$ |
| Dobrushin, sharp quarter [16] | $< \frac{2}{9}$ | |
| Curvature, forest gauge | $< \frac{4}{29}$ | $< \frac{9}{29}$ |
| Dobrushin, forest gauge | $< \frac{1}{3}$ | $\leq \frac{3}{4}$ |
| Plaquette blocks, forest gauge | $\leq 0.351$ | $\leq 0.77$ |

**Theorem 7** ($\mathrm{SU}(3)$ with plaquette blocks)**.** *Let $G = \mathrm{SU}(3)$ and $0 \leq \beta_W \leq \frac{77}{100} = 0.77$. Then $\alpha_3(\beta_W) < 1$ for the number of (15), with $\alpha_3(0.72) \leq 0.9182$, $\alpha_3(\frac{3}{4}) \leq 0.9653$ and $\alpha_3(0.76) \leq 0.99950$ and $\alpha_3(0.77) \leq 0.99978$. The conclusions of Theorem 1 hold with decay at every rate below $-\frac{1}{2}\log\alpha_3(\beta_W)$ and a transfer-matrix gap*

$$m \geq -\tfrac{1}{2}\log\alpha_3(\beta_W) > 0;$$

*at $\beta_W = \frac{3}{4}$ this gives $m \geq 0.0177$, against $1.1\cdot10^{-4}$ from Theorem 6, and at $\beta_W = 0.76$ and $0.77$, where Theorem 6 no longer applies, $m \geq 2.5\cdot10^{-4}$ and $m \geq 1.1\cdot10^{-4}$.*

All seven theorems are statements at fixed lattice spacing. They do not touch the continuum limit, the crossover to Bałaban's regime, or a gap in physical units, which together are the Clay problem; Sec. IX says where each method stops and what remains of the four open problems of [16]. The obstruction proved in [17], that a lattice gap uniform in lattice units forces trivial continuum two-point functions, applies to all of them.

*The continuum step.* Sec. X goes past that obstruction in one respect, through three implications, each proved.

(I1) A Dobrushin–Shlosman condition on one cube of fixed physical size, uniform along the continuum tuning, gives a mass gap in physical units for every continuum limit (Theorems 43 and 45), and so does exponential decay with constants of the renormalised correlations of a spanning set of observables (Theorem 46); this holds also for

[8] The dual Coxeter number is $h^\vee = N$ for $\mathrm{SU}(N)$, $N-2$ for $\mathrm{SO}(N)$, $n+1$ for $\mathrm{Sp}(n)$, $4,9,12,18,30$ for $G_2, F_4, E_6, E_7, E_8$, and the Dynkin index of a representation is normalised to one on the defining representation of $\mathrm{SU}(N)$ [58, Ch. 6]. The identity $\kappa_G = h^\vee/(2\ell_\rho)$ is the statement that the Ricci tensor of a compact simple group with a bi-invariant metric is one quarter of the negative Killing form, $\mathrm{Ric}(X,X) = -\frac{1}{4}B(X,X)$ [70], together with $B(X,X) = 2h^\vee\,\mathrm{Tr}(\rho(X)^2)/\ell_\rho$. Lower Ricci bounds of the same kind, for special holonomy and for Riemannian submersions, are studied in [20, 45].

[9] With the computer-assisted bound $h_F \leq 14.420001$ of Theorem 267 in place of $\frac{29}{2}$, every range of Table I grows by the factor $29/28.840002 = 1.0055$. Appendix L derives that bound from a certificate of T. Romanov.

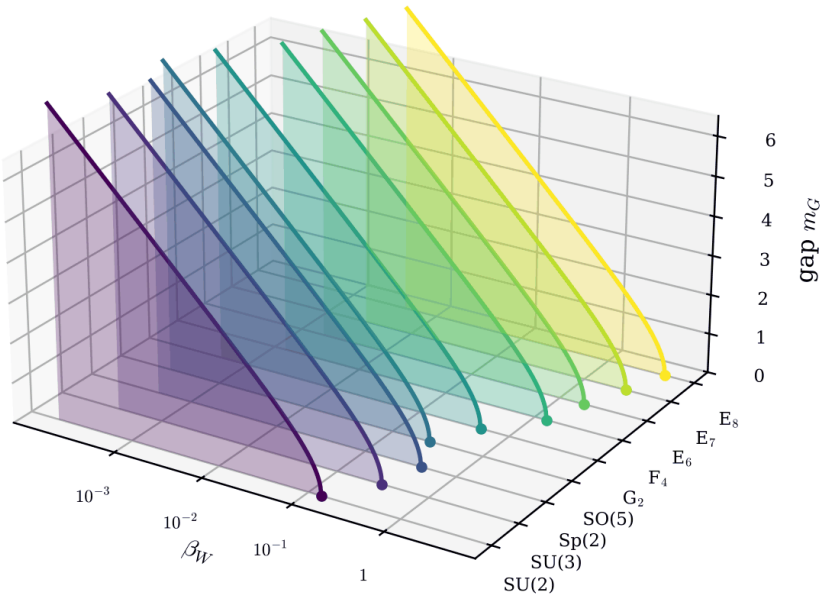



FIG. 1. The gap $m_G(\beta)$ of Theorem 5 as a function of $\beta_W = N^2\beta$ on a logarithmic scale, for nine groups in the representations of Table I. Each curve ends at the edge $\beta_W = 2N\kappa_G/29$ of the range, marked by a dot, where the gap tends to zero; as $\beta_W \to 0$ it grows like $\log(\kappa_G/(6N|\beta|))$. The curves differ only by the factor $N\kappa_G$, since $m_G$ depends on $N|\beta|/\kappa_G$ alone.

observables smoothed by the spatial gradient flow (Proposition 47).

(I2) The flowed observables, renormalised by one condition at a single physical separation, have reflection positive continuum limits along subsequences, and such a limit is nontrivial exactly when one ratio of lattice two-point functions stays away from zero (Theorem 48); that ratio is bounded above and below by ratios of correlations of one slice field at fixed physical separations (Proposition 51).

(I3) Tree bounds on the truncated correlations give all the higher Schwinger functions as reflection positive distributions, imply the ratio bound, and carry the conditional mass gap to every polynomial in the smeared fields (Theorem 57); for a smooth packet they give the ratio with an explicit rate and a reference vector of finite energy (Proposition 61).

*What is open.* The inputs of (I1)–(I3), namely the cube condition along the tuning, the slice ratio in four dimensions and the tree bounds, are not proved, and so the nontrivial existence of the continuum limit and the gap in physical units remain open. Figure 3 shows how the proved statements, the implications and the open inputs fit together.

Sec. XI measures the open inputs. The cube condition holds at strong coupling, fails at every fixed cube size at weak coupling, forces the gauge-fixed variables to be disordered on the scale of the cube, and cannot be reached from estimates on single lines of links; for SU($N$) it needs cubes of side at least of order $(\beta_W/\log\beta_W)^{1/4}$ (Propositions 84 and 85). For a Gaussian field it holds at a fixed physical scale when the field has a mass and fails exactly, at every size, when it is massless (Proposition 63); it also fails for the quadratic approximation of the forest gauge itself (Proposition 64), and of every gauge fixed on an invariant forest with infinite trees (Proposition 65). The order-two cases of the tree bounds force the lattice gap below a fixed multiple of the lattice spacing, so no strong coupling expansion gives them (Proposition 60). The slice ratio holds in two dimensions, where it is the exact solution of lattice gauge theory (Remark 54), and fails for a field whose mass is fixed in lattice units; for the free field, conditioning an energy packet on averages over blocks leaves a fixed fraction of its ordinary variance at every lattice spacing, so a reduction to block averages has to be measured in the reflected norm (Proposition 55).

*Block averages.* Sec. XII poses the cube condition for block averages in a gauge whose fixed links are trees inside blocks of side two. Its quadratic approximation then holds at every coupling once the cube is large (Theorem 98), and integrating out one scale reduces it to a cube condition for the averages, which carry the coarse quadratic form at every scale up to a factor between 1 and 256 that does not depend on the scale, with an exponentially decaying kernel whose constants do not depend on the scale either, and the constrained cube condition holds at every scale with one cube side (Corollary 106, Proposition 99, Theorems 102, 104 and 108); the averages along the lines of the forest gauge, by contrast, leave one class of lines invisible and fail at every size (Proposition 91). For the group-valued links, the straight path between block roots gives a coarse link whose fibres satisfy a coercivity bound for every compact group and every field (Theorem 111), but this map does not compose across scales (Proposition 114), and for every com-

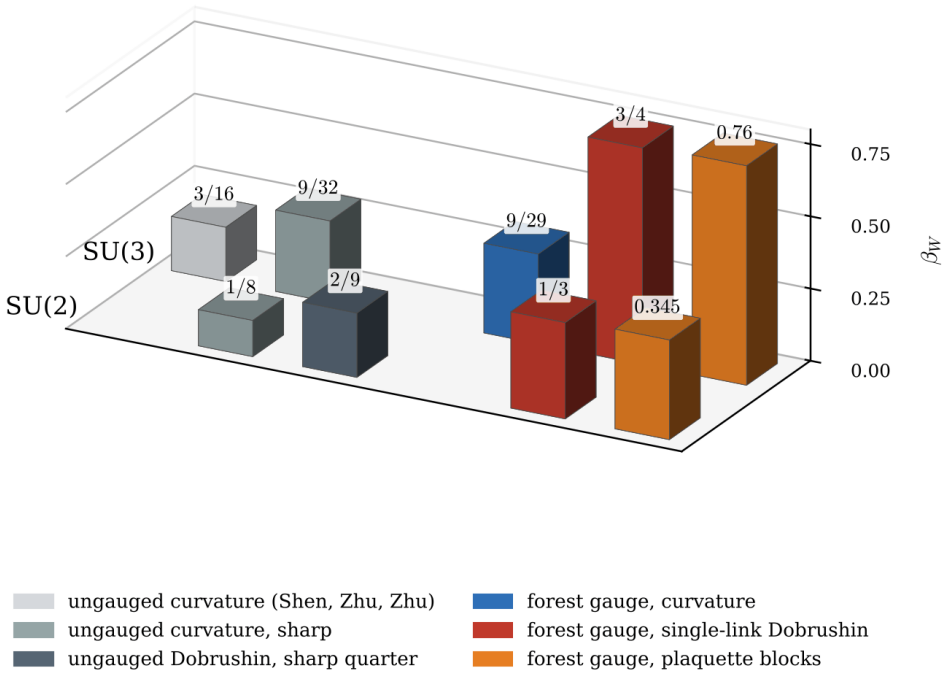



FIG. 2. Ranges of Table II. The height of each bar is the largest inverse coupling $\beta_W$ covered by the method. Grey bars are methods for the ungauged measure; blue, red and orange bars use the forest gauge. For SU(2) the red bar ends at the ceiling of Theorem 2 and the orange bar, plaquette blocks, passes it (Theorem 3); for SU(3) the red bar is Theorem 6 and the orange bar, plaquette blocks, gives a larger gap there and passes it (Theorem 7).

pact connected group no continuous covariant block map treats the positions of a block alike (Proposition 115). A cut-off of Bałaban's averaging operation is continuous, covariant and coercive for every field and agrees with the block averages to first order; it breaks the symmetry only at large fields, as it must, and its constants are those of one step (Theorem 117). On fields with values in a maximal torus whose curvature stays small at every scale it composes, with the energy nonincreasing and the thin paths contracted (Theorem 119), and for every compact group it composes on fields whose curvatures at the successive scales have a small sum (Theorem 123). Under the SU(2) Wilson measure the plaquettes that are not small lie in clusters whose physical size tends to zero in probability, and the cut-off acts only next to them (Theorem 147, Corollaries 148 and 149); such plaquettes are nevertheless present along the one-loop tuning (Proposition 150). The cut-off average composes through these clusters, which spread by at most three spacings of the current scale per step, up to a coarse spacing of order $(\beta_W/\log\beta_W)^{1/4}a$, the order of the cube side that the cube condition needs (Theorem 152 and Corollary 153). Under every exterior, excess action in a box is rare, restricting one link to a fixed set leaves its Dobrushin response bounded, and one-point rarity of a bad set gives approximate uniqueness of the Gibbs measures but no decay (Lemmas 162 and 163, Proposition 164). The curvature of the block averages costs action independent of the scale (Lemma 165, Proposition 166); the chessboard estimate applies to coarse events and is lost after $O(\log\beta_W)$ scales in the normalisation alone (Proposition 167); a right translation of the Haar measures bounds the tails of the force along any fixed direction by the second variation of the action, at every scale and under every exterior (Theorem 168), which for the abelian theory makes coarse defects rare at the rate $\beta_W x^2$ at every scale (Corollary 169), while for SU(2) that second variation grows like $4^k$ at scale $k$, on the gauge orbit of the vacuum, and the same happens when the direction is transported along a forest, which keeps the Haar measures but leaves the residual gauge freedom of the blocks (Propositions 170, 171); this locates the estimate that the cube condition still needs. In a box whose boundary field has small plaquettes and either lies in a maximal torus or, for SU(2), has plaquette angles $\rho_s$ with $l^2\rho_s$ below a fixed constant, the fraction of unit cubes with a large plaquette is small with probability exponentially close to one in the volume, uniformly over such boundary fields (Lemmas 154 and 155 and Proposition 156). The second class allows cube sides up to a constant times $(\beta_W/\log\beta_W)^{1/4}$ at the level of Corollary 153, which is the scale $a_{k_*}$ up to a constant factor; for larger boxes with general boundary fields this conditional estimate is open (Remark 157). On the fibres of the cut-off average, at a background whose plaquettes are at most $\kappa \le 10^{-8}$, the gap $\frac{1}{52}$ of the averages[10] persists up to a factor $1+O(\kappa)$, and the Hessian of the action along the fibre is at least $\lambda(\frac{1}{26}-c\kappa)$, however

[10] By the certificate of Proposition 95 this gap is at least $\frac{537}{5000} = 0.1074$, within 0.09 per cent of the best constant on $\mathbb{Z}^4$, a floating-point value.

far the links are from $I$ (Theorem 125). On a chart of the fibre with convex domain the constrained law is log-concave for large $\lambda$, coarea factor included, and satisfies a Brascamp–Lieb inequality with constant $60/\lambda$ (Theorem 127); on a chart whose domain is a product in the coordinates of the links other than the path links, the covariances of local observables decay exponentially at a rate that depends only on $G$ (Theorem 131). Off the chart the fibre has only points with a large plaquette, and under the SU(2) Wilson measure their mass is small on most fibres (Theorem 134). Along its private links the cut-off average is a submersion with a constant that depends on nothing (Lemma 135), and its law under Haar measure on them has a bounded density, so a Peierls argument inside the fibre bounds that mass on every fibre through a field whose plaquettes near the blocks are small (Proposition 138). On every fibre, whatever the boundary links, the same argument concentrates the law within any $t$ of the least action of the fields whose averaging sites are near $I$ (Proposition 141); where that action is not small the decay is open (Remark 145). Whether some block map is coercive for all fields and composes across the remaining scales is open.

*Plan.* Sec. II constructs the line field. Sec. III shows that a Dobrushin bound for the gauge-fixed specification gives the full conclusion of Theorem 1. Secs. IV, V, VI, VII and VIII prove Theorems 1–7, and Sec. IX measures what is left. Sec. X proves (I1)–(I3), Sec. XI states and measures their open inputs, and Sec. XII poses the cube condition for block averages, composes the cut-off average across scales at small fields and locates the large fields under the Wilson measure. Sec. XIII proves what a labelling of the blocks into good and bad gives for the cube condition, where the natural labels fail, the explicit coupling past which the bad islands of one scale are rarer than their entropy (Theorem 199), the fold across which the path-link law of a bad pair jumps (Proposition 180) and the bounded response of its site (Theorem 183), the factorisation of two bad islands through one kernel (Theorem 207) and the exponential decay of their comparison in the width of the good layer between them (Theorem 209, Corollary 210), and what the single-scale and the multiscale parts of (D) still need. The appendices contain the residue table of the line field, the comparison lemma and its version for blocks, the blocks of the local Hessian inequality, the details of the certified computation, the reconstruction of the transfer matrix with the equivalence of a gap and exponential clustering (Appendix F), the facts on spectral measures and on the character expansion that the continuum step uses (Appendices G and H), and a complete proof of the sharp quarter for the single-link law of SU(2) (Appendices I–K), so that the paper does not depend on our earlier one for any step.

## II. THE LINE FIELD

Write $e_0,\dots,e_3$ for the unit vectors of $\mathbb{Z}^4$ and $(x,\mu)$ for the link from $x$ to $x+e_\mu$. A *direction field* is a map $\sigma:\mathbb{Z}^4\to\{0,1,2,3\}$; its links are $F=\{(x,\sigma(x)):x\in\mathbb{Z}^4\}$. At each site put $\pi_x(\mu)=\sigma(x+e_\mu)$. The plaquette at $x$ in the plane $\{\mu,\nu\}$ has the links $(x,\mu)$, $(x+e_\mu,\nu)$, $(x+e_\nu,\mu)$, $(x,\nu)$, so it contains

$$[\sigma(x)\in\{\mu,\nu\}]+[\pi_x(\mu)=\nu]+[\pi_x(\nu)=\mu] \tag{1}$$

links of $F$.

**Lemma 8.** *Every plaquette of $\mathbb{Z}^4$ contains exactly one link of $F$ if and only if, at every site $x$, $\pi_x$ fixes $\sigma(x)$ and permutes the other three directions cyclically.*

*Proof.* Fix $x$, let $s=\sigma(x)$ and let $R$ be the set of the other three directions. For a plane $\{s,\nu\}$ the count (1) is $1+[\pi_x(s)=\nu]+[\pi_x(\nu)=s]$; it equals one for all $\nu\in R$ exactly when $\pi_x(s)=s$ and $\pi_x(\nu)\neq s$ for $\nu\in R$, that is, when $\pi_x$ fixes $s$ and maps $R$ into $R$. For a plane $\{\mu,\nu\}\subset R$ the count is $[\pi_x(\mu)=\nu]+[\pi_x(\nu)=\mu]$. Draw an arrow $\mu\to\pi_x(\mu)$ for every $\mu\in R$ with $\pi_x(\mu)\in R\setminus\{\mu\}$. Each vertex of $R$ has at most one outgoing arrow, and the condition says that each of the three pairs of $R$ carries exactly one arrow. Three arrows on three vertices with out-degree at most one give out-degree exactly one everywhere, and a tournament on three vertices with all out-degrees one is a directed three-cycle. □

From now on $\sigma$ is the field of period two given in Table III of Appendix A, read from $x$ modulo two. The table lists $\pi_x$ at each of the sixteen residue classes; in each row $\pi_x$ fixes $\sigma(x)$ and is a three-cycle on the rest, so every plaquette contains exactly one link of $F$ by Lemma 8.[11]

**Lemma 9.** (i) *$F$ is a disjoint union of bi-infinite straight lines, one through every site: every site has exactly one outgoing and one incoming link of $F$, and both point in the direction $\sigma(x)$.*

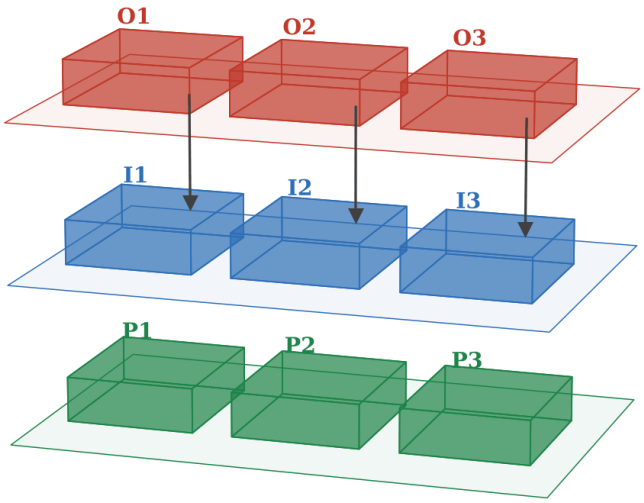
(ii) *For every box $\Lambda$ (a product of four finite intervals, with the links and plaquettes it contains), $F\cap E(\Lambda)$ is a disjoint union of segments, hence a forest.*

(iii) *A link $e=(x,\mu)\notin F$ lies in six plaquettes. Four of them contain a link of $F$ that shares an end point with $e$, and these lie in the planes $\{\mu,\sigma(x)\}$ and $\{\mu,\pi_x(\mu)\}$. In the other two, which lie in the remaining plane $\{\mu,\rho\}$, the link of $F$ is opposite to $e$.*

(iv) *Every link outside $F$ shares a plaquette with exactly twelve other links outside $F$, two in each of its plaquettes.*

*Proof.* (i) Since $\pi_x$ fixes $\sigma(x)$, $\sigma(x+e_{\sigma(x)})=\sigma(x)$. Because $\sigma$ has period two, $x-e_\mu\equiv x+e_\mu$, so also $\sigma(x-e_{\sigma(x)})=\sigma(x)$, and the link $(x-e_{\sigma(x)},\sigma(x))$ is in $F$. Conversely, if $(x-e_\nu,\nu)\in F$, then $\sigma(x-e_\nu)=\nu$, and straightness at $x-e_\nu$ gives $\sigma(x)=\nu$. So the incoming link is unique and continues the outgoing one.

[11] There are exactly eight period-two direction fields with this property, all equivalent to $\sigma$ under the symmetries of the lattice. A field with one fixed link at every site and one in every plaquette exists only in four dimensions: a link lies in $2(d-1)$ plaquettes and a site carries $d(d-1)/2$ of them, and $2(d-1)=d(d-1)/2$ forces $d=4$. The statements of this section are also checked in the Lean certificate of the archive.




**open inputs**

O1 cube estimate (D) along the tuning

O2 slice ratio in four dimensions

O3 tree bounds (T1)-(T3)

**proved implications**

I1 (D) gives a gap for every limit

I2 ratio iff nontrivial flowed limit

I3 tree bounds give the ratio

**proved at fixed spacing**

P1 gap for every simple group

P2 SU(2), SU(3) with blocks

P3 (D) at strong coupling

arrows: an open input and the implication that consumes it

FIG. 3. The logical structure of the paper. Green: theorems at fixed lattice spacing (P1 the gap for every compact simple group, Theorem 5, P2 the plaquette blocks for SU(2) and SU(3), P3 the cube condition at strong coupling). Blue: the proved implications (I1)–(I3) of the continuum step. Red: their inputs, none of which is proved along the continuum tuning. Each arrow joins an open input to the implication that consumes it; no implication feeds its own input.

(ii) A line meets a box in a segment, and distinct lines are disjoint.

(iii) The plaquettes of $e$ are, for each $\nu \neq \mu$, the one at $x$ and the one at $x - e_\nu$ in the plane $\{\mu,\nu\}$. A link of $F$ sharing an end point with $e$ is the outgoing or the incoming link of $F$ at $x$ or at $x+e_\mu$. At $x$ both point in the direction $s = \sigma(x) \neq \mu$, and they lie in the plaquettes at $x$ and at $x - e_s$ in the plane $\{\mu,s\}$. At $x+e_\mu$ both point in the direction $t = \pi_x(\mu)$, which lies in $R \setminus \{\mu\}$ because $\pi_x$ is a three-cycle on $R$, and they lie in the plaquettes at $x$ and at $x - e_t$ in the plane $\{\mu,t\}$. These four plaquettes are distinct because $s \neq t$. Each plaquette contains exactly one link of $F$, so in the two plaquettes of the third plane $\{\mu,\rho\}$ the link of $F$ is a link of direction $\mu$ or $\rho$ not touching $e$; the links of direction $\rho$ in such a plaquette touch $e$, so it is the opposite link.

(iv) Each of the six plaquettes contains $e$, one link of $F$ and two other links, and two distinct links share at most one plaquette. □

## III. GAUGE FIXING AND THE FREE-BOUNDARY STATE

Let $G \subset \mathrm{U}(N)$ be compact with Haar measure $dU$, and let $\Lambda$ be a box. The Wilson measure with free boundary is

$$W_\Lambda(dQ) = \frac{1}{Z_\Lambda} \exp\Big(N\beta \sum_{p \in P(\Lambda)} \mathrm{Re}\,\mathrm{Tr}\, Q_p\Big) \prod_{e \in E(\Lambda)} dQ_e .$$

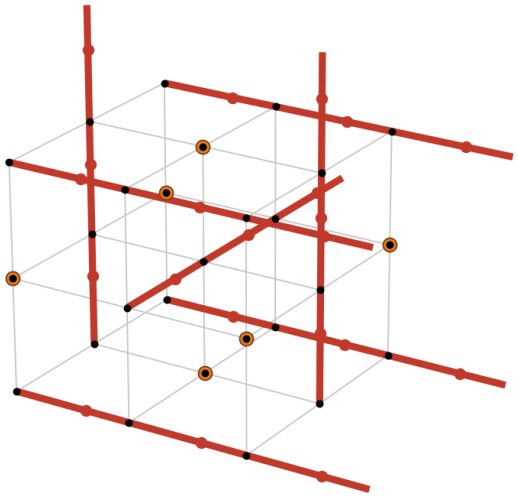
The measure is invariant under the gauge transformations $Q_{(x,y)} \mapsto g_x Q_{(x,y)} g_y^{-1}$ with $g \in G^\Lambda$. Call the links outside $F$ *active* and write $A$ for their set. For a function $O$ of the links, $O^F$ is $O$ with every link of $F$ set to the identity.

**Lemma 10** (Gauge fixing)**.** *Let $\mu_\Lambda^F$ be the probability measure on $G^{A \cap E(\Lambda)}$ whose density with respect to Haar measure is proportional to $\exp(N\beta \sum_{p \in P(\Lambda)} \mathrm{Re}\,\mathrm{Tr}\, Q_p^F)$. For every gauge-invariant $O$ on $G^{E(\Lambda)}$,*

$$W_\Lambda(O) = \mu_\Lambda^F(O^F).$$

*Proof.* Each segment of $F \cap E(\Lambda)$ is a path $x_0, x_1, \dots, x_k$ along one direction. Given the variables on $F$, put $g_{x_0} = 1$ and $g_{x_{i+1}} = g_{x_i} Q_{(x_i,x_{i+1})}$ on every segment. The transformed field has $Q_e^g = 1$ on $F \cap E(\Lambda)$, and for fixed values on $F$ the map $Q_A \mapsto Q_A^g$ is a product of left and right translations, which preserves Haar

(a) the line field in the slice $x_3 = 0$



(b) one plaquette in the gauge

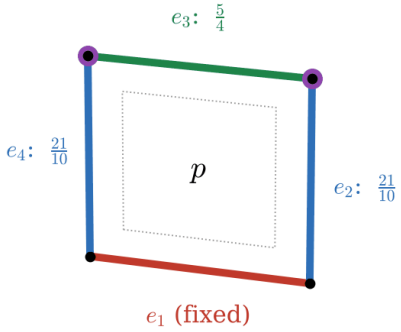



FIG. 4. (a) The line field in the slice $x_3 = 0$ of a box of side three. Thick red segments are the links of $F$ in the directions $0, 1, 2$, with a dot marking the direction of travel; an orange ring marks a site whose line leaves the slice in direction 3. Every line is straight (Lemma 9(i)), and the segments that leave the box are its boundary stubs. (b) One plaquette with its fixed link $e_1$. The weights are those of the local inequality of Lemma 20 for the three free links: $\frac{21}{10}$ on the two links adjacent to $e_1$ and $\frac{5}{4}$ on the opposite one. The two purple corners are the ones that enter the corner form (7) once $Y_1 = 0$.

measure. Gauge invariance of the action and of $O$ and the change of variables give the claim. □

No leftover holonomy appears because $F \cap E(\Lambda)$ is a forest. The measures $\mu_\Lambda^F$ are finite-volume versions of the specification $\gamma^F$ on $G^A$: for $e \in A$, $\gamma_e^F$ resamples $Q_e$ from the law with density proportional to $\exp(N\beta \sum_{p \ni e} \operatorname{Re} \operatorname{Tr} Q_p^F)$ given all other active links. No infinite-volume gauge transformation is performed; $\gamma^F$ is a specification in its own right. Fix a metric $\delta$ on $G$ of diameter $D$ that is bi-invariant (for $\mathrm{SU}(2)$ the chordal metric on $S^3$, $D = 2$; for $\mathrm{SU}(N)$ the Hilbert–Schmidt distance). The Kantorovich coefficient of $e'$ on $e$ is

$$C_{ee'} = \sup \frac{W_1\big(\gamma_e^F(\cdot \mid Q), \gamma_e^F(\cdot \mid Q')\big)}{\delta(Q_{e'}, Q'_{e'})},$$

the supremum over pairs of configurations that differ only at $e'$. It vanishes unless $e$ and $e'$ share a plaquette.

**Proposition 11.** *Suppose that every row and every column of $(C_{ee'})$ sums to at most $\alpha < 1$. Then:*

(i) *$\gamma^F$ has exactly one Gibbs measure $\mu_F$;*

(ii) *for every gauge-invariant local $O$ the limit $\omega(O) = \lim W_\Lambda(O)$ exists along every sequence of boxes exhausting $\mathbb{Z}^4$ and equals $\mu_F(O^F)$;*

(iii) *$\omega$ is invariant under translations and the hypercubic group;*

(iv) *if $\beta \geq 0$, $\omega$ is reflection positive for reflections in hyperplanes through sites and in hyperplanes bisecting links;*

(v) *for gauge-invariant local $O, O'$ with $O$ Lipschitz in each link, $|\operatorname{Cov}_\omega(O, O')| \leq C_{O,O'} e^{-\lambda \operatorname{dist}(O,O')}$ for every $\lambda < -\log \alpha$;*

(vi) *if $\beta \geq 0$, the transfer matrix of $\omega$ has spectral gap at least $-\log \alpha$.*

*Proof.* (i) Gibbs measures exist by compactness, and the comparison lemma of Appendix B with $S = \emptyset$ shows that any two coincide.

(ii) Extend $\mu_\Lambda^F$ by Haar measure on the active links outside $\Lambda$, and let $S$ be the set of active links with a plaquette outside $P(\Lambda)$; this includes every active link outside $E(\Lambda)$. For an active link $e \notin S$ all six plaquettes of $e$ lie in $P(\Lambda)$, so the conditional law of $Q_e$ under the extended measure is $\gamma_e^F$. Let $T$ be the support of $O^F$ and $\lambda < -\log \alpha$. The comparison lemma with weights $e^{-\lambda \operatorname{dist}(\cdot, T)}$ gives

$$|\mu_\Lambda^F(O^F) - \mu_F(O^F)| \leq L \frac{\theta D}{1 - \theta} \sum_{z \in S} e^{-\lambda \operatorname{dist}(z,T)},$$

with $\theta = \alpha e^\lambda < 1$ and $L$ a Lipschitz constant of $O^F$; the sum is at most $C_T R^3 e^{-\lambda R}$ when $R$ is the distance from $T$ to the complement of $\Lambda$. For $O$ continuous but not Lipschitz, approximate uniformly. Lemma 10 gives $W_\Lambda(O) = \mu_\Lambda^F(O^F)$.

(iii) $W_\Lambda(\tau_v O) = W_{\Lambda - v}(O)$, and the boxes $\Lambda - v$ also exhaust $\mathbb{Z}^4$; by (ii) the limit does not depend on the sequence. The line field is a device for computing $W_\Lambda$ and is never shifted. The same argument applies to the hypercubic group.

(iv) For $\beta \geq 0$ the measure $W_\Lambda$ is reflection positive for every box symmetric under the reflection [75], and reflection positivity passes to the limit along such boxes, which is $\omega$ by (ii).

(v) $\mathrm{Cov}_\omega(O,O') = \mathrm{Cov}_{\mu_F}(O^F, O'^F)$, and setting the links of $F$ to the identity keeps the Lipschitz property. The bound is Corollary 242.

(vi) Let $\vartheta$ be a time reflection and $\tau_t$ a time translation. For gauge-invariant local $O$ that are Lipschitz in each link, (v) gives $|\omega(\vartheta(\bar O)\tau_t O) - |\omega(O)|^2| \le C_O e^{-\lambda t}$. On the support of $O$ these functions are dense in the gauge-invariant part of $L^2(\omega)$: gauge-invariant functions are functions on the compact quotient of $G^E$ by the gauge group, Lipschitz functions for the quotient metric are dense in $L^2$ of every finite measure on it, and they lift to gauge-invariant Lipschitz functions. By Proposition 245 the spectrum of the transfer matrix on the complement of the vacuum lies in $[0, e^{-\lambda}]$, for every $\lambda < -\log\alpha$. □

The proposition replaces the torus limits and the uniqueness of the ungauged Gibbs measure used in [16, Thm. F] by the free-boundary limit, which has all the properties the reconstruction of the transfer matrix needs.[12]

## IV. SU(2) UP TO ONE THIRD

*Proof of Theorem 1.* Identify $\mathrm{SU}(2)$ with the unit quaternions and use the chordal metric. For an active link $e$ and a plaquette $p \ni e$, the staple $s_p$ is the product of the other three link variables in the order that makes $\mathrm{Re}(Q_e s_p) = \frac12 \mathrm{Tr}\, Q_p$, with links of $F$ equal to one. Since $N\beta\,\mathrm{Re}\,\mathrm{Tr}\, Q_p = 4\beta\langle Q_e, \bar s_p\rangle$ in $\mathbb{R}^4$, the conditional law of $Q_e$ is the von Mises–Fisher law $\nu_b(dx) \propto e^{\langle x,b\rangle}dx$ on $S^3$ with $b = 4\beta\sum_{p\ni e}\bar s_p$. Changing a partner $e'$ by $\delta$ in the chordal metric changes one staple by an isometric image of that displacement, so $b$ moves by $4|\beta|\delta$, and the sharp quarter (Theorem 254), $W_1(\nu_b,\nu_{b'}) \le \frac14|b-b'|$, gives $C_{ee'} \le |\beta|$. By Lemma 9(iv) every row and column has twelve nonzero entries, so $\alpha \le 12|\beta| = 3|\beta_W|$. Proposition 11 applies for $|\beta_W| < \frac13$. □

*Proof of Theorem 2.* For the line field, every staple of an active link contains two active links, so the six staples can be chosen with sum zero while the staple through a given partner $e'$ moves freely. At $b = 0$ the derivative of $\nu_b$ in the direction $h$ has Kantorovich norm $\frac14|h|$ by Lemma 253(i), so $C_{ee'} = |\beta|$ exactly, and every row sum is $12|\beta| = 3|\beta_W|$.

Now let $F'$ be any forest and $C'$ the matrix of the statement. Suppose a positive weight $w$ satisfies $C'w \le \alpha w$. For a box $\Lambda_n$ of side $n$ let $C_n$ be the principal submatrix on the active links of $E(\Lambda_n)$; then $C_n w \le \alpha w$ on these links, and the Collatz–Wielandt inequality for nonnegative matrices gives $\rho(C_n) \le \alpha$. Since $C_n$ is symmetric, $\rho(C_n)$ is at least the mean row sum. The box has $4n^4 - O(n^3)$ links and $n^4$ sites, and a forest has fewer links than sites, so the number $a$ of active links is at least $3n^4 - O(n^3)$. Let $a_p$ be the number of active links of the plaquette $p$; the box has $P = 6n^4 - O(n^3)$ plaquettes and $\sum_p a_p = 6a - O(n^3)$. Two links share at most one plaquette, so the row sums add up to $|\beta|\sum_p a_p(a_p-1)$, and by convexity

$$\frac{|\beta|}{a}\sum_p a_p(a_p-1) \ge \frac{|\beta|P}{a}\,\bar a(\bar a-1) = 6|\beta|(\bar a-1) - o(1),$$

with $\bar a = \sum_p a_p/P = 6a/P - o(1) \ge 3 - o(1)$. Hence $\alpha \ge 12|\beta| - o(1)$ for every $n$, and $\alpha \ge 12|\beta| = 3|\beta_W|$. □

Without gauge fixing the same count gives eighteen partners and the ceiling $\beta_W = \frac29$ of [16, Prop. 13]; the temporal gauge, a family of lines in one direction, leaves fourteen. The line field is optimal among forests: twelve is the mean number of partners forced by the number of free links, and it is attained at every link.

## V. SU(2) PAST ONE THIRD: PLAQUETTE BLOCKS

Throughout this section $G = \mathrm{SU}(2)$ is the group of unit quaternions with the chordal metric $|x-y|$, $\bar x$ is the conjugate, $\langle x,y\rangle = \mathrm{Re}(x\bar y)$ is the inner product of $\mathbb{R}^4$, and $dx$ is the normalised Haar measure on $S^3$. Left and right multiplication by unit quaternions and conjugation are isometries, and $\mathrm{Re}(xyz) = \mathrm{Re}(yzx) = \langle y, \bar z\bar x\rangle$. Put

$$Z_0(c) = \int_{S^3} e^{\langle c,x\rangle}dx = \sum_{j\ge0}\frac{(|c|^2/4)^j}{j!\,(j+1)!}, \qquad c\in\mathbb{R}^4,$$

let $\nu_c(dx) \propto e^{\langle c,x\rangle}dx$ be the von Mises–Fisher law, and write $g = \log Z_0$, so that $\nabla g(c)$ is the mean and $\mathrm{Hess}\, g(c)$ the covariance of $\nu_c$.

### A. The blocks

Let $\mathscr{B}$ be the set of plaquettes at $x$ in the plane $\{0,1\}$ with $x_0 \equiv x_1$ and in the plane $\{2,3\}$ with $x_2 \equiv x_3$, congruences modulo two (Table IV of Appendix C lists them modulo $2\mathbb{Z}^4$, and Fig. 5 draws them). A *block plaquette* is a plaquette of $\mathscr{B}$, and its *block* is the set of its three active links.

**Lemma 12.** *The set $\mathscr{B}$ is invariant under translations by $2\mathbb{Z}^4$ and:*

(i) *the blocks partition the active links;*

(ii) *every plaquette $p \notin \mathscr{B}$ meets three distinct blocks, one in each of its active links;*

(iii) *every active link lies in exactly five plaquettes outside $\mathscr{B}$, so it shares a plaquette outside $\mathscr{B}$ with exactly ten active links, none of them in its own block;*

(iv) *if an active link $j$ and a link of a block $B$ lie in a common plaquette outside $\mathscr{B}$, then every link of $B$ has its midpoint within distance 2 of the midpoint of $j$.*

[12] On a torus the line field closes up into cycles, and the holonomies along them cannot be removed by a gauge transformation; the argument never uses a torus. The state $\omega$ is defined only on gauge-invariant observables, which is all that the Osterwalder–Schrader reconstruction [73–75] uses.

*Proof.* Invariance under $2\mathbb{Z}^4$ is clear from the congruences. We use one fact throughout: two distinct links lie in at most one common plaquette, since they span it.

Every link lies in exactly one plaquette of $\mathscr{B}$. Indeed, a link $(y,\mu)$ with $\mu\in\{0,1\}$ lies in no plaquette of the plane $\{2,3\}$, and in exactly two of the plane $\{0,1\}$, those at $y$ and at $y-e_\nu$, where $\nu$ is the other element of $\{0,1\}$. The base points $y$ and $y-e_\nu$ differ only in the coordinate $\nu$, so $y_0-y_1$ has opposite parities at the two, and exactly one of them satisfies $x_0\equiv x_1$. The directions $2,3$ are treated in the same way. Every plaquette of $\mathbb{Z}^4$ contains exactly one link of $F$ (Lemma 8), so a block plaquette contains three active links, and every active link lies in exactly one block. This is (i).

(ii) Let $p\notin\mathscr{B}$ have active links $e_1,e_2,e_3$, and let $q_i$ be the block plaquette of $e_i$. Then $q_i\neq p$, so $q_i\cap p=\{e_i\}$ by the fact above. If $q_i=q_k$ with $i\neq k$, then $e_i,e_k$ lie in the two distinct plaquettes $p$ and $q_i$, which is impossible. So the three blocks are distinct, and each meets $p$ in one link.

(iii) An active link $e$ lies in six plaquettes (Lemma 9), and by (i) exactly one of them is in $\mathscr{B}$. Each of the other five contains two further active links. These ten links are distinct, because a link other than $e$ shares at most one plaquette with $e$, and none lies in the block of $e$, because a link $f$ of that block sharing the plaquette $p\notin\mathscr{B}$ with $e$ would put $e,f$ in the two distinct plaquettes $p$ and the block plaquette of $e$.

(iv) Let $f\in B$ share the plaquette $p\notin\mathscr{B}$ with $j$. Two links of one unit square have midpoints at distance $2^{-1/2}$ or 1, so the midpoint of $j$ is within distance 1 of that of $f$, and the midpoint of $f$ is within distance 1 of that of every link of its block plaquette. The triangle inequality gives the bound 2. □

Fix a block plaquette $p$ with active links $e_1,e_2,e_3$. Reading the plaquette product cyclically and deleting its fixed link, $\frac12\operatorname{Tr}Q_p=\operatorname{Re}(y_1y_2y_3)$, where $y_i$ is $Q_{e_i}$ or $Q_{e_i}^{-1}$. Every other plaquette containing $e_i$ meets the block only in $e_i$ by Lemma 12(ii) and contributes $4\beta\langle y_i,\bar s\rangle$ to the exponent, where $s$ is the staple of $y_i$ in it. Hence the conditional law of the block given all other active links is

$$\mu_b(dy)\propto\exp\Big(\sum_{i=1}^3\langle y_i,b_i\rangle+4\beta\operatorname{Re}(y_1y_2y_3)\Big)\,dy_1\,dy_2\,dy_3,\quad(2)$$

with $b_i$ the sum of $4\beta\bar s$ over the five staples, so that $|b_i|\le 20|\beta|$. Inversion is an isometry of $S^3$ and preserves Haar measure, so we may work with $y$ instead of $Q$. On $(S^3)^3$ we use the metric $d(y,y')=\sum_i|y_i-y_i'|$.

### B. The sensitivity of one block

Let $\lambda_*$ and $\tau$ be a lower and an upper bound for the eigenvalues of $\operatorname{Hess}g(c)$ on the ball $|c|\le 24|\beta|$, and $M$ an upper bound there for the norm of the third derivative $D^3g(c)$ as a trilinear form. Put

$$\begin{aligned}&w=8\beta^2(\tau-\lambda_*),\qquad s=\tfrac14+(e^w-1)e^w\sqrt{\tau/3},\\&C=4|\beta|sA+\tfrac{320}{3}|\beta|^3Me^w,\qquad K=\frac{(1+|\beta|)s}{1-C},\end{aligned}\quad(3)$$

assuming $C<1$, where $A$ is an upper bound for $|\nabla g(c)|$ on the ball $|c|\le 20|\beta|$.

**Lemma 13.** *Let $\nu$ be a probability measure and $\lambda=e^R\nu/\nu(e^R)$ with $\operatorname{osc}R\le\omega$. For bounded $f,h$,*

$$\operatorname{Cov}_\lambda(f,h)\le\operatorname{Cov}_\nu(f,h)+(e^\omega-1)e^\omega\operatorname{sd}_\nu(f)\operatorname{sd}_\nu(h).$$

*Proof.* We may assume $0\le R\le\omega$. The density $\rho=e^R/\nu(e^R)$ lies in $[e^{-\omega},e^\omega]$, so $\delta=\|\rho-1\|_\infty\le e^\omega-1$. With $\tilde f=f-\nu f$ and $\tilde h=h-\nu h$,

$$\begin{aligned}\operatorname{Cov}_\lambda(f,h)-\operatorname{Cov}_\nu(f,h)&=\nu\big((\rho-1)\tilde f\tilde h\big)\\&\quad-\nu\big((\rho-1)\tilde f\big)\,\nu\big((\rho-1)\tilde h\big),\end{aligned}$$

and the right side is at most $(\delta+\delta^2)\operatorname{sd}_\nu(f)\operatorname{sd}_\nu(h)$ by the Cauchy–Schwarz inequality. □

**Lemma 14.** *Let $\beta\ge0$, $a,c_0\in\mathbb{R}^4$ with $|a|,|c_0|\le 20\beta$, and*

$$\kappa(dx)\propto\exp\big(\langle x,a\rangle+g(c_0+4\beta\bar x)\big)dx\quad\text{on } S^3.$$

*For every 1-Lipschitz $f$ and every $h\in\mathbb{R}^4$:*

(i) $\operatorname{Cov}_\kappa(f,\langle x,h\rangle)\le s|h|$;

(ii) *if $\kappa'$ is the law with $c_0'$ in place of $c_0$ and $|c_0'|=|c_0|=\rho$, then $|\kappa(f)-\kappa'(f)|\le(8\beta s\varphi'(\rho^2)+\frac{16}{3}\beta^2Me^w)|c_0-c_0'|$.*

*Proof.* (i) Taylor's formula at $c_0$ gives $g(c_0+4\beta u)=g(c_0)+4\beta\langle\nabla g(c_0),u\rangle+R(u)$ with $R(u)=8\beta^2\operatorname{Hess}g(\xi)(u,u)$ for some $\xi$ on the segment, $|\xi|\le 24\beta$. For $u=\bar x$ of norm one, $R(\bar x)\in[8\beta^2\lambda_*,8\beta^2\tau]$, so $\operatorname{osc}R\le w$. Since $\langle\nabla g(c_0),\bar x\rangle=\langle x,\overline{\nabla g(c_0)}\rangle$, $\kappa$ is the tilt $e^R\nu/\nu(e^R)$ of $\nu=\nu_c$ with $c=a+4\beta\overline{\nabla g(c_0)}$. The sharp quarter, Theorem 254, gives $\operatorname{Cov}_\nu(f,\langle x,h\rangle)=\frac{d}{dt}\nu_{c+th}(f)\le\frac14|h|$. A function that is 1-Lipschitz for the chordal metric is 1-Lipschitz for the geodesic one, so $|\nabla f|\le1$ almost everywhere and the Poincaré inequality of Lemma 251 gives $\operatorname{sd}_\nu(f)\le1/\sqrt3$; and $\operatorname{sd}_\nu\langle x,h\rangle\le\sqrt\tau|h|$ because $|c|\le|a|+4\beta\le24\beta$. Lemma 13 gives (i). The same argument applies to every law $e^{R'}\nu_{c'}/\nu_{c'}(e^{R'})$ with $|c'|\le24\beta$ and $\operatorname{osc}R'\le w$.

(ii) We interpolate the exponents rather than the field. For $t\in[0,1]$ let $\kappa_t\propto\exp\big(\langle x,a\rangle+(1-t)g(c_0+4\beta\bar x)+tg(c_0'+4\beta\bar x)\big)dx$, so that $\kappa_0=\kappa$, $\kappa_1=\kappa'$ and $\frac{d}{dt}\kappa_t(f)=\operatorname{Cov}_{\kappa_t}(f,D)$ with $D(x)=g(c_0'+4\beta\bar x)-g(c_0+4\beta\bar x)$. Write $g(c+4\beta u)=g(c)+4\beta\langle\nabla g(c),u\rangle+R_c(u)$ as in (i), for $c\in\{c_0,c_0'\}$. Then $\kappa_t$ is the tilt of $\nu_{c_t}$, $c_t=a+4\beta\,\overline{(1-t)\nabla g(c_0)+t\nabla g(c_0')}$, by $R_t=(1-t)R_{c_0}+tR_{c_0'}$, with $\operatorname{osc}R_t\le w$ and $|c_t|\le24\beta$ because $|\nabla g|\le1$; so (i) holds for $\kappa_t$, and the density of $\kappa_t$ with respect to $\nu_{c_t}$ is at most $e^w$. Up to a constant, $D(x)=\langle x,h\rangle+N(\bar x)$ with $h=4\beta\,\overline{\nabla g(c_0')-\nabla g(c_0)}$ and $N=R_{c_0'}-R_{c_0}$. Since $\nabla g(c)=2\varphi'(|c|^2)c$ and $|c_0|=|c_0'|=\rho$, $|h|=8\beta\varphi'(\rho^2)|c_0-c_0'|$, and by (i) the linear part contributes at most $s|h|$. For the rest, $\nabla N(u)=16\beta^2\int_0^1\big(\operatorname{Hess}g(c_0'+4\beta\sigma u)-\operatorname{Hess}g(c_0+4\beta\sigma u)\big)u\,d\sigma$, both arguments in the convex ball $|c|\le24\beta$, so $N$ is $L$-Lipschitz on the unit ball with $L=16\beta^2M|c_0-c_0'|$. With $\nu=\nu_{c_t}$ and the Poincaré inequality of Lemma 251 again, $\operatorname{Var}_{\kappa_t}(N)\le\kappa_t((N-\nu N)^2)\le e^w\operatorname{Var}_\nu(N)\le e^wL^2/3$, and in the same way $\operatorname{Var}_{\kappa_t}(f)\le e^w/3$. So $\operatorname{Cov}_{\kappa_t}(f,N)\le e^wL/3$. Integrating over $t$ gives (ii). □

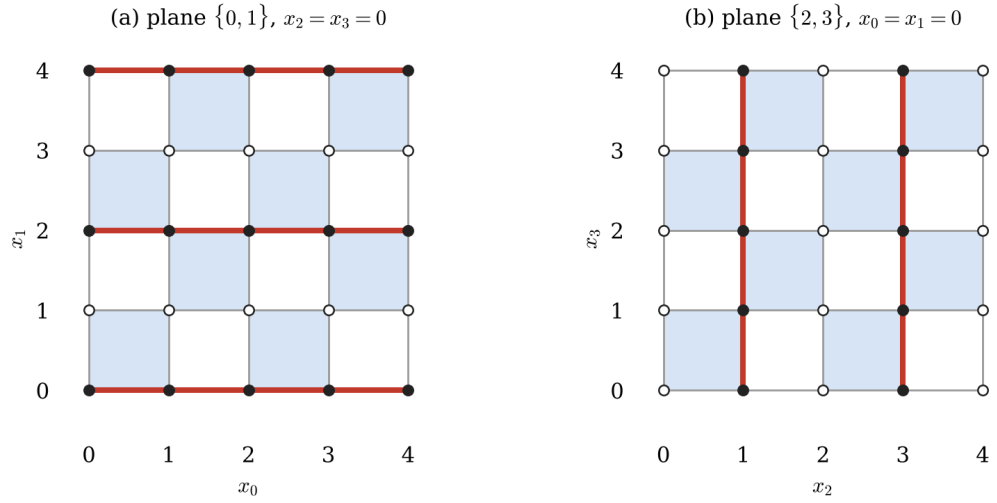



FIG. 5. Block plaquettes, shaded, in (a) a plane $\{0,1\}$ and (b) a plane $\{2,3\}$: the plaquettes at $x$ with $x_0 \equiv x_1$, respectively $x_2 \equiv x_3$, modulo two. Red links belong to the line field $F$; a filled dot marks a site whose line runs in the plane, an open dot one whose line leaves it. The shaded squares form a checkerboard, so every link of the plane borders exactly one of them, and every square contains exactly one red link (Lemma 8); hence each block plaquette carries three active links and the blocks partition the active links (Lemma 12(i)).

**Lemma 15.** *Let $C < 1$. If $b$ and $b'$ satisfy $|b_i|, |b'_i| \le 20|\beta|$ and differ only in one coordinate $b_i$, then*

$$W_1(\mu_b, \mu_{b'}) \le K\,|b_i - b'_i|$$

*in the metric $d$.*

*Proof.* The map $y_1 \mapsto -y_1$ turns $\mu_b$ for $\beta$ into the law for $-\beta$ with $b_1$ replaced by $-b_1$ and is an isometry, so we may take $\beta \ge 0$; the cyclic symmetry of $\mathrm{Re}(y_1y_2y_3)$ lets us take $i = 1$. Let $f$ be 1-Lipschitz for $d$ and $\varepsilon = |b_1 - b'_1|$.

*The middle link.* Given $y_1, y_3$, the law of $y_2$ is $\nu_c$ with $c = b_2 + 4\beta\,\overline{y_3 y_1}$, which does not involve $b_1$. Changing $y_1$ by $\eta$ moves $c$ by $4\beta\eta$, and the sharp quarter moves $\nu_c$ by at most $\beta\eta$ in $W_1$. Hence $F(y_1, y_3) = \mathbb{E}[f \mid y_1, y_3]$ is $(1+\beta)$-Lipschitz in each variable, and $\mu_b(f) = \mu_b^{13}(F)$, where $\mu_b^{13}$ is the law of $(y_1, y_3)$:

$$\mu_b^{13} \propto e^{\langle y_1, b_1\rangle + \langle y_3, b_3\rangle} Z_0\big(b_2 + 4\beta\,\overline{y_3 y_1}\big).$$

Since $|b_2 + 4\beta\overline{y_3y_1}| = |b_2y_3 + 4\beta\bar{y}_1| = |y_1b_2 + 4\beta\bar{y}_3|$, its conditional laws are

$$\kappa_1(dy_1 \mid y_3) \propto e^{\langle y_1, b_1\rangle + g(b_2y_3 + 4\beta\bar{y}_1)}\,dy_1,$$
$$\kappa_3(dy_3 \mid y_1) \propto e^{\langle y_3, b_3\rangle + g(y_1b_2 + 4\beta\bar{y}_3)}\,dy_3,$$

both of the form of Lemma 14, with $|c_0| = |b_2|$.

*Two sites.* Integrating Lemma 14(i) along the segment from $b_1$ to $b'_1$ gives $W_1(\kappa_1, \kappa'_1) \le s\varepsilon$ for the same $y_3$. Moving $y_3$ to $y'_3$ replaces $c_0 = b_2y_3$ by $c'_0 = b_2y'_3$, of the same norm $\rho = |b_2| \le 20\beta$ and with $|c_0 - c'_0| = \rho|y_3 - y'_3|$. Since $\rho \cdot 2\varphi'(\rho^2) = |\nabla g(c_0)| \le A$, Lemma 14(ii) gives $W_1(\kappa_1(\cdot \mid y_3), \kappa_1(\cdot \mid y'_3)) \le (4\beta sA + 20\beta \cdot \frac{16}{3}\beta^2 M e^w)|y_3 - y'_3| = C|y_3 - y'_3|$, and the same for $\kappa_3$, which does not depend on $b_1$. Let $\pi$ be a coupling of $\mu_b^{13}$ and $\mu_{b'}^{13}$ minimising $d_1 + d_3$, $d_k = \mathbb{E}_\pi|y_k - y'_k|$. Replacing $(y_1, y'_1)$ by an optimal coupling of $\kappa_1(\cdot \mid y_3)$ and $\kappa'_1(\cdot \mid y'_3)$ keeps a coupling, so minimality gives $d_1 \le s\varepsilon + Cd_3$; in the same way $d_3 \le Cd_1$. Hence $d_1 \le s\varepsilon/(1-C^2)$ and

$$|\mu_b(f) - \mu_{b'}(f)| \le (1+\beta)(d_1 + d_3) \le \frac{(1+\beta)s}{1-C}\,\varepsilon = K\varepsilon. \quad \square$$

### C. The constants

**Lemma 16.** *Write $g(c) = \varphi(|c|^2)$ and $r = |c|$. For $c \ne 0$ the eigenvalues of $\mathrm{Hess}\,g(c)$ are $2\varphi'(r^2)$, three times, and $2\varphi'(r^2) + 4r^2\varphi''(r^2)$, and the norm of $D^3g(c)$ is*

$$\max\Big(|A + \Gamma|, \tfrac{2}{3}|A|\Big), \qquad A = 12r\varphi''(r^2), \quad \Gamma = 8r^3\varphi'''(r^2).$$

*On the ball $r \le 2.106$ every eigenvalue of $\mathrm{Hess}\,g$ lies between $0.156381$ and $0.25000001$, and the norm of $D^3g$ is at most $0.0615172$; on the ball $r \le 1.755$, $|\nabla g| \le 0.391443$.*

*Proof.* Differentiating $\varphi(|c|^2)$ gives $D^2g(v,v) = 2\varphi'|v|^2 + 4\varphi''\langle c, v\rangle^2$ and $D^3g(v,v,v) = 12\varphi''\langle c, v\rangle|v|^2 + 8\varphi'''\langle c, v\rangle^3$. For $|v| = 1$ and $t = \langle c/r, v\rangle \in [-1, 1]$ the cubic form is $At + \Gamma t^3$, whose largest absolute value is attained at $t = \pm 1$ or at an interior critical point, where it equals $\frac{2}{3}|A||t| \le \frac{2}{3}|A|$; by Banach's theorem [9] the supremum of a symmetric trilinear form on the diagonal of the unit sphere is its norm. The numerical bounds are obtained in ball arithmetic [57] from the series of $Z_0$, truncated after forty-eight terms with a proved bound on the tail, on $2 \cdot 10^5$ subintervals of $[0, 2.106]$, with $|\nabla g(c)| = r \cdot 2\varphi'(r^2)$; the program is described in Appendix C. $\square$

### D. The proof of Theorem 3

Let $\gamma^{\mathscr{B}}$ be the block specification: for a block $B$, $\gamma^{\mathscr{B}}_B$ resamples the three links of $B$ from (2). It comes from the same interaction as $\gamma^F$, so the two specifications have the same Gibbs measures. Define the block coefficients

$$C_{Bj} = \sup \frac{W_1\big(\gamma^{\mathscr{B}}_B(\cdot \mid Q), \gamma^{\mathscr{B}}_B(\cdot \mid Q')\big)}{|Q_j - Q'_j|}$$

for an active link $j \notin B$, the supremum over pairs of configurations that differ only at $j$, with $W_1$ for the metric $d$.

**Proposition 17.** *Suppose that $\sum_B C_{Bj} \le \alpha < 1$ for every active link $j$. Then the conclusions (i)–(vi) of Proposition 11 hold with $-\frac{1}{2}\log\alpha$ in place of $-\log\alpha$.*

*Proof.* The proof of Proposition 11 goes through with Lemma 243 of Appendix B in place of Lemma 241, and with the set $S$ of part (ii), and the support of $H$ in Corollary 242, enlarged to the union of the blocks that meet it. The enlargement multiplies the size of these sets by at most three and moves them by at most one, which changes only the constants. □

*Proof of Theorem 3.* Let $|\beta| \le 0.351/4$ and take the constants of Lemma 16, whose balls $r \le 2.106$ and $r \le 1.755$ contain $|c| \le 24|\beta|$ and $|c| \le 20|\beta|$. With these fixed constants every quantity in (3) is increasing in $|\beta|$, and so is

$$\alpha(\beta_W) = 40|\beta|K. \tag{4}$$

Changing an active link $j$ by $\eta$ changes one staple of each of its ten partners outside its block by an isometric image of the displacement, so the field $b_i$ of each such partner $i$ moves by at most $4|\beta|\eta$, and Lemma 15, applied once for each of these ten fields, gives $\sum_B C_{Bj} \le 40|\beta|K$. Evaluating (3) in ball arithmetic at $\beta_W = 0.351$ gives $w \le 0.0057670$, $s \le 0.2516793$, $C \le 0.0390392$, $K \le 0.2848858$ and $\alpha \le 0.99995$; at $\beta_W = 0.345$ and $\beta_W = \frac{1}{3}$ it gives $\alpha \le 0.98044$ and $\alpha \le 0.94279$. At $\beta_W = 0.3511$ the same evaluation exceeds one. Proposition 17 applies. □

*Remark* 18. Most of the range is kept. A second bound for the same sensitivity, which keeps the law of the middle link instead of linearising it, evaluated in floating point on a quadrature grid and maximised over the fields $b$ by a random search followed by the Nelder–Mead method, gives 0.2783 at $\beta_W = 0.34$, against 0.2837 from Lemma 15, and $40|\beta|$ times it crosses one near $\beta_W = 0.357$. Most of the difference is the two-site constant $C \approx 0.038$, which the floating-point value suggests is about twice the true feedback between $y_1$ and $y_3$. No bound of this form passes $\beta_W \approx 0.3666$: at $b = 0$ the function $f = \langle y_1, e\rangle + \mathrm{Re}(y_2 y_3 e)$ is 1-Lipschitz for $d$ and has $\mathrm{Cov}_{\mu_b}(f, \langle y_1, e\rangle) = \frac{1}{4}(1 + I_2(\beta_W)/I_1(\beta_W))$ exactly (Proposition 192), which is 0.2718 at $\beta_W = 0.351$ against $K \le 0.2849$, so the sensitivity is at least this number and $40|\beta|$ times it crosses one at $\beta_W = 0.3666$. Past this point the gain from integrating out one link is spent, and larger blocks are needed, such as the free links of a cube of the line field, whose sensitivity no longer reduces to two sites.

## VI. CURVATURE IN THE GAUGE

In this section $G \subset \mathrm{U}(N)$ is a closed connected subgroup with Lie algebra $\mathfrak{g}$ and the inner product $\langle X, Y\rangle = -\mathrm{Tr}(XY)$; the bound of Theorem 4 is proved first for $\mathrm{U}(N)$. A tangent vector at a field $Q$ is $v = (X_e Q_e)$, $|v|^2 = \sum_e |X_e|^2$. For the plaquette at $x$ in the plane $\mu < \nu$ number its links $e_1 = (x, \mu)$, $e_2 = (x + e_\mu, \nu)$, $e_3 = (x + e_\nu, \mu)$, $e_4 = (x, \nu)$, transport their vectors to $x$,

$$Y_1 = X_{e_1}, \quad Y_2 = Q_{e_1} X_{e_2} Q_{e_1}^*, \quad Y_3 = Q_{e_4} X_{e_3} Q_{e_4}^*, \quad Y_4 = X_{e_4}, \tag{5}$$

and write $W = Q_p$. The second derivative of $\mathrm{Re}\,\mathrm{Tr}\,Q_p$ along the geodesic $Q_e(t) = e^{tX_e} Q_e$ is

$$f_p = \sum_{i=1}^{4} \mathrm{Re}\,\mathrm{Tr}(Y_i^2 W) + 2\,\mathrm{Re}\,\mathrm{Tr}(Y_1 Y_2 W) + 2\,\mathrm{Re}\,\mathrm{Tr}(Y_3 Y_4 W) - 2 \sum_{i \in \{1,2\},\, j \in \{3,4\}} \mathrm{Re}\,\mathrm{Tr}(Y_i W Y_j), \tag{6}$$

so that $\mathrm{Hess}\, S_0(v, v) = \sum_p f_p$. To see this, write $Q_p = Q_{e_1} Q_{e_2} Q_{e_3}^* Q_{e_4}^*$. Since $Q_{e_1} e^{tX_{e_2}} = e^{tY_2} Q_{e_1}$ and $e^{-tX_{e_3}} Q_{e_4}^* = Q_{e_4}^* e^{-tY_3}$, the plaquette variable along the geodesic is $Q_p(t) = e^{tY_1} e^{tY_2} W e^{-tY_3} e^{-tY_4}$. Differentiating twice at $t = 0$ gives the four squares, the two products of neighbouring factors and the four products across $W$, and the squares and the product $WY_3Y_4$ are moved cyclically under the trace. The corner form of $p$ is

$$c_p = 2\langle Y_1, Y_4\rangle - 2\langle Y_1, Y_2\rangle + 2\langle Y_2, W Y_3 W^*\rangle - 2\langle Y_3, Y_4\rangle. \tag{7}$$

It collects the inner products of the transported vectors of two links that meet at a right angle at a corner of $p$.

**Lemma 19.** *On $\mathbb{Z}^4$, or on a box with free boundary, for every field $Q$ and every square-summable $X$,*

$$\sum_y \Big[ |D_y X|^2 + \sum_\mu \big| X_{y,\mu} + Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu} \big|^2 \Big] = 4|X|^2 + \sum_p c_p,$$

*where $D_y X = \sum_\mu [X_{y,\mu} - Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu}]$ is the covariant divergence and terms with a link outside the box are omitted.*

*Proof.* At a site $y$ write $o_\mu$ for the vector of the outgoing link in direction $\mu$ and $i_\mu$ for the transported vector of the incoming one, so that $D_y X = \sum_\mu (o_\mu - i_\mu)$. Every link is outgoing at one end and incoming at the other, so the squares $|o_\mu|^2 + |i_\mu|^2$, which appear once in $|D_y X|^2$ and once in $|o_\mu + i_\mu|^2$, add up to $4|X|^2$. The collinear cross term $-2\langle o_\mu, i_\mu\rangle$ of $|D_y X|^2$ cancels the term $+2\langle o_\mu, i_\mu\rangle$ of $|o_\mu + i_\mu|^2$. The remaining cross terms of $|D_y X|^2$ pair two perpendicular ends at $y$, and each such pair is a corner of exactly one plaquette, with the sign and the transport of (7) (this is the proof of [16, Lemma 1]). In a box two perpendicular ends at $y$ span a plaquette of the box, because a box is a product of intervals. □

**Lemma 20.** *Let $k \in \{1,2,3,4\}$, let $k \pm 1$ (mod 4) be the positions adjacent to $k$ and $k+2$ the opposite one. For every $N$, every $W \in \mathrm{U}(N)$ and all $Y_1, \dots, Y_4 \in \mathfrak{u}(N)$ with $Y_k = 0$,*

$$\pm f_p + \tfrac{9}{10} c_p \le \tfrac{21}{10}\big(|Y_{k-1}|^2 + |Y_{k+1}|^2\big) + \tfrac{5}{4}|Y_{k+2}|^2.$$

*Proof.* Replacing $W$ by $-W$ changes the sign of $f_p$ and fixes $c_p$, so the sign $+$ suffices. Conjugation by a unitary matrix preserves both sides, so we may take $W = \mathrm{diag}(w_a)$. The form $\mathscr{F} = \mathrm{RHS} - f_p - \frac{9}{10} c_p$ then splits into a block for each diagonal position $a$, in the real variables $y_i = -i(Y_i)_{aa}$, and a block for each pair $a \neq b$, in the complex variables $z_i = (Y_i)_{ab}$, since $(Y_i)_{ba} = -\bar{z}_i$. Appendix D shows that every block is positive definite. □

*Proof of the bound in Theorem 4.* Every plaquette has exactly one link of $F$, and $X$ vanishes on it. By Lemma 9(iii) each active link is adjacent to the fixed link in four of its plaquettes and opposite to it in two. Summing Lemma 20 over the plaquettes,

$$\sum_p \big(\pm f_p + \tfrac{9}{10} c_p\big) \le \big(4 \cdot \tfrac{21}{10} + 2 \cdot \tfrac{5}{4}\big)|X|^2 = \tfrac{109}{10}|X|^2.$$

By Lemma 19, $\frac{9}{10}\sum_p c_p = \frac{9}{10}\Sigma - \frac{36}{10}|X|^2$, where $\Sigma \ge 0$ is the left side of the lemma. Hence

$$\pm \mathrm{Hess}\, S_0(v,v) + \tfrac{9}{10}\Sigma \le \big(\tfrac{109}{10} + \tfrac{36}{10}\big)|X|^2 = \tfrac{29}{2}|X|^2.$$

In a box, links near the boundary lie in fewer plaquettes and receive less. □

The factor $\frac{9}{10}$ is chosen so that the corners are paid by the divergence identity at the price $\frac{36}{10}$ while the local blocks stay positive; Lemma 19 is the divergence identity of [16, Lemma 1] with the collinear term written as a square, so nothing is lost by Cauchy–Schwarz.

The two lemmas concern vectors in $\mathfrak{u}(N)$ and fields in $\mathrm{U}(N)$, so the bound holds without change on every closed subgroup $G \subset \mathrm{U}(N)$, the vectors $X_e$ ranging over its Lie algebra $\mathfrak{g}$. It is also the Hessian of $S_0$ on $G^E$: the metric is bi-invariant, the geodesics of $G$ through $Q_e$ are the curves $e^{tX} Q_e$ with $X \in \mathfrak{g}$, which are geodesics of $\mathrm{U}(N)$, and the Hessian is read off from second derivatives along geodesics.

**Lemma 21.** *Let $G \subset \mathrm{U}(N)$ be closed and connected with simple Lie algebra $\mathfrak{g}$, let $\mathfrak{t} \subset \mathfrak{g}$ be a maximal abelian subalgebra, and write the eigenvalues of $\mathrm{ad}\, H$ on $\mathfrak{g} \otimes \mathbb{C}$ and of $H$ on $\mathbb{C}^N$, for $H \in \mathfrak{t}$, as $i\alpha(H)$ and $i\mu(H)$, with $\alpha$ running over the roots and $\mu$ over the weights of $\mathbb{C}^N$ counted with multiplicity. Then $\mathrm{Ric}_G = \kappa_G \langle \cdot, \cdot \rangle$ with*

$$\kappa_G = \frac{\sum_\alpha \alpha(H)^2}{4 \sum_\mu \mu(H)^2} \qquad (0 \neq H \in \mathfrak{t}).$$

*Proof.* For a bi-invariant metric the curvature is $R(X,Y)Z = -\frac{1}{4}[[X,Y],Z]$ on left-invariant fields [70, Sect. 7], so for an orthonormal basis $(E_i)$ of $\mathfrak{g}$,

$$\begin{aligned} \mathrm{Ric}(X,X) &= \sum_i \langle R(E_i,X)X, E_i \rangle \\ &= \tfrac{1}{4}\sum_i |[X,E_i]|^2 = -\tfrac{1}{4}\,\mathrm{Tr}(\mathrm{ad}\, X)^2, \end{aligned}$$

since $\mathrm{ad}\, X$ is skew. The Killing form $B(X,Y) = \mathrm{Tr}(\mathrm{ad}\, X\, \mathrm{ad}\, Y)$ and $\langle \cdot, \cdot \rangle$ are invariant under the adjoint action, and $\mathfrak{g}$ is simple, so $B = -4\kappa\langle \cdot, \cdot \rangle$ for a constant $\kappa$ and $\mathrm{Ric} = \kappa \langle \cdot, \cdot \rangle$. On $H \in \mathfrak{t}$ the two forms are $B(H,H) = -\sum_\alpha \alpha(H)^2$ and $\langle H, H \rangle = -\,\mathrm{Tr}\, H^2 = \sum_\mu \mu(H)^2$. □

Table I evaluates the ratio of Lemma 21 for every compact simple Lie algebra and a faithful representation of least dimension, on the root systems of [55, Sect. 12.1]; the weights of the representations of dimensions 27 and 56 of $E_6$ and $E_7$ are the roots of $E_8$ with prescribed inner products with a subsystem $A_2$, respectively $A_1$, projected to its orthogonal complement, by the decompositions $248 = (78,1) + (27,3) + (\overline{27},\bar{3}) + (1,8)$ and $248 = (133,1) + (56,2) + (1,3)$ [85]. Taking for $H$ the coroot $\theta^\vee$ of the highest root gives $\sum_\alpha \alpha(\theta^\vee)^2 = 4h^\vee$ and $\sum_\mu \mu(\theta^\vee)^2 = 2\ell_\rho$, so $\kappa_G = h^\vee/(2\ell_\rho)$, with the Dynkin index normalised to one on the defining representation of $\mathrm{SU}(N)$. The ratio depends on $\mathfrak{g}$ and $\rho$ only, so it is the same for every compact group with Lie algebra $\mathfrak{g}$, simply connected or not.

**Lemma 22.** *Let $e \neq e'$ be two links of a plaquette $p$, and write*

$$\mathrm{Hess}\, \mathrm{Re}\, \mathrm{Tr}\, Q_p(v,v) = \sum_{e,e' \in p} H^p_{ee'}(X_e, X_{e'})$$

*with bilinear forms satisfying $H^p_{ee'}(a,b) = H^p_{e'e}(b,a)$. Then $|H^p_{ee'}(a,b)| \le |a|\,|b|$ for all $a,b \in \mathfrak{u}(N)$ and every field.*

*Proof.* By (6) the terms of $f_p$ that mix two positions $i \neq j$ are $2\,\mathrm{Re}\,\mathrm{Tr}(Y_1 Y_2 W)$, $2\,\mathrm{Re}\,\mathrm{Tr}(Y_3 Y_4 W)$ and $-2\,\mathrm{Re}\,\mathrm{Tr}(Y_i W Y_j)$, so $H^p_{ee'}$ is $\pm\,\mathrm{Re}\,\mathrm{Tr}(abW)$ or $\pm\,\mathrm{Re}\,\mathrm{Tr}(aWb)$ after an isometric transport of $a$ and $b$, and $|\,\mathrm{Tr}(abW)| \le \|a\|_{\mathrm{HS}} \|bW\|_{\mathrm{HS}} = |a|\,|b|$. □

**Corollary 23.** *Let $G \subset \mathrm{U}(N)$ be closed and connected with semisimple Lie algebra and $N|\beta| < \frac{2}{29}\kappa_G$. For every box the measure $\mu^F_\Lambda$ satisfies the Bakry–Émery condition with $K = \kappa_G - \frac{29}{2}N|\beta| > 0$, uniformly in the box, and hence a logarithmic Sobolev inequality with constant $2/K$ and a Poincaré inequality with constant $1/K$.*

*Proof.* The set $\{Q_F = 1\}$ is a product factor, so its Ricci curvature is at least $\kappa_G$ on every link, and the Hessian of the restricted action is the restriction of $N\beta\,\mathrm{Hess}\, S_0$ to $X_F = 0$. Apply the bound and [5, Cor. 5.7.2]. □

### A. Decay with an explicit rate

Let $A'$ be a finite set of active links and $P'$ a finite set of plaquettes whose active links lie in $A'$. An action $S = \sum_{p \in P'} c_p\, \mathrm{Re}\,\mathrm{Tr}\, Q^F_p$ with $0 \le c_p \le 1$ is *admissible* if $|\,\mathrm{Hess}\, S(v,v)| \le \frac{29}{2}|v|^2$ for all $v$ tangent to $G^{A'}$. Two links lie in at most one common plaquette, so by Lemma 22 and Lemma 9(iv) the mixed blocks of its Hessian satisfy $|\,\mathrm{Hess}\, S_{ee'}(a,b)| \le |a|\,|b|$, and each link has at most twelve partners. The action $S_\Lambda = \sum_{p \in P(\Lambda)} \mathrm{Re}\,\mathrm{Tr}\, Q^F_p$ of a box is admissible by the bound proved above, and so is every convex combination of two admissible actions. Let $\mu$ be the probability measure on $G^{A'}$ with density proportional to $e^{N\beta S}$ with respect

to Haar measure, let $L=\Delta+N\beta\langle\nabla S,\nabla\cdot\rangle$ be its Langevin generator, which is self-adjoint in $L^2(\mu)$, and $P_t=e^{tL}$. Write $\nabla_e$ for the gradient in the factor of the link $e$, and for weights $w:A'\to(0,\infty)$ put $\Gamma^w(f)=\sum_e w_e|\nabla_e f|^2$; $\Gamma$ is the case $w\equiv1$.

**Lemma 24.** *Let $\lambda\ge0$ and $w_e=e^{\lambda d_e}$, where $|d_e-d_{e'}|\le1$ whenever $e$ and $e'$ share a plaquette, and put*

$$K_\lambda=\kappa_G-N|\beta|\Big(\tfrac{29}{2}+12\big(\cosh\tfrac{\lambda}{2}-1\big)\Big).$$

*Then $\Gamma^w(P_tf)\le e^{-2K_\lambda t}P_t\Gamma^w(f)$ for every smooth $f$ and $t\ge0$.*

*Proof.* Put $V=-N\beta S$. On the product manifold $M=G^{A'}$ the projection to the tangent space of one factor is parallel, the rough Laplacian commutes with it, and the Ricci tensor is block diagonal. Hence the Bochner–Weitzenböck formula $\nabla\Delta g=\bar\Delta\nabla g+\mathrm{Ric}(\nabla g)$, applied factor by factor, together with the derivative of the drift, gives for smooth $g$

$$\begin{aligned}&\tfrac12L\Gamma^w(g)-\Gamma^w(g,Lg)\\&\quad=\sum_e w_e\big(|\nabla\nabla_e g|^2+\mathrm{Ric}_G(\nabla_e g,\nabla_e g)\big)\\&\qquad+\sum_{e,e'}w_e\,\mathrm{Hess}\,V_{ee'}(\nabla_e g,\nabla_{e'}g),\end{aligned}$$

where $\Gamma^w(g,h)$ is the polarised form.[13] Put $u_e=\nabla_e g$ and $v_e=\sqrt{w_e}\,u_e$. By the symmetry of $\mathrm{Hess}\,V$,

$$\begin{aligned}\sum_{e,e'}w_e\,\mathrm{Hess}\,V_{ee'}(u_e,u_{e'})&=\mathrm{Hess}\,V(v,v)\\&+\sum_{e\ne e'}\Big(\cosh\tfrac{\lambda(d_e-d_{e'})}{2}-1\Big)\,\mathrm{Hess}\,V_{ee'}(v_e,v_{e'}),\end{aligned}$$

because $\frac{w_e+w_{e'}}{2\sqrt{w_ew_{e'}}}=\cosh\frac{\lambda(d_e-d_{e'})}{2}$. The first term is at least $-\frac{29}{2}N|\beta|\,|v|^2$. In the second only partners contribute, each with a coefficient at most $\cosh\frac{\lambda}{2}-1$ and a block of norm at most $N|\beta|$, and $2|v_e|\,|v_{e'}|\le|v_e|^2+|v_{e'}|^2$ with twelve partners per link bounds it by $12(\cosh\frac{\lambda}{2}-1)N|\beta|\,|v|^2$. Since $\mathrm{Ric}_G\ge\kappa_G$ and $|v|^2=\Gamma^w(g)$,

$$\tfrac12L\Gamma^w(g)-\Gamma^w(g,Lg)\ge K_\lambda\Gamma^w(g).$$

For fixed $t$ let $\varphi(s)=P_s\Gamma^w(P_{t-s}f)$, $0\le s\le t$. Then $\varphi'(s)=P_s\big(L\Gamma^w(g_s)-2\Gamma^w(g_s,Lg_s)\big)\ge2K_\lambda\varphi(s)$ with $g_s=P_{t-s}f$, and $\varphi(0)\le e^{-2K_\lambda t}\varphi(t)$ is the claim. □

**Lemma 25.** *Suppose $K_\lambda>0$. Let $f$ and $g$ be smooth functions on $G^{A'}$ that depend only on the links of $A_f$ and of $B_g$, and let $D$ be the least number of steps from $A_f$ to $B_g$ in the graph on $A'$ whose edges join links that share a plaquette. Then*

$$|\mathrm{Cov}_\mu(f,g)|\le\frac{e^{-\lambda D/2}}{K_\lambda}\,\|\nabla f\|_\infty\sum_{e\in B_g}\|\nabla_e g\|_\infty,$$

*where $\|\nabla f\|_\infty^2=\sup\sum_e|\nabla_e f|^2$.*

[13] For $w\equiv1$ this is Bochner's identity $\Gamma_2=|\nabla^2g|^2+(\mathrm{Ric}+\mathrm{Hess}\,V)(\nabla g,\nabla g)$; the weights enter only through the factor index, which the connection, the Laplacian and the Ricci tensor preserve, so only the off-diagonal blocks of $\mathrm{Hess}\,V$ see them.

*Proof.* Since $K_0\ge K_\lambda>0$, Lemma 24 with $\lambda=0$ gives $\Gamma(P_tf)\le e^{-2K_0t}\|\nabla f\|_\infty^2$, so $P_tf$ converges uniformly to a constant, which is $\mu(f)$ by invariance, $M$ being connected. Since $\frac{d}{dt}\mu(gP_tf)=\mu(gLP_tf)=-\mu(\Gamma(g,P_tf))$,

$$\mathrm{Cov}_\mu(f,g)=\int_0^\infty\mu\big(\Gamma(P_tf,g)\big)\,dt.$$

Let $d_e$ be the smaller of $D$ and the graph distance from $e$ to $A_f$, so that $|d_e-d_{e'}|\le1$ for partners, $w_e=1$ on $A_f$, $w_e=e^{\lambda D}$ on $B_g$, and $\Gamma^w(f)\le\|\nabla f\|_\infty^2$. For $e\in B_g$, Lemma 24 gives $|\nabla_eP_tf|^2\le w_e^{-1}\Gamma^w(P_tf)\le e^{-\lambda D}e^{-2K_\lambda t}\|\nabla f\|_\infty^2$. Hence $|\Gamma(P_tf,g)|\le e^{-\lambda D/2}e^{-K_\lambda t}\|\nabla f\|_\infty\sum_{e\in B_g}\|\nabla_e g\|_\infty$, and integration in $t$ gives the bound. □

*Proof of Theorem 5.* Let $\lambda^*>0$ solve $K_{\lambda^*}=0$, so that $\frac{\lambda^*}{2}=m_G(\beta)$, and let $0<\lambda<\lambda^*$; then $K_\lambda>0$.

*Decay in boxes.* Lemma 25 with $S=S_\Lambda$ gives

$$|\mathrm{Cov}_{\mu^F_\Lambda}(f,g)|\le C_{f,g}e^{-\lambda D/2},$$

uniformly in the box.

*Limit.* For boxes $\Lambda\subset\Lambda'$ let $S_s=(1-s)S_\Lambda+sS_{\Lambda'}$ on the active links of $\Lambda'$, $s\in[0,1]$, and $\mu_s$ the corresponding measure; $S_s$ is admissible, $\mu_0$ is $\mu^F_\Lambda$ times Haar measure on the active links outside $\Lambda$, and $\mu_1=\mu^F_{\Lambda'}$. For smooth $f$ of the active links of $\Lambda$,

$$\frac{d}{ds}\mu_s(f)=N\beta\sum_{p\in P(\Lambda')\setminus P(\Lambda)}\mathrm{Cov}_{\mu_s}\big(f,\mathrm{Re}\,\mathrm{Tr}\,Q_p^F\big),$$

and $|\nabla_e\,\mathrm{Re}\,\mathrm{Tr}\,Q_p|\le\|Q_p\|_{\mathrm{HS}}=\sqrt{N}$. A step in the graph of Lemma 25 moves a link by at most one lattice unit, so a plaquette at graph distance $k$ from the support $T$ of $f$ lies within distance $k+1$ of $T$, there are at most $C_T(k+1)^4$ of them, and every plaquette outside $P(\Lambda)$ is at graph distance at least $R-1$, $R$ the distance from $T$ to the complement of $\Lambda$. Lemma 25 gives

$$|\mu^F_{\Lambda'}(f)-\mu^F_\Lambda(f)|\le C_fR^4e^{-\lambda R/2}.$$

The limit therefore exists along every increasing sequence of boxes, and two sequences are compared through their intersections, which are boxes. Continuous gauge-invariant $O$ are uniform limits of smooth gauge-invariant ones, for instance of $e^{\tau\Delta}O$, since the heat semigroup of $G^E$ commutes with left and right translations; with Lemma 10 the limit $\omega(O)$ of $W_\Lambda(O)$ exists for all of them.

*Properties.* Invariance and, for $\beta\ge0$, reflection positivity follow as in Proposition 11(iii)–(iv). For Lipschitz $O$ the functions $e^{\tau\Delta}O$ are smooth with the same Lipschitz constant, since $\mathrm{Ric}_G\ge0$ [5, Sect. 3.3], so the bound of the first step passes to $\omega$ and to Lipschitz observables. Two links of one plaquette have time coordinates that differ by at most one, so observables at time separation $t$ are at graph distance at least $t-c$, and truncated correlations decay at every rate below $m_G(\beta)$. The gap follows from Proposition 245 as in Proposition 11(vi). □

*Proof of the rest of Theorem 4.* By Lemma 21 and Table I, $\kappa_{\mathrm{SU}(N)} = \frac{N}{2}$ and $\kappa_{\mathrm{SO}(N)} = \frac{N-2}{4}$; for SO(4), whose Lie algebra is $\mathfrak{so}(3)\oplus\mathfrak{so}(3)$ acting on $\mathbb{R}^4\otimes\mathbb{C} = \mathbb{C}^2\otimes\mathbb{C}^2$, each ideal gives $\frac{1}{2}$ by the same computation. The conditions of Theorem 4 are those of Theorem 5. □

*Remark* 26. Let $h_F$ be the supremum of $\pm\mathrm{Hess}\, S_0$ on the gauge-fixed directions, so that uniform curvature reaches $|\beta| < 1/(2h_F)$ and no further. At the identity field $-\mathrm{Hess}\, S_0$ restricted to $X_F = 0$ has the eigenvalue $9+\sqrt{17} = 13.123$ on the torus of side two, so $9+\sqrt{17} \le h_F \le \frac{29}{2}$. A gradient search over SU(2) fields of periods two, four and six finds nothing above 13.97401, the value at an abelian field; certificates made of plaquette forms and vertex forms on the period-two cell cannot go below 14.384. One of us (T. Romanov) [79] constructed such a certificate, with sixteen positive definite vertex forms of size eight and nonnegative plaquette weights, rational with denominator $10^6$. It gives $h_F \le 14.420001$ (Theorem 267; Appendix L gives the derivation and the checks). Like $\frac{29}{2}$ this is a statement about $\mathrm{U}(N)$, so it can replace $\frac{29}{2}$ in Theorems 4 and 5, and the curvature method then reaches $|\beta| < 1/28.840002$, that is $\beta_W < 0.1387$ for SU(2) and $\beta_W < 0.3121$ for SU(3). The bound is computer-assisted and concerns the lattice at fixed spacing, not the continuum limit. The theorems of this paper are stated with $\frac{29}{2}$, which is proved by hand. With this gauge the curvature method stops between $|\beta| = 1/28.840002$ and $\frac{1}{26.25}$.

## VII. SU(3) UP TO 3/4

In this section $G = \mathrm{SU}(3)$, $\beta \ge 0$ and $\beta_W = 9\beta$. The distance on SU(3) is $d(U,V) = \|U-V\|_{\mathrm{HS}}$, which is at most the geodesic distance for $\langle X,Y\rangle = -\mathrm{Tr}(XY)$; $K = \mathrm{conv}\,\mathrm{SU}(3)$; $A(M) = \frac{1}{2}(M-M^*)$ and $P$ removes the trace. With this inner product $\mathrm{Ric} = \frac{3}{2}$, the Casimir of the fundamental representation is $\lambda_F = \frac{8}{3}$ and that of the adjoint representation is 6.

### A. Reduction to one link

The conditional law of an active link is

$$\nu_b(dU) \propto e^{\mathrm{Re}\,\mathrm{Tr}(Ub)}dU, \qquad b = N\beta\sum_{p\ni e} S_p,$$

where $S_p$ is the staple of $e$ in $p$, a product of SU(3) matrices and identities, or its adjoint. Six staples give $b \in rK$ with $r = 6N\beta = 2\beta_W$. For a matrix $h$ put $\varphi_h(U) = \mathrm{Re}\,\mathrm{Tr}(Uh)$ and

$$L_h(b) = \sup\{\mathrm{Cov}_{\nu_b}(g,\varphi_h) : g \text{ 1-Lipschitz for } d\},$$

$$\Lambda(r) = \sup_{b\in rK}\ \sup_{\|h\|_{\mathrm{HS}}=1} L_h(b).$$

Changing a partner $e'$ changes one staple by a matrix of norm $d(Q_{e'}, Q'_{e'})$. Along the segment $b_t$ from $b$ to $b'$, which stays in $rK$, $\frac{d}{dt}\nu_{b_t}(g) = \mathrm{Cov}_{\nu_{b_t}}(g, \mathrm{Re}\,\mathrm{Tr}(U(b'-b)))$, so $W_1(\nu_b,\nu_{b'}) \le N\beta\Lambda(r)\,d(Q_{e'},Q'_{e'})$. With twelve partners per link,

$$\alpha \le 12N\beta\Lambda(r) = 2r\Lambda(r) = 4\beta_W\Lambda(2\beta_W). \tag{8}$$

Since $K$ is convex and contains 0, the mean of $I, \omega I, \omega^2 I$ with $\omega^3 = 1$, $r'K \subset rK$ for $r' \le r$ and $\Lambda$ is nondecreasing.

### B. Gradients and one-forms

**Lemma 27.** *In the left trivialisation,* $\nabla\varphi_c(U) = -PA(cU)$*; if* $V = \varphi_b$ *then* $\mathrm{Hess}\, V(X,X) = \mathrm{Re}\,\mathrm{Tr}(UX^2b)$*; and* $\Delta\varphi_c = -\lambda_F\varphi_c$.

*Proof.* $\frac{d}{dt}\mathrm{Re}\,\mathrm{Tr}(Ue^{tX}c)|_{t=0} = \mathrm{Re}\,\mathrm{Tr}(cUX) = -\langle PA(cU), X\rangle$ for $X \in \mathfrak{su}(3)$. On a group with a bi-invariant metric $\nabla_X X = 0$ for left-invariant $X$, so the Hessian is the second derivative along $e^{tX}$. The function $\varphi_c$ is a matrix coefficient of the fundamental representation, on which the Casimir acts by $\lambda_F$. □

**Lemma 28.** *The bottom of the spectrum of the rough Laplacian* $\nabla^*\nabla$ *on* $L^2$ *one-forms of* SU(3) *is* $\mu_0 = \frac{7}{6}$.

*Proof.* Trivialise one-forms on the left, so that a one-form is a function $f : \mathrm{SU}(3) \to \mathfrak{su}(3)$. For an orthonormal basis $X_a$ acting as left-invariant vector fields, $\nabla_{X_a}Y = \frac{1}{2}[X_a,Y]$ and $\nabla_{X_a}X_a = 0$, so $\nabla^*\nabla = -\sum_a(\rho(X_a) + \frac{1}{2}\,\mathrm{ad}\,X_a)^2$. On the isotypic part $V_\pi\otimes V_\pi^*\otimes\mathfrak{su}(3)$ of $L^2\otimes\mathfrak{su}(3)$ put $A_a = \rho_\pi(X_a)$, $B_a = \mathrm{ad}\,X_a$ and $T_a = A_a + B_a$, the action on the tensor product. Since $A_a$ and $B_a$ commute, $A_a + \frac{1}{2}B_a = \frac{1}{2}(A_a + T_a)$ and $A_aT_a + T_aA_a = A_a^2 + T_a^2 - B_a^2$, so on a component $\sigma \subset \pi\otimes\mathrm{ad}$

$$\nabla^*\nabla = \tfrac{1}{2}(C_\pi + C_\sigma) - \tfrac{1}{4}C_{\mathrm{ad}} = \tfrac{1}{2}(C_\pi + C_\sigma) - \tfrac{3}{2},$$

with $C(p,q) = \frac{2}{3}(p^2+q^2+pq+3p+3q)$ for the representation of highest weight $(p,q)$ [40]. The components of $(p,q)\otimes(1,1)$ are $(p+1,q+1)$, $(p+2,q-1)$, $(p-1,q+2)$, $(p+1,q-2)$, $(p-2,q+1)$, $(p-1,q-1)$ and $(p,q)$, the last with multiplicity two, one or zero as both, one or none of $p,q$ are positive, and only those with nonnegative labels occur. For $\pi = (0,0)$ the only component is $(1,1)$, with value $\frac{3}{2}$. For $\pi = (1,0)$ the components are $(2,1)$, $(0,2)$, $(1,0)$, with values $\frac{31}{6}$, $\frac{19}{6}$, $\frac{7}{6}$, and $(0,1)$ is conjugate. For $p+q \ge 2$ the smallest values occur at $(p-1,q-1)$, $(p,q)$, $(p-2,q+1)$ or $(p+1,q-2)$, and each is at least $\frac{1}{3}(2(p^2+q^2+pq)+3(p+q)-3) - \frac{3}{2} \ge \frac{3}{2}$. □

### C. The gap of the one-link measure

**Lemma 29.** *For* $b \in rK$ *the spectral gap of* $L = \Delta + \langle\nabla V, \nabla\cdot\rangle$*,* $V = \varphi_b$*, on* $L^2(\nu_b)$ *is at least*

$$\gamma(r) = \tfrac{39}{64}\mu_0 + \tfrac{3}{2} - \tfrac{r}{2} = \tfrac{283}{128} - \tfrac{r}{2}.$$

*Proof.* Let $-Lf = \lambda f$ with $\lambda > 0$, $\omega = df$ and $\eta = e^{V/2}\omega$; norms are in $L^2(dU)$. Bochner's formula integrated against

$\nu_b$ gives $\lambda\nu_b|\omega|^2 = \nu_b|\nabla\omega|^2 + \nu_b((\mathrm{Ric}-\mathrm{Hess}\,V)(\omega,\omega))$, and since $e^{V/2}\nabla\omega = \nabla\eta - \frac12 dV\otimes\eta$,

$$\lambda\|\eta\|^2 = \|\nabla\eta - \tfrac12 dV\otimes\eta\|^2 + \int(\mathrm{Ric}-\mathrm{Hess}\,V)(\eta,\eta). \quad (9)$$

Put $\mathscr{A} = \nabla\eta$ and $\mathscr{B} = \frac12 dV\otimes\eta$. Then

$$\langle\mathscr{A},\mathscr{B}\rangle = \tfrac14\int\langle\nabla V,\nabla|\eta|^2\rangle = \tfrac{\lambda_F}{4}\int V|\eta|^2,$$

because $-\Delta V = \lambda_F V$. Writing $\|\mathscr{A}-\mathscr{B}\|^2 = \|\frac58\mathscr{A}-\mathscr{B}\|^2 + \frac{39}{64}\|\mathscr{A}\|^2 - \frac34\langle\mathscr{A},\mathscr{B}\rangle$ and dropping the first term,

$$\|\mathscr{A}-\mathscr{B}\|^2 \ge \tfrac{39}{64}\mu_0\|\eta\|^2 - \tfrac12\int V|\eta|^2$$

by Lemma 28. With $\mathrm{Ric} = \frac32$, (9) gives $\lambda\|\eta\|^2 \ge (\frac{39}{64}\mu_0 + \frac32)\|\eta\|^2 - \int(\mathrm{Hess}\,V(\eta,\eta) + \frac12 V|\eta|^2)$, and the claim follows from the pointwise bound

$$\mathrm{Re}\,\mathrm{Tr}(MX^2) + \tfrac12\,\mathrm{Re}\,\mathrm{Tr}\,M \le \tfrac{r}{2}|X|^2, \qquad M = bU \in rK, \quad (10)$$

since $\mathrm{Hess}\,V(X,X) = \mathrm{Re}\,\mathrm{Tr}(MX^2)$ and $V = \mathrm{Re}\,\mathrm{Tr}\,M$. The maximum over $|X|=1$ of the left side is convex in $M$, so it suffices to take $M = rW$ with $W\in\mathrm{SU}(3)$ and, after a conjugation, $W = \mathrm{diag}(e^{i\theta_k})$ with $\sum_k\theta_k = 0$; put $c_k = \cos\theta_k$. Write $X = iH$ with $H$ Hermitian, traceless, $\mathrm{Tr}\,H^2 = 1$, with diagonal $d_k$ and entries $H_{jk}$. Then

$$\mathrm{Re}\,\mathrm{Tr}(WX^2) = -\sum_k c_k d_k^2 - \sum_{j<k}(c_j+c_k)|H_{jk}|^2,$$

with $\sum_k d_k^2 + 2\sum_{j<k}|H_{jk}|^2 = 1$, so the maximum is the larger of $q_{\mathrm{off}} = \max_{j<k} -\frac12(c_j+c_k)$ and $q_{\mathrm{diag}} = \max\{-\sum c_k d_k^2 : \sum d_k = 0,\ \sum d_k^2 = 1\}$. If $l$ is the third index, $q_{\mathrm{off}} + \frac12\sum_k c_k = \frac12 c_l \le \frac12$. The value $q_{\mathrm{diag}}$ is minus the smaller eigenvalue of $\mathrm{diag}(c)$ compressed to the plane $\sum d_k = 0$; the eigenvalues of the compression solve $\sum_k (c_k-\mu)^{-1} = 0$, that is $3\mu^2 - 2e_1\mu + e_2 = 0$ with $e_1, e_2$ the elementary symmetric functions of the $c_k$, so $q_{\mathrm{diag}} = \frac13(\sqrt{e_1^2 - 3e_2} - e_1)$. The inequality $q_{\mathrm{diag}} + \frac12 e_1 \le \frac12$ is $\sqrt{e_1^2-3e_2} \le \frac12(3-e_1)$, that is $e_1^2 - 4e_2 + 2e_1 - 3 \le 0$. When $\sum\theta_k = 0$, $\sum c_k^2 = 1 + 2c_1c_2c_3$, and with this identity the left side equals $-2(1-c_1)(1-c_2)(1-c_3) \le 0$. □

The weight $\frac58$ is the one that gives the coefficient $\frac12$ of $V$, and (10) is an equality at $W = I$. At $r = \frac32$, $\gamma = \frac{187}{128}$.[14]

[14] The direct Bochner bound, with Minkowski's inequality in place of the cross term, gives $(\sqrt{\mu_0} - \frac{g}{2}r)^2 + \frac32 - r$ with $g^2 = \max_W \|PA(W)\|^2 = 2.0655$; it is exact at $b = 0$, where the gap is $\frac83$, but falls below $\gamma$ for $r > 0.24$ and vanishes before $r = 1$.

### D. A corrector of degree two

**Lemma 30.** *For $b\in rK$, $\|h\| = 1$ and every smooth real function $u$ on* SU(3),

$$L_h(b) \le \sqrt{\nu_b|\nabla u|^2} + \sqrt{\mathrm{Var}_{\nu_b}(\varphi_h + Lu)/\gamma(r)}.$$

*Proof.* Let $-L\psi = \varphi_h - \nu_b\varphi_h$. For a 1-Lipschitz $g$, $|\nabla g| \le 1$ almost everywhere, because $d$ is at most the geodesic distance, so $\mathrm{Cov}(g,\varphi_h) = \nu_b\langle\nabla g,\nabla\psi\rangle \le \nu_b|\nabla\psi|$. Write $\psi = u + \chi$. Since $\nu_b(Lu) = 0$, $\chi$ solves $-L\chi = X - \nu_b X$ with $X = \varphi_h + Lu$, and

$$\nu_b|\nabla\chi|^2 = \mathrm{Cov}(\chi,X) \le \sqrt{\mathrm{Var}\,\chi\ \mathrm{Var}\,X} \le \sqrt{\nu_b|\nabla\chi|^2\,\mathrm{Var}\,X/\gamma}$$

by Lemma 29. Hence $\nu_b|\nabla\chi|^2 \le \mathrm{Var}\,X/\gamma$, and $\nu_b|\nabla\psi| \le \sqrt{\nu_b|\nabla u|^2} + \sqrt{\nu_b|\nabla\chi|^2}$. □

The best $u$ is $\psi$ itself, and then the second term vanishes. With $u = s\varphi_h$, Lemma 27 gives $\varphi_h + Lu = (1 - s\lambda_F)\varphi_h + s\Gamma$ with $\Gamma = \langle\nabla V,\nabla\varphi_h\rangle$; the choice $s = 1/\lambda_F$ removes $\varphi_h$ and leaves the drift. This blended corrector reaches $\beta_W = \frac{45}{64}$ and stops near 0.715 (Sec. IX). We take $u$ in a space of polynomials of degree two.

Let $\mathscr{P}_2$ be the real vector space of real polynomials of degree one or two in the entries $u_{ij}$ of $U$ and $\bar u_{ij}$ of $\bar U$. It has dimension 189: a basis $(B_k)$ consists of $\frac12(m+\bar m)$ and $\frac{1}{2i}(m-\bar m)$ for each pair $\{m,\bar m\}$ of conjugate monomials with $m\ne\bar m$, and $m$ itself when $m = \bar m$. Identify $h\in\mathbb{C}^{3\times3}$ with $\mathbb{R}^{18}$ through the real and imaginary parts of its entries. For a real $189\times18$ matrix $C$ put

$$q_h = \sum_k\sum_{m=1}^{18} C_{km}h_m B_k \in \mathscr{P}_2,$$

linear in $h$. For diagonal $b = \mathrm{diag}(a_1,a_2,x_3+iy_3)$ we have $V = \sum_k b_k\varphi_{F_k}$ with $(b_k) = (a_1,a_2,x_3,y_3)$ and $F = (e_{11},e_{22},e_{33},ie_{33})$, so

$$X_h = \varphi_h + Lq_h = \varphi_h + \Delta q_h + \sum_k b_k\langle\nabla\varphi_{F_k},\nabla q_h\rangle \quad (11)$$

is a polynomial of degree at most three, affine in $b$, while $\nabla q_h$ does not depend on $b$. The squares of these functions have degree at most six, so the quadratic forms $h\mapsto\nu_b|\nabla q_h|^2$ and $h\mapsto\nu_b[(X_h-\kappa)^2]$ are combinations of moments of $\nu_b$ of degree at most six.

To compute $\Delta$ and $\langle\nabla f,\nabla g\rangle$ exactly we use the basis of $\mathfrak{su}(3)$

$$e_{jk}-e_{kj},\quad i(e_{jk}+e_{kj})\quad (j<k),$$
$$i(e_{11}-e_{22}),\quad i(e_{11}+e_{22}-2e_{33}),$$

acting as left-invariant fields $X_a f(U) = \frac{d}{dt}f(Ue^{tX_a})|_{t=0}$, with weights $w_a = \frac12$ for the first seven and $\frac16$ for the last. Then $\sum_a w_a X_a\otimes X_a$ is the Casimir tensor of $\langle X,Y\rangle = -\mathrm{Tr}(XY)$, so

$$\langle\nabla f,\nabla g\rangle = \sum_a w_a X_a f\,X_a g, \qquad \Delta = \sum_a w_a X_a^2.$$

The fields map monomials to combinations of monomials of the same degree with coefficients in $\mathbb{Z}[i]$; for instance $\Delta\varphi_h = -\frac83\varphi_h$, as in Lemma 27.

### E. Orbits of staple sums

**Lemma 31.** *Let $V,W \in \mathrm{SU}(3)$.*

(i) *$L_h(VbW) = L_{V^{-1}hW^{-1}}(b)$ and $L_h(\bar b) = L_{\bar h}(b)$.*

(ii) *Every $b \in \mathbb{C}^{3\times 3}$ is in the orbit of some $\mathrm{diag}(a_1,a_2,z_3)$ with $a_1 \ge a_2 \ge |z_3|$ and $\operatorname{Im} z_3 \ge 0$.*

(iii) *A diagonal matrix lies in $K$ if and only if it lies in the convex hull of the diagonal torus $T$ of $\mathrm{SU}(3)$.*

*Proof.* (i) If $U$ has law $\nu_{VbW}$ then $WUV$ has law $\nu_b$; the map preserves $d$ and Haar measure, and $\varphi_h(U) = \varphi_{V^{-1}hW^{-1}}(WUV)$. For $\bar b$ use $U \mapsto \bar U$. (ii) Take a singular value decomposition $b = V_1 \Sigma W_1$ and move the determinants of $V_1, W_1$ into the last entry; a complex conjugation makes $\operatorname{Im} z_3 \ge 0$. (iii) Let $c$ be diagonal with nonzero entries of distinct moduli and let $U$ maximise $\operatorname{Re}\operatorname{Tr}(cU)$ on $\mathrm{SU}(3)$. The derivative along $\mathfrak{su}(3)$ vanishes, so the anti-Hermitian part of $cU$ is a multiple $i\lambda I$ of the identity and $cU = H + i\lambda I$ with $H$ Hermitian. Then $H^2 + \lambda^2 I = cc^*$ is diagonal with distinct entries, $H$ commutes with it and is diagonal, and $U \in T$. Hence $\max_{\mathrm{SU}(3)} \operatorname{Re}\operatorname{Tr}(cU) = \max_T \operatorname{Re}\operatorname{Tr}(ct)$ for such $c$, and by continuity for every diagonal $c$. A diagonal matrix outside $\operatorname{conv} T$ is separated from it by such a functional, which then separates it from $K$. □

Since every $b \in rK$ has operator norm at most $r$,

$$\Lambda(r) = \sup\Big\{ \sup_{\|h\|=1} L_h(b) : b \in \mathscr{R}_r \Big\},$$

where $\mathscr{R}_r$ is the set of $\mathrm{diag}(a_1,a_2,x_3+iy_3) \in r\operatorname{conv} T$ with $a_1 \ge a_2 \ge |x_3+iy_3|$ and $y_3 \ge 0$, a subset of the box $[0,r]^2 \times [-r,r] \times [0,r]$ in the coordinates $(a_1,a_2,x_3,y_3)$.

### F. Exact moments and interpolation

**Lemma 32.** *For $D, D' \in \mathbb{C}^{3\times 3}$ let $\mathbb{E}_H$ be the Haar mean on $\mathrm{SU}(3)$ and*

$$Z(D,D') = \mathbb{E}_H \exp\big(\tfrac12 \operatorname{Tr}(UD) + \tfrac12 \operatorname{Tr}(\bar U D')\big).$$

*Then*

$$Z(D,D') = \sum_{p\ge q\ge 0} \frac{s_{(p,q)}(DD'^{\top})\, G_{p,q}(\det D)\, G_{p,q}(\det D')}{d_{p,q}},$$

*with $G_{p,q}(z) = \sum_{k\ge 0} c_{(p+k,q+k,k)} z^k$, $c_\lambda = 2^{-|\lambda|}/\prod \mathrm{hooks}(\lambda)$, $d_{p,q} = \tfrac12 (p-q+1)(q+1)(p+2)$ and $s_{(p,q)}$ the Schur polynomial. The moments of degree $j$ in $U$ and $k$ in $\bar U$ are $\mathbb{E}_H[u^\alpha \bar u^\beta e^{\operatorname{Re}\operatorname{Tr}(Ub)}] = 2^{|\alpha|+|\beta|} \partial_D^\alpha \partial_{D'}^\beta Z(b,\bar b)$, the entry $u_{ij}$ pairing with $D_{ji}$.*

*Proof.* We use $(\operatorname{Tr} X)^n = \sum_{\lambda\vdash n} f^\lambda s_\lambda(X)$, where $f^\lambda$ is the number of standard tableaux of shape $\lambda$, given by the hook length formula [86, Cor. 7.21.6]. Expanding the exponential gives

$$e^{\frac12 \operatorname{Tr} X} = \sum_{n\ge 0} \frac{(\operatorname{Tr} X)^n}{2^n n!} = \sum_\lambda c_\lambda s_\lambda(X),$$

the sum over partitions with at most three parts. For $U \in \mathrm{SU}(3)$ one has $s_{\lambda+(k,k,k)}(UD) = (\det D)^k s_\lambda(UD)$, so

$$e^{\frac12 \operatorname{Tr}(UD)} = \sum_{p\ge q} G_{p,q}(\det D)\, \chi_{p,q}(UD),$$

where $\chi_{p,q}$ is the irreducible character of highest weight $(p,q,0)$ and $d_{p,q}$ its dimension. Since $\bar U = (U^{-1})^\top$, we have $\chi_{p,q}(\bar U D') = \chi_{p,q}(U^{-1} D'^{\top})$, and Schur orthogonality [26, Ch. 2],

$$\mathbb{E}_H\big[\chi_\rho(UA)\chi_{\rho'}(U^{-1}B)\big] = \delta_{\rho\rho'} \frac{\chi_\rho(AB)}{d_\rho},$$

gives the expansion. The bound below gives absolute convergence. Finally $\operatorname{Re}\operatorname{Tr}(Ub) = \tfrac12 \operatorname{Tr}(Ub) + \tfrac12 \operatorname{Tr}(\bar U \bar b)$. □

The computation keeps $p \le 34$ and $k \le 11$. Every omitted partition has more than 34 boxes, and since $|s_\lambda(X)| \le \dim(\lambda)\|X\|^{|\lambda|}$ and $\sum_{\lambda\vdash n} f^\lambda \dim\lambda = 3^n$ by Schur–Weyl duality, the omitted part of each factor is at most $\sum_{n>34} (\frac32\|D\|)^n/n!$; Cauchy's estimate on a polydisc around $b$ turns this into a bound for every derivative (Appendix E).

**Lemma 33** (Interpolation on a box)**.** *Let $Q = \prod_k [l_k, l_k + w_k]$ in the coordinates $(a_1,a_2,x_3,y_3)$, with corners $b_i$ and multilinear weights $\omega_i(b) \ge 0$, $\sum_i \omega_i(b) = 1$, and let $\varsigma = w_1^2 + w_2^2 + \max(w_3^2, w_4^2)$. For $b \in Q$:*

(i) *$Z(b) \ge e^{-\varsigma/8} \sum_i \omega_i(b) Z(b_i)$;*

(ii) *$\nu_b(f) \le e^{\varsigma/8} \max_i \nu_{b_i}(f)$ for every $f \ge 0$ that does not depend on $b$;*

(iii) *if $Y_b = (\mathscr{A} + \sum_k (b_k - \beta_k)\mathscr{B}_k)^2$ with functions $\mathscr{A}, \mathscr{B}_k$ of $U$ and a fixed point $\beta$, then*

$$\nu_b(Y_b) \le e^{\varsigma/8} \Big( \max_i \nu_{b_i}(Y_{b_i}) + \sum_k \frac{w_k^2}{4} \frac{\max_i Z(b_i)\nu_{b_i}(\mathscr{B}_k^2)}{\min_i Z(b_i)} \Big).$$

*Proof.* Write $E(b) = e^{\operatorname{Re}\operatorname{Tr}(Ub)} = \prod_k e^{c_k b_k}$, where

$$c = (\operatorname{Re} u_{11}, \operatorname{Re} u_{22}, \operatorname{Re} u_{33}, -\operatorname{Im} u_{33}),$$

so that $|c_k| \le 1$ and $c_3^2 + c_4^2 \le 1$. In each coordinate the chord of $e^{c_k x}$ over $[l_k, l_k + w_k]$ lies above the function and, by Hoeffding's lemma [54] for a two-point variable, below $e^{c_k x} e^{c_k^2 w_k^2/8}$. Multiplying,

$$E(b) \le \sum_i \omega_i(b) E(b_i) \le E(b)\, e^{\sum_k c_k^2 w_k^2/8} \le E(b)\, e^{\varsigma/8}. \tag{12}$$

Integrating the right inequality gives (i), and (ii) follows from the left one and (i). For (iii) let $\mathscr{F}(b) = Y_b E(b)$ and $I_k$ the linear interpolation in the $k$-th coordinate, so that $\mathscr{F} - I_1 I_2 I_3 I_4 \mathscr{F} = \sum_k I_1 \cdots I_{k-1}(\mathscr{F} - I_k \mathscr{F})$. Along the $k$-th coordinate $\mathscr{F}$ is $\ell^2 e^{c_k x}$ times a positive factor, with $\ell$ affine of slope $\mathscr{B}_k$, and $\partial_k^2 \mathscr{F} = ((c_k \ell + 2\mathscr{B}_k)^2 - 2\mathscr{B}_k^2) E \ge -2\mathscr{B}_k^2 E$. With the Green kernel of the interval, whose integral is at most $w_k^2/8$,

$$\mathscr{F} - I_k \mathscr{F} \le 2\mathscr{B}_k^2 \int G E \le \tfrac{w_k^2}{4} \mathscr{B}_k^2 \sum_i \omega_i^{(k)} E(b_i)$$

by the left inequality of (12), with weights $\omega_i^{(k)} \ge 0$ of sum one that do not depend on $U$. The operators $I_j$ have nonnegative weights of sum one; integrate over $U$ and divide by $Z(b)$, bounded below by (i). □

### G. The cover and the theorem

Fix a box $Q$ and a matrix $C$. For any constant $\kappa$, $\operatorname{Var}(X_h) \le \nu_b[(X_h-\kappa)^2]$; take $\kappa = \langle \kappa(b), h\rangle$ with $\kappa(b)$ affine in $b$, fitted to the means at the corners. By (11), $X_h - \kappa = \mathscr{A} + \sum_k (b_k-\beta_k)\mathscr{B}_k$ with $\beta$ the centre of $Q$ and $\mathscr{B}_k = \langle \nabla\varphi_{F_k}, \nabla q_h\rangle - \langle \partial_k\kappa, h\rangle$, and Lemmas 30 and 33 give, for all $b \in Q \cap \mathscr{R}_r$,

$$\sup_h L_h(b) \le \sqrt{e^{\varsigma/8}\max_i \lambda_A(b_i)} + \sqrt{\frac{e^{\varsigma/8}}{\gamma(r)}\Big(\max_i \lambda_V(b_i) + \frac{e_N}{\min_i Z(b_i)}\Big)}, \tag{13}$$

where $e_N = \sum_k \frac{w_k^2}{4}\max_i Z(b_i)\lambda_k(b_i)$ and $\lambda_A$, $\lambda_V$, $\lambda_k$ are the largest eigenvalues of the quadratic forms $h \mapsto \nu_{b_i}|\nabla q_h|^2$, $h \mapsto \nu_{b_i}[(X_h-\kappa)^2]$ at $b=b_i$, and $h \mapsto \nu_{b_i}[\mathscr{B}_k^2]$ on $\mathbb{R}^{18}$. Their entries are moments of $\nu_{b_i}$ of degree at most six, computed with rigorous error bounds (Appendix E); each largest eigenvalue is bounded above by an $LDL^\top$ factorisation of $tI$ minus the computed form with certified positive pivots, plus the norm of its error matrix by Weyl's inequality [51, 57].

The matrix $C$ is chosen in each box in floating point. At the mean of the corner moments and at $b=\beta$, Lemma 30 for $u = q_h$ is a quadratic problem in $C$: for $\theta > 0$ we minimise the trace of $\nu|\nabla q|^2 + \theta \operatorname{Var}(X)/\gamma$ over the 18 coordinate directions $h$, a least-squares problem of size 189, keep the $\theta$ on a geometric grid of 48 values from 0.05 to 80 that gives the smallest bound, and round $C$ to multiples of $2^{-30}$. Any choice gives a valid bound.

A box is discarded when it contains no point with $a_1 \ge a_2 \ge |z_3|$, or when a linear functional $\ell(b) = \operatorname{Re}\operatorname{Tr}(c^*b)$ satisfies $\min_Q \ell > r\max_T \ell$, which separates it from $r\operatorname{conv} T$ and hence, by Lemma 31(iii), from $\mathscr{R}_r$.

**Proposition 34.** *$2r\Lambda(r) \le 0.999887$ at $r = \frac{3}{2}$.*

*Proof.* The box $[0,r]^2 \times [-r,r] \times [0,r]$ is cut into 8192 boxes of side $r/8$. Of these, 841 survive the two tests, and 581 satisfy $2rB_Q < 1$ for the right side $B_Q$ of (13), with $\gamma(r) = \frac{187}{128}$. Each of the other 260 is cut into sixteen boxes of side $r/16$; 1760 of these survive the tests, and all satisfy $2rB_Q < 1$. The 2341 certified boxes cover $\mathscr{R}_r$. The largest bound is 0.999887, on the box $[1.3125, 1.5] \times [0.9375, 1.125] \times [0.5625, 0.75] \times [0, 0.1875]$, and the median is 0.983. Figure 6 shows the cover, and Appendix E the error analysis. □

The same cover at a larger $r$ bounds $\Lambda(r)$ where $2r\Lambda(r)$ is no longer below one; the plaquette blocks of Section VIII use that bound.

**Proposition 35.** *$2r\Lambda(r) \le 1.029967$ at $r = \frac{38}{25}$ and $2r\Lambda(r) \le 1.027468$ at $r = \frac{77}{50}$.*

*Proof.* As in the proof of Proposition 34, with $\gamma(r) = \frac{4643}{3200}$ and the threshold $2rB_Q < 1.03$: of the 8192 boxes of side $r/8$, 784 survive the two tests and 690 satisfy the threshold; each of the other 94 is cut into sixteen boxes of side $r/16$, of which 373 survive the tests, and all satisfy it. The 1063 certified boxes cover $\mathscr{R}_r$, and the largest bound is 1.029967. At $r = \frac{77}{50}$, with $\gamma(r) = \frac{4611}{3200}$ and the threshold $2rB_Q < 1.0275$: 826 boxes of side $r/8$ survive the tests and 513 satisfy the threshold; the other 313 are cut into sixteen, of which 2311 survive and 2281 satisfy it; the other 30 are cut into sixteen, of which 31 survive, and all satisfy it. The 2825 certified boxes cover $\mathscr{R}_r$, and the largest bound is 1.027468. □

*Proof of Theorem 6.* Proposition 34 gives

$$\Lambda\big(\tfrac{3}{2}\big) \le \frac{0.999887}{3} < 0.333296.$$

For $\beta_W \le \frac{3}{4}$ we have $2\beta_W \le \frac{3}{2}$, and (8) with the monotonicity of $\Lambda$ gives

$$\alpha \le 4\beta_W \Lambda\big(\tfrac{3}{2}\big) < 1.333183\,\beta_W \le \tfrac{3}{4}\cdot 1.333183 < 1.$$

Proposition 11 applies with the Hilbert–Schmidt metric. □

## VIII. SU(3) WITH PLAQUETTE BLOCKS

In this section $G = \mathrm{SU}(3)$, $\beta \ge 0$, $\beta_W = 9\beta$, $N\beta = \beta_W/3$, and the notation is that of Sec. VII: $d$ is the Hilbert–Schmidt distance, $K = \operatorname{conv}\mathrm{SU}(3)$, $\nu_c(dU) \propto e^{\operatorname{Re}\operatorname{Tr}(Uc)}dU$, $\varphi_h(U) = \operatorname{Re}\operatorname{Tr}(Uh)$ and $\Lambda(r)$ is the modulus of (8). Put $Z(c) = \int e^{\operatorname{Re}\operatorname{Tr}(Uc)}dU$ and $g = \log Z$ on $\mathbb{C}^{3\times 3}$, viewed as $\mathbb{R}^{18}$ with the inner product $\langle x,y\rangle = \operatorname{Re}\operatorname{Tr}(xy^*)$. The gradient of $g$ at $c$ is the conjugate mean $m(c)^*$, $m(c) = \nu_c(U) \in K$, and $\operatorname{Hess} g(c)(h,h) = \operatorname{Var}_{\nu_c}(\varphi_h)$. Since $Z(VcW) = Z(c)$ for $V, W \in \mathrm{SU}(3)$, the same holds for $g$.

The blocks are those of Lemma 12, which concerns the line field only and holds for every group. For a block plaquette with active links $e_1, e_2, e_3$, $\operatorname{Re}\operatorname{Tr} Q_p = \operatorname{Re}\operatorname{Tr}(y_1y_2y_3)$ with $y_i = Q_{e_i}$ or $Q_{e_i}^{-1}$, and the conditional law of the block is

$$\mu_b(dy) \propto \exp\Big(\sum_i \operatorname{Re}\operatorname{Tr}(y_ib_i) + N\beta\operatorname{Re}\operatorname{Tr}(y_1y_2y_3)\Big)dy, \tag{14}$$

where $b_i$ is $N\beta$ times the sum of the five staples of $y_i$ outside the block plaquette, so $b_i \in 5N\beta K$ and $\|b_i\|_{\mathrm{op}} \le 5N\beta$. On $\mathrm{SU}(3)^3$ we use $d(y,y') = \sum_i \|y_i - y_i'\|_{\mathrm{HS}}$.

### A. The sensitivity of one block

Let $r = 2\beta_W = 6N\beta$. Let $\lambda_*$ and $\tau$ be a lower and an upper bound for the eigenvalues of $\operatorname{Hess} g$ on $rK$, $M$ an upper bound on $rK$ for $|D^3g(c)(u,v,v)|$ over unit vectors $u, v$, and $\gamma = \gamma(r)$

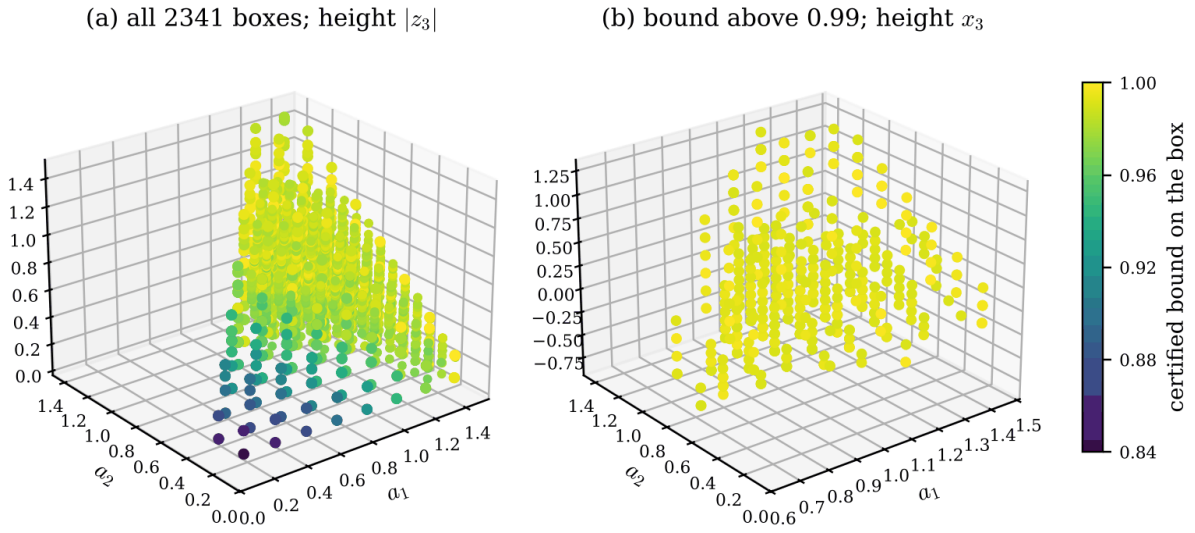



FIG. 6. The 2341 certified boxes at $r=\frac{3}{2}$, plotted at their centres in the coordinates $(a_1,a_2)$ of the staple sum $\mathrm{diag}(a_1,a_2,x_3+iy_3)$, coloured by the certified bound on $2r\sup_h L_h$ over the box; the larger points are the boxes of side $r/8$. (a) All boxes, with height $|x_3+iy_3|$. (b) The 419 boxes with bound above 0.99, with height $x_3$; the largest bounds sit in the interior of $\mathscr{R}_r$, not at its extreme points.

the gap of Lemma 29. Put

$$w=\tfrac{3}{2}(N\beta)^2(\tau-\lambda_*),\qquad s=\Lambda(r)+(e^w-1)e^w\sqrt{\tau/\gamma},$$
$$C=5(N\beta)^2\Big(\tau s+\frac{\sqrt{3}N\beta M e^w}{\gamma}\Big),$$
$$K_3=\frac{(1+N\beta\Lambda(r))\,s}{1-C},\qquad \alpha_3=10N\beta K_3, \tag{15}$$

assuming $C<1$.

**Lemma 36.** *Let $a,c_0\in 5N\beta K$ and*

$$\kappa(dx)\propto\exp\big(\mathrm{Re\,Tr}(xa)+g(c_0+N\beta x)\big)dx\quad \text{on } \mathrm{SU}(3).$$

*For every 1-Lipschitz $f$ and every $h\in\mathbb{C}^{3\times3}$:*

(i) $\mathrm{Cov}_\kappa(f,\varphi_h)\le s\|h\|_{\mathrm{HS}}$;

(ii) *if $c_0$ moves in the direction $k$ inside $5N\beta K$, then*

$$\big|\tfrac{d}{dt}\kappa(f)\big|\le N\beta\big(\tau s+\sqrt{3}N\beta M e^w/\gamma\big)\|k\|_{\mathrm{HS}}.$$

*Proof.* (i) The segment from $c_0$ to $c_0+N\beta x$ lies in $6N\beta K=rK$, and Taylor's formula gives

$$g(c_0+N\beta x)=g(c_0)+N\beta\,\mathrm{Re\,Tr}(x\,m(c_0))+R(x)$$

with $R(x)=\frac{1}{2}(N\beta)^2\mathrm{Hess}\,g(\xi)(x,x)$. Since $\|x\|_{\mathrm{HS}}^2=3$, $R\in[\frac{3}{2}(N\beta)^2\lambda_*,\frac{3}{2}(N\beta)^2\tau]$ and $\mathrm{osc}\,R\le w$. Hence $\kappa$ is the tilt $e^R\nu/\nu(e^R)$ of $\nu=\nu_{c'}$ with $c'=a+N\beta m(c_0)\in rK$. By definition of $\Lambda$, $\mathrm{Cov}_\nu(f,\varphi_h)\le\Lambda(r)\|h\|$. The function $f$ is 1-Lipschitz for the geodesic distance, which dominates $d$, so $\mathrm{sd}_\nu f\le\gamma^{-1/2}$ by Lemma 29, and $\mathrm{sd}_\nu\varphi_h\le\sqrt{\tau}\|h\|$. Lemma 13 gives (i).

(ii) The derivative is $\mathrm{Cov}_\kappa(f,\langle\nabla g(c_0+N\beta x),k\rangle)$. Write $\nabla g(c_0+N\beta u)=\nabla g(c_0)+N\beta\,\mathrm{Hess}\,g(c_0)u+N(u)$. The linear term is $\varphi_h$ with $\|h\|\le N\beta\tau\|k\|$ and contributes at most $N\beta\tau s\|k\|$ by (i). On the segment from $c_0$ the Hessian moves by at most $N\beta M\|u\|\le\sqrt{3}N\beta M$, so $x\mapsto\langle N(x),k\rangle$ is $\sqrt{3}(N\beta)^2M\|k\|$-Lipschitz for $d$. The law $\kappa$ is a tilt of $\nu$ by a function of oscillation at most $w$, so its spectral gap is at least $e^{-w}\gamma$ by the Holley–Stroock lemma [52], and the covariance of two Lipschitz functions under $\kappa$ is at most $e^w/\gamma$ times the product of their constants. □

**Lemma 37.** *Let $C<1$. If $b,b'$ have all fields in $5N\beta K$ and differ only in $b_i$, then $W_1(\mu_b,\mu_{b'})\le K_3\|b_i-b_i'\|_{\mathrm{HS}}$ for the metric $d$.*

*Proof.* By the cyclic symmetry of $\mathrm{Re\,Tr}(y_1y_2y_3)$ we may take $i=1$. Given $y_1,y_3$, the law of $y_2$ is $\nu_c$ with $c=b_2+N\beta y_3y_1\in rK$. Changing $y_1$ by $\eta$ in $d$ moves $c$ by $N\beta\eta$ and $\nu_c$ by at most $N\beta\Lambda(r)\eta$ in $W_1$, as in (8); so $F(y_1,y_3)=\mathbb{E}[f\mid y_1,y_3]$ is $(1+N\beta\Lambda(r))$-Lipschitz in each variable. The law of $(y_1,y_3)$ is proportional to $e^{\mathrm{Re\,Tr}(y_1b_1)+\mathrm{Re\,Tr}(y_3b_3)}Z(b_2+N\beta y_3y_1)$, and $Z(b_2+N\beta y_3y_1)=Z(y_3^{-1}b_2+N\beta y_1)=Z(b_2y_1^{-1}+N\beta y_3)$, so its two conditional laws have the form of Lemma 36, with $c_0=y_3^{-1}b_2$ and $c_0=b_2y_1^{-1}$. Moving $y_3$ to $y_3'$ moves $y_3^{-1}b_2$ along a segment in $5N\beta K$ of length at most $\|b_2\|_{\mathrm{op}}\|y_3-y_3'\|\le 5N\beta\|y_3-y_3'\|$. The coupling argument of Lemma 15 now gives $d_1\le s\varepsilon+Cd_3$, $d_3\le Cd_1$ and $|\mu_b(f)-\mu_{b'}(f)|\le(1+N\beta\Lambda(r))(d_1+d_3)\le K_3\varepsilon$. □

### B. The constants and the proof of Theorem 7

**Lemma 38.** *At $r=\frac{3}{2}$ every eigenvalue of* Hess $g$ *on $rK$ lies in* $[0.056015, 0.247316]$*, and* $|D^3 g(c)(u,v,v)| \le 0.182126$ *for $c \in rK$ and unit vectors $u, v$. At $r=\frac{38}{25}$ the same holds with* $[0.054978, 0.248280]$ *and* $0.183546$*, and at $r=\frac{77}{50}$ with* $[0.053955, 0.249240]$ *and* $0.184989$.

*Proof.* For $V, W \in \mathrm{SU}(3)$, $g(VcW) = g(c)$ and $x \mapsto VxW$ is an isometry, so the eigenvalues of Hess $g$ and the norm of $D^3 g$ are constant on the orbits, and by Lemma 31 it suffices to bound them on $\mathscr{R}_r$. The derivatives of $g$ are the cumulants of $\varphi$ under $\nu_c$. Cut the box $[0,r]^2 \times [-r,r] \times [0,r]$ into boxes of side $r/8$ and keep the 841 (784 at $r=\frac{38}{25}$, 826 at $r=\frac{77}{50}$) that survive the two tests of Sec. VII G. On each box $Q$ we compute, with rigorous error bounds, the moments of $\varphi$ of degree at most three at the sixteen corners $b_i$ and at the centre $b_0$, and put $k = m(b_0)$, the mean of $\varphi$ at the centre.

*Upper bound.* For $b \in Q$ and every $h$, $\mathrm{Var}_{\nu_b}(\varphi_h) \le \nu_b[(\varphi_h - \langle k,h\rangle)^2] \le e^{\varsigma/8} \max_i \nu_{b_i}[(\varphi_h - \langle k,h\rangle)^2]$ by Lemma 33(ii).

*Lower bound.* Integrating the right inequality of (12) against $f \ge 0$ and dividing by $Z(b) \le \sum_i \omega_i Z(b_i)$, the left one, gives $\nu_b(f) \ge e^{-\varsigma/8} \min_i \nu_{b_i}(f)$. With $f = (\varphi_h - \nu_b\varphi_h)^2$ this gives $\mathrm{Var}_{\nu_b}(\varphi_h) \ge e^{-\varsigma/8} \min_i \mathrm{Var}_{\nu_{b_i}}(\varphi_h)$.

*Third cumulant.* Put $\delta = m(b) - k$. The mean value theorem and the upper bound give $\|\delta\| \le \tau\rho + e$, with $\rho$ the largest distance from $b_0$ in $Q$ and $e$ the error of the computed mean. Expanding the central moment about $k$,

$$D^3 g(b)(u,v,v) = T_b(u,v,v) - \delta_u \mathrm{Var}(\varphi_v) - 2\delta_v \mathrm{Cov}(\varphi_u, \varphi_v) - \delta_u \delta_v^2,$$

with $T_b = \nu_b[(\varphi - k)^{\otimes 3}]$, $\delta_u = \langle \delta, u\rangle$ and the moments under $\nu_b$. For a function $G$ of either sign, writing $G = G^+ - G^-$ and applying the two bounds above to $G^\pm$ gives $|\nu_b(G)| \le (\max_i |\nu_{b_i}(G)| + \varepsilon \max_i \nu_{b_i}|G|)/(1-\varepsilon)$ with $\varepsilon = 1 - e^{-\varsigma/8}$. For $G = (\varphi_u - k_u)(\varphi_v - k_v)^2$ we use $|\nu_{b_i}(G)| \le \|T_{b_i}\|$, the largest singular value of the $18 \times 324$ unfolding of $T_{b_i}$, and $\nu_{b_i}|G| \le 2\sqrt{3}\, \nu_{b_i}[(\varphi_v - k_v)^2]$, since $\|U - k\|_{\mathrm{HS}} \le 2\sqrt{3}$. The largest eigenvalues and singular values are bounded as in Sec. VII G, and the maxima over these boxes are the numbers stated (Appendix C). □

*Proof of Theorem 7.* Let first $\beta_W \le \frac{3}{4}$, so that $r = 2\beta_W \le \frac{3}{2}$. Every field met in Lemmas 36 and 37 lies in $\frac{3}{2}K$, so the constants of Lemma 38 at $\frac{3}{2}$, $\Lambda(\frac{3}{2}) \le \frac{0.999887}{3}$ from Proposition 34 and $\gamma = \gamma(\frac{3}{2}) = \frac{187}{128}$ apply; with these fixed constants every quantity in (15) is increasing in $\beta_W$. For $\frac{3}{4} < \beta_W \le \frac{19}{25}$ the fields lie in $\frac{38}{25}K$, and the constants of Lemma 38 at $\frac{38}{25}$, $\Lambda(\frac{38}{25}) \le \frac{1.029967}{3.04}$ from Proposition 35 and $\gamma(\frac{38}{25}) = \frac{4643}{3200}$ apply in the same way; for $\frac{19}{25} < \beta_W \le \frac{77}{100}$ the same holds at $\frac{77}{50}$, with $\Lambda(\frac{77}{50}) \le \frac{1.027468}{3.08}$ and $\gamma(\frac{77}{50}) = \frac{4611}{3200}$. Changing an active link $j$ by $\eta$ changes one staple of each of its ten partners outside its block by a matrix of norm $\eta$, so the field $b_i$ of each such partner moves by $N\beta\eta$, and Lemma 37 gives $\sum_B C_{Bj} \le 10N\beta K_3 = \alpha_3(\beta_W)$. In ball arithmetic, at $\beta_W = \frac{3}{4}$, $w \le 0.017935$, $s \le 0.34088$, $C \le 0.043520$, $K_3 \le 0.38609$ and $\alpha_3 \le 0.96521$; at $\beta_W = 0.72$, $\alpha_3 \le 0.91815$; at $\beta_W = 0.76$, with the second set of constants, $w \le 0.018609$, $s \le 0.34673$, $C \le 0.045770$, $K_3 \le 0.39454$ and $\alpha_3 \le 0.99950$; at $\beta_W = 0.77$, with the third set, $w \le 0.019298$, $s \le 0.34186$, $C \le 0.047232$, $K_3 \le 0.38953$ and $\alpha_3 \le 0.99978$. Proposition 17, whose proof does not use the group, applies with the Hilbert–Schmidt metric. □

*Remark* 39. Theorem 6 ends where $2r\Lambda(r)$ reaches one. The blocks remove the factor $\frac{12}{10}$ of the single-link sum and keep most of it after paying for the three links: at $\beta_W = \frac{3}{4}$ the gap bound rises from $1.1 \cdot 10^{-4}$ to 0.0177, and the block bound is the better one for $\beta_W \ge 0.715$. Before the corrections for the three links their sum is $\frac{10}{12}$ of the single-link sum $2r\Lambda(r)$, so it can stay below one while $2r\Lambda(r)$ is slightly above one; Proposition 35 certifies $2r\Lambda(r) \le 1.03$ at $r = 1.52$ and $\le 1.0275$ at $r = 1.54$, which is enough at $\beta_W = 0.76$ and 0.77, with little room. Evaluated in floating point with the constants of a scan of $\mathscr{R}_r$, the block sum stays below one up to $\beta_W \approx 0.78$; that computation is not certified and is not used.

## IX. WHERE THE METHODS STOP

SU(2). Theorem 2 closes the single-link route in every forest gauge at $\beta_W = \frac{1}{3}$. Blocks of several links, where the influence of a boundary link on a whole block is bounded at once, in the spirit of the Dobrushin–Shlosman criterion [36], pass it: plaquette blocks reach $\beta_W = 0.351$ (Theorem 3), and the sharper floating-point bound of Remark 18 stops near 0.357. Larger blocks are the next step.

*Curvature.* Remark 26 locates the end of the curvature method in this gauge between $|\beta| = 1/28.840002$, by the certificate of Appendix L, and $\frac{1}{26.25}$.

SU(3). The route of Sec. VII bounds the $H^{-1}(\nu_b)$ norm of $\varphi_h - \nu_b\varphi_h$, which is at least $L_h(b)$, and needs it uniformly on $rK$. Its Galerkin values (polynomials of degree two) at matrices of $rK$ give $2r$ times the norm equal to 0.79 at $r = 1.25$, 0.89 at $r = 1.4$, 0.97 at $r = 1.5$ and 1.009 at $r = 1.55$, so the route stops at or below $\beta_W \approx 0.77$ even with exact constants. With $u = s\varphi_h$, Lemma 30 stops earlier: in floating point its largest values on $\mathscr{R}_r$ lie on the face $\mathrm{diag}(r, \rho, \rho)$ with $\rho$ between $0.75r$ and $0.8r$, and $2r$ times the bound is 0.9458 at $r = 1.36$, 0.9767 at $r = 1.40$ and 1.0080 at $r = 1.44$, so this corrector ends near $\beta_W \approx 0.715$. With $u = q_h$ of degree two the largest values on the same face are 0.9305 at $r = 1.44$, 0.9729 at $r = 1.50$ and 1.0018 at $r = 1.54$ (Fig. 7), close to the Galerkin values. Lemma 29 therefore costs little at these couplings, and most of the distance between the floating-point value at $r = 1.5$ and the certified 0.999887 of Proposition 34 is the interpolation over boxes. Between $\beta_W = \frac{3}{4}$ and 0.77 there is room only for smaller boxes. The plaquette blocks of Theorem 7 use $\Lambda(2\beta_W)$ as well, with a little more room: they reach 0.77 with the certified cover of Proposition 35 and near 0.78 in floating point (Remark 39); past that one needs larger blocks, whose sensitivity does not go through the one-link modulus.

*Other groups.* For groups other than SU(2) and SU(3) only the curvature route is carried out here (Theorem 5), and it stops

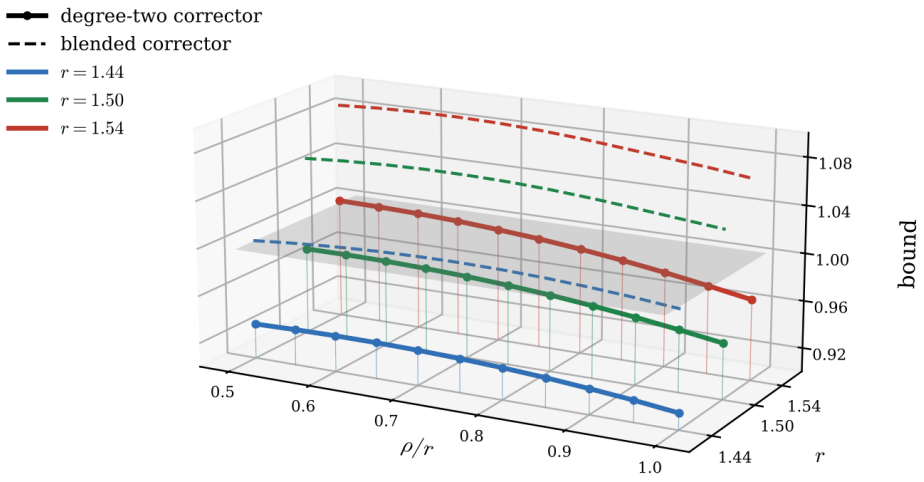



FIG. 7. The floating-point value of $2r\sup_h$ of the bound of Lemma 30 on the face $b = \mathrm{diag}(r,\rho,\rho)$, as a function of $\rho/r$, for $r = 1.44$, 1.50 and 1.54: dashed with the corrector $u = s\varphi_h$ for the best $s$, solid with $u = q_h$ of degree two for the best $C$. The grey plane is the Dobrushin threshold one. The first corrector is above it already at $r = 1.44$; the second stays below it at $r = 1.50$, where Proposition 34 certifies the bound on all of $\mathscr{R}_r$, and crosses it by $r = 1.54$.

where the bound $\frac{29}{2}$ uses up the curvature of $G$, at $N|\beta| = \frac{2}{29}\kappa_G$. The Dobrushin route needs a bound for the Kantorovich modulus of the one-link measure of each group, which is not attempted here.

*The four open problems.* Our earlier paper [16] listed four problems between the strong-coupling results and the Clay problem [56]: (1) uniqueness and a gap at intermediate fixed coupling; (2) the crossover from strong coupling to Bałaban's weak-coupling regime; (3) the continuum limit with the Osterwalder–Schrader axioms; and (4) a gap in physical units. The present paper moves the boundary of (1): for SU(2) from $\beta_W = \frac{2}{9}$ to 0.351, for SU(3) from $\frac{9}{32}$ to 0.77, with a gap at $\frac{3}{4}$ of 0.0177 in lattice units, and for every compact simple group, the exceptional groups included, past the curvature ceiling of the ungauged measure, with an explicit gap. It does not close (1); the region between these values and the scaling region, near $\beta_W \approx 2.2$ for SU(2) and $\beta_W \approx 6$ for SU(3), is open, and with it problem (2). Problems (3) and (4) are not solved here. Every bound proved here is uniform in lattice units, so by [17] none of them can by itself produce a continuum mass gap; that requires control of the gap in physical units along the renormalisation group flow [6, 19].[15]

*The continuum problem.* Two general facts limit what can be imported into problems (3) and (4) above from constructions of continuum measures. Both concern a random distribution $F$ on $\mathbb{R}^d$, a Borel probability measure on the tempered distributions with finite moments, and the growth condition

$$\mathbb{E}\, e^{F(\chi)} \le e^{C(\|\chi\|_1 + \|\chi\|_2^2)} \qquad \text{for all real } \chi \in \mathscr{S}(\mathbb{R}^d), \quad (16)$$

which is the growth condition of the regularity axiom of Glimm and Jaffe at $p = 2$ [46, Sect. 6.1, (6.1.5)]. The next proposition shows that it rules out every field whose two-point function is too singular at short distance.

**Proposition 40.** *Let $F$ satisfy (16), and suppose that its connected two-point function is, off the diagonal, a locally integrable function $S_2(x,y)$ with $S_2(x,y) \ge c|x-y|^{-2\Delta}\big(\log\frac{1}{|x-y|}\big)^{-k}$ for $0 < |x-y| \le r_0$, where $c > 0$, $k \ge 0$ and $2\Delta > d$. Then $F$ does not exist: no random distribution has both properties.*

*Proof.* Applying (16) to $\chi$ and $-\chi$ and using $\cosh x \ge 1 + \frac{x^2}{2}$ gives $\mathbb{E}F(\chi)^2 \le 2(e^{C(\|\chi\|_1+\|\chi\|_2^2)} - 1)$. Fix nonnegative $\varphi, \psi \in C_c^\infty(\mathbb{R}^d)$, not zero, with supports in a ball of radius $\frac{R}{2}$ and at distance $\delta > 0$ from each other, and put $\varphi_\varepsilon(x) = \varepsilon^{-d/2}\varphi(x/\varepsilon)$, and likewise $\psi_\varepsilon$. Then $\|\varphi_\varepsilon\|_2 = \|\varphi\|_2$ and $\|\varphi_\varepsilon\|_1 = \varepsilon^{d/2}\|\varphi\|_1 \le \|\varphi\|_1$ for $\varepsilon \le 1$, so the covariance of $F(\varphi_\varepsilon)$ and $F(\psi_\varepsilon)$ is bounded uniformly in $\varepsilon$ by the Cauchy–Schwarz inequality. On the other hand the supports of $\varphi_\varepsilon$ and $\psi_\varepsilon$ are disjoint, and for $u, v$ in the supports $\varepsilon\delta \le |\varepsilon u - \varepsilon v| \le \varepsilon R$. For $\varepsilon R \le r_0$ and $\varepsilon R < 1$ the lower bound for $S_2$ therefore gives

$$\begin{aligned}\mathrm{Cov}\big(F(\varphi_\varepsilon), F(\psi_\varepsilon)\big) &= \varepsilon^d \iint \varphi(u)\psi(v) S_2(\varepsilon u, \varepsilon v)\, du\, dv \\ &\ge c R^{-2\Delta} \varepsilon^{d-2\Delta} \Big(\log\frac{1}{\varepsilon\delta}\Big)^{-k} \|\varphi\|_1 \|\psi\|_1,\end{aligned}$$

which tends to infinity as $\varepsilon \to 0$. □

[15] Every one of these approaches needs an input at physical length scales. Renormalisation group approaches to the ultraviolet behaviour of gauge and gravitational theories, of which [6, 32] are the rigorous ones for Yang–Mills theory, have also been proposed for the effective dimension of spacetime [80] and for completions of quantum gravity [12, 18]; they do not bear on the infrared estimates needed here.

The action density $\operatorname{Tr} F_{\mu\nu}F_{\mu\nu}$ has dimension $\Delta = 4$ in four dimensions. A two-sided short-distance law of the form $|x-y|^{-8}$ times the fourth power of the running coupling, which is what asymptotic freedom[16] predicts, has $2\Delta = 8 > 4$ and $k = 2$; so that field cannot satisfy (16), and a proof of rotation invariance for it has to use something other than this condition. In three dimensions the free part of the action density already has $2\Delta = 6 > 3$. Condition (16) suits fields of dimension at most $\frac{d}{2}$, such as the field strength of the free Maxwell field, whose covariance is bounded in momentum space.

Positivity, symmetry and regularity do not produce a gap by themselves.

**Proposition 41.** *On $\mathbb{R}^4$ let $F$ be the centred Gaussian random distribution with covariance*

$$\widehat{C}(p) = \int_0^1 \frac{m^2}{|p|^2 + m^2}\, dm.$$

*Then $F$ is invariant under all Euclidean motions, reflection positive, satisfies (16) with $C = \frac{1}{2}$, and its Osterwalder–Schrader Hamiltonian has no gap above the vacuum.*

*Proof.* Since $0 \le \widehat{C} \le 1$, $\mathbb{E}e^{F(\chi)} = e^{\frac{1}{2}\langle \chi, C\chi\rangle} \le e^{\frac{1}{2}\|\chi\|_2^2}$, with the unitary Fourier transform. Invariance is clear. The covariance is the integral over $m \in (0,1)$, with the weight $m^2 dm$, of the covariances of free fields of mass $m$, each of which is reflection positive [46, Sect. 6.2]; a positive combination of reflection positive covariances is reflection positive, and so is the Gaussian measure with such a covariance [46, Sect. 6.2]. For $\chi(t,\mathbf{x}) = h(t)g(\mathbf{x})$ with $h \ge 0$ supported in $t > 0$, not zero, and $\hat{g}(0) \neq 0$, the vector $v = F(\chi)$ is orthogonal to the vacuum, and after the Fourier transform in $\mathbf{x}$,

$$\langle v, e^{-sH} v\rangle = \int_0^1 m^2\, dm \int \frac{|\hat{g}(\mathbf{k})|^2}{2\omega} e^{-\omega s} \times \Big(\int_0^\infty h(t) e^{-\omega t} dt\Big)^2 d\mathbf{k},$$

with $\omega = \sqrt{|\mathbf{k}|^2 + m^2}$ and constants absorbed in $\hat{g}$. If $H$ had a gap $m_0 > 0$, the left side would be at most $e^{-m_0 s}\|v\|^2$. For $0 < \eta < m_0$ the part of the integral with $m, |\mathbf{k}| \le \eta/2$ is at least $c_\eta e^{-\eta s}$ with $c_\eta > 0$, a contradiction as $s \to \infty$. □

The Gaussian field of Proposition 41 is not a gauge theory; it shows only that an infrared estimate, such as a bound $\langle v_a, T_a^{\lfloor t/a\rfloor} v_a\rangle \le C_v e^{-m_* t}$ uniform in the spacing $a$ along the continuum tuning, has to be proved in addition to the axioms. The lattice bounds of this paper are of this form only with $m_*$ proportional to $1/a$, and by [17] such bounds give trivial limits. The routes we know to a continuum gap therefore all stop at the same two estimates: the construction of the infinite-volume continuum measure past Bałaban's small-field analysis [6, 8], and a decay estimate in physical units at large scales.

[16] For SU($N$) the renormalisation group gives, with $b_0 = 11N/(48\pi^2)$ and the invariant scale $\Lambda$,

$$g(a)^{-2} = 2b_0 \log \tfrac{1}{a\Lambda} + O\big(\log\log \tfrac{1}{a\Lambda}\big),$$

so the inverse coupling $\beta_W = 2N/g(a)^2$ grows like $\frac{11N^2}{12\pi^2}\log\frac{1}{a\Lambda}$ along the continuum tuning.

## X. THE CONTINUUM STEP AT ONE PHYSICAL SCALE

The bounds of the preceding sections hold at fixed coupling and give a gap that is at least a fixed number in lattice units, and [17] shows that no such bound can produce a continuum gap. This section replaces the single-link condition of Proposition 11 by a condition on one cube of $n^4$ sites, averaged over its translates in the manner of Dobrushin and Shlosman [36], and proves that the condition gives a gap of order $1/n$ in lattice units. If it holds with $n$ proportional to $1/a$, that is, in a cube of fixed physical size, the gap survives in physical units and passes to every continuum limit (Theorem 45). The two statements that remain unproved, the cube estimate and the existence of the limit, are stated in Sec. XI.

### A. Averaged cubes

For a finite set $V \subset A$ let $\gamma_V(\cdot \mid Q)$ be the probability measure on $G^V$ with density proportional to $\exp\big(N\beta \sum_{p\cap V\neq\emptyset} \operatorname{Re}\operatorname{Tr} Q_p^F\big)$ with respect to Haar measure, the links outside $V$ being those of $Q$. Its conditional law of $Q_e$, $e \in V$, given the other links of $V$ is $\gamma_e^F$. Write $d_V(Q_V, Q'_V) = \sum_{i\in V} \delta(Q_i, Q'_i)$ and, for $j \in A \setminus V$,

$$C_{V,j} = \sup \frac{W_1^{d_V}\big(\gamma_V(\cdot \mid Q), \gamma_V(\cdot \mid Q')\big)}{\delta(Q_j, Q'_j)},$$

the supremum over pairs that differ only at $j$; $W_1^{d_V}$ is the Kantorovich distance for $d_V$. The coefficient vanishes unless $j$ shares a plaquette with a link of $V$.

Let $n \ge 2$ be even and let $V$ be the set of active links $(x,\mu)$ with $x \in [0,n)^4$. The direction field has period two, so the translations by $2\mathbb{Z}^4$ preserve $F$, $A$ and $\gamma^F$, and the translates $V_x = V + x$, $x \in 2\mathbb{Z}^4$, satisfy $C_{V_x, j+x} = C_{V,j}$. Every active link lies in exactly $N_n = (n/2)^4$ of them. A *class* is an orbit of $2\mathbb{Z}^4$ on $A$, that is, a residue of the base point modulo two together with a direction; there are at most 64 classes. For a class $c$ put

$$K_c = \sum_{j\in c\setminus V} C_{V,j}.$$

**Definition 42.** For $0 < \alpha < 1$, the condition $D(n,\alpha)$ at the coupling $\beta$ is $K_c \le \alpha N_n$ for every class $c$.

Since $\sum_{x\in 2\mathbb{Z}^4,\, j\notin V_x} C_{V_x,j} = \sum_x C_{V,j-x}[j - x \notin V] = K_{c(j)}$ for the class $c(j)$ of $j$, the condition says that the total influence of a link on the translates that do not contain it is at most the fraction $\alpha$ of the number of translates that do. Figure 8 shows the cubes.

For $n = 2$ it is a bound on blocks of 48 links. Distances between links are those of their midpoints; a link of $V$ and a link sharing a plaquette with $V$ are at distance at most $\rho_n = 2n$, and $c_n$ denotes the number of active links within distance $\rho_n$ of a given one.

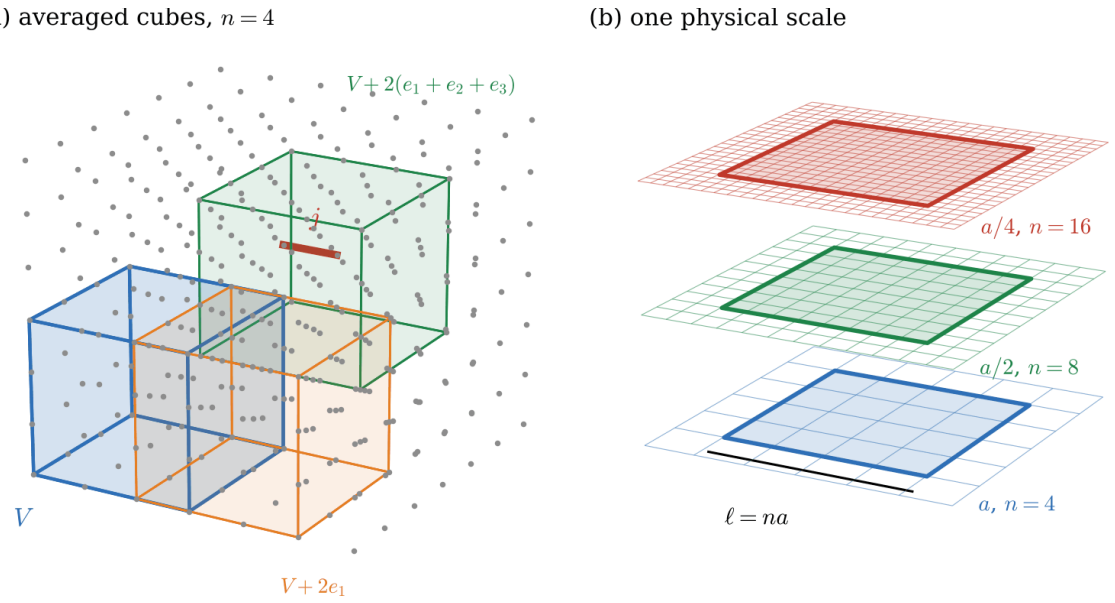



FIG. 8. (a) The cube $V$ of side $n=4$ and two of its translates by $2\mathbb{Z}^4$, drawn in three of the four dimensions. A link $j$ outside $V$ enters $K_c$ through every translate whose boundary it touches, and $D(n,\alpha)$ compares that total with the $N_n=(n/2)^4$ translates that contain $j$. (b) The cube of Theorem 45: its physical side $\ell=na$ is held fixed while the spacing is halved, so $n$ doubles at each step and the lattice gap $\log(1/\alpha)/(2n)$ is of order $a/\ell$.

**Theorem 43.** *Let $\beta\geq 0$ and suppose that $D(n,\alpha)$ holds. Then statements* (i)–(iv) *of Proposition 11 hold,* (v) *holds for every $\lambda<\log(1/\alpha)/\rho_n$, and the transfer matrix of $\omega$ has spectral gap at least* $\log(1/\alpha)/(2n)$.

*Proof.* The proof of Proposition 11 uses the coefficients $C_{ee'}$ only through Lemma 241 and Corollary 242. Lemma 244 of Appendix B gives the same conclusions under $D(n,\alpha)$, for every $\lambda>0$ with $\alpha e^{\lambda\rho_n}<1$, with the constant $\theta D/(1-\theta)$ replaced by $D(1+c_n e^{\lambda\rho_n})/(1-\theta)$ and $\theta=\alpha e^{\lambda\rho_n}$. The measures used there are invariant under every $\gamma_{V_x}$ with $V_x$ disjoint from the exceptional set: a Gibbs measure of $\gamma^F$ and its conditional measures given the links of a finite set by Lemma 244(a), and the extended finite-volume measure of part (ii) because every plaquette that meets such a $V_x$ lies in $P(\Lambda)$. Part (vi) then follows from Proposition 245 as before, with every $\lambda<\log(1/\alpha)/(2n)$. □

### B. The continuum limit

Let $a_k=2^{-k}a_0$ and let $\beta_k\geq 0$ be couplings, typically with $\beta_k\to\infty$. Suppose that the lattice states $\omega_k$ at $\beta_k$ exist as in Proposition 11, with transfer matrices $\mathbb{T}_k$ on the reconstructed spaces $\mathscr{H}_k$ and vacua $\Omega_k$ (Appendix F). Let $\mathscr{T}$ be a complex vector space of test functions on $\mathbb{R}^4$ supported in $\{t>0\}$ and, for each $k$, let $f\mapsto O_k(f)$ be a linear map from $\mathscr{T}$ to gauge-invariant local observables supported in positive lattice times; the example to keep in mind is $O_k(f)=Z_k\sum_x a_k^4 f(a_k x)\Phi(x)$ for a local gauge-invariant field $\Phi$, a renormalisation constant $Z_k$ and $\Phi(x)$ the translate of $\Phi$ to $x$. Write $\vartheta$ for the time reflection and $\tau_t$ for the time translation by $t$ lattice steps.

**Definition 44.** A *continuum limit* of $(\omega_k,O_k)$ is a Hilbert space $\mathscr{H}$ with a unit vector $\Omega$, a self-adjoint operator $H\geq 0$ with $H\Omega=0$, and a linear map $f\mapsto\psi(f)$ from $\mathscr{T}$ to $\mathscr{H}$ with dense range, such that for all $f,g\in\mathscr{T}$ and all dyadic $\tau\geq 0$,

$$\langle\psi(f),e^{-\tau H}\psi(g)\rangle=\lim_{k\to\infty}\omega_k\big(\vartheta(\overline{O_k(f)})\,\tau_{\tau/a_k}O_k(g)\big),$$
$$\langle\Omega,\psi(f)\rangle=\lim_{k\to\infty}\omega_k\big(O_k(f)\big),$$

the limits taken over the $k$ for which $\tau/a_k$ is an integer. It is *nontrivial* if $\mathscr{H}\neq\mathbb{C}\Omega$.

The Osterwalder–Schrader reconstruction [73, 74] produces such a limit from convergent renormalised Schwinger functions that satisfy the axioms; the definition records only what the next theorem uses.

**Theorem 45.** *Suppose that there are $\alpha<1$, $\ell>0$ and even $n_k$ with $n_k a_k\leq\ell$ such that $D(n_k,\alpha)$ holds at $\beta_k$ for every $k$. Then every continuum limit of $(\omega_k,O_k)$ satisfies*

$$\sigma(H)\subset\{0\}\cup[m_*,\infty),\qquad m_*=\frac{\log(1/\alpha)}{2\ell},$$

*and 0 is a simple eigenvalue. If the limit is nontrivial, $\sigma(H)\cap[m_*,\infty)\neq\emptyset$, so $H$ has a mass gap of at least $m_*$.*

*Proof.* By Theorem 43 the state $\omega_k$ exists and $\mathbb{T}_k$ has gap at least $\log(1/\alpha)/(2n_k)$ on the complement of $\Omega_k$. Let $f\in\mathscr{T}$, put $O=O_k(f)$ and $\varphi_k=\Psi_O-\omega_k(O)\Omega_k$. In the reconstruction of Appendix F, $\langle\Psi_F,\mathbb{T}_k^t\Psi_H\rangle=\omega_k(\vartheta(\bar F)\tau_t H)$, so $\langle\Omega_k,\Psi_O\rangle=\omega_k(O)$, and $\mathbb{T}_k\Omega_k=\Omega_k$; hence, for integers $t\geq 0$,

$$\langle\varphi_k,\mathbb{T}_k^t\varphi_k\rangle=\omega_k\big(\vartheta(\bar O)\,\tau_t O\big)-|\omega_k(O)|^2$$
$$\leq e^{-t\log(1/\alpha)/(2n_k)}\,\|\varphi_k\|^2.$$

Take $\tau \geq 0$ dyadic and $k$ large, so that $t = \tau/a_k$ is an integer; then $t/(2n_k) = \tau/(2n_k a_k) \geq \tau/(2\ell)$, and the right side is at most $e^{-m_*\tau}\|\varphi_k\|^2$. Both sides converge by Definition 44. With $\varphi = \psi(f) - \langle\Omega, \psi(f)\rangle\Omega$ and $e^{-\tau H}\Omega = \Omega$ the limit reads

$$\langle \varphi, e^{-\tau H}\varphi\rangle \leq e^{-m_*\tau}\|\varphi\|^2.$$

The maps $f \mapsto O_k(f)$ and $f \mapsto \psi(f)$ are linear, so the vectors $\varphi$ form a linear subspace, and it is dense in $\Omega^\perp$ because the range of $\psi$ is dense. The inequality extends to $\Omega^\perp$ by continuity, and from dyadic $\tau$ to all $\tau \geq 0$ by the strong continuity of $e^{-\tau H}$. For $\varphi \in \Omega^\perp$ with spectral measure $\mu_\varphi$ of $H$ this says $\int e^{-\tau E} d\mu_\varphi(E) \leq e^{-m_*\tau}\mu_\varphi([0,\infty))$ for all $\tau \geq 0$; if $\mu_\varphi([0, m_* - \eta]) > 0$ for some $\eta > 0$, the left side is at least $\mu_\varphi([0, m_* - \eta])e^{-(m_*-\eta)\tau}$, which exceeds the right side for large $\tau$. So $\mu_\varphi$ lives on $[m_*, \infty)$ for every $\varphi \in \Omega^\perp$, which gives both claims about 0, and the spectrum on $\Omega^\perp \neq \{0\}$ lies in $[m_*, \infty)$ and is not empty. □

The proof uses $D(n_k, \alpha)$ only through the decay of the two-point functions, and the same argument accepts any decay with constants that depend on the observable.

**Theorem 46.** *Let $m_0 > 0$ and let $\mathscr{T}_0$ span $\mathscr{T}$. Suppose that for each $f \in \mathscr{T}_0$ there are $C_f < \infty$, $\tau_f \geq 0$ and $k_f$ such that*

$$\begin{aligned} C_k^f(\tau) &= \omega_k\big(\vartheta(\overline{O_k(f)})\,\tau_{\tau/a_k} O_k(f)\big) - |\omega_k(O_k(f))|^2 \\ &\leq C_f e^{-m_0\tau} \end{aligned}$$

*for all $k \geq k_f$ and all dyadic $\tau \geq \tau_f$ with $\tau/a_k \in \mathbb{Z}$. Then every continuum limit of $(\omega_k, O_k)$ satisfies $\sigma(H) \subset \{0\} \cup [m_0, \infty)$, and 0 is a simple eigenvalue. If the limit is nontrivial, $H$ has a mass gap of at least $m_0$.*

*Proof.* For $f \in \mathscr{T}$ put $\varphi(f) = \psi(f) - \langle\Omega, \psi(f)\rangle\Omega$, which is linear in $f$. Since $e^{-\tau H}$ is self-adjoint and fixes $\Omega$, Definition 44 gives $\langle\varphi(f), e^{-\tau H}\varphi(f)\rangle = \lim_k C_k^f(\tau)$. For $f \in \mathscr{T}_0$ let $\mu$ be the spectral measure of $H$ at $\varphi(f)$. Then $\int e^{-\tau E} d\mu(E) \leq C_f e^{-m_0\tau}$ for dyadic $\tau \geq \tau_f$. If $\mu([0, m_0 - \eta]) > 0$ for some $\eta \in (0, m_0)$, the left side is at least $\mu([0, m_0 - \eta])e^{-(m_0-\eta)\tau}$, which is impossible for large $\tau$. Hence the spectral projection $E_{[0,m_0)}$ of $H$ vanishes on $\varphi(\mathscr{T}_0)$. By linearity it vanishes on $\varphi(\mathscr{T})$, and since its kernel is closed, it vanishes on the closure of $\varphi(\mathscr{T})$, which is $\Omega^\perp$ because $\psi(\mathscr{T})$ is dense. So $E_{[0,m_0)} = \langle\Omega, \cdot\rangle\Omega$, which gives both claims. □

Theorem 45 is the case $m_0 = m_*$, $\tau_f = 0$ and $C_f = \sup_{k \geq k_f}\|\varphi_k\|^2$, which is finite because $\|\varphi_k\|^2 = C_k^f(0)$ converges. Theorem 46 needs neither a lattice gap nor the gauge of this paper. It applies to the output of a cluster expansion of any coarse effective theory, provided the expansion bounds the correlations of a spanning set of renormalised observables in physical units.

Theorem 45 bypasses the obstruction of [17]: under the hypothesis the lattice gap is at least $\log(1/\alpha)/(2n_k)$, of order $a_k/\ell$ in lattice units, which tends to zero, and only its value in physical units is bounded below. In the language of [17] the hypothesis implies the gap part (R2) of Hypothesis R, with $m = m_*$, and the conclusion holds for every continuum limit without a separate assumption on the gap.

### C. Smoothed observables

The observables $O_k(f)$ of Definition 44 need not be local. The renormalised fields that one expects to converge are smoothed by the gradient flow of the Wilson action [65][17]; at positive flow time their correlation functions are finite to all orders of perturbation theory once the coupling is renormalised [66]. Flow in all four directions mixes the two sides of the reflection plane, but the spatial flow of [81, Sec. II.B] does not: it flows the links of each time slice under the Wilson action of that slice and leaves the temporal links alone.

Fix a bi-invariant Riemannian metric on $G$. For a configuration $U$ of all links of $\mathbb{Z}^4$ (for the states of this paper, the links of $F$ equal to 1), let $\Phi_s(U)$, $s \in \mathbb{R}$, solve

$$\frac{d}{ds}U_e = -\nabla_e \sum_{p \ni e,\ p \text{ spatial}} \big(N - \operatorname{Re}\operatorname{Tr} U_p\big)$$

for every spatial link $e$, the sum running over the four plaquettes of the time slice of $e$ that contain it, with the temporal links held fixed and $\nabla_e$ the gradient in the variable $U_e$.

**Proposition 47.** (i) *$\Phi_s$ is defined on every configuration of the infinite lattice, $\Phi_s \circ \Phi_{s'} = \Phi_{s+s'}$, and $\Phi_s$ commutes with gauge transformations, with spatial and temporal translations and with the reflections in time. The link $\Phi_s(U)_e$ depends only on the spatial links of the time slice of $e$, and there are constants $\kappa, c$ depending only on $G$ such that, if $\Phi_s^R$ denotes the flow of the links of that slice within distance $R$ of $e$ with all other links held fixed,*

$$\delta\big(\Phi_s(U)_e, \Phi_s^R(U)_e\big) \leq D e^{\kappa|s|}\frac{(c|s|)^{R+1}}{(R+1)!}$$

*for every configuration $U$.*

(ii) *Let $\omega$ be a state as in Appendix F and $O$ a bounded gauge-invariant function of finitely many links at times in $[t_0, t_1]$, continuous in them. Then $O \circ \Phi_s$ is a bounded gauge-invariant function of the links at times in $[t_0, t_1]$ and a uniform limit of gauge-invariant local functions at those times.*

(iii) *Theorem 45 holds with $O_k(f)$ replaced by $O_k(f) \circ \Phi_{s_k}$ for any $s_k \in \mathbb{R}$, in particular for $s_k = \tau/a_k^2$ at a fixed flow time $\tau$ in physical units.*

(iv) *The image measure $(\Phi_s)_*\omega$ is reflection positive, and its transfer matrix is unitarily equivalent to that of $\omega$, the unitary fixing the vacuum.*

*Proof.* (i) Embed $G$ in a matrix algebra. The right side of the equation is a smooth function of the links of the four plaquettes containing $e$, tangent to $G$ at $U_e$, so on the Banach space of bounded assignments of matrices to the spatial links it is a vector field that is bounded and Lipschitz on a neighbourhood of

[17] The flow moves the gauge field along the negative gradient of the Wilson action. At positive flow time the flowed fields are smooth on the scale of the square root of the flow time, and in perturbation theory their gauge-invariant composite fields need no further renormalisation [65, 66].

the configurations with values in $G$, with a Lipschitz constant that does not depend on the volume because every link meets finitely many others. The Picard–Lindelöf theorem gives a unique local solution; it stays on $G$ because the field is tangent there, and since the field is bounded on $G$ the solution exists for all $s$; uniqueness gives the group law. The action of a time slice and the metric are invariant under gauge transformations, translations and the reflections, so the flow commutes with them by uniqueness. Since the equation of a spatial link involves only links of its own slice, so does the solution. For the comparison, let $d_{e'}(s)$ be the distance between the two flows at a link $e'$ of the slice. Smoothness on the compact group gives constants $\kappa_0, c_1$ with $d_{e'}(s) \le \int_0^{|s|} \big(\kappa_0 d_{e'} + c_1 \sum_{e''\sim e'} d_{e''}\big)$ for $e'$ within distance $R$ of $e$, where $e'' \sim e'$ means that the two links share a spatial plaquette (twelve neighbours), while $d_{e'} \le D$ beyond. Iterating the integral form of this inequality, a term that reaches the links beyond distance $R$ needs at least $R+1$ steps of $\sim$, and $\sum_{k>R} x^k/k! \le e^x x^{R+1}/(R+1)!$; this gives the bound with $c = 12c_1$ and $\kappa = \kappa_0 + c$.

(ii) The truncated flow $\Phi_s^R$ commutes with gauge transformations for the same reason as $\Phi_s$, so $O\circ\Phi_s^R$ is a gauge-invariant function of finitely many links at times in $[t_0,t_1]$. By (i) and the uniform continuity of $O$, it converges to $O\circ\Phi_s$ uniformly as $R\to\infty$.

(iii) The proof of Theorem 45 uses $O_k(f)$ only through the vectors $\Psi_{O_k(f)}$ and the identity $\langle \Psi_F, \mathbb{T}^t \Psi_H\rangle = \omega(\vartheta(\bar F)\tau_t H)$. By (ii) and the footnote in Appendix F, both extend to uniform limits of local functions at positive times, and $f \mapsto O_k(f)\circ\Phi_{s_k}$ is linear.

(iv) By (i), $\Phi_s$ and $\Phi_{-s}$ preserve the functions of the links at positive times and commute with $\vartheta$ and $\tau_t$. Hence $(\Phi_s)_*\omega\big(\vartheta(\bar F)\tau_t H\big) = \omega\big(\vartheta(\overline{F\circ\Phi_s})\,\tau_t(H\circ\Phi_s)\big)$, which for $t = 0$ and $H = F$ is nonnegative, and $F \mapsto F\circ\Phi_s$ induces an isometry between the reconstructed spaces that intertwines the transfer matrices and fixes the vacuum; $\Phi_{-s}$ gives its inverse. □

Part (iv) says that smoothing changes the variables and not the theory: it neither creates nor removes a gap. Its use is in the choice of the observables. With $O_k(f)$ the spatially flowed fields at a fixed physical flow time, for instance $Z_k \sum_x a_k^4 f(a_k x)\big(\mathscr{E}_\tau(x) - \omega_k(\mathscr{E}_\tau(x))\big)$ with $\mathscr{E}_\tau(x)$ the flowed action density of the slice, statement (E) asks for the convergence of correlation functions that are known to be finite in perturbation theory [66], and Theorem 45 applies to them. Theorem 48 below constructs their continuum limits along subsequences; their nontriviality is not proved here.

### D. Renormalised flowed observables

The spatially flowed fields can be renormalised by one condition at a single physical separation, and their correlation functions then have continuum limits along subsequences without further assumptions. What remains open is whether such a limit is trivial, and that question reduces to one ratio of lattice two-point functions.

Fix a bounded gauge-invariant function $\Phi$ of finitely many spatial links of the slice $\{x_0 = 0\}$, continuous in them, a flow time $\tau > 0$ in physical units, and a nonzero test function $f_0 \in C_c(\mathbb{R}^4)$ supported in $\{t > 0\}$. Put $s_k = \tau/a_k^2$, write $\Phi_x$ for the translate of $\Phi$ to $x \in \mathbb{Z}^4$, and let

$$A_k = \sum_{x\in\mathbb{Z}^4} a_k^4 f_0(a_k x)\big(\Phi_x\circ\Phi_{s_k} - \omega_k(\Phi_x\circ\Phi_{s_k})\big),$$

$$C_k(\sigma) = \omega_k\big(\vartheta(\bar A_k)\,\tau_{\sigma/a_k} A_k\big)$$

for dyadic $\sigma \ge 0$ (the sum is finite, and $\sigma/a_k$ is an integer for large $k$). By reflection positivity $C_k(0) \ge 0$; assume $C_k(0) > 0$ and set $Z_k = C_k(0)^{-1/2}$. Let $\mathscr{D}$ be the set of points $y = (t, y') \in \mathbb{R}^4$ with dyadic coordinates (in units of $a_0$) and $t > 0$, and for $y \in \mathscr{D}$ and $k$ with $y/a_k \in \mathbb{Z}^4$ let $A_k^y$ be the translate of $A_k$ by $y/a_k$.

**Theorem 48.** (i) *For $y, y' \in \mathscr{D}$ and every $k$ with $y/a_k, y'/a_k \in \mathbb{Z}^4$,*

$$Z_k^2\,\big|\omega_k\big(\vartheta(\bar A_k^y) A_k^{y'}\big)\big| \le 1.$$

(ii) *Every sequence of $k$ has a subsequence along which $K(y,y') = \lim Z_k^2\,\omega_k(\vartheta(\bar A_k^y)A_k^{y'})$ exists for all $y, y' \in \mathscr{D}$ and $\rho(\sigma) = \lim C_k(\sigma)/C_k(0)$ exists for all dyadic $\sigma > 0$. Along it there is a probability measure $\mu$ on $[0,\infty]$ with*

$$\rho(\sigma) = \int_{[0,\infty]} e^{-\sigma E}\, d\mu(E) \qquad (\sigma > 0),$$

*the integrand being $0$ at $E = \infty$.*

(iii) *Along such a subsequence, let $\mathscr{T}$ be spanned by the translates $f_0(\cdot - y)$, $y \in \mathscr{D}$, and one more function $f_* \in C_c(\mathbb{R}^4)$ supported in $\{t > 0\}$ outside their span, and let $O_k$ be the linear map with $O_k(f_0(\cdot - y)) = Z_k A_k^y$ and $O_k(f_*) = 1$. Then $(\omega_k, O_k)$ has a continuum limit in the sense of Definition 44, on which the dyadic spatial translations act by unitary operators commuting with $H$.*

(iv) *This limit is nontrivial if and only if $\mu([0,\infty)) > 0$, if and only if $\rho(\sigma) > 0$ for one dyadic $\sigma > 0$, and then $\rho(\sigma) > 0$ for all of them. Consequently a nontrivial limit exists along some subsequence if and only if $\limsup_k C_k(\sigma)/C_k(0) > 0$ for some dyadic $\sigma > 0$.*

(v) *If the hypothesis of Theorem 45 holds, then $\mu$ is carried by $[m_*,\infty]$ and $\rho(\sigma) \le e^{-m_*\sigma}$.*

*Proof.* By Proposition 47(ii) each $\Phi_x\circ\Phi_{s_k}$, and hence each $A_k^y$, is a uniform limit of gauge-invariant local functions at times $\ge 1$, so the reconstruction of Appendix F applies to them (its footnote). Write $\Psi_k = Z_k \Psi_{A_k}$. Since $\langle \Omega_k, \Psi_{A_k}\rangle = \omega_k(A_k) = 0$ and $\|\Psi_k\|^2 = Z_k^2 C_k(0) = 1$, $\Psi_k$ is a unit vector orthogonal to $\Omega_k$. The relation $\langle \Psi_F, \mathbb{T}_k^t \Psi_H\rangle = \omega_k(\vartheta(\bar F)\tau_t H)$ for all $F$ gives $\Psi_{\tau_t H} = \mathbb{T}_k^t \Psi_H$. Spatial translations commute with $\vartheta$ and with $\tau_t$ and leave $\omega_k$ invariant, so they act on the reconstructed space by unitaries $U_k(x)$ that commute with $\mathbb{T}_k$. Hence, for $y = (t, y')$,

$$Z_k \Psi_{A_k^y} = U_k(y'/a_k)\,\mathbb{T}_k^{t/a_k}\Psi_k.$$

(i) The left side of the inequality is $|\langle Z_k\Psi_{A_k^y}, Z_k\Psi_{A_k^{y'}}\rangle| \le 1$, because $\|U_k\| = 1$ and $0 \le \mathbb{T}_k \le 1$.

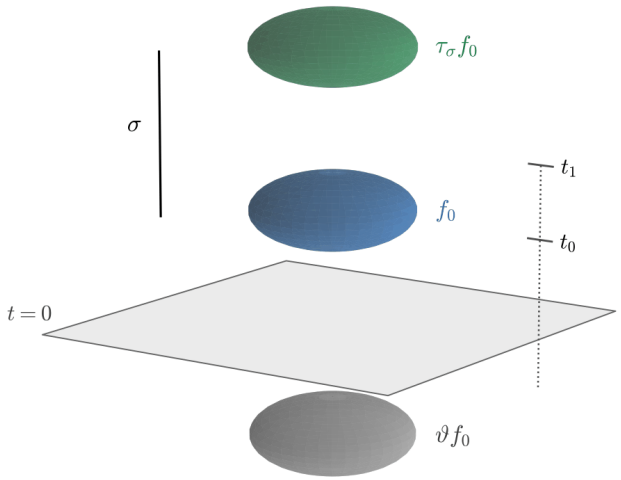



FIG. 9. The packet $f_0$ of Theorem 48, supported in the time window $[t_0,t_1]$ above the reflection plane $t=0$, its reflection $\vartheta f_0$ and its translate $\tau_\sigma f_0$ by $\sigma$ in time. The correlation $C_k(\sigma)$ pairs the reflected packet with the translated one; $C_k(0)$ is the reflected norm that renormalises the packet.

(ii) The set $\mathscr{D}\times\mathscr{D}$ is countable and the numbers are bounded by (i), so a diagonal subsequence gives all the limits $K$; the limits $\rho(\sigma)$ are among them, since $C_k(\sigma)/C_k(0)=\langle\Psi_k,\mathbb{T}_k^{\sigma/a_k}\Psi_k\rangle$ is the value of $K$ at $y=y'=(\sigma/2,0)$. Let $\nu_k$ be the spectral measure (Lemma 246) of $\mathbb{T}_k$ at $\Psi_k$, a probability measure on $[0,1]$, and $\mu_k$ its image under $\lambda\mapsto -a_k^{-1}\log\lambda\in[0,\infty]$. Then $C_k(\sigma)/C_k(0)=\int e^{-\sigma E}d\mu_k(E)$. The space $[0,\infty]$ is compact, so a further subsequence has $\mu_k\to\mu$ weakly (Lemma 247), and $E\mapsto e^{-\sigma E}$ is continuous on $[0,\infty]$ for $\sigma>0$.

(iii) A finite combination $\sum_j c_j f_0(\cdot-y_j)$ with distinct $y_j$ that vanishes has Fourier transform $\hat f_0(p)\sum_j c_j e^{-ip\cdot y_j}=0$; since $\hat f_0$ is analytic and not zero, the exponential sum vanishes on an open set and every $c_j$ is zero. So the translates are linearly independent and $O_k$ is well defined. On the space $V$ of finite sums $v=\sum_y c_y\delta_y$ over $\mathscr{D}$ put $B(v,w)=\sum\bar c_y c'_{y'}K(y,y')$. It is the limit of $\langle v_k,w_k\rangle$ with $v_k=\sum_y c_y Z_k\Psi_{A_k^y}$, hence positive semidefinite; let $\mathscr{H}_0$ be the completion of $V$ modulo its null space. For dyadic $\sigma\geq 0$ and dyadic spatial $x$ define $T(\sigma)\delta_y=\delta_{y+(\sigma,0)}$ and $U(x)\delta_y=\delta_{y+(0,x)}$. At level $k$ these are $\mathbb{T}_k^{\sigma/a_k}$ and $U_k(x/a_k)$, so in the limit $B(T(\sigma)v,w)=B(v,T(\sigma)w)$, $B(T(\sigma)v,T(\sigma)v)\leq B(v,v)$, $B(v,T(\sigma)v)\geq 0$, and $U(x)$ preserves $B$. Thus $T(\sigma)$ extends to commuting positive contractions on $\mathscr{H}_0$ with $T(\sigma)T(\sigma')=T(\sigma+\sigma')$, and $U(x)$ to unitaries commuting with them. For dyadic $\sigma>0$ the kernel of $T(\sigma)=T(\sigma/2)^2$ equals that of $T(\sigma/2)$, so all these kernels coincide; and every $\delta_y$ with $y=(t,y')$ equals $T(t/2)\delta_{(t/2,y')}$, a vector in the range of a positive operator and so orthogonal to its kernel. Hence each $T(\sigma)$ is injective on $\mathscr{H}_0$, and $H_0=-\log T(1)$ is a self-adjoint operator with $H_0\geq 0$ and $e^{-\sigma H_0}=T(\sigma)$ for dyadic $\sigma$, by the uniqueness of positive roots. Put $\mathscr{H}=\mathbb{C}\Omega\oplus\mathscr{H}_0$, $H=0\oplus H_0$, $\psi(f_0(\cdot-y))=\delta_y$ and $\psi(f_*)=\Omega$. The range of $\psi$ is dense, and the identities of Definition 44 hold: for two translates they are the limits defining $K$, with $\tau_{\sigma/a_k}A_k^{y'}=A_k^{y'+(\sigma,0)}$; and for $f_*$ they read $\omega_k(1)=1$ and $\omega_k(\vartheta(A_k^y))=\omega_k(\tau_{\sigma/a_k}A_k^y)=0$.

(iv) The limit is nontrivial exactly when $\mathscr{H}_0\neq 0$, that is when $\|\delta_y\|^2=K(y,y)=\rho(2t)$ is positive for some $y=(t,y')$. By (ii) and Lemma 248(b), $\rho(\sigma)>0$ for one $\sigma>0$ if and only if $\mu([0,\infty))>0$, and then $\rho(\sigma)>0$ for every $\sigma>0$. For the last statement choose a subsequence along which $C_k(\sigma)/C_k(0)$ tends to its lim sup, and refine it by (ii).

(v) Under that hypothesis the proof of Theorem 45 gives $\nu_k([0,e^{-\log(1/\alpha)/(2n_k)}])=1$ because $\Psi_k\perp\Omega_k$, so $\mu_k$ is carried by $[\log(1/\alpha)/(2n_k a_k),\infty]\subset[m_*,\infty]$, and so is the weak limit $\mu$. □

The measure $\mu$ is the spectral measure of the renormalised flowed field in the limit: its part on $[0,\infty)$ is what survives as a state of the continuum theory, and its atom at $\infty$ is the weight that the field puts on energies that diverge in physical units. The theorem constructs the renormalised continuum observables and their reconstruction for every tuning, and it reduces their nontriviality to the lattice inequality

$$\limsup_{k\to\infty}\frac{C_k(\sigma)}{C_k(0)}>0$$

for one physical separation $\sigma$: the flowed two-point function of the smeared packet at a fixed physical distance must not

become negligible against its reflected norm $C_k(0)$. If the lattice gap stayed above some $m>0$ in lattice units, the ratio would be at most $e^{-m\sigma/a_k}$ and the limit would be trivial, so the inequality is a condition on the tuning $\beta_k\to\infty$; it is not proved here. Nor are the independence of the limit from the subsequence, Euclidean invariance, or bounds on products of more than two renormalised fields, for which (i) gives nothing because $Z_k$ may diverge (Theorem 57 reduces them to tree bounds); (E) requires all of these.

The denominator $C_k(0)$ is the reflected norm of the packet smeared by $f_0$, not a coincidence limit of point fields, and the distinction matters. For the free massless scalar field in four dimensions with spatial momenta cut off at $\Lambda$, the covariance at spatial separation 0 is $(1-(1+\Lambda t)e^{-\Lambda t})/(4\pi^2t^2)$ at time $t>0$ and $\Lambda^2/(8\pi^2)$ at coincidence, so its ratio $2(1-(1+\Lambda t)e^{-\Lambda t})/(\Lambda t)^2$ tends to zero as $\Lambda\to\infty$, although the limiting field is the nontrivial Gaussian field; for packets smeared by $f_0$ the reflected norms of that field have finite positive limits and the ratio stays positive. The criterion of Theorem 48(iv) is about the smeared packets $A_k$ only.

When the ratio is bounded below, renormalising at the separation $\sigma$ instead of 0 removes the escape to infinite energy altogether, and one slab of the transfer matrix gives a sufficient condition for the bound. Write $\mu_k$ for the measures in the proof of Theorem 48.

**Proposition 49.** (i) *Suppose $C_k(\sigma)\ge cC_k(0)$ for a dyadic $\sigma>0$, a $c>0$ and every $k$ of a sequence. Let $\mu_k'$ be the spectral measure, in energies $E=-a_k^{-1}\log\lambda$, of the unit vector $C_k(\sigma)^{-1/2}\Psi_{A_k'}$ with $A_k'=\tau_{\sigma/2a_k}A_k$. Then $\mu_k'$ is a probability measure on $[0,\infty)$,*

$$\mu_k'([\Lambda,\infty))\le c^{-1}e^{-\sigma\Lambda},\qquad \int E^m\,d\mu_k'\le c^{-1}\Big(\frac{m}{e\sigma}\Big)^m,$$

*and $t\mapsto C_k(\sigma+t)/C_k(\sigma)$ is Lipschitz on $[0,\infty)$ with constant $(e\sigma c)^{-1}$. A subsequence of the $\mu_k'$ converges weakly to a probability measure on $[0,\infty)$ with the same bounds.*

(ii) *Suppose $\langle\Psi_k,(1-\mathbb{T}_k)\Psi_k\rangle\le Ca_k$ for a constant $C$ and every $k$ of a sequence. Then $C_k(\sigma)/C_k(0)\ge e^{-2C\sigma}$ whenever $Ca_k\le\frac12$ and $\sigma/a_k$ is an integer, and $\mu_k([\Lambda,\infty])\le C(a_k+\Lambda^{-1})$ for every $\Lambda>0$.*

*Proof.* (i) We have $\Psi_{A_k'}=\mathbb{T}_k^{\sigma/2a_k}\Psi_{A_k}$, so the measure $\mu_k'$ is $e^{-\sigma E}d\mu_k(E)$ divided by its mass $C_k(\sigma)/C_k(0)\ge c$, and $e^{-\sigma E}$ vanishes at $E=\infty$. For $E\ge\Lambda$ we have $e^{-\sigma E}\le e^{-\sigma\Lambda}$, and $E^me^{-\sigma E}\le(m/e\sigma)^m$ for all $E\ge0$. The ratio in question is the integral of $e^{-tE}$ against $\mu_k'$, and the inequality $|e^{-tE}-e^{-sE}|\le|t-s|E$ with the moment bound for $m=1$ gives the Lipschitz bound. The tail bound makes the sequence tight on $[0,\infty)$, so Prokhorov's theorem gives the weak limit, and the bounds pass to it because $\{E>\Lambda\}$ is open and $E^m$ is a monotone limit of bounded continuous functions.

(ii) Let $\nu_k$ be the spectral measure of $\mathbb{T}_k$ at $\Psi_k$ on $[0,1]$, a probability measure, and $n=\sigma/a_k$. By Jensen's inequality for the convex function $\lambda^n$,

$$\frac{C_k(\sigma)}{C_k(0)}=\int\lambda^n d\nu_k\ge\Big(\int\lambda\,d\nu_k\Big)^n\ge(1-Ca_k)^n\ge e^{-2Ca_kn},$$

since $\log(1-x)\ge-2x$ on $[0,\frac12]$. Moreover $Ca_k\ge\int(1-e^{-a_kE})d\mu_k\ge(1-e^{-a_k\Lambda})\mu_k([\Lambda,\infty])$, and $1-e^{-x}\ge x/(1+x)$. □

Part (ii) asks for an upper bound on a one-slab Dirichlet form, the opposite direction to a mass gap, which bounds such forms from below. In the box of Proposition 50, with $g$ smooth, the time smearing supplies the factor $a_k$ up to one ratio: in the notation of its proof, $\mathbb{T}_k=A$ and $\Psi_{A_k}=\sum_n w_n\mathbb{T}_k^nu$ with $w_n=a_kg(a_kn)$ and $u=a_k^3F^c\psi_0$, and the identity $(1-\mathbb{T}_k)\Psi_{A_k}=\sum_n(w_n-w_{n-1})\mathbb{T}_k^nu$ gives $\langle\Psi_k,(1-\mathbb{T}_k)\Psi_k\rangle\le a_k\|g'\|_{L^1}\|u\|/\|\Psi_{A_k}\|$, so the hypothesis of (ii) holds when the vector $u$ of a single slice stays comparable to the time-smeared packet. That comparison is not proved; Proposition 51 below bounds the ratio without it.

A one-slab bound of this kind also limits the mass that $D(n_k,\alpha)$ can supply. Let $e^{-m_k}$ be the norm of $\mathbb{T}_k$ on $\Omega_k^\perp$. If a unit vector $\varphi_k\perp\Omega_k$ has $\langle\varphi_k,(1-\mathbb{T}_k)\varphi_k\rangle\le\kappa a_k<1$, then the spectral measure of $\mathbb{T}_k$ at $\varphi_k$ lives on $[0,e^{-m_k}]$, so $1-e^{-m_k}\le\kappa a_k$ and $m_k\le-\log(1-\kappa a_k)$. If in addition $D(n_k,\alpha)$ holds with $n_ka_k\le\ell$, Theorem 43 gives $m_k\ge\log(1/\alpha)/(2n_k)\ge a_k\log(1/\alpha)/(2\ell)$, and therefore

$$\frac{\log(1/\alpha)}{2\ell}\le\frac{-\log(1-\kappa a_k)}{a_k}=\kappa+O(a_k).$$

So the physical mass $m_*$ of Theorem 45 is at most the constant $\kappa$ of any one-slab bound. The two hypotheses are compatible because both are statements in physical units. The one-slab bound forces $m_k\to0$, so it cannot come from an expansion whose convergence gives a lattice gap bounded below in lattice units.

In a periodic spatial box the assumption $C_k(0)>0$ is a theorem. Let $G=\mathrm{SU}(N)$ with its defining representation, and replace $\mathbb{Z}^4$ by $\mathbb{Z}\times(\mathbb{Z}/L_k\mathbb{Z})^3$ with $L_k\ge3$ and $\omega_k$ by the infinite-time Wilson state of the box. The proof of Theorem 48 uses only reflection positivity and the transfer matrix once the spatial translations are dropped, and parts (i)–(iv) hold in the box with $\mathscr{D}$ replaced by its points $(t,0)$, the time translates of $f_0$ being independent by the same Fourier argument in $t$.

**Proposition 50.** *In the box, let $f_0(t,x)=g(t)h(x)$ with $g,h\ge0$, and let $\Phi=\sum_{i<j}(N-\mathrm{Re}\,\mathrm{Tr}\,Q_{ij})$ be the energy of the three spatial plaquettes at the origin. If $g(a_kn)h(a_kx)>0$ for some $n\ge1$ and some $x$, then $C_k(0)>0$.*

*Proof.* Lüscher proved that the transfer matrix is strictly positive [64]; we recall the argument because the proof uses it in this form.

*The one-link kernel.* Write $\chi$ for the character of the defining representation and $b=\beta_W>0$. Then

$$e^{\frac{b}{N}\mathrm{Re}\,\chi}=\sum_{m\ge0}\frac{(b/2N)^m}{m!}(\chi+\bar\chi)^m,$$

and $(\chi+\bar\chi)^m$ is the character of $(\rho\oplus\bar\rho)^{\otimes m}$. Every irreducible representation $\sigma$ occurs in one of these tensor powers: their matrix coefficients span the algebra of polynomials in the entries of $q$ and $\bar q$, which contains the constants and separates points, so it is dense in $C(G)$ by the Stone–Weierstrass theorem, while

the matrix coefficients of a $\sigma$ that never occurred would be orthogonal to it by Schur orthogonality. So the coefficient of every irreducible character in $e^{\frac{b}{N}\operatorname{Re}\chi}$ is positive, and convolution with this function acts on each isotypic part of $L^2(G)$ by a positive scalar.

*The transfer matrix.* In the temporal gauge the transfer matrix of the box acts on the gauge-invariant part of $L^2(G^E)$, $E$ the spatial links of one slice, as $MKM$, where $K$ is the tensor product over $E$ of the convolutions above and $M$ is multiplication by the exponential of half the spatial plaquette weight [64, 75]. Both commute with gauge transformations, $K$ is positive and injective by the first step, and $M$ is positive and invertible, so $\langle v, MKMv\rangle = \langle Mv, KMv\rangle > 0$ for $v \neq 0$. Its kernel is a continuous positive function, and $K$ is compact, so its largest eigenvalue $\lambda_0$ is simple with an eigenfunction $\psi_0 > 0$, and with $A = MKM/\lambda_0$, a function $F$ of the slice at time $n \geq 0$ has $\Psi_F = A^n F\psi_0$ in the reconstruction, $\Omega$ being $\psi_0$.

*Conclusion.* Let $F = \sum_x h(a_k x)\,\Phi_x \circ \Phi_{s_k}$ and $F^c = F - \omega_k(F)$. Then $\Psi_{A_k} = a_k^4\, p(A) F^c \psi_0$ with $p(z) = \sum_{n\geq 1} g(a_k n) z^n$, a polynomial with nonnegative coefficients, not all zero, so $p > 0$ on $(0,1]$ and $p(A)$ is injective. The function $\sum_x h(a_k x)\Phi_x$ vanishes at the configuration with every link 1 and is positive at one with a single nontrivial link at a site where $h(a_k x) > 0$; the flow $\Phi_{s_k}$ is a bijection, so $F$ is not constant, and as $F$ is continuous and $\psi_0 > 0$, $F^c\psi_0 \neq 0$. Hence $C_k(0) = \|\Psi_{A_k}\|^2 > 0$. □

The proposition removes the one assumption of Theorem 48 in a box; it says nothing about the ratio of part (iv), on which nontriviality depends.

The ratio of Theorem 48(iv) can be compared with correlations of a single slice field. Suppose that $f_0(t,x) = g(t)h(x)$ with $g \geq 0$ supported in $[t_0, t_1]$, $t_0 < t_1$ positive and dyadic, let

$$U_k = \sum_{x\in\mathbb{Z}^3} a_k^3 h(a_k x)\big(\Phi_{(0,x)} \circ \Phi_{s_k} - \omega_k(\Phi_{(0,x)} \circ \Phi_{s_k})\big)$$

be the centred field of the slice at time 0, smeared in space only, and put $u_k = \Psi_{U_k}$ and

$$D_k(r) = \langle u_k, \mathbb{T}_k^{r/a_k} u_k\rangle = \omega_k\big(\vartheta(\bar U_k)\,\tau_{r/a_k} U_k\big)$$

for $r \in a_k\mathbb{Z}$, $r \geq 0$. Let $n_0 = \lceil t_0/a_k\rceil$ and $n_1 = \lfloor t_1/a_k\rfloor$.

**Proposition 51.** *Assume $C_k(0) > 0$, and let $\sigma \geq 0$ with $\sigma/a_k \in \mathbb{Z}$.*

(i) *$D_k(r) > 0$ for every $r$, and*

$$\frac{D_k(2a_k n_1 + \sigma)}{D_k(2a_k n_0)} \leq \frac{C_k(\sigma)}{C_k(0)} \leq \frac{D_k(2a_k n_0 + \sigma)}{D_k(2a_k n_1)}.$$

(ii) *$\log D_k$ and $\log C_k$ are convex and nonincreasing on $a_k\mathbb{Z}_{\geq 0}$. If $D_k(r') \geq c\,D_k(r)$ with $r < r'$ and $c \leq 1$, then $D_k(\beta) \geq c^{\max(1,(\beta-\alpha)/(r'-r))} D_k(\alpha)$ whenever $r \leq \alpha < \beta$; if $C_k(\sigma) \geq c\,C_k(0)$ with $\sigma > 0$, then $C_k(\sigma') \geq c^{\sigma'/\sigma} C_k(0)$ for $\sigma' \geq \sigma$.*

(iii) *Let $k$ run through a sequence. If $D_k(r') \geq c\,D_k(r)$ for dyadic $0 < r < r'$ with $r \leq 2t_0$, a $c \in (0,1]$ and every large $k$, then for every dyadic $\sigma \geq 0$ and every large $k$*

$$C_k(\sigma) \geq c^{\max(1,(2t_1 - 2t_0 + \sigma)/(r'-r))}\, C_k(0).$$

*Conversely, if $C_k(\sigma) \geq c\,C_k(0)$ for a dyadic $\sigma > 0$ and every $k$, then $D_k(2t_0 + \sigma') \geq c^{\sigma'/\sigma} D_k(2t_1)$ for every dyadic $\sigma' \geq \sigma$ with $\sigma' > 2(t_1 - t_0)$ and every large $k$.*

*Proof.* For a function $F$ of the slice at time 0 and an integer $n \geq 0$ we have $\Psi_{\tau_n F} = \mathbb{T}_k^n \Psi_F$, since both have inner product $\omega_k(\vartheta(\bar G)\tau_n F)$ with every $\Psi_G$. By time translation invariance of $\omega_k$, $A_k = \sum_n w_n \tau_n U_k$ with $w_n = a_k g(a_k n) \geq 0$, and $w_n = 0$ unless $n_0 \leq n \leq n_1$; so $\Psi_{A_k} = \sum_n w_n \mathbb{T}_k^n u_k$. Let $\nu$ be the spectral measure of $\mathbb{T}_k$ at $u_k$, on $[0,1]$, and $W(\lambda) = \sum_n w_n \lambda^n$. Then

$$C_k(\sigma) = \int W(\lambda)^2 \lambda^{\sigma/a_k}\, d\nu(\lambda), \qquad D_k(r) = \int \lambda^{r/a_k}\, d\nu(\lambda).$$

(i) Since $t_0 > 0$, $n_0 \geq 1$ and $W(0) = 0$; so $C_k(0) > 0$ forces $\nu((0,1]) > 0$, and then $D_k(r) > 0$. For $0 \leq \lambda \leq 1$, $\|w\|_1 \lambda^{n_1} \leq W(\lambda) \leq \|w\|_1 \lambda^{n_0}$; inserting these bounds in $C_k(\sigma)$ and $C_k(0)$ gives the two inequalities.

(ii) Convexity of $j \mapsto \log \int \lambda^j\, d\rho$ for a positive measure $\rho$ on $[0,1]$ is Hölder's inequality, and the integrals decrease in $j$ because $\lambda \leq 1$; take $\rho = \nu$ and $\rho = W^2\nu$. If $\beta \leq r'$, monotonicity gives $D_k(\beta) \geq D_k(r') \geq c D_k(r) \geq c D_k(\alpha)$. If $\beta > r'$, the difference quotients of the convex function $\log D_k$ increase in both end points, so $(\log D_k(\beta) - \log D_k(\alpha))/(\beta - \alpha) \geq (\log D_k(r') - \log D_k(r))/(r' - r) \geq \log c/(r' - r)$. The statement on $C_k$ is the same comparison of quotients on $[0,\sigma]$ and $[0,\sigma']$.

(iii) For large $k$ the times $2t_0$, $2t_1$ are multiples of $a_k$, and $2a_k n_0 \geq 2t_0$, $2a_k n_1 \leq 2t_1$. Since $D_k$ is nonincreasing, (i) gives

$$\frac{D_k(2t_1 + \sigma)}{D_k(2t_0)} \leq \frac{C_k(\sigma)}{C_k(0)} \leq \frac{D_k(2t_0 + \sigma)}{D_k(2t_1)}.$$

The first claim is the left inequality with (ii) for $\alpha = 2t_0 \geq r$ and $\beta = 2t_1 + \sigma$. For the second, (ii) gives $C_k(\sigma') \geq c^{\sigma'/\sigma} C_k(0)$, and the right inequality at $\sigma'$ gives the claim, the two times being ordered because $\sigma' > 2(t_1 - t_0)$. □

So for the packet $g \otimes h$ the open input of Theorem 48(iv) follows from an inequality between two correlations of one slice field,

$$\liminf_k D_k(r')/D_k(r) > 0 \quad \text{for one pair } 0 < r < r' \text{ with } r \leq 2t_0,$$

and by (iii) a uniform bound for one packet gives one for every packet $g' \otimes h$ with $g'$ supported in $[t_1, \infty)$. The condition $r \leq 2t_0$ cannot be dropped: a bound at a later pair says nothing about an earlier packet, as $D_k(r) = (1 - 4^{-k})2^{-kr} + 4^{-k}2^{-r}$ shows, with $D_k(4)/D_k(3) \to \frac{1}{2}$ and $D_k(2)/D_k(1) \to 0$. A bound at a later pair does apply to the packet shifted forward in time, renormalised by its own reflected norm. The left inequality of (i) bounds the ratio without the comparison of $u$ with the packet that was needed after Proposition 49(ii). Every bound on such ratios proved in this paper, at strong coupling or through Theorem 45, is an upper bound of the form $e^{-m(r'-r)/a_k}$; the inequality above is the opposite one and says that the slice field does not put its weight on energies that diverge in physical units. Reflection positivity and positivity of the transfer matrix do not give it, since a lattice theory with a fixed mass in lattice units has both and its ratio tends to zero. It holds for the free field.

*Remark* 52. Let $D_k$ be defined as above for the massless free lattice field on $a_k\mathbb{Z}^4$ in place of $\Phi$, with $h \in C_c^\infty(\mathbb{R}^3)$ and the lattice heat flow for a physical time $t_f > 0$ in place of the gradient flow. With $\hat p^2 = \sum_i (2a_k^{-1}\sin(a_k p_i/2))^2$, $\cosh(a_k\omega) = 1 + a_k^2\hat p^2/2$ and $\hat h_k(p) = \sum_x a_k^3 h(a_k x) e^{-ia_k p\cdot x}$,

$$D_k(r) = \int_{[-\pi/a_k,\pi/a_k]^3} |\hat h_k(p)|^2 e^{-2t_f\hat p^2} \frac{e^{-r\omega}}{2a_k^{-1}\sinh(a_k\omega)} \frac{d^3p}{(2\pi)^3}.$$

On the Brillouin zone $\hat p \ge 2|p|/\pi$ and $a_k^2\hat p^2/2 \le 6$; as $\operatorname{arccosh}(1+y)/\sqrt{2y}$ decreases on $(0,6]$, $a_k\omega \ge \kappa a_k\hat p$ with $\kappa = \operatorname{arccosh}7/\sqrt{12} > 0.76$, and $a_k^{-1}\sinh(a_k\omega) \ge \omega$. Since $|\hat h_k|$ is bounded uniformly in $k$, the integrands are dominated by a multiple of $e^{-8t_f|p|^2/\pi^2}/|p|$, which is integrable on $\mathbb{R}^3$, and they converge pointwise to $|\hat h(p)|^2 e^{-2t_f|p|^2} e^{-r|p|}/(2|p|)$. By dominated convergence $D_k(r')/D_k(r)$ tends to the corresponding positive ratio of continuum integrals; for $h = e^{-|x|^2}$, $t_f = 0.05$, $r = \frac12$ and $r' = 1$ it is 0.6337.

*Remark* 53. The slice field of Proposition 51 is not linear: it is a gauge-invariant function of the spatial links of one slice. Its free analogue is the magnetic energy, and the conclusion of Remark 52 survives for it. Take the abelian free lattice field on $a_k\mathbb{Z}\times a_k\mathbb{Z}^3_{L_k}$ with $a_kL_k = \ell$ fixed, Feynman gauge, the lattice heat flow of the spatial links for a physical time $t_f \ge 0$, and $U_k = \sum_x a_k^3 \sum_{i<j}(F_{ij}(0,x)^2 - \omega_k(F_{ij}^2))$, the zero spatial momentum part of $h$ (for $\mathrm{SU}(N)$ at tree level $D_k$ is multiplied by $N^2-1$). Since $F_{ij}$ is Gaussian, $\operatorname{Cov}(F^2,F'^2) = 2\operatorname{Cov}(F,F')^2$, and the mixed propagator of the magnetic field at spatial momentum $p$ is $N_{ij,lm}(p)K_p(n)$, with a numerator that does not depend on the time momentum and $K_p(n) = e^{-\omega|n|a_k}/(2\sinh(a_k\omega))$. Hence

$$D_k(r) \propto \sum_{p\in\frac{2\pi}{\ell}\mathbb{Z}^3_{L_k}\setminus 0} S_k(p)\, e^{-4t_f\hat p^2} \frac{e^{-2r\omega}}{4a_k^{-2}\sinh^2(a_k\omega)},$$

with $S_k(p) = \sum_{i<j,l<m}|N_{ij,lm}(p)|^2 \le 36\,\hat p^4$, and the bounds of Remark 52 dominate the summands by $(9/\kappa^2)\,\hat p^2 e^{-4t_f\hat p^2 - 2r\kappa\hat p}$, summable over $\frac{2\pi}{\ell}\mathbb{Z}^3$ for $r>0$ uniformly in $k$, also for $t_f = 0$. The summands converge to the continuum ones, so for every $0<r<r'$ the ratio $D_k(r')/D_k(r)$ converges to a positive limit, with or without the flow. For $\ell = 6$ the computed values of $D_k(\frac12)/D_k(\frac14)$ at $t_f = 0.05$ and $a_k^{-1} = 8, 12, 16, 24$ are 0.27341, 0.27239, 0.27201, 0.27174, against 0.27152 in the continuum; at $t_f = 0$ they are 0.0224, 0.0265, 0.0290, 0.0304, against $(r/r')^5 = \frac{1}{32}$ in infinite volume.

The behaviour at $r = 0$ separates the cases. With the flow, $D_k(0)$ stays bounded and the first-step energy $-a_k^{-1}\log(D_k(a_k)/D_k(0))$ is 6.37, 6.50, 6.56, 6.62 at the four spacings, bounded. Without the flow, $D_k(0) \sim a_k^{-5}$ and $D_k(a_k)/D_k(0) = 0.0198$ at every spacing, so the first-step energy grows like $3.92/a_k$ although every ratio at $r>0$ converges: the condition of Proposition 51 at $r>0$ is weaker than a bound on the first step. For the full energy density, which contains $F_{0i}$ and is not a function of one slice, the electric numerator depends on the time momentum, and the second time difference of $K_p$ is $-\delta_{n0}$ plus a multiple of $K_p$, a contact term: $D_k(0) \sim a_k^{-2}$ even with the flow, $D_k(a_k)/D_k(0) = 0.042$, 0.023, 0.0145 at $a_k^{-1} = 8, 12, 16$, while the ratios at $r>0$ again converge to the same continuum values (0.27092, 0.27126, 0.27137). The computation is `slice/composite_free.py`.

At next order the spatial flow, being nonlinear, fuses two hard modes into a soft one, and the state $\Psi_{U_k}$ acquires a component of energy about $2q$ for each $q \lesssim a_k^{-1}$. Its norm is of order $g^2$ and ultraviolet finite, dominated by $q \approx \tau^{-1/2}$, but its mean energy is not: in the free vacuum the first-step energy is $E_{\mathrm{LO}}(1 + g^2(c + \frac{N}{3\pi^2}\log(\sqrt{\tau}/a_k)))$, and the logarithm has coefficient $\frac{8}{11}$ of $2b_0$, $b_0 = 11N/48\pi^2$. A first-step bound at fixed coupling therefore fails logarithmically, while the ratio at a fixed $r>0$ only loses the norm of that component, damped by $e^{-2qr}$. This is why Proposition 51 is stated at $r>0$; the decomposition $\|\mathbb{T}_k^{n/2}u\| \ge c^{n/2}\|v\| - \|w\|$ for $u = v+w$ with a first-step bound on the soft part $v$ alone is the form a four-dimensional argument would use.

*Remark* 54. The inequality also holds along the right tuning in two-dimensional $\mathrm{SU}(N)$ lattice gauge theory, which is interacting and nonabelian but has no propagating gluons. Take the lattice $a_k\mathbb{Z}\times a_k\mathbb{Z}_{L_k}$ with $a_kL_k = \ell$ fixed, the weight $\prod_p \exp(\frac{\beta_k}{N}\operatorname{Re}\operatorname{Tr}U_p)$ with $\beta_k = 2N/(g^2a_k^2)$, and the temporal gauge. A gauge-invariant function of the links of one slice depends only on the conjugacy class of their holonomy $V$ around the circle; this includes every spatially flowed and spatially smeared slice field, since the flow is gauge covariant. Let $w_\beta$ be the probability density proportional to $e^{\frac{\beta}{N}\operatorname{Re}\operatorname{Tr}U}$ with respect to Haar measure and, for an irreducible representation $\lambda$ of dimension $d_\lambda$ and character $\chi_\lambda$, $\hat c_\lambda(\beta) = d_\lambda^{-1}\int \chi_\lambda w_\beta\, dU$. Between two slices put $W_x = V_x'V_x^{-1}$; the $W_x$ are independent with density $w_{\beta_k}$, which is invariant under conjugation and inversion, and

$$V_1'\cdots V_L' = W_1\,(V_1W_2V_1^{-1})\cdots \\ \cdots(V_1\cdots V_{L-1}W_L(V_1\cdots V_{L-1})^{-1})V_1\cdots V_L,$$

so the new holonomy is $QV$ with $Q$ distributed as the $L_k$-fold convolution of $w_{\beta_k}$, independent of $V$. By Schur's lemma (Lemma 249) the transfer matrix acts on class functions by $\chi_\lambda \mapsto \hat c_\lambda(\beta_k)^{L_k}\chi_\lambda$. The coefficients are real, since $w_\beta$ is invariant under $U\mapsto\bar U$, nonnegative, since the exponential of $\frac{\beta}{2N}(\chi_{\mathrm F} + \bar\chi_{\mathrm F})$ is a sum of characters with nonnegative coefficients, and below 1 for $\lambda \ne 0$. For a class function $f$ with $\int f\,dU = 0$ and isotypic components of norms $|\hat f_\lambda|$, the slice correlation is therefore

$$D_k(r) = \sum_{\lambda\ne0} |\hat f_\lambda|^2 e^{-rE_{\lambda,k}}, \qquad E_{\lambda,k} = -\frac{\ell}{a_k^2}\log\hat c_\lambda(\beta_k).$$

Laplace's method[18] at the unique maximum $U = 1$ of $\operatorname{Re}\operatorname{Tr}U$, with $U = e^X$, $\operatorname{Re}\operatorname{Tr}e^X = N + \frac12\operatorname{Tr}X^2 + O(|X|^4)$,

[18] For a smooth function with a unique nondegenerate maximum, $\int e^{\beta\varphi}h$ has an expansion in powers of $\beta^{-1}$ whose coefficients are Gaussian moments at the maximum; Lemma 250 carries it out here.

$\operatorname{Re}\operatorname{Tr}X^3=0$, $\operatorname{Tr}\rho_\lambda(X)=0$ and $\sum_a\rho_\lambda(T_a)^2=-2C_2(\lambda)$ for an orthonormal basis of $\mathfrak{su}(N)$ in the norm $-\operatorname{Tr}X^2$, gives $\hat c_\lambda(\beta)=1-NC_2(\lambda)/\beta+O(\beta^{-2})$, with the Casimir normalised by $C_2=(N^2-1)/(2N)$ on the fundamental representation (Lemma 250). Hence $E_{\lambda,k}\to g^2\ell C_2(\lambda)/2$, and since $D_k(r)\le\|f\|^2$,

$$\liminf_k\frac{D_k(r')}{D_k(r)}\ge\frac{|\hat f_\mu|^2}{\|f\|^2}e^{-r'g^2\ell C_2(\mu)/2}>0$$

for every $\mu\neq0$ with $\hat f_\mu\neq0$; for $f=\operatorname{Re}\operatorname{Tr}V$ the ratio is exactly $e^{-(r'-r)E_{\mathrm{F},k}}\to e^{-(r'-r)g^2\ell(N^2-1)/(4N)}$. This is the exact solution of two-dimensional lattice gauge theory, whose continuum limit is the measure of [37, 47, 62], written as the inequality. In two dimensions $g^2$ has the dimension of a mass squared and there is no ultraviolet problem, so the argument has no four-dimensional analogue. Nor does the spatial flow give the inequality by itself: it damps the spectral weight at large spatial momentum and leaves the weight at zero spatial momentum unchanged. For the free lattice field of Remark 52 with a mass $M_k$ and the same flow, every energy is at least $m_k=a_k^{-1}\operatorname{arccosh}(1+a_k^2M_k^2/2)$, so $D_k(r')\le e^{-m_k(r'-r)}D_k(r)$, which tends to zero relative to $D_k(r)$ when $a_kM_k$ is fixed.

A route to the slice ratio in four dimensions replaces the packet by its conditional expectations given averages of the field over blocks of increasing size and controls the error of each step. The free field shows in which norm such errors can be small. In the ordinary $L^2$ norm the first step already fails: conditioning on the averages over blocks of side two leaves a fixed fraction of the variance of an energy packet, at every lattice spacing.

**Proposition 55.** *Let $d\ge2$, let $\Lambda$ be a periodic box in $\mathbb{Z}^d$ with even sides, let $\mu>0$, and let $\phi$ be the Gaussian field on $\Lambda$ with density proportional to $\exp\bigl(-\frac12\sum_{x,\nu}(\nabla_\nu\phi(x))^2-\frac{\mu^2}{2}\sum_x\phi(x)^2\bigr)$, where $\nabla_\nu\phi(x)=\phi(x+e_\nu)-\phi(x)$. Let $\mathscr{Z}$ be a set of blocks $B=2z+\{0,1\}^d$, let $\mathfrak{B}$ be the $\sigma$-algebra generated by the averages of $\phi$ over the blocks in $\mathscr{Z}$, let $f\ge0$ on $\Lambda$, and let*

$$F=\sum_{x\in\Lambda}f(x)\sum_{\nu=1}^{d}{:}(\nabla_\nu\phi(x))^2{:},$$

*where ${:}X^2{:}=X^2-\mathbb{E}X^2$. For $B=2z+\{0,1\}^d$ put $x_B=2z$, $y_B=2z+e_1+e_2$ and*

$$\alpha_B=\frac14\sum_{w\in\{x_B,y_B\}}\Bigl(d\,f(w)+\sum_{\nu=1}^{d}f(w-e_\nu)\Bigr),$$

*and $\sigma^2=2/(2d+\mu^2)$. (i) $\mathbb{E}\operatorname{Var}(F\mid\mathfrak{B})\ge2\sigma^4\sum_{B\in\mathscr{Z}}\alpha_B^2$.*

*(ii) $\operatorname{Var}F\le2d^2\sum_x f(x)^2$.*

*(iii) Let $g\ge0$ be a Lipschitz function on $\mathbb{R}^d$ with compact support and $\int g^2>0$. For $n\ge1$ let $\Lambda_n$ contain $n\operatorname{supp}g$ in a fundamental domain, $f_n(x)=g(x/n)$, let $\mathscr{Z}_n$ contain every block that meets $n\operatorname{supp}g$, and let $\mu=\mu_n\to0$. Then for the packets $F_n$ of $f_n$,*

$$\liminf_{n\to\infty}\frac{\mathbb{E}\operatorname{Var}(F_n\mid\mathfrak{B}_n)}{n^d\int g^2}\ge2^{1-d},\qquad\limsup_{n\to\infty}\frac{\operatorname{Var}F_n}{n^d\int g^2}\le2d^2,$$

*so $\mathbb{E}\operatorname{Var}(F_n\mid\mathfrak{B}_n)\ge(2^{-d}d^{-2}-o(1))\operatorname{Var}F_n$.*

*Proof.* (i) No two of the sites $x_B,y_B$, $B\in\mathscr{Z}$, are neighbours: their coordinate sums are even, and a neighbour of a site has a coordinate sum of the other parity. Let $\mathfrak{H}\supseteq\mathfrak{B}$ be generated by $\mathfrak{B}$ and by $\phi(w)$ for every $w$ outside $\{x_B,y_B:B\in\mathscr{Z}\}$. By the law of total variance, $\operatorname{Var}(F\mid\mathfrak{B})=\mathbb{E}[\operatorname{Var}(F\mid\mathfrak{H})\mid\mathfrak{B}]+\operatorname{Var}(\mathbb{E}[F\mid\mathfrak{H}]\mid\mathfrak{B})$, so $\mathbb{E}\operatorname{Var}(F\mid\mathfrak{B})\ge\mathbb{E}\operatorname{Var}(F\mid\mathfrak{H})$. In the exponent of the density, a site $w$ enters through $(d+\frac{\mu^2}{2})\phi(w)^2$ and through $-\phi(w)\phi(w')$ for each of its $2d$ neighbours $w'$. Given $\mathfrak{H}$, the block average fixes $s_B=\phi(x_B)+\phi(y_B)$, and with $u_B=\phi(x_B)-\phi(y_B)$ we have $\phi(x_B)^2+\phi(y_B)^2=\frac12(s_B^2+u_B^2)$. Since the free sites have only $\mathfrak{H}$-measurable neighbours, the conditional density of $(u_B)_{B\in\mathscr{Z}}$ is proportional to $\prod_B\exp(-\frac14(2d+\mu^2)u_B^2+c_Bu_B)$ with $\mathfrak{H}$-measurable $c_B$: the $u_B$ are conditionally independent Gaussians of variance $\sigma^2$. A term $(\nabla_\nu\phi(w))^2$ contains at most one free site, and if it contains $x_B$, then $\phi(x_B)=\frac12(s_B+u_B)$ shows that it is $\frac14u_B^2$ plus a term linear in $u_B$ plus an $\mathfrak{H}$-measurable term; the same holds for $y_B$ with $-u_B$. The terms containing $x_B$ are those with $w=x_B$ and those with $w=x_B-e_\nu$, $d$ of each kind, so

$$F=\sum_{B\in\mathscr{Z}}\bigl(\alpha_Bu_B^2+\beta_Bu_B\bigr)+F_0$$

with $\mathfrak{H}$-measurable $\beta_B$ and $F_0$. Writing $u_B=m_B+v_B$ with $m_B$ the conditional mean, $\alpha_Bu_B^2+\beta_Bu_B=\alpha_Bv_B^2+(2\alpha_Bm_B+\beta_B)v_B+\text{const}$, and for a centred Gaussian $v$ of variance $\sigma^2$, $\operatorname{Var}(\alpha v^2+\gamma v)=2\alpha^2\sigma^4+\gamma^2\sigma^2$. By conditional independence, $\operatorname{Var}(F\mid\mathfrak{H})\ge2\sigma^4\sum_B\alpha_B^2$.

(ii) For centred jointly Gaussian $X,Y$, $\operatorname{Cov}({:}X^2{:},{:}Y^2{:})=2\operatorname{Cov}(X,Y)^2$. With $G_{\nu\rho}(x,y)=\mathbb{E}[\nabla_\nu\phi(x)\nabla_\rho\phi(y)]$,

$$\begin{aligned}\operatorname{Var}F&=2\sum_{x,y}f(x)f(y)\sum_{\nu,\rho}G_{\nu\rho}(x,y)^2\\&\le\sum_{x,y}\bigl(f(x)^2+f(y)^2\bigr)\sum_{\nu,\rho}G_{\nu\rho}(x,y)^2.\end{aligned}$$

On the dual torus $\Lambda^*$, with $\hat p^2=\sum_\nu|e^{ip_\nu}-1|^2$,

$$G_{\nu\rho}(x,y)=\frac{1}{|\Lambda|}\sum_{p\in\Lambda^*}\frac{(e^{ip_\nu}-1)(e^{-ip_\rho}-1)}{\hat p^2+\mu^2}e^{ip\cdot(x-y)},$$

and by Parseval $\sum_yG_{\nu\rho}(x,y)^2=|\Lambda|^{-1}\sum_p|e^{ip_\nu}-1|^2|e^{ip_\rho}-1|^2(\hat p^2+\mu^2)^{-2}\le1$, since each factor $|e^{ip_\nu}-1|^2$ is at most $\hat p^2$. The same holds for the sum over $x$, and (ii) follows.

(iii) If $\operatorname{Lip}g=K$, every site entering $\alpha_B$ lies within distance 3 of $2z$, so $|\alpha_B-d\,g(2z/n)|\le3dK/n$. Hence $\sum_{B\in\mathscr{Z}_n}\alpha_B^2\ge\sum_z(d\,g(2z/n)-3dK/n)_+^2$, the sum running over $z$ with $2z\in n\operatorname{supp}g$, and this is $d^2(n/2)^d\int g^2\,(1+o(1))$ as a Riemann sum on the grid $(2/n)\mathbb{Z}^d$. Since $\sigma^4=4/(2d+\mu_n^2)^2\to d^{-2}$, (i) gives the first bound. By (ii) and $\sum_xg(x/n)^2=n^d\int g^2(1+o(1))$, the second follows. □

For $d=4$ and the energy density, $\operatorname{Var}F_n$ grows like $(\ell/a)^4$ at a fixed physical size $\ell$ of the packet, and no step of conditioning on block averages removes more than a fraction $1-2^{-d}d^{-2}+o(1)$ of it. The reflected norm $C_k(0)=\omega_k(\vartheta(\bar A_k)A_k)$, which renormalises the packet in Theorem 48, involves the covariance

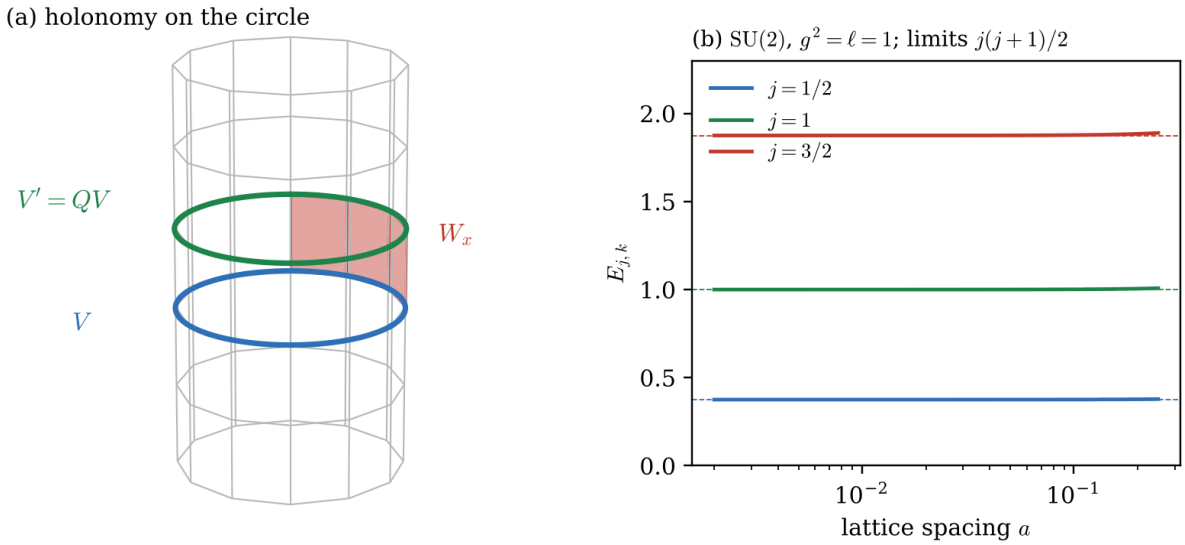



FIG. 10. Two-dimensional SU($N$) (Remark 54). (a) The lattice $a_k\mathbb{Z} \times a_k\mathbb{Z}_{L_k}$ drawn as a cylinder: the holonomy $V$ of one slice, the holonomy $V' = QV$ of the next, and the plaquette variables $W_x$ between them. (b) The energies $E_{j,k} = -(\ell/a_k^2)\log \hat{c}_j(\beta_k)$ of SU(2) for $g^2 = \ell = 1$, with $\hat{c}_j(\beta) = I_{2j+1}(\beta)/I_1(\beta)$, against the lattice spacing; the dashed lines are the limits $j(j+1)/2$.

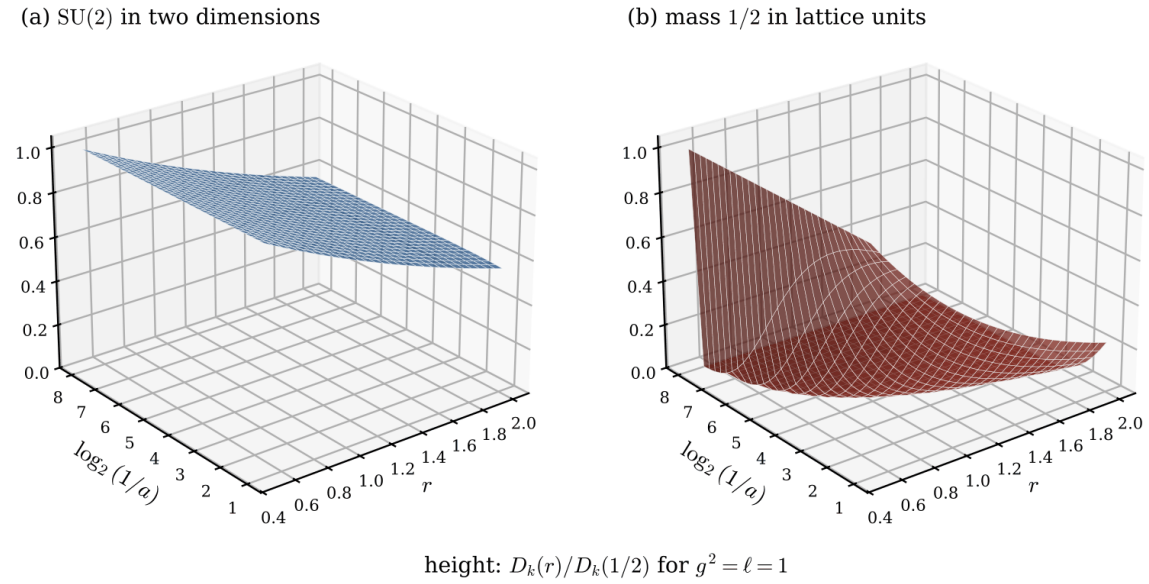
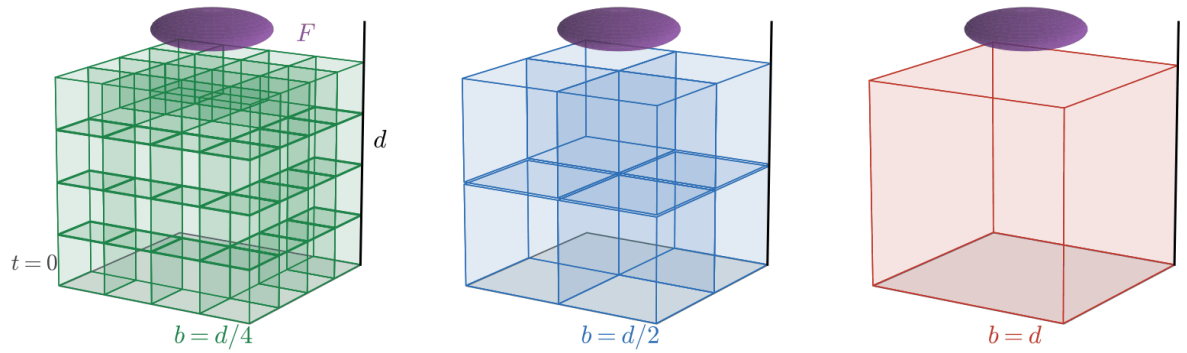



FIG. 11. The ratio $D_k(r)/D_k(\frac{1}{2})$ over the separation $r$ and the lattice spacing $a_k$. (a) Two-dimensional SU(2) with $f = \mathrm{Re}\,\mathrm{Tr}\,V$ and $g^2 = \ell = 1$, where the ratio is $e^{-(r-1/2)E_{1/2,k}}$ and converges as $a_k \to 0$. (b) The term $e^{-(r-1/2)m_k}$ of the lowest energy $m_k$ of a free field with mass $\frac{1}{2}$ in lattice units, which bounds the ratio from above and tends to zero at every $r > \frac{1}{2}$.

only between the support of $\vartheta f_0$ and that of $f_0$, at a fixed physical distance. The errors of a blocking route have therefore to be measured in that norm; Figure 12 shows the blocks between the reflection plane and the packet.

### E. Higher correlations

Theorem 48 controls two renormalised fields at a time, and part (i) of it gives nothing for more: $Z_k$ may diverge while the fields $\Phi_x \circ \Phi_{s_k}$ stay bounded, so a bound on the norm of the renormalised packet as an operator is not available. A reflection positive process with a fixed gap and a two-point function independent of the cutoff can have divergent four-




block averages of side $b$ between the plane $t=0$ and the packet $F$ at height $d$

FIG. 12. Block averages between the reflection plane $t=0$ and a packet $F$ at physical height $d$, for blocks of side $d/4$, $d/2$ and $d$. By Proposition 55 the first conditioning step already leaves a fixed fraction of the ordinary variance of the packet, so the errors of such steps have to be measured in the reflected norm.

point functions at separated times (Remark 59), so no statement about the two-point function alone can replace the following hypothesis on the truncated correlations. In this subsection $\Phi$ is real-valued, as the energy of the plaquettes is; for a complex field the conjugate field has to be adjoined and (T1) imposed on all mixed cumulants. Write $\varphi_k(x)=\Phi_x\circ\Phi_{s_k}$, $\omega_k^T$ for the truncated correlations (joint cumulants) under $\omega_k$, and for $f\in C_c(\mathbb{R}^4)$

$$A_k(f)=\sum_{x\in\mathbb{Z}^4} a_k^4 f(a_k x)\big(\varphi_k(x)-\omega_k(\varphi_k(x))\big),$$

so that $A_k=A_k(f_0)$ and $\bar A_k(f)=A_k(\bar f)$. Let $w_k:\mathbb{Z}^4\to[0,\infty)$ be even functions with $W_k=\sum_x w_k(x)<\infty$, let $r_k>0$ be amplitude scales, and consider:

(T1) *Tree bounds.* For every $n\ge 2$ there is $K_n$, independent of $k$, with

$$\big|\omega_k^T\big(\varphi_k(x_1);\dots;\varphi_k(x_n)\big)\big|\le K_n r_k^n\sum_T\prod_{\{i,j\}\in T} w_k(x_i-x_j)$$

for all $x_1,\dots,x_n\in\mathbb{Z}^4$, the sum running over the trees $T$ on $\{1,\dots,n\}$.

(T2) *Saturation.* $C_k(0)\ge c\,r_k^2 a_k^4 W_k$ for a $c>0$ and every $k$.

(T3) *Physical volume.* $B_*=\sup_k a_k^4 W_k<\infty$.

The amplitude $r_k$ separates the size of the field from the volume over which it is correlated; Remark 58 shows why it is needed. For a compact $Q\subset\mathbb{R}^4$ let $|Q|_k=a_k^4\#\{x\in\mathbb{Z}^4: a_k x\in Q\}$; for a cube of side $L$ it is at most $(L+a_0)^4$.

**Lemma 56.** *Under* (T1)*, for $n\ge 2$ and bounded $f_1,\dots,f_n$ with compact supports, $f_1$ supported in $Q$,*

$$\big|\omega_k^T\big(A_k(f_1);\dots;A_k(f_n)\big)\big|\le K_n n^{n-2} r_k^n\,|Q|_k\,(a_k^4 W_k)^{n-1}\prod_{j=1}^n\|f_j\|_\infty.$$

*Proof.* Joint cumulants of order $n\ge 2$ are multilinear and do not change when constants are added to the variables, so the left side is at most $a_k^{4n}\sum_{x_1,\dots,x_n}\prod_j|f_j(a_k x_j)|$ times the absolute value of the truncated correlation, the sums being finite. Insert (T1) and fix a tree $T$. With $x_1$ fixed, sum over the other points one at a time, each time over a leaf of what remains of $T$ other than 1; each such sum is at most $W_k$. The remaining sum over $x_1$ has at most $|Q|_k a_k^{-4}$ nonzero terms. There are $n^{n-2}$ trees. □

**Theorem 57.** *Assume* (T1)–(T3).

(i) *For $n\ge 2$, $f_j$ as in the lemma,*

$$\big|\omega_k^T\big(Z_kA_k(f_1);\dots;Z_kA_k(f_n)\big)\big|\le K_n n^{n-2}c^{-n/2}B_*^{n/2-1}\,|Q|_k\prod_{j=1}^n\|f_j\|_\infty,$$

*and the moments $\omega_k\big(\prod_j Z_kA_k(f_j)\big)$ with all $f_j$ supported in a cube $Q$ are bounded by $M_n(Q)\prod_j\|f_j\|_\infty$, with $M_n(Q)$ independent of $k$.*

(ii) *Every sequence of $k$ has a subsequence along which*

$$S_n(f_1,\dots,f_n)=\lim_k\omega_k\Big(\prod_{j=1}^n Z_kA_k(f_j)\Big)$$

*exists for all $n$ and all $f_j\in C_c(\mathbb{R}^4)$. Each $S_n$ is symmetric, multilinear, bounded by $M_n(Q)\prod_j\|f_j\|_\infty$ on functions supported in $Q$, hence a distribution on $(\mathbb{R}^4)^n$; it is invariant*

*under all translations of $\mathbb{R}^4$ if the $\omega_k$ are translation invariant, and the kernel $K$ of Theorem 48 along the subsequence is $K(y,y') = S_2\big(\overline{f_0(\cdot - y)} \circ \theta, f_0(\cdot - y')\big)$, with $\theta$ as in* (iii).

(iii) *Put $S_0 = 1$, $S_1 = 0$, and let $\theta$ be the reflection $t \mapsto -t$ of $\mathbb{R}^4$. For every finite family of $c_\alpha \in \mathbb{C}$ and $f_{\alpha,j} \in C_c(\mathbb{R}^4)$ supported in $\{t > 0\}$,*

$$\sum_{\alpha,\beta} \bar{c}_\alpha c_\beta \, S\Big(\prod_j \overline{f_{\alpha,j}} \circ \theta \cdot \prod_j f_{\beta,j}\Big) \geq 0,$$

*where $S$ of a product of $n$ test functions means $S_n$. Hence the Osterwalder–Schrader construction gives a Hilbert space $\mathscr{H}$ with vacuum $\Omega$ and a dense set of vectors $\Psi(P)$ indexed by polynomials $P$ in the smeared fields at positive times, with $\langle \Psi(P), \Psi(P') \rangle = S(\theta \bar{P} \cdot P')$; in particular the separated higher correlations of the renormalised fields are finite. The vector $\Psi(P_0)$ of the field smeared by $f_0$ is a unit vector orthogonal to $\Omega$.*

(iv) *Suppose the $\omega_k$ are translation invariant. Then the time translations by $t \geq 0$ define on $\mathscr{H}$ a strongly continuous semigroup $e^{-tH}$ with $H \geq 0$ self-adjoint, and the spatial translations a strongly continuous unitary group commuting with $H$. Moreover there is a dyadic $\sigma > 0$ with $C_k(\sigma) \geq \frac{1}{2} C_k(0)$ for all large $k$, so the limit of Theorem 48 is nontrivial.*

(v) *If $a_k^4 W_k \to 0$ along the subsequence, every $S_n$ is given by Wick's rule from $S_2$: the limit is the Gaussian field with covariance $S_2$.*

(vi) *If the $\omega_k$ are translation invariant and the hypothesis of Theorem 45 holds, then $0$ is a simple eigenvalue of $H$ and $\sigma(H) \subset \{0\} \cup [m_*, \infty)$ with $m_* = \log(1/\alpha)/(2\ell)$, and $\Omega^\perp \neq 0$.*

*Proof.* (i) By (T2), $Z_k^n = C_k(0)^{-n/2}$ is at most $(c\, r_k^2 a_k^4 W_k)^{-n/2}$; multiply the lemma by it, which cancels $r_k^n$, and use $(a_k^4 W_k)^{n/2-1} \leq B_*^{n/2-1}$ by (T3) for $n \geq 2$. A moment is the sum over partitions of $\{1,\dots,n\}$ of the products of the cumulants of the blocks; blocks of size one vanish because $A_k(f)$ is centred, and each block contributes the bound just proved with $|Q|_k \leq (L + a_0)^4$ for a cube of side $L$. This gives $M_n(Q)$.

(ii) For each cube $Q_L = [-L, L]^4$, $L \in \mathbb{N}$, the continuous functions supported in $Q_L$ form a separable space under the supremum norm. Choose a countable dense set $\mathscr{F}$ of functions; by (i) the countably many sequences $\omega_k(\prod Z_k A_k(f_j))$, $f_j \in \mathscr{F}$, are bounded, and a diagonal subsequence makes all of them converge. By multilinearity,

$$\Big|\omega_k\Big(\prod_j Z_k A_k(f_j)\Big) - \omega_k\Big(\prod_j Z_k A_k(g_j)\Big)\Big| \leq M_n(Q) \sum_i \|f_i - g_i\|_\infty \prod_{j \neq i} \max(\|f_j\|_\infty, \|g_j\|_\infty)$$

for functions supported in $Q$, so the limits exist for all $f_j \in C_c(\mathbb{R}^4)$ and obey the same bound. Symmetry and multilinearity pass to the limit. A multilinear form bounded in this way is separately continuous on $C_c^\infty(\mathbb{R}^4)$, so the Schwartz kernel theorem makes it a distribution. For dyadic $y$ and large $k$, $y/a_k \in \mathbb{Z}^4$ and translation invariance of $\omega_k$ and of the flow (Proposition 47(i)) give $\omega_k(\prod Z_k A_k(f_j(\cdot - y))) = \omega_k(\prod Z_k A_k(f_j))$. For a real $y$ take dyadic $y_m \to y$; the functions $f_j(\cdot - y_m)$ converge uniformly to $f_j(\cdot - y)$ with supports in one cube, so the bound above carries the invariance to $y$. The formula for $K$ follows from $A_k^y = A_k(f_0(\cdot - y))$ and the argument of (iii) with $n = 2$.

(iii) Let $\vartheta$ be the lattice reflection of Appendix F and $\theta_k$ the corresponding reflection of $a_k \mathbb{Z}^4$; it differs from $\theta$ by a translation of length at most $a_k$. Since $\Phi$ is real and $\vartheta$ commutes with the flow, $\vartheta(\bar{A}_k(f)) = A_k(\bar{f} \circ \theta_k)$. Let $P_k = \sum_\beta c_\beta \prod_j Z_k A_k(f_{\beta,j})$ and $R_k = \sum_\alpha \bar{c}_\alpha \prod_j Z_k A_k(\overline{f_{\alpha,j}} \circ \theta)$, so that $\vartheta(\bar{P}_k)$ is $R_k$ with $\theta$ replaced by $\theta_k$. Reflection positivity gives $\omega_k(\vartheta(\bar{P}_k) P_k) \geq 0$. By (i), applied in a cube containing all the supports and their reflections, and by multilinearity, $|\omega_k(\vartheta(\bar{P}_k) P_k) - \omega_k(R_k P_k)|$ is at most a constant independent of $k$ times $\max_{\alpha,j} \|f_{\alpha,j} \circ \theta_k - f_{\alpha,j} \circ \theta\|_\infty$, which tends to zero by uniform continuity, while $\omega_k(R_k P_k)$ tends to the left side of (iii) by (ii). So the limit of the nonnegative numbers is the left side of (iii). The construction of $\mathscr{H}$ is the usual quotient of the polynomials at positive times by the null space of this form [73]. By the same approximation of $\theta$ by $\theta_k$, $\|\Psi(P_0)\|^2 = \lim_k Z_k^2 C_k(0) = 1$, and $\langle \Omega, \Psi(P_0) \rangle = S_1(f_0) = 0$.

(iv) For a polynomial $P$ at positive times let $\tau_t P$ be its time translate by $t \geq 0$, again at positive times. By translation invariance and $\theta \circ \tau_t = \tau_{-t} \circ \theta$, $\langle \Psi(\tau_t P), \Psi(P') \rangle = \langle \Psi(P), \Psi(\tau_t P') \rangle$, and $q_P(t) = \|\Psi(\tau_t P)\|^2 = \langle \Psi(P), \Psi(\tau_{2t} P) \rangle$. If $P$ has degree at most $d$, every partition in (i) of the products in $q_P(t)$ has at most $d$ blocks, each bounded by a multiple of $|Q|_k$ for a cube $Q$ of side $O(1+t)$, so $q_P(t) \leq C_P (1+t)^{4d}$. The Cauchy–Schwarz inequality gives $q_P(t) \leq q_P(0)^{1/2} q_P(2t)^{1/2}$, hence by induction $q_P(t) \leq q_P(0)^{1-2^{-m}} q_P(2^m t)^{2^{-m}}$ for every $m$, and letting $m \to \infty$, $q_P(t) \leq q_P(0)$. So $\tau_t$ maps null vectors to null vectors and extends to a symmetric contraction $\mathsf{T}(t)$ of $\mathscr{H}$; $\mathsf{T}(s)\mathsf{T}(t) = \mathsf{T}(s+t)$, and $\mathsf{T}(t) = \mathsf{T}(t/2)^2 \geq 0$. By the bound of (ii) and uniform continuity of the translated test functions, $\|\Psi(\tau_t P) - \Psi(P)\| \to 0$ as $t \to 0$, so the semigroup is strongly continuous on a dense set, hence everywhere, and $\mathsf{T}(t) = e^{-tH}$ with $H \geq 0$ self-adjoint [73]. Spatial translations preserve the form, have the opposite translations as inverses, commute with $\tau_t$, and are strongly continuous by the same argument.

For the ratio let $f_{0,t} = f_0(\cdot - (t,0))$. For dyadic $t > 0$ and large $k$, translation invariance gives $\tau_{t/a_k} A_k = A_k(f_{0,t})$, so by (iii)

$$C_k(t) - C_k(0) = \omega_k\big(A_k(\bar{f}_0 \circ \theta_k) A_k(f_{0,t} - f_0)\big),$$

a covariance of two centred fields. The case $n = 2$ of (i) bounds $Z_k^2$ times it by $L_0 \|f_{0,t} - f_0\|_\infty$ with $L_0 = (K_2/c) \sup_k |Q|_k \|f_0\|_\infty$, $Q$ a cube containing the supports of $f_0 \circ \theta_k$ for all $k$. By uniform continuity of $f_0$ there is a dyadic $\sigma > 0$ with $L_0 \|f_{0,\sigma} - f_0\|_\infty \leq \frac{1}{2}$; then $C_k(\sigma)/C_k(0) \geq \frac{1}{2}$, and Theorem 48(iv) applies.

(v) For $n \geq 3$ the exponent $n/2 - 1$ in (i) is positive, so with $(a_k^4 W_k)^{n/2-1}$ in place of $B_*^{n/2-1}$ the cumulants of order $n \geq 3$ of the renormalised fields tend to zero. The moments are the sums over partitions of products of cumulants, so in the limit only the partitions into pairs remain, which is Wick's rule.

(vi) As in the proof of Theorem 45, $\mathbb{T}_k$ has gap at least $\log(1/\alpha)/(2n_k)$ on the complement of $\Omega_k$. For a polynomial $P$ at positive times let $v_k = \Psi_{P_k} - \omega_k(P_k)\Omega_k$ with $P_k$ its lattice version. For dyadic $t$ and large $k$,

$$\omega_k\big(\vartheta(\bar{P}_k) \tau_{t/a_k} P_k\big) - |\omega_k(P_k)|^2 = \langle v_k, \mathbb{T}_k^{t/a_k} v_k \rangle \leq e^{-m_* t} \|v_k\|^2,$$

and both sides are finite combinations of moments, which converge by (ii) and the approximation of $\theta$ in (iii). The limit is $\langle v, e^{-tH}v\rangle \le e^{-m_* t}\|v\|^2$ for $v = \Psi(P) - \langle\Omega,\Psi(P)\rangle\Omega$. These $v$ are dense in $\Omega^\perp$, so the inequality holds on $\Omega^\perp$ for all $t \ge 0$, and the spectral theorem gives the spectral statement as in Theorem 45. By (iii), $\Psi(P_0)$ is a unit vector in $\Omega^\perp$. □

Part (iv) shows that the order-two cases of (T1) and (T2) already contain the lower bound on the ratio that Theorem 48(iv) needs; they are not harmless combinatorial assumptions, and proving them along the continuum tuning is at least as hard as that lower bound. Part (vi) passes the conditional mass gap of Theorem 45 to every polynomial in the smeared fields. Neither part proves Euclidean invariance, independence of the limit from the subsequence, or the reconstruction of the field operators, for which the growth of $M_n(Q)$ in $n$ would also be needed.

*Remark* 58. The amplitude scale cannot be dropped for a bounded field. Suppose $\operatorname{Var}\varphi_k(0) \to 0$, as for the energy of the flowed plaquettes in lattice units when the flowed links approach the identity. Reflection invariance of $\omega_k$, the Cauchy–Schwarz inequality and stationarity give

$$C_k(0) \le \omega_k(|A_k|^2) \le \Big(a_k^4 \sum_x |f_0(a_k x)|\Big)^2 \operatorname{Var}\varphi_k(0) \to 0.$$

With $r_k = 1$, (T2) then forces $a_k^4 W_k \to 0$, and by part (v) the limit is Gaussian. With $r_k^2$ of the order of $\operatorname{Var}\varphi_k(0)$ this is no longer forced. Choosing $r_k$ does not prove (T1) or (T2).

*Remark* 59. Let $0 < \varepsilon < 1$, $m > 0$, let $\pi$ be the probability measure on $\{-1,0,1\}$ with weights $\varepsilon/2$, $1-\varepsilon$, $\varepsilon/2$, and let $P_t = e^{-mt}I + (1-e^{-mt})\Pi$ on $L^2(\pi)$, where $\Pi f = \pi(f)$ is the projection onto the constants. This is a symmetric Markov semigroup with $P_t \ge 0$ and gap $m$, so its stationary process is reflection positive. For $X(s) = s/\sqrt{\varepsilon}$ we have $\pi(X) = \pi(X^3) = 0$, $\pi(X^2) = 1$, $\pi(X^4) = 1/\varepsilon$ and $P_t X = e^{-mt}X$, so the two-point function is $e^{-mt}$ for every $\varepsilon$, while for $t_1 \le t_2 \le t_3 \le t_4$ and $q_i = e^{-m(t_{i+1}-t_i)}$ the four-point function is $q_3\,\pi\big(XP_{t_2-t_1}(XP_{t_3-t_2}X^2)\big) = q_1q_3(q_2/\varepsilon + 1 - q_2)$, with truncated part $q_1q_3(q_2/\varepsilon - q_2 - 2q_2^2)$; both diverge as $\varepsilon \to 0$. By part (i), no choice of amplitude scales makes (T1)–(T3) hold for this family with constants independent of $\varepsilon$.

The hypotheses (T1)–(T3) are uniform cluster estimates for the continuum tuning $\beta_k \to \infty$, and the next proposition shows that they are estimates at a critical point: with the order-two cases alone the lattice gap is at most a fixed multiple of $a_k$, and the tree majorant carries a fixed fraction of its weight at a fixed physical distance. Let $m_k \in [0,\infty]$ be the gap of $\mathbb{T}_k$ on the complement of $\Omega_k$, so that $\|\mathbb{T}_k v\| \le e^{-m_k}\|v\|$ for $v \perp \Omega_k$, and let $t_0 > 0$ be the least time in the support of $f_0$.

**Proposition 60.** (i) *For every dyadic $\sigma > 0$ and every large $k$, $C_k(\sigma) \le e^{-m_k\sigma/a_k}C_k(0)$. If $m_k/a_k \to \infty$, every limit of Theorem 48 is trivial; this is the case for $\beta_k = \beta$ fixed in the range of Theorem 5, where $m_k \ge m_G(\beta)$.*

(ii) *Suppose the $\omega_k$ are translation invariant and* (T1) *for $n = 2$ and* (T2) *hold. Then $m_k \le a_k \log 2/\sigma$ for every large $k$, where $\sigma$ is the dyadic number of Theorem 57*(iv)*, which depends only on $K_2/c$ and $f_0$. If also the hypothesis of Theorem 45 holds, then $m_* a_k \le m_k \le a_k \log 2/\sigma$, and in particular $m_* \le \log 2/\sigma$.*

(iii) *Under* (T1) *for $n = 2$ and* (T2)*, let $Q$ be a cube containing the support of $f_0$ and $\gamma = c/(K_2\|f_0\|_\infty^2 \sup_k |Q|_k)$. Then for every $k$*

$$\sum_{z\in\mathbb{Z}^4,\ |z_0|\ge R_k} w_k(z) \ge \gamma W_k, \qquad R_k = \frac{2t_0}{a_k} - 1.$$

*If $w_k(z) = e^{-\mu_k|z_0|}u_k(\bar z)$ with $\bar z$ the spatial part of $z$ and $u_k \ge 0$ summable on $\mathbb{Z}^3$, then $\mu_k R_k \le \log(2/\gamma)$, so the physical time decay rate of the majorant is at most $\log(2/\gamma)/(2t_0 - a_k)$.*

*Proof.* (i) In the proof of Theorem 48, $C_k(\sigma)/C_k(0) = \langle\Psi_k, \mathbb{T}_k^{\sigma/a_k}\Psi_k\rangle$ with $\Psi_k$ a unit vector orthogonal to $\Omega_k$. The operator $\mathbb{T}_k$ is self-adjoint and fixes $\Omega_k$, so it preserves the complement, and the inequality follows. If $m_k/a_k \to \infty$ then $\rho(\sigma) = 0$ for every dyadic $\sigma > 0$ along every subsequence, and the limit is trivial by Theorem 48(iv). At fixed $\beta$ in the range of Theorem 5 the gap is at least $m_G(\beta) > 0$ in lattice units, and $a_k \to 0$; the spatial flow does not change the transfer matrix (Proposition 47(iv)).

(ii) Theorem 57(iv) uses (T1) and (T2) only for $n = 2$, since the exponent $n/2 - 1$ in part (i) vanishes there, and gives $C_k(\sigma) \ge \frac12 C_k(0)$ for large $k$. With (i), $e^{-m_k\sigma/a_k} \ge \frac12$. Under the hypothesis of Theorem 45, the proof of that theorem gives $m_k \ge \log(1/\alpha)/(2n_k) \ge m_* a_k$.

(iii) $C_k(0) = \omega_k(A_k(\bar f_0 \circ \theta_k)A_k(f_0))$, a covariance of two centred fields, by the proof of Theorem 57(iv). A point $x$ in the support of the first sum has $a_k x_0 \le -t_0 + a_k$, and a point $y$ in the support of the second has $a_k y_0 \ge t_0$, so $|x_0 - y_0| \ge R_k$. By (T1) for $n = 2$,

$$\begin{aligned} C_k(0) &\le K_2 r_k^2 \|f_0\|_\infty^2 a_k^8 \sum_{a_k y\in Q} \sum_{|x_0-y_0|\ge R_k} w_k(x-y) \\ &\le K_2 r_k^2 \|f_0\|_\infty^2 |Q|_k a_k^4 \sum_{|z_0|\ge R_k} w_k(z), \end{aligned}$$

and (T2) bounds the left side below by $c\,r_k^2 a_k^4 W_k$. For the product form let $R' = \lceil R_k \rceil$; if $R' \le 0$ there is nothing to prove, and otherwise

$$\frac{\sum_{|j|\ge R'} e^{-\mu_k|j|}}{\sum_{j\in\mathbb{Z}} e^{-\mu_k|j|}} = \frac{2e^{-\mu_k R'}}{1+e^{-\mu_k}} \le 2e^{-\mu_k R_k},$$

while the factor $\sum_{\bar z} u_k(\bar z)$ cancels. So $2e^{-\mu_k R_k} \ge \gamma$. □

So reflection positivity and decay do not supply (T2): at fixed $\beta$ in the range of Theorem 5 the measures are reflection positive with exponential clustering, and $C_k(\sigma)/C_k(0) \to 0$, so no choice of $w_k$, $r_k$ and constants satisfies the order-two cases of (T1) and (T2) together there. For the same reason no estimate whose constants give a lattice gap bounded below, as a convergent strong coupling expansion does [75], can give the tree bounds along the tuning: (T1) and (T2) hold, if at all, only along couplings where the lattice gap closes at least as fast as $a_k$. A mass gap in physical units, which bounds the decay of the two-point function, does not give the constants

$K_n$ either. By Theorem 57(v) the limit can be non-Gaussian only if $\liminf_k a_k^4 W_k > 0$; for the product majorants of part (iii) the time factor of $W_k$ is $\coth(\mu_k/2) \geq 2/\mu_k$, which is of order $1/a_k$ at least, so the time direction alone is compatible with this. At a fixed physical flow time the flowed fields are smooth on the scale $\sqrt{\tau}$; majorants of product form for them are not proved here.

The order-two cases of (T1) and (T2) say more about the reference vector than part (iv) of Theorem 57 records: for a smooth $f_0$ they bound every time difference of the packet, which makes its energy distribution tight with finite moments and gives the ratio with an explicit rate. Recall the spectral measures $\mu_k$ on $[0,\infty]$ of the proof of Theorem 48, so that $C_k(\sigma)/C_k(0) = \int e^{-\sigma E} d\mu_k(E)$.

**Proposition 61.** *Suppose the $\omega_k$ are translation invariant,* (T1) *holds for $n = 2$ and* (T2) *holds, and $f_0$ has $m \geq 1$ continuous time derivatives. Let $Q$ be a cube containing the supports of $f_0(\cdot - (s,0))$ for $0 \leq s \leq ma_0$, and $B_m = (K_2/c)\sup_k |Q|_k \, \|\partial_t^m f_0\|_\infty^2$.*

(i) *$\|(1-\mathbb{T}_k)^m \Psi_k\|^2 \leq B_m a_k^{2m}$ for every $k$, where $\Psi_k = Z_k \Psi_{A_k}$.*

(ii) *$\int \big((1-e^{-a_k E})/a_k\big)^{2m} d\mu_k(E) \leq B_m$ and $\mu_k([\Lambda,\infty]) \leq B_m(a_k + \Lambda^{-1})^{2m}$ for $\Lambda > 0$. Every weak limit $\mu$ of the $\mu_k$ satisfies $\mu(\{\infty\}) = 0$ and $\int E^{2m} d\mu(E) \leq B_m$. If* (T1)–(T3) *hold, the unit vector $\Psi(P_0)$ of Theorem 57*(iii)–(iv) *lies in the domain of $H^m$, with $\|H^m \Psi(P_0)\|^2 \leq B_m$.*

(iii) *If $a_k\sqrt{B_1} \leq \frac{1}{2}$, then $C_k(\sigma) \geq e^{-2\sqrt{B_1}\sigma} C_k(0)$ for every $\sigma \in a_k \mathbb{Z}_{\geq 0}$, and the gap $m_k$ of Proposition 60 satisfies $m_k \leq -\log(1 - a_k\sqrt{B_1}) \leq 2\sqrt{B_1}\, a_k$.*

*Proof.* (i) Let $D_k f_0 = \sum_{j=0}^m (-1)^j \binom{m}{j} f_0(\cdot - (ja_k, 0))$. By translation invariance $\mathbb{T}_k^j \Psi_{A_k} = \Psi_{\tau_j A_k}$ and $\tau_j A_k = A_k(f_0(\cdot - (ja_k,0)))$, so $(1-\mathbb{T}_k)^m \Psi_{A_k} = \Psi_{A_k(D_k f_0)}$, and as in the proof of Theorem 57(iv) its squared norm is the covariance $\omega_k(A_k(\overline{D_k f_0} \circ \theta_k) A_k(D_k f_0))$. The function $D_k f_0$ is supported in $Q$ and is an $m$-th difference with step $a_k$, so $\|D_k f_0\|_\infty \leq a_k^m \|\partial_t^m f_0\|_\infty$. Lemma 56 with $n = 2$ bounds the covariance by $K_2 r_k^2 |Q|_k \, a_k^4 W_k \|D_k f_0\|_\infty^2$, and (T2) bounds $C_k(0) = Z_k^{-2}$ below by $c\, r_k^2 a_k^4 W_k$.

(ii) By the spectral theorem the left side of (i) is $\int (1 - e^{-a_k E})^{2m} d\mu_k$, the integrand being 1 at $E = \infty$. For $E \geq \Lambda$, $(1-e^{-a_k E})/a_k \geq (1 - e^{-a_k \Lambda})/a_k \geq \Lambda/(1 + a_k \Lambda)$, because $1 - e^{-x} \geq x/(1+x)$; this gives the tail bound. The set $(\Lambda,\infty]$ is open in $[0,\infty]$, so $\mu((\Lambda,\infty]) \leq \liminf_k \mu_k((\Lambda,\infty]) \leq B_m \Lambda^{-2m}$, and $\mu(\{\infty\}) = 0$. For a continuous $\chi$ with $0 \leq \chi \leq 1$ and support in $[0,L]$, the same inequality gives $E \leq (1 + a_k L)(1 - e^{-a_k E})/a_k$ on that support, so $\int \chi E^{2m} d\mu = \lim_k \int \chi E^{2m} d\mu_k \leq B_m$; letting $\chi$ increase to 1 gives the moment bound. The vector $\Psi(P_0)$ has $\langle \Psi(P_0), e^{-\sigma H} \Psi(P_0)\rangle = \lim_k C_k(\sigma)/C_k(0) = \int e^{-\sigma E} d\mu$ for dyadic $\sigma > 0$ along the subsequence, so its spectral measure for $H$ is $\mu$.

(iii) By (i) with $m = 1$ and the Cauchy–Schwarz inequality, $\langle \Psi_k, (1-\mathbb{T}_k)\Psi_k\rangle \leq a_k\sqrt{B_1}$, which is the hypothesis of Proposition 49(ii) with $C = \sqrt{B_1}$. Since $\Psi_k$ is a unit vector orthogonal to $\Omega_k$, the same quantity is at least $1 - e^{-m_k}$. □

Part (iii) makes Proposition 60(ii) explicit: the order-two tree and saturation bounds pin the lattice gap below $2\sqrt{B_1}\, a_k$, a physical mass of at most $2\sqrt{B_1}$, and give the ratio of Theorem 48(iv) at every separation. The open input of Theorem 48 is therefore implied by these two bounds for one smooth $f_0$; nothing here proves them for the Wilson action.

## XI. THE OPEN INPUTS

This section states the two unproved inputs of Sec. X, the cube estimate and the limit, and measures the first: what can be controlled uniformly in the coupling, and what any proof of the cube estimate has to produce.

### A. The two open statements

Theorem 45 reduces a continuum mass gap to two statements, neither of which is proved here or, to our knowledge, anywhere else.

(D) *The cube estimate.* There are $\alpha < 1$ and $\ell > 0$ such that $D(n,\alpha)$ holds with $na \leq \ell$ along the continuum tuning $\beta = \beta(a) \to \infty$, uniformly in $a$.

(E) *The limit.* The renormalised Schwinger functions converge along $a_k = 2^{-k} a_0$ to Schwinger functions that satisfy the Osterwalder–Schrader axioms and are not those of the vacuum alone.

(D) and (E) together give the mass gap of the continuum theory; the remaining Clay requirements, Euclidean invariance and the local structure of the limit [56], belong to (E). (D) is a statement about a single cube of fixed physical size and a single coupling at each spacing; (E) is the construction problem, on which Bałaban's renormalisation group analysis [6–8] gives the ultraviolet part in finite volume.

*Remark* 62. The following count relates (D) to a correlation length. Suppose that the gauge-fixed field responds to a change of one link on the boundary of a cube through a Poisson kernel with decay length $\xi$ in lattice units. Then the response summed over the cube is of order $\xi$, so $K_c$ is of order $n^3\xi$ and $D(n,\alpha)$ holds once $n \gtrsim \xi/\alpha$, that is, once $\ell \gtrsim \xi a/\alpha$, a fixed multiple of the physical correlation length. The hypothesis is that the gauge-fixed links decorrelate on the scale $\xi$, and it is a statement about the gauge, not about the gauge-invariant observables: the gauge here is built on straight lines, and in gauges of axial type the gauge-fixed variables can fluctuate on all scales while gauge-invariant observables cluster. Propositions 63 and 64 carry the count out exactly for Gaussian fields, and Proposition 64 shows that the quadratic approximation of the forest gauge fluctuates in this way. Where it does, (D) fails in this gauge and has to be posed for block-averaged variables, where it becomes part of the renormalisation group problem. Sec. XII A carries this out in the quadratic approximation.

The count of Remark 62 can be carried out exactly for a Gaussian field, the quadratic approximation of the gauge-fixed theory at weak coupling, and shows what (D) asks of the gauge-fixed variables.

**Proposition 63.** *Let $d \geq 1$, $b > 0$, $t \geq 0$, let $n \geq 2$ be even and $V = [0,n)^d \cap \mathbb{Z}^d$. Given $\eta \in \mathbb{R}^{\mathbb{Z}^d \setminus V}$, let $\gamma_V(\cdot \mid \eta)$ be the Gaussian law on $\mathbb{R}^V$ with density proportional to*

$$\exp\Big(-\frac{b}{2}\Big[\sum_{\{x,y\}\cap V\neq\emptyset}(\phi_x-\phi_y)^2+t\sum_{x\in V}\phi_x^2\Big]\Big),$$

*the first sum over nearest-neighbour pairs, with $\phi = \eta$ outside $V$. Define $C_{V,j}$, the classes $c$ (residues modulo two) and $K_c$ as above, with $d_V(\phi,\phi') = \sum_{x\in V}|\phi_x - \phi'_x|$ and $\delta(\eta_j,\eta'_j) = |\eta_j - \eta'_j|$, and put $N_n = (n/2)^d$. Then $K_c = \alpha_n(t)N_n$ for every class $c$, where*

$$\alpha_n(t) = 1 - \frac{t}{n^d}\,\mathbf{1}^{\mathsf{T}}(t+L_n)^{-1}\mathbf{1}$$

*and $L_n$ is the Dirichlet Laplacian of $V$. In particular:*

(i) *for $t = 0$, $K_c = N_n$ for every $n$ and every $b$, so $D(n,\alpha)$ fails for every $\alpha < 1$;*

(ii) *for $t > 0$, $1 - \alpha_n(t) \geq (5/6)^d\, tn^2/(tn^2+10d)$; for $d = 4$, $t = m^2a^2$ and $\ell/2 \leq na \leq \ell$ this gives $D(n,\alpha)$ with $\alpha = 1 - (5/6)^4(m\ell)^2/((m\ell)^2+160)$, uniformly in $a$.*

*Proof.* Let $A = t + L_n$ and $G = A^{-1}$, and let $B$ be the matrix with $B_{xj} = 1$ when $x \in V$ and $j \notin V$ are neighbours. The law $\gamma_V(\cdot \mid \eta)$ has covariance $b^{-1}G$ and mean $GB\eta$. The matrix $A$ has nonpositive off-diagonal entries and is positive definite, so $G \geq 0$ entrywise. Changing $\eta_j$ by $\Delta$ translates the law by $\Delta GB_j$, where $B_j$ is the column of $j$. The coupling by translation gives $W_1^{d_V} \leq |\Delta|\sum_x (GB_j)_x$, and the 1-Lipschitz function $\sum_x \phi_x$ gives the reverse inequality, so $C_{V,j} = \mathbf{1}^{\mathsf{T}}GB_j$. Since $B\mathbf{1} = L_n\mathbf{1}$ (both count the exterior neighbours of each site),

$$\sum_{j\notin V} C_{V,j} = \mathbf{1}^{\mathsf{T}}GL_n\mathbf{1} = \mathbf{1}^{\mathsf{T}}\mathbf{1} - t\,\mathbf{1}^{\mathsf{T}}G\mathbf{1} = n^d\alpha_n(t).$$

The reflection $x_i \mapsto n-1-x_i$ preserves $V$ and the coefficients and, as $n$ is even, exchanges the two residues of the coordinate $x_i$; so the $2^d$ class sums are equal, and each is $n^d\alpha_n(t)/2^d = \alpha_n(t)N_n$. This proves the formula and (i).

For (ii) let $v(i) = (i+1)(n-i)$ for $0 \leq i < n$ and $\psi(x) = \prod_i v(x_i)$. The one-dimensional Dirichlet Laplacian maps $v$ to the constant 2, so with $s_1 = \sum_i v(i) = n(n+1)(n+2)/6$ and $s_2 = \sum_i v(i)^2 = n(n+1)(n+2)(n^2+2n+2)/30$,

$$\mathbf{1}^{\mathsf{T}}\psi = s_1^d, \qquad \psi^{\mathsf{T}}A\psi = ts_2^d + 2ds_1s_2^{d-1}.$$

By the Cauchy–Schwarz inequality for the form of $A^{-1}$, $\mathbf{1}^{\mathsf{T}}G\mathbf{1} \geq (\mathbf{1}^{\mathsf{T}}\psi)^2/\psi^{\mathsf{T}}A\psi$, hence

$$1-\alpha_n(t) \geq \Big(\frac{s_1^2}{ns_2}\Big)^d \frac{t}{t+2ds_1/s_2} \geq \Big(\frac{5}{6}\Big)^d \frac{tn^2}{tn^2+10d},$$

since $s_1^2/(ns_2) = \frac{5}{6}(n^2+3n+2)/(n^2+2n+2) \geq \frac{5}{6}$ and $2ds_1/s_2 = 10d/(n^2+2n+2) \leq 10d/n^2$. With $d = 4$ and $tn^2 = m^2(na)^2 \geq (m\ell)^2/4$ the last fraction is at least $(m\ell)^2/((m\ell)^2+160)$. □

The Gaussian field is not Yang–Mills theory, and the proposition is a benchmark for the condition, not a statement about the Wilson measure. It shows that the averaged cube condition is met at a fixed physical scale by a field with a mass and fails exactly, at every size and every stiffness, for a massless one. Small local fluctuations alone therefore give nothing: in the massless Gaussian field on $\mathbb{Z}^d$, $d \geq 3$, the variance of $\phi_x - \phi_{x+e_1}$ is $1/(db)$, which tends to zero as $b \to \infty$, and still $K_c = N_n$. A proof of (D) along the tuning has to produce in the gauge-fixed variables the loss that the parameter $t$ supplies here, from the Wilson action alone.

The quadratic approximation of the forest gauge itself, with the plaquette action in place of the Laplacian, is no better than the massless field of Proposition 63(i).

**Proposition 64.** *Let $b > 0$, let $n$ be even and $V$ the cube of Definition 42. For real functions $\varphi$ on the links of $\mathbb{Z}^4$ that vanish on $F$, let $(d\varphi)_p$ be the sum of $\varphi$ around the plaquette $p$ with the signs of its orientation. Given $\eta \in \mathbb{R}^{A\setminus V}$, let $\gamma_V(\cdot \mid \eta)$ be the law on $\mathbb{R}^V$ with density proportional to*

$$\exp\Big(-\frac{b}{2}\sum_{p\cap V\neq\emptyset}(d\varphi)_p^2\Big),$$

*with $\varphi = \eta$ on $A \setminus V$ and $\varphi = 0$ on $F$, and define $C_{V,j}$ and $K_c$ as above with $d_V(\varphi,\varphi') = \sum_{i\in V}|\varphi_i - \varphi'_i|$ and $\delta(\eta_j,\eta'_j) = |\eta_j - \eta'_j|$. Then*

$$\sum_c K_c \geq |V| = 48N_n,$$

*so $K_c \geq N_n$ for some class $c$, and $D(n,\alpha)$ fails for every $\alpha < 1$, at every $n$ and every $b$.*

*Proof. The laws.* The quadratic form $\sum_{p\cap V\neq\emptyset}(d\varphi)_p^2$ with $\varphi = 0$ outside $V$ vanishes only at $\varphi = 0$: the argument of Corollary 81, written additively, gives a function $h$ on the sites with $\varphi_{(x,\mu)} = h(x) - h(x+e_\mu)$, constant outside the box and along the lines of $F$, hence constant. So $\gamma_V(\cdot \mid \eta)$ is a Gaussian law whose covariance does not depend on $\eta$ and whose mean is $\sum_j \eta_j u_j$ for some vectors $u_j \in \mathbb{R}^V$. Two exteriors that differ in finitely many links give translates of one law by $t = \sum_j(\eta'_j - \eta_j)u_j$; the coupling by translation and the 1-Lipschitz function $\sum_i \mathrm{sign}(t_i)\varphi_i$ show that their distance is $\|t\|_1$. In particular $C_{V,j} = \|u_j\|_1$, and

$$W_1^{d_V}\big(\gamma_V(\cdot \mid \eta), \gamma_V(\cdot \mid \eta')\big) \leq \sum_j C_{V,j}\,|\eta'_j - \eta_j|.$$

*Lines.* By Lemma 9(i) the lines of $F$ are straight, disjoint and cover the sites, and the two ends of an active link $(x,\mu)$ lie on different lines, since the line through $x$ has direction $\sigma(x) \neq \mu$. For a line $\ell$ let $h_\ell$ be the indicator of its sites and $(dh_\ell)_{(x,\mu)} = h_\ell(x) - h_\ell(x+e_\mu)$. Then $dh_\ell$ vanishes on $F$ and around every plaquette, so the density is invariant under $\varphi \mapsto \varphi + dh_\ell$, and $\gamma_V(\cdot \mid dh_\ell)$ is $\gamma_V(\cdot \mid 0)$ translated by the restriction of $dh_\ell$ to $V$. Hence

$$\sum_{i\in V}|(dh_\ell)_i| \leq \sum_{j\in A\setminus V} C_{V,j}\,|(dh_\ell)_j|.$$

Here $|(dh_\ell)_e| = 1$ when exactly one end of $e$ lies on $\ell$ and 0 otherwise, so $\sum_\ell |(dh_\ell)_e| = 2$ for every active link $e$, the sum running over the finitely many lines that meet a link entering the inequality. Summing over these lines gives $2|V| \le 2\sum_j C_{V,j} = 2\sum_c K_c$. □

The proof uses only that gauge transformations constant on the lines of $F$ move the whole cube while changing few boundary links, which is the behaviour of gauges of axial type anticipated in Remark 62. The class sums themselves, computed exactly for $n \le 8$, lie between 1.57 and 11.24 times $N_n$ and increase with $n$. Below the correlation length the theory is weakly coupled, and the quadratic approximation is its leading order there. A proof of (D) along the tuning therefore has to show that the nonperturbative regime reduces the cube coefficients of the forest gauge below $N_n$, or (D) has to be posed for other variables.

The lines of $F$ are not what makes the argument work. A constant field, corrected by a periodic gauge transformation so that it vanishes on the fixed links, costs no action and changes every exterior link by a bounded amount, and this happens in every gauge fixed on an invariant forest.

**Proposition 65.** *Let $b > 0$ and let $F$ be a set of links of $\mathbb{Z}^4$, invariant under the translations by $2\mathbb{Z}^4$, that contains no closed path and whose trees are infinite and contain every site. Let $A$ be the set of the other links, $V$ the set of links $(x,\mu) \in A$ with $x \in [0,n)^4$, $n \ge 2$ even, and define the classes, $\gamma_V(\cdot \mid \eta)$, $C_{V,j}$ and $K_c$ as in Proposition 64 with this $F$. Then there are numbers $a_c \ge 0$, not all zero, with*

$$\sum_c a_c K_c \ge N_n \sum_c a_c,$$

*so $K_c \ge N_n$ for some class $c$, and $D(n,\alpha)$ fails for every $\alpha < 1$, at every $n$ and every $b$.*

*Proof. The laws.* Let $\varphi$ vanish outside $V$ and on $F$, with $(d\varphi)_p = 0$ for every plaquette $p$ that meets $V$. The other plaquettes contain no link of $V$, so $d\varphi = 0$ on every plaquette, and $\varphi = dh$ for a function $h$ on the sites. Since $\varphi$ vanishes outside a box and the complement of a box is connected, $h$ is constant there, and we take that constant to be 0. As $\varphi$ vanishes on $F$, $h$ is constant on every tree of $F$; each tree is infinite and meets the complement of the box, so $h = 0$ and $\varphi = 0$. Hence, as in the proof of Proposition 64, $\gamma_V(\cdot \mid \eta)$ is a nondegenerate Gaussian law with mean $\sum_j \eta_j u_j$, where only the links $j$ that share a plaquette with $V$ contribute, and $C_{V,j} = \|u_j\|_1$.

*A flat field.* For $c \in \mathbb{R}^4$ and a function $h$ on $\mathbb{Z}^4$ of period two in each coordinate put $\psi_{(x,\mu)} = c_\mu + h(x) - h(x+e_\mu)$. Then $d\psi = 0$ on every plaquette and $\psi$ is invariant under $2\mathbb{Z}^4$; if $\psi = 0$, adding $\psi$ at $(x,\mu)$ and $(x+e_\mu,\mu)$ gives $2c_\mu = 0$, and then $h$ is constant. These fields form a space of dimension $16 - 1 + 4 = 19$. Let $f$ be the number of orbits of $2\mathbb{Z}^4$ in $F$. The box $[0,2m)^4$ has $16m^4$ sites, and the links of $F$ with both ends in it form a forest, so there are at most $16m^4 - 1$ of them, while each orbit contributes at least $(m-1)^4$; letting $m \to \infty$ gives $f \le 16$. Vanishing on $F$ is therefore at most 16 linear conditions, and some $\psi \ne 0$ of this form vanishes on $F$. It is nonzero on some link of $A$. Put $a_c = |\psi_j|$ for $j \in c$.

*The inequality.* Let $\eta$ be the restriction of $\psi$ to $A \setminus V$. The field equal to $\psi$ on $V$ and to $\eta$ outside gives the quadratic form its least value 0, so by nondegeneracy the mean of $\gamma_V(\cdot \mid \eta)$ is the restriction of $\psi$ to $V$, and $\psi_i = \sum_j \psi_j (u_j)_i$ for $i \in V$. Hence

$$\sum_{i \in V} |\psi_i| \le \sum_{j \in A \setminus V} C_{V,j} |\psi_j| = \sum_c a_c K_c.$$

A class consists of the active links with a given direction and a given residue of the base point modulo two, so it meets $V$ in $N_n$ links, and the left side is $N_n \sum_c a_c > 0$. So $K_c \ge N_n$ for some class with $a_c > 0$. □

For the direction field of Table III the flat fields vanishing on $F$ form a space of dimension 7, and for the axial field $\sigma = 0$ one of dimension 10. The proposition is weaker than Proposition 64 for the forest gauge, which bounds the sum over all classes, and it says nothing about two kinds of condition. A mass gives the constant fields an action, which is why Proposition 63 distinguishes $t > 0$ from $t = 0$. The constrained condition $D_c$ of Theorem 98 holds the block averages fixed, and a constant field changes them. Within the quadratic approximation no gauge fixed on an invariant forest gives (D); the loss has to come from the Wilson action beyond its quadratic part, or (D) has to be posed for variables, such as the block averages, that a constant field moves.

*Remark* 66. Five other routes stop at (D) or (E), or earlier.

(a) *Disjoint blocks.* The block criterion of Lemma 243 with a partition into cubes of side $n$ requires, for a link on a face, that its influence on the neighbouring cube be less than one. In the count of Remark 62 that influence is of order $\xi$, which grows like $1/a$. Averaging over the $N_n$ translates divides it by the number of translates on whose boundary the link lies, a factor of order $1/n$, and this is why (D) is stated for averaged cubes.

(b) *Rescaling a fixed lattice gap.* Any bound uniform in lattice units, such as those of Theorems 1–7, gives trivial continuum two-point functions [17]; rescaling it produces no gap.

(c) *Assuming the gap.* Hypothesis R of [17] assumes the gap in physical units together with the limit. Theorem 45 replaces the gap part by the cube estimate (D), which concerns one box and one coupling at each spacing.

(d) *Computation.* At a fixed coupling and a fixed small $n$, $D(n,\alpha)$ is a finite set of Kantorovich bounds and could in principle be certified as in Propositions 34 and 35; for $n = 2$ and SU(2) the block already has 48 links and 144 real variables. (D) asks for it at infinitely many couplings with $n \to \infty$, which no finite computation covers.

(e) *Axioms alone.* Proposition 41 shows that positivity, invariance and regularity do not force a gap, and Proposition 40 shows that the growth route to rotation invariance is closed for the action density. The infrared input therefore has to be an estimate such as (D).

### B. The ratio, the strong-coupling range and the cube estimate

This subsection states the open input of Theorem 48 exactly, proves that it fails along every sequence of couplings in the range where the paper has a gap, shows that the hypothesis of Theorem 45 holds along some of these sequences and therefore cannot imply the input, and replaces the input by a condition that is necessary and sufficient and asks only for an upper bound on an energy and a lower bound on one overlap.

The notation is that of Theorem 48: translation invariant states $\omega_k$ at couplings $\beta_k$, with transfer matrices $\mathbb{T}_k$ and vacua $\Omega_k$ (Appendix F), the centred flowed packet $A_k$ of a nonzero $f_0 \in C_c(\mathbb{R}^4)$ supported in $\{t>0\}$, and $C_k(t) = \omega_k(\vartheta(\bar{A}_k)\tau_{t/a_k}A_k)$ for $t \in a_k\mathbb{Z}_{\geq 0}$, with $C_k(0)>0$. For $u \geq 0$ and $\sigma > 0$ in $a_k\mathbb{Z}$ put

$$\rho_k(u,\sigma) = \frac{C_k(2u+\sigma)}{C_k(2u)},$$

so that $\rho(\sigma)$ of Theorem 48 is a limit of $\rho_k(0,\sigma)$. Let $f_{0,u} = f_0(\cdot-(u,0))$; for $u/a_k \in \mathbb{Z}$ its packet is $\tau_{u/a_k}A_k$, and $\Psi_{k,u} = C_k(2u)^{-1/2}\Psi_{\tau_{u/a_k}A_k}$ is the corresponding unit vector. For dyadic $u \geq 0$ consider

$(\mathrm{N}_u)$ $\liminf_{k\to\infty}\rho_k(u,\sigma) > 0$ for some dyadic $\sigma > 0$.

**Proposition 67.** (i) *$C_k(t) > 0$ for every $t \in a_k\mathbb{Z}_{\geq 0}$, and $\log C_k$ is convex and nonincreasing on $a_k\mathbb{Z}_{\geq 0}$.*

(ii) *$0 < \rho_k(u,\sigma) \leq 1$; $\rho_k(u,\sigma)$ is nondecreasing in $u$ and nonincreasing in $\sigma$, and $\rho_k(u,\sigma') \geq \rho_k(u,\sigma)^{\sigma'/\sigma}$ for $\sigma' \geq \sigma$.*

(iii) *For $u \geq 0$ dyadic, Theorem 48 applies to $f_{0,u}$, with reflected norm $C_k(2u)$ and ratio $\rho_k(u,\sigma)$. Every limit it produces along subsequences is nontrivial if and only if $(\mathrm{N}_u)$ holds, and some limit is nontrivial if and only if $\limsup_k \rho_k(u,\sigma) > 0$. If $(\mathrm{N}_u)$ holds for one $\sigma$, it holds for every dyadic $\sigma > 0$, and $(\mathrm{N}_u)$ implies $(\mathrm{N}_{u'})$ for $u' \geq u$.*

(iv) *Let $\mu_{k,u}$ be the spectral measure, in energies $E = -a_k^{-1}\log\lambda$, of $\mathbb{T}_k$ at $\Psi_{k,u}$. If $\rho_k(u,\sigma) \geq c > 0$, then $\mu_{k,u}([0,\Lambda]) \geq c/2$ with $\Lambda = \sigma^{-1}\log(2/c)$; if $\mu_{k,u}([0,\Lambda]) \geq v$, then $\rho_k(u,\sigma) \geq ve^{-\sigma\Lambda}$. So $(\mathrm{N}_u)$ holds if and only if a fixed fraction of the spectral mass of $\Psi_{k,u}$ stays at bounded physical energy, uniformly in $k$.*

*Proof.* (i) The function $f_0$ is supported in $\{t>0\}$, so the lattice points with $f_0(a_k x) \neq 0$ have $x_0 \geq 1$, and $A_k = \sum_{n\geq 1}\tau_n B_n$ is a finite sum in which $B_n$ is a function of the spatial links of the slice at time 0, by Proposition 47(i). In the reconstruction [73–75], as in the proof of Theorem 48, $\Psi_{\tau_n B} = \mathbb{T}_k^n\Psi_B$, hence $\Psi_{A_k} = \mathbb{T}_k\xi$ with $\xi = \sum_n \mathbb{T}_k^{n-1}\Psi_{B_n}$. The transfer matrix is self-adjoint and positive [64, 75, 82], and its range is orthogonal to its kernel, so the spectral measure $\nu_k$ of $\mathbb{T}_k$ at $\Psi_k = C_k(0)^{-1/2}\Psi_{A_k}$ (Lemma 246) has no atom at 0. Therefore $C_k(t)/C_k(0) = \int \lambda^{t/a_k}d\nu_k > 0$. Convexity and monotonicity are Proposition 51(ii).

(ii) The first claims follow from (i). On a lattice, a convex function has chord slopes that increase in both end points; this gives monotonicity in $u$, and, comparing the chords $[2u,2u+\sigma]$ and $[2u,2u+\sigma']$, the power bound. Monotonicity in $\sigma$ is that of $C_k$.

(iii) Translation invariance and $\vartheta\tau_u = \tau_{-u}\vartheta$ give, with times written in lattice steps,

$$\omega_k\big(\vartheta(\tau_u\bar{A}_k)\,\tau_{t+u}A_k\big) = \omega_k\big(\vartheta(\bar{A}_k)\,\tau_{t+2u}A_k\big),$$

which gives the reflected norm and the ratio. The function $f_{0,u}$ is nonzero and supported in $\{t>u\}$. If $\liminf_k\rho_k(u,\sigma) = 0$, a subsequence has $\rho_k(u,\sigma)\to 0$, and a refinement given by Theorem 48(ii) has a trivial limit by part (iv) of that theorem; if the liminf is at least $c>0$, every limit has $\rho(\sigma)\geq c$ and is nontrivial. The statement with limsup is the last part of Theorem 48(iv). Independence of $\sigma$ follows from (ii), by the power bound for $\sigma' \geq \sigma$ and by monotonicity for $\sigma' \leq \sigma$, and the last claim from monotonicity in $u$.

(iv) The vector $\Psi_{k,u}$ has no spectral weight at $\lambda = 0$, by the proof of (i), and $\rho_k(u,\sigma) = \int e^{-\sigma E}d\mu_{k,u}$. This integral is at most $\mu_{k,u}([0,\Lambda]) + e^{-\sigma\Lambda}$ and at least $e^{-\sigma\Lambda}\mu_{k,u}([0,\Lambda])$. □

The open input of Theorem 48 is therefore exactly $(\mathrm{N}_u)$ for one packet $f_0$ and one dyadic $u \geq 0$. It is a uniformity statement and nothing else: at each fixed $k$ the ratio is positive by (i). The required uniformity is in $k$ along the tuning, at fixed physical separations $u$ and $\sigma$, so that $\sigma/a_k \to \infty$ lattice steps; the constant may depend on $G$, $\Phi$, $\tau$, $f_0$, $u$, $\sigma$ and on the tuning, and nothing is required uniformly in $f_0$ or in $\tau$. The ratio does not change when $A_k$ is multiplied by a constant, so the renormalisation $Z_k$ plays no role in it. The case $u = 0$ is the strongest and implies all the others. Proposition 49(i) assumes $(\mathrm{N}_0)$ and renormalises the packet shifted by $\sigma/2$. The hypothesis of Lemma 196 is $(\mathrm{N}_{s/4})$ at $\sigma = s/2$; it is weaker than $(\mathrm{N}_0)$, as the example after Proposition 51 shows, and it suffices for a nontrivial limit of the packet shifted by $s/4$. Granted the convergence asked for in (E), the clause of (E) that the limit is not that of the vacuum alone is, for the flowed packets, $(\mathrm{N}_u)$ for one $f_0$ and one $u$; the convergence of the whole sequence, Euclidean invariance and the higher correlations remain [56]. No bound is required at one lattice step: the first-step energy $-a_k^{-1}\log\rho_k(u,a_k)$ may diverge while $(\mathrm{N}_u)$ holds, as Remark 53 shows for the slice field of the unflowed magnetic energy.

*The strong-coupling range.* In the range where the paper proves a gap, the ratio has two-sided bounds in lattice units, and the upper one rules out every continuum tuning there. Put

$$m_0 = -\tfrac{1}{2}\log 0.99995,$$

so that $m_0 \geq 2.5\cdot 10^{-5}$ by $-\log(1-x)\geq x$; interval arithmetic gives $m_0 \in [2.50006250\cdot 10^{-5}, 2.50006251\cdot 10^{-5}]$.

**Theorem 68.** *Let $G = \mathrm{SU}(2)$, let $\beta_{W,k} \in [0,0.351]$ be any couplings, and let $\omega_k$ be the states of Theorems 1 and 3. Then for every $\Phi$, $\tau$ and $f_0$ with $C_k(0)>0$, every dyadic $u\geq 0$ and $\sigma > 0$, and every $k$ with $u/a_k, \sigma/a_k \in \mathbb{Z}$,*

$$\rho_k(0,a_k)^{\sigma/a_k} \leq \rho_k(u,\sigma) \leq e^{-m_0\sigma/a_k}.$$

*Consequently every limit of Theorem 48 along these couplings, for every packet $f_{0,u}$, is trivial, and if $(\mathrm{N}_u)$ holds for some packet along a tuning $(\beta_{W,k})$ of $\mathrm{SU}(2)$, then $\beta_{W,k} > 0.351$ for all large $k$.*

*Proof.* By Theorem 3 the transfer matrix of $\omega_k$ has gap $m_k \geq -\frac{1}{2}\log\alpha(\beta_{W,k})$ on $\Omega_k^\perp$, and $\alpha$ is increasing with $\alpha(0.351) \leq 0.99995$, so $m_k \geq m_0$. The vector $\Psi_{k,u}$ is a unit vector orthogonal to $\Omega_k$, because $A_k$ is centred, and $\mathbb{T}_k$ preserves $\Omega_k^\perp$, so $\rho_k(u,\sigma) = \langle \Psi_{k,u}, \mathbb{T}_k^{\sigma/a_k}\Psi_{k,u}\rangle \leq e^{-m_0\sigma/a_k}$. For the lower bound, $\rho_k(u,\sigma) \geq \rho_k(0,\sigma)$ by Proposition 67(ii), and Jensen's inequality for the convex function $\lambda^{\sigma/a_k}$ and the probability measure $\nu_k$ of the proof of Proposition 67(i) gives $\rho_k(0,\sigma) = \int \lambda^{\sigma/a_k} d\nu_k \geq (\int \lambda\, d\nu_k)^{\sigma/a_k} = \rho_k(0,a_k)^{\sigma/a_k}$, with $\rho_k(0,a_k) > 0$ by Proposition 67(i). The upper bound tends to zero as $a_k \to 0$, which gives triviality by Proposition 67(iii). If $(\mathrm{N}_u)$ held along a tuning with $\beta_{W,k} \leq 0.351$ for infinitely many $k$, the upper bound along those $k$ would contradict it. □

Writing $\rho_k(u,\sigma) = e^{-\varepsilon_k\sigma/a_k}$, the theorem says $m_0 \leq \varepsilon_k \leq -\log\rho_k(0,a_k)$: both ends are rates in lattice units, and $(\mathrm{N}_u)$ asks for $\varepsilon_k = O(a_k)$. Reflection positivity gives the positivity of the lower end, through the range of $\mathbb{T}_k$, and the transfer-matrix gap gives the upper bound; neither bears on the size of $\varepsilon_k/a_k$ except to exclude the range. The size of $\rho_k(0,a_k)$ is not estimated here; any estimate of it at strong coupling is in lattice units, and it is at most $e^{-m_0}$. The same holds for every compact simple $G$ and couplings in $[0,\beta_1]$ with $N\beta_1 < \frac{2}{29}\kappa_G$, with $m_0 = m_G(\beta_1)$ of Theorem 5; for fixed $\beta$ this is Proposition 60(i). So the strong-coupling range bears on the continuum tuning only negatively: a tuning with a nontrivial limit leaves it for good. That the tuning has to go to $\beta_W \to \infty$ is the expectation from asymptotic freedom and is not proved here.

*The cube estimate in the same range.* The hypothesis of Theorem 45 holds at fixed strong coupling with cubes of fixed physical size, and with every $\alpha$.

**Proposition 69.** (i) *Suppose $D(n_k,\alpha_k)$ holds at $\beta_k$ with $n_k a_k \leq \ell$. Then $\rho_k(u,\sigma) \leq \alpha_k^{\sigma/(2\ell)}$ for every $u$ and $\sigma$. Hence $(\mathrm{N}_u)$ with constant $c$ at $\sigma$ forces $\alpha_k \geq c^{2\ell/\sigma}$ for large $k$, and the physical mass of Theorem 45 obeys $m_* \leq \sigma^{-1}\log(1/c)$.*

(ii) *Let $G = \mathrm{SU}(2)$, $0 \leq \beta_W < \frac{1}{3}$ fixed, $\ell > 0$, and $n_k = 2\lfloor \ell/(2a_k)\rfloor$. For every $\alpha \in (0,1)$, $D(n_k,\alpha)$ holds at $\beta_W$ for all $k$ with $n_k \geq 2$ and $5\beta_W \leq \alpha(1-3\beta_W)n_k$, and $n_k a_k \leq \ell$. So the hypothesis of Theorem 45 holds along the constant tuning, from some $k$ on, with every $\alpha \in (0,1)$, while every limit of Theorem 48 is trivial.*

*Proof.* (i) By Theorem 43 the gap of $\mathbb{T}_k$ on $\Omega_k^\perp$ is at least $\log(1/\alpha_k)/(2n_k) \geq a_k\log(1/\alpha_k)/(2\ell)$, and $\rho_k(u,\sigma) \leq e^{-m_k\sigma/a_k}$ as in the proof of Theorem 68. The bound on $m_*$ is the case $\alpha_k = \alpha$, since then $\alpha^{\sigma/(2\ell)} = e^{-m_*\sigma}$.

(ii) Proposition 78 gives $D(n,\alpha)$ for every $\alpha \geq \hat\alpha_n$, and $\hat\alpha_n \leq 5\beta_W/((1-3\beta_W)n)$. Triviality is Theorem 68. □

Part (i) shows that (D) is itself a triviality statement when its constant improves along the tuning: if $\alpha_k \to 0$ with $n_k a_k \leq \ell$, every limit is trivial. Along the constant tuning of part (ii) with $\beta_W > 0$ this gives $\rho_k(u,\sigma) \leq (5\beta_W/((1-3\beta_W)n_k))^{\sigma/(2\ell)}$, a power of $a_k$, from the cube estimate alone. Part (ii) shows that no argument that uses (D) only through the hypothesis of Theorem 45, or through Theorem 43, can give $(\mathrm{N}_u)$: that hypothesis is satisfied by Wilson measures whose continuum limits are trivial. Statement (D) of Sec. XI adds that $\beta \to \infty$ along the tuning, and an implication that uses this and the weak-coupling structure besides (D) is not excluded; it would use information that (D) does not contain.

*Trial states.* The input is equivalent to the existence of states of bounded physical energy with a fixed overlap with the packet.

**Lemma 70.** *Let $\mathbb{T}$ be self-adjoint with $0 \leq \mathbb{T} \leq 1$, let $a > 0$ and let $\Psi$ be a unit vector.*

(a) *Let $\chi$ be a unit vector with $\langle\chi,(1-\mathbb{T})\chi\rangle \leq \kappa a$ and $|\langle\chi,\Psi\rangle| \geq v > 0$. Then for $\sigma \geq 2a$,*

$$\langle\Psi,\mathbb{T}^{\sigma/a}\Psi\rangle \geq \Big(v - \sqrt{\kappa\sigma/2}\Big)_+^2,$$

*and if $a\Lambda \leq 1$ with $\Lambda = 8\kappa/v^2$, the spectral projection $P = \mathbf{1}_{[e^{-a\Lambda},1]}(\mathbb{T})$ has $\|P\Psi\| \geq v/2$.*

(b) *Conversely, if $\langle\Psi,\mathbb{T}^{\sigma/a}\Psi\rangle \geq c > 0$ with $\sigma/a \in \mathbb{Z}_{>0}$, put $\Lambda = \sigma^{-1}\log(2/c)$ and $P = \mathbf{1}_{[e^{-a\Lambda},1]}(\mathbb{T})$. Then $\chi = P\Psi/\|P\Psi\|$ is a unit vector with $\langle\chi,(1-\mathbb{T})\chi\rangle \leq a\Lambda$ and $\langle\chi,\Psi\rangle = \|P\Psi\| \geq (c/2)^{1/2}$. If $\mathbb{T}\Omega = \Omega$ and $\Psi \perp \Omega$, then $\chi \perp \Omega$.*

*Proof.* (a) Let $m = \sigma/(2a) \geq 1$ and let $\nu_\chi$ be the spectral measure at $\chi$. For $\lambda \in [0,1]$, $0 \leq 1-\lambda^m \leq m(1-\lambda)$ by Bernoulli's inequality, so

$$\|(1-\mathbb{T}^m)\chi\|^2 = \int (1-\lambda^m)^2 d\nu_\chi \leq m\int(1-\lambda)\,d\nu_\chi \leq \frac{\kappa\sigma}{2}.$$

Hence $\|\mathbb{T}^m\Psi\| \geq |\langle\mathbb{T}^m\chi,\Psi\rangle| \geq |\langle\chi,\Psi\rangle| - \|(1-\mathbb{T}^m)\chi\| \geq v - \sqrt{\kappa\sigma/2}$, and $\langle\Psi,\mathbb{T}^{\sigma/a}\Psi\rangle = \|\mathbb{T}^m\Psi\|^2$. For the projection, $1-\lambda > 1-e^{-a\Lambda} \geq a\Lambda/(1+a\Lambda) \geq a\Lambda/2$ off $[e^{-a\Lambda},1]$, so $\|(1-P)\chi\|^2 \leq 2\kappa/\Lambda = v^2/4$, and $\|P\Psi\| \geq |\langle\chi,P\Psi\rangle| \geq |\langle\chi,\Psi\rangle| - \|(1-P)\chi\| \geq v/2$.

(b) With $\nu$ the spectral measure at $\Psi$, $c \leq \int\lambda^{\sigma/a}d\nu \leq \|P\Psi\|^2 + e^{-\sigma\Lambda} = \|P\Psi\|^2 + c/2$. On the range of $P$, $1-\mathbb{T} \leq 1-e^{-a\Lambda} \leq a\Lambda$. Since $P\Omega = \Omega$, $\langle\Omega,P\Psi\rangle = \langle\Omega,\Psi\rangle = 0$. □

**Corollary 71.** $(\mathrm{N}_u)$ *holds if and only if there are $\kappa, v > 0$ and unit vectors $\chi_k$ in the reconstructed spaces of the $\omega_k$ such that, for all large $k$,*

$$\langle\chi_k,(1-\mathbb{T}_k)\chi_k\rangle \leq \kappa a_k, \qquad |\langle\chi_k,\Psi_{k,u}\rangle| \geq v.$$

*Then $\rho_k(u,\sigma) \geq (v-\sqrt{\kappa\sigma/2})_+^2$ for $\sigma \geq 2a_k$, and the $\chi_k$ may be taken of the form $\Psi_{X_k}/\|\Psi_{X_k}\|$ for gauge-invariant local functions $X_k$ at positive times, at the cost of replacing $(\kappa,v)$ by $(2\kappa,v/2)$. If moreover $D(n_k,\alpha)$ holds with $n_k a_k \leq \ell$, then $m_* = \log(1/\alpha)/(2\ell) \leq 8\kappa/v^2$.*

*Proof.* Sufficiency is Lemma 70(a) with $\Psi = \Psi_{k,u}$, $a = a_k$, and Proposition 67(ii) extends the bound from small $\sigma$ to every $\sigma$. Necessity is Lemma 70(b) with $\kappa = \sigma^{-1}\log(2/c)$ and $v = (c/2)^{1/2}$. The vectors $\Psi_X$ with $X$ local are dense (Appendix F), and both conditions are continuous in $\chi_k$ for fixed $k$, $1-\mathbb{T}_k$ being bounded. Under $D(n_k,\alpha)$, Theorem 43 puts the spectral measure of $\Psi_{k,u} \perp \Omega_k$ in $[0,e^{-a_k m_*}]$; if $m_* > 8\kappa/v^2$, the projection of Lemma 70(a) annihilates $\Psi_{k,u}$ for large $k$, which is impossible. □

For $\chi_k = \Psi_{X_k}/\|\Psi_{X_k}\|$ the two conditions are statements about lattice correlations:

$$\omega_k\big(\vartheta(\bar X_k)X_k\big) - \omega_k\big(\vartheta(\bar X_k)\tau_1 X_k\big) \le \kappa a_k\, \omega_k\big(\vartheta(\bar X_k)X_k\big),$$
$$\big|\omega_k\big(\vartheta(\bar X_k)\tau_{u/a_k}A_k\big)\big|^2 \ge v^2\, \omega_k\big(\vartheta(\bar X_k)X_k\big)\, C_k(2u).$$

The first is an upper bound on a one-slab Dirichlet form, which is the direction of variational and perturbative estimates and the opposite of a mass gap; the second is a lower bound on one correlation. The choice $\chi_k = \Psi_{k,u}$ is Proposition 49(ii), and the order-two tree and saturation bounds give it by Proposition 61(iii); the corollary shows that the trial vector need not be the packet itself.

*A soft component of the slice field.* For product packets the trial vector can be built from a part of the slice field of Proposition 51, and the conditions become one first-step bound for that part and one bound for the rest at the earliest time of the packet. Let $f_0 = g \otimes h$ with $g \ge 0$ supported in $[t_0, t_1]$, and let $u_k$, $n_0$, $n_1$ be as in Proposition 51.

**Theorem 72.** *Suppose that for all large $k$ there is a decomposition $u_k = u_k^{\mathrm{s}} + u_k^{\mathrm{h}}$ with $\mathbb{T}_k^{n_0} u_k^{\mathrm{s}} \ne 0$ and, for constants $\kappa > 0$ and $\theta \in [0,1)$ with $\kappa a_k \le \frac12$,*

(S1) $\langle u_k^{\mathrm{s}}, (1-\mathbb{T}_k)u_k^{\mathrm{s}}\rangle \le \kappa a_k \|u_k^{\mathrm{s}}\|^2$;

(S2) $\|\mathbb{T}_k^{n_0} u_k^{\mathrm{h}}\| \le \theta e^{-2\kappa(t_1-t_0)} \|\mathbb{T}_k^{n_0} u_k^{\mathrm{s}}\|$.

*Then the hypothesis of Corollary 71 holds for $u = 0$ with $\kappa e^{4\kappa(t_1-t_0)}$ in place of $\kappa$ and $v = (1-\theta)/(1+\theta)$; in particular* $(\mathrm{N}_0)$ *holds and*

$$\frac{C_k(\sigma)}{C_k(0)} \ge \Big(\frac{1-\theta}{1+\theta} - \sqrt{\tfrac12 \kappa\sigma e^{4\kappa(t_1-t_0)}}\Big)_+^2$$

*for $\sigma \ge 2a_k$.*

*Proof.* By the proof of Proposition 51, $\Psi_{A_k} = W(\mathbb{T}_k)u_k$ with $W(\lambda) = \sum_n w_n \lambda^n$, $w_n \ge 0$, $w_n = 0$ unless $n_0 \le n \le n_1$, and $\|w\|_1 \lambda^{n_1} \le W(\lambda) \le \|w\|_1 \lambda^{n_0}$ on $[0,1]$, with $\|w\|_1 > 0$ because $C_k(0) > 0$. Write $\nu$ for the spectral measure at $u_k^{\mathrm{s}}$ and $y = \mathbb{T}_k^{n_0} u_k^{\mathrm{s}}$. For a probability measure, a nondecreasing and a nonincreasing function have $\int fg \le \int f \int g$, because $\iint (f(x)-f(x'))(g(x)-g(x')) \le 0$; with $f = \lambda^{2n_0}$, $g = 1-\lambda$ and (S1),

$$\int \lambda^{2n_0}(1-\lambda)\, d\nu \le \kappa a_k \|y\|^2.$$

So $y$ satisfies (S1) as well, and Jensen's inequality for $\lambda^{2j}$, $j = n_1 - n_0$, gives $\|\mathbb{T}_k^j y\|^2 \ge \|y\|^2(1-\kappa a_k)^{2j} \ge \|y\|^2 e^{-4\kappa(t_1-t_0)}$, since $\log(1-x) \ge -2x$ on $[0, \frac12]$ and $a_k j \le t_1 - t_0$. Hence

$$\|W(\mathbb{T}_k)u_k^{\mathrm{s}}\| \ge \|w\|_1 \|\mathbb{T}_k^{n_1} u_k^{\mathrm{s}}\| \ge \|w\|_1 e^{-2\kappa(t_1-t_0)} \|y\|,$$

while $\|W(\mathbb{T}_k)u_k^{\mathrm{h}}\| \le \|w\|_1 \|\mathbb{T}_k^{n_0} u_k^{\mathrm{h}}\| \le \theta \|W(\mathbb{T}_k)u_k^{\mathrm{s}}\|$ by (S2). Let $\chi_k = W(\mathbb{T}_k)u_k^{\mathrm{s}}/\|W(\mathbb{T}_k)u_k^{\mathrm{s}}\|$. Its energy is

$$\frac{\int W^2(1-\lambda)\, d\nu}{\int W^2 d\nu} \le \frac{\|w\|_1^2 \kappa a_k \|y\|^2}{\|w\|_1^2 e^{-4\kappa(t_1-t_0)} \|y\|^2} = \kappa a_k e^{4\kappa(t_1-t_0)},$$

and with $r = \|W(\mathbb{T}_k)u_k^{\mathrm{h}}\|/\|W(\mathbb{T}_k)u_k^{\mathrm{s}}\| \le \theta$ its overlap with $\Psi_k = \Psi_{A_k}/\|\Psi_{A_k}\|$ is at least $(1-r)/(1+r) \ge (1-\theta)/(1+\theta)$. Corollary 71 gives the rest. □

Condition (S1) asks for a soft component of the slice field whose first-step energy is bounded in physical units. Remark 53 shows why the whole slice field is not a candidate: at next order its first-step energy grows like $\log(\sqrt{\tau}/a_k)$, through a hard component of norm of order $g^2$. Condition (S2) asks that this hard component, propagated to the earliest time $t_0$ of the packet, be a fixed fraction of the soft part. For the free field of Remark 52, (S1) holds with $u_k^{\mathrm{h}} = 0$: since $(1-e^{-a_k\omega})/a_k \le \omega \le a_k^{-1}\sinh(a_k\omega)$, the first-step energy $\langle u_k, (1-\mathbb{T}_k)u_k\rangle/(a_k\|u_k\|^2)$ has a numerator whose integrand is at most a multiple of $e^{-8t_f|p|^2/\pi^2}$, and a denominator $D_k(0)$ with a positive limit, as in that remark. For the unflowed magnetic energy the whole slice field violates (S1) by a factor of order $1/a_k$, and the decomposition has to separate its hard modes. Both conditions concern energies of order $1/a_k$ at physical times below $t_0$; they are ultraviolet statements, the subject of a renormalisation group analysis between the scales $a_k$ and $t_0$ [6–8], and they are not proved here. By Corollary 71 the trial-state condition is the least hypothesis that gives $(\mathrm{N}_u)$, since it is also necessary; by Proposition 69(ii) it is not a consequence of the cube estimate in the form used by Theorem 45; and (S1)–(S2) are its form for the slice field.

*Remark* 73. There is no circularity between $(\mathrm{N}_u)$ and (D). The proofs of Proposition 67, Lemma 70, Corollary 71 and Theorem 72 use reflection positivity, the transfer matrix and the spectral theorem only, and Theorem 48(i)–(iv), including the renormalisation $Z_k = C_k(0)^{-1/2}$, does not use (D). Conversely, Theorems 43 and 45 use no lower bound on any correlation. The two statements are opposite inequalities on the same spectral measures: (D) bounds $\rho_k$ from above by $e^{-m_*\sigma}$, $(\mathrm{N}_u)$ from below. Their joint validity requires only the compatibility relations $\alpha \ge c^{2\ell/\sigma}$, $m_* \le \sigma^{-1}\log(1/c)$ and $m_* \le 8\kappa/v^2$ of Proposition 69 and Corollary 71, all of them statements in physical units. Neither implies the other. Strong-coupling Wilson measures satisfy the hypothesis of Theorem 45 and violate $(\mathrm{N}_u)$ (Proposition 69(ii)). In the free abelian theory the quadratic approximation of the forest gauge violates (D) at every size and stiffness (Proposition 64), while the zero-momentum packet of the magnetic energy has slice ratios with positive limits (Remark 53) and so satisfies $(\mathrm{N}_0)$ by Proposition 51(iii); (D) depends on the gauge and $(\mathrm{N}_u)$ does not. Theorem 57(iv) derives $(\mathrm{N}_0)$ from the order-two cases of (T1) and (T2), and Proposition 60 shows that those cases fail wherever the lattice gap is bounded below, in particular in the strong-coupling range, where the hypothesis of Theorem 45 holds.

### C. Estimates uniform in the coupling

Some local quantities of the Wilson measure can be controlled uniformly in $\beta_W$, the volume and the boundary condition, by integrating by parts in a single link. Such estimates hold along every tuning, but they concern local fields in lattice units, and the three results below show how far they are from the open inputs: the force has bounded moments but its centred form is a null vector of the reflected form, the plaquette has fluctuations of order $\beta_W^{-2}$ but this is a single-site variance, and

bare links over a fixed physical distance become independent Haar variables.

Let $G=\mathrm{SU}(N)$, write $b=\beta_W$, so that the Wilson weight is $e^{bS}$ with $S=\sum_p N^{-1}\operatorname{Re}\operatorname{Tr}Q_p$, and give $\mathfrak{g}$ the metric $\langle T,R\rangle=-\operatorname{Tr}(TR)$. For a link $e$ let $S_e$ be the sum of the six plaquette terms containing $e$, $X_T$ the derivative along $U_e\mapsto e^{uT}U_e$ at $u=0$, $\nabla_e$ and $\Delta_e$ the gradient and Laplacian in $U_e$, $q=N^2-1$, $c_N=(N^2-1)/N$, and

$$E_e=b|\nabla_e S_e|^2,\qquad W_e=E_e-c_N S_e=b|\nabla_e S_e|^2+\Delta_e S_e.$$

Below, $\omega$ is a Wilson Gibbs state at $b>0$: a finite-volume measure in which $e$ is not fixed by the boundary condition, or an infinite-volume state whose conditional law of $U_e$ given the other links is proportional to $e^{bS_e}dU_e$ with $dU_e$ the Haar measure.

**Proposition 74.** (i) *For every unit $T\in\mathfrak{g}$, $t\in\mathbb{R}$ and $0\le\eta<N/12$,*

$$\mathbb{E}_\omega\big[e^{t\sqrt{b}X_TS_e}\;\big|\;U_{e^c}\big]\le e^{3t^2/N},$$
$$\mathbb{E}_\omega\big[e^{\eta E_e}\;\big|\;U_{e^c}\big]\le(1-12\eta/N)^{-q/2},$$

*so $\mathbb{E}_\omega E_e^m\le m!\,(24/N)^m2^{q/2}$ for $m\ge1$, uniformly in $b$, the volume and the boundary condition. Moreover $\mathbb{E}_\omega E_e=c_N\mathbb{E}_\omega S_e$.*

(ii) *For a bounded function $F$ of finitely many links, continuously differentiable in $U_e$, $\mathbb{E}_\omega[W_eF]=-\mathbb{E}_\omega\langle\nabla_eS_e,\nabla_eF\rangle$. In particular $\mathbb{E}_\omega[W_eF]=0$ if $F$ does not depend on $U_e$, and $\mathbb{E}_\omega[W_eW_f]=0$ if no plaquette contains both $e$ and $f$.*

(iii) *Let $\omega$ be reflection positive as in Appendix F, and let $A=\sum_e h_eW_e$ be a finite sum over links whose endpoints have times at least 2. Then $\omega(\vartheta(\bar A)A)=0$, so $\Psi_A=0$: in the reconstructed Hilbert space the packets $\sum_e h_eE_e$ and $c_N\sum_e h_eS_e$ define the same vector.*

*Proof.* (i) Along $u\mapsto e^{uT}U_e$ a plaquette term containing $e$ is $N^{-1}\operatorname{Re}\operatorname{Tr}(e^{uT}V)$ or $N^{-1}\operatorname{Re}\operatorname{Tr}(Ve^{-uT})$ with $V$ unitary, whose second derivative is $N^{-1}\operatorname{Re}\operatorname{Tr}(T^2e^{\pm uT}V)$. Since $-T^2\ge0$ and $\operatorname{Tr}(-T^2)=1$, $|\operatorname{Tr}(T^2M)|\le1$ for unitary $M$, so the second derivative of $S_e$ along the curve is at most $6/N$ in absolute value. Taylor's formula with $u=t/\sqrt{b}$ gives $b\big(S_e(e^{tT/\sqrt{b}}U_e)-S_e(U_e)\big)\ge t\sqrt{b}X_TS_e(U_e)-3t^2/N$. The conditional density is $e^{bS_e}/Z$ with respect to Haar measure, and the left invariance of Haar measure gives $\mathbb{E}_\omega[e^{b(S_e(e^{tT/\sqrt{b}}U_e)-S_e(U_e))}\mid U_{e^c}]=1$, which with the Taylor bound proves the first inequality. For the second let $G_0$ be a standard Gaussian vector in $\mathfrak{g}$, independent of $\omega$; then $e^{\eta|Y|^2}=\mathbb{E}e^{\sqrt{2\eta}\langle G_0,Y\rangle}$ for $Y=\sqrt{b}\nabla_eS_e$, and the first inequality in the direction of $G_0$ with $t=\sqrt{2\eta}|G_0|$ gives the bound $\mathbb{E}e^{6\eta|G_0|^2/N}=(1-12\eta/N)^{-q/2}$. With $\eta=N/24$ and $x^m\le m!\,\eta^{-m}e^{\eta x}$ the moment bound follows. For the mean, $\sum_A T_A^2=-c_NI$ in an orthonormal basis gives $\Delta_eS_e=-c_NS_e$, and the integral of the divergence of $e^{bS_e}\nabla_eS_e$ over $G$ vanishes.

(ii) The integral over $U_e$ of the divergence of $e^{bS_e}F\nabla_eS_e$ vanishes, and that divergence is $e^{bS_e}(W_eF+\langle\nabla_eS_e,\nabla_eF\rangle)$; integrate the conditional identity. The function $W_f$ depends only on the links of the plaquettes containing $f$.

(iii) Each $W_e$ in the sum depends on links with times at least 1, so $\vartheta(\bar A)$ depends only on links with times at most 0, none of which is a link $e$ of the sum. By (ii), $\omega(\vartheta(\bar A)W_e)=0$ for each $e$. □

**Proposition 75.** *Let $G=\mathrm{SU}(2)$, $b\ge0$, $\omega$ a Wilson Gibbs state at $b$ and $p$ a plaquette with a link that is not fixed by the boundary condition. Then*

$$\operatorname{Var}_\omega\big(\tfrac12\operatorname{Tr}Q_p\big)\ge\frac{1}{2^{19}(1+6b)^2},\quad\mathbb{E}_\omega\big[1-\tfrac12\operatorname{Tr}Q_p\big]\ge\frac{1}{4b+1}.$$

*Proof.* Let $e$ be the link and identify $\mathrm{SU}(2)$ with the unit quaternions. Given the other links, $U_e=x$ has density proportional to $e^{rx\cdot u}$ with $|u|=1$ and $0\le r\le6b$, since the six staples have norm one, and $\frac12\operatorname{Tr}Q_p=x\cdot v$ for a unit vector $v$ fixed by the conditioning. Let $X=x\cdot u$ and $T=1-X$; the law of $T$ has density proportional to $e^{-rt}\sqrt{t(2-t)}$ on $[0,2]$. Put $n=1+r$. Since $e^{-rt}\le e^2e^{-nt}$, the normalisation is at most $e^2\int_0^\infty\sqrt{2t}e^{-nt}dt=e^2\sqrt{\pi/2}\,n^{-3/2}<14n^{-3/2}$. On $I=[\frac{1}{4n},\frac{1}{2n}]$ the density is at least $\frac12n^{-1/2}e^{-1/2}$ and on $J=[\frac1n,\frac{3}{2n}]$ at least $\frac12n^{-1/2}e^{-3/2}$, so $P(T\in I)\ge\frac{1}{224}$ and $P(T\in J)\ge\frac{1}{336}$. With an independent copy $T'$, $\operatorname{Var}T=\frac12\mathbb{E}(T-T')^2\ge P(T\in I)P(T\in J)(2n)^{-2}>2^{-19}n^{-2}$. By symmetry the conditional covariance of $x$ is diagonal in $u$ and $u^\perp$, with eigenvalues $\operatorname{Var}X$ and $\mathbb{E}(1-X^2)/3$; on $I$, $1-X^2=T(2-T)\ge\frac{3}{8n}$, so $\mathbb{E}(1-X^2)/3\ge\frac{1}{1792n}\ge2^{-19}n^{-2}$. Hence the conditional variance of $x\cdot v$ is at least $2^{-19}(1+6b)^{-2}$, and so is the variance, by the law of total variance. For the mean, integrating $\frac{d}{dX}\big(e^{rX}(1-X^2)^{3/2}\big)$ over $[-1,1]$ gives $3\mathbb{E}X=r\mathbb{E}(1-X^2)\le r(1-(\mathbb{E}X)^2)$, whence $\mathbb{E}X\le(\sqrt{9+4r^2}-3)/(2r)\le2r/(2r+3)$ for $r>0$, and $\mathbb{E}[1-x\cdot v]\ge1-\mathbb{E}X\ge3/(2r+3)\ge1/(4b+1)$. □

For the blocking statement let $G=\mathrm{SU}(2)$, $B\ge4$ an integer and, for a coarse edge $(x,\mu)$, $V_{(x,\mu)}=\prod_{r=0}^{B-1}U_{(Bx+r\hat\mu,\mu)}$ the ordered product of the fine links along it.

**Proposition 76.** *For every Wilson Gibbs state $\omega$ of $\mathrm{SU}(2)$ at $b>0$ and every set $\mathscr{E}$ of $K$ coarse edges,*

$$\big\|\operatorname{Law}_\omega\big((V_e)_{e\in\mathscr{E}}\big)-m^{\otimes K}\big\|_{\mathrm{TV}}\le\frac{K}{2}e^{3b}\Big(\frac{4b}{4b+1}\Big)^{B-3}\le\frac{K}{2}\exp\Big(3b-\frac{B-3}{4b+1}\Big),$$

*where $m$ is the Haar measure. If $F$ is a function of the $V_e$ of a set of coarse edges at positive times, $|F|\le1$, and $\varepsilon$ is this bound for those edges and their reflections, then $0\le\omega(\vartheta(\bar F^c)F^c)\le6\varepsilon$ with $F^c=F-\omega(F)$.*

*Proof.* Select on each coarse edge the fine links with $1\le r\le B-2$. No plaquette contains two selected links: collinear links do not share a plaquette, parallel coarse edges are $B$ apart, and a plaquette containing a selected link of $(x,\mu)$ has its links in direction $\nu\ne\mu$ at $\mu$-offsets $r,r+1\in[1,B-1]$ from $Bx$, while coarse edges in direction $\nu$ lie at offsets in $B\mathbb{Z}$. Given the other links, the selected ones are therefore independent, each with density $e^{r_ix\cdot u_i}/Z(r_i)$, $0\le r_i\le6b$. Convolution with such a density acts on the representation of spin $l\ge1$ by $D^{(l)}(u_i)$ times $I_{l+1}(r_i)/I_1(r_i)$, whose absolute value is at most $I_2(r_i)/I_1(r_i)$ because $I_\nu(r)$ decreases in $\nu\ge0$, and $I_2(r_i)/I_1(r_i)=\mathbb{E}X\le$

$4b/(4b+1)$ by the proof of Proposition 75, so it contracts the mean-zero part of $L^2$ by that factor. The density is at most $e^{6b}$ because $Z \geq 1$ by Jensen's inequality, so its distance to 1 in $L^2$ is at most $e^{3b}$. The conditional law of $V_e$ is that of a product of $B-2$ such independent variables between two fixed ones, so its total variation distance to $m$ is at most $\frac{1}{2}e^{3b}(4b/(4b+1))^{B-3}$, and the coarse edges are conditionally independent. Total variation distance from a product is at most the sum over the factors, and averaging over the conditioning does not increase it. For the last statement, under $m^{\otimes K}$ the covariance vanishes because $F$ and its reflection depend on disjoint edges; replacing $\omega$ by $m^{\otimes K}$ changes $\omega(\vartheta(\bar{F})F)$ by at most $2\varepsilon$ and $|\omega(F)|^2$ by at most $4\varepsilon$. □

With $B = \ell/a_k$ for a fixed physical length $\ell$ and $b_k = O(\log(1/a_k))$ the bound tends to zero: the bare coarse links of a fixed coarse graph become independent Haar variables, their reflected form has rank one, and a renormalisation $Z_k$ that keeps the reflected norm of $Z_k F^c$ at least $c > 0$ must grow at least like $(c/6\varepsilon)^{1/2}$. The renormalised flowed fields of Theorem 48 are not bounded functions of bare coarse links, and none of the three propositions bears on the slice ratio, the tree bounds or (D).

### D. What the cube estimate requires

This subsection proves seven facts about $D(n,\alpha)$. It holds at small coupling for every even $n$ above an explicit size (Proposition 78), and it survives small perturbations of the action (Proposition 79), although gauge-invariant perturbations still need strong coupling at the scale of the cube (Proposition 86). It forces the gauge-fixed variables of a cube with trivial exterior to stay a fixed distance from the identity on average (Theorem 80, and Proposition 82 for SU(2)), so that for every fixed $n$ it fails once $\beta$ is large (Corollary 81). And it cannot be obtained from bounds on the conditional law of a single line of links that are uniform in the configuration around the line (Proposition 90). For the compact abelian theory with the Villain action it fails at every size once $\beta$ is large (Proposition 87). In Sec. XII A the estimate is posed for block averages, where its quadratic approximation holds along the tuning. None of these decides (D) along the continuum tuning, where $n$ grows while $\beta$ grows.[19]

For a finite $V \subset A$ and $j \in A \setminus V$ let $b(V,j)$ be the number of links of $V$ that share a plaquette with $j$, and write $k \sim i$ when two active links share a plaquette.

**Lemma 77.** *Suppose that $C_{ee'} \leq c_0$ for all active $e, e'$, with $12c_0 < 1$. Then $C_{V,j} \leq c_0\, b(V,j)/(1-12c_0)$ for every finite $V \subset A$ and every $j \in A \setminus V$.*

*Proof.* Fix $Q, Q'$ that differ only at $j$, put $d = \delta(Q_j, Q'_j)$, $\mu = \gamma_V(\cdot \mid Q)$ and $\nu = \gamma_V(\cdot \mid Q')$. The couplings of $\mu$ and $\nu$ form a weakly compact set on which $\pi \mapsto \int d_V \, d\pi$ is continuous, so one of them, $\pi$, minimises it; let $\rho_i = \int \delta(X_i, Y_i)\, d\pi$. Fix $i \in V$. Under $\mu$ the conditional law of $X_i$ given the other links of $V$ is $\gamma_i^F(\cdot \mid X)$, and likewise for $\nu$. Replace $(X_i, Y_i)$ by a pair drawn from an optimal coupling of $\gamma_i^F(\cdot \mid X)$ and $\gamma_i^F(\cdot \mid Y)$, chosen measurably [90, Cor. 5.22], and keep the other coordinates. The new coupling still has marginals $\mu$ and $\nu$, and changing the links in which $X$ and $Y$ differ one at a time gives

$$W_1\big(\gamma_i^F(\cdot \mid X), \gamma_i^F(\cdot \mid Y)\big) \leq \sum_{k \in V, k \neq i} C_{ik}\, \delta(X_k, Y_k) + C_{ij}\, d.$$

Minimality of $\pi$ therefore gives $\rho_i \leq c_0 \sum_{k \in V, k \sim i} \rho_k + c_0 [i \sim j]\, d$. Sum over $i \in V$: every $\rho_k$ occurs at most twelve times by Lemma 9(iv), so $(1 - 12c_0)\sum_i \rho_i \leq c_0\, b(V,j)\, d$. Since $W_1^{d_V}(\mu,\nu) \leq \sum_i \rho_i$, dividing by $d$ proves the lemma. □

**Proposition 78.** *Let $n = 2m$ and let $V$ be the cube of Definition 42. For every class $c$,*

$$\sum_{j \in c \setminus V} b(V,j) \leq 10m^3 - 2m^2,$$

*with equality for three classes. Consequently, if $C_{ee'} \leq c_0$ for all $e, e'$ with $12c_0 < 1$, then $D(n,\alpha)$ holds for every $\alpha \in (0,1)$ with*

$$\alpha \geq \hat{\alpha}_n = \frac{c_0}{1-12c_0}\Big(\frac{20}{n} - \frac{8}{n^2}\Big).$$

*For* SU(2) *and $|\beta_W| < \frac{1}{3}$ this reads $\hat{\alpha}_n = \frac{|\beta_W|}{1-3|\beta_W|}\big(\frac{5}{n} - \frac{2}{n^2}\big)$, which is below one for every even $n > 5|\beta_W|/(1-3|\beta_W|)$.*

*Proof.* Let $j = (y, \mu)$ lie in the class $c$, with $y \equiv r$ modulo two. A link of $V$ that shares a plaquette with $j$ has base point $y + d$ with $d \in \{-1, 0, 1\}^4$ and at most two nonzero coordinates, and the twelve offsets $d$ depend only on $c$. For a fixed offset we count the $y \equiv r$ with $y + d \in [0,n)^4$ and $y \notin [0,n)^4$. In the coordinate $l$ there are $m$ values of $y_l \equiv r_l$ with $y_l + d_l \in [0,n)$, and one of them lies outside $[0,n)$ exactly when $d_l = 1$ and $r_l = 1$ (the value $y_l = -1$) or $d_l = -1$ and $r_l = 0$ (the value $y_l = n$). If $q$ is the number of such coordinates, the count is $m^4 - m^{4-q}(m-1)^q$, that is 0, $m^3$ or $2m^3 - m^2$ for $q = 0, 1, 2$. Summing over the twelve offsets, the class sum is $a_c m^3 - b_c m^2$, where $b_c$ is the number of offsets with $q = 2$ and $a_c$ is the number with $q = 1$ plus twice $b_c$. Reading the offsets from Table III, the pairs $(a_c, b_c)$ over the 48 classes are $(2,0)$, $(7,0)$, $(7,1)$, $(9,1)$, $(9,2)$ and $(10,2)$ three times each and $(3,0)$, $(4,0)$, $(5,0)$, $(6,0)$ and $(8,1)$ six times each. In every case $a_c \leq 10$ and $8 - a_c + b_c \geq 0$, so

$$\begin{aligned}(10m^3 - 2m^2) - (a_c m^3 - b_c m^2) &= m^2\big((10 - a_c)m + b_c - 2\big) \\ &\geq m^2(8 - a_c + b_c) \geq 0,\end{aligned}$$

with equality for the three classes with $(a_c, b_c) = (10, 2)$. By Lemma 77, $K_c \leq c_0(10m^3 - 2m^2)/(1-12c_0)$, and $N_n = m^4$ gives $K_c/N_n \leq \hat{\alpha}_n$. For SU(2) the proof of Theorem 1 gives $c_0 = |\beta| = |\beta_W|/4$. □

[19] The four statements answer the question of which parts of (D) can be settled with the methods of this paper. The first places (D) in the range where the single-link method already works, the second and third show what any proof of (D) at weak coupling has to produce, and the fourth excludes the most direct one-dimensional route.

At $\beta_W = \frac{3}{10}$ and $n = 16$, for instance, $\hat\alpha_{16} = \frac{117}{128}$, and Theorem 43 gives the gap $\log\frac{128}{117}/32$, much weaker than Theorem 1. The proposition shows that the cube condition is not empty; it does not extend the range, because the absorption $12c_0 < 1$ is the single-link condition itself.

The cube condition is also stable under small changes of the action, which is what a block renormalisation would need at its last scale, where the effective action is no longer of Wilson form. Let $r \ge 0$ and $\varepsilon \ge 0$, and let $(w_X)$ be real functions on $G^A$ indexed by finite sets $X \subset A$ of diameter at most $r$, with $w_X$ depending only on the links of $X$, such that the family is invariant under the translations by $2\mathbb{Z}^4$, every active link lies in at most $M$ of the sets $X$, every $X$ has at most $s$ links, and

$$|w_X(Q) - w_X(Q')| \le \varepsilon \sum_{i\in X} \delta(Q_i, Q'_i)$$

for all $Q, Q'$. Let $\gamma_V^W(\cdot \mid Q)$ be the probability measure with density proportional to $\exp\sum_{X\cap V\neq\emptyset} w_X$ with respect to $\gamma_V(\cdot \mid Q)$, and define $C^W_{V,j}$, $K^W_c$ and the condition $D^W(n,\alpha)$ as in Definition 42, with $\gamma_V^W$ in place of $\gamma_V$.

**Proposition 79.** *Let $n$ be even and suppose that $D(n,\alpha)$ holds at $\beta$. Then $D^W(n,\alpha')$ holds with*

$$\alpha' = \alpha\, e^{\Omega}(1+2\Omega) + 96\, s\,\Omega, \qquad \Omega = 48\, M\, D\, N_n\, \varepsilon.$$

*In particular $\alpha' < 1$ for all $\varepsilon$ below an explicit $\varepsilon_0(n,\alpha,M,s) > 0$.*

*Proof.* The cube has $|V| = 48N_n$ links, so $d_V$ has diameter $\Delta_V = 48N_nD$. For fixed exterior links put $u = \sum_{X\cap V\neq\emptyset} w_X$ as a function on $G^V$. Each link of $V$ lies in at most $M$ sets $X$, so $u$ is $M\varepsilon$-Lipschitz for $d_V$ and its oscillation is at most $M\varepsilon\Delta_V = \Omega$; we normalise $u$ so that $\sup u = 0$, which does not change the tilted measures, and put $h = e^u$, so that $e^{-\Omega} \le h \le 1$ and $h$ is $M\varepsilon$-Lipschitz.

*A common tilt.* Let $\mu, \nu$ be probability measures on $G^V$ and $\mu_h = h\mu/\mu(h)$, $\nu_h = h\nu/\nu(h)$. Let $f$ be 1-Lipschitz for $d_V$; subtracting a constant we may take $|f| \le \Delta_V$. Then $fh$ is $(1+\Delta_V M\varepsilon)$-Lipschitz, $|\nu(fh)| \le \Delta_V \nu(h)$, $|\mu(h) - \nu(h)| \le M\varepsilon W_1^{d_V}(\mu,\nu)$ and $\mu(h) \ge e^{-\Omega}$, so

$$\begin{aligned}\mu_h(f) - \nu_h(f) &= \frac{\mu(fh) - \nu(fh)}{\mu(h)} + \nu(fh)\,\frac{\nu(h) - \mu(h)}{\mu(h)\nu(h)} \\ &\le e^{\Omega}(1+2\Omega)\, W_1^{d_V}(\mu,\nu),\end{aligned}$$

since $\Delta_V M\varepsilon = \Omega$. By Kantorovich duality $W_1^{d_V}(\mu_h,\nu_h) \le e^{\Omega}(1+2\Omega) W_1^{d_V}(\mu,\nu)$.

*Two tilts.* If $u'$ is another weight with $|u - u'| \le \eta$, the density of $\nu_{h'}$ with respect to $\nu_h$ lies in $[e^{-2\eta}, e^{2\eta}]$, so the total variation distance of the two is at most $1 - e^{-2\eta} \le 2\eta$, and $W_1^{d_V}(\nu_h, \nu_{h'}) \le 2\eta\Delta_V$.

*The coefficients.* Let $Q, Q'$ differ only at $j \notin V$, with $\delta = \delta(Q_j, Q'_j)$, and let $u, u'$ be the weights for the two exteriors. They differ only through the sets $X$ that contain $j$ and meet $V$; if there are $m_j$ of them, $|u - u'| \le m_j\varepsilon\delta$. With $\mu = \gamma_V(\cdot \mid Q)$ and $\nu = \gamma_V(\cdot \mid Q')$, the triangle inequality and the two steps give

$$C^W_{V,j} \le e^{\Omega}(1+2\Omega)\, C_{V,j} + 2\Delta_V\, m_j\, \varepsilon.$$

The pairs $(X, j)$ with $j \in X \setminus V$ and $X \cap V \neq \emptyset$ number at most $s$ times the number of sets meeting $V$, hence at most $sM|V|$. Summing over $j \in c \setminus V$,

$$\begin{aligned}K^W_c &\le e^{\Omega}(1+2\Omega)\,\alpha N_n + 2\Delta_V\, sM|V|\varepsilon \\ &= \big(\alpha e^{\Omega}(1+2\Omega) + 96\, s\,\Omega\big) N_n.\end{aligned}$$

The family is invariant under $2\mathbb{Z}^4$, so the coefficients are, and this is $D^W(n,\alpha')$. □

The admissible perturbation is of order $1/N_n = (2/n)^4$, so the proposition is useful in a box of a fixed number of lattice spacings, which is how a renormalisation argument would use it: after enough block steps that the cube of physical size $\ell$ has become a cube of fixed side, the effective action has to lie within $\varepsilon_0$ of an action that satisfies the condition there. Whether the effective actions along the tuning do so is not decided here, and that is where the problem lies: at the scale where the blocked theory is strongly coupled enough for a condition of this kind, the running coupling is of order one, which is outside the range of the existing ultraviolet analysis. The proposition shows only that the condition itself is not the obstacle.

The next theorem uses the configuration in which every link outside $V$ is the identity. Write $\gamma_V^1 = \gamma_V(\cdot \mid 1)$ and

$$\varepsilon_V(\beta) = \frac{1}{|V|}\sum_{i\in V} \int \delta(1, Q_i)\, d\gamma_V^1(Q)$$

for the mean distance of the gauge-fixed variables of the cube from the identity. Recall that $\delta$ is bi-invariant of diameter $D$.

**Theorem 80.** *Let $n$ be even, $V$ the cube of Definition 42, and suppose that $D(n,\alpha)$ holds at $\beta$. Then*

$$\varepsilon_V(\beta) \ge \frac{(1-\alpha)\, D}{3}.$$

*Proof. A flat configuration.* By Lemma 9(i) and the period two of $\sigma$, the sixteen residues modulo two fall into eight pairs $\{r, r + e_{\sigma(r)}\}$, the residues met by one line of $F$. The eight residues 0010, 0011, 0100, 0110, 1001, 1011, 1100, 1101 form four of these pairs. Choose $z, w \in G$ with $\delta(z,w) = D$, put $g(x) = w$ if $x$ is congruent to one of those eight residues and $g(x) = z$ otherwise, and let $Q^g_{(x,\mu)} = g(x)g(x+e_\mu)^{-1}$ for every link of $\mathbb{Z}^4$. Then $g$ is constant on every line of $F$, so $Q^g$ equals 1 on $F$, every plaquette variable of $Q^g$ is 1, and $Q^g$ has period two. By bi-invariance $\delta(1, Q^g_{(x,\mu)}) = \delta(g(x+e_\mu), g(x))$, which is $D$ when $x$ and $x+e_\mu$ receive different values and 0 otherwise; from Table III the first case occurs for exactly 32 of the 48 classes. Let $\delta_c$ be the common value of $\delta(1, Q^g_e)$ for $e$ in the class $c$, so that $\sum_c \delta_c = 32D$.

*Gauge covariance.* Let $\Phi(Q)_{(x,\mu)} = g(x) Q_{(x,\mu)} g(x+e_\mu)^{-1}$ on $G^V$. The gauge transformation by $g$ maps the configuration equal to $Q_V$ on $V$ and to 1 elsewhere to the one equal to $\Phi(Q_V)$

on $V$ and to $Q^g$ elsewhere; it leaves every plaquette trace unchanged and every link of $F$ equal to 1, and $\Phi$ preserves Haar measure. Hence $\gamma_V(\cdot \mid Q^g) = \Phi_* \gamma_V^1$.

*Upper bound.* Only the finitely many links outside $V$ that share a plaquette with $V$ enter $\gamma_V$. Changing them one at a time from 1 to $Q^g$ and applying the definition of $C_{V,j}$ at each step,

$$W_1^{d_V}\big(\gamma_V^1, \gamma_V(\cdot \mid Q^g)\big) \le \sum_{j \in A \setminus V} C_{V,j}\, \delta(1, Q_j^g) = \sum_c \delta_c K_c \le \alpha N_n \sum_c \delta_c = 32\,\alpha D N_n.$$

*Lower bound.* The function $f(Q) = -\sum_{i \in V} \delta(Q_i, Q_i^g)$ is 1-Lipschitz for $d_V$. By gauge covariance and bi-invariance, $\int f\, d\gamma_V(\cdot \mid Q^g) = -\int \sum_i \delta(Q_i, 1)\, d\gamma_V^1 = -|V|\varepsilon_V$, and by the triangle inequality $\int f\, d\gamma_V^1 \le -\sum_i \delta(1, Q_i^g) + |V|\varepsilon_V$. Every class has $N_n$ links in $V$, so $\sum_i \delta(1, Q_i^g) = 32 D N_n$ and

$$W_1^{d_V}\big(\gamma_V^1, \gamma_V(\cdot \mid Q^g)\big) \ge 32 D N_n - 2|V|\varepsilon_V.$$

With $|V| = 48 N_n$ the two bounds give $96 N_n \varepsilon_V \ge 32(1-\alpha) D N_n$. □

**Corollary 81.** *For every even $n$ and every $\alpha < 1$ there is $\beta_0(n,\alpha)$ such that $D(n,\alpha)$ fails at every $\beta \ge \beta_0(n,\alpha)$. In particular, if $D(n_k,\alpha)$ holds at couplings $\beta_k \to \infty$, then $n_k \to \infty$.*

*Proof.* For $\beta > 0$, $\gamma_V^1$ has density proportional to $e^{-N\beta S}$ with respect to Haar measure, where $S = \sum_{p \cap V \neq \emptyset} (N - \operatorname{Re}\operatorname{Tr} Q_p^F)$ is continuous and nonnegative on $G^V$ and vanishes exactly when $Q_p^F = 1$ for every plaquette $p$ that meets $V$. We show that this forces $Q_V = 1$. The other plaquettes have all their links equal to 1, so every plaquette variable is 1; since the plaquettes of $\mathbb{Z}^4$ fill every closed lattice path, there is a function $h$ on the sites with $Q_{(x,\mu)} = h(x)h(x+e_\mu)^{-1}$ on every link, and $h$ takes equal values at the two ends of every link on which $Q = 1$. Every link with base point outside $[0,n)^4$ is such a link. From a site with a coordinate at least $n$ one can move in every positive direction along links of this kind, and from a site with a negative coordinate $x_\nu$ one can move in the positive directions other than $\nu$ until another coordinate reaches $n$; so $h$ is constant on the sites outside the box. Every line of $F$ leaves the box, and $h$ is constant along it, so $h$ is constant and $Q_V = 1$.

Let $U$ be a neighbourhood of 1 in $G^V$, $s = \min_{G^V \setminus U} S > 0$, and $U' \subset U$ a neighbourhood of 1 on which $S < s/2$. Then $\gamma_V^1(G^V \setminus U) \le e^{-N\beta s}/(|U'| e^{-N\beta s/2})$, which tends to zero, so $\varepsilon_V(\beta) \to 0$ as $\beta \to \infty$, and Theorem 80 excludes $D(n,\alpha)$ once $\varepsilon_V(\beta) < (1-\alpha)D/3$. □

For SU(2) the constant of Theorem 80 can be improved: a flat configuration with four values instead of two moves every class of active links, and the means of the gauge-fixed links are scalar.

**Proposition 82.** *Let $G = \mathrm{SU}(2)$ with the chordal metric, let $n$ be even, $V$ the cube of Definition 42, and suppose that $D(n,\alpha)$ holds at $\beta$. Then*

$$\frac{1}{|V|} \sum_{i \in V} \left| \int \operatorname{Re}\operatorname{Tr} Q_i\, d\gamma_V^1 \right| \le 2\alpha,$$

$$\frac{1}{|V|} \sum_{i \in V} \int \delta(1, Q_i)^2\, d\gamma_V^1 \ge 2(1-\alpha),$$

*and $\varepsilon_V(\beta) \ge 1 - \alpha$.*

*Proof.* Write $\langle \cdot, \cdot \rangle$ for the Euclidean product of quaternions; then $\operatorname{Re}\operatorname{Tr} q = 2\langle q, 1\rangle$ and $\delta(p,q) = |p - q|$.

*Four colours.* Split the sixteen residues modulo two into the sets

$$X_1 = \{0000, 1000, 0111, 1111\},$$
$$X_2 = \{0001, 0101, 1010, 1110\},$$
$$X_3 = \{0010, 0011, 1100, 1101\},$$
$$X_4 = \{0100, 0110, 1001, 1011\}.$$

Each is the union of two of the eight pairs $\{r, r + e_{\sigma(r)}\}$, and Table III shows that $r$ and $r + e_\mu$ lie in different sets for every residue $r$ and every $\mu \neq \sigma(r)$. (The eight residues used in the proof of Theorem 80 form $X_3 \cup X_4$.) Let

$$z_1 = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}, \quad z_2 = (\mathbf{i} - \mathbf{j} - \mathbf{k})/\sqrt{3},$$
$$z_3 = (-\mathbf{i} + \mathbf{j} - \mathbf{k})/\sqrt{3}, \quad z_4 = (-\mathbf{i} - \mathbf{j} + \mathbf{k})/\sqrt{3},$$

unit quaternions with $\langle z_k, z_l \rangle = -\frac{1}{3}$ for $k \neq l$, so that $\delta(z_k, z_l) = d_* = (8/3)^{1/2}$. Put $g(x) = z_k$ when the residue of $x$ lies in $X_k$, and $Q_{(x,\mu)}^g = g(x) g(x + e_\mu)^{-1}$. As in the proof of Theorem 80, $Q^g$ equals 1 on $F$, every plaquette variable of $Q^g$ is 1, and $\gamma_V(\cdot \mid Q^g) = \Phi_* \gamma_V^1$ with $\Phi(Q)_{(x,\mu)} = g(x) Q_{(x,\mu)} g(x + e_\mu)^{-1}$; now $\delta(1, Q_e^g) = d_*$ for every active link $e$.

*Upper bound.* Changing the exterior links one at a time,

$$W_1^{d_V}\big(\gamma_V^1, \gamma_V(\cdot \mid Q^g)\big) \le d_* \sum_c K_c \le 48\,\alpha\, d_* N_n = \alpha\, d_* |V|.$$

*Lower bound.* Conjugating every link of $V$ by one unit quaternion leaves the exterior 1, the links of $F$, Haar measure and every plaquette trace unchanged, so it preserves $\gamma_V^1$. The mean $m_i = \int Q_i\, d\gamma_V^1$, taken in the space of quaternions, therefore commutes with every unit quaternion, and $m_i = a_i 1$ with $a_i = \frac{1}{2} \int \operatorname{Re}\operatorname{Tr} Q_i\, d\gamma_V^1$ real. Since $\Phi$ acts on each link by a fixed left and right multiplication, the mean of $Q_i$ under $\gamma_V(\cdot \mid Q^g)$ is $a_i Q_i^g$. Choose unit quaternions $w_i$ with $\langle w_i, a_i Q_i^g - a_i 1 \rangle = |a_i|\,|Q_i^g - 1|$. The function $f(Q) = \sum_{i \in V} \langle w_i, Q_i \rangle$ is 1-Lipschitz for $d_V$, and Kantorovich duality gives

$$W_1^{d_V}\big(\gamma_V^1, \gamma_V(\cdot \mid Q^g)\big) \ge \sum_{i \in V} |a_i|\, \delta(1, Q_i^g) = d_* \sum_{i \in V} |a_i|.$$

*Conclusion.* Comparing the two bounds, $\sum_i |a_i| \le \alpha |V|$, which is the first inequality. Since $\delta(1,q)^2 = 2 - 2\langle q, 1\rangle$, $\int \delta(1, Q_i)^2\, d\gamma_V^1 = 2(1 - a_i) \ge 2(1 - |a_i|)$, and averaging gives the second. As $\delta \le 2$, $\delta \ge \delta^2/2$, and the bound on $\varepsilon_V$ follows. □

The four colours need nothing from SU(2) except four elements whose pairwise quotients are conjugate, and every compact simple group contains such elements. For a metric that comes from a representation the bound then compares the cube with Haar measure.

**Proposition 83.** *Let $G$ be a compact simple group, $\rho$ a nontrivial irreducible unitary representation of $G$ of dimension $d$, and $\delta(p,q)=\kappa^{-1/2}\|\rho(p)-\rho(q)\|_{\mathrm{HS}}$ for some $\kappa>0$. Let $n$ be even, $V$ the cube of Definition 42, and suppose that $D(n,\alpha)$ holds at $\beta$. Then*

$$\frac{1}{|V|}\sum_{i\in V}\Big|\int \operatorname{Tr}\rho(Q_i)\,d\gamma_V^1\Big|\le \alpha\, d,$$

$$\frac{1}{|V|}\sum_{i\in V}\int \delta(1,Q_i)^2\,d\gamma_V^1\ge(1-\alpha)\int_G\delta(1,q)^2\,dq,$$

*where $dq$ is Haar measure, and $\varepsilon_V(\beta)\ge(1-\alpha)D/2$.*

*Proof. Four elements.* Let $\varphi\colon \mathrm{SU}(2)\to G$ be the homomorphism of the $\mathrm{SU}(2)$ subgroup of a root, and $z_1,\dots,z_4$ the tetrahedral quaternions of Proposition 82. For $k\neq l$ the quaternion $z_kz_l^{-1}$ has real part $\langle z_k,z_l\rangle=-\frac13$; unit quaternions with equal real parts are conjugate in $\mathrm{SU}(2)$, so the twelve elements $\varphi(z_kz_l^{-1})$ are conjugate in $G$, and by bi-invariance $\delta(\varphi(z_k),\varphi(z_l))=d_*$ for all $k\neq l$, with one number $d_*$. If $d_*=0$, then $\varphi(z_kz_l^{-1})$ lies in $\ker\rho$, a proper normal subgroup of the simple group $G$ and therefore central; it then commutes with $\varphi(\mathrm{SU}(2))$, so it is central there, and $z_kz_l^{-1}$ lies in $\{\pm1\}\ker\varphi\subset\{\pm1\}$, which its real part $-\frac13$ excludes. So $d_*>0$.

*The two bounds.* Put $g(x)=\varphi(z_k)$ when the residue of $x$ lies in $X_k$. The proof of Proposition 82 applies word for word: $Q^g$ equals 1 on $F$ and on every plaquette, $\delta(1,Q_e^g)=d_*$ on every active link, and $W_1^{d_V}(\gamma_V^1,\gamma_V(\cdot\mid Q^g))\le\alpha d_*|V|$. Simultaneous conjugation of the links of $V$ preserves $\gamma_V^1$, so the mean $\int\rho(Q_i)\,d\gamma_V^1$ commutes with $\rho(G)$, and by Schur's lemma it is $c_iI$ with $c_i=d^{-1}\int\operatorname{Tr}\rho(Q_i)\,d\gamma_V^1$. Its mean under $\gamma_V(\cdot\mid Q^g)$ is $c_i\rho(Q_i^g)$. With matrices $W_i$ of Hilbert–Schmidt norm $\kappa^{-1/2}$ chosen so that $\operatorname{Re}\operatorname{Tr}(W_i^*c_i(\rho(Q_i^g)-I))=|c_i|\,\delta(1,Q_i^g)$, the function $f(Q)=\sum_i\operatorname{Re}\operatorname{Tr}(W_i^*\rho(Q_i))$ is 1-Lipschitz for $d_V$, and Kantorovich duality gives $W_1^{d_V}\ge d_*\sum_i|c_i|$. Hence $\sum_i|c_i|\le\alpha|V|$, the first inequality.

*Comparison with Haar measure.* Since $\kappa\,\delta(1,q)^2=2d-2\operatorname{Re}\operatorname{Tr}\rho(q)$ and $\int_G\operatorname{Tr}\rho(q)\,dq=0$ for a nontrivial irreducible $\rho$, $\int_G\delta(1,q)^2dq=2d/\kappa$, while $\int\delta(1,Q_i)^2d\gamma_V^1=2d(1-\operatorname{Re}c_i)/\kappa\ge2d(1-|c_i|)/\kappa$. Averaging gives the second inequality. Finally $\kappa D^2\le4d$, because $\|\rho(p)-\rho(q)\|_{\mathrm{HS}}\le2\sqrt{d}$, so $\delta\ge\delta^2/D$ gives $\varepsilon_V\ge2d(1-\alpha)/(\kappa D)\ge(1-\alpha)D/2$. □

For $\mathrm{SU}(N)$ with the Hilbert–Schmidt distance this improves the constant $\frac13$ of Theorem 80 to $\frac12$, and for $\mathrm{SU}(3)$, where $D^2=9$ and the Haar mean of $\delta^2$ is 6, to $\frac23$: in a cube with the identity outside, $D(n,\alpha)$ leaves the gauge-fixed links at least the fraction $1-\alpha$ of the disorder of independent Haar-distributed links.

The size $\beta_0(n,\alpha)$ in Corollary 81 comes from a compactness argument. For $\mathrm{SU}(2)$ it can be made explicit, and then reads as a lower bound on the side of any cube that satisfies the condition.

**Proposition 84.** *Let $G=\mathrm{SU}(2)$ with the chordal metric, so that $D=2$, let $b=\beta_W\ge\frac19$, let $n\ge2$ be even and put $L(b)=1+\log24+\frac32\log(9b)$. Then*

$$\frac{1}{|V|}\sum_{e\in V}\int\delta(1,Q_e)^2\,d\gamma_V^1\le\frac{n^2(n+2)(2n+1)\,L(b)}{b},$$

*so $\varepsilon_V(\beta)^2$ is at most the same quantity. Consequently, if $D(n,\alpha)$ holds at $b$, then*

$$n^4\ge\frac{2(1-\alpha)\,b}{5\,L(b)}.$$

*Proof.* Put $m=n/2$. Under $\gamma_V^1$ every link outside $V$ equals 1: the links of $F$ always, the others by the choice of exterior. A plaquette that does not meet $V$ therefore has plaquette variable 1.

*Links as loops.* Let $e=(x,\mu)\in V$, $y=x+e_\mu$, $\sigma=\sigma(x)$ and $\tau=\sigma(y)$. The directions $\sigma,\tau,\mu$ are distinct: $\sigma\neq\mu$ because $e$ is active; $\tau\neq\mu$ because otherwise the line of $F$ through $y$ would contain $x$, and the lines of $F$ are disjoint; and $\tau\neq\sigma$ by Table III, in which $\sigma(x+e_\mu)\neq\sigma(x)$ whenever $\mu\neq\sigma(x)$. Since $0\le x_\sigma<n$, one of $n-x_\sigma$ and $x_\sigma+1$ is at most $m$; call it $s$ and let $\varepsilon=\pm1$ be the sign for which $x'=x+s\varepsilon e_\sigma$ has $\sigma$-coordinate outside $[0,n)$. Choose $t\le m$ and $\varepsilon'$ in the same way for $y$ and $\tau$ (note $y_\tau=x_\tau$). Let $R_1$ be the $s$ plaquettes of the rectangle with corners $x$, $x'$, $x'+e_\mu$, $y$ in the plane $\{\sigma,\mu\}$, and $R_2$ the $st$ plaquettes of the rectangle with corners $y$, $x'+e_\mu$, $x'+e_\mu+t\varepsilon'e_\tau$, $y+t\varepsilon'e_\tau$ in the plane $\{\sigma,\tau\}$. The union $R(e)=R_1\cup R_2$ is a disc, and its boundary runs from $x$ along the line of $F$ through $x$ to $x'$, then to $x'+e_\mu$, along $e_\tau$ to $x'+e_\mu+t\varepsilon'e_\tau$, along $e_\sigma$ to $y+t\varepsilon'e_\tau$, back along the line of $F$ through $y$ to $y$, and over $e$ to $x$. The links of the first and fifth pieces lie in $F$. The links of the second and third pieces have base points with $\sigma$-coordinate $x'_\sigma\notin[0,n)$, and those of the fourth have base points with $\tau$-coordinate outside $[0,n)$; none of them lies in $V$. So the holonomy of the boundary of $R(e)$ is $Q_e^{\pm1}$. Adding the plaquettes of a disc one at a time along its boundary shows that this holonomy is a product of conjugates of the plaquette variables $(Q_p^F)^{\pm1}$, $p\in R(e)$, one for each plaquette. By bi-invariance and the triangle inequality,

$$\delta(1,Q_e)\le\sum_{p\in R(e)}\delta(1,Q_p^F),\qquad |R(e)|=s(t+1)\le m(m+1).$$

*Multiplicity.* If a plaquette $p$ lies in $R_2(e)$, its plane is $\{\sigma,\tau\}$, $\mu$ is one of the two other directions, and the base point of $p$ is $y+ae_\sigma+ce_\tau$ with $-m\le a,c\le m-1$. Given $p$, there are two ways to name the directions of its plane, two choices of $\mu$ and $(2m)^2$ of $(a,c)$, and they determine $y$, hence $e$; so $p$ lies in at most $16m^2$ of the sets $R_2(e)$. Likewise $p\in R_1(e)$ forces its plane to be $\{\sigma,\mu\}$ and its base point to be $x+ae_\sigma$ with $-m\le a\le m-1$, which leaves at most $4m$ links $e$. Every plaquette therefore lies in at most $16m^2+4m=4n^2+2n$ of the discs $R(e)$.

*The energy.* For the chordal metric $\delta(1,q)^2=2-\operatorname{Re}\operatorname{Tr}q$. Let $S=\sum_{p\cap V\neq\emptyset}\delta(1,Q_p^F)^2$; then $\gamma_V^1$ has density $e^{-bS/2}/Z$ with $Z=\int e^{-bS/2}$, the integral over $\mathrm{SU}(2)^V$ with Haar measure. The

function $t \mapsto \log \int e^{-tS}$ is convex and vanishes at $t=0$, so its derivative at $t=b/2$, which is $-\int S\,d\gamma_V^1$, is at least $2\log Z/b$. A plaquette that meets $V$ contains at most three links of $V$ and the other links are 1, so $\delta(1,Q_p^F)^2 \le 3\sum_{e\in p\cap V}\delta(1,Q_e)^2$; every link lies in six plaquettes, so $S \le 18\sum_{e\in V}\delta(1,Q_e)^2$ and $Z \ge I^{|V|}$ with $I=\int e^{-9b\,\delta(1,q)^2}dq$. The chordal distance is at most the geodesic one, and the geodesic ball of radius $r=(9b)^{-1/2}\le 1$ has measure greater than $r^3/24$ (proof of Proposition 90), so $I \ge e^{-1}(9b)^{-3/2}/24$ and

$$\int S\,d\gamma_V^1 \le \frac{2|V|\,L(b)}{b}.$$

*Conclusion.* The Cauchy–Schwarz inequality inside each disc squares the loop bound:

$$\begin{aligned}\delta(1,Q_e)^2 &\le |R(e)| \sum_{p\in R(e)} \delta(1,Q_p^F)^2 \\ &\le m(m+1)\sum_{p\in R(e)}\delta(1,Q_p^F)^2.\end{aligned}$$

Summing over $e\in V$ and using the multiplicity,

$$\sum_{e\in V}\delta(1,Q_e)^2 \le m(m+1)(4n^2+2n)\,S = \frac{n^2(n+2)(2n+1)}{2}\,S,$$

and integrating against $\gamma_V^1$ gives the first bound; the bound on $\varepsilon_V$ follows by the Cauchy–Schwarz inequality. If $D(n,\alpha)$ holds, Proposition 82 gives $2(1-\alpha)\le n^2(n+2)(2n+1)L(b)/b$, and $(n+2)(2n+1)\le 5n^2$ for $n\ge 2$. □

The loops and their multiplicity use nothing of $\mathrm{SU}(2)$ but the bi-invariance of $\delta$, and the energy uses only the form of the action and the measure of small balls. The bound therefore holds for every $\mathrm{SU}(N)$, with the same power of $n$.

**Proposition 85.** *Let $G=\mathrm{SU}(N)$, $N\ge 2$, with the weight $\exp(\frac{b}{N}\sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p)$, $b=\beta_W \ge N/9$, and the Hilbert–Schmidt distance $\delta(p,q)=\|p-q\|_{\mathrm{HS}}$. Let $n\ge 2$ be even and put*

$$L_N(b) = 1+\frac{N^2-1}{2}\log(81\pi^2 b)+N(N-1)\log\frac{12}{11}.$$

*Then*

$$\frac{1}{|V|}\sum_{e\in V}\int \delta(1,Q_e)^2\,d\gamma_V^1 \le \frac{N\,n^2(n+2)(2n+1)\,L_N(b)}{b}.$$

*Consequently, if $D(n,\alpha)$ holds at $b$, then*

$$n^4 \ge \frac{2(1-\alpha)\,b}{5\,L_N(b)}.$$

*Proof. Loops.* The discs $R(e)$ of the proof of Proposition 84 and the bound $4n^2+2n$ on their multiplicity do not depend on the group, and the loop bound uses only bi-invariance and the triangle inequality. With $S=\sum_{p\cap V\neq\emptyset}\delta(1,Q_p^F)^2$ the same Cauchy–Schwarz step gives $\sum_{e\in V}\delta(1,Q_e)^2 \le \frac12 n^2(n+2)(2n+1)\,S$.

*The energy.* Since $\delta(1,q)^2 = 2N-2\,\mathrm{Re}\,\mathrm{Tr}\,q$, the measure $\gamma_V^1$ has density $e^{-bS/(2N)}/Z$ with respect to Haar measure. As in the proof of Proposition 84, the convexity of $t\mapsto\log\int e^{-tS}$ gives $\int S\,d\gamma_V^1 \le 2N\log(1/Z)/b$, and $S\le 18\sum_{e\in V}\delta(1,Q_e)^2$ gives $Z\ge I^{|V|}$ with $I=\int e^{-9b\,\delta(1,q)^2/N}dq$.

*Small balls.* Give $\mathfrak{su}(N)$ the inner product $\mathrm{Tr}(X^*Y)$ and $\mathrm{SU}(N)$ the corresponding bi-invariant metric, of volume $v_N$. In exponential coordinates the Riemannian measure is $j(X)\,dX$, where $dX$ is Lebesgue measure and

$$j(X) = \Big|\det\frac{1-e^{-\mathrm{ad}\,X}}{\mathrm{ad}\,X}\Big| = \prod_{k<l}\Big(\frac{\sin(\varphi_{kl}/2)}{\varphi_{kl}/2}\Big)^2$$

if $X$ has eigenvalues $i\theta_1,\dots,i\theta_N$ and $\varphi_{kl}=\theta_k-\theta_l$ [50, Ch. II, Thm. 1.7]; in particular $j\le 1$. Every $q\in\mathrm{SU}(N)$ is conjugate to a diagonal matrix with entries $e^{i\psi_k}$, $-\pi\le\psi_k<\pi$; then $\sum_k\psi_k = 2\pi k_0$ with $|k_0|\le N/2$, and changing $|k_0|$ of the $\psi_k$ by $\mp 2\pi$ gives angles $\theta_k$ with sum zero and $|\theta_k|\le 3\pi$. So $q=e^X$ with $X\in\mathfrak{su}(N)$ and $\|X\|_{\mathrm{HS}}\le 3\pi\sqrt{N}$, and $v_N \le \omega_{N^2-1}(3\pi\sqrt{N})^{N^2-1}$, where $\omega_k$ is the volume of the unit ball of $\mathbb{R}^k$. Now let $\|X\|_{\mathrm{HS}}\le r\le 1$. The eigenvalues have $|\theta_k|\le 1<\pi$, so exp is injective on this ball; $\delta(1,e^X)^2=\sum_k|e^{i\theta_k}-1|^2 \le \|X\|_{\mathrm{HS}}^2$; and $|\varphi_{kl}|/2 \le (\theta_k^2+\theta_l^2)^{1/2}/\sqrt{2}\le 2^{-1/2}$, so each factor of $j(X)$ is at least $(1-\frac{1}{12})^2$ because $\sin x/x \ge 1-x^2/6$. Dividing by $v_N$, the Haar measure of $\{q:\delta(1,q)\le r\}$ is at least

$$\Big(\frac{11}{12}\Big)^{N(N-1)}\frac{\omega_{N^2-1}r^{N^2-1}}{v_N} \ge \Big(\frac{11}{12}\Big)^{N(N-1)}\Big(\frac{r}{3\pi\sqrt{N}}\Big)^{N^2-1}.$$

With $r=(N/9b)^{1/2}\le 1$ this gives $I\ge e^{-1}$ times the right-hand side, and since $3\pi\sqrt{N}/r = 9\pi\sqrt{b}$, $\log(1/I)\le L_N(b)$. Hence $\int S\,d\gamma_V^1 \le 2N|V|L_N(b)/b$, and the first bound follows from the loops.

*Conclusion.* Proposition 83 with the defining representation and $\kappa=1$, for which the Haar mean of $\delta(1,q)^2$ is $2N$, gives $2N(1-\alpha)\le N\,n^2(n+2)(2n+1)L_N(b)/b$ when $D(n,\alpha)$ holds, and $(n+2)(2n+1)\le 5n^2$. □

For every $N$, a cube on which the condition holds therefore has side at least of order $(\beta_W/\log\beta_W)^{1/4}$, with a constant that depends on $N$ only through $L_N$, which grows like $\frac{N^2}{2}\log\beta_W$.

The cube condition is a statement about strong coupling at its own scale, and this persists under the perturbations of Proposition 79 when they respect the gauge. Call $(w_X)$ *gauge invariant* if $w_X(Q^h)=w_X(Q)$ for every function $h$ on the sites that is constant on every line of $F$, where $Q^h_{(x,\mu)}=h(x)Q_{(x,\mu)}h(x+e_\mu)^{-1}$. Such transformations keep the links of $F$ equal to 1, so they act on $G^A$, and the restriction to the gauge of an action invariant under all gauge transformations is gauge invariant in this sense. Write $\varepsilon_V^W$ for $\varepsilon_V$ with $\gamma_V^W(\cdot\mid 1)$ in place of $\gamma_V^1$.

**Proposition 86.** *Let $(w_X)$ be as in Proposition 79 and gauge invariant, let $n=2m$, and suppose that $D^W(n,\alpha)$ holds.*

(i) $\varepsilon_V^W \ge (1-\alpha)D/3$.

(ii) *If $\eta \geq 0$, $0 \leq \kappa \leq 1$, and the event that $\delta(1, Q_p^F) \leq \eta$ for every plaquette $p$ that meets $V$ has probability at least $1-\kappa$ under $\gamma_V^W(\cdot \mid 1)$, then*

$$m(m+1)\,\eta + \kappa D \geq \frac{(1-\alpha)D}{3}.$$

*Proof.* (i) The proof of Theorem 80 uses the action only through the identity $\gamma_V(\cdot \mid Q^g) = \Phi_* \gamma_V^1$ and through the coefficients $C_{V,j}$. The function $g$ of that proof is constant on every line of $F$, so the gauge transformation by $g$ leaves every $w_X$ unchanged, as it leaves every plaquette trace, and the same identity holds for $\gamma_V^W$. Only finitely many links outside $V$ enter $\gamma_V^W$, because the sets $X$ have bounded diameter, so the upper bound holds with $C_{V,j}^W$ and $K_c^W$ in place of $C_{V,j}$ and $K_c$; the lower bound is unchanged.

(ii) Under $\gamma_V^W(\cdot \mid 1)$ every link outside $V$ equals 1, so a plaquette that does not meet $V$ has variable 1. The loop bound of the proof of Proposition 84, which uses only the bi-invariance of $\delta$, gives $\delta(1, Q_e) \leq \sum_{p \in R(e)} \delta(1, Q_p^F) \leq m(m+1)\eta$ for every $e \in V$ on the event of the statement, and $\delta(1, Q_e) \leq D$ off it. Averaging over $e \in V$ and integrating, $\varepsilon_V^W \leq m(m+1)\eta + \kappa D$, and (i) completes the proof. □

In the small-field region of a gauge-covariant block renormalisation the blocked plaquette variables lie within a distance $\eta$ that tends to zero with the running coupling, and large fields have small probability $\kappa$. If $\kappa < (1-\alpha)/3$, part (ii) allows the cube condition on the blocked lattice only when $m(m+1) \geq x$, where $x = ((1-\alpha)/3 - \kappa)D/\eta$, so the side must grow like $\eta^{-1/2}$. On such a cube $N_n = m^4 \geq x^2/4$, and since $\alpha' \geq \alpha + 96 s \Omega$, Proposition 79 gives $\alpha' < 1$ only for

$$\begin{aligned} \varepsilon &< \frac{1-\alpha}{4608\, s M D N_n} \\ &\leq \frac{(1-\alpha)\,\eta^2}{1152\, s M D^3 \big((1-\alpha)/3 - \kappa\big)^2}, \end{aligned}$$

which is of order $\eta^2$. At a fixed side the condition is therefore a statement about strong coupling, and a renormalisation argument has to carry the blocked theory out of the small-field regime before the condition can be checked. That passage, at a running coupling of order one, is the step that remains open.

The next statement shows that the cube condition separates a theory with a massless photon from the expected behaviour of $\mathrm{SU}(N)$: for the compact abelian theory in its Coulomb phase it fails at every scale. Let $G = \mathrm{U}(1)$ with the chordal metric, $D = 2$, and replace the Wilson weight by the Villain weight

$$\varphi_\beta(\theta) = \sum_{k \in \mathbb{Z}} e^{-\beta(\theta + 2\pi k)^2/2},$$

so that the measure of a box $\Lambda$ has density proportional to $\prod_{p \in P(\Lambda)} \varphi_\beta(\theta_p)$, where $e^{i\theta_p}$ is the plaquette variable. Define $\gamma^F$, $C_{V,j}$, $K_c$ and $D(n,\alpha)$ as before with $\varphi_\beta$ in place of the Wilson weight.

**Proposition 87.** *There is $\beta_0$ such that for the Villain theory at every $\beta > \beta_0$ the condition $D(n,\alpha)$ fails for every even $n \geq 2$ and every $\alpha < 1$. In particular, for couplings $\beta_k \to \infty$ and any even $n_k$, the hypothesis of Theorem 45 fails for this theory.*

*Proof.* Lemma 10, Proposition 11(i)–(iii) and (v), Lemma 241, Corollary 242 and Lemma 244 use the Wilson weight only as a positive continuous class function of the plaquette variable, with one exception: the Doeblin bound in Lemma 244(a), which for the Villain weight reads that each single-link density is at least $(\min \varphi_\beta / \max \varphi_\beta)^6 > 0$ times its maximum. Suppose that $D(n,\alpha)$ holds. As in the proof of Theorem 43, the limit $\omega(O) = \lim W_\Lambda(O)$ of the measures of boxes with free boundary conditions then exists along every sequence of boxes exhausting $\mathbb{Z}^4$, and for gauge-invariant local $O, O'$ with $O$ Lipschitz in each link,

$$|\operatorname{Cov}_\omega(O, O')| \leq \|O'\|_\infty L_O \frac{D(1 + c_n e^{\lambda \rho_n})}{1 - \theta}\, |S_{O'}|\, e^{-\lambda \operatorname{dist}(O, O')}$$

for every $\lambda > 0$ with $\theta = \alpha e^{\lambda \rho_n} < 1$, where $L_O$ is a Lipschitz constant of $O$ and $S_{O'}$ the support of $O'$.

Put $\Psi_p = \varphi_\beta'(\theta_p)/\varphi_\beta(\theta_p)$; up to the factor $-\mathbf{i}$ this is the lattice field strength of Fröhlich and Spencer [43, Sect. 2] for the Villain action. It is real, odd and smooth in $\theta_p$, so it is gauge invariant, local and Lipschitz in each of its four links, with constants that do not depend on $p$, and $\omega(\Psi_p) = 0$ by the symmetry $\theta \mapsto -\theta$. The bound above gives $\sum_{p'} |\omega(\Psi_p \Psi_{p'})| < \infty$, the sum over the plaquettes $p'$ parallel to $p$. The state of [43] is the limit of the same free-boundary measures over boxes, which is $\omega$. Fröhlich and Spencer prove, for $\beta$ larger than some $\beta_0$, that the Fourier transform of the two-plaquette function of the field strength is discontinuous at zero momentum, so that this sum is infinite [43, Sect. 2]; see also [48]. This contradiction proves the first statement, and the second follows because $\beta_k > \beta_0$ for all large $k$. □

At large $\beta$ the plaquette variables of the abelian theory are Gaussian to leading order, as are those of $\mathrm{SU}(N)$, and every property of the action used in this paper (compactness, locality, positivity and smoothness of the weight, reflection positivity, the forest gauge) is shared by the two theories. A proof of (D) along the continuum tuning therefore has to use the non-abelian structure of $G$; Proposition 87 holds at every cube size, so letting the side grow with $\beta$, which Corollary 81 already requires, does not remove the need. Fröhlich and Spencer indicate how their result extends to the Wilson action without giving the details [43, Sect. 2], so the proposition is stated for the Villain action only.

*Remark* 88. The proof loses the area of a disc, of order $n^2$, in passing from plaquettes to links, and on loops that loss is of the right order. On the periodic lattice $(\mathbb{Z}/L\mathbb{Z})^4$ with $L$ even and coordinates $0 \leq x_j < L$, let $\phi = 4\pi/L^2$, $q(\theta) = \cos\theta + \mathbf{i}\sin\theta$, and put $U_{(x,2)} = q(\phi x_1)$, $U_{(x,1)} = q(-\phi L x_2)$ when $x_1 = L-1$, and $U = 1$ on every other link. Every plaquette of the plane $\{1,2\}$ equals $q(\phi)$, including those on the seam $x_1 = L-1$ and at the corner $x_1 = x_2 = L-1$, where the angle is $\phi - 4\pi$; all other plaquettes equal 1. So every plaquette variable lies within $\phi$ of the identity, while the boundary of the square $[0, L/2]^2$ of the plane $\{1,2\}$ has holonomy $q(\phi L^2/4) = -1$, at the largest possible distance $D = 2$, against the bound $\phi L^2/4 = \pi$ given by the disc. Every staple sum of this configuration is parallel to its link, so it is a stationary point of the Wilson action,

and gradient flow [65] does not change it. Plaquette variables within $O(n^{-2})$ of the identity therefore do not force loops of length $n$ near the identity, and an argument of the kind above that places gauge-fixed links near the identity needs plaquette bounds of order $o(n^{-2})$.

Up to the factor $L(b)^{1/2}$ the rate $b^{-1/2}$ of Proposition 84 is the right one, and the lower bound below holds for every $n$ and every exterior.

**Proposition 89.** *Let $G=\mathrm{SU}(2)$ with the chordal metric and $b=\beta_W>0$. For every Gibbs specification of the gauge-fixed theory, every active link $e$ and every configuration of the other links, the conditional law of $Q_e$ satisfies*

$$\int \delta(1,Q_e)\,d\mu_e \geq \frac{1}{9\max(6b,1)^{1/2}}.$$

*In particular $\varepsilon_V(\beta)\geq \frac{1}{9}\max(6b,1)^{-1/2}$ for every $n$; at every fixed $n$, $\varepsilon_V(\beta)$ lies between constant multiples of $b^{-1/2}$ and $(L(b)/b)^{1/2}$.*

*Proof.* Write $\langle\cdot,\cdot\rangle$ for the Euclidean product of quaternions, so that $\mathrm{Re}\,\mathrm{Tr}\,q=2\langle q,1\rangle$. The six plaquettes that contain $e$ contribute $\exp(b\langle Q_e,W_p\rangle)$ to the conditional density, with $|W_p|=1$; so $\mu_e$ has density $e^{\langle u,v\rangle}/Z(|v|)$ with respect to normalised Haar measure, for some $v$ with $a=|v|\leq 6b$. Put $a'=\max(a,1)$ and $\eta=a'^{-1/2}\leq 1$. On the geodesic ball of radius $\eta$ about $v/|v|$ the exponent is at least $a(1-\eta^2/2)\geq a-\frac{1}{2}$, and that ball has measure greater than $\eta^3/24$ (proof of Proposition 90); hence $Z(a)\geq e^{a-1/2}\eta^3/24$ and the density is at most $24e^{1/2}a'^{3/2}$ (for $a<1$ directly, since $Z\geq 1$ by Jensen's inequality). A chordal ball of radius $r$ is a geodesic ball of radius $\theta$ with $2\sin(\theta/2)=r$, so $\theta\leq \pi r/2$, and its measure is $(\theta-\sin\theta\cos\theta)/\pi\leq 2\theta^3/(3\pi)\leq \pi^2 r^3/12$. With $r=\frac{2}{9}a'^{-1/2}$,

$$\mu_e\big(\delta(1,Q_e)<r\big)\leq 24e^{1/2}a'^{3/2}\,\frac{\pi^2 r^3}{12}=2e^{1/2}\pi^2\Big(\frac{2}{9}\Big)^3<\frac{1}{2},$$

so the mean of $\delta(1,Q_e)$ is at least $r/2=\frac{1}{9}a'^{-1/2}$. Averaging over $e\in V$ and over the configuration gives the bound on $\varepsilon_V$. □

Along the continuum tuning Theorem 80 says the following. If (D) holds, then in a cube of fixed physical size $\ell$ with the identity outside, the forest-gauge variables stay on average at distance at least $(1-\alpha)D/3$ from the identity, and at least $1-\alpha$ for SU(2), at every spacing. This is not a contradiction: $n=\ell/a$ grows like $1/a$, while asymptotic freedom makes $\beta(a)$ grow only like $\log(1/a)$, and the corollary concerns fixed $n$. The size bound of Proposition 84 is likewise met along the tuning, since its right side grows only like $\log(1/a)$. It turns the hypothesis of Remark 62, that the gauge-fixed links decorrelate on the scale of the cube, into a necessary condition: (D) requires the forest-gauge variables to be disordered at a fixed physical scale. For gauge-invariant observables of a massive theory that disorder is clustering; for the gauge-fixed variables it is not proved.

The last statement concerns the following strategy: control the conditional law of one line of links, given everything else, uniformly in the configuration around it, and assemble the cube from lines. Fix a site $x$, let $s=\sigma(x)$ and $\mu\neq s$, and let $c_k=(x+ke_s,\mu)$, $k\in\mathbb{Z}$. All $c_k$ are active, since $\sigma$ is constant along the line through $x$.

**Proposition 90.** *Let $G=\mathrm{SU}(2)$, identified with the unit quaternions, and $b=\beta_W\geq 1$. There is a configuration of the active links outside $\{c_k\}$, independent of $b$, for which the conditional law of $(Q_{c_k})_k$ has a ten-step transfer operator, similar to a positive self-adjoint operator $K_b$ on $L^2(S^3)$ with simple top eigenvalue $\Lambda_b$, whose gap satisfies*

$$0<1-\frac{\Lambda_{b,1}}{\Lambda_b}\leq e^{20}\,24^{10}\,b^{15}\,e^{-b/1280},$$

*where $\Lambda_{b,1}$ is the next point of the spectrum.*

*Proof. The interactions.* By Lemma 9(iii), $c_k$ lies in two plaquettes of the plane $\{\mu,s\}$, which contain $c_{k-1}$ or $c_{k+1}$, the link $(x+ke_s,s)$ or $(x+(k-1)e_s,s)$ of $F$, and an active twist link of direction $s$; and in four plaquettes of the planes $\{\mu,t\}$ and $\{\mu,\rho\}$, each containing $c_k$, one link of $F$ and two active links not on the line. The four plaquettes at $c_k$ lie in the hyperplane of $s$-coordinate $x_s+k$, so these plaquettes at different $k$ share no link; two plaquettes through $c_k$ share no other link; and their links have directions other than $s$, so they are not twists. Set every twist to 1; then the plaquette at $x+ke_s$ in the plane $\{\mu,s\}$ contributes $b\langle q_k,q_{k+1}\rangle$ to the exponent, with $q_k=Q_{c_k}$. In each of the four other plaquettes at $c_k$ set one of its two links to 1 and the other to $X$; after a cyclic rotation and possibly an inversion the plaquette variable is $q_kX^{\pm1}$ or $q_k^{-1}X^{\pm1}$, and since $\mathrm{Re}(q\bar u)=\mathrm{Re}(\bar q u)=\langle q,u\rangle$ for unit quaternions, $X$ can be chosen so that the plaquette contributes $b\langle q_k,u\rangle$ for any prescribed unit vector $u$. Every other active link is set to 1. Four unit vectors add up to any vector of length at most 4 (two copies each of $t\pm(1-|t|^2)^{1/2}\nu$ with $t=v/4$ and $\nu\perp t$ a unit vector), so the conditional density of the line is proportional to $\exp\big(b\sum_k\langle q_k,q_{k+1}\rangle+\sum_k\langle h_k,q_k\rangle\big)$ for any fields $h_k$ with $|h_k|\leq 4b$.

*The fields.* Let $a=\frac{1}{2}$, $c=\frac{\sqrt{3}}{2}$, let $e_1,\dots,e_4$ be the standard basis of $\mathbb{R}^4$, and let $r_k$ have period ten with $(r_0,\dots,r_9)=a(e_1,-e_1,e_2,-e_2,e_3,-e_3,e_3,-e_2,e_2,-e_1)$. Put $h_k=b(2r_k-r_{k-1}-r_{k+1})$, so $|h_k|\leq 4ab=2b$, and the configuration is independent of $b$ because $h_k/b$ is. With $v_k=q_k-r_k$ and $|q_k|=1$, $|r_k|=a$, summation by parts gives, over $0\leq k\leq 9$,

$$\frac{1}{2}\sum_k|v_{k+1}-v_k|^2=\sum_k\big(1-\langle q_k,q_{k+1}\rangle\big)-\frac{1}{b}\sum_k\langle h_k,q_k\rangle+\frac{1}{2}\sum_k|r_{k+1}-r_k|^2+B(q_0)-B(q_{10}),$$

with $B(q)=\langle q,r_0-r_9\rangle$. The boundary terms only conjugate the ten-step transfer kernel by a multiplication operator, so up to similarity it is

$$K_b(x,y)=\int_{(S^3)^9}\exp\Big(-\frac{b}{2}\sum_{k=0}^{9}|v_{k+1}-v_k|^2\Big)\,dq_1\cdots dq_9,$$

with $q_0=x$, $q_{10}=y$ and normalised Haar measure. The sequence $r_0,\dots,r_{10}$ is a palindrome, so $K_b=B_bB_b^*$ for the five-step kernel $B_b$; $K_b$ is compact, positive and self-adjoint with a

continuous positive kernel, its top eigenvalue $\Lambda_b$ is simple, and its eigenfunction $\psi_b > 0$ is even under the reflection $x_4 \mapsto -x_4$, which commutes with $K_b$.

*A barrier.* Let $E = \frac{1}{2}\sum_{k=0}^{9} |v_{k+1} - v_k|^2$ and suppose that $(q_0)_4 \geq 0 \geq (q_{10})_4$ and $E \leq \frac{1}{1280}$. Put $\Delta = (20E)^{1/2} \leq \frac{1}{8}$ and $w = v_0$; by Cauchy–Schwarz $|w - v_k| \leq \Delta$ for all $k$, and $0 \leq w_4 \leq \Delta$ because the $r_k$ have no fourth coordinate. From $|v_k + r_k| = 1$, $\big||w + r_k|^2 - 1\big| \leq \Delta(2+\Delta) =: \eta$ for $0 \leq k \leq 5$. The pairs $r_k = \pm a e_l$ give $|w_l| \leq \eta/(2a) = \eta$ for $l = 1,2,3$ and $|w|^2 \geq \frac{3}{4} - \eta$, so $\frac{3}{4} \leq 3\eta^2 + \Delta^2 + \eta \leq 3(\frac{17}{64})^2 + \frac{1}{64} + \frac{17}{64} = \frac{2019}{4096}$, a contradiction. Hence $K_b(x,y) \leq e^{-b/1280}$ whenever $x_4 \geq 0 \geq y_4$.

*The top eigenvalue.* Let $\varepsilon = b^{-1/2}$. The geodesic ball of radius $\varepsilon \leq 1$ in $S^3$ has normalised measure $\frac{2}{\pi}\int_0^{\varepsilon} \sin^2\theta \, d\theta \geq \frac{8\varepsilon^3}{3\pi^3} > \frac{\varepsilon^3}{24}$. If each $q_k$ lies in the ball of radius $\varepsilon$ about $r_k + c e_4$, then $|v_k - c e_4| \leq \varepsilon$ and $E \leq 20\varepsilon^2$, so the integrand is at least $e^{-20}$. Testing $K_b$ on the normalised indicator of the ball about $r_0 + c e_4$ gives $\Lambda_b \geq e^{-20}(\varepsilon^3/24)^{10} = e^{-20} 24^{-10} b^{-15}$.

*The gap.* Let $\phi(x) = \operatorname{sign}(x_4)\psi_b(x)$, a unit vector orthogonal to $\psi_b$. Then

$$\begin{aligned}\Lambda_b - \langle \phi, K_b \phi \rangle &= 4 \int_{x_4 > 0 > y_4} \psi_b(x) K_b(x,y) \psi_b(y)\, dx\, dy \\ &\leq e^{-b/1280},\end{aligned}$$

because $\int_{x_4>0} \psi_b \leq \frac{1}{2}$ by Cauchy–Schwarz. By the variational principle $\Lambda_{b,1} \geq \langle \phi, K_b \phi \rangle$, and dividing by the lower bound on $\Lambda_b$ proves the estimate; the gap is positive because $\Lambda_b$ is simple. □

The exterior of Proposition 90 is legal at every coupling, so a bound on the conditional law of single lines that is uniform in the exterior cannot be better than $e^{-b/1280}$ up to powers of $b$, and an argument of this kind cannot give (D) along the tuning. A proof of (D) has to use what such a bound discards: the joint resampling of the transverse links inside the cube, which the averaged coefficients $C_{V,j}$ retain, or the smallness of the probability of such exteriors under the Wilson measure.

## XII. BLOCK AVERAGES

Proposition 64 locates the failure of (D) in the quadratic approximation: gauge transformations constant on the lines of $F$ move the whole cube and change few links outside it. By Proposition 65 every gauge fixed on an invariant forest fails in the same way, through the constant fields. This section takes up the programme of the report [63] of the third author, which keeps the cube condition but changes the gauge and conditions the cube on block averages of the field, as in Bałaban's renormalisation group [6, 7]. The natural block averages of the line field leave one class of lines invisible and fail at every size (Proposition 91). In a gauge whose fixed links are trees inside blocks of side two, the averages over whole blocks remove every zero mode, the constrained quadratic form has a gap that does not depend on the volume (Theorem 94), and the constrained cube condition holds at every coupling once the cube is large (Theorem 98). Integrating out one scale bounds the cube coefficients by the constrained ones plus those of the averages (Corollary 106), and the averages carry the coarse quadratic form at every scale up to a factor between 1 and 256, through forms that are local uniformly in the scale (Proposition 99 and Theorems 102 and 104), and the constrained cube condition holds at every scale with one cube side (Theorem 108). For the group-valued links themselves, the straight paths between block roots satisfy a coercivity bound for all fields but do not compose, a continuous covariant map cannot treat the transverse positions alike, and a cut-off of Bałaban's averaging operation is coercive for all fields and agrees with the block averages to first order (Sec. XII B). It composes across all scales on fields of small curvature for every compact group (Sec. XII C), and under the SU(2) Wilson measure the fields that are not small form clusters of vanishing physical size (Sec. XII E).

### A. Block-constrained cubes

In this subsection everything concerns the quadratic approximation: real fields $\varphi$ on links, $(d\varphi)_p$ the signed sum around the plaquette $p$, and $(dh)_{(x,\mu)} = h(x) - h(x + e_\mu)$ for a function $h$ on sites.

**Proposition 91** ([63])**.** *In the setting of Proposition 64, condition the cube in addition on any family of the face averages*

$$V^{\mathrm{face}}_{y,\mu}(\varphi) = \frac{1}{8}\sum_{x} \big(\varphi_{(x,\mu)} + \varphi_{(x+e_\mu,\mu)}\big),$$

*where $x$ runs over the eight sites of the block $2y + \{0,1\}^4$ with $x_\mu = 2y_\mu$. Then $\sum_{c \in C_I} K_c \geq 12 N_n$ for the twelve classes*

$$\begin{aligned}C_I = \{&(0111,\mu), (1111,\mu), (0111 + e_\mu, \mu), (1111 + e_\mu, \mu) \\ &: \mu = 1,2,3\},\end{aligned}$$

*written as residue of the base and direction. Hence $K_c \geq N_n$ for some class, and the constrained condition fails for every $\alpha < 1$, at every even $n$ and every $b$.*

*Proof.* By Table III, $\sigma(0111) = \sigma(1111) = 0$, so the lines of $F$ through the sites with $x_1, x_2, x_3$ odd have direction 0; call them invisible. Let $\ell$ be one of them and $h_\ell$ its indicator. The sum of $dh_\ell$ along a path of length two from $x$ is $h_\ell(x) - h_\ell(x + 2e_\mu)$. For $\mu \neq 0$ both end points have even $\mu$-coordinate and are off $\ell$; for $\mu = 0$ they have the same transverse coordinates and are both on $\ell$ or both off it. So $dh_\ell$ changes no face average, and the translation argument of Proposition 64 applies to the constrained laws: $\sum_{i \in V} |(dh_\ell)_i| \leq \sum_{j \notin V} C_{V,j} |(dh_\ell)_j|$. An active link $e = (x,\mu)$ has $|(dh_\ell)_e| = 1$ for at most one invisible $\ell$, since its two ends lie on different lines and both cannot be invisible (for $\mu \neq 0$ they differ in the coordinate $\mu$, and for $\mu = 0$ the link would lie in $F$). It has one such $\ell$ exactly when its base or its end has residue 0111 or 1111, which by Table III are the classes of $C_I$: the six classes with base residue 0111 or 1111 and direction $\mu \neq 0$, and the six with base residue $r + e_\mu$, $r \in \{0111, 1111\}$, whose directions $\mu$ differ from $\sigma(r + e_\mu)$ (for instance $\sigma(0011) = 3 \neq 1$). Summing over the invisible lines gives $12 N_n \leq \sum_{c \in C_I} K_c$. □

**Definition 92.** The blocks are $B(y) = 2y + \{0,1\}^4$, $y \in \mathbb{Z}^4$, and $o = x \bmod 2$ is the offset of a site. A link $(x,\mu)$ is a *tree link* if $o_\mu = 0$ and $o_\nu = 0$ for every $\nu > \mu$, and *active* otherwise; fields vanish on tree links. For a coarse link $(y,\mu)$ the *block average* is

$$V_{y,\mu}(\varphi) = \frac{1}{16} \sum_{x \in B(y)} \big(\varphi_{(x,\mu)} + \varphi_{(x+e_\mu,\mu)}\big),$$

the mean of $\varphi$ summed along the sixteen paths of length two in the direction $\mu$ that start in $B(y)$.

**Lemma 93.** (i) *The fifteen tree links of a block form a spanning tree of its sixteen sites. Each block carries* 49 *active links: the* 17 *active links inside it, with $o_\mu = 0$, and the* 32 *links with $o_\mu = 1$, which join $B(y)$ to $B(y+e_\mu)$.*

(ii) *If $dh$ vanishes on the tree links, $h$ is constant on every block, and $V_{y,\mu}(\varphi + dh) = V_{y,\mu}(\varphi) + h_{B(y)} - h_{B(y+e_\mu)}$.*

(iii) *With $S^\mu_y = \sum_{x \in B(y),\, o_\mu = 0} \varphi_{(x,\mu)}$,*

$$V_{y,\mu}(\varphi) = \frac{1}{8} \sum_{x \in B(y),\, o_\mu = 1} \varphi_{(x,\mu)} + \frac{1}{16}\big(S^\mu_y + S^\mu_{y+e_\mu}\big).$$

*Proof.* (i) A tree link of direction $\mu$ has $o_\mu = 0$ and stays in its block; there are $2^\mu$ of them for each $\mu$, fifteen in all. From any site, the tree links of direction 3 lead to $o_3 = 0$, then those of direction 2 to $o_2 = 0$, and so on to the corner, so the tree links connect the sixteen sites, and fifteen edges connecting sixteen vertices form a tree. Of the 64 links with base in the block, the 32 with $o_\mu = 0$ stay inside it. (ii) $h$ is constant along the edges of a spanning tree. The path of length two from $x \in B(y)$ ends at $x + 2e_\mu \in B(y+e_\mu)$, and the sum of $dh$ along it is $h(x) - h(x+2e_\mu)$. (iii) The path from $x$ consists of $(x,\mu)$ and $(x+e_\mu,\mu)$; a link with $o_\mu = 1$ is the second link of one path and the first of another, and the links of $S^\mu_{y+e_\mu}$ are the second links of the paths that start with $o_\mu = 1$. □

By (ii) the residual gauge freedom is the coarse gauge group, which acts on the block averages as on the links of the coarse lattice. Let $W$ be a set of blocks and let $\Phi_W$ be the space of finitely supported real $\varphi$ on the active links with base in $\bigcup W$, with $V_{y,\mu}(\varphi) = 0$ for every $y \in W$ and every $\mu$. Write $Q(\varphi) = \sum_p (d\varphi)^2_p$ over all plaquettes, $\varphi$ being zero off its support.

**Theorem 94.** *For every set of blocks $W$ and every $\varphi \in \Phi_W$,*

$$\sum_e \varphi_e^2 \le 52\, Q(\varphi).$$

*Proof. Inside a block.* Let $e = (x,\mu)$ be an active link with $o_\mu = 0$, and let $\nu_1 > \cdots > \nu_k$ be the directions above $\mu$ in which $o$ equals one; $1 \le k \le 3$. Put $x_0 = x$ and $x_i = x_{i-1} - e_{\nu_i}$. The plaquette $p_i$ at $x_i$ in the plane $\{\mu,\nu_i\}$ has the links $(x_i,\mu)$, $(x_{i-1},\mu)$ and the tree links $(x_i,\nu_i)$, $(x_i+e_\mu,\nu_i)$, whose offsets vanish in $\nu_i$ and above it; and $(x_k,\mu)$ is a tree link. Hence $\varphi_e = \pm(d\varphi)_{p_1} \pm \cdots \pm (d\varphi)_{p_k}$ and $\varphi_e^2 \le k \sum_i (d\varphi)^2_{p_i}$. The plaquette at $x'$ in the plane $\{\mu,\nu\}$ is used only by the links $(x' + e_\nu + \sum_{\rho \in R} e_\rho, \mu)$ with $R \subset \{\nu+1,\dots,3\}$, each with $k = c + 1 + |R|$, where $c$ is the number of ones of the offset of $x'$ strictly between $\mu$ and $\nu$. Its total weight $2^{3-\nu}(c+1) + (3-\nu)2^{2-\nu}$ is at most 8, the value at $\mu = 0$, $\nu = 1$. So the sum $I$ of $\varphi_e^2$ over the active links inside the blocks is at most $8Q_{\text{in}}$, where $Q_{\text{in}}$ is the sum over the plaquettes inside a block.

*Between blocks.* Fix $y \in W$ and $\mu$, and label the eight links $(x,\mu)$, $x \in B(y)$, $o_\mu = 1$, by the transverse offset $a \in \{0,1\}^3$. If $a$ and $a'$ differ in the coordinate $\nu$, with $o_\nu = 0$ at $a$, the plaquette at $x$ in the plane $\{\mu,\nu\}$ contains these two links, $(x,\nu)$ in $B(y)$ and $(x+e_\mu,\nu)$ in $B(y+e_\mu)$, the last two inside their blocks. So $\varphi_a - \varphi_{a'} = \pm(d\varphi)_p \pm \psi \pm \psi'$ with $\psi, \psi'$ inside blocks, and these twelve face plaquettes correspond to the twelve edges of the cube $\{0,1\}^3$. By Lemma 93(iii) and $V_{y,\mu}(\varphi) = 0$, the mean of the $\varphi_a$ is $m = -(S^\mu_y + S^\mu_{y+e_\mu})/16$, so $8m^2 \le \frac{1}{2}$ times the sum of the squares of the sixteen links of $S^\mu_y$ and $S^\mu_{y+e_\mu}$. The Laplacian of the graph $\{0,1\}^3$ has smallest nonzero eigenvalue 2, so

$$\begin{aligned}\sum_a \varphi_a^2 &= 8m^2 + \sum_a (\varphi_a - m)^2 \\ &\le 8m^2 + \frac{3}{2} \sum_{\text{edges}} \big((d\varphi)_p^2 + \psi^2 + \psi'^2\big).\end{aligned}$$

Summed over $y \in W$ and $\mu$, a link $(z,\nu)$ inside a block occurs as $\psi$ or $\psi'$ once for each $\mu \ne \nu$ (as $\psi$ if $o_\mu = 1$, as $\psi'$ of the face of the block below if $o_\mu = 0$) and in two of the sums $S^\nu$, so it receives weight at most $3 \cdot \frac{3}{2} + 2 \cdot \frac{1}{2} = \frac{11}{2}$. The face plaquettes are distinct and lie in no block, so the links between blocks contribute at most $\frac{3}{2} Q_{\text{face}} + \frac{11}{2} I$, and

$$\sum_e \varphi_e^2 \le \frac{13}{2} I + \frac{3}{2} Q_{\text{face}} \le 52\, Q_{\text{in}} + \frac{3}{2} Q_{\text{face}} \le 52\, Q(\varphi). \quad \square$$

The constant 52 is not sharp. A certificate of one of us (T. Romanov) lowers it to within one part in a thousand of the bulk value.

**Proposition 95.** *For every set of blocks $W$ and every $\varphi \in \Phi_W$,*

$$\sum_e \varphi_e^2 \le \frac{5000}{537}\, Q(\varphi) = 9.31098\ldots\, Q(\varphi).$$

*The same holds on every periodic lattice of even side.*

*Proof.* Let $c = \frac{537}{5000}$, and let $S(y) = B(y) \cup B(y+e_0) \cup \cdots \cup B(y+e_3)$, whose active links number $5 \cdot 49 = 245$. The certificate consists of 196 rational numbers $l_{\mu,b}$, $\mu = 0,\dots,3$, indexed by the 49 active links $b$ of a block, and a rational symmetric matrix $G$ of size 245 on the active links of $S(0)$. With $L_{y,\mu}(\varphi) = \sum_b l_{\mu,b} \varphi_{(y,b)}$, where $(y,b)$ is the link $b$ of $B(y)$, and $G_y$ the translate of $G$ to $S(y)$, these satisfy the identity of quadratic forms in finitely supported $\varphi$

$$Q(\varphi) - c \sum_e \varphi_e^2 - \sum_{y,\mu} V_{y,\mu}(\varphi) L_{y,\mu}(\varphi) = \sum_y G_y(\varphi|_{S(y)}), \quad (17)$$

and $G$ is positive definite. Both sides of (17) are invariant under translations by $2\mathbb{Z}^4$, so the identity amounts to finitely many equalities of coefficients, one for each pair of active links of

$B(0)$ and of a block near it. Now take $\varphi \in \Phi_W$. If $y \in W$ then $V_{y,\mu}(\varphi) = 0$; if $y \notin W$ then $\varphi$ vanishes on the links with base in $B(y)$, so $L_{y,\mu}(\varphi) = 0$. Every term of the middle sum vanishes, and $Q(\varphi) - c\sum_e \varphi_e^2 \geq 0$. On a periodic lattice of even side, both sides of (17) are the periodisations of the same forms, so the identity and the argument hold there. □

The certificate was found by semidefinite programming and rounded to rationals: the $l_{\mu,b}$ have denominators dividing $2^{24}$ and the entries of $G$ the denominator $2^{30}\cdot 5^5$. The coefficient identity of (17) and the positivity of all 245 leading principal minors of the integer matrix $2^{30}5^5 G$ are checked in exact arithmetic. The check is done by the program of T. Romanov and independently by our program in the archive, which rebuilds the four forms from Definition 92 alone. With $W$ the set of all blocks the best constant is the inverse of the Bloch value 0.1074878 quoted below, that is 9.3034, so $\frac{5000}{537}$ is within 0.09 per cent of it; the semidefinite optimum for this window and this class of multipliers is $c^* = 0.107488$. These floating-point values are not used. The theorems below are stated with 52, which is proved by hand. Where they use only Theorem 94, they hold with 52 replaced by $\frac{5000}{537}$.

With $W$ all blocks the bound extends to $\ell^2$, so the block-constrained quadratic form on $\mathbb{Z}^4$ has a spectral gap at least $\frac{1}{52}$; it removes in particular the four flat connections that are zero modes of the unconstrained form. In the Bloch decomposition the exact gap is $0.10748776948684\ldots$, attained at the momentum $(0,\pi,\pi,\pi)$, a floating-point value computed in [63] and reproduced in the archive; the theorem does not use it. The cube conditions now take the following form. Let $n$ be even, $V$ the active links with base in $[0,n)^4$, $W$ its blocks, and for an exterior $\eta$ and values $v = (v_{y,\mu})_{y\in W}$ let $\gamma_V^{\mathrm{c}}(\cdot \mid \eta, v)$ be the law on $\mathbb{R}^V$ with density proportional to $\exp(-\frac{b}{2}\sum_{p\cap V\neq\emptyset}(d\varphi)_p^2)$ on the affine set $\{V_{y,\mu}(\varphi) = v_{y,\mu}\}$, with $\varphi = \eta$ off $V$. Define $C^{\mathrm{c}}_{V,j}$, $K_c^{\mathrm{c}}$ and $D_{\mathrm{c}}(n,\alpha)$ as $C_{V,j}$, $K_c$ and $D(n,\alpha)$, with $v$ held fixed and the 49 classes of active links modulo two.

**Lemma 96.** *Let $H \geq 0$ be symmetric on $\mathbb{R}^m$ with $\|H\| \leq h$, let $B : \mathbb{R}^m \to \mathbb{R}^k$ with $BB^{\top} \geq \beta > 0$ and $\langle x, Hx\rangle \geq a|x|^2$ on $\ker B$, and $s > 0$. Then $K = \left(\begin{smallmatrix} H & sB^{\top} \\ sB & 0 \end{smallmatrix}\right)$ is invertible, and the solution of $K(x,y) = (f,g)$ has $|x| \leq (|f| + h|x_1|)/a + |x_1|$ with $|x_1| \leq |g|/(s\sqrt{\beta})$, and $|y| \leq (|f| + h|x|)/(s\sqrt{\beta})$.*

*Proof.* Put $x_1 = B^{\top}(BB^{\top})^{-1} g/s$, so $sBx_1 = g$ and $|x_1| \leq |g|/(s\sqrt{\beta})$. On $\ker B$ the compression of $H$ is at least $a$, so there is a unique $x_0 \in \ker B$ with $Hx_0 - f + Hx_1 \perp \ker B$, and $a|x_0| \leq |f| + h|x_1|$. Then $f - H(x_0 + x_1)$ lies in $(\ker B)^{\perp} = \operatorname{ran} B^{\top}$, and $y = (BB^{\top})^{-1}B(f - Hx)/s$ solves $sB^{\top}y = f - Hx$. For $f = g = 0$ the construction gives zero, so $K$ is injective. □

**Lemma 97** (Combes–Thomas [31])**.** *Let $K$ be an invertible symmetric matrix indexed by a finite set $X$, with $\|K^{-1}\| \leq M_0$ and $\sum_b |K_{ab}| \leq S$ for every $a$, and let $d : X \to \mathbb{R}$ satisfy $|d(a) - d(b)| \leq 1$ whenever $K_{ab} \neq 0$. If $(e^{\lambda} - 1)SM_0 \leq \frac{1}{2}$, then $\|e^{\lambda d} K^{-1} e^{-\lambda d}\| \leq 2M_0$.*

*Proof.* $e^{\lambda d} K e^{-\lambda d} = K + E$ with $|E_{ab}| \leq (e^{\lambda} - 1)|K_{ab}|$, so $\|E\| \leq (e^{\lambda} - 1)S$ by the Schur test, $\|K^{-1}E\| \leq \frac{1}{2}$, and $(K + E)^{-1} = (1 + K^{-1}E)^{-1}K^{-1}$ has norm at most $2M_0$. □

**Theorem 98.** *For every $b > 0$, every even $n$ and every exterior link $j$:*

(i) *$\gamma_V^{\mathrm{c}}(\cdot \mid \eta, v)$ is a Gaussian law whose covariance depends on neither $\eta$ nor $v$ and whose mean is affine in $(\eta, v)$, and $C^{\mathrm{c}}_{V,j} = \|u_j\|_1$, where $u_j \in \mathbb{R}^V$ is the change of the mean when $\eta_j$ increases by one; $u_j$ does not depend on $b$;*

(ii) *$\sum_{e\in V} e^{2\lambda_0 d(e)} u_j(e)^2 \leq 3500^2$ and $\|u_j\|_1 \leq 10^{14}$, where $\lambda_0 = 2.4\cdot 10^{-5}$ and $d(e)$ is the $\ell^{\infty}$ distance between the midpoints of $e$ and $j$;*

(iii) *$D_{\mathrm{c}}(n,\alpha)$ holds for every $\alpha \in (0,1)$ and every even $n \geq 2\cdot 10^{15}/\alpha$, at every $b > 0$.*

*Proof.* (i) Let $H$ be the matrix of $\sum_{p\cap V\neq\emptyset}(d\varphi)_p^2$ on $\mathbb{R}^V$ and $B$ that of the averages $V_{y,\mu}$, $y \in W$, restricted to $V$. By Theorem 94 with this $W$, $H \geq \frac{1}{52}$ on $\ker B$, so the density is that of a non-degenerate Gaussian on the affine set, its covariance is $b^{-1}$ times the inverse of $H$ on $\ker B$, and its mean $m$ minimises the quadratic form on the affine set. The mean is affine in $(\eta, v)$, and two exteriors that differ at $j$ give translates by $u_j$, which minimises $Q(u + \delta_j)$ subject to $Bu = g$, $g$ being minus the coefficients of $j$ in the averages. The translation argument of Proposition 64 gives $C^{\mathrm{c}}_{V,j} = \|u_j\|_1$.

(ii) The pair $(u_j, y)$, with $y$ the Lagrange multipliers, solves $K(u_j, y) = (f, sg)$ with $K$ as in Lemma 96, $s = 96$ and $f = -\sum_{p\ni j}(d\delta_j)_p\, d_p|_V$, where $d_p|_V$ is the row of the plaquette $p$ restricted to $V$. Each link lies in six plaquettes of four links, so the row sums of $|H|$ are at most 24 and $h \leq 24$; $|f| \leq 6\sqrt{3}$; and $|g| \leq \frac{1}{16}$, since an exterior link enters at most one average of $W$, with weight $\frac{1}{16}$. By Lemma 93(iii) each average has squared norm at least $8\cdot\frac{1}{64} = \frac{1}{8}$, two averages share links only if they are $(y,\mu)$ and $(y \pm e_\mu, \mu)$, and then their inner product is at most $8\cdot\frac{1}{256} = \frac{1}{32}$; so $BB^{\top} \geq \frac{1}{16}$ by Gershgorin's theorem. With $a = \frac{1}{52}$, Lemma 96 gives $\|K^{-1}\| \leq M_0 = 105$. The rows of $K$ have absolute sums at most $S = \max(24 + \frac{96}{8}, 96\cdot 2) = 192$, because an average has eight coefficients $\frac{1}{8}$ and sixteen of $\frac{1}{16}$ and a link enters averages with total weight at most $\frac{1}{8}$. Index the rows of $B$ by the centre $2y + \frac{1}{2}(1,1,1,1) + e_\mu$ of the two blocks of the average and put $d$ equal to the $\ell^{\infty}$ distance to the midpoint of $j$. Links of one plaquette, and an average and its links, are within distance one, so Lemma 97 applies with $\lambda = \log(1 + 1/40320) > \lambda_0$, and

$$\sum_e e^{2\lambda d(e)} u_j(e)^2 \leq \big(2M_0 e^{\lambda}(|f| + s|g|)\big)^2 \leq 3500^2,$$

since $f$ and $g$ live at distance at most one from $j$. At most $4(64k^3 + 16k)$ links have their base at $\ell^{\infty}$ distance $k \geq 1$ from the base of $j$, and their midpoints are at distance at least $k - 1$, so Cauchy–Schwarz gives

$$\|u_j\|_1^2 \leq 3500^2 \cdot 4e^{2\lambda}\Big(1 + \sum_{k\geq 1}(64k^3 + 16k)e^{-2\lambda k}\Big) \leq (10^{14})^2.$$

(iii) $C^{\mathrm{c}}_{V_x,j} = 0$ unless $j$ shares a plaquette with $V_x$ or enters an average of $V_x$, which forces the base $z$ of $j$ into $x + [-1, n+1]^4$,

while $z \notin x+[0,n)^4$ when $j \notin V_x$. For each coordinate at most $\frac{n}{2}+2$ even $x_i$ satisfy the first condition and exactly $\frac{n}{2}$ the second, so

$$K_c^{\mathrm{c}} \le \Big(\Big(\frac{n}{2}+2\Big)^4 - \Big(\frac{n}{2}\Big)^4\Big)10^{14} \le \frac{17}{n}\,10^{14}N_n$$

for $n \ge 10^4$, which is at most $\alpha N_n$ when $n \ge 2\cdot 10^{15}/\alpha$. □

The constants are crude. The computations of [63] give $C^{\mathrm{c}}_{V,j} \le 12.48$ at $n=4$, and an exact face expansion $K_c^{\mathrm{c}}/N_n = a_c/n + O(n^{-2})$ with $\max_c a_c = 61.18$, so that $D_{\mathrm{c}}(n, 0.95)$ first holds at $n = 64$; they are floating-point values, reproduced in the archive for $n \le 4$, and the theorem does not use them. The coefficients are large because the constrained responses obey no maximum principle: an exterior link changes the curvature of up to six plaquettes, and keeping the curvature small moves a dozen links inside the cube with alternating signs. Re-weighting the classes does not help much: if $\sum_{j\in c\setminus V}\sum_{c'} w_{c'}C^{c'}_{V,j} \le \alpha N_n w_c$ for positive weights $w$, where $C^{c'}_{V,j}$ is the part of $\|u_j\|_1$ on the links of the class $c'$, then $\alpha N_n$ is at least the spectral radius of the non-negative matrix of these sums, hence at least $\min_c K_c^{\mathrm{c}}$, by the Collatz–Wielandt inequality [63].

The block averages can be integrated out one scale at a time. For a field $v$ on the coarse links put $Q_1(v) = \sum_P (dv)_P^2$ over the plaquettes $P$ of the coarse lattice, and write $V(\varphi) = (V_{y,\mu}(\varphi))_{y,\mu}$. The averages of averages are formed in the same way on the coarse lattice, and $V^{(k)}$, $Q_k$ denote the $k$-fold averages and the plaquette form of the lattice of spacing $2^k$.

**Proposition 99.** *Let $\varphi$ be a finitely supported field on $\mathbb{Z}^4$ that vanishes on the tree links.*

(i) *For the coarse plaquette $P$ at $y$ in the plane $\{\mu,\nu\}$, $16\,(dV(\varphi))_P$ is the sum over $x \in B(y)$ of the curvatures of the squares of side two spanned by $x$, $x+2e_\mu$, $x+2e_\nu$, and every plaquette of $\mathbb{Z}^4$ lies in exactly four of these squares. Hence $Q_1(V(\varphi)) \le Q(\varphi)$, and $Q_k(V^{(k)}(\varphi)) \le Q(\varphi)$ for every $k \ge 1$.*

(ii) *For every finitely supported $v$ on the coarse links, the field $\varphi^*(v)$ equal to $v_{y,\mu}$ on the eight links $(x,\mu)$, $x \in B(y)$, $o_\mu = 1$, and to zero on the other links, vanishes on the tree links, has $V(\varphi^*(v)) = v$ and $Q(\varphi^*(v)) = 4\,Q_1(v)$.*

(iii) *The effective form $S_1(v) = \inf\{Q(\varphi) : V(\varphi) = v\}$ satisfies $Q_1 \le S_1 \le 4\,Q_1$, and $S_k(w) = \inf\{Q(\varphi) : V^{(k)}(\varphi) = w\} \ge Q_k(w)$ for every $k$, the infima over finitely supported fields that vanish on the tree links.*

(iv) *For every $k$, every set $W$ of blocks of the lattice of spacing $2^k$ and every $w$ in the space $\Phi_W$ of that lattice, $\sum_e w_e^2 \le 52\,S_k(w)$.*

*Proof.* (i) The path of length two from $x \in B(y)$ in the direction $\mu$ ends at $x+2e_\mu \in B(y+e_\mu)$, at the same offset, and so on around the square; so the four averages of $(dV)_P$ pair the sixteen paths into sixteen closed squares, and the sum of $\varphi$ around a square is the sum of the curvatures of its four plaquettes. A plaquette at $z$ in the plane $\{\mu,\nu\}$ lies in the square from $x$ exactly when $z-x \in \{0,1\}e_\mu + \{0,1\}e_\nu$, which gives four points $x$, each in one block. By Cauchy–Schwarz, $(dV)_P^2 \le \frac{1}{16}\sum_x \ell_x^2 \le \frac{1}{4}\sum_x\sum_{p\subset\square_x}(d\varphi)_p^2$, where $\ell_x$ is the curvature of the square from $x$; summing over $P$ gives $Q_1(V(\varphi)) \le Q(\varphi)$. The argument uses neither the tree links nor the spacing, so it applies on every lattice, and induction gives the bound for $V^{(k)}$.

(ii) The tree links have $o_\mu = 0$. By Lemma 93(iii) the average of $\varphi^*$ is $\frac{1}{8}\cdot 8 v_{y,\mu}$. A plaquette inside a block has only links with $o_\mu = 0$ in its direction $\mu$, and so vanishes. A plaquette at $x$ in the plane $\{\mu,\nu\}$ with $o_\mu = 1$, $o_\nu = 0$ has the two links $(x,\mu)$, $(x+e_\nu,\mu)$ of the same block, both equal to $v_{y,\mu}$, and two links with offset zero in the direction $\nu$; its curvature vanishes, and so does that of the plaquettes with $o_\mu = 0$, $o_\nu = 1$. If $o_\mu = o_\nu = 1$, its four links carry $v_{y,\mu}$, $v_{y+e_\mu,\nu}$, $v_{y+e_\nu,\mu}$, $v_{y,\nu}$, so its curvature is $(dv)_P$, and there are four such plaquettes for each $P$.

(iii) The lower bound is (i) and the upper bound is (ii); for $S_k$ apply (i) to the $k$-fold averages.

(iv) Theorem 94 holds on every lattice, so $\sum_e w_e^2 \le 52\,Q_k(w) \le 52\,S_k(w)$ by (iii). □

In the Bloch decomposition the eigenvalues of $S_1$ relative to $Q_1$ fill $[1,4]$: they tend to one as the momentum tends to zero and equal four at $(\pi,\pi,0,0)$, a floating-point computation reproduced in the archive. So at this order the averages carry the coarse quadratic form with no loss at long wavelengths; the coupling does not run here, and its running is the order-$g^2$ effect of Remark 109(c). Part (iv) is a gap for the averages at every scale, with the constant of Theorem 94; it does not give the localisation of Theorem 98(ii) for $S_k$, since $S_k$ is not a nearest-neighbour form.

The effective forms are equivalent to the coarse plaquette forms uniformly in the scale. Write $L = 2^k$ and $B_k(y) = Ly + \{0,\dots,L-1\}^4$.

**Lemma 100.** (i) *For every $k \ge 1$, every coarse link $(y,\mu)$ and every field $\varphi$,*

$$V^{(k)}_{y,\mu}(\varphi) = \frac{1}{L^4}\sum_{x\in B_k(y)}\sum_{t=0}^{L-1}\varphi_{(x+te_\mu,\mu)}.$$

(ii) *For a function $h$ on sites, $V^{(k)}_{y,\mu}(dh) = \bar h(y) - \bar h(y+e_\mu)$, where $\bar h(y)$ is the mean of $h$ over $B_k(y)$.*

(iii) *$S_k(w)$ equals the infimum of $Q(\varphi)$ over all finitely supported fields with $V^{(k)}(\varphi) = w$, whether or not they vanish on the tree links.*

*Proof.* (i) For $k=1$ this is the definition. If it holds for $k$, then $V^{(k)}_{y',\mu} + V^{(k)}_{y'+e_\mu,\mu}$ is $L^{-4}$ times the sum over $x \in B_k(y')$ of the sums of $\varphi$ along the straight paths of length $2L$ from $x$, because $B_k(y'+e_\mu) = B_k(y') + Le_\mu$. The blocks $B_k(y')$, $y' \in 2y+\{0,1\}^4$, tile $B_{k+1}(y)$, and averaging over $y'$ with weight $\frac{1}{16}$ gives (i) for $k+1$. (ii) The sum of $dh$ along the path of length $L$ from $x$ is $h(x) - h(x+Le_\mu)$; average over $x \in B_k(y)$. (iii) Let $\varphi$ be finitely supported. On each block $B(y')$ of side two the tree links form a spanning tree (Lemma 93(i)), so there

is $h_0$ on $B(y')$ with $(dh_0)_e = -\varphi_e$ on its tree links, unique up to a constant, which we take to be zero where $\varphi$ vanishes on the tree links of $B(y')$. Each $B_k(y)$ is a union of blocks of side two, since $k \geq 1$; adding a constant to $h_0$ on one of them makes the mean over $B_k(y)$ vanish, and this changes $h_0$ on finitely many blocks. The resulting $h$ is finitely supported, $\varphi + dh$ vanishes on the tree links, $Q(\varphi + dh) = Q(\varphi)$, and $V^{(k)}(dh) = 0$ by (ii). □

The extension uses the lowest-order Whitney forms of the coarse cubical complex whose vertices are the centres

$$c_z = Lz + \tfrac{L-1}{2}(1,1,1,1), \qquad z \in \mathbb{Z}^4,$$

of the blocks $B_k(z)$, rather than the coarse sites. Let $\lambda(t) = \max(0, 1-|t|)$, let $\chi_z^\mu$ be the indicator of $\{\xi \in \mathbb{R}^4 : c_{z,\mu} \leq \xi_\mu < c_{z,\mu} + L\}$, and $\lambda_z^\nu(\xi) = \lambda((\xi_\nu - c_{z,\nu})/L)$. For a square-summable field $u$ on the coarse links put

$$\omega_\mu(\xi) = \frac{1}{L} \sum_z u_{(z,\mu)}\, \chi_z^\mu(\xi) \prod_{\nu \neq \mu} \lambda_z^\nu(\xi),$$

$$(\Pi u)_{(x,\mu)} = \int_0^1 \omega_\mu(x + te_\mu)\, dt.$$

Fig. 13 shows the construction along one direction.

**Lemma 101.** *Let $k \geq 1$, $u$ a square-summable field on the coarse links, and let $T$ act on each component of $u$ by $T = \tau_1\tau_2\tau_3\tau_4$, where $(\tau_\nu f)(z) = \frac{3}{4} f(z) + \frac{1}{8}\big(f(z-e_\nu) + f(z+e_\nu)\big)$.*

(i) *For the plaquette $p$ at $x$ in the plane $\{\mu,\nu\}$,*

$$(d\Pi u)_p = \int_0^1 \int_0^1 \Omega_{\mu\nu}(x + se_\mu + te_\nu)\, ds\, dt,$$

$$\Omega_{\mu\nu} = \frac{1}{L^2} \sum_z (du)_{z,\mu\nu}\, \chi_z^\mu \chi_z^\nu \prod_{\rho \neq \mu,\nu} \lambda_z^\rho,$$

*where $(du)_{z,\mu\nu}$ is the curvature of $u$ at the coarse plaquette at $z$ in the plane $\{\mu,\nu\}$. In particular $\Pi(db)$ is a gradient.*

(ii) $Q(\Pi u) \leq Q_k(u)$.

(iii) $V^{(k)}(\Pi u) = Tu$.

*Proof.* (i) The function $\omega_\mu$ is continuous and piecewise linear in every coordinate $\xi_\nu$, $\nu \neq \mu$, and $\omega_\nu$ in $\xi_\mu$. Hence

$$(d\Pi u)_p = \int_0^1 \big(\omega_\mu(x + te_\mu) - \omega_\mu(x + e_\nu + te_\mu)\big)dt + \int_0^1 \big(\omega_\nu(x + e_\mu + te_\nu) - \omega_\nu(x + te_\nu)\big)dt$$

is the integral over the unit square of $\partial_\mu \omega_\nu - \partial_\nu \omega_\mu$, which exists almost everywhere. Since $\partial_\mu \lambda_z^\mu = \frac{1}{L}(\chi_{z-e_\mu}^\mu - \chi_z^\mu)$ and $\chi_z^\nu$ and $\lambda_z^\rho$, $\rho \neq \mu$, do not change when $z$ is shifted by $e_\mu$, a shift of the summation index gives

$$\partial_\mu \omega_\nu = \frac{1}{L^2} \sum_z \big(u_{(z+e_\mu,\nu)} - u_{(z,\nu)}\big) \chi_z^\mu \chi_z^\nu \prod_{\rho \neq \mu,\nu} \lambda_z^\rho,$$

and the same with $\mu, \nu$ exchanged. The difference is $\Omega_{\mu\nu}$, with $(du)_{z,\mu\nu} = u_{(z,\mu)} + u_{(z+e_\mu,\nu)} - u_{(z+e_\nu,\mu)} - u_{(z,\nu)}$, the sign convention of the fine plaquettes. If $u = db$ then $\Omega = 0$, every curvature of $\Pi u$ vanishes, and $\Pi u$ is a gradient.

(ii) By Cauchy–Schwarz, $(d\Pi u)_p^2$ is at most the integral of $\Omega_{\mu\nu}^2$ over the unit square of $p$. At a point $\xi$ exactly one pair $(z_\mu, z_\nu)$ has $\chi_z^\mu \chi_z^\nu = 1$, and the hat functions form a partition of unity, $\sum_{z_\rho, z_\sigma} \lambda_z^\rho \lambda_z^\sigma = 1$ for $\{\rho,\sigma\}$ the other two directions; so $\Omega_{\mu\nu}^2 \leq L^{-4} \sum_{z_\rho, z_\sigma} \lambda_z^\rho \lambda_z^\sigma (du)_{z,\mu\nu}^2$. Summing over the plaquettes of the plane $\{\mu,\nu\}$ integrates over $\xi_\mu, \xi_\nu$, which gives the factor $L^2$ for each pair $(z_\mu, z_\nu)$, and sums over the integers $x_\rho, x_\sigma$. Since $c_{z,\rho} - x_\rho$ lies in $\mathbb{Z} + \frac{1}{2}$, $\sum_{x_\rho \in \mathbb{Z}} \lambda_z^\rho = 2\sum_{j=0}^{L-1}\big(1 - \frac{j+1/2}{L}\big) = L$, and the same for $\sigma$. So the plaquettes of the plane $\{\mu,\nu\}$ contribute at most $L^{-4} \cdot L^2 \cdot L^2 \sum_z (du)_{z,\mu\nu}^2$.

(iii) By Lemma 100(i), $V_{y,\mu}^{(k)}(\Pi u)$ is the mean over $x = Ly + s$, $s \in \{0,\dots,L-1\}^4$, of $\int_0^L \omega_\mu(x + te_\mu)\, dt$. The term of $u_{(z,\mu)}$ in it is a product of a factor of the coordinate $\mu$ and one of each $\nu \neq \mu$, and the mean over $s$ is the product of the means over the single coordinates. Put $\delta = s_\mu - \frac{L-1}{2}$. The interval $[x_\mu, x_\mu + L)$ meets $[c_{z,\mu}, c_{z,\mu} + L)$ in length $L - |\delta|$ if $z_\mu = y_\mu$, in $\max(0, -\delta)$ if $z_\mu = y_\mu - 1$, in $\max(0, \delta)$ if $z_\mu = y_\mu + 1$, and not otherwise. As $L$ is even, $\sum_s |\delta| = 2\sum_{j=0}^{L/2-1}(j + \frac{1}{2}) = \frac{L^2}{4}$, so the means of these lengths divided by $L$ are $\frac{3}{4}$, $\frac{1}{8}$, $\frac{1}{8}$. For $\nu \neq \mu$ put $\delta = s_\nu - \frac{L-1}{2}$; then $\lambda_z^\nu(x)$ is $1 - |\delta|/L$ if $z_\nu = y_\nu$, $\max(0, \mp\delta)/L$ if $z_\nu = y_\nu \mp 1$, and zero otherwise, with the same means. The product of the four means is the coefficient of $u_{(z,\mu)}$ in $(Tu)_{(y,\mu)}$. □

**Theorem 102.** *For every $k \geq 1$ and every finitely supported field $w$ on the links of the lattice of spacing $2^k$,*

$$Q_k(w) \leq S_k(w) \leq 256\, Q_k(w).$$

*Proof.* The lower bound is Proposition 99(iii). The operator $T$ of Lemma 101 is a convolution, the same on every component, so it commutes with $d$; its symbol is $\prod_\nu (\frac{3}{4} + \frac{1}{4}\cos\theta_\nu) \geq \frac{1}{16}$, so it is invertible on square-summable fields with $\|T^{-1}\| \leq 16$. Let $u = T^{-1}w$. By Lemma 101, $V^{(k)}(\Pi u) = w$ and

$$Q(\Pi u) \leq Q_k(u) = \|T^{-1} dw\|^2 \leq 256\, Q_k(w).$$

It remains to replace $\Pi u$ by a finitely supported field. Let $u_R$ be $u$ on the coarse links with base in $[-R,R]^4$ and zero elsewhere, and $e_R = w - Tu_R = T(u - u_R)$, which is finitely supported and tends to zero. Let $\psi_R$ equal $e_R(y,\mu)$ on the links $(x,\mu)$ with $x \in B_k(y)$ and $x_\mu \equiv L-1$ modulo $L$, and zero elsewhere. Each straight path of length $L$ from a site of $B_k(y)$ in the direction $\mu$ contains exactly one such link of $B_k(y)$, so $V^{(k)}(\psi_R) = e_R$ by Lemma 100(i), and $Q(\psi_R) \leq 24\|\psi_R\|^2 = 24L^3\|e_R\|^2$, since a link lies in six plaquettes of four links. The field $\Pi u_R + \psi_R$ is finitely supported, its $k$-fold averages are $w$, and $Q(\Pi u_R + \psi_R)^{1/2} \leq Q_k(u_R)^{1/2} + (24L^3)^{1/2}\|e_R\|$, which tends to $Q_k(u)^{1/2}$. Lemma 100(iii) gives $S_k(w) \leq Q_k(u) \leq 256\, Q_k(w)$. □

So at every scale the averages carry the coarse plaquette form up to a factor at most 256, independent of $k$. The construction

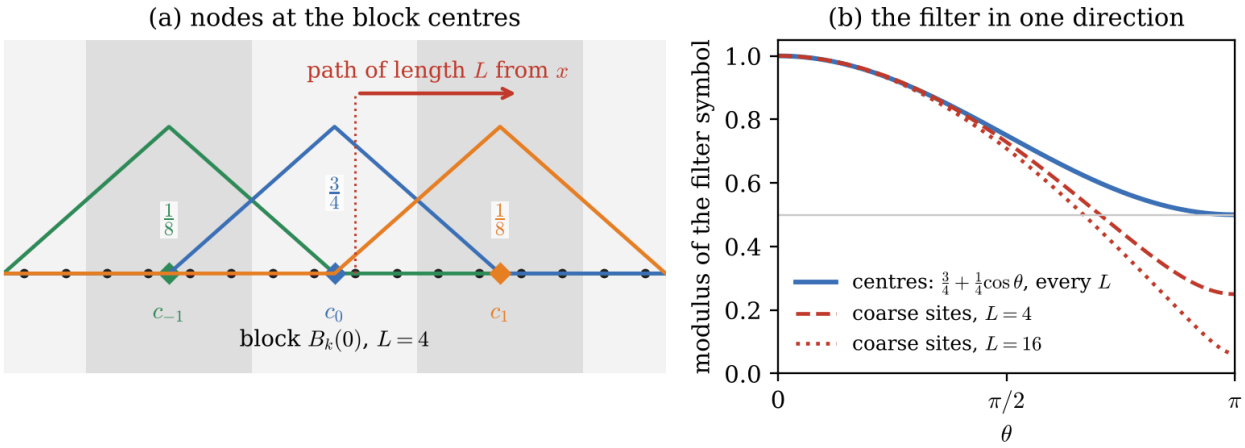



FIG. 13. The extension of Theorem 102 along one direction. (a) A block of side $L=4$: the nodes are the block centres $c_z$, the hat functions $\lambda_z$ interpolate between them, and the straight path of length $L$ from a site $x$ meets the segments of three centres; averaged over the block, their weights are $\frac{1}{8}$, $\frac{3}{4}$, $\frac{1}{8}$ at every scale. (b) The symbol of this filter stays at least $\frac{1}{2}$, while the filter obtained with the coarse sites as nodes has symbol of modulus $1/L$ at $\theta=\pi$.

inverts the averages by a fixed convolution: the filter $(\frac{1}{8},\frac{3}{4},\frac{1}{8})$ does not depend on the scale because the nodes are the block centres, and its symbol is at least $\frac{1}{2}$. With the coarse sites as nodes the corresponding filter is $\frac{L+1}{2L}\delta_0+\frac{L-1}{2L}\delta_1$, whose symbol has modulus $1/L$ at $\theta=\pi$ (Fig. 13(b)). In floating point, on a grid of $12^4$ momenta, the largest ratio of $S_k$ to $Q_k$ on curvatures is 3.9, 7.8 and 10.2 for $k=1,2,3$, and the kernel of $S_k$ decreases by the factors 0.17, 0.23 and 0.25 per coarse spacing; these values are reproduced in the archive. The theorem gives no decay. The decay comes from an exact formula for $S_k$ in Fourier variables: in a Feynman gauge for the fine field the effective form is diagonal in the six planes, and the pole of the fine propagator at zero momentum cancels between numerator and denominator.

For $t\in\mathbb{C}$ put $D_L(t)=\frac{1}{L}\sum_{j=0}^{L-1}e^{ijt}$, and for $\vartheta\in\mathbb{C}^4$ put $\Psi(\vartheta)=\prod_\nu D_L(\vartheta_\nu)$, $E_\mu(\vartheta)=e^{i\vartheta_\mu}-1$ and $\varepsilon(\vartheta)=\sum_\nu(2-2\cos\vartheta_\nu)$. For an entire function $f$ let $f^\sharp(\vartheta)=\overline{f(\bar\vartheta)}$, so that $ff^\sharp=|f|^2$ on real arguments. For $\theta\in\mathbb{C}^4$ and $n\in\{0,\dots,L-1\}^4$ let $\theta^{(n)}=(\theta+2\pi n)/L$; every function below is $2\pi$-periodic in each coordinate of $\theta^{(n)}$, and we take the real parts of these coordinates in $[-\pi,\pi]$. Write $\Psi_n=(\Psi\Psi^\sharp)(\theta^{(n)})$, $D_{n,\sigma}=(D_LD_L^\sharp)(\theta_\sigma^{(n)})$, and put

$$A_\sigma=\Psi_0D_{0,\sigma},\qquad \hat\varepsilon=L^2\varepsilon(\theta^{(0)}),\qquad \tilde a_\sigma=A_\sigma+\hat\varepsilon r_\sigma,$$
$$r_\sigma=\sum_{n\neq0}\frac{\Psi_nD_{n,\sigma}}{L^2\varepsilon(\theta^{(n)})},$$

$$Y=\Psi_0^3\prod_\tau D_{0,\tau}+\sum_\sigma E_\sigma E_\sigma^\sharp\sum_{\emptyset\neq J\subset\{1,2,3,4\}\setminus\{\sigma\}}\hat\varepsilon^{|J|-1}\prod_{\tau\in J}r_\tau\prod_{\tau\notin J\cup\{\sigma\}}A_\tau,\tag{18}$$

all functions of $\theta$, and for a plane $P=\{\lambda,\rho\}$

$$K_P=\frac{\prod_{\sigma\notin P}\tilde a_\sigma}{Y}.$$

For $L=1$ these are $\tilde a_\sigma=Y=K_P=1$. For a coarse field $w$ let $\hat w(\theta)=\sum_y e^{-i\theta\cdot y}w_y$, and let $F_P=(dw)_P$ be its curvature in the plane $P$, a function of the coarse sites.

**Lemma 103.** *Let $\gamma_1,\dots,\gamma_4>0$, $\Gamma=\mathrm{diag}(\gamma_\mu)$ and $0\neq E\in\mathbb{C}^4$, and for $P=\{\lambda,\rho\}$, $\lambda<\rho$, let $c_P$ be the row vector with $c_Pv=E_\lambda v_\rho-E_\rho v_\lambda$. Then*

$$\Gamma-\frac{\Gamma EE^*\Gamma}{E^*\Gamma E}=\frac{1}{E^*\Gamma E}\sum_P\gamma_\lambda\gamma_\rho\,c_P^*c_P.$$

*Proof.* Both sides are Hermitian. The diagonal entry $(\mu,\mu)$ of the right side is $\sum_{\lambda\neq\mu}\gamma_\mu\gamma_\lambda|E_\lambda|^2/E^*\Gamma E$, which is $\gamma_\mu-\gamma_\mu^2|E_\mu|^2/E^*\Gamma E$; the entry $(\mu,\nu)$, $\mu\neq\nu$, comes only from $P=\{\mu,\nu\}$, where $c_P$ has $-E_\nu$ in place $\mu$ and $E_\mu$ in place $\nu$, and equals $-\gamma_\mu\gamma_\nu\bar E_\mu E_\nu/E^*\Gamma E$. □

**Theorem 104.** *Let $k\geq1$ and $L=2^k$.*

(i) *The functions $K_P$ are real-analytic, even, $2\pi$-periodic and positive on $\mathbb{R}^4$, $K_P(0)=1$, and for every finitely supported field $w$ on the coarse links*

$$S_k(w)=\sum_P\int_{[-\pi,\pi]^4}K_P(\theta)\,|\hat F_P(\theta)|^2\,\frac{d\theta}{(2\pi)^4}.$$

(ii) *There are $\eta>0$ and $C<\infty$, independent of $k$, such that every $K_P$ extends analytically to $\{|\mathrm{Im}\,\theta_\nu|\leq\eta\}$ and is bounded there by $C$.*

(iii) *Hence $S_k(w)=\sum_P\sum_{y,y'}\mathscr{K}_P(y-y')F_P(y)F_P(y')$ with real kernels $\mathscr{K}_P(y)=(2\pi)^{-4}\int K_P(\theta)e^{i\theta\cdot y}\,d\theta$ that satisfy $|\mathscr{K}_P(y)|\leq Ce^{-\eta|y|_1}$ for every $k$.*

*Proof. Step 1: the form in Fourier variables.* Let $c_\mu(q) = L\Psi(q)D_L(q_\mu)$. Lemma 100(i) and the Poisson summation formula give, for every square-summable field $\varphi$,

$$\widehat{V^{(k)}(\varphi)}_\mu(\theta) = L^{-4}\sum_n c_\mu(\theta^{(n)})\,\hat\varphi_\mu(\theta^{(n)}),$$

with $\hat\varphi(q) = \sum_x e^{-iq\cdot x}\varphi_x$. Write $x_n = \hat\varphi(\theta^{(n)}) \in \mathbb{C}^4$ and $e_n = (e^{i\theta_\mu^{(n)}} - 1)_\mu$, so that $|e_n|^2 = \varepsilon(\theta^{(n)}) =: \varepsilon_n$. By the Lagrange identity the plaquette form at the momentum $\theta^{(n)}$ is $\varepsilon_n|x_n|^2 - |e_n^* x_n|^2$, and the Plancherel formula gives

$$Q(\varphi) = \int_{[-\pi,\pi]^4} L^{-4}\sum_n \big(\varepsilon_n|x_n|^2 - |e_n^* x_n|^2\big)\,\frac{d\theta}{(2\pi)^4}.$$

Since $LD_L(t)(e^{it}-1) = e^{iLt}-1$ and $e^{iL\theta_\mu^{(n)}} = e^{i\theta_\mu}$, the averages map the gauge direction at every alias to the coarse gauge direction: $c_\mu(\theta^{(n)})(e_n)_\mu = \Psi(\theta^{(n)})E_\mu(\theta)$. For $|t| \le \pi/L$ we have $|D_L(t)| = |\sin(Lt/2)|/(L|\sin(t/2)|) \ge 2/\pi$, so $\Psi(\theta^{(0)}) \ne 0$ for real $\theta$.

*Step 2: the minimum at one momentum.* Fix real $\theta \notin 2\pi\mathbb{Z}^4$; then $E = E(\theta) \ne 0$ and every $\varepsilon_n > 0$. Let $z \in \mathbb{C}^4$. Minimising $L^{-4}\sum_n \varepsilon_n|y_n|^2$ under $L^{-4}\sum_n c_\mu(\theta^{(n)})(y_n)_\mu = z_\mu$ separates in $\mu$, and the Cauchy–Schwarz inequality gives the minimum $\sum_\mu |z_\mu|^2/a_\mu$, attained at $(y_n)_\mu = z_\mu\overline{c_\mu(\theta^{(n)})}/(\varepsilon_n a_\mu)$, where

$$a_\mu = L^{-4}\sum_n \frac{|c_\mu(\theta^{(n)})|^2}{\varepsilon_n} = \frac{\tilde a_\mu}{\hat\varepsilon}.$$

If $x$ satisfies the constraint with right side $\hat w(\theta)$, write $x_n = x_n^T + \beta_n e_n$ with $e_n^* x_n^T = 0$. Then $L^{-4}\sum_n(\varepsilon_n|x_n|^2 - |e_n^*x_n|^2) = L^{-4}\sum_n \varepsilon_n|x_n^T|^2$, and by Step 1 the vectors $x_n^T$ satisfy the constraint with right side $\hat w(\theta) + tE$ for some $t \in \mathbb{C}$. Hence the plaquette form at $\theta$ is at least

$$\min_{t\in\mathbb{C}}\sum_\mu \frac{|\hat w_\mu + tE_\mu|^2}{a_\mu} = \hat w^*\Big(\Gamma - \frac{\Gamma E E^*\Gamma}{E^*\Gamma E}\Big)\hat w$$

with $\Gamma = \operatorname{diag}(a_\mu^{-1})$. Conversely, take $t$ optimal, $y$ optimal for $z = \hat w + tE$, and replace $y_0$ by $y_0 - \beta e_0$ with $\beta = L^4 t/\Psi(\theta^{(0)})$: the constraint becomes $\hat w(\theta)$ and the plaquette form does not increase, since $e_0$ is a gauge direction. By Lemma 103 the minimum is

$$\sum_P \frac{\gamma_\lambda\gamma_\rho}{E^*\Gamma E}|c_P\hat w|^2, \qquad \frac{\gamma_\lambda\gamma_\rho}{E^*\Gamma E} = \frac{\prod_{\sigma\notin P} a_\sigma}{\sum_\sigma |E_\sigma|^2\prod_{\tau\ne\sigma} a_\tau}.$$

We multiply numerator and denominator by $\hat\varepsilon^3$ and expand $\tilde a_\tau = A_\tau + \hat\varepsilon r_\tau$. The terms without $r$ add up to $\Psi_0^3\prod_\tau D_{0,\tau}\sum_\sigma |E_\sigma|^2/D_{0,\sigma}$, and $|E_\sigma|^2 = L^2 D_{0,\sigma}|e^{i\theta_\sigma^{(0)}} - 1|^2$ by the identity of Step 1, so they equal $\hat\varepsilon\,\Psi_0^3\prod_\tau D_{0,\tau}$; every other term carries a factor $\hat\varepsilon$. So the denominator is $\hat\varepsilon Y$, the numerator $\hat\varepsilon\prod_{\sigma\notin P}\tilde a_\sigma$, and the quotient is $K_P(\theta)$. Since $|c_P\hat w| = |\hat F_P|$, the minimum at $\theta$ is $\sum_P K_P(\theta)|\hat F_P(\theta)|^2$.

*Step 3: integration.* Integrating the lower bound over $\theta$ gives $Q(\varphi) \ge \sum_P \int K_P|\hat F_P|^2$ for every square-summable $\varphi$ with $V^{(k)}(\varphi) = w$. The minimisers of Step 2 are bounded by a constant times $|\hat w(\theta)|$: for $n \ne 0$ because $\varepsilon_n \ge 4/L^2$ and $a_\mu \ge A_\mu/\hat\varepsilon$; for $n = 0$ because $|c_\mu(\theta^{(0)})|/(\varepsilon_0 a_\mu) \le L^3/A_\mu$; and $|t||E| \le |\hat w|(\max\gamma/\min\gamma)^{1/2}$ by the Cauchy–Schwarz inequality for the inner product $\Gamma$, while $|e_0| \le |\theta|/L \le \frac{\pi}{2}|E|/L$. Here $A_\mu \ge (2/\pi)^{10}$, and $\gamma_\mu/\gamma_\nu = \tilde a_\nu/\tilde a_\mu$ is bounded because the $\tilde a$ are continuous and at least $A$. So the minimisers define a square-summable field $\varphi^*$ with $V^{(k)}(\varphi^*) = w$ and $Q(\varphi^*) = \sum_P\int K_P|\hat F_P|^2$. Since $V^{(k)}$ and $Q$ are bounded on square-summable fields, truncating $\varphi^*$ and correcting with $\psi_R$ as in the proof of Theorem 102 gives finitely supported fields with the same averages and plaquette forms tending to $Q(\varphi^*)$, and Lemma 100(iii) gives (i) for $S_k$. The functions are even since $|D_L|^2$ and $\varepsilon$ are, and positive since $\tilde a_\sigma \ge A_\sigma > 0$ and $Y \ge \Psi_0^3\prod_\tau D_{0,\tau} \ge (2/\pi)^{32}$ on $\mathbb{R}^4$, all other terms of (18) being non-negative there. At $\theta = 0$, $\Psi_0 = D_{0,\tau} = 1$ and $\hat\varepsilon = 0$, so $K_P(0) = 1$.

*Step 4: the strip.* Let $|\operatorname{Im}\theta_\nu| \le \frac12$, and for $n$ let $s = L\theta^{(n)}$, so $|\operatorname{Re}s_\nu| \le \pi L$ and $|\operatorname{Im}s_\nu| \le \frac12$. For $q = x + iy$ with $|x| \le \pi$, $|e^{iq}-1|^2 = (e^{-y}-1)^2 + 2e^{-y}(1-\cos x) \ge e^{-|y|}\frac{4}{\pi^2}x^2$, so

$$|D_L(q)| \le \min\Big(e^{1/2}, \frac{c_1}{|\operatorname{Re}Lq|}\Big)$$

for $|\operatorname{Im}q| \le 1/(2L)$, where $c_1 = (e^{1/2}+1)e^{1/4}\frac{\pi}{2} < 5.35$, and the same holds for $D_L^\sharp$. Also $\operatorname{Re}(2-2\cos q) \ge (2-2\cos x) - 2(\cosh y - 1)$, so $\operatorname{Re}L^2\varepsilon(\theta^{(n)}) \ge \frac{4}{\pi^2}|\operatorname{Re}s|^2 - 1.2$. For $n \ne 0$ some $|\operatorname{Re}s_\nu| \ge \pi$, and then $|L^2\varepsilon(\theta^{(n)})| \ge 0.28\,|\operatorname{Re}s|^2$. Writing $s = \theta + 2\pi m$ with $m \in \mathbb{Z}^4\setminus\{0\}$, $|\operatorname{Re}s_\nu| \ge \pi|m_\nu|$, and

$$|r_\sigma| \le \frac{c_1^{10}}{0.28\,\pi^2}\sum_m\prod_\nu f(m_\nu) = \frac{c_1^{10}}{0.28\,\pi^2}\Big(\frac43\Big)^4 < 2.2\cdot 10^7,$$

with $f(0) = 1$ and $f(j) = 1/(\pi j)^2$. The $r_\sigma$ are finite sums of functions analytic on the strip, since no $\varepsilon(\theta^{(n)})$, $n \ne 0$, vanishes there. Further $|A_\sigma| \le e^5$, $|\hat\varepsilon| \le 4(\pi^2 + \frac14)e^{1/2} < 70$ and $|E_\sigma E_\sigma^\sharp| \le (e^{1/2}+1)^2 < 7.1$, so on the strip $|Y| \le M = 10^{28}$ and $|\tilde a_\sigma| \le 1.6\cdot 10^9$, uniformly in $k$. By the Cauchy estimate on discs of radius $\frac14$ in each variable, $|\partial_\nu Y| \le 4M$ where $|\operatorname{Im}\theta| \le \frac14$, so $|Y(\theta) - Y(\operatorname{Re}\theta)| \le 16\eta M \le \frac12(2/\pi)^{32}$ for $|\operatorname{Im}\theta_\nu| \le \eta = 10^{-36}$. There $|Y| \ge \frac12(2/\pi)^{32}$, $K_P = \prod_{\sigma\notin P}\tilde a_\sigma/Y$ is analytic, and $|K_P| \le C = 10^{26}$, which is (ii). Moving the contour of each $\theta_\nu$ to $\operatorname{Im}\theta_\nu = \pm\eta$, according to the sign of $y_\nu$, gives (iii); the kernels are real because $K_P$ is real and even. □

The proof makes $\eta$ explicit but very small. In floating point the kernels decrease much faster: $\max|\mathscr{K}_P(y)|$ over $|y|_1 = d$ falls by factors 0.17, 0.24 and 0.26 per unit of $d$ for $k = 1,2,3$, the largest value of $K_P$ on $\mathbb{R}^4$ is 4.4, 8.7 and 11.3, and its smallest value is 1, attained at $\theta = 0$ (Fig. 14); these computations are reproduced in the archive. Each $K_P$ is a Riemann sum in $n$ of a function of $\theta + 2\pi n$, so the kernels have a limit as $k \to \infty$; the theorem does not use it. Uniform decay of the propagators of the averaging operations of lattice gauge theories is the subject of Bałaban's work [6]; here the hard constraint and the Feynman gauge make the effective form itself explicit. Together with Theorem 102, the effective forms of the quadratic

approximation are equivalent to the coarse plaquette forms and local, with constants independent of the scale.

**Lemma 105.** *Let $(X,d_X)$ and $(Y,d_Y)$ be Polish spaces, $\rho,\rho'$ Borel probability measures on $Y$, and $\kappa,\kappa'$ probability kernels from $Y$ to $X$, all with finite first moments. If $W_1(\kappa(\cdot\mid v),\kappa'(\cdot\mid v))\le\varepsilon$ and $W_1(\kappa'(\cdot\mid v),\kappa'(\cdot\mid v'))\le d_Y(v,v')$ for all $v,v'$, then the mixtures $\pi=\int\kappa(\cdot\mid v)\,\rho(dv)$ and $\pi'=\int\kappa'(\cdot\mid v)\,\rho'(dv)$ satisfy*

$$W_1(\pi,\pi')\le\varepsilon+W_1^{d_Y}(\rho,\rho').$$

*Proof.* Let $F$ be 1-Lipschitz on $X$ and $\kappa F(v)=\int F\,d\kappa(\cdot\mid v)$. Then $\int F\,d\pi-\int F\,d\pi'=\int(\kappa F-\kappa' F)\,d\rho+\big(\int\kappa' F\,d\rho-\int\kappa' F\,d\rho'\big)$. The first term is at most $\varepsilon$. The function $\kappa' F$ is 1-Lipschitz for $d_Y$, so the second is at most $W_1^{d_Y}(\rho,\rho')$. The Kantorovich–Rubinstein duality gives the claim. □

A law $\gamma$ on $\mathbb{R}^V$ *decomposes along the averages* if $\gamma(\cdot\mid\eta)=\int\gamma_V^{\mathrm c}(\cdot\mid\eta,v)\,\rho_V(dv\mid\eta)$, with $\rho_V(\cdot\mid\eta)$ the law of $(V_{y,\mu})_{y\in W}$. Every law with density proportional to $\exp(-\frac{b}{2}\sum_{p\cap V\neq\emptyset}(d\varphi)_p^2)\,g(V(\varphi))$, on a set cut out by conditions on the averages, decomposes in this way, because Lebesgue measure disintegrates along the fibres of a surjective linear map into multiples of Lebesgue measure on the fibres. Here $g$ can carry any action of the averages, in particular a coarse gauge fixing and a coarse effective action, which is the form a law takes after the fine scale has been integrated out.

**Corollary 106.** *Let $r_w\in\mathbb{R}^V$ minimise $\sum_{p\cap V\neq\emptyset}(d\varphi)_p^2$, with zero exterior, subject to $V_{y',\mu'}(r_w)=\delta_{w,(y',\mu')}$ for all $(y',\mu')$, $y'\in W$; put $L_w=\|r_w\|_1$ and $d_L(v,v')=\sum_w L_w|v_w-v'_w|$. For every law that decomposes along the averages,*

$$C_{V,j}\le C_{V,j}^{\mathrm c}+C_{V,j}^{\rho},\qquad C_{V,j}^{\rho}=\sup\frac{W_1^{d_L}(\rho_V(\cdot\mid\eta),\rho_V(\cdot\mid\eta'))}{|\eta_j-\eta'_j|},$$

*the supremum over exteriors that differ only at $j$, and $L_w\le 4\cdot10^{14}$. Hence $K_c\le K_c^{\mathrm c}+K_c^{\rho}$ with $K_c^{\rho}=\sum_{j\in c\setminus V}C_{V,j}^{\rho}$, and if $K_c^{\rho}\le\frac{\alpha}{2}N_n$ for every class and $n\ge4\cdot10^{15}/\alpha$, then $D(n,\alpha)$ holds for the law, with the 49 classes of active links.*

*Proof.* Apply Lemma 105 with $\kappa=\gamma_V^{\mathrm c}(\cdot\mid\eta,\cdot)$, $\kappa'=\gamma_V^{\mathrm c}(\cdot\mid\eta',\cdot)$, $\rho=\rho_V(\cdot\mid\eta)$, $\rho'=\rho_V(\cdot\mid\eta')$. The first condition holds with $\varepsilon=C_{V,j}^{\mathrm c}|\eta_j-\eta'_j|$. By Theorem 98(i) the laws $\gamma_V^{\mathrm c}(\cdot\mid\eta',v)$ and $\gamma_V^{\mathrm c}(\cdot\mid\eta',v')$ are translates by $\sum_w(v'_w-v_w)r_w$, so their distance is at most $d_L(v,v')$. For the bound on $L_w$, the pair $(r_w,y)$ solves $K(r_w,y)=(0,se_w)$ with $K$ as in the proof of Theorem 98. Measure $d$ from the centre of the row $w$; Lemma 97 gives $\sum_e e^{2\lambda d(e)}r_w(e)^2\le(2M_0s)^2=20160^2$. With $z_0=2y+e_\mu$ for $w=(y,\mu)$, a link whose base is at distance $k$ from $z_0$ has its midpoint at distance at least $k-1$ from the centre, and the count in that proof gives $L_w^2\le20160^2\cdot4e^{2\lambda}\big(1+\sum_{k\ge1}(64k^3+16k)e^{-2\lambda k}\big)\le(4\cdot10^{14})^2$. Summing over the translates as in Definition 42 gives $K_c\le K_c^{\mathrm c}+K_c^{\rho}$, and Theorem 98(iii) with $\alpha/2$ in place of $\alpha$ gives $K_c^{\mathrm c}\le\frac{\alpha}{2}N_n$. □

The responses are much smaller than the bound: in floating point, $\max_w L_w$ is 23.8, 44.1 and 50.9 for the block $(m,m,m,m)$, $m=\lfloor n/4\rfloor$, of the cubes $n=4$, 6, 8. Since Theorem 98 controls the first term at every coupling, the corollary reduces $D(n,\alpha)$ for a law that decomposes along the averages to the cube condition $K_c^{\rho}$ for its averages, a problem of the same form one scale up, in the metric $d_L$. Iterating it needs Theorem 98 and the corollary with $S_k$ in place of the plaquette form, which Theorem 108 below gives with constants independent of $k$, and ends in a cube condition for the coarsest averages, which is where a gap has to come from: the quadratic approximation describes a massless field and has none. The corollary is one step of the composition of Remark 109(a), not its conclusion.

On the lattice of spacing $2^k$ the blocks, the tree links, the active links and the averages are those of Definition 92, and after $k$ steps the averages of the quadratic approximation carry the form $S_k$. Let $(S_{ab})$ be its matrix, $S_k(w)=\sum_{a,b}S_{ab}w_aw_b$ for finitely supported $w$, and $d(a,b)$ the $\ell^\infty$ distance between the midpoints of the links $a$ and $b$. Each link lies in six plaquettes of four links, and the midpoints of the links of the plaquettes at $y$ and $y'$ are at distance at most $|y-y'|_1+1$, so Theorem 104(iii) gives, for every non-decreasing $\phi\ge0$,

$$\sum_b|S_{ab}|\,\phi(d(a,b))\le24C\sum_{z\in\mathbb{Z}^4}e^{-\eta|z|_1}\phi(|z|_1+1),\qquad(19)$$

with $C$ and $\eta$ of that theorem. In particular $\sum_b S_{ab}\eta_b$ converges for every bounded field $\eta$. For $n$, $V$, $W$ as before, a bounded exterior $\eta$ and values $v$, let $\gamma_V^{\mathrm c,k}(\cdot\mid\eta,v)$ be the law on $\mathbb{R}^V$ with density proportional to

$$\exp\Big(-\frac{b}{2}\Big(\sum_{a,b\in V}S_{ab}w_aw_b+2\sum_{a\in V,b\notin V}S_{ab}w_a\eta_b\Big)\Big)$$

on the affine set $\{V_{y,\mu}(w)=v_{y,\mu},\ y\in W\}$, the conditional law of the Gaussian with the form $bS_k$ given the exterior and the averages. Define $C_{V,j}^{\mathrm c,k}$, $K_c^{\mathrm c,k}$, $D_{\mathrm c}^k(n,\alpha)$, $r_w^{(k)}$ and $L_w^{(k)}$ as in Theorem 98 and Corollary 106, with $S_k$ in place of the plaquette form; for $k=0$ they are the former ones.

**Lemma 107.** *Let $K$ be an invertible symmetric matrix indexed by a finite set $X$, with $\|K^{-1}\|\le M_0$, and let $d:X\to\mathbb{R}$ and $\lambda>0$. If $\sum_b|K_{ab}|(e^{\lambda|d(a)-d(b)|}-1)\le\frac{1}{2M_0}$ for every $a$, then $\|e^{\lambda d}K^{-1}e^{-\lambda d}\|\le2M_0$.*

*Proof.* $e^{\lambda d}Ke^{-\lambda d}=K+E$ with $|E_{ab}|\le|K_{ab}|(e^{\lambda|d(a)-d(b)|}-1)$, a symmetric bound, so $\|E\|\le\frac{1}{2M_0}$ by the Schur test, and the proof of Lemma 97 applies. □

**Theorem 108.** *Let $\lambda_1=10^{-215}$. For every $k\ge1$, every $b>0$, every even $n$ and every exterior link $j$:*

(i) *$\gamma_V^{\mathrm c,k}(\cdot\mid\eta,v)$ is a Gaussian law whose covariance depends on neither $\eta$ nor $v$ and whose mean is affine in $(\eta,v)$, and $C_{V,j}^{\mathrm c,k}=\|u_j\|_1$, where $u_j\in\mathbb{R}^V$ minimises $S_k(u+\delta_j)$ subject to $Bu=g$, with $B$ and $g$ as in Theorem 98;*

(ii) *$\sum_{e\in V}e^{2\lambda_1 d(e)}u_j(e)^2\le10^{350}$ and $\|u_j\|_1\le10^{607}e^{-\lambda_1 r_j/2}$, where $d(e)=d(e,j)$ and $r_j=\min_{e\in V}d(e)$; and $L_w^{(k)}\le10^{438}$ for every $w$;*

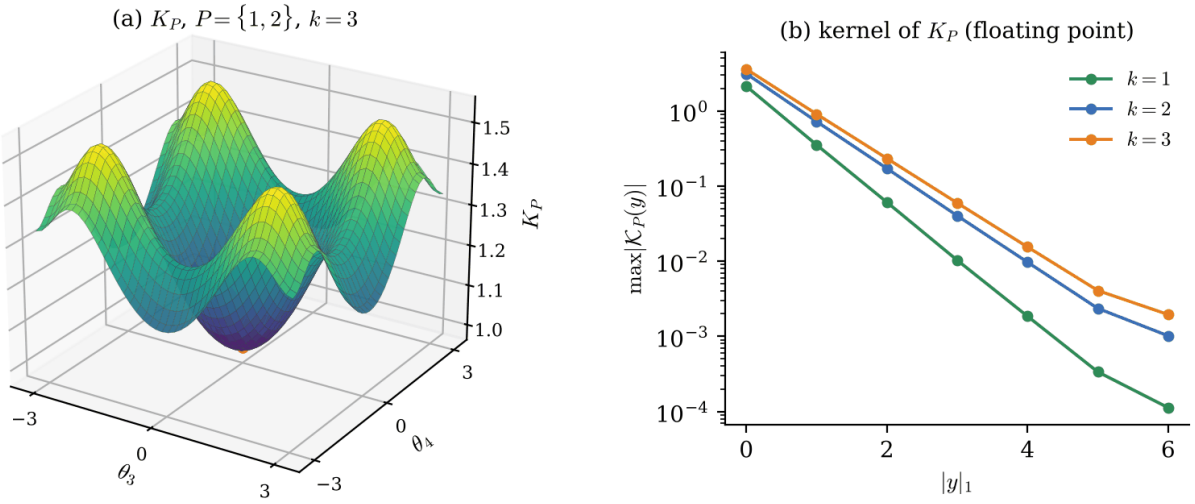



FIG. 14. The effective forms of Theorem 104. (a) The multiplier $K_P$ for the plane $P=\{1,2\}$ at $k=3$, as a function of $(\theta_3,\theta_4)$ with $\theta_1=\theta_2=0$; it equals 1 at $\theta=0$ and stays bounded. (b) The largest value of the kernel $|\mathcal{K}_P(y)|$ on $|y|_1=d$ for $k=1,2,3$, in logarithmic scale: the decay per coarse spacing hardly changes with $k$.

(iii) *$D^k_{\mathrm{c}}(n,\alpha)$ holds for every $\alpha\in(0,1)$ and every even $n\geq 10^{827}/\alpha$;*

(iv) *Corollary 106 holds on the lattice of spacing $2^k$, with $S_k$ in place of the plaquette form, $L_w^{(k)}\leq 10^{438}$ and $n\geq 2\cdot 10^{827}/\alpha$.*

*None of these constants depends on k or on b.*

*Proof.* (i) Let $H=(S_{ab})_{a,b\in V}$. A vector $w\in\ker B$, extended by zero, lies in the space $\Phi_W$ of the lattice of spacing $2^k$, so Proposition 99(iv) gives $\langle w,Hw\rangle=S_k(w)\geq\frac{1}{52}|w|^2$ on $\ker B$. The proof of Theorem 98(i) applies, the exterior entering through the linear term $\sum_{b\notin V}S_{ab}\eta_b$; changing $\eta_j$ by one translates the mean by the minimiser of $\langle u,Hu\rangle+2\sum_{a\in V}u_aS_{aj}=S_k(u+\delta_j)-S_k(\delta_j)$ subject to $Bu=g$.

(ii) The pair $(u_j,y)$ solves $K(u_j,y)=(f,sg)$ with $K$ built from $H$ and $B$ as in Lemma 96 and $f=-(S_{aj})_{a\in V}$. Since $Q_k(w)\leq 24|w|^2$, Theorem 102 gives $\|H\|\leq h=6144$; take $s=4h=24576$. The bounds of Lemma 96 for $(h,s)$ are at most those for $(24,96)$, because $|x_1|$ and $|f|/s$ decrease while $h|x_1|$ and $h/s$ do not change, so $\|K^{-1}\|\leq M_0=105$ as in the proof of Theorem 98. Index the rows of $B$ by centres as there. For $\phi(r)=e^{\lambda r}-1\leq\lambda re^{\lambda r}$ and $\lambda\leq\eta/2$, (19) bounds the row sums of Lemma 107 over $H$ by

$$24C\lambda e^{\lambda}\sum_z(|z|_1+1)e^{-\eta|z|_1/2}\leq 24C\lambda e^{\lambda}\Big(1+16\cdot 24\,e^{\eta/2}\Big(\frac{2}{\eta}\Big)^5\Big),$$

since at most $8r^3$ points have $|z|_1=r\geq 1$ and $\sum_{r\geq1}r^pe^{-xr}\leq e^xp!/x^{p+1}$. The entries of $sB^{\top}$ add at most $\frac{s}{8}(e^{\lambda}-1)$ and a row of $sB$ gives $2s(e^{\lambda}-1)$. At $\lambda=\lambda_1$ the sums are below $3\cdot10^{-4}<\frac{1}{2M_0}$, so Lemma 107 gives $\|e^{\lambda_1 d}K^{-1}e^{-\lambda_1 d}\|\leq 210$. By (19), $\|e^{\lambda_1 d}f\|\leq 24Ce^{\lambda_1}\sum_z e^{-(\eta-\lambda_1)|z|_1}=24Ce^{\lambda_1}\coth^4\frac{\eta-\lambda_1}{2}<4\cdot10^{172}$, and $\|e^{\lambda_1 d}sg\|\leq e^{\lambda_1}s/16$, which gives the first bound. The links with base at $\ell^\infty$ distance $m$ from the base of $j$ number at most $4(64m^3+16m)$, with midpoints at distance at least $m-1$, so

$$Z(x)=\sum_e e^{-xd(e)}\leq 4e^x\Big(1+\frac{384e^x}{x^4}+\frac{16e^x}{x^2}\Big),$$

and $Z(\lambda_1)<10^{864}$. As $d\geq r_j$ on $V$, the Cauchy–Schwarz inequality gives $\|u_j\|_1^2\leq 10^{350}e^{-\lambda_1 r_j}Z(\lambda_1)$. Finally $(r_w^{(k)},y)$ solves $K(r_w^{(k)},y)=(0,se_w)$; measuring $d$ from the centre of the row $w$, we get $\sum_e e^{2\lambda_1 d(e)}r_w^{(k)}(e)^2\leq(210s)^2$ and $L_w^{(k)}\leq 210sZ(2\lambda_1)^{1/2}<10^{438}$.

(iii) The exterior links of a class have their bases at $\ell^\infty$ distance $m\geq1$ from $[0,n-1]^4$; at most $4((n+2m)^4-(n+2m-2)^4)\leq128(n^3+8m^3)$ of them have distance $m$, and for them $r_j\geq m-1$. With $q=\lambda_1/2$, (ii) gives

$$K_c^{\mathrm{c},k}\leq 10^{607}\cdot128\,e^{2q}\Big(\frac{n^3}{q}+\frac{48}{q^4}\Big),$$

which is at most $\alpha N_n=\alpha(n/2)^4$ when $n\geq10^{827}/\alpha$.

(iv) The proof of Corollary 106 uses the plaquette form only through Theorem 98, the translation property of (i) and the bound on $L_w$, which (i)–(iii) provide. □

The constants are poor because the strip of Theorem 104 has width $10^{-36}$; they are checked in exact arithmetic in the archive. What the theorem adds is that none of them depends on the scale: the constrained cube condition holds with one cube side at every scale, and the weights $L_w^{(k)}$ of the metrics $d_L$ are bounded uniformly. It does not compose the scales. The cube condition $K_c^{\rho}$ that Corollary 106 leaves at scale $k$ concerns the law of the averages given the fine exterior, in the

metric $d_L$, and it has to be turned into the cube condition of the law one scale up; that step and the cube condition for the coarsest averages are open, and in the quadratic approximation the coarsest averages carry a massless Gaussian, so a gap has to come from beyond it.

*Remark* 109. Theorem 98 is the weak-coupling half of the constrained route at leading order, and Theorem 45 cannot use it as it stands. Four steps remain, each an open problem.

(a) *Composition.* $D_{\mathrm{c}}$ holds the block averages fixed. A gap in physical units needs a theorem that combines the constrained conditions at all scales with a cube condition for the coarsest averages, in the place of Theorem 45. Corollary 106 is one step of it, Proposition 99(iv) gives the gap at every scale, and Theorem 102 the equivalence of $S_k$ and $Q_k$ uniformly in $k$, Theorem 104 the locality of $S_k$ uniformly in $k$, and Theorem 108 the constrained cube condition and the bounds on the weights $L_w$ at every scale with constants independent of $k$. Turning the cube condition for the averages into that of the law one scale up, and a cube condition for the coarsest averages, are open.

(b) *Beyond the quadratic approximation.* For the Wilson measure the block averages are products of group elements along paths, their level sets are curved, and the Bakry–Émery tensor of Sec. VI is negative at large fields; Theorem 94 controls only the small fields, and the large fields have to be split off as in Bałaban's analysis [8]. For the straight paths between block roots the group-valued estimate holds for all fields (Theorem 111), but that map does not compose across scales (Proposition 114), and for every compact connected group no continuous covariant map treats the transverse positions of a block alike (Proposition 115). The cut-off average of Theorem 117 is coercive for all fields and agrees with the averages to first order, with constants for one step; on torus-valued fields of small curvature it composes (Theorem 119), and for every compact group it composes on fields whose curvature is small at every scale with a small sum (Theorem 123); under the Wilson measure the remaining large fields form microscopic clusters (Corollary 148), but the conditional estimate needed near them is open. Away from them, on the fibres of the cut-off average, the Hessian of the action along the fibre is at least $\lambda(\frac{1}{26} - c_G\kappa)$ at every background with plaquettes at most $\kappa$ (Theorem 125), and on a convex chart of the fibre the constrained law satisfies a Brascamp–Lieb inequality for large $\lambda$ (Theorem 127), and on a product chart its covariances decay exponentially (Theorem 131), and the mass off that chart is small on most fibres of the SU(2) Wilson measure (Theorem 134) and on every fibre through a field with small plaquettes (Corollary 139), and on every fibre the law concentrates near the least action of its regular fields (Proposition 141); the remaining small-field steps are listed in Remark 145.

(c) *Second order.* Let $b_0 = 11N/(48\pi^2)$ for SU($N$), $t = 2b_0 \log 2$, and let $g_m^{-2} = g_*^{-2} + tm$ be the one-loop coupling $m$ steps above a matching scale. Comparison with integrals gives $\sum_{m\le M} g_m^2 = t^{-1}\log M + O(1)$ and $\sum_m g_m^4 \le \pi^2/(6t^2)$ [63]. A loss $1 + cg_m^2$ per step therefore compounds to a power of $M$, and a step from one scale to the next has to be exact at order $g^2$, with that term absorbed into the running coupling and the normalisation of the field, and remainders of order $g^4$.

(d) *Crossover and existence.* The passage from weak coupling to the strong-coupling range of Sec. IV–VIII and the existence of the limit, (E), are untouched.

The renormalisation group needed for (a) and (b) is the ultraviolet half of (E), so the constrained route places both residues on one construction.

### B. Path blocks for the group-valued links

Remark 109(b) asks for estimates on the actual group-valued links. They can be proved for a coarser block map, the straight path between block roots, and this map exhibits the second half of the difficulty: it does not compose across scales. A topological argument shows why no continuous map that treats the transverse positions alike can take its place.

Let $G \subset \mathrm{U}(r)$ be a compact matrix group, $\delta(A,B) = \|A - B\|_{\mathrm{HS}}$, which is invariant under left and right multiplication, and work on the periodic lattice of side $2L$ with the block tree of Definition 92, all fifteen tree links of each block equal to $I$. Write $\mathscr{B}$ for the number of blocks. The *path map* is

$$V^{\mathrm{p}}_{y,\mu}(U) = U_{2y+e_\mu,\mu},$$

the product along the straight path from $2y$ to $2y+2e_\mu$, whose first link is a tree link; it transforms as a link under the gauge transformations at the block roots. Fixing $V^{\mathrm{p}} = v$ fixes four of the 49 active links of each block, so the fibre is $G^{45\mathscr{B}}$, and the Wilson law in the tree gauge restricts to it with product Haar measure as reference, since the tree gauge has unit Jacobian. The *reconstruction* $R(v)$ puts $R_{2y+o,\mu} = v_{y,\mu}$ if $o_\mu = 1$ and $R_{2y+o,\mu} = I$ otherwise. For $U$ in the fibre of $v$ put

$$D_v(U)^2 = \sum_e \delta(U_e, R_e(v))^2, \qquad Q(U) = \sum_p \delta(U_p, I)^2,$$

the first sum having zero terms on the tree and path links, and let $Q_1(v) = \sum_P \delta(v_P, I)^2$ on the coarse lattice.

**Lemma 110.** *For vectors $z_a$ of a real Hilbert space indexed by $a \in \{0,1\}^3$, with $z_0 = 0$,*

$$\sum_a \|z_a\|^2 \le \frac{3+\sqrt{7}}{2} \sum_{\{a,a'\}} \|z_a - z_{a'}\|^2 \le 3 \sum_{\{a,a'\}} \|z_a - z_{a'}\|^2,$$

*the sums on the right over the twelve edges of the cube.*

*Proof.* It suffices to treat real numbers. The right side is $\langle z, L_0 z\rangle$, where $L_0$ is the Laplacian of the cube with the vertex 0 deleted, and the claim is that its least eigenvalue is $3 - \sqrt{7}$. The permutations of the coordinates commute with $L_0$. On the functions $f_k$ of the level $k = |a|$, $L_0$ acts by $f_1 \mapsto 3f_1 - 2f_2$, $f_2 \mapsto 3f_2 - 2f_1 - f_3$, $f_3 \mapsto 3f_3 - 3f_2$, whose characteristic

polynomial in $t = 3-\lambda$ is $t^3-7t$; the eigenvalues are 3 and $3\pm\sqrt{7}$. On the functions of levels one and two whose sum over each level vanishes, indexed by the coordinate that is one at level one and zero at level two, $L_0$ acts by the matrix $\left(\begin{smallmatrix}3&1\\1&3\end{smallmatrix}\right)$, with eigenvalues 2 and 4; at level three these functions vanish. The two spaces have dimensions 3 and 4 and span $\mathbb{R}^7$. Finally $1/(3-\sqrt{7}) = (3+\sqrt{7})/2 < 3$.[20] □

**Theorem 111.** *For every $G \subset \mathrm{U}(r)$, every $L \geq 1$, every $v$ and every $U$ in the fibre of $v$,*

$$D_v(U)^2 \leq 224\, Q(U).$$

*Proof.* Let $I_0$ be the sum of $\delta(U_e, I)^2$ over the active links inside the blocks and $J_0$ the sum of $\delta(U_e, v_{y,\mu})^2$ over the links that leave $B(y)$ in the direction $\mu$, so $D_v^2 = I_0 + J_0$. For an active link inside a block, the chain of plaquettes in the proof of Theorem 94 applies to group elements: each plaquette of the chain has its two transverse links on the tree, so the holonomy is the product of two consecutive links of the chain, and invariance of $\delta$ with the triangle inequality gives $\delta(U_e, I)^2 \leq k \sum_i \delta(U_{p_i}, I)^2$ with $k \leq 3$. The loads are those of that proof, so $I_0 \leq 8 Q_{\mathrm{in}}$. Fix $y, \mu$ and label the eight links leaving $B(y)$ by the transverse offset $a \in \{0,1\}^3$; the link $a = 0$ is the path link and equals $v_{y,\mu}$. For a transverse edge $\{a, a'\}$ the face plaquette has holonomy $U_a \psi' U_{a'}^{-1} \psi^{-1}$ with $\psi, \psi'$ links inside the two blocks, so $\delta(U_a, U_{a'}) \leq \delta(U_p, I) + \delta(\psi, I) + \delta(\psi', I)$. Lemma 110 with $z_a = U_a - v_{y,\mu}$ in the real Hilbert–Schmidt space and Cauchy–Schwarz bound the eight terms by 9 times the sum of the three squares over the twelve edges. Each face plaquette occurs once, and each link inside a block occurs once for each of the three directions other than its own, as in the proof of Theorem 94. Hence $J_0 \leq 9 Q_{\mathrm{face}} + 27 I_0$ and $D_v^2 \leq 28 I_0 + 9 Q_{\mathrm{face}} \leq 224\, Q$. □

The bound holds for all fields, with no small-field condition and no logarithmic chart; Theorem 94 is its linearisation for the averages, with the better constant 52.

**Proposition 112.** *$Q_1(V^{\mathrm{p}}(U)) \leq 4Q(U)$, $Q(R(v)) = 4Q_1(v)$ and $Q(U) \leq 8Q_1(v) + 48 D_v(U)^2$ for $U$ in the fibre of $v$; if $Q_1(v) = 0$, then $Q(U) \leq 24 D_v(U)^2$.*

*Proof.* The holonomy of a coarse plaquette is that of the square of side two at $2y$, a product of four conjugated fine plaquette holonomies, so $\delta(v_P, I) \leq \sum_{p \subset P} \delta(U_p, I)$; these squares tile each coordinate plane, which gives the first bound. In $R(v)$ only the plaquettes with offset one in both directions carry holonomy, equal to that of the coarse plaquette, four for each $P$. For any plaquette, telescoping gives $\delta(U_p, R_p) \leq \sum_{e \in p} \delta(U_e, R_e)$, so $\delta(U_p, I)^2 \leq 2\delta(R_p, I)^2 + 8 \sum_{e \in p} \delta(U_e, R_e)^2$, and each link lies in six plaquettes. If $Q_1(v) = 0$, every $R_p$ is $I$ and the factor 2 is not needed. □

[20] The constant $(3+\sqrt{7})/2$ is attained by a function of the level $|a|$ alone, so it is the best constant in the lemma.

Let $m = 45\mathscr{B}$, let the action be $\lambda Q$ with $\lambda \geq 0$ (for $\mathrm{SU}(N)$, $\lambda = \beta/(2N)$ up to a constant), and put

$$z_G(t) = \int_G e^{-t\delta(g,I)^2} dg, \qquad Z_\lambda(v) = \int_{G^m} e^{-\lambda Q(U)} dU,$$

with normalised Haar measures, the integral over the fibre of $v$.

**Theorem 113.** *For every $v$,*

$$e^{-8\lambda Q_1(v)} z_G(48\lambda)^m \leq Z_\lambda(v) \leq e^{-\lambda Q_1(v)/8} z_G(\lambda/448)^m,$$

*and if $Q_1(v) = 0$, then $z_G(24\lambda)^m \leq Z_\lambda(v) \leq z_G(\lambda/224)^m$. Let $A_\lambda = z_G(\lambda/448)/z_G(24\lambda)$. Under the conditional law at such a $v$,*

$$\mathbb{E}\, e^{\lambda D_v^2/448} \leq A_\lambda^m, \qquad \mathbb{E}\, \delta(U_e, I)^2 \leq \frac{20160}{\lambda} \log A_\lambda$$

*for each link $e$ when $v \equiv I$, and $A_\lambda \to 10752^{q/2}$ as $\lambda \to \infty$ if $G$ has dimension $q \geq 1$.*

*Proof.* By Theorem 111 and Proposition 112, $Q \geq \frac{1}{2}(D_v^2/224 + Q_1/4)$ and $Q \leq 8Q_1 + 48 D_v^2$. Each term of $D_v^2$ involves one free link, and right multiplication by $R_e(v)^{-1}$ preserves Haar measure, so $\int e^{-tD_v^2} dU = z_G(t)^m$; this gives the two bounds, and the flat case is the same with $Q \leq 24 D_v^2$. Next, $-\lambda Q + \lambda D_v^2/448 \leq -\lambda D_v^2/448$, and integration with the lower bound gives the exponential moment; Jensen's inequality gives $\mathbb{E} D_v^2 \leq (448m/\lambda)\log A_\lambda$. When $v \equiv I$ the law is invariant under translation by blocks, so the $\mathscr{B}$ links of each of the 45 classes have equal expectations, and $448 \cdot 45 = 20160$. Finally, in exponential coordinates $\delta(e^X, I)^2 = \|X\|^2 + O(\|X\|^4)$ and the Haar density is smooth and positive at 0, while $\delta(\cdot, I)$ is bounded below away from $I$; so $z_G(t) \sim c_G t^{-q/2}$, and the ratio at $\lambda/448$ and $24\lambda$ tends to $(24 \cdot 448)^{q/2}$. □

So the fluctuations of the free links around a flat path field are of order $\lambda^{-1/2}$ uniformly in the volume, for the group variables themselves. The path map cannot be iterated, however, and this is where it differs from the averages. Write $V^{\mathrm{p},k}$ for the $k$-fold path map, the straight path of length $s = 2^k$, and $Q_k$ for the plaquette energy of the lattice of spacing $s$.

**Proposition 114.** *On the periodic lattice of side $3s$, $s = 2^k$, there are $\mathrm{SU}(2)$ fields in the tree gauge with*

$$\frac{Q(U)}{Q_k(V^{\mathrm{p},k}(U))} = s \sin^2 \frac{\pi}{2s} \leq \frac{\pi^2}{4s}.$$

*For the averages, by contrast, $Q_k(V^{(k)}(\varphi)) \leq Q(\varphi)$ for every $k$ in the quadratic approximation (Proposition 99(i)).*

*Proof.* Let $T = \mathrm{diag}(i\pi/2, -i\pi/2)$. Put the $s/2$ links in the direction 0 at $(j,0,0,0)$, $j$ odd, $j < s$, equal to $e^{2T/s}$ and all other links equal to $I$. These links have offset one in the direction 0, so the tree gauge holds, and the links at even $j$ are tree links. The path from 0 to $se_0$ carries $e^T$ and every other path carries $I$, so one coarse link is $e^T$, and it lies in six coarse plaquettes, each with $\delta^2 = \|e^T - I\|_{\mathrm{HS}}^2 = 4$: $Q_k = 24$. No fine plaquette contains two of the changed links, since the

two links of a plaquette in the direction 0 differ in a transverse coordinate, so there are $3s$ nonzero fine plaquettes, each with $\delta^2 = 8\sin^2(\pi/2s)$, and $Q = 24s\sin^2(\pi/2s)$. Gauge covariance of the path map makes the ratio gauge invariant. □

A field concentrated on one thin path carries a coarse curvature of order one at a fine cost of order $1/s$. In the averages it is one transverse position among $s^3$ and enters with the weight $s^{-3}$, which is why the averages compose and the paths do not. The two properties needed for Remark 109(a) and (b) are therefore held by two different maps: the paths give a global estimate for the group-valued links on fibres that do not compose, and the averages compose on linear fibres that exist only in the quadratic approximation.

A block map with both properties cannot treat the transverse positions alike and be continuous, for any compact connected gauge group. Let $s = 2^k$ and let $V$ be a map of the links with both ends in $B(y)\cup B(y+e_\mu)$, blocks of side $s$, to $G$. Call it *covariant* if $V(U^h) = h_1 V(U) h_2^{-1}$ for every gauge transformation $h$ equal to $h_1$ on $B(y)$ and to $h_2$ on $B(y+e_\mu)$, and *normalised* if $V(I) = I$. The path map and its iterates are covariant, normalised and continuous. Let $T$ be the set of the $s^3$ links in the direction $\mu$ from $B(y)$ to $B(y+e_\mu)$, and for $u \in G^T$ let $U(u)$ carry $u_t$ on the link $t \in T$ and $I$ on every other link (Fig. 15). Let $S \subset G$ be a maximal torus, of dimension $\ell$; a basis of the kernel of its exponential map identifies $S$ with $(\mathbb{R}/2\pi\mathbb{Z})^\ell$ and its first homology with $\mathbb{Z}^\ell$.

**Proposition 115.** *Let $G \subset \mathrm{U}(r)$ be a compact connected Lie group of positive dimension, let $V$ be covariant and normalised, and let $F(u) = V(U(u))$ be continuous on $G^T$. Then $F$ maps $S^T$ into $S$, and the integer matrices $W_t$ by which the maps $u_t \mapsto F(u)$, $u_t \in S$, $u_{t'} = I$ for $t' \neq t$, act on $H_1(S) = \mathbb{Z}^\ell$ satisfy*

$$\sum_{t\in T} W_t = I_\ell .$$

*Hence some $W_t \neq 0$, so that the map $u_t \mapsto V(U(u))$ on fields equal to $I$ off the link $t$ is not homotopic to a constant on $S$, and, if $|T| \geq 2$, no transitive group of permutations of $T$ leaves $F$ invariant. For $G = \mathrm{SU}(2)$, $\ell = 1$, the $W_t$ are integers adding up to one, and some field equal to $I$ off one link of $T$ has $V = -I$.*

*Proof.* The gauge transformation equal to $g$ on $B(y)\cup B(y+e_\mu)$ maps $U(u)$ to $U(gug^{-1})$, since the links equal to $I$ stay $I$, and the one equal to $g$ on $B(y)$ and to $I$ on $B(y+e_\mu)$ maps it to $U(gu)$. So $F(gug^{-1}) = gF(u)g^{-1}$ and $F(gu) = gF(u)$, and with $F(I) = I$ the second gives $F(g,\dots,g) = g$. If every $u_t$ lies in $S$, every $g \in S$ fixes $u$ under conjugation, so $F(u)$ commutes with $S$; in a compact connected Lie group a maximal torus is its own centraliser [50], so $F(u) \in S$. Therefore $F$ restricts to a continuous map $f : S^T \to S$. The first homology of $S^T$ is $\bigoplus_{t\in T}\mathbb{Z}^\ell$, the $t$-th summand coming from the inclusion of the $t$-th factor with the identity elsewhere, so $f$ acts on it by $(x_t)_t \mapsto \sum_t W_t x_t$. The diagonal map $S \to S^T$ acts by $x \mapsto (x,\dots,x)$, and $f$ composed with it is the identity of $S$ because $F(g,\dots,g) = g$. Hence $\sum_t W_t = I_\ell$. If all $W_t$ vanished the sum would vanish, and a map of tori that is homotopic to a constant acts by zero on homology. A transitive permutation group preserving $F$ would make all $W_t$ equal to one matrix $W$, and $|T|W = I_\ell$ has no integer solution for $|T| \geq 2$. For $\mathrm{SU}(2)$, $S$ is a circle through $-I$, and a loop of nonzero degree in a circle is onto. □

For the path map, $W_t$ is the identity at the position of the path and zero elsewhere. Averages of the path holonomies weigh the positions alike, so by the proposition they have no continuous covariant extension to all of $G^T$, for $\mathrm{SU}(N)$ and for every compact simple group. Bałaban's averages are defined only on fields whose plaquettes are small, and the large fields are split off [6, 8]. A block map with both properties for all fields is therefore discontinuous or, like the path map, not invariant under the transverse symmetries of the block. Theorem 117 below gives a continuous one whose departure from the symmetry is confined to large fields.[21]

The proposition leaves room for a map that treats the positions alike to first order and breaks the symmetry only where the field is large. For a coarse link $(y,\mu)$ and $x \in B(y)$ put

$$P_x = U_{x,\mu} U_{x+e_\mu,\mu},$$

the product along the straight path of length two from $x$ to $x+2e_\mu$. In the tree gauge it is the parallel transport from $2y$ to $2y+2e_\mu$ through $x$ and $x+2e_\mu$, it transforms as a link under the gauge transformations at the block roots, and $P_{2y} = V^{\mathrm{p}}_{y,\mu}$, since the first link of that path is a tree link. Let $\mathfrak{g}$ be the Lie algebra of $G$, $\|X\|$ the Hilbert–Schmidt norm and log the principal logarithm of unitary matrices without the eigenvalue $-1$.

**Lemma 116.** *If $g$ is unitary and $\delta(g,I) \leq 1$, then $\|\log g\| \leq \frac{\pi}{3}\,\delta(g,I)$. If $X$ is skew-Hermitian, then $\delta(e^X,I) \leq \|X\|$.*

*Proof.* Let $e^{i\theta_j}$, $|\theta_j| \leq \pi$, be the eigenvalues of $g$. Then $\delta(g,I)^2 = \sum_j 4\sin^2(\theta_j/2)$ and $\|\log g\|^2 = \sum_j \theta_j^2$. Each term of the first sum is at most 1, so $|\theta_j| \leq \pi/3$, and $\theta/(2\sin(\theta/2))$ increases on $(0,\pi)$ and equals $\pi/3$ at $\theta = \pi/3$. The second statement is $|e^{i\theta} - 1| \leq |\theta|$ for the eigenvalues of $X$. □

Let $\rho_G \in (0,1]$ be such that $\log g \in \mathfrak{g}$ for every $g \in G$ with $\delta(g,I) \leq \rho_G$.[22] Let $\phi(t) = \min\{1, \max\{0, 2-2t\}\}$, which equals 1 on $[0,\frac12]$ and 0 on $[1,\infty)$, put

$$\chi_{y,\mu}(U) = \phi\Big(\rho_G^{-1} \max_{x\in B(y)} \delta(P_x, P_{2y})\Big),$$

and define the *cut-off average*

$$V^\sharp_{y,\mu}(U) = P_{2y} \exp\Big(\frac{\chi_{y,\mu}(U)}{16} \sum_{x\in B(y)} \log\big(P_{2y}^{-1} P_x\big)\Big), \qquad (20)$$

---

[21] That spheres and tori carry no continuous symmetric mean goes back to Aumann [2] and Eckmann [38]; the proof above is their homology argument in gauge-covariant form, with the links of the slice $T$ as the arguments of the mean and a maximal torus in place of the circle.

[22] The exponential map of $G$ is a diffeomorphism from a ball of $\mathfrak{g}$ of operator-norm radius less than $\pi$ onto a neighbourhood $N$ of $I$ in $G$, on which it inverts the principal logarithm, and $N$ contains $\{g \in G : \delta(g,I) \leq \rho\}$ for small $\rho > 0$. For $\mathrm{U}(r)$ one may take $\rho_G = 1$. For $\mathrm{SU}(N)$ one may take $\rho_G = \min\{1, 5/\sqrt{N}\}$: by Lemma 116 the trace of $\log g$ has modulus at most $\sqrt{N}\,\pi\rho_G/3 < 2\pi$, and it lies in $2\pi i\mathbb{Z}$. Some restriction is needed: on the circle $\{\mathrm{diag}(z,z^7)\} \subset \mathrm{U}(2)$ the point $z = e^{2\pi i/7}$ has $\delta(g,I) < 1$ but $\log g \notin \mathfrak{g}$.

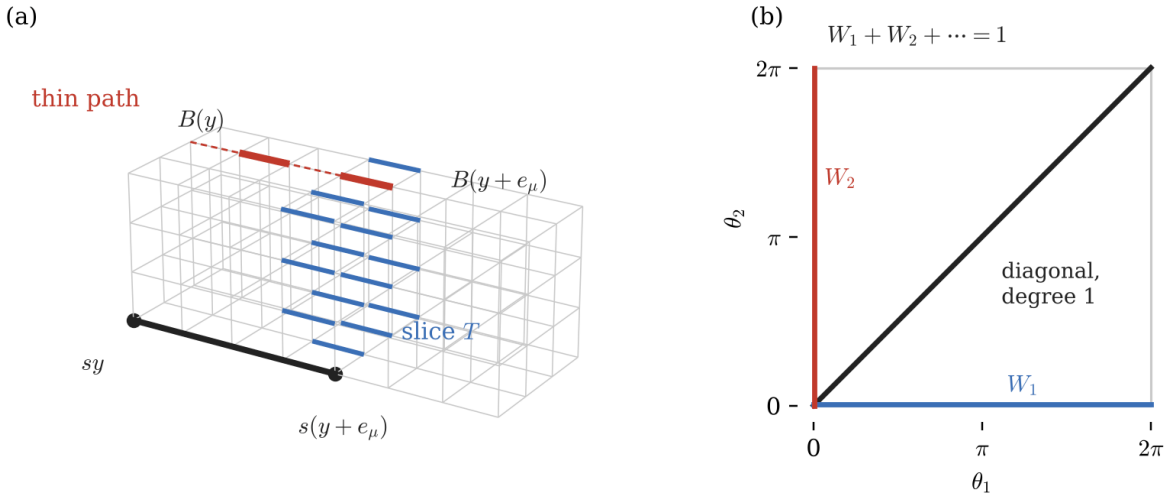



FIG. 15. (a) Two adjacent blocks $B(y)$, $B(y+e_\mu)$ of side $s=4$, drawn in three of the four dimensions. The black segment is the straight path between the roots, the coarse link of the path map. The blue links form the slice $T$ of Proposition 115. The red links carry the field of Proposition 114: a path whose holonomy is that of a coarse link of order one, at a fine energy of order $1/s$. (b) For SU(2), two coordinates of the torus $(\mathbb{R}/2\pi\mathbb{Z})^T$ of slice fields with values in a maximal torus. The diagonal is pure gauge and is sent to a loop of degree one, so the degrees $W_t$ of the coordinate loops add up to one.

the exponential factor being $I$ when $\chi_{y,\mu}(U)=0$ (Fig. 16). When every path is close to the straight one it is the average of the sixteen paths taken in the chart at $P_{2y}$; when some path is far from it, it is the path map.

**Theorem 117.** *For every compact $G\subset \mathrm{U}(r)$ the cut-off average is continuous, covariant and normalised, and the following hold, with $c=\pi/3$.*

(i) *If $\delta(P_x,P_{2y})\le \rho_G/2$ for every $x\in B(y)$, then $V^\sharp_{y,\mu}(U)=P_{2y}\exp\big(\frac{1}{16}\sum_x \log(P_{2y}^{-1}P_x)\big)$. For $\varphi$ with values in $\mathfrak{g}$, vanishing on the tree links,*

$$V^\sharp_{y,\mu}(e^{t\varphi})=I+t\,V_{y,\mu}(\varphi)+O(t^2),$$

*with $V_{y,\mu}$ the block average of Definition 92.*

(ii) *For every field $U$ in the tree gauge,*

$$\sum_{y,\mu}\sum_{x\in B(y)}\delta(P_x,P_{2y})^2\le 896\,Q(U),$$

$$\sum_{y,\mu}\delta\big(V^\sharp_{y,\mu}(U),V^{\mathrm{p}}_{y,\mu}(U)\big)^2\le 56c^2\,Q(U),$$

$$\sum_e\delta\big(U_e,R_e(V^\sharp(U))\big)^2\le(448+896c^2)\,Q(U),$$

$$Q_1\big(V^\sharp(U)\big)\le(8+2688c^2)\,Q(U),$$

*and $448+896c^2<1431$, $8+2688c^2<2956$.*

(iii) *In the setting of Proposition 115 with $s=2$, $W_t$ is the identity at the position of the straight path and 0 at the seven others.*

(iv) *If $U$ equals $I$ except on one link from $B(y)$ to $B(y+e_\mu)$, where it equals $g$ with $\delta(g,I)\le\rho_G/2$, then $V^\sharp_{y,\mu}(U)=\exp(\frac{1}{8}\log g)$, at each of the eight transverse positions of the link; the path map gives $g$ at one position and $I$ at the seven others.*

*Proof.* Write $P=P_{2y}$, $v_x=P^{-1}P_x$ and $m=\max_x\delta(v_x,I)=\max_x\delta(P_x,P)$. Where $m<\rho_G$, every $\log v_x$ lies in $\mathfrak{g}$, depends continuously on $U$ by Lemma 116, which keeps the eigenvalues of $v_x$ away from $-1$, and the exponent of (20) lies in $\mathfrak{g}$, so $V^\sharp\in G$. Where $m\ge\rho_G$, $V^\sharp=P$. As $m$ increases to $\rho_G$ the exponent has norm at most $\chi c\rho_G\to 0$, so $V^\sharp$ is continuous. Let $h$ be a gauge transformation equal to $h_1$ on $B(y)$ and to $h_2$ on $B(y+e_\mu)$. The middle point $x+e_\mu$ of each path lies in one of the two blocks, so $P_x\mapsto h_1P_xh_2^{-1}$ and $v_x\mapsto h_2v_xh_2^{-1}$; $\delta$ is invariant and log is equivariant under unitary conjugation, so $\chi$ is unchanged and $V^\sharp\mapsto h_1V^\sharp h_2^{-1}$. At $U=I$ every $v_x$ is $I$, so $V^\sharp(I)=I$.

(i) The first statement is $\chi=1$. If $U=e^{t\varphi}$ with $\varphi=0$ on the tree, then $P=I+t\varphi_{2y+e_\mu,\mu}+O(t^2)$ and $\log v_x=t(\varphi_{x,\mu}+\varphi_{x+e_\mu,\mu}-\varphi_{2y+e_\mu,\mu})+O(t^2)$, and for small $t$ the condition of the first statement holds. The mean over the sixteen $x$ is $t(V_{y,\mu}(\varphi)-\varphi_{2y+e_\mu,\mu})+O(t^2)$, since $\varphi_{2y,\mu}=0$, and multiplying by $P$ gives the claim.

(ii) In the tree gauge $P=v_{y,\mu}:=V^{\mathrm{p}}_{y,\mu}(U)$. If $o_\mu=0$, then $U_{x+e_\mu,\mu}$ is the link leaving $B(y)$ at the transverse offset of $x$, and invariance of $\delta$ with the triangle inequality gives $\delta(P_x,P)\le\delta(U_{x,\mu},I)+\delta(U_{x+e_\mu,\mu},v_{y,\mu})$; if $o_\mu=1$, the roles of the two links are exchanged. Each link leaving a block in the direction $\mu$ occurs for two sites $x$ of that block, and each link in the direction $\mu$ with $o_\mu=0$ occurs once for the block containing it and once for the block before it. With $(a+b)^2\le 2a^2+2b^2$ and the notation of the proof of Theorem 111, the sum is at most $2(2I_0+2J_0)=4D_v^2\le 896Q$. Next, if $\chi>0$, Lemma 116

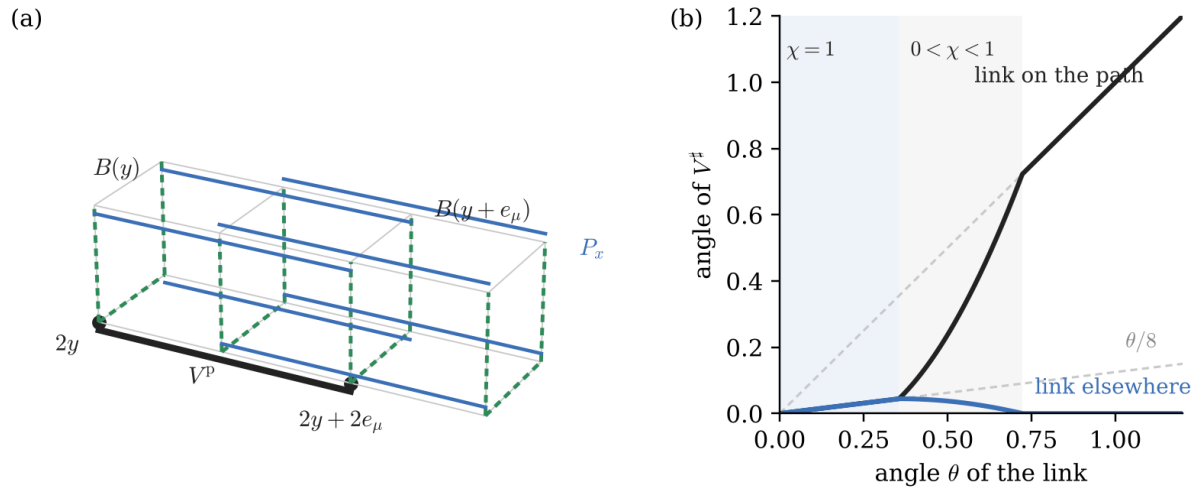



FIG. 16. (a) A block $B(y)$ of side two and the next block in the direction $\mu$, drawn in three of the four dimensions. Green dashed: the tree links, equal to $I$. Blue: the paths $P_x$ of length two from the sites of $B(y)$, each joining $2y$ to $2y+2e_\mu$ through the trees; black: the straight path $P_{2y}$, the coarse link of the path map. The cut-off average is the mean of the sixteen paths in the chart at $P_{2y}$ while every path is close to it. (b) For SU(2) and one link $e^{\theta X}$, $X=\mathrm{diag}(i,-i)$, the angle of $V^\sharp$ when the link lies on the straight path (black) or at another transverse position (blue), with $\rho_G=1$. Both equal $\theta/8$, the block average, while $\chi=1$; past the cut-off they are the path map, $\theta$ and 0, so the slice windings are 1 and 0.

and the Cauchy–Schwarz inequality give

$$\delta(V^\sharp,P)\le\Big\|\frac{\chi}{16}\sum_x\log v_x\Big\|\le\frac{c}{16}\sum_x\delta(v_x,I)$$
$$\le\frac{c}{4}\Big(\sum_x\delta(P_x,P)^2\Big)^{1/2},$$

and the second bound follows from the first, $896c^2/16=56c^2$. The reconstruction $R(V^\sharp)$ agrees with $R(v)$ on the links with $o_\mu=0$, and on the eight links with $o_\mu=1$ of each block and direction $\delta(U_e,V^\sharp)^2\le 2\delta(U_e,v)^2+2\delta(V^\sharp,v)^2$; so the third sum is at most $2D_v^2+16\cdot 56c^2Q\le(448+896c^2)Q$. Finally, write $V^\sharp=vM$. Removing the four factors $M$ from the holonomy of a coarse plaquette one at a time changes $\delta(\cdot,I)$ by at most $\delta(M,I)$ each time, so $\delta(V_P^\sharp,I)^2\le 2\delta(v_P,I)^2+8\sum_{\ell\in P}\delta(M_\ell,I)^2$. Each coarse link lies in six coarse plaquettes, so $Q_1(V^\sharp)\le 2Q_1(v)+48\cdot 56c^2Q$, and $Q_1(v)\le 4Q$ by Proposition 112. The numerical bounds use $c^2<1.0967$.

(iii) For $s=2$ the slice $T$ consists of the eight links leaving $B(y)$, and the links inside the blocks are $I$, so $P_x=u_a$ for the transverse offset $a$ of $x$, two sites for each $a$. Let $t$ be a position other than the straight path and $u_{t'}=I$ for $t'\neq t$. Then $P=I$, two of the $v_x$ equal $u_t$ and the others $I$, so $F(u)=\exp(\frac{1}{8}\chi\log u_t)$ with $\chi=\phi(\delta(u_t,I)/\rho_G)$. On $S$ the maps $u_t\mapsto\exp(\frac{\lambda}{8}\chi\log u_t)$, $0\le\lambda\le 1$, are continuous by the argument above and take values in $S$, since $\log u_t$ commutes with $S$ and the centraliser of a maximal torus in $\mathfrak{g}$ is its Lie algebra; they join $F$ to the constant $I$, so $W_t=0$. By Proposition 115 the $W_t$ add up to the identity, which is therefore the matrix at the straight path.

(iv) At the straight path, $P_x=g$ for the two sites with transverse offset 0 and $P_x=I$ for the others, so $V^\sharp=g\exp(-\frac{14}{16}\log g)=\exp(\frac{1}{8}\log g)$; at another position, $P=I$ and two of the $P_x$ equal $g$, so $V^\sharp=\exp(\frac{2}{16}\log g)$. □

The small-field part of (20) is Bałaban's averaging operation [7]; the cut-off extends it to all fields with the bounds of (ii). The theorem shows how far Proposition 115 reaches. The cut-off average is continuous, covariant and coercive for every field, and it agrees with the block averages to first order. Its slice windings are those of the path map, as the proposition demands of any such map, but the departure from the transverse symmetry happens only where some path differs from the straight one by more than $\rho_G/2$. Below that the field of a single link enters with the weight $\frac{1}{8}$ of the averages, whichever position it occupies.

What the theorem does not give is composition. The constants of (ii) are those of one step, and iterating them gives a factor that grows exponentially in the number of steps; on its large-field branch the map is the path map, for which Proposition 114 rules out any bound uniform in that number. Whether the $k$-fold cut-off average obeys $Q_k(V^{\sharp,k}(U))\le CQ(U)$ and the corresponding coercivity bound with $C$ independent of $k$ is open for general fields; it is the form in which Remark 109(b) remains. Section XII C proves it, with $C=1.001$, for fields with values in a maximal torus whose curvature stays small at every scale, and with $C=e^{12060\sigma}$ for fields of any compact group whose curvatures at the successive scales are small with sum $\sigma$.

### C. Composition across scales

This subsection proves that the cut-off average composes across all scales on fields with values in a maximal torus whose curvature stays small, which includes the thin paths that defeat the path map, and then that it composes for every compact group on fields whose curvature is small, with a summable loss over the scales.

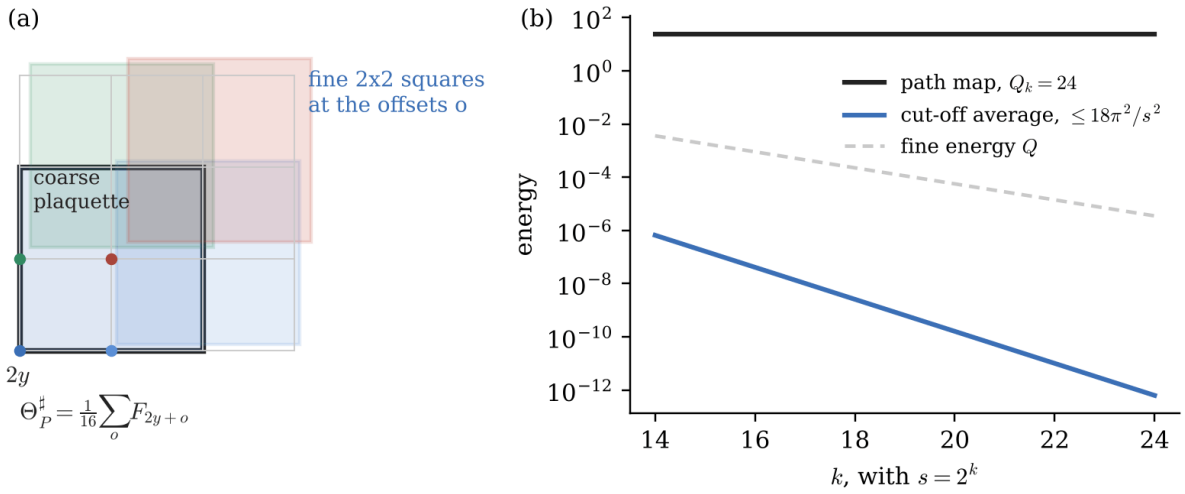



FIG. 17. (a) In the torus, the coarse plaquette of the cut-off average is the mean of the fluxes $F_{2y+o}$ of the sixteen fine squares of side two at the offsets $o$ (four of them drawn), whatever the gauge of the coarse links (Lemma 118). (b) For the thin-path fields of Proposition 114, the coarse energy after $k$ steps: 24 for the path map, at most $18\pi^2/s^2$ for the cut-off average (Corollary 120); the fine energy is about $6\pi^2/s$.

Let $S$ be a maximal torus of $G \subset \mathrm{U}(r)$ and $\mathfrak{s}$ its Lie algebra. The exponential maps $\mathfrak{s}$ onto $S$, and its kernel $\Lambda$ consists of diagonalisable matrices with eigenvalues in $2\pi i\mathbb{Z}$, so every nonzero element of $\Lambda$ has norm at least $2\pi$. For an $S$-valued field with $\delta(U_p, I) \le \rho_G$ on every plaquette put $\Theta_p = \log U_p$; it lies in $\mathfrak{g}$, commutes with $S$, and so lies in $\mathfrak{s}$. For a fine site $x$ and a coordinate plane $\{\mu,\nu\}$ let $\square_x$ be the square of side two at $x$ in that plane and $F_x = \sum_{p\subset\square_x} \Theta_p$, the sum of its four plaquettes, oriented as the square. Put

$$\eta_G = \rho_G/12.$$

The $k$-fold cut-off average is defined by putting the coarse field back into the tree gauge, by an $S$-valued gauge transformation, before each step. The tree gauge fixes such a transformation up to one that is constant on blocks, under which $V^\sharp$ is covariant, so the coarse plaquette holonomies $U_P^{(j)}$ of the $j$-th iterate $U^{(j)}$ are well defined; we write $\Theta_P^{(j)} = \log U_P^{(j)}$ when $\delta(U_P^{(j)}, I) \le \rho_G$.

**Lemma 118.** *Let $U$ be an $S$-valued field in the tree gauge with $\|\Theta_p\| \le \eta_G$ for every plaquette. Then $\chi_{y,\mu}(U) = 1$ for every coarse link, and the coarse plaquette $P$ at $y$ in the plane $\{\mu,\nu\}$ has*

$$V^\sharp(U)_P = \exp \bar F_P, \qquad \bar F_P = \frac{1}{16}\sum_{o\in\{0,1\}^4} F_{2y+o},$$

*with $\|\bar F_P\| \le 4\max_p\|\Theta_p\|$, so that $\log V^\sharp(U)_P = \bar F_P$.*

*Proof.* Let $\eta = \max_p\|\Theta_p\|$. Fix $(y,\mu)$ and $x = 2y+o$. The tree path $\tau$ from $2y$ to $x$ sets the coordinates of $o$ one at a time in increasing order, so it has at most four edges, at most three of them transverse to $\mu$. In the tree gauge its links are $I$, so $P_{2y}^{-1}P_x$ is the holonomy of the loop $\tau$, then the straight path from $x$ to $x+2e_\mu$, then $\tau+2e_\mu$ backwards, then the straight path from $2y+2e_\mu$ back to $2y$. This loop is the boundary of the surface swept by the transverse edges of $\tau$ along two steps in the direction $\mu$: at most six distinct plaquettes, each with coefficient $\pm1$. The group is abelian, so $P_{2y}^{-1}P_x = \exp d_x$ with $d_x$ the signed sum of their $\Theta_p$, and $\|d_x\| \le 6\eta \le \rho_G/2$. By Lemma 116, $\delta(P_x, P_{2y}) \le \rho_G/2$, so $\chi = 1$, and since the eigenvalues of $d_x$ have modulus at most $\frac12$, $\log(P_{2y}^{-1}P_x) = d_x$. Hence $V^\sharp_{y,\mu} = \exp(X_{y,\mu} + \bar d_{y,\mu})$, where $X_{y,\mu} \in \mathfrak{s}$ is any logarithm of $P_{2y}$ and $\bar d_{y,\mu}$ is the mean of the $d_x$.

Let the sides of $P$ be the coarse links $\ell$ with signs $\varepsilon_\ell$, and write $d_{o,\ell}$ for the $d_x$ of the side $\ell$ at the offset $o$. For each $o$ the four straight paths of length two at the offset $o$ of the four sides bound the square $\square_{2y+o}$, so

$$Y_o = \sum_\ell \varepsilon_\ell\big(X_\ell + d_{o,\ell}\big) - F_{2y+o} \in \Lambda.$$

Since $d_{0,\ell} = 0$, $\|Y_o - Y_0\| \le 4\cdot 6\eta + 8\eta = 32\eta \le \frac{8}{3} < 2\pi$, so $Y_o = Y_0$ for every $o$. Averaging over $o$ gives $\sum_\ell \varepsilon_\ell(X_\ell + \bar d_\ell) = \bar F_P + Y_0$, and $\exp Y_0 = I$. Finally $\|\bar F_P\| \le \max_o\|F_{2y+o}\| \le 4\eta < \pi/3$, so $\bar F_P$ is the principal logarithm. □

**Theorem 119.** *Let $U$ be an $S$-valued field in the tree gauge on the periodic lattice of side divisible by $2^k$, with $\delta(U_p, I) \le \rho_G$ on every plaquette, and put $\eta_0 = \max_p\|\Theta_p\|$, $A = \sum_p\|\Theta_p\|$. Suppose that*

$$\max_P \|\Theta_P^{(j)}\| \le \eta_G \quad \text{for } 0 \le j < k, \tag{21}$$

*which holds in particular if $\eta_0 A \le \eta_G^2$. Then the cut-off is inactive at every step, $\Theta^{(j+1)} = \bar F(\Theta^{(j)})$, and for $j < k$*

$$\sum_P \|\Theta_P^{(j+1)}\|^2 \le \sum_P \|\Theta_P^{(j)}\|^2, \qquad \sum_P \|\Theta_P^{(j+1)}\| \le \frac14 \sum_P \|\Theta_P^{(j)}\|,$$

$$\max_P \|\Theta_P^{(j+1)}\| \le 4 \max_P \|\Theta_P^{(j)}\|.$$

*Consequently*

$$Q_k\big(V^{\sharp,k}(U)\big) \le \sum_p \|\Theta_p\|^2 \le 1.001\, Q(U),$$
$$Q_k\big(V^{\sharp,k}(U)\big) \le 16^{-k} A^2.$$

*Proof.* The first assertions follow from Lemma 118 by induction, the $S$-valued gauge fixing between steps leaving the plaquette holonomies unchanged. The coarse plaquettes $P$ in a plane together with the offsets $o$ are in bijection with the fine squares $\square_x$ in that plane, $x = 2y+o$, and a fine plaquette lies in four such squares. With $\|\bar F_P\|^2 \le \frac{1}{16}\sum_o \|F_{2y+o}\|^2$ and $\|F_x\|^2 \le 4\sum_{p\subset\square_x}\|\Theta_p\|^2$ this gives $\sum_P \|\bar F_P\|^2 \le \frac{1}{16}\cdot 4\cdot 4\sum_p\|\Theta_p\|^2$; the same count without Cauchy–Schwarz gives the factor $\frac14$ in the second inequality, and the third is $\|F_x\| \le 4\eta$.

For the sufficient condition, let $\eta_j$ and $A_j$ be the maximum and the sum of $\|\Theta_P^{(j)}\|$. While (21) holds up to $j-1$, the inequalities give $\eta_j \le 4^j\eta_0$ and $\eta_j \le A_j \le 4^{-j}A$. The product of the two bounds is $\eta_0 A \le \eta_G^2$, so the smaller one is at most $\eta_G$, and (21) holds at $j$.

Finally $\delta(\exp\Theta, I) \le \|\Theta\|$ (Lemma 116), so $Q_k(V^{\sharp,k}(U)) \le \sum_P \|\Theta_P^{(k)}\|^2$, which is at most both $\sum_p\|\Theta_p\|^2$ and $\eta_k A_k \le A_k^2 \le 16^{-k}A^2$. The eigenvalues of each $\Theta_p$ are $i\theta$ with $|\theta| \le \eta_0 \le \eta_G \le \frac{1}{12}$, since $\eta_0^2 \le \eta_0 A$ in the sufficient condition and $\eta_0 \le \eta_G$ is the case $j=0$ of (21); and $\theta^2 \le 1.001\cdot 4\sin^2(\theta/2)$ there, which gives $\sum_p\|\Theta_p\|^2 \le 1.001\, Q(U)$. □

**Corollary 120.** *For $G = \mathrm{SU}(2)$ and the fields of Proposition 114 with $s = 2^k \ge 2^{14}$,*

$$Q_k\big(V^{\sharp,k}(U)\big) \le \frac{18\pi^2}{s^2} \le \frac{4}{s}\,Q(U),$$
$$Q_k\big(V^{\mathrm{p},k}(U)\big) = 24 \ge \frac{4s}{\pi^2}\,Q(U).$$

*Proof.* Here $\rho_G = 1$ and $S$ is the diagonal torus. The $3s$ nonzero plaquettes have $\Theta_p = \pm 2T/s$, of norm $\sqrt{2}\pi/s$, so $\eta_0 A = 6\pi^2/s \le \frac{1}{144}$ when $s \ge 864\pi^2$, which holds for $s \ge 2^{14}$, and $A^2 = 18\pi^2$. Proposition 114 gives $Q(U) = 24s\sin^2(\pi/2s)$, which lies between $(6\pi^2/s)(1-\pi^2/(12s^2))$ and $6\pi^2/s$. □

So on torus-valued fields of small curvature the cut-off average behaves like the linear averages: the curvature is averaged over the sixteen offsets at every step, the energy never increases, and a thin path is spread out and contracted instead of being carried to the coarse lattice (Fig. 17).

For general fields the paths do not commute, and the logarithms carry commutators from the second order on. The next two lemmas bound these commutators by the square of the local curvature. The same composition then holds for every compact group, with a loss of at most $1+O(\eta_j)$ at scale $j$.

**Lemma 121.** *Let $Z_1,\dots,Z_n \in \mathfrak{g}$ with $r = \sum_i\|Z_i\| \le \min\{\frac14, \rho_G\}$, and let $g = e^{Z_1}\cdots e^{Z_n}$. Then $\delta(g,I) \le r$, $\log g \in \mathfrak{g}$, and*

$$\Big\|\log g - \sum_i Z_i\Big\| \le 1.6\, r^2.$$

*Proof.* The distance $\delta$ is invariant under multiplication by unitary matrices on either side, so $\delta(g,I) \le \sum_i \delta(e^{Z_i},I) \le r$ by Lemma 116, and $\log g \in \mathfrak{g}$ by the choice of $\rho_G$. Expanding the exponentials, $g - I - \sum_i Z_i$ is a sum of products of at least two factors. Since the Hilbert–Schmidt norm is submultiplicative, its norm is at most $e^r - 1 - r \le \frac12 r^2 e^r \le 0.643\, r^2$, and $M = g - I$ has $\|M\| \le e^r - 1 \le 1.137\, r \le 0.285$. The operator norm of $M$ is smaller than 1. The principal logarithm is holomorphic on the disc $|z-1|<1$, which contains the spectrum of $g$, so it is given there by its power series, $\log g = \sum_{k\ge1}(-1)^{k+1}M^k/k$. Hence

$$\|\log g - M\| \le \frac{\|M\|^2}{2(1-\|M\|)} \le 0.905\, r^2,$$

and the two bounds add up to less than $1.6\, r^2$. □

Let $U$ be a $G$-valued field in the tree gauge with $\delta(U_p, I) \le \rho_G$, and put $\Theta_p = \log U_p$. A change of base point conjugates $\Theta_p$ and does not change its norm. Let $P$ be a coarse plaquette at $y$ in the plane $\{\mu,\nu\}$. Its sides are $\ell_1 = (y,\mu)$, $\ell_2 = (y+e_\mu,\nu)$, $\ell_3 = (y+e_\nu,\mu)$ and $\ell_4 = (y,\nu)$, with initial points $y_i$. For an offset $o \in \{0,1\}^4$ let $P_{i,o}$ be the path product of $\ell_i$ through $2y_i + o$, and put

$$K_{P,o} = P_{1,o}P_{2,o}P_{3,o}^{-1}P_{4,o}^{-1}, \qquad \bar F_P = \frac{1}{16}\sum_o \log K_{P,o}.$$

The tree segments between consecutive sides cancel. Hence $K_{P,o}$ is the holonomy of the square $\square_{2y+o}$, carried to $2y$ along the tree. In a torus, $\log K_{P,o} = F_{2y+o}$, and $\bar F_P$ is the mean of Lemma 118. Let $\sigma_{P,o}$ be the sum of $\|\Theta_p\|$ over the four plaquettes of $\square_{2y+o}$.

**Lemma 122.** *Let $U$ be a $G$-valued field in the tree gauge with $\delta(U_p,I) \le \rho_G$ and $\|\Theta_p\| \le \eta \le \rho_G/1000$ for every plaquette. Then $\chi_{y,\mu}(U) = 1$ for every coarse link. Every coarse plaquette has $\delta(V^\sharp(U)_P, I) \le 0.03\,\rho_G$ and*

$$\big\|\log V^\sharp(U)_P - \bar F_P\big\| \le 3.2\, r_P^2,$$
$$\|\bar F_P\| \le \frac{1+6.4\eta}{16}\sum_o \sigma_{P,o}, \tag{22}$$

*with $r_P \le 28.26\,\eta$ defined in the proof. Let $E$, $A$ be the sum of $\|\Theta_p\|^2$ and of $\|\Theta_p\|$, and let $E'$, $A'$, $\eta'$ be the sum of squares, the sum and the maximum of $\|\log V^\sharp(U)_P\|$. Then*

$$\sqrt{E'} \le (1+6030\eta)\sqrt{E}, \qquad A' \le (1+9147\eta)\frac{A}{4},$$
$$\eta' \le 4(1+647\eta)\,\eta.$$

*Proof. Paths.* Fix a coarse link $(y,\mu)$ and $x = 2y+o$. As in the proof of Lemma 118, $P_{2y}^{-1}P_x$ is the holonomy of a loop bounding a surface of at most six plaquettes, swept by the transverse edges of the tree path. Build the surface one plaquette at a time. This writes the holonomy as a product of at most six factors $hU_p^{\pm1}h^{-1} = \exp(\pm h\Theta_p h^{-1})$. Let $\sigma_{(y,\mu),o} \le 6\eta$ be the sum of their $\|\Theta_p\|$. By Lemma 121, $\delta(P_x, P_{2y}) \le 6\eta \le \rho_G/2$, so $\chi = 1$. The logarithm $D_{(y,\mu),o} = \log(P_{2y}^{-1}P_x)$ lies in $\mathfrak{g}$, with

$$\|D_{(y,\mu),o}\| \le (1+9.6\eta)\,\sigma_{(y,\mu),o}.$$

In the same way $K_{P,o}$ is a product of four such factors, so $\|\log K_{P,o}\| \le (1+6.4\eta)\sigma_{P,o}$. Averaging over $o$ gives the second inequality of (22).

*The plaquette.* Write $D_{i,o}$ for $D_{\ell_i,o}$. Then $P_{i,o} = P_{i,0}e^{D_{i,o}}$ and $D_{i,0} = 0$, and $V^\sharp_{\ell_i} = P_{i,0}e^{m_i}$ with $m_i = \frac{1}{16}\sum_o D_{i,o}$. The paths and $V^\sharp$ transform as links under gauge transformations constant on blocks. These transformations preserve the tree gauge and conjugate $D_{i,o}$, $m_i$, $K_{P,o}$ and $V^\sharp_P$ by fixed elements. Choose one with $P_{1,0} = P_{2,0} = P_{3,0} = I$. Then $P_{4,0} = K_{P,0}^{-1}$, and with $c = \log K_{P,0}$,

$$V^\sharp_P = e^{m_1}e^{m_2}e^{-m_3}e^{-m_4}e^{c},$$
$$K_{P,o} = e^{D_{1,o}}e^{D_{2,o}}e^{-D_{3,o}}e^{-D_{4,o}}e^{c}.$$

Put $r_P = \sum_i \max_o\|D_{i,o}\| + \|c\|$. Since $\|m_i\| \le \max_o\|D_{i,o}\|$, the five exponents in each product have norms adding up to at most $r_P$, and

$$r_P \le 24(1+9.6\eta)\eta + 4(1+6.4\eta)\eta \le 28.26\,\eta.$$

Lemma 121 gives $\delta(V^\sharp_P, I) \le r_P \le 0.03\,\rho_G$ and

$$\log V^\sharp_P = m_1 + m_2 - m_3 - m_4 + c + R,$$
$$\log K_{P,o} = D_{1,o} + D_{2,o} - D_{3,o} - D_{4,o} + c + R_o,$$

with $\|R\|, \|R_o\| \le 1.6\,r_P^2$. Average the second line over $o$ and subtract it from the first; since $m_i$ is the mean of the $D_{i,o}$, this is the first inequality of (22).

*Sums.* Each fine plaquette lies in four squares $\square_x$, and the pairs $(P,o)$ are in bijection with these squares. The second inequality of (22) and Cauchy–Schwarz therefore give $\sum_P\|\bar F_P\|^2 \le (1+6.4\eta)^2 E$; without Cauchy–Schwarz they give $\sum_P\|\bar F_P\| \le (1+6.4\eta)A/4$. For the remainders, let $\mathscr{U}_\ell$ be the set of plaquettes of the surfaces of the side $\ell = (y,\mu)$ for all sixteen offsets. These lie in planes $\{\mu,\alpha\}$, $\alpha \ne \mu$, at $\mu$-positions $2y_\mu$, $2y_\mu+1$, $2y_\mu+2$, and there are at most 36 of them. A plaquette lies in $\mathscr{U}_\ell$ for at most four coarse links. With $m'_\ell = \sum_{p\in\mathscr{U}_\ell}\|\Theta_p\|$,

$$r_P \le 1.0096\Big(\sum_{\ell\subset P} m'_\ell + \sigma_{P,0}\Big).$$

A coarse link is a side of six coarse plaquettes, and a plaquette lies in $\square_{2y}$ for at most one coarse plaquette. Therefore

$$\sum_P r_P \le 1.0096\,(6\cdot 4A + A) \le 25.24\,A,$$
$$\sum_P r_P^2 \le 1.0096^2\cdot 5\,(6\cdot 36\cdot 4E + 4E) \le 4424\,E.$$

Hence, with $r_P \le 28.26\,\eta$,

$$\sum_P 3.2\,r_P^2 \le 3.2\cdot 28.26\cdot 25.24\,\eta A \le 2283\,\eta A,$$
$$\Big(\sum_P \big(3.2\,r_P^2\big)^2\Big)^{1/2} \le 3.2\cdot 28.26\,\eta\,(4424\,E)^{1/2} \le 6015\,\eta\sqrt{E}.$$

The triangle inequality, for the Euclidean norm over $P$ in the first case, gives $\sqrt{E'} \le (1+6.4\eta+6015\eta)\sqrt{E}$ and $A' \le (1+6.4\eta)A/4 + 2283\,\eta A$. For the maximum, $\eta' \le (1+6.4\eta)4\eta + 3.2\,(28.26\,\eta)^2 \le 4(1+647\eta)\eta$. □

For a $G$-valued field, the $k$-fold cut-off average is defined as before, with $G$-valued gauge transformations. The tree gauge fixes them up to transformations constant on blocks, under which $V^\sharp$ is covariant. These transformations conjugate the plaquette holonomies and leave the norms $\|\Theta_P^{(j)}\|$ unchanged.

**Theorem 123.** *Let $G \subset \mathrm{U}(r)$ be compact and let $U$ be a $G$-valued field on the periodic lattice of side divisible by $2^k$, with $\delta(U_p, I) \le \rho_G$ on every plaquette. Let $\eta_j$, $A_j$, $E_j$ be the maximum, the sum and the sum of squares of $\|\Theta_P^{(j)}\|$ over the plaquettes of the $j$-th iterate.*

(i) *If $\eta_j \le \rho_G/1000$ for $0 \le j < k$, the cut-off is inactive at every step. With $\sigma = \sum_{j<k}\eta_j$,*

$$E_k \le e^{12060\sigma}E_0, \qquad A_k \le e^{9147\sigma}4^{-k}A_0,$$
$$\eta_k \le e^{647\sigma}4^k\eta_0.$$

(ii) *If $\eta_0 A_0 \le 2\cdot 10^{-10}\rho_G^2$, then (i) applies for every $k$ with $\sigma \le 7.25\sqrt{\eta_0 A_0}$, and*

$$Q_k\big(V^{\sharp,k}(U)\big) \le e^{8.8\cdot 10^4\sqrt{\eta_0 A_0}}E_0 \le 1.0001\,e^{8.8\cdot 10^4\sqrt{\eta_0 A_0}}\,Q(U),$$
$$Q_k\big(V^{\sharp,k}(U)\big) \le e^2\,16^{-k}A_0^2.$$

*Proof.* (i) Apply Lemma 122 at each step to the field in the tree gauge; the gauge fixing does not change the norms. The lemma gives $\delta(U_P^{(j+1)}, I) \le \rho_G$, so $\Theta^{(j+1)}$ is defined, and with $1+t \le e^t$ the three inequalities of the lemma multiply over the steps.

(ii) Suppose that for $i < j$ we have $\eta_i \le \rho_G/1000$ and $\sum_{i<j}\eta_i \le 7.25\sqrt{\eta_0 A_0} \le 1/9747$. Then (i) gives $A_j \le e\,4^{-j}A_0$ and $\eta_j \le e\,4^j\eta_0$, and $\eta_j \le A_j$. So $\eta_j \le e\min\{4^j\eta_0, 4^{-j}A_0\} \le e\sqrt{\eta_0 A_0} \le \rho_G/1000$. Let $i_*$ be the last index with $4^{i_*}\eta_0 \le 4^{-i_*}A_0$. The terms of $\min\{4^i\eta_0, 4^{-i}A_0\}$ increase by the factor 4 up to $i_*$ and decrease by the factor 4 after it. The terms at $i_*$ and $i_*+1$ are at most $\sqrt{\eta_0 A_0}$, so the sum over all $i$ is at most $\frac{8}{3}\sqrt{\eta_0 A_0}$, and $\sum_{i\le j}\eta_i \le \frac{8e}{3}\sqrt{\eta_0 A_0} \le 7.25\sqrt{\eta_0 A_0}$. This completes the induction. Then $E_k \le e^{12060\cdot 7.25\sqrt{\eta_0 A_0}}E_0$. By Lemma 116, $Q_k(V^{\sharp,k}(U)) \le E_k$, and $E_k \le \eta_k A_k \le A_k^2 \le e^2\,16^{-k}A_0^2$. Finally the eigenvalues $i\theta$ of each $\Theta_p$ have $|\theta| \le \eta_0 \le \sqrt{\eta_0 A_0} < 10^{-4}$, and $\theta^2 \le 1.0001\cdot 4\sin^2(\theta/2)$ there. □

For a torus Theorem 119 is sharper, and its hypothesis is weaker; the content of Theorem 123 is the non-abelian case. Neither theorem gives the supremum over boundary conditions in (D). Theorem 117 has the constants of one step. Theorems 119 and 123 need the curvature to be small at every scale: in Theorem 123 the losses $e^{6030\eta_j}$ must have a bounded product, and a field whose curvature is smooth on a scale $\ell$ grows by a factor four per step until the coarse spacing reaches $\ell$. The large fields, where the cut-off is active, and the Wilson measure itself are outside both; Sec. XII E shows where the large fields lie under that measure.

### D. Small fields on the fibres of the cut-off average

Theorem 94 is a gap on the linear fibres of the block averages. For group-valued links the fibres of the cut-off average are curved, and Remark 109(b) asks whether the gap survives on them. It does at every background whose plaquettes are small, however far its links are from $I$, with a loss linear in the curvature.

Let $G \subset \mathrm{U}(r)$ be compact, work on the periodic lattice of side $2L$ in the tree gauge of Definition 92, and let $W$ be a set of blocks. Write $A_W$ for the active links with base in $\bigcup W$ and $W^+$ for the blocks at $\ell^\infty$ distance at most one from $W$. Fix $U$ and let $F$ be the set of fields that agree with $U$ off $A_W$ and have the same cut-off averages $V^\sharp_{y,\mu}$ as $U$ for every $y \in W$ and every $\mu$. The group $G^{A_W}$ carries the bi-invariant metric with $|\xi|^2 = T(\xi) = \sum_e \|\xi_e\|^2$, $\xi_e \in \mathfrak{g}$, and its geodesics through $U$ are the curves $U_e \exp(t\xi_e)$; write $\mathrm{Ad}(g)X = gXg^{-1}$. For a plaquette $p$ at $x$ in the plane $\{\mu,\nu\}$ with links $e_1 = (x,\mu)$, $e_2 = (x+e_\mu,\nu)$, $e_3 = (x+e_\nu,\mu)$, $e_4 = (x,\nu)$ and $U_p = U_1U_2U_3^{-1}U_4^{-1}$, the *covariant curvature*

$$D_p(\xi) = \mathrm{Ad}(U_1)\xi_1 + \mathrm{Ad}(U_1U_2)(\xi_2 - \xi_3) - \mathrm{Ad}(U_1U_2U_3^{-1})\xi_4$$

is the derivative of $U_p(t)U_p^{-1}$ along these curves. Reversing the orientation replaces it by $-\mathrm{Ad}(U_p^{-1})D_p(\xi)$, of the same norm, so $Q_U(\xi) = \sum_p \|D_p(\xi)\|^2$ does not depend on the orientations. At $U \equiv I$, $D_p(\xi) = (d\xi)_p$ and $Q_I$ is the form $Q$ of Theorem 94. For $y \in W$ and a direction $\mu$ let $v = U_{2y+e_\mu,\mu}$ be the path link, let $\tilde\xi$ equal $\mathrm{Ad}(v^{-1})\xi_e$ on the links of $B(y)$ and $\xi_e$ on the other links, and let $N_{y,\mu}(\xi)^2$ be the sum of $\|\xi_e\|^2$ over the 24 links that enter $V_{y,\mu}$.

**Lemma 124.** *Suppose that $\delta(U_p,I) \le \kappa \le 10^{-8}\rho_G$ for every plaquette with a vertex in a block of $W^+$, and let $y \in W$.*

(i) *Every active link $e$ inside a block of $W^+$ has $\delta(U_e,I) \le 3\kappa$, and every link $e$ that leaves $B(y)$ in the direction $\mu$ has $\delta(U_e,v) \le 21\kappa$.*

(ii) *For a plaquette inside a block of $W^+$, $\|D_p(\xi) - (d\xi)_p\| \le 18\kappa \sum_{e\in p}\|\xi_e\|$.*

(iii) *Let $x \in B(y)$ with $x_\mu$ odd and $x_\nu$ even, and let $p$ be the plaquette at $x$ in the plane $\{\mu,\nu\}$, oriented as above, so that $e_1, e_3$ leave $B(y)$, $e_2$ lies in $B(y+e_\mu)$ and $e_4$ in $B(y)$. Then*

$$\big\|\mathrm{Ad}(v^{-1})D_p(\xi) - \big(\xi_1 + \xi_2 - \xi_3 - \mathrm{Ad}(v^{-1})\xi_4\big)\big\| \le 48\kappa \sum_{e\in p}\|\xi_e\|.$$

(iv) *Near $U$ the cut-off average is*

$$V^\sharp_{y,\mu} = v\exp\Big(\frac{1}{16}\sum_x \log(v^{-1}P_x)\Big),$$

*and its derivative satisfies*

$$\Big\|V^\sharp_{y,\mu}(U)^{-1}\frac{d}{dt}V^\sharp_{y,\mu}\big(Ue^{t\xi}\big)\Big|_{t=0} - V_{y,\mu}(\tilde\xi)\Big\| \le 260\,\kappa N_{y,\mu}(\xi).$$

*Proof.* We use $\|\mathrm{Ad}(g)X - X\| \le 2\delta(g,I)\|X\|$, which holds since $gXg^{-1} - X = ((g-I)X - X(g-I))g^{-1}$, and $\delta(gh,I) \le \delta(g,I) + \delta(h,I)$.

(i) The chains of plaquettes in the proof of Theorem 94 have their transverse links on the tree, so $\delta(U_e,I) \le \sum_{i\le k}\delta(U_{p_i},I) \le 3\kappa$. For the links leaving $B(y)$, the face plaquettes give $\delta(U_a,U_{a'}) \le \kappa + 3\kappa + 3\kappa$ for adjacent transverse offsets, as in the proof of Theorem 111, and every offset is joined to the straight path, which carries $v$, by at most three such steps.

(ii) The links of $p$ and the products $U_1U_2$, $U_1U_2U_3^{-1}$ are within $3\kappa$, $6\kappa$ and $9\kappa$ of $I$ by (i).

(iii) By (i), $\delta(v^{-1}U_1,I) \le 21\kappa$ and $\delta(v^{-1}U_1U_2,I) \le 24\kappa$, and $v^{-1}U_1U_2U_3^{-1} = v^{-1}U_pU_4$ with $\delta(U_pU_4,I) \le 4\kappa$.

(iv) Write $P_x = U_{x,\mu}U_{x+e_\mu,\mu}$, $w_x = v^{-1}P_x$, $X_x = \log w_x$ and $M = \frac{1}{16}\sum_x X_x$; the path $P_{2y}$ is $v$, its first link being on the tree. One of the two links of $P_x$ leaves $B(y)$ and the other lies in $B(y)$ or $B(y+e_\mu)$, so $\delta(P_x,v) \le 24\kappa < \rho_G/2$ by (i); hence $\chi_{y,\mu} = 1$ near $U$, and $\|X_x\|, \|M\| \le 8\pi\kappa$ by Lemma 116. Since $\|[X,Y]\| \le 2\|X\|\,\|Y\|$, the operators $\mathrm{ad}_{X_x}$ and $\mathrm{ad}_M$ have norm at most $a = 16\pi\kappa \le 50.3\kappa$. Along $Ue^{t\xi}$ put $\alpha = \xi_{2y+e_\mu,\mu}$ and $\beta_x = \mathrm{Ad}(U^{-1}_{x+e_\mu,\mu})\xi_{x,\mu} + \xi_{x+e_\mu,\mu}$, the right logarithmic derivatives of $v$ and $P_x$. Then $w_x^{-1}\dot w_x = \beta_x - \mathrm{Ad}(w_x^{-1})\alpha$, and the formula for the derivative of the exponential map gives $\dot X_x = \Phi(\mathrm{ad}_{X_x})(\beta_x - \mathrm{Ad}(w_x^{-1})\alpha)$ and

$$(V^\sharp)^{-1}\dot V^\sharp = \mathrm{Ad}(e^{-M})\alpha + \Psi(\mathrm{ad}_M)\dot M,$$

with $\Phi(z) = z/(1-e^{-z})$ and $\Psi(z) = (1-e^{-z})/z$. The operators $\mathrm{ad}_X$ are skew-adjoint, and from the power series, $|\Phi(z)-1| \le |z|/2 + 0.09|z|^2 \le |z|$ and $|\Psi(z)-1| \le |z|$ for $|z| \le 1$, the Bernoulli numbers giving $\sum_{k\ge1}|B_{2k}|/(2k)! < 0.09$. By Lemma 93(iii), $V_{y,\mu}(\tilde\xi) = \frac{1}{16}\sum_x(\tilde\xi_{x,\mu} + \tilde\xi_{x+e_\mu,\mu}) = \alpha + \frac{1}{16}\sum_x(\beta_x - \alpha) - \frac{1}{16}\sum_x\varepsilon_x$, where $\varepsilon_x = \mathrm{Ad}(U^{-1}_{x+e_\mu,\mu})\xi_{x,\mu} - \tilde\xi_{x,\mu}$. If $x_\mu$ is even, $U_{x+e_\mu,\mu}$ leaves $B(y)$ and $\tilde\xi_{x,\mu} = \mathrm{Ad}(v^{-1})\xi_{x,\mu}$; otherwise $U_{x+e_\mu,\mu}$ lies in $B(y+e_\mu)$ and $\tilde\xi_{x,\mu} = \xi_{x,\mu}$; so $\|\varepsilon_x\| \le 42\kappa\|\xi_{x,\mu}\|$ by (i). With $m_\beta$ and $m_\xi$ the means of $\|\beta_x\|$ and $\|\xi_{x,\mu}\|$, the four differences $\mathrm{Ad}(e^{-M})\alpha - \alpha$, $(\Psi(\mathrm{ad}_M)-1)\dot M$, $\dot M - \frac{1}{16}\sum_x(\beta_x - \alpha)$ and $\frac{1}{16}\sum_x\varepsilon_x$ have norms at most $51\kappa\|\alpha\|$, $51.6\kappa(m_\beta + \|\alpha\|)$, $51\kappa m_\beta + 99\kappa\|\alpha\|$ and $42\kappa m_\xi$, using $\|\mathrm{Ad}(w_x^{-1}) - 1\| \le 48\kappa$. The links leaving $B(y)$ occur twice among the $\xi_{x,\mu}$, $\xi_{x+e_\mu,\mu}$ and the other sixteen links once, so the Cauchy–Schwarz inequality gives $m_\beta \le \frac{\sqrt{48}}{16}N_{y,\mu} \le 0.434N_{y,\mu}$ and $m_\xi \le \frac14 N_{y,\mu}$, and the sum is at most $(202 + 103\cdot0.434 + 42/4)\kappa N_{y,\mu} \le 260\kappa N_{y,\mu}$. □

**Theorem 125.** *Under the hypothesis of Lemma 124:*

(i) *for every $\xi \in \mathfrak{g}^{A_W}$ tangent to $F$ at $U$, that is with $\frac{d}{dt}V^\sharp_{y,\mu}(Ue^{t\xi})|_{t=0} = 0$ for all $y \in W$ and all $\mu$,*

$$T(\xi) \le 52\big(1 + 1.4\cdot10^6\kappa\big)\,Q_U(\xi);$$

(ii) *for every $\xi \in \mathfrak{g}^{A_W}$,*

$$\frac{d^2}{dt^2}Q\big(Ue^{t\xi}\big)\Big|_{t=0} \ge 2Q_U(\xi) - 84\kappa T(\xi),$$

*so on the tangent space of $F$ the Hessian of $Q$ is at least $(\frac{1}{26} - 5.4\cdot 10^4\kappa)T(\xi)$; for $\mathrm{SU}(2)$ the Hessian of the Wilson action $\beta\sum_p(1-\frac{1}{2}\operatorname{Re}\operatorname{Tr}U_p) = \frac{\beta}{4}Q$ on the linearised constrained directions is at least $\frac{\beta}{2}(\frac{1}{52} - 2.7\cdot 10^4\kappa)T(\xi)$;*

(iii) *near $U$, $F$ is a smooth submanifold of $G^{A_W}$ of codimension $4|W|\dim G$, and the Hessian of $Q|_F$ for the induced metric satisfies $\operatorname{Hess}_F Q(\xi,\xi) \geq (\frac{1}{26} - c_G\kappa)T(\xi)$, where $c_G$ depends only on $G$.*

*Proof.* (i) We follow the proof of Theorem 94, using $(a+b)^2 \leq (1+\kappa)a^2 + (1+\kappa^{-1})b^2$ for each error term. *Inside a block.* For the chain $p_1,\dots,p_k$ of an active link $e=(x_0,\mu)$, $\xi_e = \sum_i \pm(d\xi)_{p_i}$ with tree links on the transverse sides, and Lemma 124(ii) gives

$$\|\xi_e\| \leq \sum_i \|D_{p_i}\| + 36\kappa \sum_{j<k} \|\xi_{x_j,\mu}\|.$$

Hence $\|\xi_e\|^2$ is at most $(1+\kappa)k\sum_i\|D_{p_i}\|^2$ plus $3888\kappa(1+\kappa)\sum_{j<k}\|\xi_{x_j,\mu}\|^2$. A link $(x',\mu)$ is a point $x_j$ of the chains of at most eight links, which differ from $x'$ by a set of directions above $\mu$, so

$$I \leq 8(1+\kappa)Q_{\mathrm{in}} + 31104\kappa(1+\kappa)I.$$

*Between blocks.* Fix $y\in W$ and $\mu$, and work in the frame of Lemma 124(iii),(iv); the norms do not depend on it. For a transverse edge $\{a,a'\}$, part (iii) gives $\|\xi_a - \xi_{a'}\| \leq \|D_p\| + \|\xi_\psi\| + \|\xi_{\psi'}\| + 48\kappa(\|\xi_a\| + \|\xi_{a'}\| + \|\xi_\psi\| + \|\xi_{\psi'}\|)$, so the square is at most $3(1+\kappa)(\|D_p\|^2 + \|\xi_\psi\|^2 + \|\xi_{\psi'}\|^2)$ plus $9216\kappa(1+\kappa)$ times the sum of the four squares. Since $\xi$ is tangent, part (iv) gives $\|V_{y,\mu}(\tilde\xi)\| \leq 260\kappa N_{y,\mu}$, and by Lemma 93(iii) the mean $m$ of the $\xi_a$ is $V_{y,\mu}(\tilde\xi)$ minus a sixteenth of the sum of the sixteen framed links of $S^\mu_y$ and $S^\mu_{y+e_\mu}$; so $8\|m\|^2 \leq (1+\kappa)\frac{1}{2}\sum_{16}\|\xi_e\|^2 + 540800\kappa(1+\kappa)N^2_{y,\mu}$. With the Laplacian of the cube as before,

$$\begin{aligned}\sum_a\|\xi_a\|^2 \leq{}& (1+\kappa)\Big(\frac{1}{2}\sum_{16}\|\xi_e\|^2 + \frac{3}{2}\sum_{\text{edges}}\Sigma_p\Big)\\ &+\kappa(1+\kappa)\Big(540800N^2_{y,\mu}\\ &+4608\big(3\sum_a\|\xi_a\|^2 + \sum_\Psi\|\xi_e\|^2\big)\Big),\end{aligned}$$

where $\Sigma_p = \|D_p\|^2 + \|\xi_\psi\|^2 + \|\xi_{\psi'}\|^2$ and $\Psi$ is the set of the 24 links $\psi,\psi'$. A link lies in at most two of the sets of $N_{y,\mu}$ and in at most three of the sets $\Psi$, so summing over $y\in W$ and $\mu$ as in the proof of Theorem 94, with $J = T - I$,

$$\begin{aligned}J \leq{}& (1+\kappa)\Big(\frac{11}{2}I + \frac{3}{2}Q_{\text{face}}\Big)\\ &+1.1\cdot 10^6\kappa(1+\kappa)T.\end{aligned}$$

Adding $I$ and inserting the bound for $I$ gives $T \leq 52(1+\kappa)^2Q_U(\xi) + 1.31\cdot 10^6\kappa(1+\kappa)^2T$, and the claim follows for $\kappa \leq 10^{-8}$.

(ii) Write $D_p(\xi) = B_1+B_2+B_3+B_4$ with

$$\begin{aligned}B_1 &= \operatorname{Ad}(U_1)\xi_1, \quad B_2 = \operatorname{Ad}(U_1U_2)\xi_2,\\ B_3 &= -\operatorname{Ad}(U_1U_2)\xi_3, \quad B_4 = -\operatorname{Ad}(U_1U_2U_3^{-1})\xi_4.\end{aligned}$$

Then $U_p(t) = e^{tB_1}e^{tB_2}e^{tB_3}e^{tB_4}U_p$, so $\ddot U_p = (D_p^2 + C_p)U_p$ with $D_p = \sum_i B_i$ and $C_p = \sum_{i<j}[B_i,B_j]$. Since $\delta(U_p,I)^2 = 2r - 2\operatorname{Re}\operatorname{Tr}U_p$, $\operatorname{Tr}C_p = 0$ and $-\operatorname{Tr}D_p^2 = \|D_p\|^2$,

$$\begin{aligned}\frac{d^2}{dt^2}\delta(U_p,I)^2 = 2\|D_p\|^2 &- 2\operatorname{Re}\operatorname{Tr}\big(D_p^2(U_p-I)\big)\\ &- 2\operatorname{Re}\operatorname{Tr}\big(C_p(U_p-I)\big),\end{aligned}$$

and $\|C_p\| \leq \sum_{i<j}2\|B_i\|\|B_j\| \leq 3\sum_{e\in p}\|\xi_e\|^2$. So the second derivative is at least $2(1-\kappa)\|D_p\|^2 - 6\kappa\sum_{e\in p}\|\xi_e\|^2$ when $\delta(U_p,I)\leq\kappa$. Every plaquette that contains a link of $A_W$ has a vertex in a block of $W^+$, each link lies in six plaquettes, and $Q_U(\xi) \leq 24T(\xi)$; this gives the bound, and with (i) $\frac{2}{52(1+1.4\cdot 10^6\kappa)} - 84\kappa \geq \frac{1}{26} - 5.4\cdot 10^4\kappa$.

(iii) Let $C_{y,\mu}(U') = \log(V^\sharp_{y,\mu}(U)^{-1}V^\sharp_{y,\mu}(U'))$ near $U$, so that $F$ is $C^{-1}(0)$ there, and let $\mathscr{G} = DC\,DC^*$ at $U$, a block matrix indexed by the pairs $k=(y,\mu)$. By Lemma 124(iv), $DC_k = \tilde B_k + R_k$, where $\tilde B_k\xi = V_{y,\mu}(\tilde\xi)$ and $\|R_k\| \leq 260\kappa$ as maps from the 24 links. Each $\operatorname{Ad}(v^{-1})$ is an isometry, so $\tilde B_k\tilde B_k^*$ is the sum of the squared coefficients of $V_{y,\mu}$ on $A_W$, at least $\frac{1}{8}$, $\|\tilde B_k\| \leq \sqrt{3/16} \leq 0.434$, and $\|\tilde B_k\tilde B_l^*\| \leq \frac{1}{32}$ for the two neighbours $l = (y\pm e_\mu,\mu)$, the other blocks being zero. So the diagonal blocks of $\mathscr{G}$ are at least $\frac{1}{8} - 227\kappa$, the two off-diagonal blocks of a row have norm at most $\frac{1}{32} + 227\kappa$, and $\mathscr{G} \geq \frac{1}{16} - 700\kappa$. Hence $DC$ is onto and $F$ is a submanifold near $U$; and the Neumann series gives $\|\mathscr{G}^{-1}\| \leq 17$ in the norm $\max_k\|z_k\|$. Let $\gamma$ be a geodesic of $F$ with $\gamma(0)=U$, $\dot\gamma(0)=\xi$, written as $Ue^{X(t)}$ link by link. The curves $Ue^{tX}$ are the geodesics of $G^{A_W}$, so $\eta = \ddot X(0)$ is the second fundamental form at $(\xi,\xi)$; it is normal to $F$, and differentiating $C(\gamma(t)) = 0$ twice gives $\eta = -DC^*\mathscr{G}^{-1}h$, where $h_k$ is the second derivative of $C_k(Ue^{t\xi})$ at $t=0$. Then

$$\operatorname{Hess}_F Q(\xi,\xi) = \frac{d^2}{dt^2}Q\big(Ue^{t\xi}\big)\Big|_{t=0} + dQ(\eta),$$

and $dQ(\eta) = -\langle\mathscr{G}^{-1}DC\nabla Q, h\rangle$. As in (ii), $dQ(\xi) = -2\sum_p\operatorname{Re}\operatorname{Tr}(D_p(U_p-I))$, so $\|\nabla_e Q\| \leq 12\kappa$, $\|(DC\nabla Q)_k\| \leq (0.434+260\kappa)\,12\kappa\sqrt{24} \leq 26\kappa$ and $\|(\mathscr{G}^{-1}DC\nabla Q)_k\| \leq 442\kappa$. The map $C_k(Ue^X)$ depends only on the 24 links of $V_{y,\mu}$ and is real-analytic in $(U,X)$ on a neighbourhood of the compact set where $\delta(P_x,P_{2y}) \leq \rho_G/4$ for every $x$, so $\|h_k\| \leq c_2N_{y,\mu}(\xi)^2$ with $c_2$ depending only on $G$. Summing, $|dQ(\eta)| \leq 884c_2\kappa T(\xi)$, and (ii) gives the claim with $c_G = 5.4\cdot 10^4 + 884c_2$. □

So on the fibres of the cut-off average the action is uniformly convex along every geodesic of the fibre on which all plaquettes stay below $\kappa$, with modulus $\lambda(\frac{1}{26} - c_G\kappa)$, and the background enters only through its curvature. The constants are crude: on the periodic lattice of side four, for $\mathrm{SU}(2)$ fields equal to block

gauge transforms of fields with $\kappa$ up to 0.08 and $W$ all sixteen blocks, the least value of $Q_U/T$ on the tangent space stays within 0.4% of the gap 0.1075 of the quadratic approximation, and with $\kappa$ up to 0.24 the frame errors of Lemma 124 stay below $1.3\kappa$, a floating-point computation reproduced in the archive.

*Remark* 126. Theorem 125 is the curvature input of the small-field half of Remark 109(b); three steps remain open.

(a) With respect to the Riemannian volume of $F$, the constrained law has the density $e^{-\lambda Q}$ times the coarea factor $\det(DC\,DC^{*})^{-1/2}$, and its Bakry–Émery tensor also contains the Ricci curvature of $F$. These terms do not depend on $\lambda$, but they have not been estimated here.

(b) The set of points of $F$ at which every plaquette is at most $\kappa$ is not known to be geodesically convex in $F$, so the theorem gives neither a Brascamp–Lieb inequality on it nor a Helffer–Sjöstrand representation of the constrained influences, and the decay through Lemma 97 is not reached.

(c) The mass of the constrained law outside that set is bounded by Theorem 113 only for the fibres of the path map.

The hypothesis $\kappa \le 10^{-8}\rho_G$ concerns every plaquette near $W$; under the SU(2) Wilson measure the plaquettes that are not small form the clusters of Sec. XII E, and near them the theorem does not apply.

The first two steps can be taken on a chart of $F$ whose domain is convex. Write $T_U \subset \mathfrak{g}^{A_W}$ for the tangent space of $F$ at $U$, $N_U$ for its orthogonal complement, $\|X\|_\infty = \max_e \|X_e\|$ and $K_s = \{\eta \in T_U : \|\eta\|_\infty \le s\}$, a convex set. The *constrained law* is the conditional law of $e^{-\lambda Q}\prod_{e\in A_W} dU_e$, the other links being those of $U$, given the averages $V^{\sharp}_{y,\mu}$, $y \in W$, in the version given by the coarea formula.

**Theorem 127.** *Assume the hypothesis of Lemma 124. There are constants $c_3 \ge 1$ and $c_6$ depending only on $G$ such that, if $0 < s \le 1/(6000c_3)$ and $\kappa + 6s \le 10^{-8}\rho_G$:*

(i) *there is a smooth map $h : K_s \to N_U$ with $\|h(\eta)\|_\infty \le s/2$ and $\Phi(\eta) = Ue^{\eta+h(\eta)} \in F$; $\Phi$ is an embedding, its image contains every $Ue^X \in F$ with $\|X\|_\infty \le s/40$, and every point of $\Phi(K_s)$ satisfies the hypothesis of Lemma 124 with $\kappa + 6s$ in place of $\kappa$;*

(ii) *for $\eta$ in the interior of $K_s$ and $\zeta \in T_U$,*

$$\frac{d^2}{dt^2} Q\big(\Phi(\eta + t\zeta)\big)\Big|_{t=0} \ge \Big(\frac{1}{26} - 4\cdot 10^5 c_3(\kappa + s)\Big) T(\zeta);$$

(iii) *on $\Phi(K_s)$ the constrained law has, in the coordinate $\eta$, a density proportional to $e^{-\lambda Q(\Phi(\eta)) - \psi(\eta)}$ with respect to Lebesgue measure on $T_U$, where $\psi$ is smooth, does not depend on $\lambda$, and $|\frac{d^2}{dt^2}\psi(\eta + t\zeta)| \le c_6 T(\zeta)$ on the interior of $K_s$;*

(iv) *if moreover $4\cdot 10^5 c_3(\kappa + s) \le \frac{1}{195}$ and $\lambda \ge 60c_6$, the constrained law conditioned on $\Phi(K_{s/2})$, written as a law $\mu$ on $K_{s/2}$, satisfies*

$$\mathrm{Var}_\mu(f) \le \frac{60}{\lambda}\int |\nabla f|^2\, d\mu$$

*for every smooth $f$ on $T_U$.*

*The constants do not depend on $L$, $W$, $U$ or $\lambda$.*

For SU(2) and the Wilson action, $\lambda = \beta/4$, so (iv) is a Brascamp–Lieb inequality [21] with constant $240/\beta$ on a chart of the fibre around every background with small plaquettes, for $\beta \ge 240c_6$.

*Proof. Local constants.* For $k = (y,\mu)$ let $c_k(X) = C_{y,\mu}(Ue^X)$, which depends only on the 24 links of $V_{y,\mu}$, and let $j(X_e) = \det\Psi(\mathrm{ad}_{X_e})$, the density of Haar measure in the coordinate $U_e e^{X_e}$, with $\Psi$ as in Lemma 124. Both are real-analytic on a neighbourhood of the compact set of backgrounds with $\delta(P_x, P_{2y}) \le 10^{-6}\rho_G$ for every $x$ and of $X$ with $\|X\|_\infty \le 10^{-3}\rho_G$, and $j > 0$ there. So there is $c_3 \ge 1$ depending only on $G$, computed in Proposition 128, such that there the $m$-th derivative of $c_k$ in $X$ is at most $c_3\prod_{i\le m} N_{y,\mu}(Y^i)$ on $Y^1,\dots,Y^m$ for $m = 1,2,3$, and the first two derivatives of $\log j$ are at most $c_3$ in norm. By Lemma 124(i) the background $U$ lies in this set.

Let $B = Dc(0)$, a map from $\mathfrak{g}^{A_W}$ to $\mathfrak{g}^K$, $K$ the set of pairs $k$, so that $T_U = \ker B$ and $N_U = B^*\mathfrak{g}^K$, and $\mathscr{G} = BB^*$. By the proof of Theorem 125(iii), $\mathscr{G} \ge \frac{1}{17}$ and $\|\mathscr{G}^{-1}\| \le 17$ in the norms $\max_k\|z_k\|$ and, $\mathscr{G}$ being symmetric, $\sum_k\|z_k\|$; each $B_k$ has norm at most 0.44 on the 24 links of $V_{y,\mu}$, and every link lies in at most two of these sets. Hence $\|B\| \le 0.63$, $\|BX\|_\infty \le 2.2\|X\|_\infty$, $\|B^*z\|_\infty \le 0.88\max_k\|z_k\|$ and $\|B^*z\|_1 \le 10.6\sum_k\|z_k\|$, where $\|X\|_1 = \sum_e\|X_e\|$.

(i) Write $c(X) = BX + R(X)$. For $\|X\|_\infty, \|\hat X\|_\infty \le 2s$, Taylor's formula gives $\|R_k(X)\| \le \frac{c_3}{2}N_{y,\mu}(X)^2 \le 12c_3\|X\|_\infty^2$ and $\|R_k(X) - R_k(\hat X)\| \le 48c_3 s\|X - \hat X\|_\infty$. For $\eta \in K_s$ the equation $c(\eta + B^*z) = 0$ reads $z = \Theta(z) = -\mathscr{G}^{-1}R(\eta + B^*z)$, and on $\max_k\|z_k\| \le s$ the map $\Theta$ has values of norm at most $17\cdot 12c_3\cdot 4s^2 = 816c_3s^2 \le s$ and Lipschitz constant at most $17\cdot 48\cdot 0.88\,c_3 s \le 720c_3 s \le \frac12$. So it has a unique fixed point $z(\eta)$, smooth in $\eta$ by the implicit function theorem, and $h = B^*z$ has $\|h\|_\infty \le 0.88\cdot 816c_3s^2 \le s/2$. Since $h(\eta) \perp T_U$, $\eta$ is the orthogonal projection of $\eta + h(\eta)$, so $\Phi$ is injective; its differential $1 + Dh$ is injective, and $\dim T_U = \dim F$. If $Ue^X \in F$ with $\|X\|_\infty \le s/40$, then $X = \eta + B^*z$ with $z = \mathscr{G}^{-1}BX$, $\max_k\|z_k\| \le 17\cdot 2.2\,s/40 < s$ and $\|\eta\|_\infty \le \|X\|_\infty + 0.88\max_k\|z_k\| < s$, and $z$ is a fixed point of $\Theta$, so $Ue^X = \Phi(\eta)$. Finally $\|\eta + h\|_\infty \le \frac32 s$, and by bi-invariance $\delta(U'_p, U_p) \le \sum_{e\in p}\delta(e^{X_e}, I) \le 6s$ for $U' = Ue^X$; the links off $A_W$ are unchanged, so $U'$ is in the tree gauge and $F$ is its fibre.

(ii) Let $X(t) = \eta + t\zeta + h(\eta + t\zeta)$, $\gamma(t) = Ue^{X(t)}$, and let $A = Dc(X)$. Differentiating $c(X(t)) = 0$ gives $X' = \zeta - B^*(AB^*)^{-1}A\zeta$ and $X'' = -B^*(AB^*)^{-1}D^2c(X)[X', X']$. As above $\|(A - B)Y\| \le c_3 N_{y,\mu}(X)N_{y,\mu}(Y) \le 7.4c_3 s N_{y,\mu}(Y)$ in each component, so $\|A - B\| \le 10.5c_3 s$ and $\sum_k\|((A - B)Y)_k\| \le 14.8c_3 s\|Y\|_1$. Since $A\zeta = (A - B)\zeta$ and $AB^* = \mathscr{G} + (A - B)B^*$, the Neumann series gives $\|(AB^*)^{-1}\| \le 34$ in the Euclidean norm and in the norm $\sum_k\|z_k\|$, the latter because

$17\cdot 14.8\cdot 10.6\,c_3 s \le \frac12$; hence $\|Dh\| \le 0.63\cdot 34\cdot 10.5 c_3 s \le \frac16$, $T(X') \le 1.03 T(\zeta)$, the two parts of $X'$ being orthogonal, and $\|X''\|_1 \le 10.6\cdot 34\cdot 2c_3\cdot 1.03 T(\zeta) \le 744 c_3 T(\zeta)$.

Let $w = \gamma^{-1}\gamma' = \Psi(\mathrm{ad}_X)X'$, link by link. Then $\gamma(t+\tau) = \gamma(t)\exp(\tau w + \frac{\tau^2}{2}\dot w + O(\tau^3))$, so

$$\frac{d^2}{dt^2}Q(\gamma) = \frac{d^2}{d\sigma^2}Q\big(\gamma e^{\sigma w}\big)\Big|_{\sigma=0} + dQ(\dot w).$$

The vector $w$ is tangent to $F$ at $\gamma(t)$, which satisfies the hypothesis with $\kappa' = \kappa + 6s$, so by Theorem 125(i), (ii) at $\gamma(t)$ the first term is at least $(\frac{1}{26} - 5.4\cdot 10^4\kappa')T(w)$. From the series of $\Psi$ and $\|\mathrm{ad}_X\| \le 2\|X\|$, $\|w_e - X_e'\| \le 2\|X_e\|\,\|X_e'\| \le 3s\|X_e'\|$, so $T(w) \ge (1-6s)T(\zeta)$; and $\dot w = D\Psi(\mathrm{ad}_X)[\mathrm{ad}_{X'}]X' + \Psi(\mathrm{ad}_X)X''$, where the term of first order in ad vanishes because $\mathrm{ad}_{X'}X' = 0$, so $\|\dot w_e\| \le 3s\|X_e'\|^2 + (1+3s)\|X_e''\|$ and $\|\dot w\|_1 \le 750 c_3 T(\zeta)$. With $\|\nabla_e Q\| \le 12\kappa'$,

$$\frac{d^2}{dt^2}Q(\gamma) \ge \Big(\frac{1}{26} - \frac{s}{4} - (5.4\cdot 10^4 + 9000 c_3)\kappa'\Big)T(\zeta),$$

and $6(5.4\cdot 10^4 + 9000 c_3) + 1 \le 4\cdot 10^5 c_3$.

(iii) In the coordinate $X$, Haar measure is $\prod_e j(X_e)\,dX$, and the coarea formula for $c$ writes $dX$ as the integral over the values of $c$ of the area measure of its level sets divided by $\det(AA^*)^{1/2}$. On the level set $F$ the graph $X = \eta + h(\eta)$ has area element $\det(1+Dh^*Dh)^{1/2}\,d\eta$. So the density is $e^{-\lambda Q(\Phi)-\psi}$ with

$$\psi(\eta) = -\sum_e \log j(X_e) + \tfrac12 \log\det(AA^*) - \tfrac12 \log\det(1+Dh^*Dh).$$

Along $\eta + t\zeta$, the first sum has second derivative at most $c_3(T(X') + \|X''\|_1) \le 745 c_3^2 T(\zeta)$. Next, $A' = D^2c(X)[X',\cdot]$ and $A'' = D^3c(X)[X',X',\cdot] + D^2c(X)[X'',\cdot]$; their components are maps of rank at most $\dim G$ on the 24 links of $V_{y,\mu}$, of norms at most $c_3 N_{y,\mu}(X')$ and $c_3(N_{y,\mu}(X')^2 + N_{y,\mu}(X''))$. Hence the Hilbert–Schmidt norm satisfies $\|A'\|_2^2 \le 2.1\dim G\, c_3^2 T(\zeta)$, and the trace norm $\|A''\|_1 \le \dim G\, c_3(2.1T(\zeta) + 2\|X''\|_1)$. Since $AA^* \ge (17^{-1/2} - 10.5 c_3 s)^2 \ge \frac{1}{25}$ and $\|A\| \le 1$,

$$\Big|\frac{d^2}{dt^2}\log\det(AA^*)\Big| \le 25\big(2\|A''\|_1 + 2\|A'\|_2^2\big) + 2500\|A'\|_2^2,$$

by the identity

$$\frac{d^2}{dt^2}\log\det M = \mathrm{Tr}(M^{-1}M'') - \mathrm{Tr}\big((M^{-1}M')^2\big)$$

and the bounds $|\mathrm{Tr}(XY)| \le \|X\|\,\|Y\|_1$ and $\|XY\|_1 \le \|X\|_2\|Y\|_2$. Finally $Dh = -\Pi$ on $T_U$ with $\Pi = B^*(AB^*)^{-1}A$; $\Pi'$ is a sum of two terms that contain $A'$ once, and $\Pi''$ of terms that contain $A''$ once or $A'$ twice, so $\|\Pi'\|_2 \le 481\|A'\|_2$ and $\|\Pi''\|_1 \le c(\|A''\|_1 + \|A'\|_2^2)$ with $c$ absolute, and the same identity with $M = 1 + Dh^*Dh \ge 1$ and $\|Dh\| \le \frac16$ bounds the last term. Each bound is a multiple of $T(\zeta)$ by a constant that depends only on $c_3$ and $\dim G$; this is $c_6$, computed in Proposition 128.

(iv) Let $H = \lambda Q\circ\Phi + \psi$ on the interior of $K_s$. By (ii) and (iii), $\mathrm{Hess}\,H \ge \lambda(\frac{1}{26} - \frac{1}{195}) - c_6 = \frac{\lambda}{30} - c_6 \ge m = \frac{\lambda}{60}$, so for $a,\eta \in K_{s/2}$, $H(\eta) \ge H(a) + \nabla H(a)\cdot(\eta - a) + \frac{m}{2}|\eta - a|^2$, the segment lying in the convex set $K_{s/2}$. The supremum $\tilde H$ over $a \in K_{s/2}$ of the right-hand sides is finite, equals $H$ on $K_{s/2}$, and $\tilde H - \frac{m}{2}|\eta|^2$ is convex. With $d$ the distance to $K_{s/2}$, a convex function, the functions $\tilde H + nd$ are $m$-convex on $T_U$, and so are their mollifications, which are smooth; for these the Bakry–Émery criterion [5] gives the Poincaré inequality with constant $1/m$. Letting the mollification and then $n\to\infty$, by dominated convergence ($\tilde H$ grows quadratically and $d = 0$ exactly on $K_{s/2}$), the inequality passes to $e^{-H}\mathbf{1}_{K_{s/2}}$, which is $\mu$ up to normalisation by (iii). □

The constants of Theorem 127 can be computed. Cauchy's estimates on a complex neighbourhood of the real chart bound the derivatives of the averages, and the rest of the proof turns these bounds into $c_6$.

**Proposition 128.** *Let $G \subset \mathrm{U}(r)$ be compact. Theorem 127 holds with*

$$c_3 = \max\{1.5\cdot 10^4\sqrt{r}, \dim G\}, \qquad c_6 = 6.81\cdot 10^5 \dim G\, c_3^2.$$

*For* SU(2) *these are $c_3 \le 2.13\cdot 10^4$ and $c_6 \le 9.3\cdot 10^{14}$, so part* (iv) *holds for $\lambda \ge 5.6\cdot 10^{16}$, that is, for the Wilson action at $\beta \ge 2.3\cdot 10^{17}$, once $\kappa + s \le 6.0\cdot 10^{-13}$.*

*Proof.* Write $\|\cdot\|_{\mathrm{op}}$ for the operator norm, which is at most the Hilbert–Schmidt norm, while $\|X\| \le \sqrt{r}\|X\|_{\mathrm{op}}$ on $\mathfrak{g}$. Since $\rho_G \le 1$, every background of the set of the proof has its sites $w_x = v^{-1}P_x$ within $a = 10^{-6}$ of $I$, and every $X$ of that set has $\|X_e\| \le x_0 = 10^{-3}$.

*Three estimates.* Let $X$ be skew-Hermitian and $Z$ complex with $\|Z\|_{\mathrm{op}} \le t$. Duhamel's formula $e^{X+Z} - e^X = \int_0^1 e^{\sigma(X+Z)}Ze^{(1-\sigma)X}\,d\sigma$, with $\|e^{\sigma(X+Z)}\|_{\mathrm{op}} \le e^{\sigma t}$ by the Lie–Trotter product formula, gives $e^{X+Z} = e^X(I+F)$ and likewise $e^{-X-Z} = (I+F')e^{-X}$, with $\|F\|_{\mathrm{op}}, \|F'\|_{\mathrm{op}} \le \varphi = e^t - 1$. The same formula gives $\|e^A - e^B\| \le \|A - B\|$ for skew-Hermitian $A,B$, in both norms. Finally, on $\|w - I\|_{\mathrm{op}} < 1$ the series $\log w = \sum_{k\ge1}(-1)^{k+1}(w-I)^k/k$ is holomorphic, equals the principal logarithm at unitary $w$, and its derivative satisfies $\|D\log(w)[H]\| \le \|H\|/(1 - \|w-I\|_{\mathrm{op}})$ in both norms.

*The complex chart.* Fix $U$ and a real $X$ in the set, put $U' = Ue^X$, and let $Z$ be complex with $\|Z_e\|_{\mathrm{op}} \le t$ on the 24 links of $V_{y,\mu}$. For a path $P_x = U_1U_2$ with path link $v$, the site at the complex field $U_e e^{X_e+Z_e}$ is

$$\begin{aligned} w_x &= e^{-X_v - Z_v}v^{-1}U_1e^{X_1+Z_1}U_2e^{X_2+Z_2} \\ &= (I + F_v')\,w_x'\,(I+\tilde F_1)(I+F_2), \end{aligned}$$

where $w_x'$ is the site at $U'$, a unitary matrix with $\delta(w_x', I) \le a + 3x_0 = e_0$ since each link moves by $\delta(e^{X_e}, I) \le x_0$, and $\tilde F_1$ is $F_1$ conjugated by the unitary $U_2e^{X_2}$. Hence $\|w_x - w_x'\|_{\mathrm{op}} \le \psi = (1+\varphi)^3 - 1$, and if $e_0 + \psi < 1$ then $\|\log w_x - \log w_x'\|_{\mathrm{op}} \le d = \psi/(1 - e_0 - \psi)$, so the mean $M$ of the sixteen logarithms is within $d$ of its value $M'$ at $U'$, which is skew-Hermitian.

The average is $V^\sharp = v e^{X_v+Z_v} e^M$, and with $\|e^M\|_{\mathrm{op}} \le e^d$ and $\|e^M - e^{M'}\|_{\mathrm{op}} \le e^d - 1$,

$$r = V^\sharp (U')^{-1} V^\sharp = e^{-M'}(I+F_v)e^M,$$
$$\|r - I\|_{\mathrm{op}} \le (1+\varphi)e^d - 1 = \eta.$$

At real points $\chi = 1$, as $e_0\rho_G < \rho_G/2$. The unitary $q' = V^\sharp(U)^{-1}V^\sharp(U') = e^{-M_0}e^{X_v}e^{M'}$ has $\delta(q',I) \le x_0 + \|M' - M_0\| \le x_0 + 3x_0/(1-e_0) = \gamma_0 < 0.0041$. So if $\gamma_0 + \eta < 1$, then $c(X+Z) = \log(q'r)$ is holomorphic in $Z$, extends $c$, and

$$\|c(X+Z) - c(X)\|_{\mathrm{op}} \le S(t) = \frac{\eta}{1-\gamma_0-\eta}.$$

*Derivatives of $c_k$.* For $N_{y,\mu}(Y) = 1$ the function $f(z) = c(X+zY)$ is holomorphic on $|z| < 0.1$, where $e_0 + \psi < 0.36$ and $\gamma_0 + \eta < 0.91$, and $\|zY_e\|_{\mathrm{op}} \le |z|$. Cauchy's estimate $\|f^{(m)}(0)\| \le m! R^{-m} \sup_{|z|=R}\|f(z) - f(0)\|$ gives

$$\|D^m c(X)[Y,\dots,Y]\| \le \sqrt{r}\, m!\, S(R)/R^m.$$

At $R = 10^{-3}$, 0.05 and 0.07 the values of $m!S(R)/R^m$ for $m = 1,2,3$ are below 4.2, 307 and 14811; at $R = 0.07$, for instance, $\varphi < 0.0726$, $\psi < 0.2337$, $d < 0.3062$, $\eta < 0.4567$ and $S < 0.847$. By Banach's theorem [9] a symmetric multilinear form on a real Hilbert space has the same norm as its restriction to the diagonal. Applied to $\langle u, D^m c(X)[\cdot]\rangle$ for unit vectors $u \in \mathfrak{g}$, with the Euclidean norm $N_{y,\mu}$ on the 24 links, this gives $\|D^m c(X)[Y^1,\dots,Y^m]\| \le 1.5\cdot 10^4 \sqrt{r}\prod_i N_{y,\mu}(Y^i)$ for $m = 1,2,3$.

*Derivatives of $\log j$.* Since $\mathfrak{g}$ is reductive, $\mathrm{Tr}\,\mathrm{ad}_X = 0$, and $\Psi(z) = e^{-z/2}\sinh(z/2)/(z/2)$, so $\log j(X) = \mathrm{Tr}\, g(\mathrm{ad}_X)$ with $g(w) = \log(\sinh(w/2)/(w/2)) = \sum_{k\ge1} g_{2k} w^{2k}$ for $|w| < 2\pi$. On $|w| = 3$, $|\sinh(w/2)/(w/2) - 1| \le \sinh(1.5)/1.5 - 1 < 0.4196$, so $|g| \le 0.5441$ and $|g_{2k}| \le 0.5441 \cdot 3^{-2k}$. For complex $X$ with $\|X\|_{\mathrm{op}} \le b$ the operator $\mathrm{ad}_X$ on $\mathfrak{g}\otimes\mathbb{C}$ has norm at most $2b$, and $|\mathrm{Tr}\, A^{2k}| \le \dim G\, \|A\|_{\mathrm{op}}^{2k}$, so $|\log j(X)| \le 0.5441 \dim G\, u^2/(1-u^2)$ with $u = 2b/3$. Along $X + zH$, $\|H\| = 1$, $|z| \le 0.8$, this is at most $0.218 \dim G$, and Cauchy's estimate bounds the first two derivatives of $\log j$ by $0.28\dim G$ and $0.69\dim G$.

*The constant $c_6$.* In the proof of Theorem 127(iii), with $c_3$ as above, $a' = \|A'\|_2^2 \le 2.1 \dim G\, c_3^2 T(\zeta)$ and $b' = \|A''\|_1 \le \dim G\, c_3(2.1 + 1488 c_3) T(\zeta) \le 1490.1 \dim G\, c_3^2 T(\zeta)$. Write $K = (AB^*)^{-1}$, so that

$$K' = -KA'B^*K, \quad K'' = 2KA'B^*KA'B^*K - KA''B^*K,$$
$$\Pi' = B^*K'A + B^*KA', \quad \Pi'' = B^*K''A + 2B^*K'A' + B^*KA''.$$

With $\|B\| \le 0.63$, $\|K\| \le 34$, $\|A\| \le 1$ and $k = 34\cdot 0.63 = 21.42$ these give $\|\Pi'\|_2 \le (k+k^2)\|A'\|_2 \le 481\|A'\|_2$ and $\|\Pi''\|_1 \le (2k^3+2k^2)\|A'\|_2^2 + (k^2+k)\|A''\|_1 \le 20576\, a' + 481\, b'$. For $M = 1 + Dh^*Dh \ge 1$ with $\|Dh\| \le \frac16$, the identity for $\log\det$ gives $|(\log\det M)''| \le \frac13\|\Pi''\|_1 + \frac{19}{9}\|\Pi'\|_2^2$. Adding the three terms of $\psi$,

$$\Big|\frac{d^2}{dt^2}\psi\Big| \le 745 c_3^2 T + 25 b' + 1275 a' + \tfrac12\Big(\tfrac13(20576a' + 481b') + \tfrac{19}{9}\, 481^2 a'\Big),$$

which is at most $6.81\cdot 10^5 \dim G\, c_3^2 T(\zeta)$.

For SU(2), $r = 2$ and $\dim G = 3$. Then $60c_6 < 5.6\cdot 10^{16}$, and $4\cdot 10^5 c_3(\kappa + s) \le \frac{1}{195}$ holds when $\kappa + s \le 6.0\cdot 10^{-13}$; it implies $s \le 1/(6000 c_3)$ and $\kappa + 6s \le 10^{-8}$. □

The bounds are far from the true sizes: on random SU(2) backgrounds the first three derivatives of $c_k$ along unit directions stay below 0.14, 0.03 and 0.02, a floating-point computation reproduced in the archive. The loss comes from the radius 0.07 of the complex neighbourhood, which enters the third derivative as $0.07^{-3}$.

*Remark* 129. Theorem 127 removes the coarea factor and the curvature of $F$ for $\lambda \ge 60c_6$ and replaces the small-field set by the convex chart domain $K_{s/2}$. What it does not give:

(a) the mass of the constrained law outside $\Phi(K_{s/2})$, which is Remark 126(c);

(b) decay: (iv) is a variance bound for the law on the chart, not a decay of its covariances between distant links. That needs a Helffer–Sjöstrand representation on the convex set $K_{s/2}$, which has a boundary, and the locality of $\mathrm{Hess}\, H$, which through $Dh$ contains $\mathscr{G}^{-1}$, exponentially decaying by Lemma 97 but not of finite range; Theorem 131 below reaches it on a chart whose domain is a product;

(c) numbers: the constants of Proposition 128 are far from optimal, so for SU(2) the threshold is $\lambda \ge 5.6\cdot 10^{16}$ and $\kappa$ and $s$ are at the scale $10^{-13}$.

On the periodic lattice of side four, for SU(2) backgrounds as in the check of Theorem 125, two to sixteen blocks and $s$ up to 0.03, the chart Hessian of $Q$ is at least 0.21 and $|\psi''|/T$ stays below one, a floating-point computation reproduced in the archive.

The obstacle in Remark 129(b) is the shape of $K_{s/2}$, which is not a product in any coordinates adapted to the links. A slightly different chart domain is. For $k = (y,\mu)$ let $d_k = (2y + e_\mu, \mu)$ be the path link of $V_{y,\mu}$. It has $o_\mu = 1$, so it is active and lies in $A_W$; it enters $V_{y,\mu}$ with weight $\frac18$ by Lemma 93(iii); and it enters no other average, since the links with $o_\mu = 1$ that enter $V_{y',\mu}$ have their base in $B(y')$. Let $L' = A_W \setminus \{d_k\}$ and, for $r > 0$,

$$\Omega_r = \{\eta \in T_U : \|\eta_e\| \le r \text{ for every } e \in L'\}.$$

Give every link the position of its midpoint and every pair $k$ that of $d_k$, and let $d(a,b)$ be the $\ell^\infty$ distance between the positions on the periodic lattice. The links that enter $V_{y,\mu}$ have midpoints within distance 1 of that of $d_k$. Let $\nabla'$ be the gradient in the coordinates $(\eta_e)_{e\in L'}$, and, for a block matrix $M = (M_{ab})$ whose indices are links or pairs and whose blocks are maps between copies of $\mathfrak{g}$, and for $\beta \ge 0$, let

$$\|M\|_\beta = \max\Big(\sup_a \sum_b e^{\beta d(a,b)}\|M_{ab}\|,\ \sup_b \sum_a e^{\beta d(a,b)}\|M_{ab}\|\Big).$$

At most two indices share a position, so the sum

$$N(\beta) = \sup_a \sum_b e^{-\beta d(a,b)}$$

is finite for $\beta > 0$ and depends only on $\beta$.

**Lemma 130.** (i) *The operator norm satisfies $\|M\| \le \|M\|_\beta$, and $\|M_1M_2\|_\beta \le \|M_1\|_\beta\|M_2\|_\beta$ and $\|M^*\|_\beta = \|M\|_\beta$.*

(ii) *If $|\varphi(a)-\varphi(b)| \le d(a,b)$ for all $a,b$ and $0 \le \alpha \le \beta$, then $\|e^{\alpha\varphi}Me^{-\alpha\varphi} - M\| \le \frac{\alpha}{\beta}\|M\|_\beta$.*

(iii) *If $M$ is invertible, $\|M^{-1}\| \le M_0$ and $\|M\|_\beta \le M_1$, then $\|M^{-1}\|_\gamma \le 2M_0N(\gamma)$ with $\gamma = \beta/(4M_0M_1)$.*

*Proof.* (i) is the Schur test and the triangle inequality for $d$. (ii) The blocks of the difference are $(e^{\alpha(\varphi(a)-\varphi(b))} - 1)M_{ab}$, and $|e^{\alpha(\varphi(a)-\varphi(b))} - 1| \le e^{\alpha d(a,b)} - 1 \le \frac{\alpha}{\beta}e^{\beta d(a,b)}$ by convexity of $\alpha \mapsto e^{\alpha d}$; apply (i) with $\beta = 0$. (iii) Let $\alpha = 2\gamma \le \beta$, since $M_0M_1 \ge \|M^{-1}\|\|M\| \ge 1$. By (ii), $e^{\alpha\varphi}Me^{-\alpha\varphi} = M+E$ with $\|M^{-1}E\| \le \frac12$, so $\|e^{\alpha\varphi}M^{-1}e^{-\alpha\varphi}\| \le 2M_0$ as in Lemma 97. With $\varphi = d(\cdot,b)$ the block $(a,b)$ gives $e^{2\gamma d(a,b)}\|(M^{-1})_{ab}\| \le 2M_0$ for all $a,b$, and summing $e^{\gamma d(a,b)}\|(M^{-1})_{ab}\| \le 2M_0e^{-\gamma d(a,b)}$ over $b$ or over $a$ gives the claim. □

**Theorem 131.** *Assume the hypotheses of Theorem 127*(iv)*, with $c_6 \ge 1$, as we may, and let $r = s/32$.*

(i) *The coordinates $(\eta_e)_{e\in L'}$ identify $T_U$ with $\mathfrak{g}^{L'}$, Lebesgue measure on $T_U$ being a constant multiple of that on $\mathfrak{g}^{L'}$; $\Omega_{r'}$ is a product of balls in them, and $K_{r'} \subset \Omega_{r'} \subset K_{15.1r'}$ for every $r' > 0$.*

(ii) *There are $\alpha_G > 0$ and $c_9$ depending only on $G$ with the following property. Let $\mu$ be the constrained law conditioned on $\Phi(\Omega_r)$, written as a law on $\Omega_r$. For every $\varphi : L' \to \mathbb{R}$ with $|\varphi(e)-\varphi(e')| \le d(e,e')$, every $0 \le \alpha \le \alpha_G$ and all smooth $F, G$ on $T_U$,*

$$|\mathrm{Cov}_\mu(F,G)| \le \frac{120}{\lambda}\left\|e^{-\alpha\varphi}\nabla' F\right\|_{L^2(\mu)}\left\|e^{\alpha\varphi}\nabla' G\right\|_{L^2(\mu)}.$$

(iii) *If $X_1, X_2 \subset A_W$, $\tilde F, \tilde G$ are smooth functions of the links in $X_1$ and in $X_2$ whose left-invariant gradients have norm at most 1 in each link, $F = \tilde F \circ \Phi$ and $G = \tilde G \circ \Phi$, then*

$$|\mathrm{Cov}_\mu(F,G)| \le \frac{c_9}{\lambda}\,|X_1|\,|X_2|\,e^{-\alpha_G d(X_1,X_2)}.$$

*The constants do not depend on $L$, $W$, $U$, $\lambda$, $s$ or $\kappa$.*

So the covariances of local observables of the field under the constrained law, conditioned on the chart, decay exponentially at a rate that depends only on $G$, with amplitude $O(1/\lambda)$; for SU(2) and the Wilson action the amplitude is $O(1/\beta)$.

*Proof.* All constants called $C$ depend only on $G$.

*The section.* By Lemma 124(iv), $B_k = \tilde B_k + R_k$ with $\|R_k\| \le 260\kappa$, and the block of $\tilde B_k$ at $d_k$ is $\frac18$ times an isometry of $\mathfrak{g}$, so $\|B_{k,d_k}^{-1}\| \le (\frac18 - 260\kappa)^{-1} \le 8.01$. For $\xi \in \mathfrak{g}^{L'}$ let $(S\xi)_e = \xi_e$ for $e \in L'$ and $(S\xi)_{d_k} = -B_{k,d_k}^{-1}\sum_{e\neq d_k} B_{k,e}\xi_e$. Since $d_k$ enters only $V_k$, $BS = 0$; $S$ is injective, and $\dim T_U = |L'|\dim G$ because $F$ has codimension $4|W|\dim G$, so $S$ is an isomorphism onto $T_U$ whose inverse is the restriction to $L'$. The other coefficients of $\tilde B_k$ add up to $\frac78 + 1 = \frac{15}{8}$, so $\|(S\xi)_{d_k}\| \le 8.01(\frac{15}{8} + \sqrt{24}\cdot 260\kappa)\max_e\|\xi_e\| \le 15.1\max_e\|\xi_e\|$. This proves (i), and $S$ has blocks of norm at most 1.1 that vanish beyond distance 1. Write $\mathrm{Box}_{r'}$ for the product of balls $S^{-1}\Omega_{r'}$. Since $15.1\cdot 2r < s$, the estimates of Theorem 127 hold on $S(\mathrm{Box}_{2r})$.

*The density.* Put $X(\xi) = S\xi + h(S\xi)$, $A = Dc(X)$, $P = 1 - B^*(AB^*)^{-1}A$ and $\bar P = PS$. In the proof of Theorem 127(ii) the derivative of $X$ along $\zeta \in T_U$ is $P\zeta$, and $P\zeta = \zeta + Dh\,\zeta$ with $Dh\,\zeta \perp T_U$; so $\partial_\xi X = \bar P$, $\bar P^*\bar P = S^*(1+Dh^*Dh)S \ge S^*S \ge 1$, and $\det\bar P^*\bar P = \det(S^*S)\det(1+Dh^*Dh)$. By Theorem 127(iii) the law $\mu$ has, in the coordinate $\xi$, the density $e^{-\bar H}\mathbf{1}_{\mathrm{Box}_r}$ up to normalisation, with

$$\bar H(\xi) = \lambda Q\big(Ue^X\big) - \sum_e \log j(X_e) + \tfrac12\log\det(AA^*) - \tfrac12\log\det(\bar P^*\bar P),$$

and $\mathrm{Hess}\,\bar H \ge m = \frac{\lambda}{60}$ on $\mathrm{Box}_{2r}$, because $|S\zeta| \ge |\zeta|$.

*Locality of the Hessian.* We show that $\|\mathrm{Hess}\,\bar H(\xi)\|_\beta \le c_8\lambda$ on $\mathrm{Box}_{2r}$ for some $\beta > 0$ and $c_8 \ge 1$ depending only on $G$, the Hessian being the block matrix of second derivatives in the coordinates $\xi_e$. The maps $B$, $A$, $S$, the second derivative of $Q(Ue^X)$ in $X$ and that of $\sum_e \log j(X_e)$ have blocks bounded by $C$ (the last two by compactness, as $c_3$) that vanish beyond distance 2, so their norms $\|\cdot\|_1$ are at most $C$. By the proof of Theorem 127, $\|(AB^*)^{-1}\| \le 34$ and $\|(AA^*)^{-1}\| \le 25$, and $\|(\bar P^*\bar P)^{-1}\| \le 1$. Applying Lemma 130(iii) to $AB^*$ and $AA^*$, then (i) to $P$, $\bar P$ and $\bar P^*\bar P$, then (iii) to $\bar P^*\bar P$, gives $\beta_0 \in (0,1]$ such that $(AB^*)^{-1}$, $(AA^*)^{-1}$, $P$, $\bar P$, $\bar P^*\bar P$ and $(\bar P^*\bar P)^{-1}$ have norm $\|\cdot\|_{\beta_0}$ at most $C$ on $\mathrm{Box}_{2r}$. Let $\gamma = \beta_0/2$, and for links $e,e'$ and $v,v' \in \mathfrak{g}$ write $v,v'$ also for the vectors of $\mathfrak{g}^{L'}$ supported at $e$, $e'$. Call a block matrix $M$, depending linearly on $v$, *concentrated at $e$* if $\|M_{ab}\| \le C|v|e^{-\gamma(d(a,e)+d(b,e))}$, and, depending bilinearly on $(v,v')$, *concentrated at $(e,e')$* if $\|M_{ab}\| \le C|v||v'|e^{-\gamma(d(a,e)+d(a,e')+d(b,e)+d(b,e'))}$; a vector is concentrated if it is so as a column. If $\|Y\|_{\beta_0} \le C$, then $YM$ and $MY$ are concentrated where $M$ is, because $e^{-\gamma(d(c,e)+d(c,e'))} \le e^{\beta_0 d(a,c)}e^{-\gamma(d(a,e)+d(a,e'))}$. For $M$ concentrated at $e$ and $M'$ at $e'$, using $d(a,e)+d(a,e') \ge d(e,e')$ in both indices,

$$|\mathrm{Tr}(MM')| \le C|v||v'|N(\gamma/2)^2e^{-\gamma d(e,e')},$$

and if $M$ is concentrated at $(e,e')$, then

$$|\mathrm{Tr}M| \le C|v||v'|N(\gamma)e^{-\gamma d(e,e')}.$$

Now $x_v = \partial_v X = \bar P v$ has $\|(x_v)_a\| \le Ce^{-\beta_0 d(a,e)}|v|$, so it is concentrated at $e$. The vector $D^2c(X)[x_v,x_{v'}]$ has components at most $c_3N_k(x_v)N_k(x_{v'}) \le C|v||v'|e^{-\beta_0(d(k,e)+d(k,e'))}$, so $x_{vv'} = \partial_v\partial_{v'}X = -B^*(AB^*)^{-1}D^2c(X)[x_v,x_{v'}]$ is concentrated at $(e,e')$. In the same way $A_v = \partial_v A = D^2c(X)[x_v,\cdot]$ is concentrated at $e$, $A_{vv'} = D^3c(X)[x_v,x_{v'},\cdot] + D^2c(X)[x_{vv'},\cdot]$ is concentrated at $(e,e')$, $\partial_v P = -B^*(AB^*)^{-1}A_vP$ is concentrated at $e$, and $\partial_v\partial_{v'}P$ is a sum of products of operators bounded in $\|\cdot\|_{\beta_0}$ with either one factor concentrated at $(e,e')$ or one factor concentrated at $e$ and one at $e'$. The same holds for the derivatives of $\mathscr{G}_1 = AA^*$

and $\mathscr{G}_2 = \bar{P}^*\bar{P}$. Then

$$\partial_v\partial_{v'}\bar{H} = \lambda\big(D^2Q[x_v,x_{v'}] + DQ[x_{vv'}]\big) - \sum_e \partial_v\partial_{v'}\log j(X_e) + \sum_{i=1,2} \pm\tfrac{1}{2}\big(\mathrm{Tr}(\mathscr{G}_i^{-1}\partial_v\partial_{v'}\mathscr{G}_i) - \mathrm{Tr}(\mathscr{G}_i^{-1}\partial_v\mathscr{G}_i\mathscr{G}_i^{-1}\partial_{v'}\mathscr{G}_i)\big),$$

where $D^2Q$ and $DQ$ are the derivatives of $Q(Ue^X)$ in $X$. By the bounds above every term is at most

$$C(\lambda+1)|v||v'|e^{-\gamma d(e,e')/2};$$

for instance

$$|D^2Q[x_v,x_{v'}]| \le C\sum_a e^{-\beta_0(d(a,e)+d(a,e')-2)}|v||v'|.$$

Hence, using $\lambda \ge 60c_6 \ge 60$, every block of the Hessian of $\bar{H}$ satisfies

$$\|(\mathrm{Hess}\,\bar{H})_{ee'}\| \le C\lambda e^{-\beta_0 d(e,e')/4},$$

and $\|\mathrm{Hess}\,\bar{H}\|_\beta \le c_8\lambda$ with $\beta = \beta_0/8$.

*Approximation.* Let $\rho$ be smooth on $[0,\infty)$ with $0 \le \rho' \le 1$, $\rho(t) = t$ for $t \le \frac{3}{2}r$, $\rho \le \frac{7}{4}r$ and $\rho$ constant on $[2r,\infty)$, and let $\tau(\xi)_e = \rho(\|\xi_e\|)\xi_e/\|\xi_e\|$, which maps $\mathfrak{g}^{L'}$ into $\mathrm{Box}_{7r/4}$. Let $v$ be smooth and convex on $[0,\infty)$, zero on $[0,r]$, with $0 < v'' \le 1$ on $(r,\infty)$ and $v'' = 1$ on $[\frac{5}{4}r,\infty)$, and $V(\xi) = \sum_e v(\|\xi_e\|)$. For $t \ge \frac{3}{2}r$, $v''(t) = 1$ and $v'(t)/t \ge 1 - \frac{5}{4}r/t \ge \frac{1}{6}$, so $\mathrm{Hess}\,V \ge \frac{1}{6}$ in the links with $\|\xi_e\| \ge \frac{3}{2}r$. Let $H_n = \bar{H}\circ\tau + nV$. Then

$$\mathrm{Hess}\,H_n = D\tau^*\big(\mathrm{Hess}\,\bar{H}\circ\tau\big)D\tau + \Delta_n,$$

where $D\tau$ and $\Delta_n$ are block diagonal, $D\tau$ is the identity in the links with $\|\xi_e\| \le \frac{3}{2}r$, $\|D\tau\| \le 1$, and $\Delta_n \ge n\frac{1}{6} - C_1$ in the other links, $C_1$ bounding $\nabla\bar{H}$ times the second derivative of $\tau$. So $\mathrm{Hess}\,H_n \ge m$ on $\mathfrak{g}^{L'}$ for $n \ge 6(m+C_1)$, and all derivatives of $H_n$ of order at least two are bounded. For $\varphi$ as in (ii) and $\alpha \le \alpha_G = \beta/(120c_8)$, the block-scalar matrix $e^{\alpha\varphi}$ commutes with $D\tau$ and $\Delta_n$, so by Lemma 130(ii)

$$\big\|e^{\alpha\varphi}\,\mathrm{Hess}\,H_n e^{-\alpha\varphi} - \mathrm{Hess}\,H_n\big\| \le \frac{\alpha}{\beta}c_8\lambda \le \frac{m}{2}.$$

*Weighted Helffer–Sjöstrand.* Let $\mu_n$ have density proportional to $e^{-H_n}$ on $\mathfrak{g}^{L'}$ and $\mathscr{L}_n = -\Delta + \nabla H_n\cdot\nabla$ on $L^2(\mu_n)$. For $H_n$ smooth with $\mathrm{Hess}\,H_n \ge m > 0$ and bounded derivatives of order at least two, and $F,G$ smooth with bounded derivatives, the Helffer–Sjöstrand representation [53, 72] reads $\mathrm{Cov}_{\mu_n}(F,G) = \langle\nabla F, \mathscr{A}^{-1}\nabla G\rangle_{L^2(\mu_n)}$ with $\mathscr{A} = \mathscr{L}_n\otimes 1 + \mathrm{Hess}\,H_n$ acting on $L^2(\mu_n;\mathfrak{g}^{L'})$. With $w = e^{\alpha\varphi}$, constant in $\xi$, $\mathscr{A}_w = w\mathscr{A}w^{-1} = \mathscr{L}_n\otimes 1 + w\,\mathrm{Hess}\,H_n w^{-1}$, and $\mathrm{Re}\langle u,\mathscr{A}_w u\rangle \ge \frac{m}{2}\|u\|^2$ because $\mathscr{L}_n \ge 0$; the adjoint satisfies the same bound, so $\|\mathscr{A}_w^{-1}\| \le \frac{2}{m} = \frac{120}{\lambda}$; this is the weighted form of the representation used in [22, 49]. Since $\mathscr{A}^{-1} = w^{-1}\mathscr{A}_w^{-1}w$,

$$|\mathrm{Cov}_{\mu_n}(F,G)| = \big|\langle w^{-1}\nabla F, \mathscr{A}_w^{-1}w\nabla G\rangle\big| \le \frac{120}{\lambda}\|w^{-1}\nabla F\|_{L^2(\mu_n)}\|w\nabla G\|_{L^2(\mu_n)}.$$

As $n\to\infty$, $e^{-H_n}$ decreases to $e^{-\bar{H}}\mathbf{1}_{\mathrm{Box}_r}$ off the boundary of $\mathrm{Box}_r$, since $\tau$ is the identity there and $V > 0$ off it, so $\mu_n \to \mu$ in total variation. For $F,G$ smooth on $T_U$, apply the bound to $(F\circ S)\chi$ and $(G\circ S)\chi$ with $\chi$ smooth, compactly supported and equal to 1 near $\mathrm{Box}_r$, and let $n\to\infty$; this proves (ii).

(iii) The left-invariant derivative of $U_a e^{X_a}$ along $\delta X_a$ is $\Psi(\mathrm{ad}_{X_a})\delta X_a$, of norm at most $2\|\delta X_a\|$, so $\|\partial_{\xi_f}F\| \le 2\sum_{a\in X_1}\|\bar{P}_{af}\|$, and similarly for $G$. Take $\varphi = d(\cdot,X_2)$ and $\alpha = \alpha_G \le \beta_0$. Then $e^{-\alpha d(f,X_2)} \le e^{\beta_0 d(a,f)}e^{-\alpha d(a,X_2)}$, so $|e^{-\alpha\varphi}\nabla' F| \le 2|X_1|\|\bar{P}\|_{\beta_0}e^{-\alpha d(X_1,X_2)}$, and $e^{\alpha d(f,X_2)} \le e^{\beta_0 d(b,f)}$ for $b\in X_2$ gives $|e^{\alpha\varphi}\nabla' G| \le 2|X_2|\|\bar{P}\|_{\beta_0}$. Insert these in (ii); with $C$ a bound for $\|\bar{P}\|_{\beta_0}$ on $\mathrm{Box}_{2r}$, $c_9 = 480C^2$. □

The rate $\alpha_G$ and the amplitude $c_9$ can be computed, and they turn out not to depend on $G$. The proof above passes through the norms $\|\cdot\|_\beta$, whose Schur sums bring in the large factors $N(\beta)$; the computation below replaces them by weighted operator norms, in which inverses and products cost nothing beyond their operator norms. For a block matrix $M$ indexed by links and pairs and for $\vartheta \ge 0$ let

$$\|M\|_{(\vartheta)} = \sup_\varphi \big\|e^{\vartheta\varphi}Me^{-\vartheta\varphi}\big\|,$$

the supremum over all $\varphi$ with $|\varphi(a) - \varphi(b)| \le d(a,b)$ for all indices $a,b$, the factors $e^{\pm\vartheta\varphi}$ acting as scalars on each block. For such $\varphi$ write $w = e^{\vartheta\varphi}$ and $M^w = wMw^{-1}$.

**Lemma 132.** (i) *For $\vartheta \le \vartheta'$, $\|M\| \le \|M\|_{(\vartheta)} \le \|M\|_{(\vartheta')}$; moreover $\|M_1M_2\|_{(\vartheta)} \le \|M_1\|_{(\vartheta)}\|M_2\|_{(\vartheta)}$, $\|M^*\|_{(\vartheta)} = \|M\|_{(\vartheta)}$, and $\|M_{ab}\| \le \|M\|_{(\vartheta)}e^{-\vartheta d(a,b)}$. If the blocks of $M$ vanish beyond distance $R$, then $\|M\|_{(\vartheta)} \le e^{\vartheta R}(R_M C_M)^{1/2}$, where $R_M$ and $C_M$ are the largest row and column sums of the norms of the blocks.*

(ii) *If $M$ is Hermitian and $0 \le \vartheta \le \vartheta'$, then $\|\mathrm{Re}\,M^w - M\| \le 2(\vartheta/\vartheta')^2\|M\|_{(\vartheta')}$ for every $\varphi$, where $\mathrm{Re}\,T = \frac{1}{2}(T+T^*)$.*

(iii) *If $\mathrm{Re}\,M^w \ge c > 0$ for every $\varphi$, then $\|M^{-1}\|_{(\vartheta)} \le 1/c$. If $\|M^{-1}\|_{(\vartheta)}\|E\|_{(\vartheta)} < 1$, then $\|(M+E)^{-1}\|_{(\vartheta)} \le \|M^{-1}\|_{(\vartheta)}/(1 - \|M^{-1}\|_{(\vartheta)}\|E\|_{(\vartheta)})$.*

*Proof.* (i) Take $\varphi = 0$ for the first inequality and replace $\varphi$ by $\vartheta\varphi/\vartheta'$ for the second. Products and adjoints follow from $(M_1M_2)^w = M_1^wM_2^w$ and $(M^*)^w = (M^{w'})^*$ with $w' = e^{-\vartheta\varphi}$. With $\varphi = d(\cdot,b)$ the block $(a,b)$ of $M^w$ is $e^{\vartheta d(a,b)}M_{ab}$, of norm at most $\|M^w\|$. The last bound is the Schur test, since the blocks of $M^w$ are $e^{\vartheta(\varphi(a)-\varphi(b))}M_{ab}$.

(ii) Let $f(z) = e^{z\varphi}Me^{-z\varphi}$, an entire function of $z\in\mathbb{C}$ with values in matrices. For $z = x+iy$, $f(z) = e^{iy\varphi}f(x)e^{-iy\varphi}$ with $e^{iy\varphi}$ unitary, so $\|f(z)\| \le K = \|M\|_{(\vartheta')}$ for $|x| \le \vartheta'$. Since $M$ is Hermitian, $f(x)^* = f(-x)$ for real $x$, so $\mathrm{Re}\,M^w - M = g(\vartheta)$ with $g(z) = \frac{1}{2}(f(z)+f(-z)) - f(0)$. The function $g$ is even, vanishes at 0 and is bounded by $2K$ on $|z| < \vartheta'$; so $g(z)/z^2$ is holomorphic there, and the maximum principle on the circles $|z| = \rho < \vartheta'$, applied to $\ell(g(z))/z^2$ for every linear functional $\ell$ of norm one, gives $\|g(z)\| \le 2K|z|^2/\vartheta'^2$.

(iii) If $\mathrm{Re}\,T \ge c$, then $\|Tu\|\|u\| \ge \mathrm{Re}\langle u,Tu\rangle \ge c\|u\|^2$, so $\|T^{-1}\| \le 1/c$; apply this to $T = M^w$, whose inverse is $(M^{-1})^w$.

The second statement is the Neumann series for $(M^w+E^w)^{-1}$. □

**Proposition 133.** *Let $G\subset \mathrm{U}(r)$ be compact. Theorem 131 holds with*

$$\alpha_G = 2.5\cdot 10^{-4}, \qquad c_9 = 1.2\cdot 10^4,$$

*and in* (iii) *the factor $|X_1||X_2|$ may be replaced by $(|X_1||X_2|)^{1/2}$.*

So the covariances of local observables under the constrained law on the chart decay at least like $e^{-d/4000}$, with amplitude $1.2\cdot 10^4/\lambda$, for every compact group; for SU(2) and the Wilson action the amplitude is $4.8\cdot 10^4/\beta$.

*Proof.* Throughout, $\xi\in \mathrm{Box}_{2r}$, $X=S\xi+h(S\xi)$, and the notation is that of the proof of Theorem 131. The hypothesis $4\cdot 10^5 c_3(\kappa+s)\le \frac{1}{195}$ of Theorem 127(iv) gives $c_3 s\le 1.29\cdot 10^{-8}$ and $c_3\kappa'\le 7.7\cdot 10^{-8}$ for $\kappa'=\kappa+6s$, and $\kappa\le 10^{-8}$. Every $\varphi$ below is defined on all indices; a function on $L'$ with the Lipschitz property extends to all indices with the same property, by $\varphi(a)=\min_{e\in L'}(\varphi(e)+d(a,e))$.

*Local operators.* The row $k$ of $B$ has blocks of norm at most $\frac18+260\kappa$, with sum at most $2+24\cdot 260\kappa$, by Lemma 124(iv) and the coefficients of $V_{y,\mu}$, eight of them $\frac18$ and sixteen $\frac{1}{16}$; every link lies in at most two rows, and the blocks vanish beyond distance 1. So $b=\|B\|_{(\vartheta)}\le e^{\vartheta}(\frac12+10^{-4})^{1/2}$ by Lemma 132(i). The row $k$ of $A-B$ has norm at most $c_3N_{y,\mu}(X)\le 7.35c_3 s=\varepsilon_A$, since $\|X\|_\infty\le\frac32 s$, so $\|A-B\|_{(\vartheta)}\le\sqrt{48}e^{\vartheta}\varepsilon_A$. The row $d_k$ of $S$ is $s_k\xi=-B^{-1}_{k,d_k}\sum_{e\neq d_k}B_{k,e}\xi_e$, of norm $\sigma\le(\frac{\sqrt{11}}{8}+260\kappa)/(\frac18-260\kappa)\le 3.32$, the squares of the other coefficients adding up to $\frac{11}{64}$; the other rows of $S$ are those of the identity. If $D_k$ is the diagonal matrix with entries $e^{\vartheta(\varphi(d_k)-\varphi(e))}$ on the links $e$ of $V_{y,\mu}$, then the row $d_k$ of $S^w$ is $s_kD_k$, and every link of $L'$ lies in at most two of these sets, so

$$\bar s=\|S\|_{(\vartheta)}\le(1+2\sigma^2e^{2\vartheta})^{1/2}.$$

Finally $Q(Ue^X)=\sum_p\|U_p-I\|^2=\sum_p(2r-2\mathrm{Re}\,\mathrm{Tr}\,U_p)$. By Duhamel's formula the derivative of $U_ee^{X_e}$ in $X_e$ is an average of $U_ee^{\theta X_e}Ye^{(1-\theta)X_e}$, and its second derivative an average, of total weight one, of products with $Y$ and $Z$ inserted between unitary factors. With $|\mathrm{Tr}(V_1YV_2ZV_3)|\le\|Y\|\|Z\|$ for unitary $V_i$ this gives

$$\big|D^2Q_p[Y,Z]\big|\le 2\sum_{e\in p}\|Y_e\|\sum_{e\in p}\|Z_e\|,$$

and, since $\mathrm{Re}\,\mathrm{Tr}(VY)=\mathrm{Re}\,\mathrm{Tr}((V-I)Y)$ for skew-Hermitian $Y$ and a conjugate $V$ of $U_p$, $|DQ_p[Y]|\le 2\kappa'\sum_{e\in p}\|Y_e\|$. A link lies in six plaquettes, so $\|D^2Q\|_{(\vartheta)}\le 48e^{\vartheta}$ and $|DQ[Y]|\le 12\kappa'\|Y\|_1$.

*Inverses.* The diagonal blocks of $\mathscr{G}=BB^*$ have norm at most $\frac{3}{16}+227\kappa$, each row has two other nonzero blocks, of norm at most $\frac{1}{32}+227\kappa$, at distance 2, and $\mathscr{G}\ge\frac{1}{16}-700\kappa$, by the proof of Theorem 125(iii). The blocks of $\mathrm{Re}\,\mathscr{G}^w$ are $\cosh(\vartheta(\varphi(k)-\varphi(l)))\mathscr{G}_{kl}$, so $\mathrm{Re}\,\mathscr{G}^w\ge\frac{1}{16}-700\kappa-(\cosh 2\vartheta-1)(\frac{1}{16}+454\kappa)$, and Lemma 132(iii) bounds $g_0=\|\mathscr{G}^{-1}\|_{(\vartheta)}$ by the inverse of the right-hand side. With $E=(A-B)B^*$, the same lemma gives

$$\|(AB^*)^{-1}\|_{(\vartheta)}\le\frac{g_0}{1-g_0\|E\|_{(\vartheta)}}=h_0,$$
$$\|(AA^*)^{-1}\|_{(\vartheta)}\le\frac{g_0}{1-g_0(2\|E\|_{(\vartheta)}+\|A-B\|^2_{(\vartheta)})}=g_1,$$

since $AB^*=\mathscr{G}+E$ and $AA^*=\mathscr{G}+E+E^*+(A-B)(A-B)^*$. Let $Y=B^*(AB^*)^{-1}$, so $y=\|Y\|_{(\vartheta)}\le bh_0$. Since $BS=0$, $\bar P=PS=S-Y(A-B)S$, so $\bar p=\|\bar P\|_{(\vartheta)}\le\bar s(1+y\|A-B\|_{(\vartheta)})$; and $M=Y(A-B)S$ takes values in $N_U$, orthogonal to the values of $S$, so $\bar P^*\bar P=S^*S+M^*M$. Here $S^*S=1+\sum_k s_k^*s_k$, and $(s_k^*s_k)^w=v^*u$ with $u=s_kD_k$, $v=s_kD_k^{-1}$, so $\mathrm{Re}(s_k^*s_k)^w=\frac14((u+v)^*(u+v)-(u-v)^*(u-v))\ge-\sigma^2\sinh^2\vartheta\,J_k$, $J_k$ the projection on the links of $V_{y,\mu}$; and $\mathrm{Re}(M^*M)^w\ge-2(\vartheta/\vartheta'')^2\|M\|^2_{(\vartheta'')}$ by Lemma 132(ii), for $\vartheta\le\vartheta''$. Hence

$$g_2=\|(\bar P^*\bar P)^{-1}\|_{(\vartheta)}\le\Big(1-2\sigma^2\sinh^2\vartheta-2\frac{\vartheta^2}{\vartheta''^2}\|M\|^2_{(\vartheta'')}\Big)^{-1}.$$

For the $\ell^1$ norms, $\|\mathscr{G}^{-1}\|\le 17$ in the norm $\sum_k\|z_k\|$ and $E$ has column sums at most $2(2+10^{-4})\varepsilon_A$, so $(AB^*)^{-1}$ has norm $h_1\le 17.001$ there, and $\|B^*z\|_1\le(2+10^{-4})\sum_k\|z_k\|$.

*The Hessian.* Let $u,u'\in\mathfrak{g}^{L'}$, $p=wu$ and $q=w^{-1}u'$. Since $\langle u,(\mathrm{Hess}\,\bar H)^wu'\rangle=\partial_p\partial_q\bar H$, it suffices to bound $|\partial_p\partial_q\bar H|$ by $K|u||u'|$ for every $\varphi$. Write $x_p=\bar Pp=w\tilde x$ and $x_q=w^{-1}\tilde y$, with $|\tilde x|\le\bar p|u|$ and $|\tilde y|\le\bar p|u'|$. The links of $V_{y,\mu}$ lie within distance 1 of the position of $k$, so the weights of $x_p$ and $x_q$ on them differ from $e^{\pm\vartheta\varphi(k)}$ by factors at most $e^{\vartheta}$, and

$$\sum_k N_{y,\mu}(x_p)N_{y,\mu}(x_q)\le 2e^{2\vartheta}|\tilde x||\tilde y|,$$

every link lying in at most two of these sets. In the same way $A_p=\partial_pA=D^2c(X)[x_p,\cdot]$ equals $w\check A_p$ and $A_q=w^{-1}\check A_q$, where $\check A_p$ and $\check A_q$ have rows of rank at most $\dim G$, so their Hilbert–Schmidt norms satisfy

$$\|\check A_p\|_2\|\check A_q\|_2\le 2\dim G\,c_3^2e^{2\vartheta}\bar p^2|u||u'|,$$

and conjugating them by $w$ multiplies these norms by at most $e^{\vartheta}$. The second derivative $x_{pq}=-YD^2c(X)[x_p,x_q]$ has $\|x_{pq}\|_1\le c_3x_1|u||u'|$ with $x_1=2(2+10^{-4})h_1e^{2\vartheta}\bar p^2$, and $A_{pq}=D^3c(X)[x_p,x_q,\cdot]+D^2c(X)[x_{pq},\cdot]$ has trace norm $\bar a\le 2\dim G\,c_3^2(e^{2\vartheta}\bar p^2+x_1)|u||u'|$, the weights cancelling in both.

The four parts of $\bar H$ are bounded as follows. First, $\partial_p\partial_q\lambda Q=\lambda(\langle\tilde x,(D^2Q)^w\tilde y\rangle+DQ[x_{pq}])$, at most $\lambda(48e^{\vartheta}\bar p^2+12c_3\kappa'x_1)|u||u'|$. Second, the Hessian of $\sum_e\log j(X_e)$ in $X$ is block diagonal and commutes with $w$, so its part is at most $c_3(\bar p^2+c_3x_1)|u||u'|$. Third, for $\mathscr{G}_1=AA^*$,

$$\partial_p\partial_q\log\det\mathscr{G}_1=\mathrm{Tr}(\mathscr{G}_1^{-1}\partial_p\partial_q\mathscr{G}_1)-\mathrm{Tr}(\mathscr{G}_1^{-1}\partial_p\mathscr{G}_1\mathscr{G}_1^{-1}\partial_q\mathscr{G}_1),$$

with $\partial_p\mathscr{G}_1=A_pA^*+AA_p^*$ and $\partial_p\partial_q\mathscr{G}_1=A_{pq}A^*+A_pA_q^*+A_qA_p^*+AA_{pq}^*$. Each term is the trace of a product in which

$w$ and $w^{-1}$ occur once each, next to $\check A_p$ and $\check A_q$, or not at all, next to $A_{pq}$. A cyclic permutation writes it as $\mathrm{Tr}(wFw^{-1}F')$, or $\mathrm{Tr}(w^{-1}FwF')$, and $|\mathrm{Tr}(wFw^{-1}F')| \le \|wFw^{-1}\|_2\|F'\|_2$ or $\|wFw^{-1}\|\|F'\|_1$. With $a = \|A\|_{(\vartheta)} \le b + \|A-B\|_{(\vartheta)}$ this gives

$$\tfrac12|\partial_p\partial_q \log\det \mathscr{G}_1| \le g_1 a\bar a + (g_1 + 2g_1^2a^2)e^{2\vartheta}\|\check A_p\|_2\|\check A_q\|_2.$$

Fourth, $\partial_p\bar P = -YA_p\bar P$ and $\partial_p\partial_q\bar P = -YA_{pq}\bar P + YA_qYA_p\bar P + YA_pYA_q\bar P$, and the same counting for $\mathscr{G}_2 = \bar P^*\bar P$ gives

$$\tfrac12|\partial_p\partial_q \log\det \mathscr{G}_2| \le g_2\bar p^2 y\bar a + (3g_2\bar p^2y^2 + 2g_2^2\bar p^4y^2)e^{2\vartheta}\|\check A_p\|_2\|\check A_q\|_2.$$

So

$$\|\mathrm{Hess}\,\bar H\|_{(\vartheta)} \le K = \lambda k_1 + \dim G\, c_3^2 k_0,$$

where $k_1 = 48e^{\vartheta}\bar p^2 + 12c_3\kappa' x_1$ and $k_0$ collects the other three parts, divided by $\dim G\, c_3^2|u||u'|$, which is possible as $c_3 \ge 1$. At $\vartheta = \vartheta' = 0.197$, with $\vartheta'' = 0.6$, these give $b \le 0.862$, $g_0, h_0, g_1 \le 17.37$, $y \le 14.96$, $\bar s, \bar p \le 5.80$, $g_2 \le 7.41$, $k_1 \le 1966$ and $k_0 \le 4.16\cdot 10^9$; and since $\lambda \ge 60c_6 = 4.086\cdot 10^7 \dim G\, c_3^2$ by Proposition 128, $K \le 2068\lambda$.

*Decay.* In the proof of Theorem 131, replace the bound on $e^{\alpha\varphi}\,\mathrm{Hess}\,H_n e^{-\alpha\varphi} - \mathrm{Hess}\,H_n$ by the following. Since $D\tau$ and $\Delta_n$ are block diagonal and symmetric and commute with $w = e^{\alpha\varphi}$, Lemma 132(ii) with $\vartheta = \alpha$ gives

$$\left\|\mathrm{Re}(\mathrm{Hess}\,H_n)^w - \mathrm{Hess}\,H_n\right\| \le 2\frac{\alpha^2}{\vartheta'^2}K \le \frac{\lambda}{120} = \frac{m}{2}$$

for $\alpha \le \vartheta'(240\cdot 2068)^{-1/2}$, which holds for $\alpha \le 2.5\cdot 10^{-4}$. Then $\mathrm{Re}\langle u, \mathscr{A}_w u\rangle \ge \frac{m}{2}\|u\|^2$ as before, and (ii) follows. For (iii), $\nabla' F = \bar P^* g$ with $g$ supported on $X_1$ and $|g_a| \le 2$, so with $\varphi = d(\cdot, X_2)$,

$$|e^{-\alpha\varphi}\nabla' F| = |(\bar P^*)^{w^{-1}}e^{-\alpha\varphi}g| \le 2\bar p_\alpha |X_1|^{1/2}e^{-\alpha d(X_1,X_2)},$$

where $\bar p_\alpha = \|\bar P\|_{(\alpha)} \le 4.8$ at $\alpha = 2.5\cdot 10^{-4}$, and $|e^{\alpha\varphi}\nabla' G| \le 2\bar p_\alpha |X_2|^{1/2}$. Insert these in (ii): $c_9 = 480\bar p_\alpha^2 \le 1.2\cdot 10^4$. □

The chart misses only points of the fibre at which some plaquette near $W$ is not small, and under the Wilson measure such points carry little mass on most fibres.

**Theorem 134.** (i) *Assume the hypotheses of Theorem 131 and $s \ge 1.24\cdot 10^5\kappa$. Every $U' \in F$ with $\delta(U'_p, I) \le \kappa$ for every plaquette $p$ with a vertex in a block of $W^+$ lies in $\Phi(\Omega_r)$. Hence the constrained law gives the complement of $\Phi(\Omega_r)$ at most the mass of the points of $F$ at which such a plaquette exceeds $\kappa$.*

(ii) *Let $G = \mathrm{SU}(2)$, let the side $2L$ be a power of two, let $\mu$ be the Wilson measure in the tree gauge at $\beta_W \ge 1$, so that $\lambda = \beta_W/4$, and let $E$ be the set of fields with $\delta(U_p, I) \le \kappa$ for every plaquette with a vertex in a block of $W^+$. For $\mu$-almost every $U$ let $\mu_U$ be the constrained law on the fibre $F(U)$ through $U$. Then*

$$\mathbb{E}_\mu\left[\mu_U(E^c)\right] = \mu(E^c) \le 10^4|W|\,q_\kappa(\beta_W),$$

*with $q_\kappa$ as in Theorem 147. If $\kappa$, $s$ and $\lambda$ satisfy the hypotheses of* (i) *and $0 < \varepsilon < 1$, the fields $U \in E$ with $\mu_U(F(U)\setminus\Phi_U(\Omega_r)) \le \varepsilon$, where $\Phi_U$ is the chart at $U$, have probability at least $1 - (1+\varepsilon^{-1})10^4|W|q_\kappa(\beta_W)$.*

(iii) *At every such $U$, for $X_1, X_2 \subset A_W$ and $\tilde F, \tilde G$ as in Theorem 131*(iii)*,*

$$|\mathrm{Cov}_{\mu_U}(\tilde F, \tilde G)| \le \Big(\frac{4c_9}{\beta_W}e^{-\alpha_G d(X_1,X_2)} + 30\varepsilon\Big)|X_1|\,|X_2|.$$

*Proof.* (i) The fields $U$ and $U'$ satisfy the hypothesis of Lemma 124 and agree off $A_W$, in particular on the tree links. An active link $e$ with base in a block $B(y)$, $y \in W$, either lies inside $B(y)$ or leaves it in a direction $\mu$. In the first case part (i) of the lemma, at $U$ and at $U'$, gives $\delta(U_e, U'_e) \le 6\kappa$. In the second, let $v$ and $v'$ be the path links of $V_{y,\mu}$ at $U$ and at $U'$. Part (iv), at both fields, writes the common cut-off average as $ve^M = v'e^{M'}$ with $\|M\|, \|M'\| \le 8\pi\kappa$, so by bi-invariance and Lemma 116

$$\delta(v,v') = \delta\left(e^M e^{-M'}, I\right) \le \|M\| + \|M'\| \le 16\pi\kappa < 50.3\kappa,$$

and part (i) gives $\delta(U_e, U'_e) \le \delta(U_e, v) + \delta(v, v') + \delta(v', U'_e) \le 92.3\kappa$. Let $X_e = \log(U_e^{-1}U'_e)$. By Lemma 116, $\|X_e\| \le \frac{\pi}{3}92.3\kappa < 96.7\kappa \le s/1280$ for every $e \in A_W$. Theorem 127(i) with $s/32$ in place of $s$, whose chart is the restriction of $\Phi$ by the uniqueness of the fixed point in its proof, puts $U' = Ue^X$ in $\Phi(K_{s/32})$, and $K_{s/32} \subset \Omega_r$ by Theorem 131(i).

(ii) The tree links span every block, so for every field there is exactly one gauge transformation, equal to $I$ at the corner of every block, that makes the tree links equal to $I$. Given the tree links, it multiplies each active link on both sides by elements that do not depend on that link, which preserves Haar measure, and $S$ is gauge invariant; so in the tree gauge the active links have density proportional to $e^{-\beta_W S}$ with respect to product Haar measure. Since $\beta_W S = \frac{\beta_W}{4}\sum_p \delta(U_p, I)^2$, conditionally on the links off $A_W$ the links of $A_W$ have density proportional to $e^{-\lambda Q}$, and conditioning further on the averages gives the constrained law. So $\mu_U$ is a version of the conditional law of $\mu$ given the links off $A_W$ and the averages, and the identity is the tower property of conditional expectation. If $U \notin E$, some plaquette with a vertex in a block $B(y')$, $y' \in W^+$, is bad for $\rho = \kappa$. Its corner $z$ differs from that vertex by an element of $\{0,1\}^4$, so $z$ lies in a block at $\ell^\infty$ distance at most two from $W$, and the plaquette lies in the unit cube $\mathscr{C}_z$. There are at most $5^4|W|$ such blocks, with 16 sites each, and $E$ is gauge invariant, so Theorem 147 with $|A| = 1$ gives $\mu(E^c) \le 10^4|W|q_\kappa(\beta_W)$. For $U \in E$, part (i) gives $\mu_U(F(U)\setminus\Phi_U(\Omega_r)) \le \mu_U(E^c)$, so the exceptional fields lie in $E^c \cup \{\mu_U(E^c) > \varepsilon\}$, which by Markov's inequality has probability at most $\mu(E^c)(1+\varepsilon^{-1})$.

(iii) Let $A = \Phi_U(\Omega_r)$, $m = \mu_U$, $m_A = m(\cdot \mid A)$ and $\varepsilon' = m(A^c) \le \varepsilon$. By Theorem 131(iii) with $\lambda = \beta_W/4$, $|\mathrm{Cov}_{m_A}(\tilde F, \tilde G)| \le \frac{4c_9}{\beta_W}|X_1||X_2|e^{-\alpha_G d(X_1,X_2)}$. The group SU(2)

has diameter $\sqrt{2}\pi$, and $\tilde F$ has gradient at most 1 in each link, so after subtracting a constant, which changes no covariance, $\|\tilde F\|_\infty \le \frac{\sqrt{2}\pi}{2}|X_1| \le 2.23|X_1|$, and likewise for $\tilde G$. Writing $m = (1-\varepsilon')m_A + \varepsilon' m_{A^c}$ gives $|m(fg) - m_A(fg)| \le 2\varepsilon'\|f\|_\infty\|g\|_\infty$ and $|m(f) - m_A(f)| \le 2\varepsilon'\|f\|_\infty$ for bounded $f, g$, so

$$|\mathrm{Cov}_m(f,g) - \mathrm{Cov}_{m_A}(f,g)| \le 6\varepsilon'\|f\|_\infty\|g\|_\infty,$$

and $6 \cdot 2.23^2 < 30$. □

So on a fixed set of blocks the constrained law of the SU(2) Wilson measure decorrelates, coarea factor and curvature of the fibre included, on every fibre outside a set of fibres whose probability tends to zero as $\beta_W \to \infty$. The bound grows linearly in $|W|$, which is harmless for the fixed cubes of the cube condition, but the thresholds are far from the one-loop range: with the constants of Proposition 128 the hypotheses force $\kappa + s \le 6.0 \cdot 10^{-13}$ and $s \ge 1.24 \cdot 10^5 \kappa$, so $\kappa < 4.9 \cdot 10^{-18}$, and $q_\kappa(\beta_W) < 1$ needs $\beta_W \kappa^2 > 96(\log \beta_W + C_0)$, so $\beta_W$ exceeds $3.8 \cdot 10^{38}$ (Fig. 18).

The mass off the chart can also be bounded on a single fibre, by a Peierls argument inside the fibre. Two properties of the cut-off average make this possible: along a fixed set of links it is a submersion with a constant that does not depend on the field, and its law under Haar measure on those links has a bounded density.

Fix $y \in W$ and a direction $\mu$. Write the offsets of the sites of $B(y)$ in the three directions other than $\mu$ as $a \in \{0,1\}^3$, and let $x_a$ and $x'_a$ be the sites of offset $a$ with $o_\mu = 0$ and $o_\mu = 1$. Let $\ell_a = (x'_a, \mu)$, which joins $B(y)$ to $B(y + e_\mu)$, and let $u_a = (x_a, \mu)$ and $w_a = (x'_a + e_\mu, \mu)$, which lie inside $B(y)$ and $B(y + e_\mu)$. The links $u_0$ and $w_0$ are tree links and $\ell_0 = d_k$ is the path link, so in the tree gauge

$$P_{2y} = \ell_0, \qquad P_{x_a} = u_a \ell_a, \qquad P_{x'_a} = \ell_a w_a,$$

and $v_x = P_{2y}^{-1} P_x$ equals $I$ at the two sites of offset 0. The path of $V^\sharp_{y',\nu}$ from a site $x$ consists of the link $(x,\nu)$ and the next link $(x + e_\nu, \nu)$, so $\ell_a$, whose base has $o_\mu = 1$, enters only the paths from $x_a$ and $x'_a$ of the pair $(y,\mu)$. We call $\ell_0, \dots, \ell_7$ the *private links* of $(y,\mu)$. The 32 links with $o_\mu = 1$ of $B(y)$ are the private links of the four pairs $(y,\mu)$; the other 17 active links of $B(y)$ lie inside it.

Let

$$\alpha = 2 \arcsin \frac{1}{2\sqrt{2}} = 0.7227\ldots, \qquad \Lambda = \frac{\alpha}{\sin \alpha} = 1.0926\ldots,$$

so that $1 - \frac{7}{8}\Lambda > 0.0439$. Two facts about $\mathrm{ad}X$ for $X \in \mathfrak{g}$, which is skew-adjoint for the Hilbert–Schmidt product, will be used throughout. If the eigenvalues $i\psi$ of $\mathrm{ad}X$ satisfy $|\psi| < 2\alpha$, then

$$\log\left(e^{tZ} e^{X}\right) = X + t L_X Z + O(t^2),$$
$$L_X = \frac{\mathrm{ad}X}{e^{\mathrm{ad}X} - 1}, \qquad \|L_X\| \le \Lambda,$$

since the eigenvalues of $L_X$ have modulus $\frac{|\psi|/2}{\sin(|\psi|/2)}$, which increases with $|\psi|$. For every $Y \in \mathfrak{g}$,

$$\frac{d}{dt} e^{Y+tZ}\Big|_{t=0} e^{-Y} = E_Y Z,$$
$$E_Y = \frac{e^{\mathrm{ad}Y} - 1}{\mathrm{ad}Y}, \qquad \|E_Y\| \le 1,$$

because $|e^{i\psi} - 1| \le |\psi|$; in particular $d(e^A, e^B) \le \|A - B\|$ for the Riemannian distance $d$ of the bi-invariant metric, along the curve $e^{A + t(B-A)}$. If $\delta(v, I) < 1$ and $e^{i\theta_j}$, $|\theta_j| \le \pi$, are the eigenvalues of $v$, then $\sum_j 4\sin^2(\theta_j/2) < 1$; each $|\theta_j|$ is less than $\frac{\pi}{3}$, where $\sin^2(t/2)$ is convex, so $|\theta_j| + |\theta_k| < 2\alpha$ for all $j, k$, and $X = \log v$ satisfies the hypothesis on $\mathrm{ad}X$.

**Lemma 135.** *Let $\sigma_0 = (1 - \frac{7}{8}\Lambda)/\sqrt{2} > 0.031$. At every field at which $V^\sharp_{y,\mu}$ is differentiable, its derivative $J : \mathfrak{g}^8 \to \mathfrak{g}$ with respect to its eight private links, right trivialised, satisfies $JJ^* \ge \sigma_0^2$. The fields at which it is not differentiable form a null set for Haar measure.*

*Proof.* Write $m = \max_x \delta(v_x, I)$ and $\chi = \phi(m/\rho_G)$, so that $V^\sharp = P_{2y} e^{\chi N}$ with $N = \frac{1}{16}\sum_x \log v_x$. Where $m > \rho_G$, the cut-off vanishes near the field, $V^\sharp = \ell_0$, and the derivative along $\ell_0$ is the identity. Suppose $m < \rho_G$, and that the maximum is attained at one path $x_*$, after identifying identical path functions, and $m \ne \rho_G/2$. Copies that coincide because $u_a, w_a$ are tree links are counted together. Other ties lie in finitely many proper real-analytic subsets, a null set. The variation below fixes both copies of an identical pair, since then $u_b = w_b = I$ and $A - 1 = 0$. Then every $\delta(v_x, I)$ is less than 1, the logarithms are defined, $\chi$ is differentiable, and every variation that fixes $v_{x_*}$ to first order fixes $\chi$ to first order. The site $x_*$ has offset $b \ne 0$.

Fix $\zeta \in \mathfrak{g}$ and put $\eta = \mathrm{Ad}(\ell_0)\zeta$. Move $\ell_0 \mapsto e^{t\eta}\ell_0$, and, if $x_* = x'_b$, $\ell_b \mapsto e^{t\eta}\ell_b$, or, if $x_* = x_b$, $\ell_b \mapsto e^{t\xi}\ell_b$ with $\xi = \mathrm{Ad}(u_b^{-1})\eta$, so that $u_b\ell_b \mapsto e^{t\eta}u_b\ell_b$. In both cases $P_{2y}$ and $P_{x_*}$ are multiplied on the left by $e^{t\eta}$, so $v_{x_*}$ and $\chi$ are fixed to first order. The other site $\bar x$ of offset $b$ has $P_{\bar x} \mapsto e^{t\,\mathrm{Ad}(u_b^{\pm 1})\eta} P_{\bar x}$, and the paths from the remaining twelve sites with $v_x \ne I$ do not move. With the isometry $A = \mathrm{Ad}(\ell_0^{-1} u_b^{\pm 1} \ell_0)$,

$$v_{\bar x} \mapsto e^{t(A-1)\zeta} v_{\bar x} + O(t^2), \qquad v_x \mapsto e^{-t\zeta} v_x + O(t^2)$$

for the twelve others, so $N$ moves by $t\,\delta N$ with $16\,\delta N = L_{X_{\bar x}}(A-1)\zeta - \sum_x L_{X_x}\zeta$, twelve terms, and $\|\delta N\| \le \frac{14}{16}\Lambda\|\zeta\|$. Since $\chi$ is fixed,

$$\frac{d}{dt} V^\sharp (V^\sharp)^{-1} = \eta + \mathrm{Ad}(\ell_0) E_{\chi N}(\chi\,\delta N) = \mathrm{Ad}(\ell_0)\, T\zeta,$$

with $\|T\zeta - \zeta\| \le \frac{7}{8}\Lambda\|\zeta\|$, as $\chi \le 1$. The variation is $K\zeta$ for a linear map $K : \mathfrak{g} \to \mathfrak{g}^8$ with two nonzero components of norm $\|\zeta\|$, so $\|K\| = \sqrt{2}$ and $JK = \mathrm{Ad}(\ell_0)T$. For a unit vector $e \in \mathfrak{g}$,

$$\|J^* e\| \ge \frac{\|K^* J^* e\|}{\sqrt{2}} = \frac{\|T^* \mathrm{Ad}(\ell_0)^* e\|}{\sqrt{2}} \ge \frac{1 - \frac{7}{8}\Lambda}{\sqrt{2}} = \sigma_0,$$

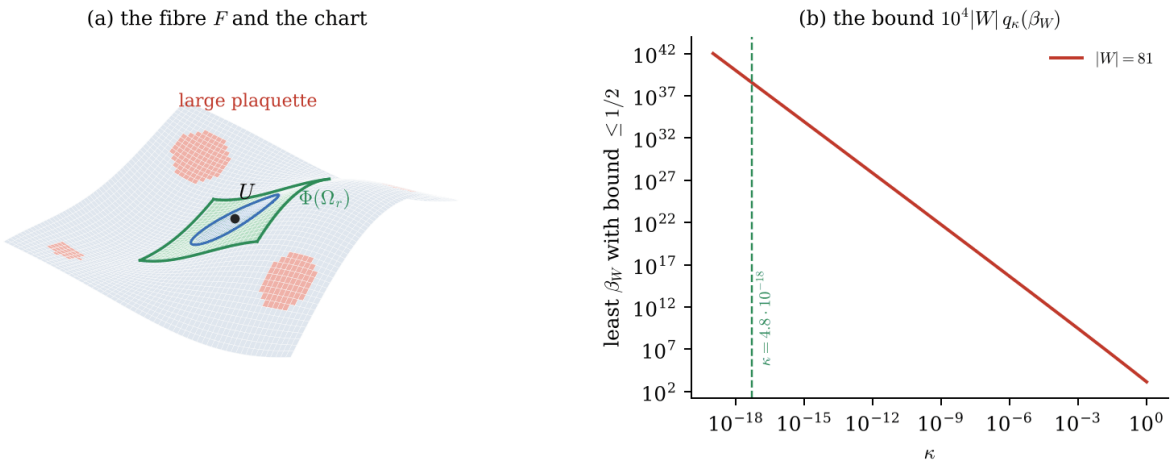



FIG. 18. Theorem 134. (a) A fibre $F$ of the cut-off average, drawn as a surface: the product chart $\Phi(\Omega_r)$ around the background $U$ (green) contains every point of $F$ whose plaquettes near $W$ are at most $\kappa$ (inside the blue curve), and the rest of $F$ meets only points with a large plaquette (red). (b) The least $\beta_W$ at which the bound $10^4|W|q_\kappa(\beta_W)$ is at most $\frac{1}{2}$, for the 81 blocks of a cube of side three; it grows like $\kappa^{-2}\log(1/\kappa)$, and at the largest $\kappa$ the hypotheses allow it is about $4.0\cdot 10^{38}$.

since the least singular value of $T$ is at least $1-\frac{7}{8}\Lambda$. At the other points of differentiability the same bound follows by continuity: the cut-off has finitely many smooth branch formulas, at such a point the formulas of the adjacent full-dimensional branches agree with its differential, and the lower bound on the singular value is preserved under limits. □

**Corollary 136.** *Let $C = (V^\sharp_{y,\mu})_{y\in W,\mu}$, as a function of the links of $A_W$, the other links being fixed. Wherever $C$ is differentiable,*

$$\det\bigl(DC\,DC^*\bigr)^{-1/2} \leq \sigma_0^{-4|W|\dim G} < 33^{4|W|\dim G}.$$

*Proof.* The private links of distinct pairs are distinct, and each enters only the average of its pair, so the columns of $DC$ that belong to private links form a block diagonal matrix with the blocks of Lemma 135. Hence $DC\,DC^* \geq \bigoplus_k J_k J_k^* \geq \sigma_0^2$. □

The coarea factor of the constrained law is therefore bounded on the whole fibre by a constant per block, for every value of the cut-off and however far the links inside the blocks are from $I$. The volume of the fibre is bounded through the law of the averages. From here on $G = \mathrm{SU}(2)$, which with the metric $\|X\|^2 = \mathrm{Tr}(X^*X)$ is the round sphere of radius $\sqrt{2}$, $\rho_G = 1$, and $h$ is normalised Haar measure.

**Lemma 137.** *Let $G = \mathrm{SU}(2)$, fix every link other than the private links of $(y,\mu)$, and let these eight links be independent and Haar distributed. Then the law of $V^\sharp_{y,\mu}$ has a density with respect to $h$ that is at most*

$$M = 1 + 2\Bigl(1-\frac{7}{8}\Lambda\Bigr)^{-3} < 2.4\cdot 10^4$$

*at every point, in the version given by the formula (23) of the proof.*

*Proof.* Let $g = \ell_0$, and let $m_1$ and $m_2$ be the largest values of $\delta(v_{x_a}, I)$ and of $\delta(v_{x'_a}, I)$. Up to the null set $m = 1$, split $G^8$ into the three regions

$$R_0 = \{m > 1\},$$
$$R_1 = \{m_1 \geq m_2,\ m_1 < 1\}, \qquad R_2 = \{m_2 > m_1,\ m_2 < 1\};$$

ties between the two families, including two path functions that coincide identically, belong to $R_1$. On $R_0$, $\chi = 0$ and $V^\sharp = g$.

On $R_1$ use the coordinates $(g, h_1, \dots, h_7)$ with $h_a = v_{x_a} = g^{-1}u_a\ell_a$. For fixed $g$ the map $\ell_a \mapsto h_a$ is a left translation, so these coordinates carry product Haar measure, and

$$v_{x'_a} = g^{-1}\ell_a w_a = g^{-1}u_a^{-1}g\,h_a w_a.$$

Fix $h$ and put $m_h = \max_a \delta(h_a, I) < 1$. The set $S_h$ of $g$ with $(g,h) \in R_1$ is the measurable set where $\delta(v_{x'_a}, I) \leq m_h$ for every $a$; on it $m = m_h$, so $\chi = \phi(m_h)$ does not depend on $g$, and $V^\sharp = g\,e^{\chi N(g)}$ with $N(g) = \frac{1}{16}\sum_a(\log h_a + \log v_{x'_a})$. The set $\{\delta(v,I) < 1\}$ is the ball of angular radius $\alpha$ about $I$ on the sphere, which is geodesically convex, and on it $\|\log v - \log v'\| \leq \Lambda\, d(v,v')$. By bi-invariance $d(g^{-1}u^{-1}g, g'^{-1}u^{-1}g') \leq 2d(g,g')$, so for $g, g' \in S_h$

$$\|N(g) - N(g')\| \leq \frac{7}{16}\,2\Lambda\, d(g,g') = \frac{7}{8}\Lambda\, d(g,g').$$

If $V^\sharp(g) = V^\sharp(g') = c$, then $g = c\,e^{-\chi N(g)}$ and $g' = c\,e^{-\chi N(g')}$, so $d(g,g') \leq \frac{7}{8}\Lambda\, d(g,g')$ and $g = g'$: the map $g \mapsto V^\sharp(g)$ is injective on $S_h$. Its derivative is computed as in the proof of Lemma 135: with $\eta = \mathrm{Ad}(g)\zeta$ the sites $x'_a$ move by $e^{t(A_a - 1)\zeta}$, $A_a = \mathrm{Ad}(g^{-1}u_a^{-1}g)$, the sites $x_a$ do not move, and the right trivialised derivative is $\mathrm{Ad}(g)T$ with $\|T - 1\| \leq \frac{7}{16}\,2\Lambda = \frac{7}{8}\Lambda$, so its determinant is at least $(1-\frac{7}{8}\Lambda)^3$ in modulus. On $R_2$ the coordinates $(g, h'_1, \dots, h'_7)$ with $h'_a = v_{x'_a} = g^{-1}\ell_a w_a$ and $v_{x_a} = g^{-1}u_a g\,h'_a w_a^{-1}$ give the same conclusions.

For fixed $h$, the smooth formula with $\chi = \phi(m_h)$ fixed extends to a neighbourhood of each point of $S_h$, where no logarithm meets $-I$; its restriction to the measurable set $S_h$ is injective and has the Jacobian bounds above. The change of variables formula for measurable restrictions therefore applies, even if part of the boundary of $S_h$ has positive measure, and likewise on $R_2$. The law of $V^\sharp$ has the density

$$\rho(c) = h^{\otimes 7}\big(\{\ell : (c,\ell) \in R_0\}\big) + \sum_{i=1,2} \int \sum_{g \in S_h^{(i)},\ V^\sharp = c} \frac{dh^{\otimes 7}}{|\det DV^\sharp(g)|}, \quad (23)$$

the inner sum having at most one term, and $\rho(c) \le 1 + 2(1 - \frac{7}{8}\Lambda)^{-3}$. Finally $1 - \frac{7}{8}\Lambda = 0.04391\ldots$ and $M = 23621\ldots$. □

The four averages of a block depend on disjoint sets of private links, and on the links inside the blocks, which are private to none. So, given the links inside the blocks of $W$, the averages $V_k^\sharp$ are independent, and the formula (23) for each pair defines, for every value $c$ of $C$, a measure $\nu_c$ on the fibre $F_c = C^{-1}(c)$: in the coordinates of the proof of Lemma 137, chosen for each pair according to its region, it is the product Haar measure of the links inside the blocks and of the seven coordinates other than $g$, divided by the product of the determinants $|\det DV_k^\sharp(g)|$. Then $h = \int \nu_c\, dh(c)$ on $G^{A_W}$, so $\nu_c$ is a version of the conditional law, and

$$\nu_c(F_c) \le M^{4|W|} \qquad \text{for every } c. \quad (24)$$

Where $C$ is differentiable,

$$\nu_c = \mathrm{vol}(G)^{-45|W|} \det\big(DC\,DC^*\big)^{-1/2}\, d\mathscr{H}.$$

Indeed, if $C(z,g) = c$ is solved for $g$ by $g = \psi(z)$, then $DC = \partial_g C\,(-D\psi,\, 1)$, so

$$\det\big(DC\,DC^*\big)^{1/2} = |\det \partial_g C|\, \det\big(1 + D\psi^* D\psi\big)^{1/2},$$

and the second factor is the Jacobian of the graph of $\psi$. So $\nu_c$ is the coarea version on the charts of Theorems 127 and 131, where $C$ is smooth, and we use it on the whole fibre.

**Proposition 138.** *Let $G = \mathrm{SU}(2)$, $\lambda > 0$ and $0 < K \le 10^{-8}$, and let $n_P \le 384|W|$ be the number of plaquettes that contain a link of $A_W$. Suppose the fibre $F$ contains a field $U$ with $\delta(U_p, I) \le \kappa$ for every plaquette $p$ with a vertex in a block of $W^+$, where $\kappa \le K/(4\sqrt{n_P})$. Then the constrained law $\mu_U$ on $F$ gives the set of fields with a plaquette $p$ with a vertex in a block of $W^+$ and $\delta(U_p', I) > K$ mass at most*

$$\varepsilon_W(K,\lambda) = \Big(\frac{2\cdot 10^9\, n_P^{3/2}}{K^3}\Big)^{45|W|} e^{-3\lambda K^2/4}.$$

*Proof.* Let $c = C(U)$, let $B$ be the set in question, and let $Q_W$ be the sum of $\delta(U_p', I)^2$ over the $n_P$ plaquettes that contain a link of $A_W$; on $F$ it differs from $Q$ by a constant, so $\mu_U = e^{-\lambda Q_W}\nu_c / \nu_c(e^{-\lambda Q_W})$. A plaquette that contains no link of $A_W$ has $U_p' = U_p$ for $U' \in F$, so on $B$ the large plaquette is one of the $n_P$ and $Q_W > K^2$. By (24),

$$\int_B e^{-\lambda Q_W}\, d\nu_c \le M^{4|W|} e^{-\lambda K^2}.$$

For the denominator we solve the constraints for the path links. Let $L' = A_W \setminus \{d_k\}$, $\rho = K/(360\sqrt{n_P})$ and $R = 44\rho$, and let $D$ be the set of $z \in G^{L'}$ with $d(z_e, U_e) \le \rho$ for every $e \in L'$. The field $U$ satisfies the hypothesis of Lemma 124, as $\kappa \le K/4$, and part (i) of the lemma gives $\delta(U_e, I) \le 3\kappa$ for the links inside the blocks and $\delta(U_e, U_{d_k}) \le 21\kappa$ for the private links of $k$, so $\delta(v_x, I) \le 24\kappa$ and, by Lemma 116, $d(v_x, I) \le 8\pi\kappa$ at every site of every pair. Fix $z \in D$ and a pair $k$, and let $\ell$ range over the ball $B_k$ of radius $R$ about $U_{d_k}$. Each $v_x$ is $\ell^{-1}$ times a product of at most two links of $L'$ and fixed links, so at the field with links $z$ and $\ell$ it lies within $R + 2\rho$ of its value at $U$, and within $8\pi\kappa + 46\rho < 0.01$ of $I$. Hence the cut-off equals 1 near this field and $V_k^\sharp = \ell\, e^{N_k(\ell)}$, with $N_k$ depending on $z$ and $\ell$. Each $v_x$ is moved by $d(\ell, \ell')$ when $\ell$ is replaced by $\ell'$, so the map $\Psi(\ell) = c_k\, e^{-N_k(\ell)}$ satisfies $d(\Psi(\ell), \Psi(\ell')) \le \frac{7}{8}\Lambda\, d(\ell, \ell')$. At $U$ the path link solves $U_{d_k} = c_k e^{-N_k}$, and the sites move by at most $R + 2\rho$ between $U$ and the field with links $z$ and $\ell$, so

$$d\big(\Psi(\ell), U_{d_k}\big) \le \frac{7}{8}\Lambda\,(R + 2\rho) < 0.9561 \cdot 46\rho < R.$$

By the contraction principle $\Psi$ has a unique fixed point $\psi_k(z) \in B_k$. The path link $d_k$ is private to $k$, so the field $\Gamma(z)$ with links $z$ on $L'$ and $\psi_k(z)$ on $d_k$ lies in $F$, and $\Gamma$ is injective.

Near $\Gamma(D)$ every cut-off equals 1, so $C$ is smooth, and $\partial_g C$ is block diagonal with the blocks $\partial_{d_k} V_k^\sharp$. Moving $d_k \mapsto e^{t\eta} d_k$ multiplies every $v_x$ of $k$ by $e^{-t\zeta}$, $\zeta = \mathrm{Ad}(d_k^{-1})\eta$, so the block is $\mathrm{Ad}(d_k)(1 - E_{N_k}\frac{1}{16}\sum_x L_{X_x})$, with determinant at most $(1 + \frac{7}{8}\Lambda)^3 < 1.957^3$ in modulus. By the identity above, $\nu_c$ restricted to $\Gamma(D)$ is the image of $h^{\otimes L'}$ on $D$ divided by $|\det \partial_g C|$, so

$$\nu_c\big(\Gamma(D)\big) \ge 1.957^{-12|W|} h(B_\rho)^{45|W|},$$
$$h(B_\rho) = \frac{2\theta - \sin 2\theta}{2\pi} \ge \frac{\rho^3}{14},$$

with $\theta = \rho/\sqrt{2}$ the angular radius, as $2\theta - \sin 2\theta \ge \frac{4}{3}\theta^3(1 - \frac{\theta^2}{5})$. A plaquette has at most two path links: two path links $(2y + e_\mu, \mu)$ and $(2y' + e_\mu, \mu)$ of one direction are never opposite sides of a plaquette, since their bases have exactly one odd coordinate, $\mu$, and opposite sides differ by $e_\rho$, $\rho \ne \mu$; two of different directions can be adjacent sides. So on $\Gamma(D)$ each plaquette moves by at most $2R + 2\rho = 90\rho$ and $\delta(U_p', I) \le \kappa + 90\rho \le K/(2\sqrt{n_P})$. Hence $Q_W \le K^2/4$ there and

$$\int_F e^{-\lambda Q_W}\, d\nu_c \ge e^{-\lambda K^2/4}\, 1.957^{-12|W|} \Big(\frac{\rho^3}{14}\Big)^{45|W|}.$$

Dividing,

$$\mu_U(B) \le \big(M^4 1.957^{12}\big)^{|W|} \Big(\frac{14\cdot 360^3 n_P^{3/2}}{K^3}\Big)^{45|W|} e^{-3\lambda K^2/4},$$

and $(M^4 1.957^{12})^{1/45} < 2.93$, $2.93 \cdot 14 \cdot 360^3 < 2 \cdot 10^9$. Finally $n_P \le 384|W|$, since each plaquette that contains a link of $A_W$ has the base of that link, a site of a block of $W$, as a vertex, and each site is a vertex of 24 plaquettes. □

**Corollary 139.** *In addition to the hypotheses of Proposition 138, assume those of Theorem 131 with $K$ in place of $\kappa$, and $s \ge 1.24 \cdot 10^5 K$. Then $\mu_U(F \setminus \Phi_U(\Omega_r)) \le \varepsilon_W(K,\lambda)$, and for $X_1, X_2 \subset A_W$ and $\tilde F, \tilde G$ as in Theorem 131*(iii)*,*

$$|\mathrm{Cov}_{\mu_U}(\tilde F, \tilde G)| \le \Big( \frac{c_9}{\lambda} e^{-\alpha_G d(X_1,X_2)} + 30\,\varepsilon_W(K,\lambda) \Big) |X_1|\,|X_2|.$$

*Proof.* The field $U$ satisfies the hypothesis of Lemma 124 with $K$, so Theorem 134(i) with $K$ in place of $\kappa$ puts every point of $F$ outside $\Phi_U(\Omega_r)$ in the set of Proposition 138. The covariance bound follows as in the proof of Theorem 134(iii). □

So the constrained law decorrelates on every fibre through a field whose plaquettes near $W$ are at most $K/(4\sqrt{n_P})$, with no average over fibres, once $\lambda K^2$ is a large multiple of $|W|\log(n_P/K)$; with $K < 4.9 \cdot 10^{-18}$ this needs $\beta_W$ above $1.4 \cdot 10^{39}|W|$ under the Wilson measure. The restriction is on the fibre: the links off $A_W$ are fixed on it, so a fibre whose boundary links carry a large plaquette near $W$ contains no such field, and neither may a fibre with small boundary links, and the proposition does not cover these fibres. It is a lower bound for the fibre measure near one good field against an upper bound for the whole fibre; it does not compare different fibres.

The hypothesis on $U$ enters the proof twice: through Lemma 124, to place the sixteen sites $v_x$ of every pair near $I$, and through the bound on $Q_W$ near $U$. Neither needs the links off $A_W$ or the links inside the blocks to be small, and every fibre contains a field at which the sites are exactly $I$.

**Lemma 140.** *For every value $c$ of $C$ and every choice of the links off $A_W$, the fibre $F_c$ contains a field at which $v_x = I$ for every site $x$ of every pair $(y,\mu)$, $y \in W$.*

*Proof.* Fix $\mu$. The links of $A_W$ in the direction $\mu$ are, for each $y \in W$, the links $u_a$ and $\ell_a$ of the pair $(y,\mu)$, and the links $w_a$ of that pair are the links $u_a$ of the pair $(y+e_\mu,\mu)$ when $y+e_\mu \in W$, and fixed links otherwise. Whether $u_a$ is a tree link depends only on the offset, so $u_a$ and $w_a$ are both tree links or neither. For a pair with path link $g$ and given links $w_a$ put $\ell_0 = g$ and, for $a \ne 0$,

$$\ell_a = g, \quad \text{or} \quad \ell_a = g\,w_a^{-1},\ u_a = g\,w_a\,g^{-1},$$

the first when $u_a$ is a tree link and the second otherwise. Then $v_{x_a} = g^{-1}u_a\ell_a = I$ and $v_{x'_a} = g^{-1}\ell_a w_a = I$, as $u_a = w_a = I$ in the first case. The blocks of $W$ along a line in the direction $\mu$ form runs $y_1,\dots,y_n$ of consecutive blocks. If $y_n + e_\mu \notin W$, choose the links of the pairs $(y_n,\mu),\dots,(y_1,\mu)$ in this order, with $g = c_{y_j,\mu}$, the links $w_a$ of $(y_j,\mu)$ being the fixed links for $j = n$ and the links $u_a$ just chosen for $(y_{j+1},\mu)$ otherwise. If the run closes up around the torus, put $u_a = I$ and $\ell_a = g$ for every pair of the run. At the resulting field every $v_x$ equals $I$, so the cut-off equals 1, $N = 0$ and $V^\sharp_{y,\mu} = \ell_0 = c_{y,\mu}$. The directions do not interact, since the pairs $(y,\mu)$ use only links in the direction $\mu$. □

Call a field $U \in F$ *regular* if $\delta(v_x, I) \le \frac14$ at every site $x$ of every pair, and let $q(F)$ be the infimum of $Q_W$ over the regular fields of $F$, finite by Lemma 140. Write $\mu_F = e^{-\lambda Q_W}\nu_c / \nu_c(e^{-\lambda Q_W})$ for the constrained law on $F = F_c$.

**Proposition 141.** *Let $G = \mathrm{SU}(2)$, $\lambda > 0$ and $0 < t \le 1$. For every fibre $F$, whatever the links off $A_W$, and every regular $U \in F$,*

$$\mu_F\big(\{U' \in F : Q_W(U') > Q_W(U) + t\}\big) \le \varepsilon'_W(t,\lambda),$$
$$\varepsilon'_W(t,\lambda) = \Big( \frac{3.5 \cdot 10^{11}\, n_P^3}{t^3} \Big)^{45|W|} e^{-3\lambda t/4}.$$

*In particular $\mu_F(\{Q_W > q(F) + t\}) \le \varepsilon'_W(t,\lambda)$ on every fibre.*

*Proof.* The numerator is bounded by (24) as in the proof of Proposition 138, by $M^{4|W|}e^{-\lambda(Q_W(U)+t)}$. For the denominator take $\rho = t/(2040\,n_P) \le 10^{-3}$, $R = 44\rho$, and $D$ and $B_k$ as in that proof. At the field with links $z \in D$ on $L'$ and $\ell \in B_k$ on $d_k$, each $v_x$ of the pair $k$ lies within $R + 2\rho = 46\rho < 0.043$ of its value at $U$, and $\delta \le d$, so $\delta(v_x, I) < \frac14 + 0.043 < \frac12$. Hence the cut-off equals 1 near this field, the logarithms are $\Lambda$-Lipschitz along the moves, and the contraction argument of that proof gives the injective map $\Gamma : D \to F$, near whose image $C$ is smooth with $|\det \partial_g C| \le 1.957^{3\cdot4|W|}$. So

$$\nu_c\big(\Gamma(D)\big) \ge 1.957^{-12|W|} \Big( \frac{\rho^3}{14} \Big)^{45|W|}.$$

On $\Gamma(D)$ each of the $n_P$ plaquettes moves by at most $90\rho$, as in that proof, and $\delta(U_p, I) \le 2\sqrt{2}$, so

$$\delta(U'_p, I)^2 \le \delta(U_p, I)^2 + 180 \cdot 2\sqrt{2}\rho + 90^2\rho^2 < \delta(U_p, I)^2 + 510\rho,$$

and $Q_W \le Q_W(U) + 510 n_P \rho = Q_W(U) + \frac{t}{4}$ there. Dividing as before,

$$\mu_F(\{Q_W > Q_W(U) + t\}) \le \big(M^4 1.957^{12}\big)^{|W|} \Big( \frac{14 \cdot 2040^3 n_P^3}{t^3} \Big)^{45|W|} e^{-3\lambda t/4},$$

and $2.93 \cdot 14 \cdot 2040^3 < 3.5 \cdot 10^{11}$. For the last statement apply this to regular fields with $Q_W(U)$ decreasing to $q(F)$ and use the continuity of $\varepsilon'_W$ in $t$. □

**Corollary 142.** *Let $G = \mathrm{SU}(2)$, $\lambda > 0$ and $0 < K \le 1$. Suppose the fibre $F$ contains a regular field $U$ with $\delta(U_p, I) \le K/(4\sqrt{n_P})$ on every plaquette $p$ that contains a link of $A_W$, whatever the plaquettes made of fixed links. Then $\mu_F$ gives the fields with such a plaquette above $K$ mass at most $\varepsilon'_W(\frac{15}{16}K^2, \lambda)$.*

*Proof.* At $U$, $Q_W \le K^2/16$, and a field with one of these plaquettes above $K$ has $Q_W > K^2 \ge Q_W(U) + \frac{15}{16}K^2$. Apply Proposition 141 with $t = \frac{15}{16}K^2 \le 1$. □

The field of Proposition 138 is regular, since its sites lie within $24\kappa$ of $I$, so Corollary 142 contains that proposition up to the constants and drops its hypothesis on the plaquettes made

of fixed links and on the links inside the blocks. It reaches every fibre whose boundary links carry large plaquettes, provided the sites of the averages can be made small there together with the plaquettes that move. Proposition 141 holds on every fibre with no hypothesis at all, but its reference value $q(F)$ need not be small. Where it is of order one, as it can be when the averages are far from what the boundary links suggest, the constrained law sits within $t$ of $q(F)$ but away from the charts of Theorem 131, which are built at fields with all plaquettes near $W$ small. The proposition also does not compare regular with irregular fields, at which some site is far from $I$.

Different fibres with the same links off $A_W$ are compared through the law of the averages. Call the links off $A_W$ the *exterior* $E$, write $h$ for product Haar measure on $G^{A_W}$, and let

$$\pi_E = \frac{e^{-\lambda Q_W} h}{h(e^{-\lambda Q_W})}$$

on $G^{A_W}$. Under the SU(2) Wilson measure in the tree gauge, $\pi_E$ with $\lambda = \beta_W/4$ is the conditional law of the links of $A_W$ given the exterior. Since $h = \int \nu_c \, dh(c)$,

$$\pi_E = \int \mu_{F_c} \frac{\nu_c(e^{-\lambda Q_W})}{h(e^{-\lambda Q_W})} \, dh(c),$$

so $\mu_{F_c}$ is the conditional law of $\pi_E$ given $C = c$, and the law of the fibre through a field drawn from $\pi_E$ is the law of $C$ under $\pi_E$. Let $Q_*(E)$ be the least value of $Q_W$ on $G^{A_W}$, which is attained since $G^{A_W}$ is compact.

**Proposition 143.** *Let $G = \mathrm{SU}(2)$, $\lambda > 0$ and $0 < t \le 1$. For every exterior $E$,*

$$\pi_E\big(\{Q_W > Q_*(E) + t\}\big) \le \varepsilon''_W(t,\lambda),$$
$$\varepsilon''_W(t,\lambda) = \Big(\frac{1.05 \cdot 10^7 \, n_P^3}{t^3}\Big)^{49|W|} e^{-3\lambda t/4}.$$

*In particular the law of the averages under $\pi_E$ gives the fibres $F$ with $\inf_F Q_W > Q_*(E) + t$ mass at most $\varepsilon''_W(t,\lambda)$.*

*Proof.* Since $h$ is a probability measure,

$$\int_{\{Q_W > Q_* + t\}} e^{-\lambda Q_W} \, dh \le e^{-\lambda(Q_* + t)}.$$

For the normalisation let $U_*$ be a field with $Q_W(U_*) = Q_*$, let $\rho = t/(90.6 n_P)$, and let $D$ be the set of fields with $d(U_e, U_{*,e}) \le \rho$ for each of the $49|W|$ links of $A_W$, the $45|W|$ links of $L'$ and the $4|W|$ path links. Then $h(D) = h(B_\rho)^{49|W|} \ge (\rho^3/14)^{49|W|}$. Each plaquette contains at most four links of $A_W$, so on $D$ it moves by at most $4\rho$, and as $\delta \le d$ and $\delta(U_p, I) \le 2\sqrt{2}$,

$$\delta(U_p,I)^2 \le \delta(U_{*,p},I)^2 + 16\sqrt{2}\rho + 16\rho^2 < \delta(U_{*,p},I)^2 + 22.63\rho,$$

because each link of $A_W$ lies in six plaquettes, so $n_P \ge 74$ and $\rho < 1.5 \cdot 10^{-4}$. Hence $Q_W \le Q_* + 22.63 n_P \rho < Q_* + \frac{t}{4}$ on $D$, and

$$\pi_E\big(\{Q_W > Q_* + t\}\big) \le \Big(\frac{14 \cdot 90.6^3 n_P^3}{t^3}\Big)^{49|W|} e^{-3\lambda t/4},$$

with $14 \cdot 90.6^3 < 1.05 \cdot 10^7$. A field $U$ with $Q_W(U) \le Q_* + t$ lies in its own fibre, so $\inf_{F(U)} Q_W \le Q_* + t$, which gives the last statement. □

Call a field that agrees with $E$ off $A_W$ a *completion* of $E$.

**Corollary 144.** *Let $G = \mathrm{SU}(2)$, $\lambda > 0$, $0 < K \le 10^{-8}$ and $\kappa = K/(4\sqrt{n_P})$. Suppose the exterior $E$ has a completion $U^0$ with $\delta(U^0_p, I) \le \kappa/(2\sqrt{n_P})$ for every plaquette $p$ with a vertex in a block of $W^+$. Let $\mathscr{E}$ be the set of completions $U$ with $\delta(U_p, I) \le \kappa$ for every such plaquette. Then:*

(i) $\pi_E(\mathscr{E}^c) \le \varepsilon''_W(\kappa^2/2, \lambda)$;

(ii) *every $U \in \mathscr{E}$ is regular and its fibre has $q(F(U)) < \kappa^2$, so the law of the averages under $\pi_E$ gives the fibres with $q(F) \ge \kappa^2$, among them every fibre where $q(F)$ is of order one, mass at most $\varepsilon''_W(\kappa^2/2, \lambda)$;*

(iii) *if $\lambda$, $s$ and $K$ satisfy the hypotheses of Corollary 139, then for every $U \in \mathscr{E}$ the constrained law on $F(U)$ gives the complement of the chart $\Phi_U(\Omega_r)$ at $U$ mass at most $\varepsilon_W(K, \lambda)$, and its covariances obey the bound of that corollary.*

*Proof.* (i) $Q_*(E) \le Q_W(U^0) \le n_P \kappa^2/(4 n_P) = \kappa^2/4$. If $Q_W(U) \le Q_* + \kappa^2/2$, then $Q_W(U) \le \frac{3}{4}\kappa^2$, so every plaquette that contains a link of $A_W$ has $\delta(U_p, I) < \kappa$, and a plaquette with a vertex in a block of $W^+$ that contains no link of $A_W$ has the value it has at $U^0$, at most $\kappa/(2\sqrt{n_P})$. So $\mathscr{E}^c \subset \{Q_W > Q_* + \kappa^2/2\}$, and Proposition 143 with $t = \kappa^2/2 \le 1$ applies. (ii) A field $U \in \mathscr{E}$ satisfies the hypothesis of Lemma 124, which places every site of every pair within $24\kappa < \frac{1}{4}$ of $I$, as in the proof of Proposition 138; so $U$ is regular and $q(F(U)) \le Q_W(U) < \kappa^2$. Hence a fibre with $q(F) \ge \kappa^2$ contains no field of $\mathscr{E}$, and the law of the averages gives these fibres at most the mass $\pi_E(\mathscr{E}^c)$. (iii) The fibre $F(U)$ contains the field $U$, whose plaquettes with a vertex in a block of $W^+$ are at most $\kappa = K/(4\sqrt{n_P})$, which is the hypothesis of Proposition 138; Corollary 139 applies at $U$. □

So on every exterior whose plaquettes near $W$ can all be made small by the links of $A_W$, the fibres where $q(F)$ is of order one, and more generally the fibres that contain no field with small plaquettes near $W$, carry total mass at most $\varepsilon''_W(\kappa^2/2, \lambda)$ in the law of the averages, and every other fibre has the chart decay of Corollary 139. This is a bound for every such exterior, not a mean over exteriors as in Theorem 134(ii). If $\lambda\kappa^2 \ge 196|W|\ln(8.4 \cdot 10^7 n_P^3/\kappa^6)$, then $\varepsilon''_W(\kappa^2/2, \lambda) \le e^{-\lambda\kappa^2/8}$; with $n_P \le 384|W|$ and $K = 4.8 \cdot 10^{-18}$ this holds under the Wilson measure for $\beta_W \ge 6.4 \cdot 10^{43}|W|^2(1 + \ln|W|)$. The fibres where $q(F)$ is of order one remain possible in two places: on an exterior with no completion whose plaquettes near $W$ are small, and as values of $c$ of small mass on an exterior with one.

*Remark* 145. Theorem 131 is the decay that Remark 129(b) asks for, on the chart $\Phi(\Omega_r)$ with $r = s/32$, which contains $\Phi(K_{s/32})$ and so every $Ue^X \in F$ with $\|X\|_\infty \le s/1280$. Its constants are explicit by Propositions 128 and 133: the rate $\alpha_G = 2.5 \cdot 10^{-4}$ is the same for every group and lies far below the decay seen in the computation described at the end of this remark. What remains open for the small fields:

(a) the mass of the constrained law outside $\Phi(\Omega_r)$ on the fibres that contain no field whose plaquettes near $W$ are small: Corollary 139 bounds it on every fibre that contains one, Corollary 144 shows that on every exterior with a completion whose plaquettes near $W$ are small these fibres, among them those where $q(F)$ is of order one, have mass at most $\varepsilon''_W$ in the law of the averages, and Theorem 134 bounds them in mean over all exteriors; the constrained cube condition of Remark 109(a) takes a supremum over boundary conditions, which include these fibres of small mass and the exteriors with no such completion, and on them no chart of Theorem 131 is available; by Proposition 161 these are the typical case on cubes of fixed physical size along the tuning;

(b) the passage from the decay of the constrained law on one fibre, at a fixed background, to the constrained cube condition of Remark 109(a), whose supremum runs over boundary conditions that include large fields.

On the periodic lattice of side twelve, for $\mathrm{SU}(2)$ backgrounds as in the check of Theorem 125 and $W$ a row of five blocks, the section $S$ has row sums at most 14, the blocks of $\bar{P}$ beyond the range of $S$ decay by a factor about 0.4 per unit of distance, and the blocks of the chart Hessian of $Q$ fall from 15 at distance zero to $10^{-8}$ at distance nine, a floating-point computation reproduced in the archive.

### E. Large fields under the Wilson measure

Theorem 123 needs every plaquette to be small. This subsection shows that, under the $\mathrm{SU}(2)$ Wilson measure, the plaquettes that are not small form clusters of microscopic size, and that the cut-off average departs from the average only next to them. It also shows that such plaquettes are present along the one-loop tuning, so the composition has to pass through them.

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$ with $L = 2^m \geq 4$, and let $\mu_L$ be the $\mathrm{SU}(2)$ Wilson measure on $\Lambda_L$, with density proportional to $e^{-\beta_W S(U)}$ with respect to product Haar measure, where

$$S(U) = \sum_p s_p, \qquad s_p = 1 - \tfrac{1}{2}\operatorname{Tr} U_p = \tfrac{1}{4}\,\delta(U_p, I)^2.$$

This is the Wilson measure of Sec. III on the torus, without gauge fixing; it has $4L^4$ links and $6L^4$ plaquettes. For $0 < \rho \leq 1$ call a plaquette *bad* if $\delta(U_p, I) > \rho$. The closed unit cube $\mathscr{C}_x = x + [0,1]^4$ contains 24 plaquettes. Let $B_x$ be the event that one of them is bad. A plaquette lies in exactly four unit cubes, one for each choice of the two transverse coordinates of the cube's corner.

**Lemma 146.** *Let $Z_L$ be the normalising constant of $\mu_L$ and $C_0 = \frac{1}{2} + \frac{2}{3}\log(24\pi^3)$. For $\beta_W \geq 1$,*

$$-\log Z_L \leq 6L^4(\log\beta_W + C_0), \qquad \mathbb{E}_{\mu_L} S \leq 6L^4\,\frac{\log\beta_W + C_0}{\beta_W}.$$

*Proof.* Write $U = \cos\theta\, I + \sin\theta\, J$ with $0 \leq \theta \leq \pi$ and $J^2 = -I$. The angle $\theta$ of $g^{-1}h$ is a bi-invariant distance, and under Haar measure $\theta$ has density $\frac{2}{\pi}\sin^2\theta$. Put $r = (4\sqrt{\beta_W})^{-1}$. If every link has angle at most $r$, every plaquette has angle at most $4r \leq 1$ and $s_p \leq (4r)^2/2 = 1/(2\beta_W)$, so $S \leq 3L^4$. Each link angle is at most $r$ with probability

$$\frac{2}{\pi}\int_0^r \sin^2 t\, dt \geq \frac{8r^3}{3\pi^3} = \frac{1}{24\pi^3\beta_W^{3/2}},$$

by $\sin t \geq 2t/\pi$. Hence $Z_L \geq e^{-3L^4}(24\pi^3\beta_W^{3/2})^{-4L^4}$, which is the first bound. The function $F(b) = -\log Z_L$ at the coupling $b$ is concave with $F(0) = 0$ and $F'(b) = \mathbb{E}S$, so $bF'(b) \leq F(b)$, which is the second. □

**Theorem 147.** *For $\beta_W \geq 1$, $0 < \rho \leq 1$ and every set $A$ of unit cubes,*

$$\mu_L\Big(\bigcap_{x\in A} B_x\Big) \leq q^{|A|},$$
$$q = q_\rho(\beta_W) = \min\big\{1, \exp\big(6(\log\beta_W + C_0) - \beta_W\rho^2/16\big)\big\}.$$

*Proof.* The measure $\mu_L$ is reflection positive in every hyperplane through sites [75]: the plaquettes are either in the hyperplane or on one side of it, so conditionally on the links in the hyperplane the weight factors into two mirror images, and reflection preserves Haar measure and $\operatorname{Tr} U_p$. On a torus of side $2^m$ this gives the chessboard estimate for functions of the links of closed unit cubes [42], [24, Section 5.3]:

$$\mu_L\Big(\bigcap_{x\in A} B_x\Big) \leq \mu_L\Big(\bigcap_{x\in\Lambda_L} B_x\Big)^{|A|/L^4},$$

since the reflection of $B_x$ into another cube is the event $B_y$ of that cube. If every cube is bad, there are at least $L^4/4$ bad plaquettes, so $S \geq \rho^2 L^4/16$ and the event has unnormalised weight at most $e^{-\beta_W\rho^2 L^4/16}$. Dividing by $Z_L$ and using Lemma 146 gives the bound. □

**Corollary 148.** *Join two unit cubes when their corners are at $\ell^\infty$ distance one on the torus. For an integer $r \geq 1$,*

$$\mu_L\{\text{some connected set of } r \text{ bad cubes}\} \leq L^4(6400\,q)^r.$$

*Hence, on a torus of fixed physical side $La$, if $a \to 0$ along $L = 2^m$ and $\beta_W(a) \to \infty$, the physical diameters of all connected clusters of bad cubes tend to 0 in probability, for every fixed $\rho$.*

*Proof.* A connected set of $r$ cubes containing a given cube is recovered from a walk of length $2(r-1)$ around a spanning tree, and each cube has 80 neighbours, so there are at most $80^{2(r-1)} \leq 6400^r$ such sets. Apply Theorem 147 and sum over the $L^4$ roots. A cluster of physical diameter at least $R$ contains a connected set of $\lceil R/a \rceil$ cubes, and once $6400\,q \leq \frac{1}{2}$ the bound is at most $(La/a)^4 2^{-R/a} \to 0$. □

The bad cubes control where the cut-off average differs from the average. For a coarse link $(y,\mu)$ let $\mathscr{Z}_{y,\mu}$ be the 24 unit cubes with corners $2y + te_\mu + w$, $t \in \{0,1,2\}$, $w \in \{0,1\}^4$, $w_\mu = 0$ (Fig. 19).

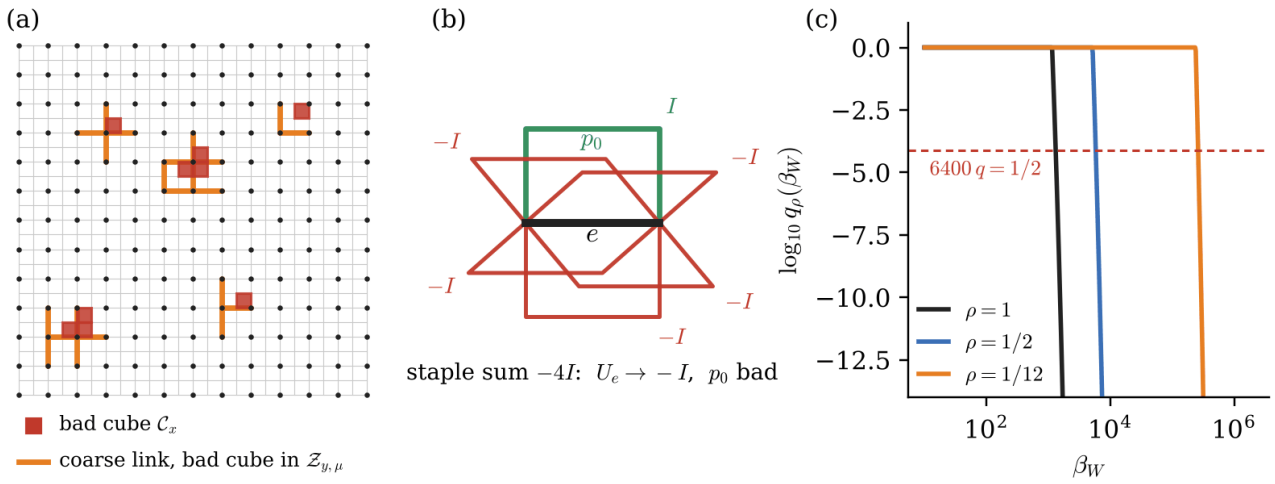



FIG. 19. Large fields under the SU(2) Wilson measure. (a) A plane section of the lattice: bad unit cubes (red) form clusters, and the cut-off average can differ from the path average only on coarse links (orange) whose 24 cubes $\mathcal{Z}_{y,\mu}$ meet a bad cube (Corollary 149). (b) The frustrated exterior of a link $e$: the staple of $p_0$ is $I$ and the other five staples are $-I$, so $U_e$ is driven to $-I$ and $p_0$ is bad. (c) The weight $q_\rho(\beta_W)$ of Theorem 147 for three values of $\rho$; below the dashed line the cluster bound of Corollary 148 is summable.

**Corollary 149.** *Let $\rho = \frac{1}{12}$. If no cube of $\mathcal{Z}_{y,\mu}$ is bad, then $\chi_{y,\mu}(U) = 1$ in the tree gauge, and $V^\sharp_{y,\mu}$ is the average of the sixteen paths in the chart at $P_{2y}$. A bad cube lies in $\mathcal{Z}_{y,\mu}$ for at most eight coarse links, so the expected fraction of coarse links with $\chi < 1$ is at most $4.5 \cdot 10^5 (\log\beta_W + C_0)/\beta_W$. Along the limits of Corollary 148, with probability tending to one these links lie within physical distance $2a$ of clusters of bad cubes of physical diameter less than any fixed $R > 0$.*

*Proof.* As in the proof of Lemma 122, $P_{2y}^{-1} P_x$ is a product of at most six conjugates of plaquette holonomies or their inverses. These plaquettes lie in planes $\{\mu, \alpha\}$ with corners $2y + te_\mu + w$ as above, and a plaquette lies in the unit cube at its corner. So $\delta(P_x, P_{2y}) \le 6\rho = \frac{1}{2}$, and $\chi = 1$. A cube with corner $z$ lies in $\mathcal{Z}_{y,\mu}$ only if $z_\mu - 2y_\mu \in \{0,1,2\}$ and $z_\alpha - 2y_\alpha \in \{0,1\}$ for $\alpha \ne \mu$, which leaves at most two coarse links in each direction. There are $L^4/4$ coarse links, the number of bad cubes is at most four times the number of bad plaquettes, and a bad plaquette has $s_p > \rho^2/4$, so by Lemma 146 the expected fraction is at most $4 \cdot 8 \cdot 4 \cdot 24 \cdot 144 (\log\beta_W + C_0)/\beta_W$. The last assertion is Corollary 148. □

Small plaquettes everywhere, which Theorem 123 assumes, are nevertheless improbable along the one-loop tuning. For SU(2), $\beta_W = 4g^{-2}$, and the one-loop coupling of Remark 109(c) at spacing $a = 1/L$ gives $\beta_W = 8b_0 \log L + O(1) = \frac{11}{3\pi^2} \log L + O(1)$.

**Proposition 150.** *Let $0 < \rho \le 1$, $v_* = \frac{2}{\pi} \int_{\arccos(3/4)}^{\pi/3} \sin^2 t\, dt$ and $p = v_* e^{-3\beta_W}$. Conditionally on all other links, a plaquette through a link $e$ is bad with probability at least $p$. Hence*

$$\mu_L\{\text{no plaquette is bad}\} \le e^{-\lfloor L/3 \rfloor^4 p},$$

*which tends to 0 when $\beta_W \le B \log L + O(1)$ with $B < 4/3$, in particular along the one-loop tuning.*

*Proof.* Identify SU(2) with the unit quaternions. Conditionally on the other links, $U_e$ has density proportional to $e^{\kappa h \cdot u}$ with $0 \le \kappa \le 6\beta_W$ and $|h| = 1$, the six staples contributing at most $\beta_W$ each. The plaquette holonomy is $U_e$ or its inverse multiplied by a fixed element, so it has a density of the same form. Writing the density as a superposition of indicators of caps centred at $h$, the probability of a cap about $I$ is largest when $h = I$, and for $h = I$ it increases with $\kappa$, because its derivative is the covariance of an increasing function of $\mathrm{Tr}\, U$ with $\frac{1}{2}\mathrm{Tr}\, U$. So the probability of being bad is at least that of the angle exceeding $\arccos(1 - \rho^2/4) \le \arccos(3/4)$ under the density proportional to $e^{6\beta_W \cos t} \sin^2 t$, which is at least $p$: bound the numerator from below on $[\arccos(3/4), \pi/3]$, where $\cos t \ge \frac{1}{2}$, and the denominator by $e^{6\beta_W} \pi/2$.

Let $M = \lfloor L/3 \rfloor$. Choose the links in direction 0 at the sites $3z$, $z \in \{0, \dots, M-1\}^4$, and their plaquettes in the plane $\{0,1\}$. No chosen plaquette contains another chosen link. Removing one chosen link at a time therefore bounds the probability that all chosen plaquettes are good by $(1-p)^{M^4}$. With $\beta_W = B \log L + O(1)$ the exponent is of order $L^{4-3B}$. □

The composition of Theorem 123 is local, so it continues across the scales away from the defects, and the defects spread by at most three spacings of the current scale at each step. For a coarse plaquette $P$ at $y$ let $\mathcal{B}_P$ be the fine box $2y + [0,4]^4$.

**Lemma 151.** *Let $U$ be a $G$-valued field in the tree gauge and $P$ a coarse plaquette. If every plaquette $p$ in $\mathcal{B}_P$ has $\delta(U_p, I) \le \rho_G$ and $\|\Theta_p\| \le \eta \le \rho_G/1000$, then $\chi = 1$ on the four sides of $P$, $\delta(V^\sharp(U)_P, I) \le 0.03\,\rho_G$, the inequalities (22) hold for $P$, and $\|\log V^\sharp(U)_P\| \le 4(1 + 647\eta)\eta$.*

*Proof.* For the plaquette $P$ the proof of Lemma 122 uses only the plaquettes of the surfaces $\mathcal{U}_\ell$ of the four sides $\ell$ of $P$ and those of the sixteen squares $\square_{2y+o}$. A plaquette of $\mathcal{U}_\ell$, $\ell = (y_i, \mu_i)$, has its corner $x$ at $x_{\mu_i} - 2(y_i)_{\mu_i} \in \{0,1,2\}$ and

$x_\alpha - 2(y_i)_\alpha \in \{0,1\}$ for $\alpha \neq \mu_i$, with $y_i \in \{y, y+e_\mu, y+e_\nu\}$; a plaquette of $\square_{2y+o}$ has $x-2y \in \{0,1,2\}^4$. So $x-2y \in \{0,1,2,3\}^4$ in every case, and the plaquette lies in $\mathscr{B}_P$. The bounds on $\chi$, on $\delta(V_P^\sharp, I)$, the inequalities (22) and the bound $\|\bar F_P\| + 3.2\, r_P^2 \le 4(1+647\eta)\eta$ in that proof use only the norms of these plaquettes. □

Let $a$ be the fine spacing and $a_j = 2^j a$. Given $\eta_0 > 0$ put $\eta_{j+1} = 4(1+647\eta_j)\eta_j$. Let $\mathfrak{B}_0$ be the set of fine plaquettes with $\delta(U_p, I) > \rho_G$ or $\|\Theta_p\| > \eta_0$, and let $\mathfrak{B}_{j+1}$ be the set of plaquettes $P$ of the lattice of spacing $a_{j+1}$ whose box $\mathscr{B}_P$ contains a plaquette of $\mathfrak{B}_j$. Let $E_j^\circ$ and $A_j^\circ$ be the sum of squares and the sum of $\|\Theta_P^{(j)}\|$ over the plaquettes $P \notin \mathfrak{B}_j$ of the $j$-th iterate.

**Theorem 152.** *Let $G \subset \mathrm{U}(r)$ be compact and $U$ a $G$-valued field on the periodic lattice of side divisible by $2^k$, and suppose that $\eta_j \le \rho_G/1000$ for $j < k$.*

(i) *For $j \le k$, every plaquette $P \notin \mathfrak{B}_j$ of the $j$-th iterate has $\delta(U_P^{(j)}, I) \le \rho_G$ and $\|\Theta_P^{(j)}\| \le \eta_j$. For $j < k$ the cut-off is inactive on the sides of every plaquette outside $\mathfrak{B}_{j+1}$, and (22) holds there.*

(ii) *Every point of a plaquette of $\mathfrak{B}_j$ lies within $\ell^\infty$ distance $3(a_j - a)$ of a plaquette of $\mathfrak{B}_0$.*

(iii) *With $\sigma = \sum_{j<k} \eta_j$,*

$$E_k^\circ \le e^{12060\sigma} E_0^\circ, \qquad A_k^\circ \le e^{9147\sigma} 4^{-k} A_0^\circ.$$

(iv) *If $4^k\eta_0 \le 10^{-4}\rho_G$, the hypothesis holds, $\eta_j \le 1.023 \cdot 4^j\eta_0$ for $j \le k$, $\sigma \le 0.35 \cdot 4^k\eta_0$, and $E_k^\circ \le 1.53\, E_0^\circ$.*

*Proof.* (i) By induction on $j$. If $P \notin \mathfrak{B}_{j+1}$, no plaquette of $\mathscr{B}_P$ lies in $\mathfrak{B}_j$, so by the induction hypothesis all of them have $\delta \le \rho_G$ and norm at most $\eta_j \le \rho_G/1000$. Lemma 151, applied to the $j$-th iterate in the tree gauge, gives $\delta(U_P^{(j+1)}, I) \le 0.03\,\rho_G$ and $\|\Theta_P^{(j+1)}\| \le \eta_{j+1}$; the gauge fixing between steps does not change these norms.

(ii) Let $P$ at $y$ lie in $\mathfrak{B}_{j+1}$, and let $p \in \mathfrak{B}_j$ lie in $\mathscr{B}_P$, with corner $x$, $x - 2y \in \{0,1,2,3\}^4$. In units of $a_j$ every coordinate of a point of $P$ lies in $[2y_c, 2y_c+2]$, and $p$ contains a point whose coordinates are $x_c$ or lie in $[x_c, x_c+1]$; so the two are at distance at most 3 in each coordinate. Hence every point of a plaquette of $\mathfrak{B}_{j+1}$ is within $3a_j$ of a plaquette of $\mathfrak{B}_j$, and $\sum_{i<j} 3a_i = 3(a_j - a)$.

(iii) Repeat the part *Sums* of the proof of Lemma 122 with the plaquettes $P \notin \mathfrak{B}_{j+1}$ only. Their squares $\square_{2y+o}$ and surfaces $\mathscr{U}_\ell$ consist of plaquettes outside $\mathfrak{B}_j$, so the same counts give $\sqrt{E_{j+1}^\circ} \le (1+6030\eta_j)\sqrt{E_j^\circ}$ and $A_{j+1}^\circ \le (1+9147\eta_j)A_j^\circ/4$, and $1 + t \le e^t$.

(iv) The recursion gives $\eta_j \le e^{647\sigma_j} 4^j \eta_0$ with $\sigma_j = \sum_{i<j}\eta_i$. Suppose $\sigma_j \le 0.35 \cdot 4^j\eta_0$, which holds at $j = 0$. Then $\sigma_j \le 3.5 \cdot 10^{-5}$, $e^{647\sigma_j} \le 1.023$, $\eta_j \le 1.023 \cdot 4^j\eta_0 \le 1.023 \cdot 10^{-4}\rho_G$, and $\sigma_{j+1} \le 1.373 \cdot 4^j\eta_0 \le 0.35 \cdot 4^{j+1}\eta_0$. Finally $e^{12060 \cdot 3.5 \cdot 10^{-5}} \le 1.53$. □

Under the Wilson measure the defects of Theorem 152 are the bad plaquettes of Theorem 147. For SU(2), $\rho_G = 1$; a plaquette with $\delta(U_p, I) = 2\sqrt{2}\sin(\theta/2) \le \rho \le \frac{1}{10}$ has $\|\Theta_p\| = \sqrt{2}\,\theta \le 1.01\,\delta(U_p, I)$, so with $\eta_0 = 1.01\rho$ every plaquette of $\mathfrak{B}_0$ is bad, and lies in a bad cube.

**Corollary 153.** *Let $G = \mathrm{SU}(2)$, $C_1 = C_0 + \frac{1}{6}\log 12800$, and for $\beta_W \ge 1$ let $\rho(\beta_W)$ be given by $\rho^2 = 96(\log\beta_W + C_1)/\beta_W$, so that $6400\, q_\rho(\beta_W) = \frac{1}{2}$. Let $k_*$ be the largest integer with $4^{k_*} \cdot 1.01\rho \le 10^{-4}$. If $1.01\rho \le 10^{-4}$, then $k_* \ge 0$,*

$$a_{k_*} \ge 1.5 \cdot 10^{-3} \Big( \frac{\beta_W}{\log\beta_W + C_1} \Big)^{1/4} a,$$

*and with $\mu_L$-probability at least $1 - L^4 2^{-r}$ every connected cluster of bad cubes has fewer than $r$ cubes, while for every $j \le k_*$ the defects of the $j$-th iterate lie within $\ell^\infty$ distance $3a_j$ of these clusters and the iterate satisfies Theorem 152(i), (iii) with $E_{k_*}^\circ \le 1.53\, E_0^\circ$. Along the limits of Corollary 148 with $a\,\rho(\beta_W)^{-1/2} \to 0$, which holds along the one-loop tuning, the defects of all iterates up to the spacing $a_{k_*}$ lie, with probability tending to one, within clusters of physical size tending to 0.*

*Proof.* With this $\rho$, $q_\rho(\beta_W) = \exp(-\log 12800)$, and Corollary 148 gives the probability of a connected set of $r$ bad cubes at most $L^4(6400q)^r = L^4 2^{-r}$. By maximality, $4^{k_*+1} \cdot 1.01\rho > 10^{-4}$, so $2^{k_*} > \frac{1}{2} \cdot 10^{-2}(1.01\rho)^{-1/2}$, and $\rho^{-1/2} = 96^{-1/4}(\beta_W/(\log\beta_W + C_1))^{1/4}$ with $96^{1/4} < 3.14$. The rest is Theorem 152 with $\eta_0 = 1.01\rho$, whose hypothesis holds by part (iv), and $\mathfrak{B}_0$ contained in the bad cubes; the bound $\|\Theta_p\| \le 1.01\,\delta(U_p, I)$ applies since $\rho \le \frac{1}{10}$. For the limits take $r = \lceil 8\log_2 L \rceil$, so that $L^4 2^{-r} \le L^{-4}$ and $ra \to 0$ for a torus of fixed physical side $La$. The defects then lie in sets of physical diameter at most $ra + 6a_{k_*}$, and $a_{k_*} \le 10^{-2}(1.01\rho)^{-1/2} a \to 0$. □

The scale $a_{k_*}$ is of the order $(\beta_W/\log\beta_W)^{1/4}a$, the order of the cube side that the condition (D) needs by Proposition 85. So the cut-off average carries the field through the defects of the Wilson measure, without spreading them, up to the scale at which (D) can first hold, but not beyond: past $a_{k_*}$ the bound $\eta_j \le 1.023 \cdot 4^j\eta_0$ no longer keeps the curvature small. Along the one-loop tuning $\beta_W$ grows like $\log L$, so $k_*$ grows like $\frac{1}{4}\log_2\log L$, a vanishing fraction of the $\log_2 L$ scales.

The joint bound of Theorem 147 is not a conditional one, and (D) needs a supremum over boundary conditions. The two differ genuinely. Fix a link $e$ and a plaquette $p_0$ through it. Put all links on the six staples of $e$ at $I$, except the middle links of the five staples other than that of $p_0$, which are $-I$. Then the conditional density of $U_e$ is proportional to $e^{-2\beta_W \mathrm{Tr}\, U_e}$, and $p_0$, whose holonomy is $U_e^{\pm 1}$, is bad with conditional probability tending to 1 as $\beta_W \to \infty$, for every $\rho < 2\sqrt{2}$. Nearby exteriors behave the same way. So the clusters of Corollary 148 are small in probability, but the boundary response of a cube to a frustrated exterior is not small, and the composition beyond the scale $a_{k_*}$ of Corollary 153 and the conditional estimate in (D) remain open.

For exteriors that are calm on the boundary of a box and lie in a maximal torus, the conditional estimate does hold. Let

$R=[0,l]^4$, $l\ge 2$, be a box of $\Lambda_L$ with $l<L$. Its boundary $\partial R$ is a cubical 3-sphere. Call a link or plaquette of $R$ a *boundary* link or plaquette if it lies in $\partial R$, and *inner* otherwise. A plaquette that contains an inner link lies in $R$, so conditionally on all other links the inner links have the law

$$\nu_\eta(dU)=Z_\eta^{-1}\exp\Big(-\beta_W\sum_{p\in P^\circ}s_p\Big)\prod_e dU_e,$$

the product over the inner links, where $P^\circ$ is the set of inner plaquettes. It depends on the exterior $\eta$ only through the boundary links. Let $T\subset \mathrm{SU}(2)$ be the diagonal torus of the elements $\exp(tJ)$, $J=\mathrm{diag}(i,-i)$, and for $u\in\mathrm{SU}(2)$ let $\angle u\in[0,\pi]$ be its angle, so that $\delta(u,I)=2\sqrt{2}\sin(\angle u/2)$.

**Lemma 154.** *Let $0\le\rho_s<\pi/3$. Suppose the boundary links of $R$ lie in $T$ and $\angle U_p\le\rho_s$ for every boundary plaquette. Then the inner links can be chosen in $T$ so that $\angle U_p\le C\rho_s$ for every plaquette of $R$, where*

$$C=\sqrt{3}\,(24\cdot 13118+9)<5.46\cdot 10^5.$$

*Proof.* Write a $T$-valued field as $U_e=\exp(a_eJ)$ with $a_e\in\mathbb{R}/2\pi\mathbb{Z}$. For a plaquette $p$ let $(da)_p$ be the signed sum of $a$ over its four links, so that $U_p=\exp((da)_pJ)$. For a boundary plaquette let $\theta_p\in(-\pi,\pi]$ be the representative of $(da)_p$; then $|\theta_p|=\angle U_p\le\rho_s$.

*Closedness.* The signed angles of the six plaquettes of a unit 3-cube of $\partial R$ add up to 0 modulo $2\pi$, since each link occurs twice with opposite signs. Their representatives add up to a number of modulus at most $6\rho_s<2\pi$, hence to 0. So $\theta$ is a closed real 2-cochain on $\partial R$.

*Lift.* We construct a real 1-cochain $\alpha$ on the boundary links with $d\alpha=\theta$ and $|\alpha_e|\le 13118\,l\rho_s$. The boundary is the union of the eight facets $F_i^0=\{x\in R: x_i=0\}$ and $F_i^l=\{x\in R: x_i=l\}$, each a cube $[0,l]^3$. Take them in the order $F_1^l$, $F_2^l$, $F_2^0$, $F_3^l$, $F_3^0$, $F_4^l$, $F_4^0$, $F_1^0$, and let $\Gamma_k$ be the union of the intersections of the $k$-th facet with the earlier ones. In the cube of the $k$-th facet, $\Gamma_k$ is empty for $k=1$, one face for $k=2,3$, three faces, two of them opposite, for $k=4,5$, five faces for $k=6,7$ and all six for $k=8$. So $\Gamma_k$ is contractible or a 2-sphere, and its first cellular cohomology vanishes.

On a facet with coordinates $t_1,t_2,t_3$, the axial gauge gives $\alpha^0$ with $d\alpha^0=\theta$ on the facet. Put $\alpha^0=0$ on the links in direction $t_3$. Determine $\alpha^0$ on the links in direction $t_2$ from the plaquettes in the planes $\{t_2,t_3\}$, starting from 0 at $t_3=0$. On the links in direction $t_1$ at $t_3=0$, determine it from the plaquettes in the plane $\{t_1,t_2\}$, starting from 0 at $t_2=0$, and then from the plaquettes in the planes $\{t_1,t_3\}$. The plaquettes in the planes $\{t_1,t_2\}$ with $t_3>0$ then satisfy $d\alpha^0=\theta$ by closedness, by induction on $t_3$. Each value is a signed sum of at most $l$, respectively $2l$, angles, so $|\alpha^0|\le 2l\rho_s$.

For $k=1$ put $\alpha=\alpha^0$, so $M_1=\max|\alpha|\le 2l\rho_s$. For $k\ge 2$, $\alpha$ is already defined on the links of $\Gamma_k$, with $|\alpha|\le M_{k-1}$, and $\omega=\alpha-\alpha^0$ is closed there. Hence $\omega=d\lambda$ for a function $\lambda$ on the vertices of $\Gamma_k$, and $|\lambda(x)-\lambda(y)|\le\|\omega\|_\infty d_\Gamma(x,y)$, where $d_\Gamma$ is the path metric of $\Gamma_k$.

We claim $d_\Gamma\le 3d_1$, where $d_1$ is the $\ell^1$ metric of the facet. If $x$ and $y$ lie in one face of $\Gamma_k$, that face contains an $\ell^1$ geodesic between them. If they lie in adjacent faces $\{t_i=s\}$ and $\{t_j=s'\}$, the path through the point with $t_i=s$, $t_j=s'$ and third coordinate that of $x$ has length $d_1(x,y)$. If they lie in opposite faces $\{t_i=0\}$ and $\{t_i=l\}$, then $d_1(x,y)\ge l$. In that case $\Gamma_k$ contains a face adjacent to both, and the path from $x$ to that face, across it and on to $y$ has length at most $d_1(x,y)+2l\le 3d_1(x,y)$.

So $\lambda$ is $3\|\omega\|_\infty$-Lipschitz for $d_1$ on $\Gamma_k$. The function

$$\tilde\lambda(x)=\min_y\big(\lambda(y)+3\|\omega\|_\infty d_1(x,y)\big)$$

extends it to the vertices of the facet with the same constant [69]. Put $\alpha=\alpha^0+d\tilde\lambda$ on the links of the facet. It agrees with the earlier values on $\Gamma_k$, satisfies $d\alpha=\theta$ on the facet, and has

$$|\alpha|\le 2l\rho_s+3(M_{k-1}+2l\rho_s).$$

So $M_k\le 3M_{k-1}+8l\rho_s$, which gives $M_8\le 13118\,l\rho_s$.

*Extension.* Let $c$ be the centre of $R$, $R_0=l/2$, $r(x)=|x-c|_\infty$ and $\varphi(x)=c+R_0(x-c)/r(x)$, the radial projection of $R\setminus\{c\}$ onto $\partial R$. Let $\hat\alpha$ and $\hat\theta=d\hat\alpha$ be the lowest-order Whitney forms of $\alpha$ and $\theta$ on the unit cubes of $\partial R$ [92]. On a unit cube with coordinates $t_1,t_2,t_3$, $\hat\alpha=\sum_i a_i\,dt_i$, where $a_i$ is the multilinear interpolation of the values of $\alpha$ on the four links in direction $t_i$; similarly $\hat\theta=\sum_{i<j}\theta_{ij}\,dt_i\wedge dt_j$. These forms have continuous tangential components across the faces of the unit cubes, satisfy $\int_e\hat\alpha=\alpha_e$ for every boundary link, and for tangent vectors

$$|\hat\alpha(u)|\le\sqrt{3}\,M_8|u|,\qquad |\hat\theta(u,w)|\le\sqrt{3}\,\rho_s|u||w|.$$

Put $\tilde B=(r/R_0)^2\varphi^*\hat\alpha$ on $R\setminus\{c\}$. Since $|\nabla r|=1$ almost everywhere and $|x-c|\le 2r$, we have $|d\varphi(v)|\le 3R_0|v|/r$. Then

$$d\tilde B=d\big(r^2/R_0^2\big)\wedge\varphi^*\hat\alpha+(r/R_0)^2\varphi^*\hat\theta$$

gives

$$|d\tilde B(v,w)|\le\big(12\sqrt{3}\,M_8/R_0+9\sqrt{3}\,\rho_s\big)|v||w|\le C\rho_s|v||w|,$$

and $|\tilde B|\le 3\sqrt{3}\,M_8 r/R_0$ is bounded.

The set $R\setminus\{c\}$ is the union of the cones from $c$ over the unit cubes of $\partial R$. These cones are convex polytopes cut out by hyperplanes through $c$. On each of them $\tilde B$ is smooth, and its tangential components are continuous across them since $\varphi$ is continuous. Put $B_e=\int_e\tilde B$ for every link $e$ of $R$. Apply Stokes' theorem on each piece of a plaquette: the boundary terms on the interior cuts cancel, and the point $c$ contributes nothing since $\tilde B$ is bounded. This gives $(dB)_p=\int_p d\tilde B$, so $|(dB)_p|\le C\rho_s$. On $\partial R$, $r=R_0$ and $\varphi$ is the identity on tangent vectors, so $B=\alpha$ on the boundary links.

*Gauge.* The cochain $a-\alpha$ on the boundary links has $(d(a-\alpha))_p\equiv\theta_p-\theta_p=0$ modulo $2\pi$ on every boundary plaquette. So it is a flat U(1)-field on the 2-skeleton of $\partial R$, which is simply connected, and $a-\alpha\equiv d\mu$ modulo $2\pi$ for a function $\mu$ on the boundary vertices. Extend $\mu$ by 0 to the inner vertices and put $U_e=\exp((B_e+(d\mu)_e)J)$ on the inner links. On the boundary links $B+d\mu\equiv a$, and every plaquette of $R$ has angle at most $|(dB)_p|\le C\rho_s$. □

For a general SU(2)-valued boundary field the curvature is not a closed cochain, and the lift in the proof of Lemma 154 is not available. A lift by gluing local gauges still works when $l^2\rho_s$ is small. Identify $\mathfrak{su}(2)$ with the imaginary quaternions and write $|X|$ for the operator norm, which is the Euclidean norm, so that $|[X,Y]| \le 2|X||Y|$, $\|e^X - I\| \le |X|$ and $\angle e^X = |X|$ for $|X| \le \pi$. With $d(g,h) = \angle(g^{-1}h)$, the map $X \mapsto e^X$ is 1-Lipschitz, since its differential has singular values 1 and $\sin|X|/|X|$, and the principal logarithm is 2-Lipschitz on $\{\angle h \le \frac{1}{2}\}$, a geodesically convex ball on which the inverse differential has singular values at most $\frac{1}{2}/\sin\frac{1}{2} < 2$. Put

$$C_1 = 2, \qquad C_k = 11\,C_{k-1} + 24 \quad (2 \le k \le 8),$$

so that $C_8 = 85\,743\,550$, and

$$K = 42\,C_8 + 20, \qquad \varepsilon_0 = \frac{1}{100\,C_8^2}.$$

**Lemma 155.** *Let the boundary links $\eta$ of $R$ take values in* SU(2)*, with $\angle\eta_p \le \rho_s$ for every boundary plaquette and $l^2\rho_s \le \varepsilon_0$. Then the inner links can be chosen so that $\angle U_p \le K\rho_s$ for every plaquette of $R$.*

*Proof.* For a vertex gauge $g$ write $\eta^g_{xy} = g_x \eta_{xy} g_y^{-1}$. A product of conjugates of $m$ elements of angle at most $\rho_s$ has angle at most $m\rho_s$, since $d$ is bi-invariant.

*One facet.* On a facet with coordinates $t_1,t_2,t_3$ let $P_x$ be the path from the corner to $x$ that moves first in direction $t_1$, then $t_2$, then $t_3$, and put $g^0_x = \eta(P_x)$, the ordered product of the links along $P_x$. Then $\eta^{g^0}_{xy}$ is the holonomy of the loop $P_x\,(xy)\,P_y^{-1}$. For a link in direction $t_3$ this loop retraces itself. For a link in direction $t_2$ at height $t_3 = c$ it bounds $c$ plaquettes of the planes $\{t_2,t_3\}$, and for a link in direction $t_1$ at $(t_2,t_3) = (b,c)$ it bounds $b$ plaquettes of a plane $\{t_1,t_2\}$ and $c$ of a plane $\{t_1,t_3\}$. So the holonomy is a product of conjugates of at most $2l$ plaquette holonomies, and every link of the facet has $\angle\eta^{g^0}_e \le 2l\rho_s$.

*Gluing.* Take the facets in the order of the proof of Lemma 154, with the same sets $\Gamma_k$; there $d_\Gamma \le 3d_1$. Two vertices of $\Gamma_k$ joined by a link of the facet lie in a common face of $\Gamma_k$, since $l \ge 2$, so that link lies in $\Gamma_k$. We construct a gauge $g$ on the boundary vertices with $\angle\eta^g_e \le M_k = C_k l\rho_s$ on the links of the first $k$ facets. For $k = 1$ take $g = g^0$. For $k \ge 2$, let $g^0$ be the gauge of the $k$-th facet and $t_x = g_x(g^0_x)^{-1}$ for $x \in \Gamma_k$. For a link $xy$ of $\Gamma_k$, $\eta^g_{xy} = t_x \eta^{g^0}_{xy} t_y^{-1}$, so

$$d(t_x,t_y) = \angle\big(t_x^{-1}t_y\big) \le \angle\eta^g_{xy} + \angle\eta^{g^0}_{xy} \le M_{k-1} + 2l\rho_s,$$

and $t$ is $3(M_{k-1}+2l\rho_s)$-Lipschitz for $d_1$ on $\Gamma_k$. Fix $x_0 \in \Gamma_k$ and put $h_x = t_{x_0}^{-1}t_x$. The facet has $d_1$-diameter $3l$, so

$$\angle h_x \le 9l(M_{k-1}+2l\rho_s) = 9(C_{k-1}+2)\,l^2\rho_s \le 9(C_8+2)\varepsilon_0 < \tfrac{1}{2},$$

and $a_x = \log h_x$ is $L$-Lipschitz for $d_1$, with $L = 6(M_{k-1}+2l\rho_s)$. Extend each of the three real coordinates of $a$ to the vertices of the facet by the formula of McShane [69], as in the proof of Lemma 154; the extension $\tilde a$ is $\sqrt{3}L$-Lipschitz. Put $g_x = t_{x_0}e^{\tilde a_x}g^0_x$ on the facet; on $\Gamma_k$ this is the old value. For a link $xy$ of the facet, $\eta^g_{xy} = \tilde t_x \eta^{g^0}_{xy} \tilde t_y^{-1}$ with $\tilde t_x = t_{x_0}e^{\tilde a_x}$, so

$$\begin{aligned}\angle\eta^g_{xy} &\le d(\tilde t_x,\tilde t_y) + 2l\rho_s \le \sqrt{3}L + 2l\rho_s \\ &= 6\sqrt{3}\,M_{k-1} + (12\sqrt{3}+2)\,l\rho_s \le M_k.\end{aligned}$$

After the eighth facet, $\angle\eta^g_e \le M = C_8 l\rho_s$ on every boundary link.

*Potential.* Since $M \le C_8\varepsilon_0/l < \frac{1}{2}$, the logarithms $A_e = \log\eta^g_e$ are defined, with $|A_e| \le M$ and $A_{-e} = -A_e$. Let $(dA)_p$ be the signed sum of $A$ over the four links of a boundary plaquette. For $X_1,\dots,X_q \in \mathfrak{su}(2)$,

$$\Big\|e^{X_1}\cdots e^{X_q} - I - \sum_i X_i\Big\| \le \tfrac{1}{2}\Big(\sum_i |X_i|\Big)^2,$$

by induction on $q$ from $e^Xe^Y - I - X - Y = (e^X - I - X) + (e^Y - I - Y) + (e^X - I)(e^Y - I)$ and $\|e^X - I - X\| \le |X|^2/2$. Applied to the four links of $p$, and since $\|\eta^g_p - I\| = 2\sin(\angle\eta_p/2) \le \rho_s$, this gives

$$|(dA)_p| \le \sigma = \rho_s + 8M^2.$$

Let $\hat A$ be the lowest-order Whitney form of $A$ on the unit cubes of $\partial R$, as in the proof of Lemma 154, now with values in $\mathfrak{su}(2)$. Its coefficients are interpolations of values of $A$, and those of $d\hat A$ are interpolations of values of $dA$, so

$$|\hat A(u)| \le \sqrt{3}\,M|u|, \qquad |d\hat A(u,w)| \le \sqrt{3}\,\sigma|u||w|.$$

With $r$, $R_0$ and $\varphi$ as there, put $\tilde B = (r/R_0)^2\varphi^*\hat A$ on $R\setminus\{c\}$ and $\tilde B(c) = 0$. The same computation gives

$$|\tilde B(v)| \le 3\sqrt{3}M\frac{r}{R_0}|v|, \qquad |d\tilde B(v,w)| \le \kappa_1|v||w|,$$

where $\kappa_1 = 12\sqrt{3}M/R_0 + 9\sqrt{3}\,\sigma$. On each cone from $c$ over a unit cube of $\partial R$, $\tilde B$ is smooth away from $c$, and its tangential components are continuous across the cones.

*Transport.* For a link $e$ of $R$, parametrised by $s \in [0,1]$, let $V_e = T(1)$, where $T' = T\,\tilde B(\dot e)$ and $T(0) = I$; the coefficient is bounded and piecewise smooth, and $V$ composes along paths in the order of the links. For a closed path $\gamma$ of length $\ell$,

$$T(1) - I - \int_\gamma \tilde B = \int_0^1 (T - I)\,\tilde B(\dot\gamma)\,ds,$$

and $\|T(s) - I\| \le \int_0^s |\tilde B(\dot\gamma)|$, so the holonomy $V_\gamma$ satisfies $\|V_\gamma - I\| \le |\int_\gamma \tilde B| + \frac{1}{2}(\int_\gamma|\tilde B|)^2$. Let $p$ be a plaquette of $R$ and divide it into $N^2$ squares $q$ of side $h = 1/N$. Its holonomy is a product of conjugates of the holonomies of the boundaries of the squares, since every inner edge of the subdivision is traversed once in each direction. For each square, Stokes' theorem on the finitely many convex polygons in which the cones cut $q$ gives $\int_{\partial q}\tilde B = \int_q d\tilde B$: the integrals over the interior cuts cancel because the tangential components agree, and if $c \in q$ a disc of radius $\delta$ around $c$ contributes $O(\delta^2)$, since $|\tilde B| \le 3\sqrt{3}Mr/R_0$. Hence

$$\|V_{\partial q} - I\| \le \kappa_1 h^2 + \tfrac{1}{2}\big(4h\cdot 3\sqrt{3}M\big)^2 = \kappa h^2,$$

with $\kappa = \kappa_1 + 216M^2$. Since $2\arcsin(x/2) \le x(1+x^2)$ for $0 \le x \le 1$, summing over the squares gives $\angle V_p \le \kappa(1+\kappa^2 N^{-4})$, and letting $N \to \infty$, $\angle V_p \le \kappa$. Now $M/R_0 = 2C_8\rho_s$ and $M^2 = C_8^2 l^2 \rho_s^2 \le \rho_s/100$, so

$$\begin{aligned}\kappa &\le \big(24\sqrt{3}C_8 + 9\sqrt{3}\big)\rho_s + \big(72\sqrt{3}+216\big)M^2 \\ &< (42C_8+20)\,\rho_s = K\rho_s.\end{aligned}$$

*Boundary values.* On $\partial R$, $r = R_0$ and $\varphi$ is the identity on tangent vectors, and along a boundary link $e$ the tangential coefficient of $\hat{A}$ is the constant $A_e$. So $V_e = e^{A_e} = \eta_e^g$. Extend $g$ by $I$ to the inner vertices and put $U_{xy} = g_x^{-1} V_{xy} g_y$. Then $U = \eta$ on the boundary links, and $U_p$ is conjugate to $V_p$, so $\angle U_p \le K\rho_s$ for every plaquette of $R$. □

The boundary fields of Lemma 155 with strict inequalities form an open set of positive Haar measure that contains the identity field; unlike those of Lemma 154, they are not a null set. The constants are far from optimal.

For $0 < \varepsilon \le 1$ let $E_\varepsilon$ be the event that at least $\varepsilon l^4$ of the $l^4$ unit cubes $x + [0,1]^4$, $x \in \{0,\dots,l-1\}^4$, of $R$ contain a bad plaquette.

**Proposition 156.** *Let $\beta_W \ge 1$ and $0 < \rho \le 1$. Suppose that no boundary plaquette of $R$ is bad under $\eta$, and that the inner links can be chosen so that every inner plaquette has angle at most $\kappa$. Then*

$$-\log \nu_\eta(E_\varepsilon) \ge \frac{\beta_W \rho^2 \varepsilon}{16}\, l^4 - 6(l+1)^4\big(\log\beta_W + C_0 + \tfrac{1}{2} + \beta_W\kappa^2\big),$$

*with $C_0$ as in Lemma 146. In particular, if the boundary links lie in $T$ with angle at most $\rho_s < \pi/3$ on every boundary plaquette and $\rho \ge \sqrt{2}\,\rho_s$, the bound holds with $\kappa = C\rho_s$ and the constant $C$ of Lemma 154. If the boundary links lie in $\mathrm{SU}(2)$ with angle at most $\rho_s$ on every boundary plaquette, $l^2\rho_s \le \varepsilon_0$ and $\rho \ge \sqrt{2}\,\rho_s$, it holds with $\kappa = K\rho_s$.*

*Proof.* On $E_\varepsilon$ the bad plaquettes are inner, and each lies in at most four of the unit cubes. So there are at least $\varepsilon l^4/4$ of them, each with $s_p > \rho^2/4$, and $\sum_{P^\circ} s_p \ge \rho^2\varepsilon l^4/16$. Hence the unnormalised weight of $E_\varepsilon$ is at most $\exp(-\beta_W\rho^2\varepsilon l^4/16)$.

For the normalisation let $\bar{U}$ be the inner links of the hypothesis, $t = (4\sqrt{\beta_W})^{-1}$, and consider the inner links $U_e = \bar{U}_e h_e$ with $\angle h_e \le t$. The angle of $g^{-1}h$ is a bi-invariant distance, so every inner plaquette has $\angle U_p \le \kappa + 4t$ and

$$s_p \le \tfrac{1}{2}(\kappa+4t)^2 \le \kappa^2 + 16t^2 = \kappa^2 + \beta_W^{-1}.$$

Each $h_e$ has probability at least $8t^3/(3\pi^3) = (24\pi^3\beta_W^{3/2})^{-1}$, as in the proof of Lemma 146. There are $n_P \le 6l^2(l+1)^2$ inner plaquettes and $n_L \le 4l(l+1)^3$ inner links, so

$$Z_\eta \ge \exp\big(-n_P(\beta_W\kappa^2+1)\big)\big(24\pi^3\beta_W^{3/2}\big)^{-n_L}.$$

Since $C_0 = \frac{1}{2} + \frac{2}{3}\log(24\pi^3)$, $n_P + n_L\log(24\pi^3\beta_W^{3/2}) \le 6(l+1)^4(\log\beta_W + C_0 + \frac{1}{2})$, which gives the bound. For the last assertion, a boundary plaquette has $\delta(U_p, I) \le \sqrt{2}\angle U_p \le \sqrt{2}\rho_s \le \rho$, so it is not bad, and Lemma 154 or Lemma 155 gives the inner links. □

So in a box whose boundary is calm at a level $\rho_s$ small compared with $\rho\sqrt{\varepsilon}/C$ and $T$-valued, or small compared with $\rho\sqrt{\varepsilon}/K$ and $\mathrm{SU}(2)$-valued with $l^2\rho_s \le \varepsilon_0$, the fraction of bad cubes exceeds $\varepsilon$ only with probability exponentially small in the volume once $\beta_W\rho^2\varepsilon$ is large compared with $\log\beta_W$, uniformly over such exteriors and over $l$. The estimate is conditional, unlike Theorem 147.

*Remark* 157. The exteriors with boundary links in $T$ form a null set for $\mu_L$. Lemma 155 removes the torus at the price of the smallness $l^2\rho_s \le \varepsilon_0$. In the continuum the corresponding condition is that of Uhlenbeck's theorem [88], which gives a Coulomb gauge on a ball when the curvature is small in $L^{n/2}$; on 3-dimensional pieces of $\partial R$ of side comparable to $l$ and for curvature of size $\rho_s$ this norm is of order $l^2\rho_s$. The lemma replaces the Coulomb gauge by the gluing of the facet gauges. With $\rho_s$ of the order of the level $\rho(\beta_W)$ of Corollary 153, $\rho_s = c_0(\log\beta_W/\beta_W)^{1/2}$, it applies for

$$l \le (\varepsilon_0/c_0)^{1/2}\,(\beta_W/\log\beta_W)^{1/4},$$

the scale $a_{k_*}$ of Corollary 153 up to a constant factor. Along the tuning $\beta_W \sim b\log(1/a)$ the physical side $al$ then tends to 0, so the lemma does not reach cubes of a fixed physical size; this is a limitation of the argument, not a proof that such fillings fail. Moreover Proposition 156 bounds the fraction of bad cubes under one exterior and does not compare the laws under two exteriors, so it gives no contraction coefficient.

The calm boundary is needed only for the small-curvature filling. If the boundary layer is paid for instead of filled, every condition on the exterior disappears, at the price of a surface term that is small against the volume on large boxes.

**Proposition 158.** *Let $\beta_W \ge 1$, $0 < \rho \le 1$ and $0 < \varepsilon \le 1$ with $\varepsilon l \ge 16$. For every boundary field $\eta$,*

$$-\log\nu_\eta(E_\varepsilon) \ge l^4\Big(\frac{\beta_W\rho^2\varepsilon}{32} - 6\log\beta_W - C_1 - \frac{48\beta_W}{l}\Big),$$

*with $C_1 = 3 + 4\log(24\pi^3) < 31$.*

*Proof.* A unit cube $x + [0,1]^4$ with $x \in \{1,\dots,l-2\}^4$ does not meet $\partial R$, so all its plaquettes are inner; the other $l^4 - (l-2)^4 \le 8l^3 \le \varepsilon l^4/2$ unit cubes meet it. On $E_\varepsilon$ at least $\varepsilon l^4/2$ cubes of the first kind contain a bad plaquette, which is inner, and a plaquette lies in at most four unit cubes. So at least $\varepsilon l^4/8$ inner plaquettes are bad, $\sum_{P^\circ} s_p \ge \rho^2\varepsilon l^4/32$, and the unnormalised weight of $E_\varepsilon$ is at most $\exp(-\beta_W\rho^2\varepsilon l^4/32)$.

For the normalisation take inner links $U_e = h_e$ with $\angle h_e \le t = (4\sqrt{\beta_W})^{-1}$. An inner plaquette whose four links are inner has $\angle U_p \le 4t$ and $s_p \le (2\beta_W)^{-1}$. An inner plaquette spans two directions and has its other two coordinates in $\{1,\dots,l-1\}$; it contains a boundary link exactly when its cell in the plane of the two spanned directions meets the boundary of $[0,l]^2$, which happens for $4(l-1)$ of the $l^2$ cells. So $24(l-1)^3 \le 24l^3$ inner plaquettes contain a boundary link, and on them $s_p \le 2$. There are $n_P = 6l^2(l-1)^2 \le 6l^4$ inner plaquettes and $n_L = 4l(l-1)^3 \le 4l^4$ inner links, so, as in the proof of Proposition 156,

$$Z_\eta \ge \exp\big(-48\beta_W l^3 - 3l^4\big)\big(24\pi^3\beta_W^{3/2}\big)^{-4l^4}.$$

Dividing gives the bound, and $24\pi^3 < e^7$ gives $C_1 < 31$. □

**Corollary 159.** *Let $\beta_W \geq e^{44}$, $l \geq 6144\beta_W^{3/4}$ and $\rho = \varepsilon = \beta_W^{-1/4}$. Then $\nu_\eta(E_\varepsilon) \leq \exp(-\beta_W^{1/4} l^4/64)$ for every boundary field $\eta$. In particular, along $\beta_W \sim b\log(1/a)$ with $l = \lceil r/a \rceil$ and fixed $r, b > 0$, on cubes of fixed physical side $r$ the fraction of unit cubes that contain a plaquette with $\delta(U_p, I) > \beta_W^{-1/4}$ is at most $\beta_W^{-1/4}$ with conditional probability tending to one, uniformly over all exteriors.*

*Proof.* Here $\varepsilon l \geq 6144\beta_W^{1/2} \geq 16$ and $\rho \leq 1$. The first term of Proposition 158 is $\beta_W^{1/4}/32$, and $48\beta_W/l \leq \beta_W^{1/4}/128$. For $x = \log\beta_W \geq 44$ we have $6x+31 \leq e^{x/4}/128$: at $x = 44$ the left side is 295 and the right side exceeds 467, and $x/4 - \log(6x+31)$ increases for $x \geq 44$. So $6\log\beta_W + C_1 \leq \beta_W^{1/4}/128$ and the bracket is at least $\beta_W^{1/4}(\frac{1}{32} - \frac{1}{128} - \frac{1}{128}) = \beta_W^{1/4}/64$. Along the tuning, $\beta_W \to \infty$ and $l/\beta_W^{3/4} \to \infty$, so the hypotheses hold for small $a$. □

Like Proposition 156, Corollary 159 bounds the bad cubes under one exterior and does not compare the laws under two, so it gives no contraction coefficient. The bad cubes are rare on cubes of fixed physical size for every exterior, and the conditional estimate in (D) remains open.

The same count locates the windows on which the fibre estimates of Sec. XII D apply. Let the side of the torus be even, let $B(y) = 2y + \{0,1\}^4$ be the blocks of Definition 92, and let $l$ be even, so that $R$ is a union of blocks.

**Corollary 160.** *Let $G = \mathrm{SU}(2)$, $\beta_W \geq 1$, let $W_0$ be a set of $m$ blocks, $0 < K \leq 10^{-8}$, $\rho = K/(3072m)$, and $0 < \varepsilon \leq 1$ with $\varepsilon l \geq 16$. For every boundary field $\eta$ of $R$, with $\nu_\eta$-probability at least*

$$1 - \exp\Big(-l^4\Big(\frac{\beta_W\rho^2\varepsilon}{32} - 6\log\beta_W - 31 - \frac{48\beta_W}{l}\Big)\Big),$$

*all but fewer than $256m\varepsilon l^4$ of the translates $W = y_0 + W_0$ whose blocks of $W^+$ lie in $[1, l-1]^4$ have the following property: in the tree gauge the exterior of $W$ has a completion that satisfies the hypothesis of Corollary 144, namely the field itself.*

*Proof.* Outside $E_\varepsilon$, which has the stated probability by Proposition 158, fewer than $\varepsilon l^4$ unit cubes of $R$ contain a bad plaquette. A plaquette with a vertex in a block of $W^+$ has all its vertices in $R$ and lies in a unit cube $x + [0,1]^4$ of $R$; if it is bad, that cube contains a bad plaquette. The vertex lies in a block $B(y')$ with $y'_i \in \{\lfloor x_i/2 \rfloor, \lfloor (x_i+1)/2 \rfloor\}$, and $y' \in \{y\}^+$ for some $y \in W$, so $y_i$ takes at most four values and $y$ lies in a set of at most 256 blocks determined by $x$; then $y_0 \in y - W_0$ takes at most $256m$ values. Hence fewer than $256m\varepsilon l^4$ translates have a bad plaquette with a vertex in a block of $W^+$. At every other translate these plaquettes have $\delta(U_p, I) \leq \rho$. The number $n_P$ of plaquettes that contain a link of $A_W$ is at most $384m$, so $\rho \leq K/(8n_P) = \kappa/(2\sqrt{n_P})$ with $\kappa = K/(4\sqrt{n_P})$. A gauge transformation conjugates every plaquette holonomy and leaves $\delta(U_p, I)$ unchanged, so the field in the tree gauge is a completion of its own exterior with this property. □

So under every exterior of a box, the windows of a fixed number of blocks on which Corollary 144 applies fill all but at most $256m\varepsilon l^4$ of the positions, a fraction of about $4096m\varepsilon$. The bound is useful only when $\beta_W\rho^2\varepsilon$ is large, that is for windows of bounded size, and the next proposition shows that this restriction cannot be removed.

**Proposition 161.** *Let $G = \mathrm{SU}(2)$, $\beta_W \geq 1$ and $0 < \rho' \leq 1$.*

(i) *Conditionally on all links but one link of a plaquette $p_0$, the event $\delta(U_{p_0}, I) \leq \rho'$ has probability at most $14(\beta_W\rho'^2)^{3/2}$ under the Wilson measure.*

(ii) *Let $W$ be a set of blocks with $W^+ \neq W$, $0 < K \leq 10^{-8}$ and $\kappa = K/(4\sqrt{n_P})$. Under the Wilson measure in the tree gauge, the exterior of $W$ has a completion as in Corollary 144 with probability at most $7 \cdot 10^{-32}\beta_W^{3/2}|W|^{-3}$, and $\varepsilon''_W(\kappa^2/2, \beta_W/4) < 1$ only if $\beta_W > 10^{24}|W|^2$.*

*Proof.* (i) As in the proof of Proposition 150, the holonomy of $p_0$ has a conditional density proportional to $e^{\kappa' h\cdot u}$ with $0 \leq \kappa' \leq 6\beta_W$ and $|h| = 1$, and the probability of a cap about $I$ is largest at $h = I$, $\kappa' = 6\beta_W$. Since $\delta(u, I) = 2\sqrt{2}\sin(\angle u/2)$ and $\sin x \geq 2x/\pi$ on $[0, \frac{\pi}{2}]$, the event forces $\angle U_{p_0} \leq \theta = \pi\rho'/(2\sqrt{2})$. With $\cos t - 1 \geq -t^2/2$ and $\sin t \geq 2t/\pi$ on $[0, \frac{\pi}{2}]$,

$$\int_0^\pi e^{6\beta_W(\cos t - 1)} \sin^2 t\, dt \geq \frac{4}{\pi^2}\int_0^{\pi/2} e^{-3\beta_W t^2} t^2\, dt \geq \frac{4 \cdot 0.44}{\pi^2(3\beta_W)^{3/2}},$$

because the integral of $e^{-s^2}s^2$ over $[0, \pi\sqrt{3}/2]$ exceeds 0.44. The integral of $e^{6\beta_W(\cos t-1)}\sin^2 t$ over $[0,\theta]$ is at most $\theta^3/3$, so the ratio is at most $9.72(\beta_W\theta^2)^{3/2}$, and $(\pi/(2\sqrt{2}))^3 < 1.371$.

(ii) Take $y' \in W^+ \setminus W$ and a plaquette $p_0$ inside $B(y')$. Its links have their base in $B(y')$, so none lies in $A_W$, and every completion of the exterior agrees with the field on $p_0$. A completion as in Corollary 144 therefore needs $\delta(U_{p_0}, I) \leq \kappa/(2\sqrt{n_P}) = K/(8n_P)$. This event is gauge invariant, so its probability is the same without gauge fixing, and by (i) it is at most $14(\beta_W K^2/(64n_P^2))^{3/2}$. Each link of $A_W$ lies in six plaquettes and a plaquette contains at most four of them, so $n_P \geq 73.5|W|$, and $14 \cdot 10^{-24}/(512 \cdot 73.5^3) < 7 \cdot 10^{-32}$. Finally, $\varepsilon''_W(\kappa^2/2, \lambda) < 1$ means $\frac{3}{8}\lambda\kappa^2 > 49|W|\ln(3.45 \cdot 10^{11} n_P^6/K^6)$, and with $\lambda = \beta_W/4$, $\kappa^2 = K^2/(16n_P)$, $n_P \geq 73.5|W|$ and $K \leq 10^{-8}$ the logarithm exceeds 162.8, so $\beta_W > 512 \cdot 49 \cdot 73.5 \cdot 162.8 \cdot 10^{16}|W|^2/3 > 10^{24}|W|^2$. □

Along the tuning $\beta_W \sim b\log(1/a)$, a cube of fixed physical side $r$ has $|W| = (r/2a)^4$ blocks, so both bounds of (ii) fail for small $a$: on such a cube the exterior has no calm completion with probability tending to one, and the fibre estimates of Sec. XII D are vacuous there. Their entropy factors are powers of $|W|$, so they are estimates for windows of bounded size, which Corollary 160 places at almost every position under every exterior. The exteriors with no calm completion and the rare fibres of Corollary 144 are thus the typical case at the physical scale and not an exceptional one. The cube condition

(D) needs estimates on the whole cube that are uniform in $|W|$, with the windows that are not calm split off as in Bałaban's analysis [8], and that remains open.

### F. Conditioning on a fixed large-field set

The estimates of Sec. XII E are statements about the Wilson measure on the torus. The cube condition (D) takes a supremum over exteriors, and three facts about conditional laws hold under every exterior; the setting is that of a Gibbs field whose potential is itself random, the uniqueness theory of which goes back to Bassalygo and Dobrushin [3]. The first bounds the excess of the action over its minimum in a box, with no reflection positivity and no torus. The second says that restricting a single-link law to a fixed set, however irregular, leaves its Kantorovich response to the neighbouring links bounded by a constant times $\beta_W$. The third turns the one-point rarity of a bad set into closeness of any two Gibbs measures, at order of that rarity, and shows at the same time why one-point rarity gives closeness and not decay.

**Lemma 162** (excess action)**.** *Let $G=\mathrm{SU}(2)$, let $\Lambda$ be a set of $N$ links of $\mathbb{Z}^4$ and fix the links outside $\Lambda$. Let $P_\Lambda$ be the plaquettes that contain a link of $\Lambda$, $S_\Lambda=\sum_{p\in P_\Lambda}s_p$, and $S_{\min}$ the minimum of $S_\Lambda$ over the links of $\Lambda$. For every $E>0$, under the Wilson measure conditioned on the links outside $\Lambda$,*

$$\Pr\big(S_\Lambda \ge S_{\min}+E\big) \le \Big(\frac{3\pi^3}{8r^3}\Big)^N e^{-\beta_W E/2},$$

*where $r=\min\{E/(6\pi N),\pi/2\}$.*

*Proof.* The plaquettes that contain no link of $\Lambda$ do not depend on the links of $\Lambda$, so the conditional law has density $Z^{-1}e^{-\beta_W S_\Lambda}$ with respect to the product of Haar measures on $\Lambda$. The unnormalised mass of the event is at most $e^{-\beta_W(S_{\min}+E)}$. For the lower bound on $Z$, let $U^*$ be a minimiser, which exists by compactness. If a link of a plaquette $p$ moves by geodesic distance $t$, the angle $\delta(U_p,I)\in[0,\pi]$ moves by at most $t$, by bi-invariance, so $s_p=\frac14\delta(U_p,I)^2$ moves by at most $\frac14 t\cdot 2\pi=\frac{\pi}{2}t$. Each link lies in six plaquettes. If every link of $\Lambda$ lies within distance $r$ of its value in $U^*$, then $S_\Lambda\le S_{\min}+6\cdot\frac{\pi}{2}rN\le S_{\min}+E/2$. The Haar measure of a geodesic ball of radius $r\le\pi/2$ is $\frac{2}{\pi}\int_0^r\sin^2 t\,dt\ge 8r^3/(3\pi^3)$, by $\sin t\ge 2t/\pi$. Hence $Z\ge(8r^3/3\pi^3)^N e^{-\beta_W(S_{\min}+E/2)}$, and the quotient of the two bounds is the claim. □

The constant depends on $N$ and $E$ alone, so excess action is rare under every exterior, at a cost that is exponential in $N$. The minimum itself is a function of the exterior, and the lemma says nothing about the size of its plaquettes: it controls the fluctuation of the field about the configuration the exterior forces, not that configuration.

**Lemma 163** (restriction to a fixed set)**.** *Let $G=\mathrm{SU}(2)$, let $e$ be a link and let $h\ge 0$ be an integrable function on $G$ with $\int h\,dU>0$ that does not depend on the other links. For a configuration $\sigma$ of the other links let $\gamma_e^h(\cdot\mid\sigma)$ be the law on $G$ with density proportional to $e^{-\beta_W S}h$. If $\sigma$ and $\sigma'$ differ only at a link $e'\neq e$, then*

$$W_1\big(\gamma_e^h(\cdot\mid\sigma),\gamma_e^h(\cdot\mid\sigma')\big)\le\frac{\pi^2}{8}\beta_W\,\delta(\sigma_{e'},\sigma'_{e'}),$$

*and the two laws coincide unless $e$ and $e'$ lie in a common plaquette.*

*Proof.* Only the plaquettes containing $e$ enter the law, and two distinct links of $\mathbb{Z}^4$ lie in at most one common plaquette $p_0$; if there is none the laws are equal. Otherwise move $\sigma_{e'}$ along a unit-speed geodesic $U(t)$, $0\le t\le T=\delta(\sigma_{e'},\sigma'_{e'})$, and let $\gamma_t$ be the corresponding law. The function $t\mapsto s_{p_0}$ has $|\frac{d}{dt}s_{p_0}|\le\frac{\pi}{2}$ as in the proof of Lemma 162, and the derivative $\dot s_{p_0}$ is a function of $U_e$ with oscillation at most $\pi$. For a 1-Lipschitz $f$ on $G$, which has oscillation at most the diameter $\pi$, $\frac{d}{dt}\gamma_t(f)=-\beta_W\operatorname{Cov}_{\gamma_t}(f,\dot s_{p_0})$ because $h$ does not depend on $t$, and $|\operatorname{Cov}(f,\dot s_{p_0})|\le\sqrt{\operatorname{Var}f\,\operatorname{Var}\dot s_{p_0}}\le\frac{\pi}{2}\cdot\frac{\pi}{4}$. Integrating over $t$ and taking the supremum over $f$ gives the bound. □

When $h$ is the indicator of a set $C$ the lemma says that conditioning a link to lie in $C$ costs nothing in its response, however small or irregular $C$ is, as long as $C$ is fixed. The responses that are not bounded are those to a set that moves with the neighbours, such as the intersection of balls centred at them; this is the difference between conditioning on a fixed set of bad links and conditioning on a plaquette being large.

**Proposition 164** (one-point rarity)**.** *Let $\mathscr{A}$ be a set of links, $D$ the diameter of $G$, and $\gamma_e$, $e\in\mathscr{A}$, single-link kernels with*

$$W_1\big(\gamma_e(\cdot\mid\sigma),\gamma_e(\cdot\mid\sigma')\big)\le\sum_{e'}c_{ee'}(\sigma,\sigma')\,\delta(\sigma_{e'},\sigma'_{e'})$$

*for all $\sigma,\sigma'$. Let $\sigma\mapsto B(\sigma)\subset\mathscr{A}$ be measurable, and suppose there are constants $a_{ee'},A_{ee'}\ge0$ with $c_{ee'}(\sigma,\sigma')\le a_{ee'}$ whenever $e\notin B(\sigma)\cup B(\sigma')$, $c_{ee'}\le A_{ee'}$ always, $\sup_e\sum_{e'}a_{ee'}\le\theta<1$ and $\sup_e\sum_{e'}A_{ee'}\le\bar A$. If $\mu_1,\mu_2$ are invariant under every $\gamma_e$ and $\mu_i(e\in B)\le q$ for every $e$ and $i=1,2$, there is a coupling of $\mu_1$ and $\mu_2$ with*

$$\sup_{e\in\mathscr{A}}\mathbb{E}\,\delta(\sigma_e,\sigma'_e)\le\frac{2\bar A D q}{1-\theta}.$$

*In particular $|\mu_1(F)-\mu_2(F)|\le L_F|S_F|\,2\bar A D q/(1-\theta)$ for every $F$ with support $S_F$ that is $L_F$-Lipschitz in each link.*

*Proof.* Fix $b_0\in\mathscr{A}$, $\lambda>0$, and let $w_b=e^{-\lambda|b-b_0|}$, which is summable. As in the proof of Lemma 241 the functional $\Psi(\pi)=\sum_b w_b\mathbb{E}_\pi\delta(\sigma_b,\sigma'_b)$ has a minimiser $\pi^*$ among the couplings, and replacing the pair at one link $e$ by a sample of an optimal coupling of $\gamma_e(\cdot\mid\sigma)$ and $\gamma_e(\cdot\mid\sigma')$ gives a coupling again. Only the term at $e$ changes, so minimality gives, with $\rho(e)=\mathbb{E}_{\pi^*}\delta(\sigma_e,\sigma'_e)$,

$$\rho(e)\le\mathbb{E}_{\pi^*}\sum_{e'}c_{ee'}(\sigma,\sigma')\,\delta(\sigma_{e'},\sigma'_{e'})$$

for every $e$, with no weight left. On the event that $e$ is good in both configurations the coefficients are at most $a_{ee'}$; on its

complement, whose probability is at most $\mu_1(e\in B)+\mu_2(e\in B)\le 2q$, they are at most $A_{ee'}$ and every distance is at most $D$. Hence $\rho(e)\le\theta\sup\rho+2\bar{A}Dq$, and $\sup\rho\le D$ is finite, which gives the bound. The last statement follows by writing $F(\sigma)-F(\sigma')$ as a telescopic sum over the links of $S_F$. □

The proposition is approximate uniqueness: all Gibbs measures of the specification lie within $2\bar{A}Dq/(1-\theta)$ of each other, link by link. It is not decay. Its proof sees the bad set only through the one-point probabilities $q$, and the term $2\bar{A}Dq$ does not shrink with the distance from the region where $\mu_1$ and $\mu_2$ are allowed to differ. The bound $D$ on the distance at a bad link is not what causes this: replacing it by Hölder's inequality gives $\rho\le\theta\rho+\bar{A}(qD)^{1/p}\rho^{1/p'}$, a recursion with the positive fixed point $\rho^*$ of order $\bar{A}^pqD/(1-\theta)^p$. To obtain decay the bad set has to be held fixed while the comparison runs, as in the disagreement percolation of van den Berg and Maes [89], where the two configurations are coupled link by link and the disagreement has to percolate through the bad set; the price is that the laws conditioned on that set, not the Wilson measure, then have to satisfy the fibre estimates of Sec. XII D.

### G. Rarity of coarse defects

Theorem 147 makes a large plaquette rare at the lattice scale. The cube condition (D) concerns a cube of fixed physical side, which along the tuning spans $m\sim\beta_W/b$ scales, and it needs the analogue of that rarity for the curvature of the block averages at every one of them. This subsection isolates the two sides of that question. On the energy side, the coarse curvature of a field is bounded by its action uniformly in the scale, for linear fields and for group-valued fields in a torus. On the entropy side, the chessboard estimate applies to coarse events, and it is the normalisation of Lemma 146, not the energy, that makes it useless beyond $O(\log\beta_W)$ scales. A right translation of the Haar measures gives rarity at every scale and under every exterior, with a rate governed by the second variation of the action along a fixed direction; for the abelian theory that second variation is the energy of the direction, while for SU(2) it grows like $4^k$ at scale $k$, and the growth is produced by the gauge orbit of the vacuum.

Throughout, $\Lambda_L$ is the torus of side $L=2^m$, the scale $k$ satisfies $2^{k+2}\le L$, and the lattice of spacing $2^k$ is the torus $\Lambda_{L'}$ with $L'=L/2^k\ge 4$. For a real field $\varphi$ on the links, $d\varphi$ is its curvature, $d^{\mathsf T}$ the adjoint of $d$, and $\Delta=d^{\mathsf T}d+dd^{\mathsf T}$, which acts on each component of a field as the lattice Laplacian $(\Delta f)(x)=8f(x)-\sum_{\pm\nu}f(x\pm e_\nu)$. The Green function $G_{L'}$ of $\Lambda_{L'}$ is the function of mean zero with $\Delta G_{L'}=\delta_0-L'^{-4}$; it is invariant under the permutations and sign changes of the coordinates, and applied to each component it inverts $\Delta$ on fields whose components have mean zero. It commutes with $d$, $d^{\mathsf T}$ and the translations.

**Lemma 165** (energy of a coarse defect)**.** *Let $P$ be a plaquette of the lattice of spacing $2^k$ and $J_{k,P}$ the field with $(d_kV^{(k)}(\varphi))_P=\langle\varphi,J_{k,P}\rangle$ for every field $\varphi$ on $\Lambda_L$, given by Lemma 100(i). Then $d^{\mathsf T}J_{k,P}=0$, every field $\varphi$ satisfies*

$$(d_kV^{(k)}(\varphi))_P^2\le\gamma_k\,Q(\varphi),$$

*where*

$$\gamma_k=\langle J_{k,P},G_LJ_{k,P}\rangle=\sup_{\varphi}\frac{(d_kV^{(k)}(\varphi))_P^2}{Q(\varphi)},$$

*and $\frac{1}{432}\le\gamma_k\le\frac12$ for every $k$. In particular $Q(\varphi)\ge 2r^2$ whenever $|(d_kV^{(k)}(\varphi))_P|\ge r$, at every scale.*

*Proof.* Lemma 100(ii) shows that $V^{(k)}(dh)$ is the gradient of $\bar{h}$ on the coarse lattice for every function $h$ on sites, so $\langle dh,J\rangle=0$, the curvature of a gradient vanishing, and $d^{\mathsf T}J=0$; $J$ is supported on the links of directions $\mu$, $\nu$ of the plane of $P$, and its component in each direction has mean zero, the two sides of $P$ entering with opposite signs. Put $u=G_LJ$. Then $\Delta u=J$, $d^{\mathsf T}u=G_Ld^{\mathsf T}J=0$, and for every field $\varphi$

$$\begin{aligned}\langle\varphi,J\rangle=\langle\varphi,\Delta u\rangle&=\langle d\varphi,du\rangle+\langle d^{\mathsf T}\varphi,d^{\mathsf T}u\rangle\\&=\langle d\varphi,du\rangle\le\sqrt{Q(\varphi)Q(u)},\end{aligned}$$

with $Q(u)=\langle u,d^{\mathsf T}du\rangle=\langle u,\Delta u\rangle=\langle u,J\rangle=\gamma_k$, and equality at $\varphi=u$. This proves the inequality and the characterisation of $\gamma_k$ as a supremum.

For the upper bound, Proposition 99(i) gives $Q_k(V^{(k)}(\varphi))\le Q(\varphi)$ for every field, its proof being a local identity, so $\gamma_k$ is at most the supremum of $(d_kw)_P^2/Q_k(w)$ over the fields $w$ of $\Lambda_{L'}$. The argument of the first paragraph on $\Lambda_{L'}$, with $j=d_k^{\mathsf T}e_P$ the field equal to $\pm1$ on the four sides of $P$, identifies that supremum with $\langle j,G_{L'}j\rangle$. The two sides of $P$ in the direction $\mu$ are the links $(y,\mu)$ and $(y+e_\nu,\mu)$ with opposite signs, so

$$\begin{aligned}\langle j,G_{L'}j\rangle&=4G_{L'}(0)-2G_{L'}(e_\nu)-2G_{L'}(e_\mu)\\&=4\big(G_{L'}(0)-G_{L'}(e_1)\big)=\tfrac12\big(1-L'^{-4}\big),\end{aligned}$$

by $(\Delta G_{L'})(0)=8G_{L'}(0)-8G_{L'}(e_1)=1-L'^{-4}$ and the symmetry of $G_{L'}$.

For the lower bound take $P$ at $y=0$ in the plane $\{1,2\}$, write $L_k=2^k$, and let $f$ be the function on $\mathbb{Z}/L\mathbb{Z}$ with $f(s)=s/L_k$ for $0\le s\le L_k$, $f(s)=1$ for $L_k\le s\le 2L_k$, $f(s)=1-(s-2L_k)/L_k$ for $2L_k\le s\le 3L_k$ and $f=0$ elsewhere; let $g$ be the indicator of $[0,2L_k)$; and let $h(s)=(1-\operatorname{dist}(s,[0,L_k))/L_k)_+$. Put $\varphi_{(x,2)}=L_k^{-1}f(x_1)g(x_2)h(x_3)h(x_4)$ and $\varphi=0$ on the other links; all supports fit in the torus since $4L_k\le L$. By Lemma 100(i), $V^{(k)}_{y,2}(\varphi)=L_k^{-4}\sum_{x\in B_k(y)}\sum_{t<L_k}\varphi_{(x+te_2,2)}$, and for $y=0$ and $y=e_1$ the factors $g$ and $h$ equal 1 on the ranges that occur, so $V^{(k)}_{0,2}(\varphi)=\bar{f}(0)$ and $V^{(k)}_{e_1,2}(\varphi)=\bar{f}(1)$, the means of $f$ over $[0,L_k)$ and $[L_k,2L_k)$, which are $(L_k-1)/(2L_k)$ and 1. Hence $(d_kV^{(k)}(\varphi))_P=\bar{f}(1)-\bar{f}(0)=(L_k+1)/(2L_k)\ge\frac12$. The curvature of $\varphi$ lives on the plaquettes of the planes $\{1,2\}$, $\{2,3\}$, $\{2,4\}$, and $g$ is never differenced, so

$$\begin{aligned}Q(\varphi)&=L_k^{-2}\big(\|\nabla f\|^2\|g\|^2\|h\|^4+2\|f\|^2\|g\|^2\|\nabla h\|^2\|h\|^2\big)\\&\le L_k^{-2}\Big(\frac{2}{L_k}2L_k(3L_k)^2+2\cdot3L_k\cdot2L_k\frac{2}{L_k}3L_k\Big),\end{aligned}$$

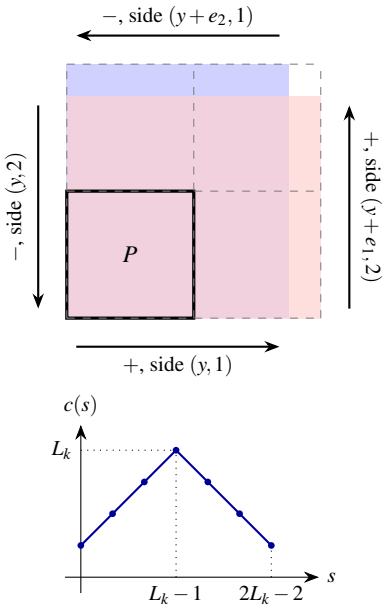



FIG. 20. The smeared current loop $J_{k,P}$ of Lemma 165 in the plane of $P$, drawn for $L_k = 4$. Left: each side of the coarse plaquette $P$ (thick square) contributes a current of its own direction on the links of a slab of transverse size $L_k^3$ and length $2L_k - 1$ (blue for the direction 1, red for the direction 2), with the signs of the four sides; the slabs of opposite sides are disjoint and the two directions are carried by different links. Right: the longitudinal profile $L_k^4 J = \pm c(s)$ on a slab, the tent $c(s) = \min\{s+1, L_k, 2L_k-1-s\}$, which counts the straight paths of Lemma 100(i) through the link at position $s$. The current is divergence free and its energy $\gamma_k = \langle J_{k,P}, G_L J_{k,P}\rangle$ lies in $[\frac{1}{432}, \frac{1}{2}]$ at every scale.

where $\nabla$ is the forward difference: $f$ and $h$ change by $1/L_k$ on $2L_k$ steps and by nothing elsewhere, and $f^2$, $g^2$, $h^2$ have sums at most $3L_k$, $2L_k$, $3L_k$. So $Q(\varphi) \le 108$, and $\gamma_k \ge \frac{1}{4}/108$. □

The fields $J_{k,P}$ are smeared current loops: by Lemma 100(i) each side of $P$ contributes $\pm L_k^{-4} c(s)$ on the $L_k^3(2L_k-1)$ links of its direction whose transverse coordinates lie in the block and whose longitudinal coordinate $s$ ranges over $[0, 2L_k-2]$, with the tent $c(s) = \min\{s+1, L_k, 2L_k-1-s\}$. A bare holonomy around a loop of $n$ links forces action only of order $r^2/n$, since the loop can be thickened to a tube of any radius; the block average smears the loop over a four-dimensional region, and this is why its energy does not decrease with the scale. The lemma is the quadratic part of Theorem 102, seen from one coarse plaquette. The same scale invariance holds for group-valued fields in a torus.

**Proposition 166** (energy of a coarse defect, torus-valued fields)**.** *Let $U$ be an $S$-valued field in the tree gauge on $\Lambda_L$, as in Theorem 119, and let $P$ be a plaquette of the lattice of spacing $2^k$ with $\delta(V^{\sharp,k}(U)_P, I) \ge r$. Then*

$$Q(U) \ge \min\Big\{\frac{r^2}{1.001}, \frac{1}{144.2}\Big\} \ge \frac{r^2}{1154}.$$

*Proof.* If some plaquette has $\delta(U_p, I) > 1 = \rho_G$ then $Q(U) > 1 \ge r^2/8$, since $r \le \operatorname{diam} \mathrm{SU}(2) = 2\sqrt{2}$. Otherwise every $\Theta_p$ is defined, and for $\Theta_p = \theta_p J_p$ with $J_p^2 = -I$ one has $\|\Theta_p\|^2 = 2\theta_p^2$ and $\delta(U_p, I)^2 = 8\sin^2(\theta_p/2)$, so $\|\Theta_p\|^2 \le 1.001\,\delta(U_p, I)^2$ when $\theta_p \le 1/(12\sqrt{2})$, that is when $\|\Theta_p\| \le \eta_G = \frac{1}{12}$. Let $j_*$ be the least $j \le k$ with $\max_P \|\Theta_P^{(j)}\| > \eta_G$, if there is one. If there is none, (21) holds for every $j < k$, and Theorem 119 gives $r^2 \le \delta(V^{\sharp,k}(U)_P, I)^2 \le Q_k(V^{\sharp,k}(U)) \le 1.001\, Q(U)$. If $j_* = 0$, some plaquette has $\theta_p > 1/(12\sqrt{2})$ and $Q(U) \ge 8\sin^2(\theta_p/2) \ge 1/144.1$. If $1 \le j_* \le k$, (21) holds for $j < j_*$, so Theorem 119 at the scale $j_*$ shows that $\Theta^{(j_*)}$ is defined and $\sum_P \|\Theta_P^{(j_*)}\|^2 \le \sum_p \|\Theta_p\|^2 \le 1.001\, Q(U)$, every fine plaquette having $\|\Theta_p\| \le \eta_G$; one term of the left side exceeds $1/144$, so $Q(U) \ge 1/144.2$. □

The $S$-valued fields form a null set for the Wilson measure, and for general $\mathrm{SU}(2)$ fields Theorem 123 controls the cut-off average only while $\sum_j \eta_j$ stays small; whether a bound of the form $Q(U) \ge c\, r^2$, uniform in $k$, holds for every $\mathrm{SU}(2)$ field is not known. The next proposition shows that, whatever the energy of a coarse defect, the chessboard estimate cannot use it beyond a bounded number of scales.

**Proposition 167** (chessboard for coarse events)**.** *Let $\beta_W \ge 1$, let $b = 2^j$ with $2b \mid L$, and let $E$ be an event determined by the links of the closed cube $[0,b]^4$ on which the action of the plaquettes of that cube is at least $A$. For $z \in (b\mathbb{Z}/L\mathbb{Z})^4$ let $E_z$ be the event obtained from $E$ by the reflections that carry $[0,b]^4$ to $z + [0,b]^4$. Then for every set $Z$ of such $z$,*

$$\mu_L\Big(\bigcap_{z\in Z} E_z\Big) \le \exp\Big(|Z|\Big(6b^4(\log\beta_W + C_0) - \frac{\beta_W A}{4}\Big)\Big).$$

*Proof.* As in the proof of Theorem 147, $\mu_L$ is reflection positive in the hyperplanes through sites, and the chessboard estimate for functions of the links of the closed blocks $z + [0,b]^4$, $z \in (b\mathbb{Z}/L\mathbb{Z})^4$, bounds the probability of $\bigcap_{z\in Z} E_z$ by that of $\bigcap_z E_z$ over all $(L/b)^4$ blocks, raised to the power $|Z|(b/L)^4$. Reflection preserves the Haar measures and $\operatorname{Tr} U_p$, so on every $E_z$ the plaquettes of the block $z + [0,b]^4$ carry action at least $A$. A plaquette lies in at most four closed blocks, one for each choice of the two transverse coordinates of the corner, so on the intersection $S \ge (L/b)^4 A/4$, and the event has unnormalised weight at most $e^{-\beta_W (L/b)^4 A/4}$. Dividing by $Z_L \ge e^{-6L^4(\log\beta_W + C_0)}$ of Lemma 146 and raising to the power $|Z|(b/L)^4$ gives the bound. □

A plaquette of the lattice of spacing $2^k$ depends, by Lemma 151 applied at each step, on the fine links of a cube of side $2^{k+2} - 4$, so the event that it carries a large cut-off average is of the kind in the proposition with $b = 2^{k+2}$. Whatever action $A$ that event forces, the bound is below one only when $6 \cdot 16^{k+2}(\log\beta_W + C_0) < \beta_W A/4$, that is $16^k \lesssim \beta_W A/\log\beta_W$. For the events of Lemma 165 and Proposition 166, $A$ is of the order of $r^2$ at every scale, and the loss is entirely in the normalisation: the lower bound of Lemma 146 charges the full fluctuation volume of the block, $6b^4(\log\beta_W + C_0)$, against a defect whose energy does not grow with the block. In the Gaussian theory the two fluctuation volumes, around the defect and around the vacuum, are the same, and the following

translation argument transfers that cancellation to the group, under every exterior.

**Theorem 168** (Haar tilt)**.** *Let $G$ be* SU(2) *or* U(1)*, let $\Lambda$ be a finite set of links, either all links of $\Lambda_L$ or the inner links of a box $R \subset \mathbb{Z}^4$ whose other links are fixed, and let $\mu$ be the law with density $Z^{-1}e^{-\beta_W S}$ with respect to the product of the Haar measures on $\Lambda$, where $S=\sum_p s_p$ runs over the plaquettes meeting $\Lambda$ and $s_p = 1-\frac{1}{N}\operatorname{Re}\operatorname{Tr}U_p$. For $r=(r_e)_{e\in\Lambda}$ with $r_e\in\mathfrak{g}$ and $\rho_e=\|r_e\|_{\mathrm{op}}$ put $U_e^t = U_e e^{tr_e}$,*

$$D_rS(U)=\partial_t S(U^t)\big|_{t=0}, \qquad K_r=\sup_U \partial_t^2 S(U^t)\big|_{t=0}.$$

(i) *$K_r \le \sum_p\big(\sum_{e\subset p,\,e\in\Lambda}\rho_e\big)^2 \le 24\sum_{e\in\Lambda}\rho_e^2$. For $G=\mathrm{U}(1)$ and $r_e=ia_e$ with $a_e\in\mathbb{R}$, $K_r\le Q(a)=\sum_p (da)_p^2$.*

(ii) *$\mathbb{E}_\mu e^{\beta_W t D_rS}\le e^{\beta_W t^2K_r/2}$ for every $t\in\mathbb{R}$.*

(iii) *For every $x>0$, uniformly in the fixed links,*

$$\mu(D_rS\ge x)\le e^{-\beta_W x^2/(2K_r)}, \qquad \mu(D_rS\le -x)\le e^{-\beta_W x^2/(2K_r)}.$$

*Proof.* (i) Order the links of a plaquette $p$ as traversed, so that $U_p=g_1^{\varepsilon_1}\cdots g_4^{\varepsilon_4}$ with $\varepsilon_i=\pm1$. Since $ge^{tr}=e^{t\,\mathrm{Ad}(g)r}g$ and $(ge^{tr})^{-1}=e^{-tr}g^{-1}$, moving the exponentials to the left gives $U_p(t)=e^{t\tilde a_1}\cdots e^{t\tilde a_4}U_p$ with $\tilde a_i=\pm\mathrm{Ad}(h_i)r_{e_i}$ for unitary $h_i$, so $\|\tilde a_i\|_{\mathrm{op}}=\rho_{e_i}$, and $\tilde a_i=0$ if $e_i\notin\Lambda$. Then

$$\partial_t^2\operatorname{Tr}U_p(t)\big|_{t=0}=\operatorname{Tr}\Big[\Big(\sum_i\tilde a_i^2+2\sum_{i<j}\tilde a_i\tilde a_j\Big)U_p\Big],$$

and $|\operatorname{Tr}(XU)|\le N\|X\|_{\mathrm{op}}$ for unitary $U$, so $|\partial_t^2 s_p|\le(\sum_i\rho_{e_i})^2\le 4\sum_{e\subset p}\rho_e^2$; each link lies in six plaquettes. The second derivative at any other $t$ is the one at 0 for the field $U^t$, so the bound holds for $K_r$. For U(1) all factors commute, $U_p(t)=e^{it(da)_p}U_p$ and $\partial_t^2 s_p=(da)_p^2\operatorname{Re}\operatorname{Tr}U_p\le(da)_p^2$.

(ii) Right multiplication by the fixed element $e^{-tr_e}$ preserves the Haar measure of each link of $\Lambda$, so

$$\int e^{-\beta_W S(Ue^{-tr})}\prod_{e\in\Lambda}dU_e=Z,$$

that is, $\mathbb{E}_\mu e^{-\beta_W[S(Ue^{-tr})-S(U)]}=1$. Taylor's formula along $s\mapsto Ue^{-sr}$ with the Lagrange remainder gives

$$\begin{aligned}S(Ue^{-tr})-S(U)&=-tD_rS(U)+\tfrac{t^2}{2}\partial_s^2S(Ue^{-sr})\big|_{s=s_*}\\&\le -tD_rS(U)+\tfrac{t^2}{2}K_r,\end{aligned}$$

the second derivative being even in $r$. Hence

$$\mathbb{E}_\mu e^{\beta_W tD_rS-\beta_W t^2K_r/2}\le 1.$$

(iii) By Chebyshev's inequality and (ii), $\mu(D_rS\ge x)\le e^{-\beta_W tx+\beta_W t^2K_r/2}$ for $t>0$; take $t=x/K_r$. The lower tail is the upper tail for $-r$. Nothing depends on the fixed links. □

The map $U\mapsto Ue^{-tr}$ is the group version of a translation of the Gaussian field; its Jacobian is one, and no smallness of the field enters. The observable $D_rS$ is the component of the force along $r$: at a plaquette, $\partial_t s_p=-\frac{1}{N}\operatorname{Re}\operatorname{Tr}\big((\sum_i\tilde a_i)U_p\big)$, the covariant derivative of $r$ in the background $U$ paired with the sine part of $U_p$. For the abelian theory the second variation is the energy of the direction, and the theorem gives the rarity of coarse defects at every scale.

**Corollary 169** (abelian theory)**.** *Let $G=\mathrm{U}(1)$, let $\mu_L$ be its Wilson measure on $\Lambda_L$, $U_e=e^{i\theta_e}$, $\theta_p=(d\theta)_p$ the angle of $U_p$ in $(-\pi,\pi]$, and let $a=G_LJ_{k,P}$ be the minimiser of Lemma 165. For every $k$, every plaquette $P$ of the lattice of spacing $2^k$ and every $x>0$,*

$$\mu_L\Big(\sum_p\sin\theta_p\,(da)_p\ge x\Big)\le e^{-\beta_W x^2}.$$

*Here $\sum_p\theta_p(da)_p=(d_kV^{(k)}(\theta))_P$ for every choice of link angles $\theta_e$ with $(d\theta)_p\equiv\theta_p$ modulo $2\pi$, up to $2\pi\sum_p n_p(da)_p$ with the integers $n_p=((d\theta)_p-\theta_p)/2\pi$. Under a fixed exterior of a box, the same bound holds with a replaced by its restriction $\tilde a$ to the inner links and $x^2$ replaced by $x^2/(2Q(\tilde a))$.*

*Proof.* By Theorem 168(i) with $r_e=ia_e$, $K_r\le Q(a)=\gamma_k\le\frac12$, and $D_rS=\sum_p\sin\theta_p(da)_p$ since $s_p(t)=1-\cos(\theta_p+t(da)_p)$. Part (iii) gives the bound. For the identity, $\sum_p(d\theta)_p(da)_p=\langle\theta,d^{\mathsf T}da\rangle=\langle\theta,\Delta a\rangle=\langle\theta,J_{k,P}\rangle$ because $d^{\mathsf T}a=0$. In a box the tilt acts on the inner links only, and $K_r\le Q(\tilde a)$. □

For the abelian theory this is the statement that a smeared coarse flux of size $x$ costs $\beta_W x^2$ at every scale, uniformly in the exterior; the four-dimensional abelian theory has no running coupling, and the bound is consistent with its known behaviour at weak coupling [43, 48]. For SU(2) the same tilt is available, but its second variation grows with the scale.

**Proposition 170** (the gauge orbit of the vacuum)**.** *Let $G=\mathrm{SU}(2)$ on $\Lambda_L$, $T\in\mathfrak{g}$ with $\|T\|_{\mathrm{op}}=1$, $a=G_LJ_{k,P}$ for a plaquette $P$ of the lattice of spacing $2^k$, and $r_e=a_eT$. Then*

$$\Big(\sqrt{2\|a\|^2-\gamma_k}-\sqrt{\gamma_k}\Big)_+^2\le K_r\le 24\|a\|^2$$

*and*

$$\|a\|^2\ge\frac{\gamma_k^2}{\|J_{k,P}\|^2}\ge\frac{4^k}{1492992},$$

*with $\gamma_k=\langle J_{k,P},G_LJ_{k,P}\rangle$ as in Lemma 165; in particular $K_r\ge 4^k/(3\cdot10^6)$ for $k\ge11$. The lower bound is attained, up to the stated loss, at a pure gauge field, where $Q(U)=0$.*

*Proof.* The upper bound is Theorem 168(i). For the lower bound let $U$ be the pure gauge field $U_e=h_xh_{x+e}^{-1}$ of a function $h$ on sites with values in SU(2). The action is invariant under gauge transformations and $U_ee^{tr_e}=h_x\exp(t\,\mathrm{Ad}(h_{x+e}^{-1})r_e)h_{x+e}^{-1}$, so $S(Ue^{tr})=S(e^{tr'})$ with $r'_e=\mathrm{Ad}(h_{x+e}^{-1})r_e$. At the identity field, $U_p(t)=\exp(tX_p+O(t^2))$ with $X_p=(dr')_p$ the signed sum of the $r'_e$ around $p$, and $s_p=1-\cos\|X_p(t)\|_{\mathrm{op}}$, so $\partial_t^2s_p|_{t=0}=$

$\|(dr')_p\|_{\mathrm{op}}^2$ and $K_r \geq \sum_p \|(dr')_p\|_{\mathrm{op}}^2$. Take $h_x = R^{x_2}$ with $R = i\sigma_1$ and $T = i\sigma_3$, which is well defined on the torus since $R^4 = I$ and $4 \mid L$, and satisfies $\mathrm{Ad}(R)T = -T$. Then $r'_e = \varepsilon_e a_e T$ with $\varepsilon_{(x,\mu)} = (-1)^{x_2}$ for $\mu \neq 2$ and $\varepsilon_{(x,2)} = (-1)^{x_2+1}$, and $K_r \geq Q(\varepsilon a)$. Since $J_{k,P}$ has components only in the directions 1, 2 of the plane of $P$ and $G_L$ acts on each component, $a_3 = a_4 = 0$; the reflection exchanging the coordinates 1 and 2 carries the component $a_1$ to $-a_2$, so $\|a_1\|^2 = \|a_2\|^2 = \frac{1}{2}\|a\|^2$. On the plaquette at $x$ in the plane $\{1,2\}$ the signs give

$$(d(\varepsilon a))_p = (-1)^{x_2}\big[(a_{(x,1)} + a_{(x+e_2,1)}) - (a_{(x+e_1,2)} - a_{(x,2)})\big] = (-1)^{x_2}\big[\sigma_x - v_x\big],$$

so that $Q(\varepsilon a) \geq \|\sigma - v\|^2 \geq (\|\sigma\| - \|v\|)_+^2$, with $\|\sigma\|^2 = 4\|a_1\|^2 - \|\nabla_2 a_1\|^2$ and $\|v\|^2 = \|\nabla_1 a_2\|^2$, where $\nabla_\nu$ is the forward difference in the direction $\nu$. All directional differences together give $\sum_{\mu,\nu} \|\nabla_\nu a_\mu\|^2 = \langle a, \Delta a\rangle = \langle G_L J, J\rangle = \gamma_k$, so $\|\sigma\|^2 \geq 2\|a\|^2 - \gamma_k$ and $\|v\|^2 \leq \gamma_k$, which is the lower bound. Finally $\gamma_k = \langle J, a\rangle \leq \|J\|\|a\|$, and by Lemma 100(i) each of the four sides of $P$ contributes $L_k^3$ rows of $2L_k - 1$ links with coefficients $L_k^{-4} c(s)$, $c \leq L_k$, so $\|J\|^2 \leq 4L_k^{-8} L_k^3 (2L_k - 1) L_k^2 \leq 8 \cdot 4^{-k}$; with $\gamma_k \geq \frac{1}{432}$ this gives $\|a\|^2 \geq 4^k/(8 \cdot 432^2)$. For $k \geq 11$ that is at least 2, so $2\|a\|^2 - \gamma_k \geq 7\gamma_k$ and $K_r \geq (1 - 7^{-1/2})^2 \cdot \frac{7}{8} \cdot 2\|a\|^2 \geq 0.67\|a\|^2 \geq 4^k/(3 \cdot 10^6)$. □

So for SU(2) the tilt of Theorem 168 proves the rarity of the force component $D_r S$ at the rate $\beta_W x^2/(2K_r)$, which is of order $\beta_W x^2/4^k$ at scale $k$, and not more: the supremum in $K_r$ is reached on the gauge orbit of the vacuum, where the field has no action and no curvature, because a fixed direction $r$ is not gauge covariant and its covariant derivative in a pure gauge background is as large as $r$ itself. The natural repair is to transport the direction with the field, $r_e \mapsto \mathrm{Ad}(h(U))r_e$. When the transport runs along a forest, the translation still preserves the Haar measures; but the residual gauge freedom of the forest then realises the field of the proof on the links between the trees, and the constant is reached at zero action in that gauge as well.

**Proposition 171** (covariant tilt on a forest)**.** *Let $G = \mathrm{SU}(2)$, let $F$ be a set of links of $\Lambda_L$ without cycles, with one root chosen in every component, and for a site $y$ let $h_y(U)$ be the ordered product of the link variables along the path in $F$ from the root of its component to $y$, with $h_y = I$ at the roots. For $r = (r_e)$ with $r_e \in \mathfrak{g}$ and $r_e = 0$ on $F$ put*

$$\Phi_t(U)_e = U_e \exp\big(t\,\mathrm{Ad}(h_y(U)^{-1})r_e\big), \quad y \text{ the end point of } e,$$

*and $\Phi_t(U)_f = U_f$ for $f \in F$; let $D_r^F S(U) = \partial_t S(\Phi_t U)|_{t=0}$ and*

$$K_r^F = \sup\big\{\partial_t^2 S(\tilde{U}e^{tr})\big|_{t=0} :\ \tilde{U}_f = I \text{ for all } f \in F\big\}.$$

(i) *$\Phi_t$ preserves the product of the Haar measures for every $t$.*

(ii) *$\mathbb{E}_\mu e^{\beta_W t D_r^F S} \leq e^{\beta_W t^2 K_r^F/2}$ for every $t \in \mathbb{R}$, and $\mu(\pm D_r^F S \geq x) \leq e^{-\beta_W x^2/(2K_r^F)}$ for every $x > 0$. $D_r^F S$ is invariant under the gauge transformations that are trivial at the roots.*

(iii) *Let $F$ be one of the sixteen translates $F_o$ of the forest of Definition 92, with blocks $2y + o + \{0,1\}^4$, let $a = G_L J_{k,P}$ and $T$ be as in Proposition 170, and let $r_e = a_e T$ on the links outside $F$. Then*

$$K_r^F \geq 2\sum_{e \notin F} a_e^2,$$

*and the bound is reached on pure gauge fields, where $Q(U) = 0$. For at least one $o$ the right side is at least $\frac{29}{16}\|a\|^2 \geq \frac{29}{16} \cdot \frac{4^k}{1492992}$.*

*Proof.* (i) Let $\Gamma(U) = (U_F, \tilde{U})$ with $\tilde{U}_e = h_x(U) U_e h_y(U)^{-1}$ for $e = (x,y) \notin F$. Given the variables on $F$, the map $U_e \mapsto \tilde{U}_e$ is a left and a right translation by elements that depend only on $U_F$, so $\Gamma$ is a bijection of the configuration space that preserves the product of the Haar measures; this is the change of variables of Lemma 10. Let $R_t$ multiply every coordinate $\tilde{U}_e$, $e \notin F$, on the right by $e^{tr_e}$ and leave $U_F$ alone. Then

$$(\Gamma^{-1} R_t \Gamma U)_e = h_x^{-1}\big(h_x U_e h_y^{-1} e^{tr_e}\big) h_y = U_e \exp\big(t\,\mathrm{Ad}(h_y^{-1}) r_e\big),$$

so $\Phi_t = \Gamma^{-1} R_t \Gamma$ is a composition of three maps that preserve the measure.

(ii) Since the action is gauge invariant, $S(\Phi_t U) = S(\tilde{U}e^{tr})$, and $\tilde{U}e^{tr}$ again has trivial links on $F$. Taylor's formula along $s \mapsto \Phi_{-s}U$ with the Lagrange remainder gives $S(\Phi_{-t}U) - S(U) \leq -tD_r^F S(U) + \frac{t^2}{2} K_r^F$, and $\int e^{-\beta_W S \circ \Phi_{-t}}\, dU = Z$ by (i); the rest is the proof of Theorem 168 (ii), (iii). Under a gauge transformation $g$ the transports change to $g_o h_y g_y^{-1}$, with $o$ the root of the component of $y$, so $\tilde{U}$ is conjugated by the root values and $D_r^F S$ is unchanged when they are trivial.

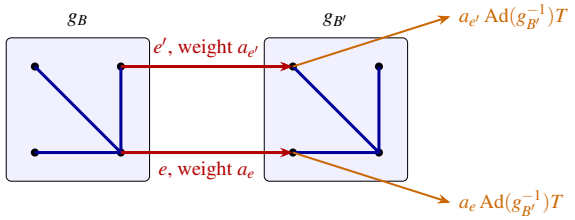
(iii) The tree links of $F_o$ join sites of one block, so a pure gauge field $U = I^g$, $U_e = g_x g_{x+e}^{-1}$, with $g$ constant on the blocks has trivial links on $F_o$, lies in the set of the supremum and has $Q(U) = 0$. By the proof of Proposition 170, $\partial_t^2 S(Ue^{tr})|_{t=0} = \sum_p \|(dr')_p\|_{\mathrm{op}}^2$ with $r'_e = \mathrm{Ad}(g_{x+e}^{-1}) r_e$, the end point of $e$ taken. Since $a_3 = a_4 = 0$, a plaquette $p$ at $x$ in a plane $\{\mu,\nu\}$ with $\mu \in \{1,2\}$ and $\nu \in \{3,4\}$ has at most two tilted sides, $e = (x,\mu)$ with end point $x + e_\mu$ and $e' = (x + e_\nu, \mu)$ with end point $x + e_\mu + e_\nu$, and

$$\|(dr')_p\|_{\mathrm{op}}^2 = a_e^2 + a_{e'}^2 - 2a_e a_{e'} \langle \mathrm{Ad}(g_{x+e_\mu}^{-1})T, \mathrm{Ad}(g_{x+e_\mu+e_\nu}^{-1})T\rangle,$$

with the inner product of $\mathfrak{g}$ in which $\|T\| = 1$ and $a_e = 0$ on $F_o$. When $(x-o)_\nu$ is odd the two end points lie in different blocks. Take the block values $g_B$ independent with Haar law. The adjoint representation of SU(2) is irreducible of dimension three, so $\mathbb{E}\,\mathrm{Ad}(g_B) = 0$, and the expectation of the inner product vanishes. Every link $e = (x,\mu) \notin F_o$ with $\mu \in \{1,2\}$ is, for each $\nu \in \{3,4\}$, a side of exactly one such plaquette, at $x$ if $(x-o)_\nu$ is odd and at $x - e_\nu$ otherwise. Dropping the other plaquettes, whose terms are nonnegative,

$$\mathbb{E}_g\, \partial_t^2 S(I^g e^{tr})\big|_{t=0} \geq \sum_{\nu \in \{3,4\}} \sum_{e \notin F_o} a_e^2 = 2\sum_{e \notin F_o} a_e^2,$$

and the supremum over the section is at least the expectation. A link $(x,\mu)$ lies in $F_o$ for one of the sixteen offsets $o$ when $\mu = 1$




tree links (blue) are trivial in the section; the residual gauge group turns the tilt direction by one SU(2) element per block

FIG. 21. The covariant tilt on the block forest (Proposition 171). Transporting the direction along the trees keeps the Haar measures, but the section still contains every pure gauge field that is constant on the blocks. On a plaquette whose tilted sides $e$, $e'$ end in different blocks the transported directions are turned by independent elements, and the average of $\|(dr')_p\|^2$ over them is $a_e^2+a_{e'}^2$ instead of the $(a_e-a_{e'})^2$ of the abelian energy.

and for two of them when $\mu=2$. Averaging over $o$ and using $\|a_1\|^2=\|a_2\|^2=\frac12\|a\|^2$ from the proof of Proposition 170,

$$\frac{1}{16}\sum_o\sum_{e\notin F_o}a_e^2=\frac{15}{16}\|a_1\|^2+\frac{14}{16}\|a_2\|^2=\frac{29}{32}\|a\|^2,$$

so one offset reaches $\frac{29}{32}\|a\|^2$; the last bound is the lower bound for $\|a\|^2$ of that proposition. □

Proposition 171 and Figure 21 show where the obstruction sits. The Haar translation survives gauge covariance along a forest, and for the block forest the constant it produces is at least $\frac{29}{16}\|a\|^2$, reached by fields with no action and no curvature at all; the chessboard estimate and the covariant tilt are therefore lost on the same fields, the gauge orbit of the vacuum, for different reasons. A bound of the form

$$\partial_t^2 S(\tilde U e^{tr})\le Q(r)+\mathscr{R}(\tilde U,r),$$

with $\mathscr{R}$ supported on the large-field region, is false for every forest section, since the left side exceeds $Q(r)$ by a fixed multiple of $\|a\|^2$ at $\tilde U=I^g$, where the large-field region is empty. A section that is smooth on the orbits, such as the Landau gauge, has no such pure gauge fields, but its Jacobian is a Faddeev–Popov determinant and not one, and part (ii) does not apply to it as it stands.[23]

Along the tuning $\beta_W\sim b\log(1/a)$ a cube of fixed physical side spans $m\sim\beta_W/b$ scales. Proposition 167 covers $O(\log\beta_W)$ of them, Propositions 170 and 171 cover $O(\log\beta_W)$ of them without the torus and uniformly in the exterior, and Corollary 169 covers all of them for the abelian theory. What the cube condition needs for SU(2) is a translation of the Haar measures whose second variation is the energy of the direction up to a constant independent of the scale, together with a non-perturbative treatment of the last $O(1)$ scales, where the running coupling is of order one and coarse defects are not rare at all. Proposition 171 shows that no translation transported along a forest does this: its constant is reached at zero action, with the growth $4^k$ of the energy of the smeared current under the residual gauge freedom, so the missing estimate is not a large-field remainder but a different section of the gauge orbits, with a Jacobian that enters the argument of part (ii). This is the content of Bałaban's large-field expansion [8], and it is not carried out here. The cube condition itself is a mixing condition of Dobrushin–Shlosman type, which for compact spins with finite range is equivalent to a logarithmic Sobolev inequality uniform in the volume [83]; the rarity of coarse defects is the part of that inequality that concerns the large fields.

[23] The Wilson measure satisfies no curvature-dimension condition at the values of $\beta_W$ considered here, so the inequality cannot be obtained from a Bakry–Émery argument either: for one link equal to $-I$ with trivial neighbours the second variation of $S$ along a unit direction on that link is $-6$, since each of its six plaquettes contributes $\partial_t^2(1+\cos t)=-1$, while the Ricci curvature of SU(2) with unit-speed geodesics $e^{tr}$, $\|r\|_{\mathrm{op}}=1$, is 2; the Hessian of $\beta_W S$ plus the curvature is negative as soon as $\beta_W>1/3$.

## XIII. LABELS, ISLANDS AND THE ENVIRONMENT

The cube condition (D) is a statement about conditional laws under every exterior, and Sec. XII F showed that one-point rarity of a bad set gives closeness of Gibbs measures but not decay. The route that remains is the one of the large-field expansions: label each block of a window good or bad, prove decay on the good windows, bound the bad islands by their rarity, and control the covariance that the labels themselves carry. This section proves what each of these four steps gives for the cut-off average of Sec. XII B, and proves, by explicit examples, where the natural choices of labels fail. The statements are exact and finite; none of them closes (D), and Sec. XIII P lists what they leave.

Throughout, $G=\mathrm{SU}(2)$ with $\rho_G=1$, the lattice is in the tree gauge of Definition 92, $d(g,h)=\|\log(g^{-1}h)\|$ is the bi-invariant geodesic distance in the Hilbert–Schmidt norm, so that $d(g,I)=\|X\|$ for $g=e^X$ with $\|X\|\le\pi\sqrt2$, and $\delta$ is the chordal distance. A *pair* $k=(y,\mu)$ consists of a block and a direction; its *path link* is $\ell_k=U_{2y+e_\mu,\mu}$, its *private links* are the eight links leaving $B(y)$ in the direction $\mu$, which enter no other cut-off average, and the seven private links other than $\ell_k$ are its *non-path* links. In the sixteen paths of $V_k^\sharp$ the path link occurs twice, as the two paths of transverse offset zero, and each non-path private link occurs twice, each time multiplied by one link inside $B(y)$ or $B(y+e_\mu)$. For a coarse field $c$ the reconstruction $R(c)$ of Sec. XII B puts $c_k$ on the private links of $k$ and $I$ on the links inside the blocks.

### A. Backgrounds and the path links

The first question is which background the chart of Theorem 131 should be centred at when a window is conditioned on its labels. The natural label of a link is its distance to the reconstruction of the coarse field, and the natural background is the reconstruction itself. The first result shows that this is not enough: a coarse field all of whose plaquettes are large has a reconstruction at which every relative link label is good and

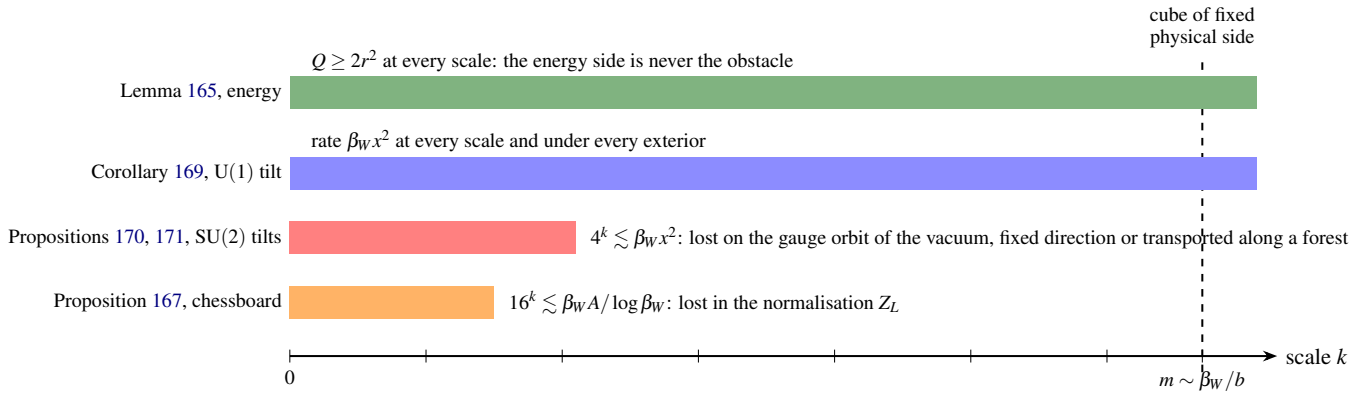



FIG. 22. Which scales the estimates of this subsection reach, along the tuning $\beta_W \sim b\log(1/a)$, where a cube of fixed physical side spans $m \sim \beta_W/b$ scales. The energy of a coarse defect is scale invariant (Lemma 165), and for the abelian theory the Haar tilt makes coarse defects rare at every scale (Corollary 169). For SU(2) the chessboard estimate stops after $O(\log\beta_W)$ scales because the normalisation charges the whole fluctuation volume of a block against the defect (Proposition 167), and the tilt stops after $O(\log\beta_W)$ scales because its second variation grows like $4^k$ on the gauge orbit of the vacuum, whether the direction is fixed or transported along a forest (Propositions 170, 171). The remaining scales are the open input of the cube condition.

the constrained action has a negative second variation along the fibre.

**Proposition 172** (a rough background)**.** *On the periodic lattice of side four, with coarse torus $(\mathbb{Z}/2\mathbb{Z})^4$, let $c_{y,\mu} = (-I)^{\sum_{\nu<\mu} y_\nu}$. Then every coarse plaquette of $c$ equals $-I$, the field $U = R(c)$ is in the tree gauge and has $V^\sharp(U) = c$, and $d(U_e, R_e(c)) = 0$ for every link. There is a curve $U(t) = U\exp(t\xi)$, $\xi$ supported on four links, such that $V^\sharp(U(t)) = c$ for all small $|t|$, $d(U_e(t), R_e(c)) \leq |t|$, and the Wilson action $S = \sum_p (1 - \frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p)$ satisfies*

$$\frac{d^2}{dt^2}\Big(\beta_W S(U(t))\Big)\Big|_{t=0} = -\beta_W, \qquad T(\xi) = \sum_e \|\xi_e\|^2 = 4.$$

*Every first derivative of $S$ at $U$ vanishes, so the intrinsic Hessian of $\beta_W S$ on the fibre of $c$ at $U$ has an eigenvalue at most $-\beta_W/4$, in any smooth chart of the fibre.*

*Proof.* The coarse plaquette at $y$ in the plane $\{\mu,\nu\}$, $\mu<\nu$, is the product of $c_{y,\mu}$, $c_{y+e_\mu,\nu}$, $c^{-1}_{y+e_\nu,\mu}$ and $c^{-1}_{y,\nu}$; these are central, and the exponent of $-I$ is $\sum_{\rho<\mu} y_\rho + \sum_{\rho<\nu}(y+e_\mu)_\rho - \sum_{\rho<\mu}(y+e_\nu)_\rho - \sum_{\rho<\nu} y_\rho = 1$ modulo two, since $\mu<\nu$ puts $e_\mu$ inside the second sum and $e_\nu$ outside the third. So every coarse plaquette is $-I$, also across the periodic seam. In $U = R(c)$ the links inside the blocks are $I$, in particular the tree links, and all sixteen paths of a pair $k$ equal $c_k$, so every $v_x = I$, the cut-off is one and $V^\sharp_k(U) = c_k$; the relative labels vanish by construction.

Let $T \in \mathfrak{su}(2)$ have $\|T\| = 1$, with eigenvalues $\pm i/\sqrt{2}$, and let $\xi_e = a_e T$ with $a_e = -1,+1,+1,-1$ on the links $((0,1,1,1),1)$, $((0,1,1,1),2)$, $((0,1,0,0),1)$, $((0,0,1,0),2)$ and $a_e = 0$ elsewhere. All four links are private links; the last two are path links. The pair $((0,0,0,0),1)$ has the non-path link $((0,1,1,1),1)$ with $a=-1$ and the path link $((0,1,0,0),1)$ with $a=+1$, the pair $((0,0,0,0),2)$ has $((0,1,1,1),2)$ with $a=+1$ and the path link $((0,0,1,0),2)$ with $a=-1$, and no other pair contains a link of the support. Since the background is central, $P_x(t) = c_k \exp(t a_x T)$ for the sixteen paths, with $a_x$ the value on the private link of the path, so $v_x(t) = \exp(t(a_x - a_0)T)$ lies in the chart for small $|t|$, the cut-off is one, and

$$V^\sharp_k(U(t)) = c_k \exp\Big(\frac{t}{16}\sum_x a_x T\Big) = c_k,$$

because each private link occurs in exactly two paths and the two nonzero values of each pair cancel. The curve stays in the fibre, and $d(U_e(t), R_e(c)) = |t a_e| \leq |t|$.

Write $U_p = s_p I$ with $s_p = \pm 1$ and let $q_p = (d\,a)_p$ be the signed curl of the scalar field $a$ around $p$. Since the perturbations commute with the background, $U_p(t) = s_p \exp(t q_p T)$ and $\operatorname{Re}\operatorname{Tr} U_p(t) = 2 s_p \cos(t q_p/\sqrt{2})$, so

$$S(U(t)) = \sum_p \big(1 - s_p \cos(t q_p/\sqrt{2})\big), \qquad S''(0) = \frac{1}{2}\sum_p s_p q_p^2.$$

The plaquettes with $q_p \neq 0$ are those containing a link of the support. A link $(x,\mu)$ lies in six plaquettes, two in each plane $\{\mu,\nu\}$, at $x$ and at $x - e_\nu$. Listing the six plaquettes of each of the four links, the plaquette at $(0,1,1,1)$ in the plane $\{1,2\}$ contains two links of the support, $((0,1,1,1),1)$ and $((0,1,1,1),2)$, with $q_p = -1-1 = -2$; the other 22 plaquettes contain one link each, with $q_p = \pm 1$. The sign $s_p$ of a plaquette is $-1$ exactly when it has an odd number of links on which $U$ equals $-I$; counting through the four links, which is a finite check, gives twelve plaquettes with $s_p = 1$, $q_p^2 = 1$, ten with $s_p = -1$, $q_p^2 = 1$, and the one with $s_p = -1$, $q_p^2 = 4$. Hence $\sum_p s_p q_p^2 = 12 - 10 - 4 = -2$, $S''(0) = -1$, and $T(\xi) = 4$.

Every plaquette of $U$ is central, so every first derivative of $S$ at $U$ vanishes and the second derivative along any curve through $U$ with velocity $\xi$ depends only on $\xi$: the intrinsic Hessian on the fibre has the value $-1$ at $\xi$, which has $T(\xi) = 4$. The Haar density of the fibre, in a chart in which the path links

are solved for, is the inverse Jacobian of the path-link equations, which at $U$ is a fixed positive number independent of $\beta_W$; its logarithmic second derivatives are bounded in any fixed chart, so no choice of chart turns the Hessian of $\beta_W S$ plus the log-density positive for all large $\beta_W$. □

The proposition does not contradict Theorem 131, which assumes small plaquettes; it shows that the hypothesis of that theorem on the coarse field cannot be dropped in favour of relative link labels. A coarse-curvature label is therefore part of any labelling, and the next lemma shows that it has a witness at the fine scale, so that it is as rare as a fine defect.

**Lemma 173** (a fine witness for coarse curvature)**.** *Let $P$ be the coarse plaquette at $y$ in the plane $\{\mu,\nu\}$ and suppose that $\delta(U_p,I)\le\kappa\le 10^{-8}$ for every fine plaquette $p$ with a vertex in one of the blocks at $\ell^\infty$ distance at most one from $B(y)$, $B(y+e_\mu)$, $B(y+e_\nu)$ or $B(y+e_\mu+e_\nu)$. Then $c=V^\sharp(U)$ satisfies*

$$\delta(c_P,I)\le(4+28\pi)\kappa<92\kappa.$$

*Proof.* For a pair $k$ among the four sides of $P$, Lemma 124(i) gives $\delta(U_e,I)\le 3\kappa$ for the link of each path inside a block and $\delta(U_e,\ell_k)\le 21\kappa$ for its private link, so $\delta(P_x,\ell_k)\le 24\kappa$ for every path, and by Lemma 116 $\|\log(\ell_k^{-1}P_x)\|\le 8\pi\kappa$. The two paths of offset zero contribute nothing, so $c_k=\ell_k M_k$ with $\|\log M_k\|\le\frac{14}{16}\,8\pi\kappa=7\pi\kappa$ and $\delta(c_k,\ell_k)\le 7\pi\kappa$. The four path links around $P$ are the holonomy of the square of side two at $2y$: the other four links of its sides are the first links of the straight paths, which are tree links and equal $I$. That square is tiled by four fine plaquettes, and inserting them with their parallel transports gives $\delta(\ell_P,I)\le 4\kappa$, where $\ell_P$ is that holonomy. Replacing the four $\ell_k$ by $c_k$ one at a time, bi-invariance of $\delta$ and the triangle inequality add $4\cdot 7\pi\kappa$. □

With the coarse field calm, the chart has a centre. The next lemma shows that for a pair whose non-path links are near the reconstruction, the constraint $V_k^\sharp=c_k$ determines the path link, on every branch of the cut-off, and the fibre of the window over its non-path links is a single graph.

**Lemma 174** (the path link is determined)**.** *Let $k$ be a pair with cut-off average $c_k$, and suppose that each of its seven non-path private links and each link inside $B(y)$ and $B(y+e_\mu)$ is within geodesic distance $r$ of its reconstruction value. Then every path other than the two copies of $\ell_k$ has $d(P_x,c_k)\le 2r$, and every $\ell_k$ with $V_k^\sharp=c_k$ satisfies*

$$d(\ell_k,c_k)\le 14r,\qquad d(v_x,I)\le 16r \text{ for every path.}$$

*If moreover $r\le 10^{-4}$, the cut-off equals one at every such $\ell_k$, and $\ell_k$ is unique: it is the only fixed point of*

$$X\mapsto F(X)=-\frac{1}{16}\sum_x\log\bigl(e^{-X}c_k^{-1}P_x\bigr),\qquad \ell_k=c_k e^X,$$

*in the ball $\|X\|\le 14r$, where the sum runs over the fourteen paths of nonzero offset, and $F$ is a contraction of that ball with constant less than $0.9$.*

*Proof.* A path of nonzero offset is the product of a link inside a block, within $r$ of $I$, and a non-path private link, within $r$ of $c_k$, in one of the two orders; bi-invariance of $d$ and the triangle inequality give $d(P_x,c_k)\le 2r$. If the cut-off is zero, $V_k^\sharp=\ell_k$, so $\ell_k=c_k$. Otherwise every $v_x=\ell_k^{-1}P_x$ has $\delta(v_x,I)<1$, its principal logarithm is defined and $\|\log v_x\|=d(v_x,I)=d(\ell_k,P_x)$. Put $D=d(\ell_k,c_k)$ and $N=\frac{1}{16}\sum_x\log v_x$, the two terms of offset zero being zero. Then $c_k=\ell_k\exp(\chi N)$ and

$$D=d(e^{\chi N},I)\le\chi\|N\|\le\frac{14}{16}\max_x d(\ell_k,P_x)\le\frac{7}{8}(D+2r),$$

so $D\le 14r$, and $d(v_x,I)\le D+2r\le 16r$. For $r\le 10^{-4}$ this is less than $\frac12$, so $\chi=1$, and $V_k^\sharp=c_k$ reads $X=F(X)$ for $\ell_k=c_k e^X$ with $\|X\|\le 14r$. On that ball each $A_x=c_k^{-1}P_x$ has $d(A_x,I)\le 2r$, so $d(e^{-X}A_x,I)\le 16r$ and $\|F(X)\|\le\frac{14}{16}\cdot 16r=14r$: $F$ maps the ball into itself. Writing $e^{-X}A_x=I+M$ with $\|M\|\le 16r$, the series of the logarithm gives $\|D\log(I+M)\|\le 1/(1-16r)$, and the differential of $X\mapsto e^{-X}A_x$ has norm at most $e^{14r}$; hence $\|DF\|\le\frac78 e^{14r}/(1-16r)<0.9$, and the fixed point is unique by the contraction principle. □

**Proposition 175** (good windows are graphs)**.** *Let $W$ be a finite set of blocks, let $c$ be a coarse field, let $K_W$ be the set of pairs based in $W$ or entering $W$ from a block outside it, and let $Z_W$ be the set of non-path active links with base in $\bigcup W$. Free the links of $A_W$ and the path links of the pairs entering $W$, fix every other link, and condition on $V_k^\sharp=c_k$ for every $k\in K_W$ and on the event that every link of $Z_W$ and every non-path link of a pair entering $W$ is within $r\le 10^{-4}$ of its reconstruction value. Then the fibre of these constraints is the graph of a smooth map $\Gamma$ from the product of the $r$-balls of the links of $Z_W$ to the path links of the pairs of $K_W$, and the conditional law is the Wilson weight times the Haar disintegration of this graph, whose density with respect to the product Haar measure on $Z_W$ is the product over $k\in K_W$ of the inverse Jacobians of $X\mapsto X-F(X)$ at the fixed point.*

*Proof.* The constraint of a pair $k\in K_W$ involves its own private links and the links inside two blocks, so it involves only its own path link, links of $Z_W$, and fixed links that are within $r$ of their reconstruction values; no other constraint involves a freed link, since the other pairs have their supports off the free set. Lemma 174 applies to every $k\in K_W$ with the same $r$ and gives one solution for each pair, a smooth function of the links of $Z_W$ by the implicit function theorem since $I-DF$ is invertible. The fibre is therefore the graph of $\Gamma$, and the coarea formula for the submersion $U\mapsto(V_k^\sharp)_k$ restricted to the product of the balls gives the stated density, since the derivative of the constraint of $k$ in its own path link is $I-DF$ in the chart at $c_k$ and the cross derivatives between different pairs vanish. □

The proposition says that, once the coarse field is calm and the labels are placed on the non-path links, the good windows carry the product structure that the proof of Theorem 131 needs, with no branch of the cut-off to discard; the Hessian estimate of that theorem then applies on the graph. Two further examples show that the two obvious alternatives fail: labelling all

physical links, including the path links, produces a nonconvex domain, and a bad pair can have several path links.

**Proposition 176** (the all-link label domain is not convex)**.** *Let $c \equiv I$, freeze every link outside the eight private links of one pair at $I$, and write the seven non-path links as $e^{z_j}$ and the path link as $e^X$. For every sufficiently small $r>0$ the set of $(z_1,\dots,z_7)$ for which all eight links lie in the geodesic ball of radius $r$ at $I$ is not convex.*

*Proof.* With the cut-off equal to one, the constraint is $H(X,z) = X + \frac{1}{8}\sum_{j=1}^{7} \log(e^{-X}e^{z_j}) = 0$, since each non-path link occurs in two of the sixteen paths and the two paths of offset zero contribute nothing. Take $A,B \in \mathfrak{su}(2)$ orthonormal for the Hilbert–Schmidt norm, so that $[A,[A,B]] = -2B$ and $[B,[B,A]] = -2A$, and put $z_1 = -\frac{a}{7}A + hB$, $z_2 = -\frac{a}{7}A - hB$, $z_3 = \cdots = z_7 = -\frac{a}{7}A$ with $a>0$ small. The inner automorphism of $\mathrm{SU}(2)$ that fixes $A$ and negates $B$ and $[A,B]$ exchanges the logarithms of $e^{-xA}e^{z_1}$ and $e^{-xA}e^{z_2}$, so their sum lies along $A$ when $X = xA$, and the equation reduces to its $A$-component; by the implicit function theorem the solution is $X = x(a,h)A$ with $x$ analytic, even in $h$ by the same symmetry, and $x(a,0) = a$ since $\log(e^{-xA}e^{bA}) = (b-x)A$ gives $x + \frac{7}{8}(b-x) = 0$ at $h=0$. Let $L(x,b,h)$ be the $A$-component of $\log(e^{-xA}e^{bA+hB})$. The Baker–Campbell–Hausdorff series through degree three has one term of degree two in $h$ along $A$, namely $\frac{1}{12}[hB,[hB,-xA]] = \frac{xh^2}{6}A$, so $\partial_h^2 L(x,b,0) = \frac{x}{3} + O((|x|+|b|)^2)$. The $A$-component of $H$ is $x + \frac{1}{8}\big(2L(x,-\frac{a}{7},h) + 5(-\frac{a}{7}-x)\big)$, and $\partial_x L(x,b,0) = -1$; differentiating $H=0$ twice in $h$ at $h=0$ gives $\frac{1}{8}\partial_h^2 x + \frac{1}{4}\partial_h^2 L = 0$, that is

$$\partial_h^2 x(a,0) = -2\,\partial_h^2 L(a,-\tfrac{a}{7},0) = -\frac{2a}{3} + O(a^2) < 0.$$

Put $\eta = \frac{r}{2}$ and take $a$ with $r < a < r(1+\frac{\eta^2}{4})$. The points $z^{\pm}$ with $h = \pm\eta$ and their midpoint $z^0$ with $h=0$ have all seven non-path links within $\sqrt{a^2/49+\eta^2} < r$ of $I$; the path link of $z^{\pm}$ is at distance $x(a,\pm\eta) = a - \frac{a}{3}\eta^2 + O(r^4) \le r - \frac{r^3}{48} + O(r^4) < r$ from $I$ for small $r$, while that of $z^0$ is at distance $x(a,0) = a > r$. So the endpoints lie in the domain and the midpoint does not. □

**Proposition 177** (a bad pair with three path links)**.** *Let $c \equiv I$, freeze every link outside the eight private links of one pair at $I$, and take the private links in the one-parameter subgroup $\mathrm{diag}(e^{iq},e^{-iq})$: one non-path link at the angle $-\frac{1}{2}$, the other six at the angle $\frac{49}{100}$. Then there are at least three values of the path-link angle $q$ with the same cut-off average, and the derivative of the cut-off average with respect to $q$ vanishes at some $q \in (0,\frac{1}{5})$.*

*Proof.* On the subgroup all logarithms are scalar and the cut-off average has the angle $f(q) = q + \frac{\chi(q)}{8}\big(-\frac{1}{2} + 6\cdot\frac{49}{100} - 7q\big)$, where $\chi(q) = \phi(m(q))$, $m(q) = \max_j 2\sqrt{2}\,|\sin((\theta_j - q)/2)|$ and $\theta_1 = -\frac{1}{2}$, $\theta_2 = \cdots = \theta_7 = \frac{49}{100}$, the factor $2\sqrt{2}|\sin(\theta/2)|$ being the chordal distance of $\mathrm{diag}(e^{i\theta},e^{-i\theta})$ from $I$. At $q=0$ the maximum is attained at $\theta_1$ alone, with $m(q) = 2\sqrt{2}\sin((q+\frac{1}{2})/2)$ near 0 and $\frac{1}{2} < m(0) < 1$, so $\chi = 2-2m$ is smooth near 0 with $\chi'(0) = -2\sqrt{2}\cos\frac{1}{4}$, and

$$\begin{aligned} f'(0) &= 1 - \frac{7}{8}\chi(0) + \frac{\chi'(0)}{8}\Big(-\frac{1}{2} + \frac{294}{100}\Big) \\ &= -\frac{3}{4} + \sqrt{2}\Big(\frac{7}{2}\sin\frac{1}{4} - \frac{61}{100}\cos\frac{1}{4}\Big) \end{aligned}$$

after inserting $\chi(0) = 2 - 4\sqrt{2}\sin\frac{1}{4}$. With $\sin\frac{1}{4} < \frac{1}{4}$, $\cos\frac{1}{4} > \frac{31}{32}$ and $1 < \sqrt{2} < \frac{3}{2}$ on the positive and the negative term separately, $f'(0) < -\frac{91}{3200} < 0$. Put $\gamma = f(0)$, which lies in $(0,1)$. For $|q| = 1$ the maximal chordal distance exceeds one, so $\chi(\pm1) = 0$ and $f(\pm1) = \pm1$. Since $f$ decreases through $\gamma$ at 0 and $f(-1) < \gamma < f(1)$, the intermediate value theorem gives one preimage of $\gamma$ in $(-1,0)$, one in $(0,1)$, and $q = 0$. On $[0,\frac{1}{5}]$ the maximiser stays $\theta_1$ and $\frac{9}{10} < m < 1$ at $q = \frac{1}{5}$, so $\chi(\frac{1}{5}) < \frac{1}{5}$ and $f'(\frac{1}{5}) > 1 - \frac{7}{40} - 3\cdot\frac{13}{100} > 0$; with $f'(0) < 0$, continuity gives a zero of $f'$ in $(0,\frac{1}{5})$. □

Hence the Jacobian of the path-link constraint, which enters the density of Proposition 175, has no positive lower bound on the bad side, and a bad pair contributes a sum over branches to any estimate of the conditional law. On the good side none of this happens: the labels of Proposition 175 are placed on the non-path links only.

### B. The path-link law on its two sides

The conditional law of the path link of a pair $k$, given its seven non-path private links $z = (\ell_a)_{a\neq 0}$, the links $u,w$ inside its two blocks and the rest of the field $\sigma$, is

$$\pi(dg \mid z,u,w,\sigma) = \frac{1}{Z}\sum_{g:\,V(g;z,u,w)=c_k} \frac{e^{-\beta_W S(g;\cdot)}}{|\det D_g V(g;z,u,w)|}\,\delta_g, \tag{25}$$

where $V = V_k^{\sharp}$ as a function of the path link, the sum runs over all its preimages of $c_k$, and $|\det D_g V|$ is the Jacobian between tangent spaces of the bi-invariant metric. This is the Haar disintegration along the Lipschitz map $V$ on the compact three-manifold $\mathrm{SU}(2)$, by the coarea formula, times the Wilson weight; on the good side it is the one-point law of Proposition 175, and on the bad side it carries the branches of Proposition 177. We determine its response to each kind of exterior link. Throughout, $W_1^{\delta}$ and $W_1^{d}$ are the Kantorovich distances for the chordal and the geodesic distance, and the facts $\|L_X\| \le \theta/\sin\theta$ for an eigenangle $\theta < \pi$ of $X/\sqrt{2}$, $\|E_Y\| \le 1$ and $d(e^A,e^B) \le \|A-B\|$ of Sec. XII D are used as stated there.

**Lemma 178** (links that enter only the Wilson weight)**.** *Let $e'$ be a link that is neither a private link of $k$ nor inside $B(y)$ or $B(y+e_\mu)$, and let $\sigma,\sigma'$ differ only at $e'$. Then for every $(z,u,w)$ and $c_k$ for which (25) is a finite nonzero measure,*

$$W_1^{\delta}\big(\pi(\cdot \mid z,u,w,\sigma), \pi(\cdot \mid z,u,w,\sigma')\big) \le \beta_W\, d(\sigma_{e'},\sigma'_{e'}),$$

*and the two laws coincide unless $e'$ lies in a plaquette with a private link of $k$. For the law of all eight private links given*

*$u, w, \sigma$ and restricted to any fixed measurable set $B$ of the non-path links, with the cost $\sum_a \delta(\ell_a, \ell'_a)$, the same holds with $8\beta_W$. Both bounds include every branch and the disintegration density.*

*Proof.* Only $S$ depends on $\sigma_{e'}$: $V$ is a function of $g, z, u, w$, and so is the measure $\nu = \sum_g \delta_g / |\det D_g V|$, respectively $\mathbf{1}_B J_8^{-1} \mathrm{vol}_M$ with $M = \{V = c_k\}$ and $J_8$ the Jacobian of Lemma 135. An exterior link lies in at most one plaquette with a private link: the private links are the eight $\mu$-links at the sites $x'_a$, whose $\mu$-coordinates agree, so the two plaquettes in a plane $\{\mu, \nu\}$ through a $\nu$-link, which sit at different $\mu$-coordinates, cannot both contain one; and a $\mu$-link $(x, \mu)$ with $x$ not a site $x'_a$ has $x + \varepsilon e_\nu = x'_a$ for at most one pair $(\nu, \varepsilon)$, since two such relations would put a coordinate of $x$ outside $\{0, 1\}$ in one direction and inside it in another. If there is no such plaquette $p_0$ the laws are equal. Otherwise move $\sigma_{e'}$ along a minimal geodesic $\sigma_{e'} e^{t\xi}$, $\|\xi\| = 1$, $0 \le t \le T = d(\sigma_{e'}, \sigma'_{e'})$, with laws $\pi_t$. For bounded $f$, $\frac{d}{dt}\pi_t(f) = -\beta_W \operatorname{Cov}_{\pi_t}(f, \dot{s}_{p_0})$, by differentiation under the integral against the fixed finite measure $\nu$. Here $\frac{d}{dt} U_{p_0}(t) = U_{p_0}(t)\xi'$ with $\|\xi'\| = 1$, $\xi'$ a conjugate of $\pm\xi$, so $\dot{s}_{p_0} = -\frac{1}{2}\operatorname{Re}\operatorname{Tr}(U_{p_0}\xi')$; in a basis in which $U_{p_0} = \operatorname{diag}(e^{i\varphi}, e^{-i\varphi})$ only the diagonal entries $\pm i\alpha$ of $\xi'$ contribute, $2\alpha^2 \le 1$, and $\dot{s}_{p_0} = \alpha \sin\varphi$. Hence $|\dot{s}_{p_0}| \le \frac{1}{\sqrt{2}}$, and a link of a plaquette moving at geodesic speed $v$ gives $|\dot{s}_p| \le v/\sqrt{2}$. A function that is 1-Lipschitz for $\delta$ has oscillation at most the chordal diameter $2\sqrt{2}$ of SU(2), and at most $8 \cdot 2\sqrt{2}$ for the sum cost on eight links. Since $|\operatorname{Cov}(f, h)| \le \frac{1}{4} \operatorname{osc} f \operatorname{osc} h$, $|\frac{d}{dt}\pi_t(f)| \le \beta_W$, respectively $8\beta_W$; integrate over $[0, T]$ and use the Kantorovich duality. □

This is Lemma 163 for the fibre law: the branches and the density are harmless for every link that enters only through the action. The links it leaves out are the seven non-path links, which are in the conditioning, and the inside links $u_a, w_a$, which enter $V$. On the good side both are harmless, and the radius of Lemma 174 and Proposition 175 can be taken as $1/32$ rather than $10^{-4}$.

**Proposition 179** (the good side to radius $1/32$)**.** *Let $r \le \frac{1}{32}$, and suppose $d(\ell_a, c_k) \le r$ for $a \ne 0$ and $d(u_a, I), d(w_a, I) \le r$. Put $\theta_0 = \frac{1}{2\sqrt{2}}$ and $\kappa_0 = \frac{7}{8}\theta_0 / \sin\theta_0 < 0.8935$. Then every solution $g$ of $V(g; z, u, w) = c_k$ has cut-off equal to one and $d(g, c_k) \le 14r$, and there is exactly one: writing $g = c_k e^X$, $X$ is the unique fixed point of $F(X) = -\frac{1}{16}\sum_x \log(e^{-X} c_k^{-1} P_x)$ in $\{\|X\| \le 14r\}$, and $F$ is a contraction of that ball with constant $\kappa_0$. At the solution the singular values of $D_g V$ lie in $[1 - \kappa_0, 1 + \kappa_0]$, so the density $1/|\det D_g V|$ lies in $[(1+\kappa_0)^{-3}, (1-\kappa_0)^{-3}] \subset [0.147, 828]$. The law (25) is the point mass at $g = g(z, u, w)$, and for two configurations satisfying the hypothesis*

$$d(g, g') \le 1.1986 \sum_{a \ne 0} d(\ell_a, \ell'_a) + 0.5993 \sum_{a \ne 0} \big( d(u_a, u'_a) + d(w_a, w'_a) \big),$$

*with no dependence on $\beta_W$, on $c_k$ or on any other link.*

*Proof.* The first part is the proof of Lemma 174, which holds for every $r$ up to the contraction: every solution has $D = d(g, c_k) \le 14r$ and $d(v_x, I) \le 16r \le \frac{1}{2}$, so $m \le \frac{1}{2}$ and $\chi = 1$, and $V = c_k$ reads $X = F(X)$, with $F$ mapping the ball into itself. For the contraction constant, on the ball each $A_x = c_k^{-1} P_x$ has $d(A_x, I) \le 2r$, so $Y_x = \log(e^{-X} A_x)$ has $\|Y_x\| \le 16r$, an eigenangle at most $16r/\sqrt{2} \le \theta_0$. Since $e^{-(X+tZ)} A_x = (I - tE_{-X} Z + O(t^2)) e^{Y_x}$ and $\log(e^{tW} e^Y) = Y + tL_Y W + O(t^2)$, $\|DF\| \le \frac{14}{16} \cdot \|E_{-X}\| \max_x \|L_{Y_x}\| \le \kappa_0$, as $\theta / \sin\theta$ increases on $[0, \pi)$; the fixed point is unique by the contraction principle. For the Jacobian, vary $g \mapsto g e^{tZ}$: then $v_x \mapsto e^{-tZ} v_x$, $\log v_x \mapsto \log v_x - t L_{Y_x} Z + O(t^2)$, and with $N = \frac{1}{16}\sum_x \log v_x$ and $\bar{L} = \frac{1}{14}\sum_x L_{Y_x}$,

$$V^{-1} \frac{dV}{dt}\Big|_{t=0} = \operatorname{Ad}(e^{-N}) \big( I - \tfrac{7}{8} E_N \bar{L} \big) Z,$$

by $\frac{d}{dt} e^{N+tW} e^{-N} = E_N W$. As $\operatorname{Ad}(e^{-N})$ is an isometry and $\|\frac{7}{8} E_N \bar{L}\| \le \kappa_0$, the singular values lie in $[1 - \kappa_0, 1 + \kappa_0]$. Finally, the law has one atom, so $W_1^d = d(g, g')$. Move one link at a time along a minimal geodesic; the geodesic ball of radius $r$ is convex, so the hypothesis holds along the way, and $d(g, g') \le \|X - X'\|$. For $\ell_a \mapsto \ell_a e^{t\zeta}$, $\|\zeta\| = 1$, the two paths through $\ell_a$ move, $A_{x_a} \mapsto A_{x_a} e^{t\zeta}$ and $A_{x'_a} \mapsto A_{x'_a} e^{t \operatorname{Ad}(w_a^{-1})\zeta}$, and $\log(e^Y e^{t\zeta'}) = Y + tL_{-Y}\zeta' + O(t^2)$, so $\|\partial_t F\| \le \frac{2}{16}\theta_0 / \sin\theta_0$ and, by the implicit function theorem, $\|dX/dt\| \le \frac{2}{16}\frac{\theta_0}{\sin\theta_0}\frac{1}{1-\kappa_0} < 1.1986$. For $u_a \mapsto u_a e^{t\zeta}$ one path moves, $A_{x_a} \mapsto A_{x_a} e^{t \operatorname{Ad}(\ell_a^{-1})\zeta}$, which gives half of it, and likewise for $w_a$. □

The only change from Lemma 174 is that the series bound $\|D \log(I + M)\| \le (1 - 16r)^{-1}$ is replaced by the exact norm $\theta / \sin\theta$ of $L_Y$. So on the good side the response of the path-link law to every exterior link is bounded, with a constant independent of $\beta_W$, and the bad side begins at $r = \frac{1}{32}$. On the bad side the response to the non-path links is not bounded at all. The configuration of Proposition 177 has a fold: a value $c$ at which two branches are born, with infinite density, and the law jumps across it.

**Proposition 180** (the fold of the path-link law)**.** *In the configuration of Proposition 177, with every link other than the eight private links equal to $I$, let $c_\lambda = \operatorname{diag}(e^{i\lambda}, e^{-i\lambda})$ and let $\pi_\lambda$ be the law (25) with $c_k = c_\lambda$. Let $f_0$ be the angle of the cut-off average on the torus, the function $f$ of Proposition 177, let $q^* \in (0.0741, 0.0742)$ be its minimiser on $[-0.005, 0.2227]$, $\gamma^* = f_0(q^*) \in [0.16978261, 0.16978263]$, $g^* = e^{iq^*}$, and $\Lambda = [\gamma^* - 10^{-3}, \gamma^* + 10^{-3}]$. Then:*

(i) *For $\lambda \in \Lambda$ every preimage of $c_\lambda$ lies on the torus. For $\lambda < \gamma^*$ there is one, $g_L(\lambda) = e^{iq_L(\lambda)}$ with $q_L \in [-0.0172, -0.0156]$; for $\lambda > \gamma^*$ there are three, $g_L(\lambda)$ and $g_\pm(\lambda) = e^{iq_\pm}$ with $q_- < q^* < q_+$ and $|q_\pm - q^*| \le \sqrt{(\lambda - \gamma^*)/2.3}$; for $\lambda = \gamma^*$ there are two, $g_L$ and $g^*$. Off $\gamma^*$ the value $c_\lambda$ is regular.*

(ii) *The weight $e^{-\beta_W S} / |\det D_g V|$ of $g_L(\lambda)$ is at most 2.78, and the weights of $g_\pm(\lambda)$ tend to infinity as $\lambda \downarrow \gamma^*$, for every $\beta_W \ge 0$.*

(iii) $\pi_\lambda \to \delta_{g^*}$ *as* $\lambda \downarrow \gamma^*$ *and* $\pi_\lambda \to \delta_{g_L^*}$ *as* $\lambda \uparrow \gamma^*$, $g_L^* = g_L(\gamma^*)$, *weakly, with* $\delta(g^*, g_L^*) \geq 0.1268$. *Hence for every* $\beta_W \geq 0$ *and* $\varepsilon > 0$ *there are* $\lambda < \gamma^* < \lambda'$ *with* $\lambda' - \lambda < \varepsilon$ *and* $W_1^\delta(\pi_\lambda, \pi_{\lambda'}) \geq 0.126$.

(iv) *Fix* $c = c_{\gamma^*}$ *and move the seven non-path links together,* $\ell_a \mapsto \ell_a e^{it}$, *with laws* $\pi^{(t)}$. *Then* $\pi^{(t)} \to \delta_{g^*}$ *as* $t \uparrow 0$ *and* $\pi^{(t)} \to \delta_{g_L^*}$ *as* $t \downarrow 0$. *So for every* $\beta_W \geq 0$ *and* $\varepsilon > 0$ *there are two configurations of the non-path links at total distance* $7\sqrt{2}\,(t'-t) < 10\varepsilon$ *whose path-link laws are at* $W_1^\delta$ *distance at least* $0.126$*: the Kantorovich response of the path-link law to its non-path links is infinite, for every* $\beta_W$.

*The two inequalities marked* $(*)$ *in the proof are verified in interval arithmetic in the archive; everything else is proved by hand.*

*Proof.* Write $g = g_0 + g_1 i + g_2 j + g_3 k$ as a unit quaternion, so that the torus $T$ is $\{e^{iq}\}$, and $\theta_a$ for the angles of the non-path links, $-\frac{1}{2}$ once and $\frac{49}{100}$ six times. Then $V(g) = g\exp(\chi(m)\frac{1}{8}\sum_{a\neq 0}\log(g^{-1}\ell_a))$ with $m = \max_a \delta(g^{-1}\ell_a, I)$, since each $\ell_a$ lies on two paths and $u = w = I$, and the Wilson weight of the path link is $e^{-\beta_W S(g)}$ with $S(g) = \frac{3}{4}[\delta(g,I)^2 + \delta(g, e^{0.49i})^2] \in [0, 12]$, the three plaquettes through the path link and its partner and the three through the path link and $I$.

*Step 1, the torus.* For $q \in [q_{lo}, q_{hi}] = [\frac{49}{100} - \alpha, -\frac{1}{2} + \alpha] \approx [-0.2327, 0.2227]$, where $\alpha = \arccos\frac{3}{4}$, the maximiser of $\delta(g^{-1}\ell_a, I)$ at $g = e^{iq}$ is the far link when $q \geq -0.005$ and the six near links when $q \leq -0.005$, $m = 2\sqrt{2}\sin(d(q)/2) \in (\frac{1}{2}, 1)$ with $d(q) = \max\{q + \frac{1}{2}, \frac{49}{100} - q\}$, and

$$f_0(q) = q + \frac{\chi(q)}{8}\Big(\frac{244}{100} - 7q\Big), \qquad \chi(q) = 2 - 4\sqrt{2}\sin\frac{d(q)}{2};$$

outside $[q_{lo}, q_{hi}]$ some $\delta(g^{-1}\ell_a, I) \geq 1$, $\chi = 0$ and $f_0(q) = q$. On $[q_{lo}, -0.005]$, $\chi' = 2\sqrt{2}\cos\frac{0.49-q}{2} > 0$ and $\frac{244}{100} - 7q > 0$, so, as $\chi(-0.005) = 2 - 4\sqrt{2}\sin 0.2475 < 0.6146$,

$$f_0' \geq 1 - \tfrac{7}{8}\chi \geq 1 - \tfrac{7}{8}\cdot 0.6146 > 0.46 :$$

$f_0$ increases from $q_{lo}$ to $f_0(-0.005) = 0.18501\ldots$. On $[-0.005, q_{hi}]$,

$$\chi' = -2\sqrt{2}\cos\tfrac{q+1/2}{2} < 0, \qquad \chi'' = \sqrt{2}\sin\tfrac{q+1/2}{2} > 0,$$

so $f_0'' = \frac{1}{8}\chi''(\frac{244}{100} - 7q) - \frac{7}{4}\chi' > 0$: $f_0$ is strictly convex, with $f_0'(-0.005^+) = -0.3858\ldots < 0$ and $f_0'(q_{hi}) = 0.7087\ldots > 0$, a unique minimiser $q^*$, and $f_0'(0.0741) < 0 < f_0'(0.0742)$ places it, while convexity gives $f_0(0.0741) + 10^{-4} f_0'(0.0741) \leq \gamma^* \leq \min\{f_0(0.0741), f_0(0.0742)\}$, the stated interval. For $\lambda \in \Lambda \subset (0.168, 0.171)$ a torus preimage $e^{iq}$ with $q \notin [q_{lo}, q_{hi}]$ would need $q \equiv \lambda$ modulo $2\pi$, impossible. So the torus preimages are: one root $q_L$ in $[q_{lo}, -0.005]$, which lies in $[-0.0172, -0.0156]$ because $f_0$ is increasing there and $f_0(-0.0172) < \gamma^* - 10^{-3}$, $f_0(-0.0156) > \gamma^* + 10^{-3}$; and in $[-0.005, q_{hi}]$ none if $\lambda < \gamma^*$, the point $q^*$ if $\lambda = \gamma^*$, and two if $\lambda > \gamma^*$, as $f_0(-0.005)$ and $f_0(q_{hi}) = q_{hi}$ exceed $\lambda$. These two lie in $[0.05, 0.1]$, where $f_0'' > 4.6$, since $f_0(0.05), f_0(0.1) > \gamma^* + 10^{-3}$, whence $\lambda - \gamma^* \geq \frac{4.6}{2}(q_\pm - q^*)^2$.

*Step 2, off the torus.* Conjugation by $h \in T$ fixes every $\ell_a$ and $c_\lambda$, so $V(hgh^{-1}) = hV(g)h^{-1}$ and the preimages of $c_\lambda$ are invariant under it; every $g \notin T$ is conjugate to $g(q,s) = \cos s\, e^{iq} + \sin s\, k$ with $s \in (0, \frac{\pi}{2}]$. Since $(g^{-1}\ell_a)_0 = \cos s\cos(\theta_a - q)$, if $\cos s \leq \frac{3}{4}$ or $q \notin (q_{lo}, q_{hi})$ some $\delta(g^{-1}\ell_a, I) \geq 1$, so $\chi = 0$ and $V(g) = g \notin T$. On the remaining strip $[q_{lo}, q_{hi}] \times (0, \alpha)$, $V_k(q,0) = 0$ and $V_k(q,s) = \int_0^s \partial_s V_k$; the certificate verifies $(*)$ that $\partial_s V_k > 0$ for $s \in [0, 0.02]$ and $V_k > 0$ for $s \in [0.02, \alpha]$, on adaptive subdivisions of the strip, so that $V(g) \notin T$ for $s > 0$. Hence $V^{-1}(T) = T$, which with Step 1 gives (i) except for regularity.

*Step 3, the Jacobian at a torus point.* $D_g V$ at $g = e^{iq}$ commutes with the differential of conjugation by $e^{i\varphi}$, which fixes the axial line $g \cdot i\mathbb{R}$ and rotates the transverse plane by $2\varphi$; a linear map commuting with all these rotations preserves the line and the plane and is a complex scalar $\lambda_\perp e^{i\omega}$ on the plane. The axial eigenvalue is $f_0'(q)$, and $\lambda_\perp = \|\partial_s V(g(q,s))|_{s=0}\| = |\partial_s V_k(q,0)|$, since $\partial_s g = k$ is a unit transverse vector and, by the reflection $g(q,s) \mapsto g(q,-s)$ given by conjugation with $i$, $\partial_s V_j(q,0) = 0$. So $|\det D_g V| = |f_0'(q)|\lambda_\perp(q)^2$. The certificate verifies $(*)$ that $\partial_s V_k \geq 0.5149$ on $q \in [-0.0172, -0.0156]$ and $\leq 0.7326$ on $q \in [0.05, 0.1]$, $s \in [0, 0.02]$. At $q_L$ the near links are the strict maximiser and at $q_\pm$ the far link is, with $m \in (\frac{1}{2}, 1)$, so $V$ is real-analytic near these points; $f_0'(q_L) \in [1.36, 1.37]$ by the formula, $f_0'(q_\pm) \neq 0$ by strict convexity and $\lambda_\perp > 0$, so $c_\lambda$ is regular for $\lambda \neq \gamma^*$, and (25) is the continuous disintegration there. The weight of $g_L$ is at most $1/(1.36 \cdot 0.5149^2) < 2.78$, and those of $g_\pm$ are at least $e^{-12\beta_W}/(|f_0'(q_\pm)|\,0.7326^2)$, with $f_0'(q_\pm) \to f_0'(q^*) = 0$. This is (ii).

*Step 4, the limits.* Let $f(g) = \delta(g, g^*)$, 1-Lipschitz for $\delta$ and at most $2\sqrt{2}$. For $\lambda \downarrow \gamma^*$, $\pi_\lambda(f) \leq 2.78 \cdot 2\sqrt{2}/(w_- + w_+) + \sqrt{2}\max_\pm |q_\pm - q^*| \to 0$, and as $\pi_\lambda$ lives on three points converging to $\{g^*, g_L^*\}$, $\pi_\lambda \to \delta_{g^*}$. For $\lambda \uparrow \gamma^*$, $\pi_\lambda = \delta_{g_L(\lambda)} \to \delta_{g_L^*}$. For $\lambda \in \Lambda$, $q^* - q_L(\lambda) \in [0.0897, 0.0914]$, so $\pi_\lambda(f) = 2\sqrt{2}\sin\frac{q^* - q_L}{2} \geq 0.1268$, and the Kantorovich duality $W_1^\delta(\pi_\lambda, \pi_{\lambda'}) \geq |\pi_\lambda(f) - \pi_{\lambda'}(f)|$ gives (iii).

*Step 5, the exterior form.* Multiplying every path on the right by $e^{it}$ gives $V(ge^{it}; z_t) = V(g; z_0)e^{it}$, the covariance of $V^\sharp$ with $h_1 = I$, $h_2 = e^{-it}$. So $g$ solves $V(g; z_t) = c_{\gamma^*}$ iff $g' = ge^{-it}$ solves $V(g'; z_0) = c_{\gamma^* - t}$, with the same Jacobian, and the action at $g = g'e^{it}$ under $z_t$ differs from $S(g')$ by at most $6|t|$. Hence $\pi^{(t)}$ is the image of $\pi_{\gamma^* - t}$ under the translation by $e^{it}$, which moves every point by $\sqrt{2}|t|$, with weights multiplied by factors in $[e^{-6\beta_W|t|}, e^{6\beta_W|t|}]$, and the limits of Step 4 persist as $t \to 0$; each non-path link moves by $\sqrt{2}|t' - t|$. This is (iv). □

The mechanism is the one Proposition 177 exhibits: at the fold the density $1/|\det D_g V|$ is infinite, two branches carry all the mass on one side and do not exist on the other, and the Wilson weight cannot repair it, since a vanishing Jacobian wins against any fixed $e^{-\beta_W S}$. The configuration is bad for every $r \leq \frac{1}{32}$. So item (i) of Sec. XIII P in its original form, a Lipschitz bound for the law (25) in the non-path links, is false; what can be asked is the response of the law of all eight private links of a bad pair, in which the non-path links are integrated

over the bad set and Lemma 135 removes both the branches and the density. Lemma 178 settles that response for every link that enters only the action; what remains is its response to the inside links $u_a, w_a$, which move the fibre $\{V = c_k\}$ while the bad set does not, and the question is the mass that the law gives to the set where the fibre is tangent to the boundary of the bad set.

*The site of a bad pair.* The response of the eight private links to the inside links $u_a, w_a$ is avoided by putting those links into the variables of the law. Let $k = (y, \mu)$, $k' = (y - e_\mu, \mu)$ and $k'' = (y + e_\mu, \mu)$. The inside links of $k'$ in $B(y)$ are the links $u_a$ of $k$, and the inside links of $k''$ in $B(y + e_\mu)$ are the links $w_a$ of $k$: consecutive pairs along $\mu$ share their inside $\mu$-links. The *site* of $k$ is the set $\mathscr{S}_k$ of the 24 links $\ell_a$, $u_a$, $w_a$ ($a \neq 0$ for the last two) and the path links $\ell'_0$, $\ell''_0$ of $k'$ and $k''$, with the cost $\sum_{e \in \mathscr{S}_k} \delta(\cdot, \cdot)$. Its links are based in $B(y - e_\mu) \cup B(y) \cup B(y + e_\mu)$, and the cut-off averages that contain a link of $\mathscr{S}_k$ are $V_k^\sharp$, which contains site links only, $V_{k'}^\sharp$, whose other links are the non-path links $z'$ of $k'$ and its inside links $u'$ in $B(y - e_\mu)$, and $V_{k''}^\sharp$, with $z''$ and $w''$ in $B(y + 2e_\mu)$. Every other link enters the law of the site only through the Wilson weight.

**Lemma 181** (the law of a site)**.** *Let $r \leq \frac{1}{32}$, suppose $d(\ell'_a, c_{k'})$, $d(u'_a, I)$, $d(\ell''_a, c_{k''})$, $d(w''_a, I) \leq r$ for $a \neq 0$, and let $E$ be a measurable set of site variables contained in $\{d(u_a, I) \leq r,\ d(w_a, I) \leq r,\ a \neq 0\}$. Then the law of the site given the exterior $\sigma$, the three constraints $V_k^\sharp = c_k$, $V_{k'}^\sharp = c_{k'}$, $V_{k''}^\sharp = c_{k''}$ and $E$ is*

$$\frac{1}{Z}\, e^{-\beta_W S}\, \mathbf{1}_E\, J_{22}^{-1}\, \mathrm{vol}_{M_k}(d\ell\, du\, dw)\, \frac{\delta_{\Gamma'}(d\ell'_0)}{|\det D_{\ell'_0} V_{k'}^\sharp|}\, \frac{\delta_{\Gamma''}(d\ell''_0)}{|\det D_{\ell''_0} V_{k''}^\sharp|},$$

*where $M_k = \{V_k^\sharp = c_k\}$ in the 22 variables $(\ell, u, w)$, $J_{22} = \det(DV_k^\sharp DV_k^{\sharp *})^{1/2} \geq \sigma_0^3$ almost everywhere, $\Gamma' = \Gamma'(z', u', u)$ and $\Gamma'' = \Gamma''(z'', w'', w)$ are the unique solutions of Proposition 179 for $k'$ and $k''$, and the two determinants lie in $[(1 - \kappa_0)^3, (1 + \kappa_0)^3]$.*

*Proof.* Only the three named constraints involve site variables. For $k'$ the hypothesis of Proposition 179 holds, its inside links in $B(y)$ being the $u_a$, which $E$ keeps within $r$ of $I$; so $\ell'_0 \mapsto V_{k'}^\sharp$ has the single regular preimage $\Gamma'$, the Haar disintegration of $\ell'_0$ along it is the atom divided by the determinant, and the determinant bounds are those of the proposition; likewise for $k''$. The Haar disintegration of $(\ell, u, w)$ along $V_k^\sharp$ is $J_{22}^{-1}\mathrm{vol}_{M_k}$ by the coarea formula. Since $DV_k^\sharp DV_k^{\sharp *}$ is the sum over the variables of the nonnegative terms $D_e V_k^\sharp (D_e V_k^\sharp)^*$, it dominates the sum over the eight private links, which is at least $\sigma_0^2$ by Lemma 135. The product of the three disintegrations is the disintegration of the product Haar measure along the three constraints, and the Wilson weight and $\mathbf{1}_E$ multiply it. □

The constraint of the site's own pair is thus a fixed measure on $M_k \cap E$: the exterior enters only through $S$, through the positions $\Gamma'$, $\Gamma''$ of two atoms, and through two densities on the good side, whose variation the next lemma bounds.

**Lemma 182** (the good-side density is log-Lipschitz)**.** *Under the hypothesis of Proposition 179, let one link of the pair move along a unit-speed geodesic. Then $\left|\frac{d}{dt}\log|\det D_g V|\right| \leq 64.94$ when the link is a non-path link and $\leq 32.47$ when it is an inside link.*

*Proof.* By the proof of Proposition 179, $|\det D_g V| = |\det T|$ with $T = I - \frac{7}{8} E_N \bar{L}$, so $\frac{d}{dt}\log|\det T| = \mathrm{tr}(T^{-1}\dot{T})$, which is at most $3\|T^{-1}\|\|\dot{T}\|$ on the three-dimensional $\mathfrak{g}$, with $\|T^{-1}\| \leq (1 - \kappa_0)^{-1}$ and $\dot{T} = -\frac{7}{8}(\dot{E}_N \bar{L} + E_N \dot{\bar{L}})$, $\|\bar{L}\| \leq \Lambda_0 := \theta_0 / \sin\theta_0 = 1.02115$, $\|E_N\| \leq 1$.

For a power series $\varphi$ whose radius of convergence exceeds the spectral radius of a normal operator $A$ with spectral projections $P_i$ and eigenvalues $\lambda_i$, the derivative is $d\varphi(A)[B] = \sum_{i,j} \varphi[\lambda_i, \lambda_j] P_i B P_j$, with the divided differences $\varphi[\lambda_i, \lambda_j]$ of Daleckii–Krein [33]; the blocks $P_i B P_j$ are orthogonal for the Hilbert–Schmidt product, so $\|d\varphi(A)[B]\| \leq \max_{i,j}|\varphi[\lambda_i, \lambda_j]|\,\|B\|_{HS}$, and by the Hermite–Genocchi formula $\varphi[\lambda_i, \lambda_j] = \int_0^1 \varphi'((1-u)\lambda_i + u\lambda_j)\, du$ this maximum is at most the maximum of $|\varphi'|$ on the convex hull of the spectrum. With $E_Y = e(\mathrm{ad}\, Y)$, $L_Y = f(\mathrm{ad}\, Y)$, $e(x) = (e^x - 1)/x$, $f(x) = x/(e^x - 1)$, and $\|\mathrm{ad}\, W\|_{HS} = 2\|W\|$, this gives $\|\dot{L}_Y\| \leq 2M_L\|\dot{Y}\|$ and $\|\dot{E}_N\| \leq 2M_E\|\dot{N}\|$, where $M_L$ and $M_E$ bound $|f'|$ and $|e'|$ on the segments $i[-a', a']$, $a' \leq \sqrt{2}\cdot 16r \leq \frac{\sqrt{2}}{2}$, and $i[-b', b']$, $b' \leq \sqrt{2}\cdot 14r \leq 0.6188$, which contain the spectra $\{0, \pm i a'\}$ of $\mathrm{ad}\, Y_x$ and $\{0, \pm i b'\}$ of $\mathrm{ad}\, N$. Since $e'(z) = \int_0^1 t e^{tz}\, dt$, $|e'(iy)| \leq \frac{1}{2}$, so $M_E = \frac{1}{2}$. Since $f'(iy) = -\frac{1}{2} - i\frac{\sin y - y}{4\sin^2(y/2)}$, and $y - \sin y \leq \frac{y^3}{6}$, $\sin\frac{y}{2} \geq \frac{y}{2}(1 - \frac{y^2}{24})$, $M_L \leq \left(\frac{1}{4} + \frac{a^2}{36}(1 - \frac{a^2}{24})^{-4}\right)^{1/2} \leq 0.5149$ at $a = \frac{\sqrt{2}}{2}$.

Along the motion the fixed point $X$ moves with $\|\dot{X}\| \leq \dot{x}$, $\dot{x} = 1.1986$ for a non-path link and 0.5993 for an inside link, by the proof of Proposition 179. For a path that does not contain the moving link, $\dot{Y}_x = -L_{Y_x} E_{-X} \dot{X}$, so $\|\dot{Y}_x\| \leq \Lambda_0 \dot{x}$; for a path that contains it, $\log(e^Y e^{t\zeta'}) = Y + tL_{-Y}\zeta' + O(t^2)$ adds $\Lambda_0$. With $n = 2$ moving paths for a non-path link and $n = 1$ for an inside link, $\|\dot{N}\| \leq \frac{\Lambda_0}{16}(14\dot{x} + n)$ and $\|\dot{\bar{L}}\| \leq 2M_L\Lambda_0(\dot{x} + \frac{n}{14})$. Hence

$$\left|\frac{d}{dt}\log|\det T|\right| \leq \frac{21}{8}\,\frac{2M_E\Lambda_0\|\dot{N}\| + \|\dot{\bar{L}}\|}{1 - \kappa_0},$$

which is at most 64.94 and 32.47 in the two cases, with $(1 - \kappa_0)^{-1} \leq 9.390$. □

**Theorem 183** (the site of a bad pair responds boundedly)**.** *Under the hypotheses of Lemma 181, let $\sigma, \sigma'$ differ only at one link $e'$ outside $\mathscr{S}_k$, and let $\pi(\cdot \mid \sigma)$ be the law of the site of that lemma. Then*

$$W_1\big(\pi(\cdot \mid \sigma), \pi(\cdot \mid \sigma')\big) \leq C_{e'}\, d(\sigma_{e'}, \sigma'_{e'}),$$

*for the cost $\sum_{e \in \mathscr{S}_k} \delta$, with $C_{e'} = 24\, n_p(e')\beta_W$ if $e'$ enters only the Wilson weight, $n_p(e') \leq 6$ the number of its plaquettes that contain a site link, $C_{e'} = 197\beta_W + 2206$ if $e'$ is a non-path link of $k'$ or $k''$, and $C_{e'} = 87\beta_W + 1103$ if $e'$ is an inside link of $k'$ in $B(y - e_\mu)$ or of $k''$ in $B(y + 2e_\mu)$. The constants do not depend on $c_k$, $c_{k'}$, $c_{k''}$, $E$ or the rest of $\sigma$, and every branch of $V_k^\sharp = c_k$ inside the site is integrated, not compared.*

*Proof.* Move $\sigma_{e'}$ along a unit-speed geodesic, $0 \le t \le T = d(\sigma_{e'}, \sigma'_{e'})$; geodesic balls of radius $r \le \frac{1}{32}$ are convex, so the hypotheses of Lemma 181 hold along the way. By that lemma, $\pi_t = (\Phi_t)_* \mu_t$ with $\Phi_t(\ell, u, w) = (\ell, u, w, \Gamma'_t, \Gamma''_t)$ and $\mu_t$ the law on $M_k \cap E$ with density $h_t = e^{-\beta_W S_t} J_t'^{-1} J_t''^{-1}$ against the fixed measure $\mathbf{1}_E J_{22}^{-1} \mathrm{vol}_{M_k}$; plaquettes without a site link are constant on $M_k$ and cancel. For $f$ Lipschitz with constant one for the site cost, whose oscillation is at most the diameter $24 \cdot 2\sqrt{2}$,

$$\pi_T(f) - \pi_0(f) = (\mu_T - \mu_0)(f \circ \Phi_T) + \mu_0(f \circ \Phi_T - f \circ \Phi_0).$$

The second term is at most $\dot{x} T$ by Proposition 179, since one atom moves, and it vanishes when $e'$ enters only the Wilson weight. For the first, $\frac{d}{dt}\mu_t(\varphi) = \mathrm{Cov}_{\mu_t}(\varphi, \partial_t \log h_t)$ and $|\mathrm{Cov}(\varphi, \psi)| \le \frac{1}{4} \operatorname{osc}\varphi \operatorname{osc}\psi$, so the first term is at most $24\sqrt{2}\, T \sup|\partial_t \log h_t|$, and $|\partial_t \log h_t| \le \beta_W |\partial_t S_t| + |\partial_t \log J'_t|$. Links of a plaquette moving at total geodesic speed $v$ give $|\dot{s}_p| \le v/\sqrt{2}$, by the proof of Lemma 178. If $e'$ enters only the Wilson weight, $\operatorname{osc} \partial_t S_t \le \sqrt{2}\, n_p(e')$ and $J'$ does not move, which gives $24 n_p \beta_W$. If $e' = \ell'_a$, it shares at most one plaquette with a site link, and the atom $\Gamma'$ moves at speed at most 1.1986 through its six plaquettes, so $|\partial_t S_t| \le (1 + 6 \cdot 1.1986)/\sqrt{2} \le 5.793$, and Lemma 182 bounds the density term by 64.94; hence $C_{e'} \le 24\sqrt{2}(5.793\beta_W + 64.94) + 1.1986 < 197\beta_W + 2206$. If $e' = u'_a$, it shares no plaquette with a site link, the atom moves at speed at most 0.5993, $|\partial_t S_t| \le 6 \cdot 0.5993/\sqrt{2} \le 2.543$, and $C_{e'} \le 24\sqrt{2}(2.543\beta_W + 32.47) + 0.5993 < 87\beta_W + 1103$. Kantorovich duality gives the bound on $W_1$. □

If the inside $\mu$-links of $B(y)$ are themselves off, so that $k'$ is not on the good side, the site is continued along $\mu$ through the private and inside links of $k'$, and so on in both directions, up to the first pair on each side whose non-path links lie within $r$ of their coarse value and whose inside links lie within $r$ of $I$; the proofs of Lemma 181 and Theorem 183 apply to this chain unchanged, with the diameter $24 \cdot 2\sqrt{2}$ replaced by $2\sqrt{2}$ times the number of its links. The chain lies in the island together with its $\mu$-neighbours. The site therefore costs one block of collar on each side of an island: with the bad blocks joined at $\ell^\infty$ distance at most $2s+3$, the islands enlarged by one block are separated by layers of good blocks of width at least $2s+1$, no constraint of a pair based in a window of such a layer contains a site link, and the witnesses $M = 11^4$, $H = 6^4$ of Lemma 184 are unchanged. Every statement of Theorem 199 then holds with $(4s+7)^4 - 1$ neighbours in place of $(4s+3)^4 - 1$: $|\log q^*| = 25777.3$ and $\beta^*(10^{-8}, 1) = 4.177 \cdot 10^{21}$ for $s = 1$, $|\log q^*| = 35087.3$ and $\beta^*(10^{-8}, 5) = 5.667 \cdot 10^{21}$ for $s = 5$, and $\beta^*(10^{-9}, 1) = 4.181 \cdot 10^{23}$. With the layer between two enlarged islands of width at least $2s+8$, as Corollary 210 needs, the joining distance is $2s+10$ and $\beta^* = 1.183 \cdot 10^{37}$ for $s = 1$ and $1.358 \cdot 10^{37}$ for $s = 5$ at $\rho = 2.17 \cdot 10^{-16}$.

### C. Counting bad labels

A label is bad when some fine plaquette in a fixed neighbourhood is bad: for the relative link labels this is the content of Theorem 117 and Lemma 124, and for the coarse-curvature labels it is Lemma 173. The chessboard estimate of Theorem 147 bounds the probability that a set of fine plaquettes is simultaneously bad by a product. The following lemma transfers such a bound to the labels, and the proposition after it shows that the loss it incurs is the true loss of this kind of argument.

**Lemma 184** (coverage counting)**.** *Let a family of events $E_y$, indexed by a finite set, be such that each $E_y$ forces at least one of a specified set $Q_y$ of at most $M$ sites to be bad, and suppose that each site lies in at most $H$ of the sets $Q_y$ and that $\Pr(\text{all sites of } T \text{ bad}) \le p^{|T|}$ for every finite set $T$ of sites. Then for every nonempty set $S$ of $n$ labels and $k = \lceil n/H \rceil$,*

$$\Pr\Big(\bigcap_{y \in S} E_y\Big) \le \binom{Mn}{k} p^k \le (eMHp)^k,$$

*and if $A = eMHp < 1$ then $\Pr(\bigcap_{y \in S} E_y) \le A^{n/H}$.*

*Proof.* On the intersection, the union of the witness sets $Q_y$, $y \in S$, contains at least $k$ distinct bad sites, since one site witnesses at most $H$ labels; the union has at most $Mn$ sites. A union bound over its $k$-subsets and the product bound for each give the first inequality, and $\binom{Mn}{k} \le (eMn/k)^k \le (eMH)^k$ since $n/k \le H$. No choice of witness is required to be measurable, and no independence is used. □

**Proposition 185** (the exponent $1/H$ cannot be improved)**.** *In the setting of Lemma 184 let $H$ labels share one site and let each be bad exactly when that site is bad, with probability $p$. Then $\Pr(\text{all } H \text{ labels bad}) = p$, so every bound of the form $q^{|S|}$ valid for this class of families has $q \ge p^{1/H}$.*

*Proof.* The event that all $H$ labels are bad is the event that the one site is bad. □

For the labels of Sec. XIII A the witness sets are boxes of side 7 for the relative link labels, $M = 7^4$, $H = 4^4$, and of side 11 when the coarse-curvature labels are included, $M = 11^4$, $H = 6^4$, with $p$ the chessboard bound of Theorem 147 at the threshold of Lemma 173. Lemma 184 then charges a bad island of $n$ blocks at the rate $A^{n/H}$; the proposition shows that the denominator $H$ is not an artefact of the union bound but the price of knowing only that each label has a witness, and that a better exponent needs joint information on the gauge field.

### D. Islands and the environment

Two bad islands are separated by good blocks, and the question is how the weight of one island depends on the state of the other. The following lemma answers it in terms of a single quantity: the cross-ratio of a kernel on the good layer between them. It integrates arbitrary measures on the two islands, including hard constraints and sums over branches, and their masses do not enter.

**Lemma 186** (integration through a common kernel)**.** *Let $K(x,y) > 0$ be measurable, let $a_0, a_1$ and $b_0, b_1$ be finite positive measures on the spaces of $x$ and of $y$, and suppose*

*that $Z_{ij}=\iint K\,da_i\,db_j$ is finite and positive for the four pairs $i,j\in\{0,1\}$. If*

$$e^{-\Delta}\le\frac{K(x_1,y_1)K(x_0,y_0)}{K(x_1,y_0)K(x_0,y_1)}\le e^{\Delta}$$

*for all $x_i$ in the support of $a_i$ and $y_j$ in the support of $b_j$, then*

$$\bigl|\log Z_{11}+\log Z_{00}-\log Z_{10}-\log Z_{01}\bigr|\le\Delta.$$

*Proof.* Multiply the upper inequality by $K(x_1,y_0)K(x_0,y_1)$ and integrate against $da_1(x_1)\,da_0(x_0)\,db_1(y_1)\,db_0(y_0)$; by Tonelli the left side integrates to $Z_{11}Z_{00}$ and the right side to $e^{\Delta}Z_{10}Z_{01}$. The lower inequality gives the reverse bound, and the four numbers are positive, so logarithms may be taken. □

**Corollary 187** (two labels)**.** *Let $(A,B)$ be a pair of binary labels with joint law $\Pr(A=i,B=j)=Z_{ij}/\sum Z$, all $Z_{ij}>0$, and let $\Delta$ bound the connected normaliser as in Lemma 186. Then*

$$\bigl|\Pr(B=1\mid A=1)-\Pr(B=1\mid A=0)\bigr|\le\tanh\frac{\Delta}{4},$$

$$|\operatorname{Cov}(A,B)|\le\frac{1}{4}\tanh\frac{\Delta}{4},$$

*and both bounds are attained.*

*Proof.* The conditional odds of $B=1$ given $A=1$ and given $A=0$ are $Z_{11}/Z_{10}$ and $Z_{01}/Z_{00}$, whose logarithms differ by $\log(Z_{11}Z_{00}/Z_{10}Z_{01})$, of modulus at most $\Delta$. For logistic probabilities $\sigma(u)=(1+e^{-u})^{-1}$ the difference $\sigma(u+s)-\sigma(u)$ is largest at $u=-s/2$ by the symmetry $\sigma(-u)=1-\sigma(u)$, where it equals $\sigma(s/2)-\sigma(-s/2)=\tanh(s/4)$, and it increases in $s$. The covariance is $\Pr(A=1)\Pr(A=0)$ times the difference of the conditional probabilities, and $\Pr(A=1)\Pr(A=0)\le\frac14$. The weights $Z_{11}=Z_{00}=e^{\Delta/2}$, $Z_{10}=Z_{01}=1$ have $\Pr(A=1)=\frac12$ and conditional probabilities $\sigma(\pm\Delta/2)$, so both bounds are equalities. □

When the kernel is the partition function of a calm layer, its cross-ratio is a connected second derivative of a log-normaliser, and that derivative is a covariance under the law of the layer.

**Lemma 188** (connected normalisers)**.** *Let $D$ be a fixed product of balls in the coordinates $z$ of a window, let $\mathscr{U}(z;\eta,\zeta)$ be smooth in all variables, with $\eta$ and $\zeta$ exterior parameters, and let $Z(\eta,\zeta)=\int_D e^{-\mathscr{U}}\,dz$. Then for directions $u$ in $\eta$ and $v$ in $\zeta$,*

$$\partial_u\partial_v\log Z=-\mathbb{E}\bigl[\partial_u\partial_v\mathscr{U}\bigr]+\operatorname{Cov}\bigl(\partial_u\mathscr{U},\partial_v\mathscr{U}\bigr),$$

*expectation and covariance under $e^{-\mathscr{U}}\,dz/Z$ on $D$. If $\mathscr{U}$ is a sum of local terms and the supports of $\eta$ and $\zeta$ are farther apart than its range, the first term vanishes. If in addition, at every point of two segments $\eta_0\to\eta_1$ and $\zeta_0\to\zeta_1$, the law on $D$ satisfies*

$$|\operatorname{Cov}(F,H)|\le\mathscr{A}\,\|\nabla F\|\,\|\nabla H\|\,e^{-\alpha\operatorname{dist}(\operatorname{supp}\nabla F,\operatorname{supp}\nabla H)}$$

*for all smooth $F$ and $H$, then*

$$\bigl|\log Z(\eta_1,\zeta_1)-\log Z(\eta_1,\zeta_0)-\log Z(\eta_0,\zeta_1)+\log Z(\eta_0,\zeta_0)\bigr|\le\mathscr{A}L_\eta L_\zeta\,\|\eta_1-\eta_0\|\,\|\zeta_1-\zeta_0\|\,e^{-\alpha d'},$$

*where $L_\eta,L_\zeta$ bound the gradients in $z$ of $\partial_u\mathscr{U}$ and $\partial_v\mathscr{U}$ per unit of $u,v$ along the segments, and $d'$ is a lower bound there for the distance between the supports of those gradients.*

*Proof.* Differentiating under the integral sign on the fixed domain $D$ gives $\partial_v\log Z=-\mathbb{E}[\partial_v\mathscr{U}]$ and then the identity, the covariance arising from the derivative of the normalised density. The left side of the integrated bound is the integral of $\partial_u\partial_v\log Z$ over the product of the two segments, in which the first term of the identity vanishes and the second is bounded by the covariance hypothesis. □

On a good window the law of Proposition 175, with the Hessian bound of Theorem 131 on the graph, supplies a covariance bound of the required form, so Lemma 188 controls the connected normaliser for continuous changes of the sources inside the chart, and Lemma 186 then integrates the hard island weights on the two sides. What the three statements do not supply is the kernel bound when a label change moves the domain $D$, enters an island, or adds a branch of Proposition 177; that is the open comparison of Sec. XIII P. The last result of this subsection shows that the comparison cannot be replaced by rarity, however strong.

**Proposition 189** (rarity does not control the environment)**.** *Fix $q\in(0,1)$. Let $S$ be a fair bit and, given $S$, let the labels $B_x$ be independent Bernoulli variables with parameter $q/2$ if $S=0$ and $q$ if $S=1$. Then $\Pr(B_x=1\ \forall x\in T)\le q^{|T|}$ for every finite $T$, and for all distinct $x,y$,*

$$\operatorname{Cov}(B_x,B_y)=\frac{q^2}{16}.$$

*If the configuration is a function of the labels, every covariance under the law conditioned on all labels vanishes, and the covariance of two observables is the covariance of their conditional expectations. More generally, for a binary label $B$ with $p=\Pr(B=1\mid\xi)\in(0,1)$ given the other labels $\xi$ and $\Delta_F=\mu(F\mid B=1,\xi)-\mu(F\mid B=0,\xi)$,*

$$\operatorname{Cov}(F,\mathbf{1}_B\mid\xi)=p(1-p)\Delta_F,$$

$$\operatorname{Cov}\bigl(\mathbb{E}[F\mid B,\xi],\mathbb{E}[H\mid B,\xi]\;\big|\;\xi\bigr)=p(1-p)\Delta_F\Delta_H.$$

*Proof.* Given $S$ the labels are independent, so $\Pr(B_x=1\ \forall x\in T)=\frac12\bigl((q/2)^{|T|}+q^{|T|}\bigr)\le q^{|T|}$, and $\operatorname{Cov}(B_x,B_y)=\operatorname{Var}(\Pr(B_x=1\mid S))=\frac14(q-q/2)^2=q^2/16$ by the conditional covariance formula. The two identities follow by expanding the expectations over the two values of $B$. □

The example satisfies every finite-intersection rarity bound at every order, yet its labels have a covariance with a positive floor at all distances. A labelling argument must therefore bound $\Delta_F\Delta_H$, the change of the conditional expectations under a label flip, and not only the probability of the flip: small $p$ does not help when $\Delta_F\Delta_H$ does not decay, and normalising by $p$ can cancel it. That bound is the connected normaliser of Lemma 186, which is where the island comparison enters.

### E. Coupling cost and the comparison lemma

The proofs of Lemmas 15 and 241 take a coupling that minimises a weighted sum of expected displacements and derive, from its minimality, one inequality per site. The following lemma isolates this step; it needs neither stationarity of the optimal coupling under the local updates nor any property of the updates beyond the two stated ones, and its conclusion is exactly the recurrence that Lemma 241 iterates.

**Lemma 190** (minimality gives the recurrence)**.** *Let $I$ be a finite index set, let $\Pi$ be a set of couplings of two probability measures on a product of metric spaces indexed by $I$, let $d_i(\pi) = \mathbb{E}_\pi \operatorname{dist}(x_i, x_i')$, let $w_i > 0$, and let $S \subset I$. Suppose $\pi \in \Pi$ minimises $\sum_i w_i d_i$ over $\Pi$, and that for each $i \notin S$ there is $\pi^{(i)} \in \Pi$ with $d_j(\pi^{(i)}) = d_j(\pi)$ for $j \neq i$ and $d_i(\pi^{(i)}) \leq \sum_j C_{ij} d_j(\pi)$. Then $d_i(\pi) \leq \sum_j C_{ij} d_j(\pi)$ for every $i \notin S$. If $\pi$ is minimal only up to $\varepsilon_i$ against $\pi^{(i)}$, then $d_i(\pi) \leq \sum_j C_{ij} d_j(\pi) + \varepsilon_i / w_i$. Consequently, if $C_{ij} \geq 0$, $\sum_i w_i C_{ij} \leq \theta w_j$ with $\theta < 1$ and $d_i(\pi) \leq D$ for $i \in S$, then $\sum_{i \notin S} w_i d_i(\pi) \leq \frac{\theta D}{1-\theta} \sum_{j \in S} w_j$.*

*Proof.* Since $\pi^{(i)}$ changes only the $i$-th displacement, minimality gives $w_i d_i(\pi) \leq w_i d_i(\pi^{(i)}) + \varepsilon_i$, and the bound on $d_i(\pi^{(i)})$ gives the recurrence. The last statement is the arithmetic of Lemma 241: with $X = \sum_{i \notin S} w_i d_i$ and $Y = \sum_{i \in S} w_i d_i \leq D \sum_{i \in S} w_i$, the recurrence and the column condition give $X \leq \sum_j d_j \sum_i w_i C_{ij} \leq \theta(X + Y)$, so $X \leq \theta Y / (1 - \theta)$. □

**Lemma 191** (minimal couplings exist)**.** *Let $\mu, \nu$ be probability measures on a compact metric space $\Omega$ and $c$ a continuous cost on $\Omega \times \Omega$. The set of couplings of $\mu$ and $\nu$ is nonempty and compact in the weak topology, and $c$ attains its minimum on it.*

*Proof.* The product $\mu \otimes \nu$ is a coupling; the marginal conditions are closed in the weak topology, and the probability measures on the compact space $\Omega \times \Omega$ form a compact set, so the couplings form a compact set on which the continuous functional $\pi \mapsto \int c \, d\pi$ attains its minimum. □

The displacements $d_i$ are continuous costs on the compact product of copies of SU(2), so the minimal couplings of Lemma 15 and of the block comparison exist by Lemma 191, and the coupled update of one site is the choice of $\pi^{(i)}$ in Lemma 190. The recurrence, the column-weighted and row-weighted comparisons of Lemma 241, the decay with a floor of Lemma 243, and the two lemmas above are also written in Lean 4 in the archive, with Mathlib as the library; the specification of SU(2) enters only through the hypotheses of Lemma 190, which the proofs of Secs. V and X supply by hand.

The block response that Lemma 15 bounds can also be computed exactly in one direction, which measures the room that the coupling argument leaves.

**Proposition 192** (the block response at zero field)**.** *Let $\mu_0$ be the law of three unit quaternions $(y_1, y_2, y_3)$ with density proportional to $\exp(\beta_W \operatorname{Re}(y_1 y_2 y_3))$ with respect to Haar measure, and $h$ a unit quaternion. Then $f = \langle y_1, h \rangle + \langle \overline{y_2 y_3}, h \rangle$ is 1-Lipschitz for the sum of the three chordal distances and*

$$\operatorname{Cov}_{\mu_0}\big(f, \langle y_1, h \rangle\big) = \frac{1}{4}\Big(1 + \frac{I_2(\beta_W)}{I_1(\beta_W)}\Big) = \frac{1}{4}\Big(1 + \frac{\beta_W}{4} - \frac{\beta_W^3}{96} + O(\beta_W^5)\Big),$$

*which equals $0.271826$ at $\beta_W = 0.351$. Hence the Kantorovich modulus of the block in the field at $y_1$ is at least this number, and $10\beta_W$ times it exceeds one for $\beta_W > 0.3666$.*

*Proof.* Given $y_1$, the product $w = y_2 y_3$ is Haar distributed a priori, and the weight is $\exp(\beta_W \langle y_1, \bar{w} \rangle)$ since $\operatorname{Re}(pq) = \langle p, \bar{q} \rangle$ for unit quaternions. So $\bar{w}$ given $y_1$ follows the von Mises–Fisher law on $S^3$ with mean direction $y_1$, whose mean resultant is $\mathbb{E}[\bar{w} \mid y_1] = A(\beta_W) y_1$ with $A = I_2 / I_1$, the ratio of the Bessel functions of orders $\frac{p}{2}$ and $\frac{p}{2} - 1$ for $p = 4$, and the marginal of $y_1$ is Haar. Multiplication and conjugation preserve the chordal distance, so $f$ changes by at most the sum of the three displacements. Since $y_1$ is uniform, $\operatorname{Var}\langle y_1, h \rangle = \frac{1}{4}$ and $\operatorname{Cov}(\langle \bar{w}, h \rangle, \langle y_1, h \rangle) = \mathbb{E}[A \langle y_1, h \rangle^2] = A/4$. The series of $A$ follows from $I_\nu(x) = \sum_m (x/2)^{2m+\nu} / (m! \, \Gamma(m + \nu + 1))$, and the derivative of $\mu_b(f)$ in the field $b_1$ at $b = 0$ in the direction $h$ is the covariance, which bounds the modulus below; the crossing is the root of $\frac{5}{2}\beta_W(1 + A(\beta_W)) = 1$. □

### F. From one scale to the next

A multiscale version of Sec. XII C would eliminate the fine coordinates of a window scale by scale. The following statements say what one elimination step preserves and what it does not, for the quadratic forms that Theorem 131 controls at one scale. They are finite matrix lemmas; no all-scale construction is made here.

**Lemma 193** (weighted Schur complements)**.** *Let $H = \left(\begin{smallmatrix} A & B \\ B^* & D \end{smallmatrix}\right)$ be positive definite in retained and eliminated coordinates, with $S = A - BD^{-1}B^*$ its constrained minimum form on the retained ones. Give both index sets a metric, and for a 1-Lipschitz function $\phi$ and $\alpha \geq 0$ write $\|M\|_\alpha$ for the supremum over $\phi$ of the operator norms of $e^{\alpha\phi} M e^{-\alpha\phi}$. If $\|A\|_\alpha \leq a$, $\|B\|_\alpha \leq b$ and $\|D^{-1}\|_\alpha \leq g$, then $\|S\|_\alpha \leq a + b^2 g$, and $|S_{ij}| \leq (a + b^2 g) e^{-\alpha \operatorname{dist}(i,j)}$. If $H_0$ has blocks $A_0, B_0, D_0$ with the same bounds and $\|A - A_0\|_\alpha, \|B - B_0\|_\alpha, \|D - D_0\|_\alpha \leq \varepsilon$, then*

$$\|S - S_0\|_\alpha \leq \varepsilon(1 + bg)^2.$$

*Proof.* Conjugation by $e^{\alpha\phi}$ respects products and inverses, so the weighted norm is submultiplicative on compatible blocks, and $\|B^*\|_\alpha = \|B\|_\alpha$ because $-\phi$ is admissible; the bound on $S$ follows from its formula. For the entry bound take $\phi(x) = \operatorname{dist}(x, j)$. For the difference, $S - S_0 = (A - A_0) - (B - B_0)D^{-1}B^* - B_0 D^{-1}(B - B_0)^* - B_0(D^{-1} - D_0^{-1})B_0^*$ with $D^{-1} - D_0^{-1} = -D^{-1}(D - D_0)D_0^{-1}$, and the four terms are bounded by $\varepsilon$, $\varepsilon bg$, $\varepsilon bg$ and $\varepsilon b^2 g^2$. □

**Proposition 194** (what form comparison preserves)**.** *Let $H_0, H$ be positive definite with $(1 - \varepsilon)H_0 \leq H \leq (1 + \varepsilon)H_0$, $0 \leq \varepsilon < 1$,*

*and for a surjective linear map $Q$ let $S(v)=\inf_{Qx=v}x^{\mathsf T}Hx$ and $S_0$ likewise. Then $(1-\varepsilon)S_0\le S\le(1+\varepsilon)S_0$. Form comparison does not preserve locality: on a path of $n$ sites, $H_0=I$ and $H=I+\varepsilon uu^{\mathsf T}$ with $u=(1,\dots,1)/\sqrt{n}$ satisfy the comparison, while $(H^{-1})_{1n}=-\varepsilon/((1+\varepsilon)n)$ decays in no exponential way. If instead, for $W=e^{\alpha\phi}$, $\|WH_0^{-1}W^{-1}\|\le G_0$ and $\|W(H-H_0)W^{-1}\|\le\varepsilon/G_0$, then $\|WH^{-1}W^{-1}\|\le G_0/(1-\varepsilon)$.*

*Proof.* The comparison of forms passes to infima over the same affine subspaces. The rank-one inverse is $H^{-1}=I-\frac{\varepsilon}{1+\varepsilon}uu^{\mathsf T}$, whose off-diagonal entries all equal $-\varepsilon/((1+\varepsilon)n)$. The last statement is the Neumann series $H^{-1}=\sum_m(-H_0^{-1}(H-H_0))^mH_0^{-1}$ in the weighted norm. □

Repeating the comparison over scales multiplies the factors $(1\pm\varepsilon_k)$, so the one-step errors must be summable, and Lemma 193 identifies the one-step amplification $(1+b_kg_k)^2$ of the locality norm. Both are conditions that an effective action built from Theorem 131 at every scale would have to meet; the theorem at one scale provides the bounds $a,b,g$ for the first step only.

An error bound that is a multiple of the Euclidean form does not survive this repetition, because the block constraint enlarges it at every step.

**Proposition 195** (absolute errors grow under the block map)**.** *Let $U$ have plaquettes at most $\kappa$, let $B$ be the derivative at $U$ of the block averages of a window, as in Theorem 125, and $\mathscr{G}=BB^*$. For every positive semidefinite form $N$ on the link coordinates, every $\varepsilon\ge0$ and every $w$ in the range of $B$,*

$$\inf_{BX=w}\big(N(X)+\varepsilon\|X\|^2\big)\ge\inf_{BX=w}N(X)+\frac{4\varepsilon}{1+2724\kappa}|w|^2.$$

*Proof.* The infimum of a sum is at least the sum of the infima, and $\inf_{BX=w}\|X\|^2=\langle w,\mathscr{G}^{-1}w\rangle$, attained at $X=B^*\mathscr{G}^{-1}w$. By the proof of Theorem 125(iii), a row of $\mathscr{G}$ has its diagonal block of norm at most $\frac{3}{16}+227\kappa$ and two further blocks of norm at most $\frac{1}{32}+227\kappa$, so the Schur test gives $\|\mathscr{G}\|\le\frac14+681\kappa$ and $\mathscr{G}^{-1}\ge4/(1+2724\kappa)$. □

A one-step bound of the form $|\mathrm{Hess}-H_{\mathrm{ref}}|\le\lambda\varepsilon\|\cdot\|^2$ on the fine links, which is the form in which Theorems 125 and 127 and Proposition 133 control every non-Gaussian term, is therefore met by $H_{\mathrm{ref}}+\lambda\varepsilon\|\cdot\|^2$, whose block infimum exceeds that of $H_{\mathrm{ref}}$ by $4\lambda\varepsilon(1+2724\kappa)^{-1}|w|^2$. Carried over $k$ blocking steps with the Gaussian fibre gap independent of the scale (Proposition 99(iv)), such a bound certifies the gap of the fibre only while $4^k\varepsilon$ stays small against it, that is, for a number of scales of order $\log_4(1/\varepsilon)$: an absolute error is a mass term and is relevant in four dimensions. The comparison of Proposition 194 is relative and is preserved with factor one, so the one-step errors that an effective action needs are relative bounds by a gauge-invariant form; since the block action $\inf_{BX=w}Q_U(X)$ vanishes along the coarse gauge directions, such a bound requires the one-step error to vanish along them as well.

### G. Spectral weight at positive time

Theorem 48 makes the nontriviality of the continuum limit equivalent to a ratio of lattice two-point functions staying away from zero. The following two lemmas show that a ratio at a fixed physical time controls the spectral weight of the renormalised observable, so that the ratio criterion is a tightness condition on its spectral measure and not on its unsmeared moments, and the example after them shows that a mass gap by itself does not give nontriviality.

**Lemma 196** (tightness at positive time)**.** *Let $\sigma_a$ be positive measures on $[0,\infty)$ with Laplace transforms $C_a(t)=\int e^{-tE}d\sigma_a(E)$, let $s>0$ be fixed, and suppose $0<C_a(s/2)<\infty$ and $C_a(s)/C_a(s/2)\ge c>0$ uniformly in $a$. Let $d\nu_a=e^{-sE}d\sigma_a/C_a(s)$. Then*

$$\int E\,d\nu_a\le M_s=\frac{2}{s}\log\frac{1}{c}\le\frac{2}{esc},\qquad \nu_a([R,\infty))\le\frac{M_s}{R},$$
$$\frac{C_a(s+t)}{C_a(s)}\ge e^{-M_st}\quad(t\ge0).$$

*Proof.* The Laplace transform $C_a$ is finite on $[s/2,\infty)$, smooth on $(s/2,\infty)$ with $C_a'=-\int Ee^{-tE}d\sigma_a$, and $\log C_a$ is convex there by the Cauchy–Schwarz inequality $C_a(\frac{t+t'}{2})^2\le C_a(t)C_a(t')$. A convex function has derivative at $s$ at least the slope of its chord over $[s/2,s]$, so

$$\int E\,d\nu_a=-(\log C_a)'(s)\le\frac{2}{s}\log\frac{C_a(s/2)}{C_a(s)}\le\frac{2}{s}\log\frac{1}{c},$$

and $\log u\le u/e$ gives $M_s\le2/(esc)$. Markov's inequality gives the tail, and convexity gives $\log C_a(s+t)-\log C_a(s)\ge t(\log C_a)'(s)\ge-M_st$. □

When $\sigma_a$ is the spectral measure of a centred observable $O_a$ for a nonnegative Hamiltonian, $\nu_a$ is the spectral law of the unit vector $e^{-sH_a/2}O_a\Omega_a/C_a(s)^{1/2}$; the lemma keeps its spectral mass from escaping to infinite physical energy. The ratio at the two separations $s/2$ and $s$ is the quantity that Proposition 51 bounds by slice correlations.

**Lemma 197** (nonvanishing limits)**.** *If $\sigma_a([0,\infty))\ge v>0$ and $\int E\,d\sigma_a\le M$ uniformly in $a$, then $C_a(t)\ge ve^{-Mt/v}$ for all $t\ge0$, and every limit of the $C_a(t)$ is positive.*

*Proof.* With $v_a=\sigma_a([0,\infty))$, Jensen's inequality for the probability $\sigma_a/v_a$ gives

$$C_a(t)\ge v_a\exp\Big(-\frac{t}{v_a}\int E\,d\sigma_a\Big)\ge v_ae^{-Mt/v_a},$$

and $x\mapsto xe^{-Mt/x}$ is increasing. □

**Proposition 198** (a gap without a limit)**.** *Let $X$ be the symmetric two-state Markov process with jump rate $m/2$ in each direction, sampled at times $na$, and let $O_a=aX$. The transfer matrix has eigenvalues $1$ and $e^{-ma}$, so the gap in physical units is $m$ for every $a$, while $C_a(t)=a^2e^{-mt}\to0$ and every moment of $O_a$ tends to zero. The sampled process is reflection positive, and the spectral measure $\sigma_a$ is the atom $a^2\delta_m$,*

*whose Laplace transform at $s/2$ and $s$ has the ratio $e^{-ms/2}$, uniform in $a$; Lemma 196 applies, and the limit vanishes because $\sigma_a([0,\infty)) = a^2 \to 0$, which Lemma 197 excludes by its hypothesis on the mass.*

*Proof.* The correlation $\mathbb{E}[X_0 X_t] = e^{-mt}$ for the two-state process, and $O_a$ scales it by $a^2$; the transfer matrix of the sampled chain is the matrix of its transition probabilities, with the stated eigenvalues. Reflection positivity of a stationary reversible Markov chain is standard. The rest is read off. □

The example separates the two conditions that Theorem 48 keeps apart: a uniform gap and a nonvanishing ratio. For a sequence of measures whose mass does not vanish, Lemma 197 converts a bound on the first moment at positive time, which Lemma 196 derives from one ratio, into a positive lower bound for the limit; the mass itself is the normalisation of Theorem 48.

### H. The island threshold at one scale

The counting of Lemma 184 and the collar of Proposition 175 determine a coupling past which the bad blocks of one scale form islands whose entropy is beaten by their rarity. This subsection computes that coupling. Fix a radius $r$ for the relative link labels and a radius $\eta$ for the coarse plaquettes: a block $y$ is *bad* when some non-path active link of a pair based at $y$ is at distance more than $r$ from its reconstruction value, or some coarse plaquette with corner $y$ has $\delta(c_P, I) > \eta$, and *good* otherwise. Fix an integer collar $s \geq 1$: a window $W$ is conditioned as in Proposition 175 when every block at $\ell^\infty$ distance at most $s$ from $W$ is good. Join two blocks when they are at $\ell^\infty$ distance at most $2s+1$, and call a maximal connected set of bad blocks an *island*; two islands are then separated by a layer of good blocks of width at least $2s+1$, whose middle blocks are at distance more than $s$ from both, and the good layer of Lemma 186 is available in the middle of it. Each block has $\Delta = (4s+3)^4 - 1$ neighbours in this graph, and the witness sets of Lemma 184 are the boxes of side 11, $M = 11^4$, $H = 6^4$.

**Theorem 199** (the island threshold)**.** *Let $G = \mathrm{SU}(2)$, let $\rho = \min\{r/43, \eta/92\} \leq 10^{-8}$, and put*

$$\log q^* = -H\big(\log 2 + 2\log\Delta\big) - \log(eMH),$$
$$f(\beta) = 6(\log\beta + C_0) - \frac{\beta\rho^2}{16},$$

*with $C_0$ as in Lemma 146. The equation $f(\beta) = \log q^*$ has a largest solution $\beta^* = \beta^*(\rho, s) > 96/\rho^2$, and for every $\beta_W \geq \beta^*$:*

(i) *every block is bad with probability at most $A = eMH\, q_\rho(\beta_W) \leq (2\Delta^2)^{-H}$;*

(ii) *for every block $y$ and every $n \geq 1$,*

$$\mu_L\{y \text{ lies in an island of at least } n \text{ blocks}\} \leq 2^{-n}\Delta^{-2},$$

*and the expected number of islands of at least $n$ blocks on a torus of $L^4$ blocks is at most $L^4 2^{-n}\Delta^{-2}/n$;*

(iii) *if an estimate charges every bad block a factor $\Gamma \geq 1$ and is summed over the islands through a given block, the sum $\sum_{n\geq 1} \Delta^{2(n-1)}(\Gamma A^{1/H})^n$ is at most $\Delta^{-2}$ as soon as $\beta_W \geq \beta^*(\rho, s, \Gamma)$, the largest solution of $f(\beta) = \log q^* - H\log\Gamma$.*

*Numerically, $\beta^*(10^{-8}, 1) = 3.427 \cdot 10^{21}$, $\beta^*(10^{-8}, 5) = 5.401 \cdot 10^{21}$, $\beta^*(10^{-9}, 1) = 3.431 \cdot 10^{23}$, $\beta^*(10^{-9}, 5) = 5.405 \cdot 10^{23}$ and $\beta^*(10^{-8}, 5, 10^{30}) = 1.973 \cdot 10^{22}$; in every case $\beta^* = 16|\log q^*|\rho^{-2}(1+\varepsilon)$ with $0 < \varepsilon < 2 \cdot 10^{-2}$, and $|\log q^*| = 21090.2$ for $s = 1$ and $33424.9$ for $s = 5$.*

*Proof. Witnesses.* Let $y$ be bad because of a non-path link $e$ of the pair $k = (y, \mu)$, and let $\kappa$ be the largest defect $\delta(U_p, I)$ over the plaquettes with a vertex in a block at $\ell^\infty$ distance at most one from $B(y)$ or $B(y + e_\mu)$. If $\kappa \leq 10^{-8}$, Lemma 124(i) with $W = \{y, y + e_\mu\}$ gives $\delta(U_e, I) \leq 3\kappa$ when $e$ lies inside a block, where $R_e(c) = I$, and $\delta(U_e, \ell_k) \leq 21\kappa$ when $e$ leaves $B(y)$, where $R_e(c) = c_k$ and $\ell_k$ is the path link; the proof of Lemma 173 gives $\delta(c_k, \ell_k) \leq 7\pi\kappa$, so $\delta(U_e, R_e(c)) \leq (21 + 7\pi)\kappa < 42.992\kappa$; since $d(g,h) \leq (1 + 10^{-12})\delta(g,h)$ when $\delta(g,h) \leq 10^{-6}$, the geodesic distance of the label obeys $d(U_e, R_e(c)) < 43\kappa$. Since the label is bad, either $\kappa > 10^{-8} \geq \rho$ or $43\kappa > r$, and in both cases some plaquette of that set has defect more than $\rho$. If $y$ is bad because of a coarse plaquette with corner $y$, Lemma 173 gives in the same way a plaquette with defect more than $\min\{10^{-8}, \eta/92\} \geq \rho$ among those with a vertex in a block at distance at most one from the four corner blocks. All these plaquettes lie in cubes whose corners are at $\ell^\infty$ distance at most five from $2y$, the box of side 11 of the fine lattice, and a cube lies in the box of at most $6^4$ blocks. With the sites of Lemma 184 the unit cubes and $p = q_\rho(\beta_W)$, Theorem 147 gives the product bound, and the lemma gives $\mu_L(\text{all blocks of } S \text{ bad}) \leq A^{n/H}$ for every set $S$ of $n$ blocks when $A = eMHp < 1$, and $\leq A$ for one block.

*The threshold.* The function $f$ is concave on $(0,\infty)$ with its maximum at $\beta = 96/\rho^2$, where $f = 6\log(96/\rho^2) + 6C_0 - 6 > 0 > \log q^*$; since $f(\beta) \to -\infty$, the equation $f = \log q^*$ has exactly two solutions and the larger one, $\beta^*$, exceeds $96/\rho^2$, with $f \leq \log q^*$ on $[\beta^*, \infty)$. For $\beta_W \geq \beta^*$ this gives $q_\rho(\beta_W) \leq e^{f(\beta_W)} \leq q^*$, hence $A \leq eMHq^* = (2\Delta^2)^{-H}$ and $\Delta^2 A^{1/H} \leq \frac{1}{2}$, which is (i).

*Islands.* A connected set of $n$ blocks containing $y$ in the graph of the collar is recovered from a walk of length $2(n-1)$ around a spanning tree, each step to one of $\Delta$ neighbours, so there are at most $\Delta^{2(n-1)}$ of them. The island of $y$ has at least $n$ blocks only if some connected set of exactly $n$ blocks containing $y$ consists of bad blocks, and a given set of $n$ blocks is bad with probability at most $A^{n/H}$, so

$$\mu_L\{\text{the island of } y \text{ has at least } n \text{ blocks}\} \leq \Delta^{2(n-1)} A^{n/H} \leq \Delta^{-2} 2^{-n}.$$

An island of at least $n$ blocks contains at least $n$ blocks that lie in islands of at least $n$ blocks, so the number of such islands is at most $1/n$ times the number of such blocks, whose expectation is at most $L^4 2^{-n}\Delta^{-2}$; on a torus the number of neighbours of a block is at most $\Delta$, which is all the count uses. For (iii), replace $A$ by $\Gamma^H A$ in the threshold: then $\Gamma\Delta^2 A^{1/H} \leq \frac{1}{2}$

and the sum is at most $\Delta^{-2}\sum_n 2^{-n} = \Delta^{-2}$. The numerical values are the roots of $f = \log q^*$ computed to thirty digits, and the two-term expression follows from $f(\beta) = \log q^*$ with $6(\log\beta + C_0) \le 0.02\,|\log q^*|$ at these values. □

The theorem gives the value asked for in item (iii) of the earlier versions of this section, but only at the block scale and in lattice units: at $\beta_W = 5\cdot 10^{21}$ the lattice spacing is far below every physical length, the correlation length of the theory in lattice units is astronomically larger than the side of a block, and the rarity of the islands at that scale says nothing yet about the islands of the effective theory at the scale of the cube, which is what condition (D) needs. The threshold therefore closes the entropy count of one scale, and leaves the comparison between scales to the effective action of Proposition 194 and Lemma 193. The numbers also show where the loss sits: the factor $16/\rho^2$ comes from the witness radius $\rho \le 10^{-8}$ of Lemma 124, and the factor $|\log q^*| \approx 2H\log\Delta$ from the exponent $1/H$, which Proposition 185 shows cannot be improved without joint information on the gauge field.

### I. The island threshold at every scale

The threshold of Theorem 199 composes with the defect propagation of Theorem 152: the labels of the $j$-th iterate of the cut-off average have fine witnesses as well, in a larger box, so that the chessboard estimate and the coverage count apply at every scale up to the one that Corollary 153 reaches, divided by 256. Write $U^{(j)}$ for the $j$-th iterate in the tree gauge, an $\mathrm{SU}(2)$ field on the lattice of spacing $a_j = 2^j a$, and call a block $y$ of that lattice *bad at scale $j$* when it is bad for the field $U^{(j)}$ in the sense of Theorem 199, with the same radii $r$, $\eta$ and $\rho = \min\{r/43, \eta/92\} \le 10^{-8}$. Let $\mathfrak{B}_j$, $\eta_j$ be as in Theorem 152 with $\eta_0 = 1.01\rho_0$ for a threshold $\rho_0 \le \rho$, so that every plaquette of $\mathfrak{B}_0$ has defect more than $\rho_0$.

**Lemma 200** (fine witnesses at scale $j$)**.** *Let $j \ge 1$ and suppose that $\eta_j \le \rho$. If the block $y$ is bad at scale $j$, then some fine plaquette $p \in \mathfrak{B}_0$, with defect more than $\rho_0$, has its corner $x$ in $2^{j+1}y + [-8\cdot 2^j + 2,\, 9\cdot 2^j - 3]^4$. The cubes of this box number $M_j = (17\cdot 2^j - 4)^4$, and a fine cube lies in the box of at most $H_j$ blocks of scale $j$, with $H_j = 8^4$ for $j = 1,2$ and $H_j = 9^4$ for $j \ge 3$.*

*Proof.* Since $\eta_{i+1} \ge 4\eta_i$, the hypothesis gives $\eta_i \le \rho \le 10^{-8}$ for $i < j$, so Theorem 152(i) and (ii) apply up to scale $j$. The witness paragraph of the proof of Theorem 199 is a statement about one field in the tree gauge and applies to $U^{(j)}$: a bad block $y$ has a plaquette $P$ of $U^{(j)}$ with corner $x' \in 2y + [-5,5]^4$ and $\delta(U_P^{(j)}, I) > \rho$. If $P \notin \mathfrak{B}_j$, Theorem 152(i) and Lemma 116 give $\delta(U_P^{(j)}, I) \le \|\Theta_P^{(j)}\| \le \eta_j \le \rho$; so $P \in \mathfrak{B}_j$, and by part (ii) some point of $P$ is within $\ell^\infty$ distance $3(2^j - 1)$, in fine units, of a plaquette $p \in \mathfrak{B}_0$ with corner $x$. In fine units the coordinates of $P$ lie in $[2^j x_c', 2^j x_c' + 2^j]$ and those of $p$ in $[x_c, x_c + 1]$, so $x_c \in [2^j x_c' - 3\cdot 2^j + 2,\, 2^j x_c' + 4\cdot 2^j - 3]$, and the range of $x_c'$ gives the box. It has $17\cdot 2^j - 4$ integer points per coordinate. A cube with corner $x$ lies in the box of $y$ exactly when $2^{j+1}y_c$ lies in an interval of length $17\cdot 2^j - 5$ for each coordinate, which contains at most $\lfloor 8.5 - 5\cdot 2^{-j-1}\rfloor + 1$ multiples of $2^{j+1}$, that is 8 for $j = 1,2$ and 9 for $j \ge 3$. □

**Theorem 201** (the island threshold at scale $j$)**.** *Let $G = \mathrm{SU}(2)$, $s \ge 1$, $j \ge 1$, $\rho = \min\{r/43, \eta/92\} \le 10^{-8}$, and put $\rho_0 = \rho/(1.01\cdot 1.023\cdot 4^j)$, $\eta_0 = 1.01\rho_0$,*

$$\log q_j^* = -H_j\big(\log 2 + 2\log\Delta\big) - \log(eM_jH_j),$$
$$f_j(\beta) = 6(\log\beta + C_0) - \frac{\beta\rho_0^2}{16},$$

*with $\Delta = (4s+3)^4 - 1$ and $M_j$, $H_j$ as in Lemma 200. Let $\beta_j^*$ be the largest solution of $f_j(\beta) = \log q_j^*$. Then for every $\beta_W \ge \beta_j^*$:*

(i) *$\eta_i \le 1.023\cdot 4^i\eta_0 \le \rho$ for $i \le j$, the iterates up to scale $j$ satisfy Theorem 152(i) and (iii) with $E_j^\circ \le 1.53\,E_0^\circ$, and every block at scale $j$ is bad with probability at most $A_j = eM_jH_jq_{\rho_0}(\beta_W) \le (2\Delta^2)^{-H_j}$;*

(ii) *for every block $y$ at scale $j$ and every $n \ge 1$, with the islands of Theorem 199 formed from the bad blocks of scale $j$,*

$$\mu_L\{y \text{ in an island of at least } n \text{ blocks}\} \le 2^{-n}\Delta^{-2},$$

*and the expected number of such islands on the torus is at most $(L/2^{j+1})^4 2^{-n}\Delta^{-2}/n$; the weighted variant (iii) of Theorem 199 holds with $\log q_j^* - H_j\log\Gamma$.*

*For $s = 5$ and $\rho = 10^{-8}$, $\beta_1^* = 2.896\cdot 10^{23}$, $\beta_2^* = 4.634\cdot 10^{24}$, $\beta_3^* = 1.186\cdot 10^{26}$, and $16^{-j}\beta_j^* \le 2.90\cdot 10^{22}$ for $1 \le j \le 12$. Hence, for $\beta_1^* \le \beta_W < \beta_{12}^*$, the conclusions hold at every scale $j \le j_{\max}$ with*

$$2^{j_{\max}} \ge \frac{1}{2}\Big(\frac{\beta_W}{2.90\cdot 10^{22}}\Big)^{1/4},$$

*while the scale $k_*$ of Corollary 153 satisfies $k_* - j_{\max} \in \{8,9\}$ at every one of 2001 couplings spaced evenly in the logarithm between $10^{22}$ and $10^{42}$: the islands of the composed average are rare up to the scale $a_{k_*}/512$.*

*Proof.* (i) $4^j\eta_0 = \rho/1.023 \le 10^{-8}$, so Theorem 152(iv) gives the bounds on $\eta_i$ and $E_j^\circ$, and $\eta_j \le \rho$ is the hypothesis of Lemma 200. That lemma makes the sets of Lemma 184 the boxes of $M_j$ cubes with overlap $H_j$, and Theorem 147 at the threshold $\rho_0$ gives the product bound $p = q_{\rho_0}(\beta_W)$; the threshold paragraph of the proof of Theorem 199, with $\rho_0$, $M_j$, $H_j$, $q_j^*$ in place of $\rho$, $M$, $H$, $q^*$, shows that $\beta_j^* > 96/\rho_0^2$ exists and that $q_{\rho_0}(\beta_W) \le q_j^*$ for $\beta_W \ge \beta_j^*$, hence $A_j \le (2\Delta^2)^{-H_j}$ and $\Delta^2 A_j^{1/H_j} \le \frac{1}{2}$. (ii) is the island paragraph of that proof on the torus of $(L/2^{j+1})^4$ blocks of scale $j$, where a block has at most $\Delta$ neighbours. The numbers are the roots of $f_j = \log q_j^*$ to thirty digits; $\beta_{j+1}^*/\beta_j^* \ge 16$ for $1 \le j \le 12$ since $\rho_0$ falls by 4 and $|\log q_j^*|$ increases, so the largest $j$ with $\beta_W \ge \beta_j^*$ satisfies $16^{j_{\max}+1} > \beta_W/(2.90\cdot 10^{22})$, which is the displayed bound; $k_*$ is computed from its definition at the same couplings. □

The theorem settles the entropy count of the islands at every scale that the fine witnesses reach. The next proposition shows that they reach no further: a coarse plaquette can be large at scale $j$ while every fine plaquette is small, because the cut-off average multiplies a constant curvature by four at every step.

**Proposition 202** (constant curvature grows by four)**.** *Let $T = \frac{1}{\sqrt{2}}\operatorname{diag}(i,-i)$, let $L = 2^m$, $n \geq 1$ and $\theta = 2\sqrt{2}\pi n/L^2$, and let $U$ be the field with $U_{x,1} = \exp(\theta x_0 T)$ for $0 \leq x_0 < L$, $U_{x,0} = \exp(-\theta L x_1 T)$ on the links with $x_0 = L-1$, and every other link $I$. Then $\Theta_p = \theta T$ on every plaquette of the plane $\{0,1\}$ and $\Theta_p = 0$ on the others, so that $\delta(U_p, I) = 2\sqrt{2}\sin(\theta/2\sqrt{2}) \leq \theta$ for every $p$, and if $4^{j+1}\theta \leq \frac{1}{12}$ then for every $i \leq j+1$*

$$\Theta_P^{(i)} = 4^i \theta\, T \quad \text{on the plane } \{0,1\},$$
$$\delta(U_P^{(i)}, I) = 2\sqrt{2}\sin\bigl(4^i\theta/2\sqrt{2}\bigr) \geq 4^i\theta\Bigl(1 - \frac{16^i\theta^2}{48}\Bigr).$$

*Consequently, with $\eta_0 \geq \theta$ the set $\mathfrak{B}_0$ is empty, while every block is bad at scale $j$ for the coarse label as soon as $4^{j+1}\theta(1 - 16^{j+1}\theta^2/48) > \eta$; for $\eta = 92\rho$ this is $4^j\theta > 23\rho$ up to a factor $1 + 10^{-3}$, whereas Theorem 201 needs $4^j\eta_0 \leq \rho/1.023$.*

*Proof.* The field takes values in the torus $S = \exp(\mathbb{R}T)$, and the plaquette exponents are the abelian curl: on the plane $\{0,1\}$ every plaquette, including those of the column $x_0 = L-1$, has exponent $\theta$, the seam links contributing $-\theta L x_1 + \theta L(x_1+1) - \theta(L-1) = \theta$ there and cancelling on the planes $\{0,2\}$, $\{0,3\}$; the plaquettes through $x_1 = L-1$ are consistent since $\exp(\theta L^2 T) = \exp(2\pi i n \cdot \operatorname{diag}(1,-1)) = I$. An $S$-valued gauge transformation puts $U$ in the tree gauge without changing the plaquettes, and Theorem 119 applies with $\eta_G = \frac{1}{12}$: by Lemma 118, $\Theta_P^{(i+1)}$ is the mean over the sixteen offsets of the sum of the four plaquettes of a square of side two, which for a constant $\Theta^{(i)} = 4^i\theta T$ on the plane $\{0,1\}$ and 0 elsewhere is $4^{i+1}\theta T$ on that plane and 0 elsewhere, and the hypothesis (21) holds by induction while $4^i\theta \leq \frac{1}{12}$. The eigenvalues of $T$ are $\pm i/\sqrt{2}$, so $\delta(\exp(tT), I) = 2\sqrt{2}|\sin(t/2\sqrt{2})|$, and $\sin u \geq u - u^3/6$. The coarse label of a block at scale $j$ is the plaquette of $U^{(j+1)}$. □

Between the two conditions lies a factor $23.5 < 4^{2.3}$, so the fine witnesses of Lemma 200 stop within three scales of where any fine witness must stop. At the radius $10^{-8}$ the remaining six scales up to $a_{k_*}$ are the ratio between the curvature margin $10^{-4}$ of Theorem 152(iv) and the witness radius of Lemma 124; Theorem 205 below removes that ratio and carries the count to $a_{k_*}/8$, and past $a_{k_*}$ the rarity of a large coarse plaquette can no longer come from the fine scale, by the proposition, but must come from the coarse energy of Lemma 165, which is known for torus-valued fields only. Theorem 201 is an annealed statement about the pushforward of $\mu_L$: it says nothing about the law of $U^{(j)}$ given its labels, which is what the effective action of item (iv) below is for.

### J. The witness radius for SU(2)

The thresholds above carry the factor $16/\rho^2$ of the witness radius $\rho \leq 10^{-8}$. For SU(2) the witness paragraph of Theorem 199 holds up to an explicit radius, which is sharp for Lemma 124(iv). Write $\angle g \in [0,\pi]$ for the angle of $g \in \mathrm{SU}(2)$, the bi-invariant geodesic distance to $I$ on $S^3$, so that $\angle(gh) \leq \angle g + \angle h$, $\angle(hgh^{-1}) = \angle g$, $\delta(g,I) = 2\sqrt{2}\sin(\angle g/2) \leq \sqrt{2}\angle g$ and $\|\log g\| = \sqrt{2}\angle g$ for $\angle g < \pi$. Put $p(\kappa) = 2\arcsin(\kappa/2\sqrt{2})$, the largest angle of a plaquette with $\delta(U_p, I) \leq \kappa$, and

$$\psi_c = \tfrac{1}{3}\arcsin\tfrac{1}{4\sqrt{2}}, \qquad \kappa_c = 2\sqrt{2}\sin\tfrac{\psi_c}{2} = 0.0837613\ldots,$$

so that $p(\kappa) < \psi_c$ exactly when $\kappa < \kappa_c$, and $\sqrt{2}\,p(\kappa) \leq 1.00015\,\kappa$ for $\kappa \leq \kappa_c$, since $\sqrt{2}\,p(\kappa)/\kappa = \frac{p/2}{\sin(p/2)}$ increases with $\kappa$.

**Lemma 203** (the ladder)**.** *Let $U$ be in the tree gauge, $v = P_{2y}$ the straight path of the pair $(y,\mu)$, and $x \in B(y)$ with $n_x \leq 3$ transverse coordinates of the offset equal to one. Then $v^{-1}P_x$ is a product of $2n_x$ conjugates of plaquette variables or their inverses, all with their vertices in $B(y) \cup B(y+e_\mu)$. If these plaquettes have $\delta(U_p, I) \leq \kappa$, then $\angle(v^{-1}P_x) \leq 2n_x p(\kappa)$, and every link $e$ leaving $B(y)$ in the direction $\mu$ has $\angle(v^{-1}U_e) \leq 9p(\kappa)$.*

*Proof.* The tree path from $2y$ to $x$ inside $B(y)$ raises the coordinates of the offset in increasing order of direction, so a step raising the coordinate $\alpha$ at $z$ uses the tree link $(z,\alpha)$, and $(z+2e_\mu,\alpha)$, with the same offset in $B(y+e_\mu)$, is a tree link as well. A step with $\alpha = \mu$ changes nothing, since then $(z,\mu)$ and $(z+2e_\mu,\mu)$ are tree links and $P_z = U_{z+e_\mu,\mu} = P_{z+e_\mu}$. For $\alpha \neq \mu$ let $p_1, p_2$ be the plaquettes at $z$ and $z+e_\mu$ in the plane $\{\mu,\alpha\}$, $g = U_{z+e_\mu,\alpha}$ and $z' = z + e_\alpha$. With the two tree links equal to $I$, $U_{p_1} = U_{z,\mu}\,g\,U_{z',\mu}^{-1}$ and $U_{p_2} = U_{z+e_\mu,\mu}\,U_{z'+e_\mu,\mu}^{-1}\,g^{-1}$, hence

$$P_z = U_{p_1}\bigl(U_{z',\mu}\,g^{-1}\,U_{p_2}\,g\,U_{z',\mu}^{-1}\bigr)P_{z'},$$

and each of the $n_x$ transverse steps multiplies by two conjugated plaquettes. The angle is subadditive and conjugation invariant. For a leaving link $e = (x,\mu)$, $P_{x-e_\mu} = U_a U_e$ with $U_a$ a link inside $B(y)$, whose angle is at most $3p$ by the chains of the proof of Lemma 124(i); so $\angle(v^{-1}U_e) \leq \angle U_a + \angle(v^{-1}P_{x-e_\mu}) \leq 9p$. □

**Proposition 204** (Lemma 124 for SU(2) to the sharp radius)**.** *Let $G = \mathrm{SU}(2)$.*

(i) *Parts (i)–(iii) of Lemma 124 hold for every $\kappa \geq 0$, and the constant 21 of (i) can be replaced by 9.*

(ii) *Part (iv) holds for every $\kappa < \kappa_c$, with $28.04\,\kappa$ in place of $260\,\kappa$.*

(iii) *For every $\kappa > \kappa_c$ there is a field with $\delta(U_p, I) \leq \kappa$ for every plaquette at which the formula of (iv) is false.*

(iv) *For $\kappa<\kappa_c$, Lemma 173 holds with $\delta(c_P,I)\le 16.003\,\kappa$, and every non-path active link $e$ of a pair at $y$ has $\delta(U_e,R_e(c))\le d(U_e,R_e(c))\le 12.002\,\kappa$.*

*Proof.* (i) The proofs of (i)–(iii) use only the triangle inequality for $\delta$, bi-invariance and $\|\mathrm{Ad}(g)X-X\|\le 2\delta(g,I)\|X\|$, none of which needs $\kappa$ small; Lemma 203 gives $9\kappa$ for the leaving links.

(ii) By Lemma 203, $\delta(P_x,v)\le 2\sqrt2\sin(3p)$, which is below $\frac12$ exactly when $p<\psi_c$. Then the cut-off equals one near $U$ and $V^\sharp=v\exp M$ with $M=\frac{1}{16}\sum_x \log w_x$, $w_x=v^{-1}P_x$. Since $\|\log w_x\|\le 2\sqrt2 n_x p$ and $\sum_x n_x=24$, $\|M\|\le 3\sqrt2\,p$, so $\|\mathrm{ad}_{\log w_x}\|\le 12p$ and $\|\mathrm{ad}_M\|\le 6p$, the spectrum of $\mathrm{ad}_X$ being $\{0,\pm i\sqrt2\|X\|\}$. The four differences of the proof of Lemma 124(iv) are then bounded with $\|\mathrm{Ad}(g)-1\|=2\sin\angle g$ for $\angle g\le\frac{\pi}{2}$ in place of $2\delta(g,I)$, and with $\sup_{|z|\le a}|z/(1-e^{-z})-1|\le\hat\varphi(a)=\frac{a}{2}+0.09a^2$ for $a\le 1$ and $\sup_{|z|\le a}|(1-e^{-z})/z-1|\le\hat\psi(a)=\frac{a}{2}e^a$. The frame changes by $e^{-M}$ and by $w_x^{-1}$ cost at most $6p$ and $12p$ times $\|\alpha\|\le N_{y,\mu}$; the factor $(1-e^{-\mathrm{ad}_M})/\mathrm{ad}_M$ and the derivatives of the logarithms cost $\hat\psi(6p)(1+\hat\varphi(12p))+\hat\varphi(12p)$ times $m_\beta+\|\alpha\|\le(1+\frac{\sqrt3}{4})N_{y,\mu}$; and the terms $\varepsilon_x$, of norm at most $2\sin(9p)\|\xi_{x,\mu}\|$ for $x_\mu$ even and $2\sin(3p)\|\xi_{x,\mu}\|$ for $x_\mu$ odd, add $\frac{1}{16}\sqrt{8(18p)^2+8(6p)^2}N_{y,\mu}$ by Cauchy–Schwarz. The sum is $pG(p)$ with $G$ increasing and $p/\kappa$ increasing, so its supremum over $\kappa\le\kappa_c$, divided by $\kappa$, is attained at $\kappa_c$, where interval arithmetic encloses it in $[28.0374,28.0375]$.

(iii) Let $L=2^m$, $\varphi=2\pi n/L^2$ and $T=\frac{1}{\sqrt2}\mathrm{diag}(i,-i)$, and multiply the three torus-valued fields of Proposition 202 that have plaquette angle $\varphi$ in the planes $\{\mu,\alpha\}$, $\alpha\ne\mu$, with a common orientation. Every plaquette of these planes has $\delta(U_p,I)=2\sqrt2\sin(\varphi/2)$ and every other plaquette is $I$. In the tree gauge the ladder identity reads $P_{z+e_\alpha}=U_{p_1}^{-1}U_{p_2}^{-1}P_z$, so for the offset $(1,1,1)$ the angle of $v^{-1}P_x$ is $6\varphi$ and $\delta(P_x,v)=2\sqrt2\sin 3\varphi$. If $\kappa>\kappa_c$, choose $L$ and $n$ with $\psi_c<\varphi\le p(\kappa)$: every plaquette has $\delta(U_p,I)\le\kappa$, the cut-off $\phi(\delta(P_x,v))=2-2\delta(P_x,v)$ is below one, and the logarithms $\log w_x$ are nonnegative multiples of one element with mean of angle $3\varphi\ne 0$, so $V^\sharp\ne v\exp M$ and (iv) fails. For $L=32$, $n=10$ one has $\kappa=0.0867614\ldots$, $\delta(P_x,v)=0.517715\ldots$ and cut-off $0.964570\ldots$.

(iv) By (ii), $c_k=\ell_k e^{M_k}$ with $\angle e^{M_k}\le 3p$. A link leaving $B(y)$ has $\angle(c_k^{-1}U_e)\le 9p+3p$, and a link inside a block has $\angle U_e\le 3p$; so $\delta(U_e,R_e(c))\le d(U_e,R_e(c))\le 12\sqrt2\,p\le 12.002\,\kappa$. The holonomy of the square of side two at $2y$ is a product of four conjugated plaquettes, of angle at most $4p$, and replacing its four sides $\ell_k$ by $c_k$ adds at most $4\cdot 3p$; so $\angle c_P\le 16p$ and $\delta(c_P,I)\le 2\sqrt2\sin 8p\le 16\sqrt2\,p\le 16.003\,\kappa$. □

**Theorem 205** (the island count at radius $10^{-4}$)**.** *Let $G=\mathrm{SU}(2)$.*

(i) *Theorem 199 holds with the hypothesis $\rho\le 10^{-8}$ replaced by $\rho<\kappa_c$, and Lemma 200 and Theorem 201 hold with it replaced by $\rho\le 1.023\cdot 10^{-4}$.*

(ii) *At $\rho=10^{-4}$,*

$$\beta^*(10^{-4},1)=3.40906\cdot 10^{13},\quad \beta^*(10^{-4},5)=5.38304\cdot 10^{13},$$

*and for $s=5$, $\beta_1^*=2.89281\cdot 10^{15}$, $\beta_2^*=4.62936\cdot 10^{16}$, $\beta_3^*=1.18541\cdot 10^{18}$, $16^{-j}\beta_j^*\le 2.90\cdot 10^{14}$ for $1\le j\le 20$, and for $\beta_1^*\le\beta_W<\beta_{20}^*$ the conclusions of Theorem 201 hold at every scale $j\le j_{\max}$ with $2^{j_{\max}}\ge\frac12(\beta_W/2.90\cdot 10^{14})^{1/4}$.*

(iii) *At every one of 2001 couplings spaced evenly in the logarithm between $10^{22}$ and $10^{42}$, $k_*-j_{\max}\in\{2,3\}$: the islands of the composed average are rare up to the scale $a_{k_*}/8$.*

*Proof.* (i) The witness paragraph of the proof of Theorem 199 uses the cap $10^{-8}$ only through Lemma 124(i) and the proof of Lemma 173. Let $\kappa$ be the largest defect of the witness region. If $\kappa<\kappa_c$, Proposition 204(iv) gives $d(U_e,R_e(c))\le 12.002\kappa<43\kappa$ and $\delta(c_P,I)\le 16.003\kappa<92\kappa$, so a bad label forces $43\kappa>r\ge 43\rho$ or $92\kappa>\eta\ge 92\rho$; otherwise $\kappa\ge\kappa_c>\rho$. In both cases some plaquette has defect more than $\rho$, and the threshold and island paragraphs do not involve the cap. In Lemma 200 the cap enters only through the hypothesis $\eta_j\le\rho_G/1000$ of Theorem 152(i), (ii), with $\rho_G=1$, and in Theorem 201(i) through the hypothesis $4^j\eta_0\le 10^{-4}$ of Theorem 152(iv), which reads $\rho\le 1.023\cdot 10^{-4}$; then $\eta_i\le\rho\le 10^{-3}$ for $i\le j$. (ii), (iii) Each $\beta^*$ is enclosed by $g(\beta_{\mathrm{lo}})>0>g(\beta_{\mathrm{hi}})$ with $g=f_j-\log q_j^*$ decreasing past $96/\rho_0^2$, to relative width $10^{-12}$, and $k_*$ by enclosing $\log_4(10^{-4}/1.01\rho(\beta))$ away from the integers; the same computation reproduces the values of Theorem 201 at $\rho=10^{-8}$. The value of $k_*-j_{\max}$ is 2 at 1731 of the couplings and 3 at 270. □

The two or three scales that remain are not a matter of radius. With $\beta_j^*\approx 16|\log q_j^*|(1.033\cdot 4^j)^2/\rho^2$ and $4^{k_*}\approx 10^{-4}/(1.01\rho(\beta))$,

$$4^{k_*-j_{\max}}\approx\frac{4.09\cdot 10^{-4}}{\rho}\Big(\frac{|\log q_j^*|}{96(\log\beta+C_0)}\Big)^{1/2},$$

which is about 20 at $\rho=10^{-4}$, so the loss is the entropy exponent $|\log q_j^*|\approx 2H_j\log\Delta$ of Proposition 185. At this radius the labels have $r=43\rho\le 4.4\cdot 10^{-3}$, beyond the hypothesis $r\le 10^{-4}$ of Lemma 174, and Proposition 179 supplies the good windows to radius $\frac{1}{32}$. The chart of Theorem 127 is a different matter: its proof bounds the second fundamental form of the fibre by $\|\nabla Q\|_\infty\|\dot w\|_1$ with a constant $c_2$ for the second derivative of the reconstruction, and no such constant is below the flat value of the next proposition.

**Proposition 206** (the flat second derivative)**.** *At the flat background $U\equiv I$,*

$$D^2c_k(0)[Y,Y]=\frac{1}{16}\sum_{x\in B(y)}[Y_{1x},Y_{2x}],$$

*where $Y_{1x}$, $Y_{2x}$ are the variables of the two links of $P_x$, and $\sup\{\|D^2c_k(0)[Y,Y]\|:N(Y)=1\}=\frac{1}{16}$.*

*Proof.* Expand to second order, by the Baker–Campbell–Hausdorff formula,

$$c = \log\bigl(e^{X_v} e^{M}\bigr), \qquad M = \frac{1}{16}\sum_{x} \log\bigl(e^{-X_v} e^{X_{1x}} e^{X_{2x}}\bigr) :$$

the terms $[X_v, X_{1x}+X_{2x}]$ from the two logarithms cancel, and $\frac{1}{32}\sum_x [X_{1x}, X_{2x}]$ remains, whose second derivative is the display. For $\mu = 0$ put $d = 7^{-1/2}$, $l = 14^{-1/2}$ and, for each of the seven nonzero transverse offsets, $Y = \frac{d}{2} T_1$ on the link of $B(y)$, $l T_2$ on the leaving link and $-\frac{d}{2} T_1$ on the link of $B(y+e_0)$, with $T_a = i\sigma_a/\sqrt{2}$ and zero elsewhere. Then $N(Y) = 1$ and, by $\|[T_1, T_2]\| = \sqrt{2}$, $\|D^2 c_k(0)[Y,Y]\| = \frac{7\sqrt{2} dl}{16} = \frac{1}{16}$. Conversely $\|[X,Y]\| \le \sqrt{2}\|X\|\|Y\|$ and $\sqrt{2}\|D\|\|L\| \le \frac{1}{2}\|D\|^2 + \|L\|^2$, with $D$ the difference of the two inside variables and $L$ the leaving one, give $\frac{1}{16}$ as the maximum. □

In the proof of Theorem 127(ii) the constant $c_3$ enters the term $9000 c_3 \kappa'$, $\kappa' = \kappa + 6s$, only as a bound for the second derivative of $c$, so by the proposition no admissible constant there is below $\frac{1}{16}$, and with $\frac{1}{16}$ that term alone exceeds $\frac{1}{195}$ once $\kappa' > 9.12 \cdot 10^{-6}$. A chart at radius $10^{-4}$ needs a different estimate of the second fundamental form; no field is known at which the chart fails.

### K. Two islands factorise through one kernel

Item (ii) of Sec. XIII P, in its original form, asks for the hypothesis of Lemma 186: a kernel on the good layer between two bad islands through which the weight of every pair of label patterns factors, a domain that the labels of the islands do not move, and a cross-ratio of the kernel controlled by Lemma 188. This subsection proves the first two unconditionally, for every exterior and every pair of patterns, once the layer is seven blocks wider than the collar, and reduces the third to one covariance bound for the law of the layer, which is the bound of Theorem 131(ii) in the coordinates of the labels.

*The data.* Let $G = \mathrm{SU}(2)$, fix a coarse field $c$ with its reconstruction $R(c)$, the radii $r \le 10^{-4}$ and $\eta$, and the labels of Sec. XIII H. Given $c$ the coarse-plaquette part of every label is a constant, and the event that a block $y$ is good is the intersection, over the non-path active links $e$ of the pairs based at $y$, of the single-link events $G_e = \{d(U_e, R_e(c)) \le r\}$; these links are based in $B(y)$ and in the four blocks $B(y+e_\mu)$. Let $\Lambda$ be a finite set of blocks, $U^{\mathrm{ext}}$ an arbitrary configuration on the active links based outside $\Lambda$, $I_a, I_b \subset \Lambda$ two disjoint sets of blocks, $s \ge 1$ the collar, and

$$w = \mathrm{dist}(I_a, I_b) - 1$$

the number of block layers strictly between them, distances between blocks being $\ell^\infty$ distances of their coarse coordinates. Every block of $\Lambda \setminus (I_a \cup I_b)$ is good; on $I_a$ a pattern $\omega_a^i$, $i \in \{0,1\}$, is prescribed, with $\omega_a^0$ the pattern in which every block is good, and likewise $\omega_b^j$ on $I_b$. The sector $(i,j)$ is the event that the labels of $\Lambda$ are these, and its weight is

$$Z_{ij} = \int \prod_e dU_e \, e^{-\beta_W S(U)} \prod_{k \in K_\Lambda} \delta\bigl(V_k^\sharp(U) - c_k\bigr)\, \mathbf{1}_{(i,j)}(U),$$

the product over the active links based in $\Lambda$, $S$ the sum of the Wilson terms of the plaquettes containing one of them, $K_\Lambda$ the pairs based in $\Lambda$, and the constraints understood through the coarea formula as in Proposition 175; the pairs entering $\Lambda$ from outside have their path link in the exterior and their constraints are not imposed. Only ratios of the $Z_{ij}$ are used.

*The coordinates.* Write every non-path active link as $U_e = R_e(c) e^{\zeta_e}$, $\zeta_e \in \mathfrak{g}$, so that on a good block the event $G_e$ is the ball $\{\|\zeta_e\| \le r\}$ and Haar measure is $j(\zeta_e)\, d\zeta_e$ with $j$ smooth and positive on it. A pair all of whose non-path links lie in good blocks has, on the sector event, exactly one path link compatible with its constraint, by Lemma 174: $\ell_k = \Gamma_k$, a smooth function of its 23 non-path links, with cut-off one. Let $d_a(y) = \mathrm{dist}(y, I_a)$ and $d_b$ likewise, and form the classes

- $L_a$: all active links based at $d_a \le s$, path links included, all free;
- $p_a$: the path links of the pairs based at $d_a = s+1$, free;
- $x$: the non-path active links based at $d_a \in \{s+1, s+2\}$ and the 17 inside active links of every block at $d_a = s+3$;
- $z$: every other non-path active link based in $\Lambda$;
- $L_b$, $p_b$, $y$: the mirror classes for $I_b$,

and solve the path link of every pair based at $d_a \ge s+2$ and $d_b \ge s+2$. For $w \ge 2s+6$ a link at $d_a \le s+3$ has $d_b \ge s+4$, so the classes are disjoint. Let $D_x$ and $D_y$ be the products of the $r$-balls in the coordinates $\zeta_e$ of $x$ and of $y$; they are fixed by $r$ and the geometry, since every block carrying an $x$- or $y$-link is good in all four sectors.

**Theorem 207** (two islands and one kernel)**.** *Let $s \ge 1$, $r \le 10^{-4}$ and $w \ge 2s+7$. There are finite positive measures $a_i$ on $D_x$ and $b_j$ on $D_y$, $i,j \in \{0,1\}$, and a smooth function $K : D_x \times D_y \to (0,\infty)$ with*

$$Z_{ij} = \int_{D_x}\int_{D_y} K(x,y)\, a_i(dx)\, b_j(dy) \qquad (i,j \in \{0,1\}),$$

*such that:*

(i) *$K$ does not depend on $i$ or $j$; it depends on $c$, $r$, $\beta_W$, $U^{\mathrm{ext}}$ and on the labels of the blocks at distance at least $s+2$ from both islands only;*

(ii) *$a_i$ depends on $\omega_a^i$, on $c$, $\beta_W$ and $U^{\mathrm{ext}}$, and on nothing else; it carries every branch of the constraints of the pairs based at $d_a \le s+1$, integrated, every label indicator that involves a link of $L_a$, and the coarea densities of the pairs at $d_a = s+2$; likewise $b_j$;*

(iii) *the supports of all four measures lie in $D_x$, $D_y$: the labels of the islands do not move the domain;*

(iv) *$K(x,y) = \int_{D_z} e^{-\mathscr{U}(z;x,y)} dz$, where $D_z$ is the product of the $r$-balls of the $z$-coordinates and $\mathscr{U} = \beta_W S_K + \psi$ is smooth on a neighbourhood of $D_z \times D_x \times D_y$, $S_K$ the sum of the Wilson terms assigned to $K$ with the solved path links substituted, $\psi$ the sum of $-\log j$ over the $z$-links*

*and of $-\log J_k$ over the solved pairs based at distance at least $s+3$ from both islands, $J_k$ the inverse Jacobian of Proposition 175;*

(v) *no term of $\mathscr{U}$ depends on both an $x$- and a $y$-coordinate, $\nabla_z\partial_x\mathscr{U}$ is supported on the $z$-links based at $d_a \le s+4$ and $\nabla_z\partial_y\mathscr{U}$ on those at $d_b \le s+4$, at coarse distance at least $w-2s-7$ and, for $w \ge 2s+8$, at fine distance at least $d' = 2(w-2s-8)$ between link midpoints;*

(vi) *$Z_{ij} > 0$ if and only if $a_i \ne 0$ and $b_j \ne 0$, that is, if and only if each pattern is realisable on its own side together with its collar; $a_0$ and $b_0$ are nonzero.*

*Proof. Step 1: the factors.* The weight of a sector is the product Haar measure times a product of factors of three kinds: the Wilson factor of a plaquette, whose four links are based in blocks at mutual distance at most one; the constraint of a pair $k = (y,\mu)$, which involves the 24 links of its paths, based in $B(y)\cup B(y+e_\mu)$; and the label indicator of a block, which for a good block is the product of the $\mathbf{1}_{G_e}$ over the non-path links of its pairs and for a bad block is one minus that product, all these links being based in $B(y)$ and $B(y+e_\mu)$.

*Step 2: solving the far path links.* A pair based at $d_a \ge s+2$ and $d_b \ge s+2$ has its non-path links based at distance at least $s+1$ from both islands, hence in good blocks, hence inside their balls on the sector event. Lemma 174 applies, the fibre of its constraint over its non-path links is the one point $\ell_k = \Gamma_k$, and since $\ell_k$ is private to $k$ the integration of $\ell_k$ against the constraint replaces $\ell_k$ by $\Gamma_k$ in every factor and multiplies the weight by $J_k$, as in Proposition 175; the cross derivatives between different pairs vanish, so the order is immaterial. After this step every factor is a function of the free coordinates and of $U^{\mathrm{ext}}$, and a solved path link based at distance $d$ depends on non-path links based at distances $d-1$, $d$, $d+1$.

*Step 3: the assignment.* We list what each factor depends on, distances being from $I_a$. A solved path link at $s+2$ depends on non-path links at $s+1, s+2, s+3$, those at $s+3$ being the second links of its paths, inside $B(y+e_\mu)$: all of them are $x$-coordinates. A solved path link at $d \ge s+3$ depends on $x$- and $z$-coordinates. Hence: a plaquette with a link at $d \le s$ depends on $L_a$, $p_a$ and $x$; one with its nearest link at $s+1$ on $p_a$ and $x$; one with its nearest link at $s+2$ on $x$ and $z$ (its path links at $s+2$ are functions of $x$, its private links at $s+3$ are $z$, its path links at $s+3$ are functions of $x$ and $z$); one with all links at $d \ge s+3$ on $x$ and $z$. A pair based at $d \le s$ has its path link in $L_a$ and its non-path links in $L_a$ and $x$; one at $s+1$ has its path link in $p_a$ and its non-path links in $L_a$ and $x$; one at $s+2$ is solved with $J_k$ a function of $x$; one at $d \ge s+3$ is solved with $J_k$ a function of $x$ and $z$, of $z$ alone for $d \ge s+5$. The bad-block indicators of $I_a$ involve links at $d \le 1 \le s$, in $L_a$; the good single-link indicators are assigned to the class of their link. Call a factor an $a$-factor if it depends on a coordinate of $L_a \cup p_a$, a $b$-factor if on $L_b \cup p_b$, and a $K$-factor otherwise; factors depending on $x$ alone go with $a$, on $y$ alone with $b$. An $a$-factor depends only on $L_a, p_a, x$ and $U^{\mathrm{ext}}$, through links at $d_a \le s+2$, which have $d_b \ge w+1-(s+2) \ge s+6$ for $w \ge 2s+7$, so it contains no coordinate of $L_b \cup p_b$ and no factor is both. The sector changes only the indicators of $I_a$ and $I_b$, which are $a$- and $b$-factors, so the product $\Phi_K$ of the $K$-factors is the same in all four sectors.

*Step 4: Tonelli.* All factors are nonnegative and the Haar measures finite, so, with $\Phi_a^i$ the product of the $a$-factors in sector $(i,j)$ and $\Phi_b^j$ that of the $b$-factors,

$$Z_{ij} = \int \Big[\int \Phi_a^i \, dL_a \, dp_a\Big]\Big[\int \Phi_K \, dz\Big]\Big[\int \Phi_b^j \, dL_b \, dp_b\Big] \times \prod_{e\in x} j\, d\zeta_e \prod_{e\in y} j\, d\zeta_e,$$

the inner integrals over $L_a, p_a$ being the coarea integrals of the unsolved pairs, in which every branch of every constraint is integrated and none discarded, and the $x$- and $y$-integrals restricted to $D_x$ and $D_y$ because every $x$-link carries its indicator $\mathbf{1}_{G_e}$. Define $a_i(dx)$ as the first bracket times $\prod_{e\in x} j\, d\zeta_e$, $b_j(dy)$ likewise and $K$ as the middle bracket with the $j$-factors of the $z$-links included. This is the factorisation; (i), (ii) and (iii) are read off Step 3.

*Step 5: the kernel.* $\Phi_K$ is a product of positive smooth functions of $(z,x,y)$ (Wilson factors, Jacobians $J_k > 0$ since $\|DF_k\| < 0.9$, densities $j > 0$) and of the indicators of the $z$-balls; the solved $\Gamma_k$ are smooth on a neighbourhood of the closed product of balls, the contraction of Lemma 174 still applying at radius $1.01r$. So $K = \int_{D_z} e^{-\mathscr{U}}\, dz$ with $\mathscr{U}$ smooth and finite near the compact set $D_z \times D_x \times D_y$, $K$ is smooth and $K \ge |D_z| e^{-\sup \mathscr{U}} > 0$. This is (iv).

*Step 6: supports.* Two facts of the tree gauge are used. First, an inside link $(a,\nu)$, $o_\nu(a) = 0$, and a path link $(2y+e_\mu,\mu)$ lie in a common plaquette only if $a \in B(y)\cup B(y+e_\mu)$: as opposite sides they would have the same direction and bases differing by a unit vector $e_\rho$, $\rho \ne \mu$, forcing $o_\mu(a) = 1$; as adjacent sides the shared corner is $2y+e_\mu$ or $2y+2e_\mu$ (then $a$ or, when $\nu = \mu$, $a+e_\mu$ is that corner, so $a \in B(y)\cup B(y+e_\mu)$), or it is $a+e_\nu$ with $\nu \ne \mu$ equal to one of them, forcing $o_\nu(a) = 1$. Second, if $d_a(y) = s+4$ and $d_a(y+e_\mu) = s+3$, then $d_a(y-e_\nu) \le s+4$ for every $\nu \ne \mu$, since some $a' \in I_a$ has $\|y+e_\mu - a'\|_\infty = s+3$, so $|y_\nu - a'_\nu| \le s+3$ for $\nu \ne \mu$. We claim that a $K$-term depending on an $x$-coordinate has all its links based at $d_a \le s+4$, one of them at $d_a \le s+3$, and its $z$-gradient supported at $d_a \le s+4$. A Jacobian $J_k$ depends on $x$ only when $k$ is based at $s+3$, or at $s+4$ pointing inward, and then every link of $k$ is at $d_a \le s+4$. A Wilson term depends on $x$ only through an $x$-link (non-path at $s+2$, inside at $s+3$) or through a solved path link at $s+2$, at $s+3$, or inward at $s+4$. In the first two cases all links of the plaquette are at $d_a \le s+4$, and a path link of it at $s+4$ lies within one block of a link at $\le s+3$: if that link is inside at $s+3$ the path link points into its block by the first fact and depends on links at $s+3, s+4$; a plaquette cannot contain links at $s+2$ and at $s+4$; and a path link $\ell'$ at $s+4$ in a plaquette with a path link at $s+3$ cannot point outward to a block at $s+5$, since its base block is adjacent to a block at $s+3$, so $\ell'$ depends on links at $d_a \le s+4$. In the third case the plaquettes through an inward path link $(2y+e_\mu,\mu)$ with $d_a(y) = s+4$ lie in the planes $(\mu,\nu)$ at $2y+e_\mu$ and at $2y+e_\mu-e_\nu$, with links based in $B(y)$, $B(y+e_\mu)$, $B(y-e_\nu)$, $B(y+e_\mu-e_\nu)$, all at $d_a \le s+4$ by the second fact, and their path links are within one block of $B(y+e_\mu)$ at $s+3$, so point to blocks at $d_a \le s+4$. This proves the claim, and its mirror for $b$. Hence $\nabla_z\partial_x\mathscr{U}$ lives at $d_a \le s+4$ and $\nabla_z\partial_y\mathscr{U}$ at $d_b \le s+4$; since $d_b \ge w+1-d_a$ their coarse

distance is at least $w-2s-7$, and two links based in blocks at coarse distance $D\geq 1$ have midpoints at fine $\ell^\infty$ distance at least $2D-2$, which gives $d'$. A $K$-term depending on both $x$ and $y$ would contain a link at $d_a\leq s+3$ and a link at $d_b\leq s+3$ at mutual block distance at most one, so $w+1\leq 2s+7$; for $w\geq 2s+7$ there is none. This is (v).

*Step 7: positivity.* Since $K>0$ on $D_x\times D_y$ and the measures live there, $Z_{ij}>0$ exactly when $a_i(D_x)>0$ and $b_j(D_y)>0$; $a_i(D_x)$ is the constrained weight of $\omega_a^i$ with its two collars, which for the all-good pattern is positive by Proposition 175. This is (vi). □

The theorem uses only Lemma 174; it does not use $\beta_W$ large, $\eta$, or Theorem 131. The three dangers named after Lemma 188 do not arise: the island labels change $a_i$ and $b_j$ only, the domain $D_x\times D_y\times D_z$ is fixed by the labels of the layer, which are good in every sector, and every branch of every bad pair is integrated inside $a_i$ or $b_j$ and never compared across sectors. The price is the width. Joining two bad blocks in Sec. XIII H at $\ell^\infty$ distance at most $2s+7$ instead of $2s+1$, so that two islands are separated by a layer of width at least $2s+7$, changes only the number of neighbours $\Delta$ of Theorem 199, from $(4s+3)^4-1$ to $(4s+15)^4-1$: $|\log q^*|$ becomes 31444.0 for $s=1$ and 37777.9 for $s=5$, and $\beta^*(10^{-8},1)=5.084\cdot 10^{21}$, $\beta^*(10^{-8},5)=6.097\cdot 10^{21}$, $\beta^*(10^{-9},1)=5.088\cdot 10^{23}$, by the same formula $\beta^*=16|\log q^*|\rho^{-2}(1+\varepsilon)$.

*The layer law.* For $(x,y)\in D_x\times D_y$ let $\mu_{x,y}$ be the law $e^{-\mathscr{U}(z;x,y)}dz/K(x,y)$ on $D_z$. It is the law of Proposition 175 for the window of the blocks at distance at least $s+3$ from both islands, in the coordinates $\zeta$ of the labels, with the inside links of the blocks at distance $s+3$ frozen and the exterior given by $(x,y,U^{\mathrm{ext}})$. The covariance bound of Theorem 131(ii) for this law reads: there are $\mathscr{A}>0$ and $\alpha>0$ such that for all $(x,y)$, every $\varphi$ on the $z$-links with $|\varphi(e)-\varphi(e')|\leq d(e,e')$ and all smooth $F,H$ near $D_z$,

$$|\mathrm{Cov}_{\mu_{x,y}}(F,H)|\leq \mathscr{A}\left\|e^{-\alpha\varphi}\nabla_z F\right\|_{L^2(\mu_{x,y})}\left\|e^{\alpha\varphi}\nabla_z H\right\|_{L^2(\mu_{x,y})}. \tag{26}$$

Let $n_x=45(N_{s+1}+N_{s+2})+17N_{s+3}$, with $N_d$ the number of blocks of $\Lambda$ at $d_a=d$, be the number of $x$-links, $n_y$ likewise, and let $\Lambda_2$ be the largest operator norm of $\partial_x\nabla_z\mathscr{U}$ and $\partial_y\nabla_z\mathscr{U}$ on $D_z\times D_x\times D_y$; it is finite, of the form $\beta_W c_W+c_\psi$ with $c_W$, $c_\psi$ depending on $r$ and the geometry only.

**Proposition 208** (the cross-ratio under the layer bound)**.** *Let $s\geq 1$, $r\leq 10^{-4}$, $w\geq 2s+8$, and assume (26). Then for all $x_0,x_1\in D_x$ and $y_0,y_1\in D_y$,*

$$\left|\log\frac{K(x_1,y_1)K(x_0,y_0)}{K(x_1,y_0)K(x_0,y_1)}\right|\leq\Delta,$$

$$\Delta:=\mathscr{A}\Lambda_2^2\|x_1-x_0\|\|y_1-y_0\|e^{-\alpha d'}\leq 4\mathscr{A}\Lambda_2^2r^2\sqrt{n_xn_y}\,e^{-\alpha d'},$$

*so that, for every pair of patterns with all $Z_{ij}>0$, $|\log Z_{11}+\log Z_{00}-\log Z_{10}-\log Z_{01}|\leq\Delta$ and the two island labels have conditional covariance at most $\frac14\tanh(\Delta/4)$ given the labels of the layer and $U^{\mathrm{ext}}$. Moreover, for every bounded $F\geq 0$ depending on the links of $L_a\cup p_a\cup x$ only, with $\mathbb{E}_{ij}$ the expectation in sector $(i,j)$,*

$$e^{-\Delta}\leq\frac{\mathbb{E}_{i1}[F]}{\mathbb{E}_{i0}[F]}\leq e^{\Delta}\qquad(i=0,1)$$

*whenever the four expectations are positive.*

*Proof.* $D_x$ and $D_y$ are convex, so the segments $x(\sigma)$, $y(\tau)$ stay in them, and $\log K$ is smooth near $D_x\times D_y$ by Theorem 207(iv), so the cross-ratio is the double integral of $\partial_\sigma\partial_\tau\log K$ over the unit square. Lemma 188 with the fixed domain $D_z$ gives $\partial_\sigma\partial_\tau\log K=-\mathbb{E}[\partial_\sigma\partial_\tau\mathscr{U}]+\mathrm{Cov}(\partial_\sigma\mathscr{U},\partial_\tau\mathscr{U})$, and the first term vanishes by Theorem 207(v). Put $F=\partial_\sigma\mathscr{U}$, $H=\partial_\tau\mathscr{U}$: $\nabla_zF$ is supported on the $z$-links at $d_a\leq s+4$, $\nabla_zH$ on those at $d_b\leq s+4$, at distance at least $d'$, with $\|\nabla_zF\|\leq\Lambda_2\|x_1-x_0\|$ and $\|\nabla_zH\|\leq\Lambda_2\|y_1-y_0\|$ pointwise. Apply (26) with $\varphi(e)=-\mathrm{dist}(e,\mathrm{supp}\,\nabla_zF)$, which is 1-Lipschitz, equal to 0 on the support of $\nabla_zF$ and at most $-d'$ on that of $\nabla_zH$; this gives the bound on the covariance uniformly in $(\sigma,\tau)$, and $\|x_1-x_0\|\leq 2r\sqrt{n_x}$ since each coordinate is a vector of $\mathfrak{g}$ of norm at most $r$. The consequences are Lemma 186 applied to $a_i,b_j,K$ and Corollary 187. For the last statement, $F\Phi_a^i$ is again a product of $a$-factors, so the factorisation holds with the measures $Fa_i$ and $a_i$, the same $K$ and the same $b_j$, and Lemma 186 for these gives $|\log Z(Fa_i,b_1)+\log Z(a_i,b_0)-\log Z(Fa_i,b_0)-\log Z(a_i,b_1)|\leq\Delta$ with $Z(Fa_i,b_j)/Z(a_i,b_j)=\mathbb{E}_{ij}[F]$. □

The last bound is the quantity that Proposition 189 says a labelling argument must control: the change of the conditional expectation of an observable of one island under a flip of the label of the other, with the branches, the hard label constraints and the disintegration densities of both islands integrated, not compared. The hypothesis (26) is Theorem 131(ii) for the layer law, but in other coordinates. The chart law of that theorem is written in the coordinates $\xi$ of $T_U$, on the product of balls $\mathrm{Box}_r$; the label coordinate of a non-path link is $\zeta_e=\xi_e+h_e(S\xi)$, so the label event $\prod_e\{\|\zeta_e\|\leq r\}$ is a deformation of $\mathrm{Box}_r$, convex in $\zeta$ and not in $\xi$, and the Hessian of the density in $\zeta$ acquires the term $\langle\nabla_\xi\bar H,D^2\xi\rangle$, which is not coordinate invariant. The next theorem carries the proof over to the coordinates $\zeta$; it needs the layer calmer than Theorem 131 needs a chart, and a coupling four times larger.

**Theorem 209** (the layer law obeys the chart bound)**.** *Let $G=\mathrm{SU}(2)$, $\lambda=\beta_W/4$, and let the data be those of Theorem 207. Let $W'$ be the set of blocks of $\Lambda$ at distance at least $s+3$ from both islands, and assume*

(a) *$r\leq 8\cdot 10^{-5}$ and $4\cdot 10^5c_3(\eta+62r)\leq\frac{1}{195}$, which holds when $\eta+62r\leq 6.0\cdot 10^{-13}$;*

(b) *$\lambda\geq 240\,c_6$, which holds when $\beta_W\geq 8.93\cdot 10^{17}$;*

(c) *every block outside $\Lambda$ at distance at most two from $W'$ is good and the constraints of its pairs hold; this is void when $\Lambda$ contains the blocks at distance at most two from $W'$.*

*Then (26) holds with $\mathscr{A}=120/\lambda=480/\beta_W$ and every $\alpha\leq\alpha_G=2.5\cdot 10^{-4}$, uniformly in $(x,y)\in D_x\times D_y$, in $U^{\mathrm{ext}}$, in $\Lambda$ and in the width $w$.*

*Proof.* All constants without a name are those of the proofs of Theorems 127 and 131 and of Proposition 133, at $\vartheta = 0$ unless a weight is written: $\mathscr{G} = BB^* \geq \frac{1}{17}$, $\|B^* z\|_\infty \leq 0.88 \max_k \|z_k\|$, $\|R_k(X)\| \leq 12 c_3 \|X\|_\infty^2$, the rows of $A-B$ of norm at most $7.35 c_3 s$, $Y = B^*(AB^*)^{-1}$ with $\|Y\| \leq 11.32$, $\bar P = S - Y(A-B)S$ with $\|\bar P\| \leq 4.80$ and $\|\bar P\|_{1\to 1} \leq 3.003$, $x_v = \bar P v$ and $x_{vv'} = -YD^2c(X)[x_v, x_{v'}]$ with $\|x_{vv'}\|_1 \leq 1564.3\, c_3 |v||v'|$. Put $s' = 32r$ for the chart radius.

*Step 1: the background is calm.* Let $U$ be the point of the fibre whose non-path links in $W'$ are $R(c)$ and whose path links there are the solutions $\Gamma_k(0)$ of Lemma 174. Take a plaquette $p$ with a vertex in a block of $W'^+$. Its links are based in blocks at distance at most two from $W'$, which are good, by the sector event inside $\Lambda$ and by (c) outside it. A plaquette contains at most two path links, one of each of its directions: a path link $(2y + e_\mu, \mu)$ has exactly one odd coordinate at its base, and two such links of one direction on one plaquette would have to be opposite sides with bases differing by $e_\rho$, $\rho \neq \mu$, which gives one of them two odd coordinates. The plaquette of $R(c)$ at $p$ is $I$ unless both coordinates of the corner in its plane are odd, and it is then the coarse plaquette of a good block, so $\delta(R(c)_p, I) \leq \eta$. Replacing the four links one at a time, with $\delta \leq d$ (Lemma 116), bi-invariance and $d(\Gamma_k, c_k) \leq 14r$, gives $\delta(U_p, I) \leq \eta + 2\cdot 14r + 2r$. So the defect $\kappa_0$ of the background satisfies $\kappa_0 + s' \leq \eta + 62r$, and (a) is the hypothesis $4\cdot 10^5 c_3(\kappa_0 + s') \leq \frac{1}{195}$ of Theorem 127(iv) at radius $s'$; by Proposition 128 it implies $s' \leq 1/(6000 c_3)$, $\kappa_0 + 6s' \leq 10^{-8}$ and $c_3 s' \leq 1.29\cdot 10^{-8}$.

*Step 2: the two coordinates.* Let $\rho_1 = \frac{5}{4} r$, which is at most $10^{-4}$ by (a). For $\zeta$ with every $\|\zeta_e\| \leq \rho_1$ let $U'$ be the point of the fibre with non-path links $R_e(c) e^{\zeta_e}$, and $X = \log(U^{-1}U')$ link by link. On the non-path links $\|X_e\| \leq \rho_1$; on a path link, by Lemma 174 at the radii $r$ and $\rho_1$, $\|X_{d_k}\| \leq 14r + 14\rho_1 = 31.5r$. So $t = \|X\|_\infty < s'$, and $BX = -R(X)$ with $\|R_k(X)\| \leq 12 c_3 t^2$. With $z = \mathscr{G}^{-1} BX$ and $\nu = X - B^* z \in T_U$ we get $\max_k \|z_k\| \leq 204 c_3 t^2 \leq 8.2\cdot 10^{-5} r$ and $\|\nu\|_\infty \leq t(1 + 180 c_3 t) \leq 31.51 r < s'$, and $z$ is the fixed point that defines $h(\nu)$, so $U' = \Phi(\nu)$ with $\|h(\nu)\|_\infty \leq 7.2\cdot 10^{-5} r$. Hence the map $\Psi(\xi) = \xi + g(\xi)$, $g(\xi) = h(S\xi)$ restricted to the non-path links, which sends the chart coordinate to the label coordinate, maps a subset of $\mathrm{Box}_{2r}$ onto $\{\|\zeta_e\| \leq \rho_1\}$; it is injective there because two chart points with the same non-path links within $\rho_1$ of $R(c)$ have the same path links by Lemma 174. Its derivative is $1 + Dg$ with $Dg = -Y(A-B)S$ restricted to the non-path links, so

$$\|Dg\| \leq 3.6\cdot 10^{-5}, \qquad \|Dg\|_{(0.197)} \leq 7.0\cdot 10^{-5}, \qquad \|Dg\|_{1\to 1} \leq 2.0\cdot 10^{-5},$$

from $\|S\|_{1\to 1} \leq 1 + 2\cdot 8.01(\frac{1}{8} + 260\kappa_0)$, the rows of $A-B$, and the weighted bounds of Proposition 133. Thus $\Psi$ is a diffeomorphism onto the $\rho_1$-balls, and $J = D\xi = (1+Dg)^{-1}$ has the norms $j_0 = 1 + 3.6\cdot 10^{-5}$, $j_\vartheta = 1 + 7.0\cdot 10^{-5}$ and $j_1 = 1 + 2.0\cdot 10^{-5}$ in the three norms.

*Step 3: the density.* By Theorem 127(iii) and the proof of Theorem 131, the constrained law on the chart has density $e^{-\bar H(\xi)}$ in $\xi$. By Step 2 and the change of variables $\zeta = \Psi(\xi)$, the law conditioned on the label event has density $e^{-\tilde U} \mathbf{1}_{\{\|\zeta_e\| \leq r\}}$ with

$$\tilde U(\zeta) = \bar H(\xi(\zeta)) + \log\det\big(1 + Dg(\xi(\zeta))\big),$$

smooth on the $\rho_1$-balls. It is the law of Proposition 175, since the disintegration of a smooth positive density along a submersion is unique up to a factor constant on each fibre, and the layer law $\mu_{x,y}$ is this law with the inside links at distance $s+3$ frozen.

*Step 4: the lower bound on the Hessian.* By the chain rule, with $a = Jv$ and $M = 1 + Dg$,

$$\partial_v^2 \tilde U = \mathrm{Hess}_\xi \bar H[a,a] + \big\langle \nabla_\xi \bar H, D^2\xi[v,v]\big\rangle + \mathrm{Tr}\big(M^{-1}M''\big) - \mathrm{Tr}\big((M^{-1}M')^2\big),$$

where $M' = D^2 g[a,\cdot]$, $M'' = D^3 g[a,a,\cdot] + D^2 g[D^2\xi[v,v],\cdot]$ and $D^2\xi[v,v] = -JD^2 g[a,a]$. The first term is at least $(\frac{\lambda}{30} - c_6)|a|^2$ by Theorem 127(ii), (iii), since $\frac{1}{26} - 4\cdot 10^5 c_3(\kappa_0 + s') \geq \frac{1}{30}$. For the second, the gradient of $\bar H$ in one link is the sum of $\lambda DQ[x_v]$, the derivative of the Haar densities and the derivatives of the two determinants; $|DQ[x_v]| \leq 12(\kappa_0 + 6s')\|x_v\|_1$, the Haar part is at most $0.84\|x_v\|_1$, and the determinants give $\mathrm{Re}\,\mathrm{Tr}(A^*(AA^*)^{-1}A_v)$ and $\mathrm{Re}\,\mathrm{Tr}(\bar P(\bar P^*\bar P)^{-1}\bar P^* Y A_v)$ with $A_v = D^2 c(X)[x_v,\cdot]$, bounded by $5\|A_v\|_1$ and $11.32\|A_v\|_1$ since $AA^* \geq \frac{1}{25}$ and the second operator is $Y$ followed by a projection; and $\|A_v\|_1 \leq 2\dim G\, c_3 \|x_v\|_1$ because each row of $D^2 c$ has rank at most $\dim G$ and each link enters two rows. So every component of $\nabla_\xi \bar H$ is at most $12\lambda(\kappa_0 + 6s')\cdot 3.003 + 296.5\, c_3$, and with $\|D^2\xi[v,v]\|_1 \leq j_1 j_0^2 \cdot 1564.3\, c_3 |v|^2$ the second term is at most $(4.35\cdot 10^{-3}\lambda + 4.64\cdot 10^5 c_3^2)|v|^2$. The derivatives of $g$ are those of $X$: $D^2 g[a,\cdot] = -YA_a\bar P$ and

$$D^3 g[a,b,\cdot] = YA_aYA_b\bar P + YA_bYA_a\bar P - YD^3c[x_a,x_b,\cdot]\bar P - YD^2c[x_{ab},\cdot]\bar P,$$

restricted to the non-path links, and the Hilbert–Schmidt bound $\|A_a\|_2^2 \leq 2\dim G\, c_3^2 \|\bar P\|^2 |a|^2$ together with $|\mathrm{Tr}(ST)| \leq \|S\|\|T\|_1$ and $\|ST\|_1 \leq \|S\|_2\|T\|_2$ bounds the two traces term by term by $(4.07 + 1.70 + 0.08 + 5.10 + 15.30)\cdot 10^5 c_3^2 |v|^2$. Since $c_6 = 6.81\cdot 10^5 \dim G\, c_3^2$, the part of $\partial_v^2\tilde U$ that does not grow with $\lambda$, beyond the $c_6$ of the first term, is at most $1.512\, c_6 |v|^2$, and by (b)

$$\mathrm{Hess}_\zeta \tilde U \geq \lambda\Big(\frac{1/30 - 1/240}{(1 + 3.6\cdot 10^{-5})^2} - 4.35\cdot 10^{-3} - \frac{1.512}{240}\Big) \geq 0.0185\,\lambda \geq \frac{\lambda}{60}.$$

*Step 5: locality of the Hessian.* The same four terms, with the vectors $p = wu$, $q = w^{-1}u'$ for $w = e^{\vartheta\varphi}$, $\vartheta = 0.197$, are bounded as in the paragraph "The Hessian" of the proof of Proposition 133: the weights of $x_p$ and $x_q$ on the links of one pair cancel up to $e^{2\vartheta}$, the conjugation of $A_a$ by $w$ costs at most $e^\vartheta$, and the other factors are bounded in $\|\cdot\|_{(\vartheta)}$ by $\bar p \leq 5.80$, $y \leq 14.96$ and $j_\vartheta$. With $\|\mathrm{Hess}_\xi \bar H\|_{(0.197)} \leq 2068\lambda$ this gives

$$\|\mathrm{Hess}_\zeta \tilde U\|_{(0.197)} \leq \lambda\Big(2068\, j_\vartheta^2 + 9.41\cdot 10^{-3} + \frac{5.88}{240}\Big) \leq 2068.4\,\lambda.$$

*Step 6: the covariance bound.* The proof of Theorem 131(ii) now applies word for word with $\mathrm{Box}_r$ replaced by the product of the $r$-balls in $\zeta$, both being products of balls in the coordinates in which the Hessian bounds of Steps 4 and 5 hold, and with the radii of the approximation $\frac{3}{2}r, \frac{7}{4}r, 2r$ replaced by $1.1r, 1.2r, 1.25r$, for which the convex penalty has Hessian at least $\frac{1}{22}$ outside $1.1r$. Lemma 132(ii) gives $\|\mathrm{Re}(\mathrm{Hess}\, H_n)^w - \mathrm{Hess}\, H_n\| \le \frac{\lambda}{120}$ for $\alpha \le 0.197(240\cdot 2068.4)^{-1/2} = 2.79\cdot 10^{-4}$, which contains $\alpha_G$, and the weighted Helffer–Sjöstrand representation gives the bound with $\mathscr{A} = 2/m = 120/\lambda$. Freezing the inside links at distance $s+3$ restricts the Hessian to a principal submatrix, which keeps both bounds, and the argument runs on the free links with the same constants. □

**Corollary 210** (two islands in a good region)**.** *Under the hypotheses of Theorems 207 and 209, with $w \ge 2s+8$, every pair of label patterns with all $Z_{ij} > 0$ satisfies*

$$\big|\log Z_{11} + \log Z_{00} - \log Z_{10} - \log Z_{01}\big| \le \Delta,$$
$$\Delta = \frac{1920}{\beta_W}\Lambda_2^2 r^2 \sqrt{n_x n_y}\, e^{-2\alpha_G(w-2s-8)},$$

*the two island labels have conditional covariance at most $\frac{1}{4}\tanh(\Delta/4)$, and every observable of one island and its collars changes by a factor in $[e^{-\Delta}, e^{\Delta}]$ when the label of the other island is flipped. In the normalisation $r = 43\rho$, $\eta = 92\rho$ of Theorem 199, hypothesis (a) of Theorem 209 holds for $\rho \le 2.17\cdot 10^{-16}$, where the island threshold with the joining distance $2s+8$ is $\beta^* = 1.122\cdot 10^{37}$ for $s = 1$ and $1.322\cdot 10^{37}$ for $s = 5$, and hypothesis (b) holds at every $\beta_W \ge \beta^*$.*

*Proof.* Proposition 208 with $\mathscr{A} = 480/\beta_W$ from Theorem 209 and $d' = 2(w-2s-8)$. For the numbers, $\eta + 62r = (92 + 62\cdot 43)\rho = 2758\rho$, and the threshold is the largest root of the equation of Theorem 199 with $(4s+17)^4 - 1$ neighbours, the count for the joining distance $2s+8$, which makes every layer between two islands of width at least $2s+8$: $|\log q^*| = 32481.7$ for $s = 1$ and 38354.1 for $s = 5$. □

**Lemma 211** (the mixed derivatives of the layer action)**.** *Let $G = \mathrm{SU}(2)$, let the data be those of Theorem 207 with $r \le 8\cdot 10^{-5}$, and let $\kappa \le 10^{-8}$ be such that $\delta(U_p, I) \le \kappa$ on $D_z \times D_x \times D_y$ for every plaquette $p$ of $S_K$ that contains the solved path link of a pair one of whose arguments, the non-path links of its paths, is a z-link. Then*

$$\Lambda_2 \le \beta_W c_W + c_\psi, \qquad c_W \le 101.11, \qquad c_\psi \le 4.94\cdot 10^6,$$

*with $c_W$ and $c_\psi$ the explicit functions of $r$ and $\kappa$ given in (28). Under hypotheses (a) and (c) of Theorem 209 the condition holds with $\kappa = \eta + 30r \le 6.0\cdot 10^{-13}$, and*

$$\Lambda_2 \le 99.0001\,\beta_W + 4.65\cdot 10^6.$$

*Proof.* Norms on $\mathfrak{g}$ are Hilbert–Schmidt norms, $\delta(g,h) = \|g - h\|$ on $\mathrm{SU}(2)$, and the coordinates are the $\zeta_e$ of Theorem 207, $U_e = R_e(c)e^{\zeta_e}$, with the reconstruction value $R_e(c) = I$ for a link inside a block and $R_e(c) = c_k$ for a private link of the pair $k$. For a function $f$ of the coordinates and two coordinates $g \ne g'$ write $\|\partial_g \partial_{g'} f\|$ for the supremum of $|\partial_s \partial_t f(\zeta + s v_g + t w_{g'})|$ at $s = t = 0$ over $v, w \in \mathfrak{g}$ of norm one placed at $g$ and $g'$: it is the operator norm of the $(g, g')$ block of the Hessian.

*Step 1: reduction to row sums.* Let $M = \partial_x \nabla_z \mathscr{U}$, with blocks $M_{ef}$ for $z$-links $e$ and $x$-links $f$. For vectors $u$ on the $z$-links and $v$ on the $x$-links, $|\langle u, Mv\rangle| \le \sum_{e,f} \|M_{ef}\|\|u_e\|\|v_f\|$, and the Cauchy–Schwarz inequality gives $\|M\| \le \max\{\max_e \sum_f \|M_{ef}\|, \max_f \sum_e \|M_{ef}\|\}$. A term $-\log j(\zeta_e)$ of $\psi$ depends on one coordinate, so $M_{ef}$ is the sum of the blocks of the Wilson terms $\beta_W W_p$, $W_p = 1 - \frac{1}{2}\mathrm{Re}\,\mathrm{Tr}\, U_p$, of $S_K$ and of the terms $-\log J_k = \log\det(1 - DF_k)$ of $\psi$, $J_k$ the inverse Jacobian of Proposition 175 and $DF_k$ the differential in $X$ of the map $F$ of Lemma 174 at the fixed point, that depend on $e$ and on $f$. It therefore suffices to prove that for every $x$- or $z$-coordinate $g$

$$R_g = \sum_{g'} \sum_{\tau} \|\partial_g \partial_{g'} \tau\| \le \beta_W c_W + c_\psi,$$

$g'$ running over the coordinates of the other class ($z$ if $g$ is an $x$-coordinate, $x$ if $g$ is a $z$-coordinate) and $\tau$ over the Wilson terms and the Jacobians that depend on $g$ and $g'$; the bound for $\partial_y \nabla_z \mathscr{U}$ is the mirror statement.

*Step 2: the geometry.* (i) Every link lies in six plaquettes. (ii) The solved path link $\Gamma_k$ of $k = (y, \mu)$ is a function of the 21 non-path links of the fourteen paths of nonzero offset (Lemma 174), its *arguments*: the seven non-path private links, each in two of these paths, and the fourteen links of direction $\mu$ in $B(y)$ and $B(y + e_\mu)$ other than $(2y, \mu)$ and $(2y + 2e_\mu, \mu)$, each in one path, the tree links among them being fixed at $I$. A path of $k$ consists of links of direction $\mu$, so a non-path private link is an argument of its own pair only, and a link of direction $\mu$ inside $B(y)$ of the two pairs $(y, \mu)$ and $(y - e_\mu, \mu)$. Let $K(g)$ be the set of solved pairs having $g$ as an argument and $n_g$ the number of paths of each containing $g$: $n_g = 2$, $|K(g)| \le 1$ for a private link, $n_g = 1$, $|K(g)| \le 2$ for an inside link. (iii) Let $\ell = (a, \mu)$, $a = 2y + e_\mu$, be a path link and $\nu \ne \mu$. The plaquette at $a$ in the plane $\{\mu, \nu\}$ has the further links $(a, \nu)$, of offset $e_\mu$, inside $B(y)$; $(a + e_\mu, \nu)$, based at the root $2y + 2e_\mu$, a tree link; and $(a + e_\nu, \mu)$, of offset $e_\mu + e_\nu$, a non-path private link of $(y, \mu)$. The plaquette at $a - e_\nu$ has the further links $(a - e_\nu, \mu)$ and $(a - e_\nu, \nu)$, both based at $2(y - e_\nu) + e_\mu + e_\nu$ and so non-path private links of $(y - e_\nu, \mu)$ and $(y - e_\nu, \nu)$, and $(a + e_\mu - e_\nu, \nu) = (2(y + e_\mu - e_\nu) + e_\nu, \nu)$, the path link of $(y + e_\mu - e_\nu, \nu)$. Hence each of the six plaquettes of a path link carries at most two non-path links, exactly three of them, those at $a - e_\nu$, carry a second path link, and no plaquette carries three path links. (iv) By (iii) a non-path link lies on a plaquette with a path link only if it is an inside link $(2y + e_\mu, \nu)$ with $\nu < \mu$, the only other case $\nu > \mu$ being a tree link, or a private link of offset $e_\mu + e_\nu$ and direction $\mu$ or $\nu$. The first lies on one such plaquette, the one at $2y + e_\mu$ in the plane $\{\mu, \nu\}$, with one path link. The second, $(b, \mu)$ with $b = 2y + e_\mu + e_\nu$, plays the role of $(a + e_\nu, \mu)$ for $a = 2y + e_\mu$, on a plaquette with one path link, and the roles of $(a - e_\nu, \mu)$ for $a = 2(y + e_\nu) + e_\mu$ and of $(a - e_\mu, \mu)$ for $a = 2(y + e_\mu) + e_\nu$, which are one plaquette, the one at $b$ in the plane $\{\mu, \nu\}$, with two path links. So the six plaquettes of a non-path link carry at most three path links in all.

*Step 3: one Wilson term.* Let $V_1,\dots,V_4$ be smooth functions of $(s,t)$ with values in SU(2), $P=V_1V_2V_3V_4$, and $a_i=\|\partial_s V_i\|$, $b_i=\|\partial_t V_i\|$, $c_i=\|\partial_s\partial_t V_i\|$. Then

$$\left|\partial_s\partial_t \tfrac12 \operatorname{Re}\operatorname{Tr} P\right| \le \tfrac12\Big(\sum_i a_i\Big)\Big(\sum_j b_j\Big) + \tfrac12\,\delta(P,I)\sum_i c_i. \quad (27)$$

Indeed $\partial_s\partial_t P$ is the sum of the products with $\partial_s V_i$ and $\partial_t V_j$ in two places $i\neq j$, whose traces have modulus at most $a_ib_j$ by $|\operatorname{Tr}(AB)|\le\|A\|\|B\|$ and unitarity of the other factors, and of the products with $\partial_s\partial_t V_i$ in place $i$, whose trace is $\operatorname{Tr}(\tilde P_i Q_i)$ with $Q_i=V_i^{-1}\partial_s\partial_t V_i$ and $\tilde P_i$ a conjugate of $P$. Differentiating $V_i^*V_i=I$ twice gives $Q_i+Q_i^*=-(\partial_sV_i)^*\partial_tV_i-(\partial_tV_i)^*\partial_sV_i$, so $|\operatorname{Re}\operatorname{Tr}Q_i|\le a_ib_i$, and $|\operatorname{Tr}((\tilde P_i-I)Q_i)|\le\delta(P,I)c_i$. The orientation of a link does not change $a_i,b_i,c_i$.

*Step 4: first derivatives.* For $W\in\mathfrak g$ the map $\operatorname{ad}W$ is skew for the trace form, with eigenvalues 0 and $\pm i\theta$, $\theta=\sqrt2\|W\|$: in a basis diagonalising $W=\operatorname{diag}(i\varphi,-i\varphi)$, $\|W\|=\sqrt2\varphi$, it multiplies the off-diagonal entries by $\pm2i\varphi$. A power series $f(\operatorname{ad}W)$ is therefore normal with norm $\max\{|f(0)|,|f(\pm i\theta)|\}$. For $\varpi(u)=(1-e^{-u})/u$ one has $\varpi(i\theta)=e^{-i\theta/2}\frac{\sin(\theta/2)}{\theta/2}$, so with $m(\theta)=\frac{\theta/2}{\sin(\theta/2)}$, increasing on $(0,2\pi)$,

$$\|\varpi(\operatorname{ad}W)\|=1,\quad \|\varpi(\operatorname{ad}W)^{-1}\|=m(\theta),$$
$$\|e^{\operatorname{ad}W}-1\|\le\theta,\quad \|\varpi(\operatorname{ad}W)-1\|\le 1-\tfrac{1}{m(\theta)}+\tfrac{\theta}{2},$$
$$\|\varpi(\operatorname{ad}W)^{-1}-1\|\le m(\theta)-1+m(\theta)\tfrac{\theta}{2}.$$

The derivative of the exponential is $\frac{d}{dt}e^{W(t)}=e^W\varpi(\operatorname{ad}W)W'$. Hence a non-path link has $\|\partial_g U_g[v]\|=\|\varpi(\operatorname{ad}\zeta_g)v\|\le\|v\|$. Let $k=(y,\mu)$ be solved, $\Gamma_k=c_ke^X$. For a path $x$ of nonzero offset, $c_k^{-1}P_x=e^{\alpha_x}e^{\beta_x}$, where $(\alpha_x,\beta_x)$ is $(\operatorname{Ad}(c_k^{-1})\zeta_u,\zeta_q)$ when $P_x=uq$ with $u$ inside $B(y)$ and $q$ private, and $(\zeta_q,\zeta_{u'})$ when $P_x=qu'$ with $u'$ inside $B(y+e_\mu)$, a tree link contributing 0; so $\|\alpha_x\|,\|\beta_x\|\le r$. Put $Z_x=\log(e^{-X}e^{\alpha_x}e^{\beta_x})$; by Lemma 174, $\|X\|\le14r$ and $\|Z_x\|\le16r$. With $Y=e^{Z_x}$ and $Y^{-1}\partial Y=\varpi(\operatorname{ad}Z_x)\partial Z_x$,

$$\begin{aligned}
\partial_X Z_x[h] &= -T_x h,\\
T_x &= \varpi(\operatorname{ad}Z_x)^{-1}\operatorname{Ad}(e^{-\beta_x}e^{-\alpha_x})\,\varpi(-\operatorname{ad}X),\\
\partial_\beta Z_x[w] &= \varpi(\operatorname{ad}Z_x)^{-1}\varpi(\operatorname{ad}\beta_x)w,\\
\partial_\alpha Z_x[v] &= \varpi(\operatorname{ad}Z_x)^{-1}\operatorname{Ad}(e^{-\beta_x})\varpi(\operatorname{ad}\alpha_x)v.
\end{aligned}$$

With $\theta_Z=16\sqrt2 r$, $\theta_X=14\sqrt2 r$, $m_1=m(\theta_Z)$, the last two have norm at most $m_1$, and writing $T_x-1$ as $(\varpi(\operatorname{ad}Z_x)^{-1}-1)\operatorname{Ad}(\cdot)\varpi(\cdot)+(\operatorname{Ad}(\cdot)-1)\varpi(\cdot)+(\varpi(-\operatorname{ad}X)-1)$,

$$\|T_x-1\|\le\varepsilon_T=m_1-1+m_1\tfrac{\theta_Z}{2}+2\sqrt2 r+1-\tfrac{1}{m(\theta_X)}+\tfrac{\theta_X}{2}.$$

Since $F=-\frac{1}{16}\sum_x Z_x$ over the fourteen paths, $1-DF_k=\frac18-\frac{1}{16}\sum_x(T_x-1)$, so $1-DF_k$ is invertible with

$$\|(1-DF_k)^{-1}\|\le\mathsf m=\frac{8}{1-7\varepsilon_T},$$

and $\det(1-DF_k)>0$, every eigenvalue of $8(1-DF_k)$ lying within $7\varepsilon_T<1$ of 1. The implicit function theorem gives $\partial_g X=(1-DF_k)^{-1}\partial_g F$ and $\|\partial_gF\|\le n_gm_1/16$, and $\|\partial_g\Gamma_k[v]\|=\|e^X\varpi(\operatorname{ad}X)\partial_gX[v]\|\le\|\partial_gX[v]\|$, so

$$\|\partial_g\Gamma_k\|\le\|\partial_gX\|\le\gamma_g=\frac{n_gm_1}{2(1-7\varepsilon_T)}:$$

$\gamma_g=\gamma$ for a private link and $\gamma_g=\frac{\gamma}{2}$ for an inside link, with $\gamma=m_1/(1-7\varepsilon_T)$. In both cases $\gamma_g|K(g)|\le\gamma$, and the sum of $\gamma_{g'}$ over the arguments of one pair is at most $7\gamma+14\frac{\gamma}{2}=14\gamma$.

*Step 5: higher derivatives.* Let $G(X,\alpha,\beta)=\log(e^{-X}e^\alpha e^\beta)$ for $X,\alpha,\beta\in\mathfrak g\otimes\mathbb C$ with $\|X\|+\|\alpha\|+\|\beta\|\le\sigma=\frac12$. The Hilbert–Schmidt norm is submultiplicative, so $\|e^{-X}e^\alpha e^\beta-I\|\le e^\sigma-1<1$, the logarithmic series converges, $G$ is holomorphic and agrees with the principal logarithm at real arguments, and $\|G\|\le M_G=-\log(2-e^{1/2})$. At a real point with $\|X\|+\|\alpha\|+\|\beta\|\le16r$ and for unit vectors $v_1,\dots,v_j$ of $\mathfrak g$ placed in any of the three slots, $t\mapsto G(\cdot+\sum_i t_iv_i)$ is holomorphic on the polydisc $|t_i|\le\rho_j=(\frac12-16r)/j$, and the Cauchy estimates give

$$\|D^jG[v_1,\dots,v_j]\|\le B_j=M_G\rho_j^{-j}\qquad(j=2,3).$$

For $g,g'$ arguments of $k$ let $s_{gg'}\le1$ be the number of paths containing both. Differentiating $X=F(X,\zeta)$ twice and using $\|D_X^2F\|\le\frac{14}{16}B_2$, $\|D_XD_gF\|\le\frac{n_g}{16}B_2$, $\|D_gD_{g'}F\|\le\frac{s_{gg'}}{16}B_2$,

$$\begin{aligned}
\|\partial_g\partial_{g'}X\| &\le \frac{\mathsf mB_2}{16}\big(14\gamma_g\gamma_{g'}+n_{g'}\gamma_g+n_g\gamma_{g'}+s_{gg'}\big)\\
&\le X_2=\frac{\mathsf mB_2}{16}(14\gamma^2+4\gamma+1).
\end{aligned}$$

The second differential of exp at $X$ has norm at most $e^{\|X\|}$ (each of the $n(n-1)$ terms of the second derivative of $X^n/n!$ has norm at most $\|X\|^{n-2}$), so $\|\partial_g\partial_{g'}\Gamma_k\|\le C_2=e^{14r}\gamma^2+X_2$. For $\mathscr M=1-DF_k$, the chain rule and the same bounds with $B_3$ for the third derivatives of $F$ give

$$\|\partial_g\mathscr M\|\le M_1=\frac{B_2}{16}(14\gamma+2),$$
$$\|\partial_g\partial_{g'}\mathscr M\|\le M_2=\frac{B_3}{16}(14\gamma^2+4\gamma+1)+\frac{14}{16}B_2X_2,$$

the terms of $\partial_g\partial_{g'}\mathscr M$ being $D_X^3F[\partial_gX,\partial_{g'}X]$, $D_X^2D_{g'}F[\partial_gX]$, $D_X^2D_gF[\partial_{g'}X]$, $D_X^2F[\partial_g\partial_{g'}X]$ and $D_XD_gD_{g'}F$.

*Step 6: the Wilson terms.* For a plaquette $p$ and a coordinate $g$ let $A_g(p)$ be the sum over the links of $p$ of the bounds of Step 4 for their derivatives in $g$: $A_g(p)=\mathbf1_{g\in p}+\gamma_g\#\{k\in K(g):\Gamma_k\in p\}$. By (27), $\|\partial_g\partial_{g'}W_p\|\le\frac12A_g(p)A_{g'}(p)+\frac12\delta(U_p,I)C_{gg'}(p)$, with $C_{gg'}(p)\le C_2$ times the number of solved path links of $p$ with both $g$ and $g'$ as arguments. Put $S_g(p)=\sum_{g'\neq g}A_{g'}(p)$. A non-path link contributes at most one to $S_g(p)$ and a solved path link at most $14\gamma$ (Step 4), so $S_g(p)\le N+14\gamma\Pi$ with $N$ the number of non-path links of $p$ other than $g$ and $\Pi$ the number of path links of $p$. If $g\in p$ then $N+\Pi\le3$ and $S_g(p)\le3+14\gamma\Pi$, and by Step 2(iv) the sum over the six plaquettes of $g$ is at most $18+42\gamma$. If $\Gamma_k\in p$ then $N\le2$ and $\Pi\le2$, with $\Pi=2$ on three of the six plaquettes of $\Gamma_k$ (Step 2(iii)), so the sum over them

is at most $12+14\gamma(6+3)=12+126\gamma$. Hence

$$\sum_p A_g(p)S_g(p) \le 18+42\gamma+\gamma_g|K(g)|(12+126\gamma)$$
$$\le 18+54\gamma+126\gamma^2.$$

The sum of the first part over all $g'\neq g$ bounds its sum over the other class. The second part of (27) enters only through a solved path link $\Gamma_k\in p$ with $g$ and $g'$ among its arguments; one of $g,g'$ is a $z$-coordinate, so $k$ has a $z$-link among its arguments and the hypothesis gives $\delta(U_p,I)\le\kappa$. There are at most $|K(g)|\le 2$ such pairs, six plaquettes for each and twenty coordinates $g'$, so this part adds at most $\frac12\kappa\cdot 240C_2$. The Wilson terms contribute at most $\beta_W c_W$ to $R_g$.

*Step 7: the Jacobians.* The term $\log\det\mathscr{M}$ of a solved pair depends on $g$ only if $k\in K(g)$, and on $g'\neq g$ only if $g'$ is one of the other twenty arguments. With $|\mathrm{Tr}A|\le 3\|A\|$ on $\mathfrak{g}$ and $\|\mathscr{M}^{-1}\|\le \mathsf{m}$,

$$\left|\partial_g\partial_{g'}\log\det\mathscr{M}\right| = \left|\mathrm{Tr}(\mathscr{M}^{-1}\partial_g\partial_{g'}\mathscr{M})\right.$$
$$\left.-\mathrm{Tr}(\mathscr{M}^{-1}\partial_{g'}\mathscr{M}\,\mathscr{M}^{-1}\partial_g\mathscr{M})\right| \le L = 3\mathsf{m}M_2+3\mathsf{m}^2M_1^2,$$

and the Jacobians contribute at most $c_\psi = 2\cdot 20\cdot L$ to $R_g$. Together,

$$c_W = 9+27\gamma+63\gamma^2+120\kappa C_2, \qquad c_\psi = 40L. \tag{28}$$

*Step 8: the numbers.* At $r=8\cdot10^{-5}$: $m_1=1+1.4\cdot10^{-7}$, $\varepsilon_T=1.924\cdot10^{-3}$, $\mathsf{m}=8.110$, $\gamma=1.01365$, $M_G=1.04618$, $B_2=16.825$, $B_3=227.72$, $X_2=165.77$, $C_2=166.80$, $M_1=17.026$, $M_2=2717.1$, $L=1.2329\cdot10^5$; every constant increases with $r$, so for $r\le 8\cdot10^{-5}$ and $\kappa\le10^{-8}$, $c_W\le 101.1001+0.0003$ and $c_\psi\le 4.94\cdot10^6$. Under hypothesis (a) of Theorem 209, $62r\le 6.0\cdot10^{-13}$, so $r\le 9.68\cdot10^{-15}$, $\varepsilon_T\le 3.0\cdot10^{-13}$, $\gamma\le 1+2.1\cdot10^{-12}$, $C_2\le 160.02$, $L\le 1.1614\cdot10^5$, and $9+27\gamma+63\gamma^2\le 99+4\cdot10^{-10}$. For the condition on $\kappa$, let $k=(y,\mu)$ be solved with a $z$-link among its arguments. A $z$-link is based in a block of $W'$, so $y$ or $y+e_\mu$ lies in $W'$, and by Step 2(iii) the links of a plaquette $p$ through $\Gamma_k$ are based in blocks at distance at most one from $B(y)$, hence at distance at most two from $W'$; they are $\Gamma_k$, non-path links, a tree link, and possibly the path link of $k'=(y+e_\mu-e_\nu,\nu)$, both of whose blocks are at distance at most two from $W'$. These blocks are good, by the sector event inside $\Lambda$, since they lie at distance at least $s+1$ from both islands, and by (c) outside it, so every non-path link of $p$ is within $r$ of its reconstruction value. The path link of $k'$ is within $14r$ of $c_{k'}$: if $k'$ is based in $\Lambda$ it is solved, since a pair based at distance at most $s+1$ from an island has its path link in $L_a\cup p_a$ or $L_b\cup p_b$ and its plaquettes are not $K$-factors, and otherwise Lemma 174 applies by (c). Step 1 of the proof of Theorem 209 gives $\delta(R(c)_p,I)\le\eta$ for these plaquettes, and replacing the links one at a time with bi-invariance and $\delta\le d$ (Lemma 116) gives $\delta(U_p,I)\le\eta+2\cdot14r+2r=\eta+30r\le 6.0\cdot10^{-13}$. Hence $120\kappa C_2\le 1.2\cdot10^{-8}$, $c_W\le 99.0001$ and $c_\psi\le 40L\le 4.65\cdot10^6$. □

With Lemma 211, $\Lambda_2^2\le\beta_W^2(c_W+c_\psi/\beta_W)^2$ and the bound of Corollary 210 becomes

$$\Delta \le 1920\,\beta_W\left(c_W+\frac{c_\psi}{\beta_W}\right)^2 r^2\sqrt{n_x n_y}\,e^{-2\alpha_G(w-2s-8)},$$

which grows linearly in $\beta_W$; at $\beta_W=1.13\cdot10^{37}$, just above $\beta^*$, $s=1$ and $r=43\rho$, $\rho=2.17\cdot10^{-16}$, where $\beta_W r^2=9.84\cdot10^8$, and for islands of one block, $n_x=n_y=45(544+1776)+17\cdot4160=175120$, it reads $\Delta\le 3.25\cdot10^{21}e^{-2\alpha_G(w-10)}$. This is not small at the threshold: the bound is at most one exactly when $w\ge 99072$, at fixed $r$ every factor $e$ in $\beta_W/\beta^*$ adds $1/(2\alpha_G)=2000$ to this width, and islands of $|I_a|$ and $|I_b|$ blocks, for which $N_d\le|I_a|((2d+1)^4-(2d-1)^4)$, add at most $1000\log(|I_a||I_b|)$.

The covariance therefore decays exponentially in the width of the good layer (Figure 23), at the rate of the chart, with the amplitude of Lemma 211, which is below one at the threshold only across good layers about $10^5$ blocks wide. If further bad blocks lie in $\Lambda$ at distance at least $s+4$ from both islands, the factorisation of Theorem 207 through $K$ still holds as an identity of measures, with the constraints of the bad pairs of the layer kept as hard factors inside $K$, but $\mu_{x,y}$ is then no longer a good-window law, and the cross-ratio of $K$ is the comparison for the remaining islands: the corollary is the inductive step for two islands, and Sec. XIII L keeps those constraints out of the layer and performs the induction.

### L. Many islands

The paragraph after Corollary 210 leaves the case in which the layer between two islands contains further islands. If the constraints of those islands are kept inside the kernel, the law of the layer carries hard constraints and branches, and Theorem 209 does not apply to it. This subsection keeps them out of the layer. Every island receives its own collar, the integral over the island and its collar becomes a positive measure on the collar coordinates of that island, as $a_i$ and $b_j$ in Theorem 207, and the layer between all the islands is a good window. The kernel of two islands is then a mixture, over the collar coordinates of the others, of good-window kernels. A mixture of kernels with small cross-ratios can have a large one, and Proposition 189 is such a mixture over one bit.[24] Theorem 215 bounds the additional cross-ratio by the product of the sensitivities of the two islands to the collars of the others, with no condition on the other islands, and by a sum over chains of islands under a Dobrushin condition on them.[25] The measures of the other islands enter only through their positivity.

*The data.* The data are those of Sec. XIII K with the following changes. Let $\mathscr{I}=\{a,b\}\cup\mathscr{D}$ be a finite index set and $I_u\subset\Lambda$, $u\in\mathscr{I}$, pairwise disjoint sets of blocks with

$$w_{uv}=\mathrm{dist}(I_u,I_v)-1\ge 2s+7 \qquad (u\neq v),$$

[24] The cross-ratio $Z_{11}Z_{00}/(Z_{10}Z_{01})$ of the four weights with the two islands in their two patterns measures how much the pattern of one island changes the odds of the pattern of the other: it is one exactly when the two islands are independent. A mixture of independent pairs need not be independent: if a hidden switch raises the chance of both patterns at once, the two islands are correlated however weakly each one depends on the switch.

[25] Dobrushin's condition asks that the conditional law of each island move by less than a fixed fraction $\theta<1$ in total when the islands around it are changed; an influence then fades by the factor $\theta$ at each island it passes through, and the sum over chains converges.

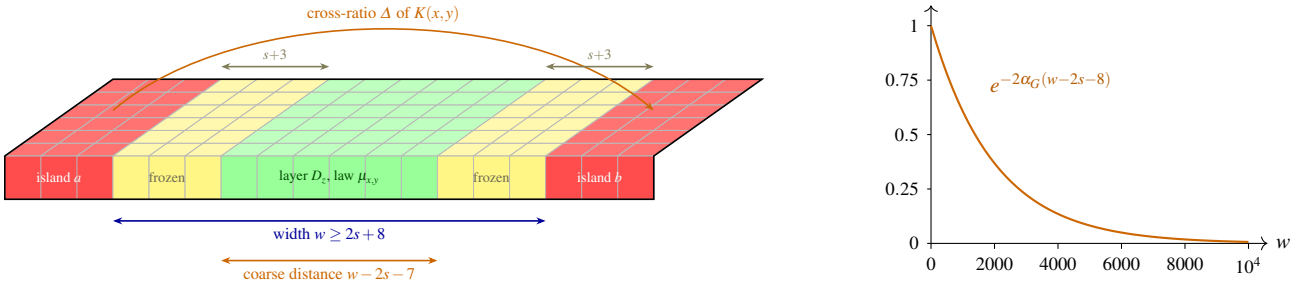



FIG. 23. Left: two bad islands at the ends of a slab, the frozen collars of width $s+3$ beside them, and the good layer between them, on which the layer law $\mu_{x,y}$ obeys the chart bound in the label coordinates (Theorem 209). The cross-ratio $\Delta$ of the kernel of Theorem 207 compares the two islands through the layer. Right: the factor $e^{-2\alpha_G(w-2s-8)}$ of Corollary 210 at $\alpha_G = 2.5\cdot 10^{-4}$; the amplitude in front of it is $1920\Lambda_2^2 r^2\sqrt{n_x n_y}/\beta_W$, of order $10^{-34}\Lambda_2^2 r^2\sqrt{n_x n_y}$ at the threshold $\beta_W = \beta^*$.

which holds for distinct islands formed with the joining distance $2s+7$ of the paragraph after Theorem 207, and with $w_{uv} \geq 2s+8$ for the joining distance $2s+8$ of Corollary 210. Every block of $\Lambda \setminus \bigcup_u I_u$ is good; on $I_a$ two patterns $\omega_a^0, \omega_a^1$ are prescribed, on $I_b$ two patterns $\omega_b^0, \omega_b^1$, and on each $I_q$, $q \in \mathscr{Q}$, one pattern $\omega_q$. The sector $(i,j)$ is the event that the labels of $\Lambda$ are these, with $\omega_a^i$ on $I_a$ and $\omega_b^j$ on $I_b$, and its weight $Z_{ij}$ is defined as in Sec. XIII K. With $d_u(y) = \mathrm{dist}(y, I_u)$, form for every $u \in \mathscr{I}$ the classes $L_u$, $p_u$, $x_u$ of Sec. XIII K, with $d_a$ replaced by $d_u$, let $z$ be every other non-path active link based in $\Lambda$, and solve the path link of every pair based at $d_u \geq s+2$ for every $u$. A link at $d_u \leq s+3$ has $d_v \geq w_{uv} + 1 - (s+3) \geq s+5$ for $v \neq u$, so the classes are disjoint, and every block at $d_u \in \{s+1, s+2, s+3\}$ is good in all four sectors. Let $D_u$ be the product of the $r$-balls in the coordinates $\zeta_e$ of $x_u$ and $D_z$ that of $z$, let $n_u = 45(N_{s+1}^u + N_{s+2}^u) + 17N_{s+3}^u$, with $N_d^u$ the number of blocks of $\Lambda$ at $d_u = d$, be the number of $x_u$-links, let $\Lambda_2$ be the largest operator norm of $\partial_{x_u}\nabla_z \mathscr{U}$ over $u \in \mathscr{I}$ and over $D_z \times \prod_u D_u$, and put

$$\Delta_{uv} = \frac{1920}{\beta_W}\Lambda_2^2 r^2 \sqrt{n_u n_v}\, e^{-2\alpha_G(w_{uv}-2s-8)_+} \qquad (u \neq v),$$

the quantity $\Delta$ of Corollary 210 for the pair $I_u, I_v$ when $w_{uv} \geq 2s+8$.

**Lemma 212** (many islands, one kernel)**.** *Let $s \geq 1$ and $r \leq 10^{-4}$. There are finite positive measures $a_i$ on $D_a$, $b_j$ on $D_b$, $i,j \in \{0,1\}$, and $\gamma_q$ on $D_q$, $q \in \mathscr{Q}$, and a smooth function $\tilde{K} : \prod_{u\in\mathscr{I}} D_u \to (0,\infty)$ with*

$$Z_{ij} = \int \tilde{K}(x)\, a_i(dx_a)\, b_j(dx_b) \prod_{q\in\mathscr{Q}} \gamma_q(dx_q) \qquad (i,j \in \{0,1\}),$$

*such that:*

(i) *$\tilde{K}$ and every $\gamma_q$ are the same in all four sectors; $a_i$ depends on $\omega_a^i$, $c$, $\beta_W$, $U^{\mathrm{ext}}$ only, and likewise $b_j$;*

(ii) *$\tilde{K}(x) = \int_{D_z} e^{-\mathscr{U}(z;x)} dz$ with $\mathscr{U} = \beta_W S_K + \psi$ as in Theorem 207(iv), smooth on a neighbourhood of $D_z \times \prod_u D_u$; no term of $\mathscr{U}$ depends on coordinates of $x_u$ and of $x_v$ for $u \neq v$, and $\nabla_z \partial_{x_u}\mathscr{U}$ is supported on the $z$-links based at $d_u \leq s+4$;*

(iii) *for $u \neq v$ the supports of $\nabla_z\partial_{x_u}\mathscr{U}$ and $\nabla_z\partial_{x_v}\mathscr{U}$ are at fine distance at least $d'_{uv} = 2(w_{uv} - 2s - 8)_+$ between link midpoints.*

*Proof.* The proof of Theorem 207 treats each island through its own distance function, and runs with one class per island. Steps 1 and 2 are unchanged, a pair being solved when it is based at distance at least $s+2$ from every island; its non-path links are then based at distance at least $s+1$ from every island, in good blocks. In Step 3 call a factor a $u$-factor if it depends on a coordinate of $L_u \cup p_u$, and assign the factors depending on $x_u$ alone to $u$. A $u$-factor depends only on $L_u$, $p_u$, $x_u$ and $U^{\mathrm{ext}}$, through links at $d_u \leq s+2$, which have $d_v \geq w_{uv} + 1 - (s+2) \geq s+6$ for every $v \neq u$; so no factor is a $u$-factor and a $v$-factor, and the product $\Phi_K$ of the remaining factors, the $K$-factors, does not depend on the sector, which changes only the indicators of $I_a$ and $I_b$. The indicators of the bad blocks of $I_q$ involve links at $d_q \leq 1 \leq s$ and are $q$-factors. Step 4 with one bracket for each island gives the identity, with $\gamma_q(dx_q)$ the integral of the product of the $q$-factors over $L_q$ and $p_q$ times $\prod_{e\in x_q} j\, d\zeta_e$ on $D_q$, and $a_i$, $b_j$, $\tilde{K}$ as there; this is (i). Step 5 gives the first clause of (ii). Step 6 is a statement about one island at a time, so it gives the support of $\nabla_z\partial_{x_u}\mathscr{U}$, and a $K$-term depending on $x_u$ and on $x_v$ would contain a link at $d_u \leq s+3$ and a link at $d_v \leq s+3$ in blocks at distance at most one, so that $w_{uv} + 1 \leq 2s+7$, which is excluded. The two supports are at coarse distance at least $w_{uv} + 1 - 2(s+4) = w_{uv} - 2s - 7$, and links based in blocks at coarse distance $D \geq 1$ have midpoints at fine distance at least $2D-2$; this is (iii) for $w_{uv} \geq 2s+8$, and (iii) is trivial for $w_{uv} = 2s+7$. □

**Lemma 213** (the layer between many islands)**.** *In addition, assume hypotheses (a)–(c) of Theorem 209, with $W'$ the set of blocks of $\Lambda$ at distance at least $s+3$ from every island. Then for every $x \in \prod_u D_u$ the law $\mu_x = e^{-\mathscr{U}(z;x)}dz/\tilde{K}(x)$ on $D_z$ satisfies (26) with $\mathscr{A} = 480/\beta_W$ and $\alpha = \alpha_G$, and for $u \neq v$ and vectors $h_u$, $h_v$ in the coordinates of $x_u$ and $x_v$,*

$$\left|\partial_{h_u}\partial_{h_v} \log \tilde{K}(x)\right| \leq \frac{480}{\beta_W}\Lambda_2^2 \|h_u\|\|h_v\| e^{-2\alpha_G(w_{uv}-2s-8)_+}.$$

*In particular the logarithm of the cross-ratio of $\tilde{K}$ in $(x_u, x_v)$,*

*between two points of $D_u$ and two points of $D_v$ with the other arguments fixed, has modulus at most $\Delta_{uv}$.*

*Proof.* The law $\mu_x$ is the law of Proposition 175 for the window $W'$ in the coordinates of the labels, with the inside links of the blocks at distance $s+3$ from some island frozen and the exterior given by $x$ and $U^{\mathrm{ext}}$, by the identification in the paragraph on the layer law after Theorem 207, which uses only that the $z$-links are the free non-path links of $W'$ and that the solved pairs are those of Lemma 212. The proof of Theorem 209 uses the islands only in Step 1, through the goodness of the blocks at distance at most two from $W'$; these are at distance at least $s+1$ from every island, hence good in all sectors inside $\Lambda$ and good by (c) outside it. Steps 2 to 6 concern the window $W'$ and are uniform in its exterior, and freezing the inside links at distance $s+3$ from any island restricts the Hessian to a principal submatrix. So (26) holds with the stated constants. For the derivative, Lemma 188 on the fixed domain $D_z$ gives $\partial_{h_u}\partial_{h_v}\log\tilde{K} = -\mathbb{E}_{\mu_x}[\partial_{h_u}\partial_{h_v}\mathscr{U}] + \mathrm{Cov}_{\mu_x}(\partial_{h_u}\mathscr{U},\partial_{h_v}\mathscr{U})$, and the first term vanishes by Lemma 212(ii). As in the proof of Proposition 208, apply (26) with $\varphi(e) = -\mathrm{dist}(e,\mathrm{supp}\,\nabla_z\partial_{h_u}\mathscr{U})$, which vanishes on that support and is at most $-d'_{uv}$ on the support of $\nabla_z\partial_{h_v}\mathscr{U}$ by Lemma 212(iii), with $\|\nabla_z\partial_{h_u}\mathscr{U}\| \le \Lambda_2\|h_u\|$ and $\|\nabla_z\partial_{h_v}\mathscr{U}\| \le \Lambda_2\|h_v\|$ pointwise. For the cross-ratio, $D_u$ and $D_v$ are convex, the logarithm of the cross-ratio is the integral of the mixed derivative over the unit square, and two points of $D_u$ differ by at most $2r\sqrt{n_u}$. □

The other islands now enter only through the measures $\gamma_q$, and the kernel of $a$ and $b$ is $K(x_a,x_b) = \int \tilde{K}(x)\prod_q \gamma_q(dx_q)$, an integral against the probability law on $\prod_q D_q$ proportional to $\tilde{K}(x_a,x_b,\cdot)\prod_q\gamma_q$. The next lemma bounds the covariances of that law in terms of the cross-ratios of its density alone.

**Lemma 214** (covariances over the islands)**.** *Let $\mathscr{Q}$ be finite, $X_q$ compact metric spaces, $\gamma_q$ nonzero finite Borel measures on $X_q$, $X = \prod_q X_q$, $k : X \to (0,\infty)$ continuous, and $P = k\prod_q\gamma_q / \int k\prod_q\gamma_q$. For $\xi \in X$ and $\eta_q \in X_q$ let $\xi^{\eta_q}$ be $\xi$ with its $q$-coordinate replaced by $\eta_q$. For $q \neq q'$ let $\kappa_{qq'}$ bound $|\log k(\xi^{\eta_q\eta_{q'}}) + \log k(\xi) - \log k(\xi^{\eta_q}) - \log k(\xi^{\eta_{q'}})|$ for all $\xi,\eta_q,\eta_{q'}$, let $C_{qq'} = \tanh(\kappa_{qq'}/4)$ and $C_{qq} = 0$, and for continuous $f$ on $X$ let $\delta_q(f) = \sup\{|f(\xi^{\eta_q}) - f(\xi)|\}$. Then for continuous $f,g$:*

(i) $|\mathrm{Cov}_P(f,g)| \le \frac{1}{4}\sum_q \delta_q(f)\sum_q\delta_q(g)$;

(ii) *if $\sum_{q'} C_{qq'} \le \theta < 1$ for every $q$, then $D = \sum_{n\ge0} C^n$ converges and*

$$|\mathrm{Cov}_P(f,g)| \le \frac{1}{4}\sum_{q,q'} \delta_q(f)\,D_{qq'}\,\delta_{q'}(g);$$

(iii) *if $\ell$ is a pseudometric on $\mathscr{Q}$, $\lambda \ge 0$ and $\sum_{q'} C_{qq'}e^{\lambda\ell(q,q')} \le \theta < 1$ for every $q$, then $0 \le D_{qq'} \le e^{-\lambda\ell(q,q')}/(1-\theta)$.*

*Proof. Conditional laws.* The law

$$P_q(d\eta \mid \xi) = \frac{k(\xi^\eta)\,\gamma_q(d\eta)}{\int k(\xi^{\eta'})\,\gamma_q(d\eta')}$$

on $X_q$ depends on $\xi$ through $\xi_{-q}$ only, its density with respect to $\gamma_q$ is continuous in $(\eta,\xi)$, and $P$ is invariant under resampling the $q$-coordinate from it.

*Tilts.* Let $\mu$ be a probability measure and $\mu' = h\mu/\mu(h)$ with $h$ measurable and $0 < \inf h \le \sup h \le e^\kappa \inf h$. For a Borel set $E$ with $0 < \mu(E) < 1$ the odds of $E$ under $\mu'$ and under $\mu$ have the ratio $\frac{\mu(h\mathbf{1}_E)/\mu(E)}{\mu(h\mathbf{1}_{E^c})/\mu(E^c)} \in [e^{-\kappa},e^\kappa]$, so the logistic bound in the proof of Corollary 187 gives $|\mu'(E) - \mu(E)| \le \tanh(\kappa/4)$, and $\mu'(E) = \mu(E)$ when $\mu(E) \in \{0,1\}$. Hence $\|\mu' - \mu\|_{\mathrm{TV}} \le \tanh(\kappa/4)$. If $\xi$ and $\xi'$ differ only in the coordinate $q' \neq q$, the density of $P_q(\cdot \mid \xi')$ with respect to $P_q(\cdot \mid \xi)$ is proportional to $k(\xi'^\eta)/k(\xi^\eta)$, whose logarithm oscillates in $\eta$ by at most $\kappa_{qq'}$, so $\|P_q(\cdot \mid \xi) - P_q(\cdot \mid \xi')\|_{\mathrm{TV}} \le C_{qq'}$.

*Comparison.* Let $g$ be continuous, $t \in \mathbb{R}$, and $P^t = e^{tg}P/P(e^{tg})$, again of the form of the lemma; its conditional law at $q$ is the tilt of $P_q(\cdot \mid \xi)$ by $e^{tg(\xi^\eta)}$, so $b_q = \sup_\xi \|P_q^t(\cdot \mid \xi) - P_q(\cdot \mid \xi)\|_{\mathrm{TV}} \le \tanh(|t|\delta_q(g)/4)$. The couplings of $P$ and $P^t$ form a weakly compact set (Lemma 191), and $\Psi(\pi) = \sum_q \rho_q(\pi)$, $\rho_q(\pi) = \pi\{\xi_q \neq \xi'_q\}$, is weakly lower semicontinuous, since $\{\xi_q \neq \xi'_q\}$ is open; let $\pi$ minimise it. For $q \in \mathscr{Q}$ replace $(\xi_q,\xi'_q)$ by a sample of the maximal coupling of $P_q(\cdot \mid \xi)$ and $P_q^t(\cdot \mid \xi')$, which, with $p,p'$ the two densities with respect to $\gamma_q$ and $m = \int \min\{p,p'\}d\gamma_q$, draws $\eta = \eta'$ from $\min\{p,p'\}\gamma_q/m$ with probability $m$ and otherwise, when $m < 1$, $\eta$ and $\eta'$ independently from $(p - \min\{p,p'\})\gamma_q/(1-m)$ and $(p' - \min\{p,p'\})\gamma_q/(1-m)$; it is measurable in $(\xi,\xi')$ by the continuity of the densities, and it gives $\{\eta \neq \eta'\}$ probability at most $1 - m = \|P_q(\cdot \mid \xi) - P_q^t(\cdot \mid \xi')\|_{\mathrm{TV}}$. Both marginals are invariant, so the result $\pi^{(q)}$ is again a coupling; it changes only $\rho_q$, and changing the coordinates of $\xi_{-q}$ to those of $\xi'_{-q}$ one at a time,

$$\rho_q(\pi^{(q)}) \le \mathbb{E}_\pi \sum_{q'} C_{qq'}\mathbf{1}_{\xi_{q'}\neq\xi'_{q'}} + b_q = \sum_{q'} C_{qq'}\rho_{q'}(\pi) + b_q.$$

By minimality $\rho_q(\pi) \le \rho_q(\pi^{(q)})$, which is the argument of Lemma 190 for the discrete metric, so $\rho \le C\rho + b$ componentwise. Under the hypothesis of (ii), $C^n\mathbf{1} \le \theta^n\mathbf{1}$ and $\rho \le \mathbf{1}$, so iterating gives $\rho \le \sum_{k<n} C^k b + \theta^n\mathbf{1}$ and, as $n \to \infty$, $\rho \le Db$. Changing the coordinates one at a time,

$$\begin{aligned}|P^t(f) - P(f)| &= \left|\mathbb{E}_\pi[f(\xi') - f(\xi)]\right| \\ &\le \sum_q \delta_q(f)\rho_q(\pi) \le \sum_{q,q'} \delta_q(f)D_{qq'}b_{q'}.\end{aligned}$$

*Covariances.* The functions $f$ and $g$ are bounded, so the derivative of $P^t(f)$ at $t = 0$ is $\mathrm{Cov}_P(f,g)$, and $b_{q'} \le |t|\delta_{q'}(g)/4$; dividing by $|t|$ and letting $t \to 0$ gives (ii). For (i), $|\mathrm{Cov}_P(f,g)| \le (\mathrm{Var}_P f\,\mathrm{Var}_P g)^{1/2}$, a variance is at most a quarter of the squared oscillation, and changing the coordinates one at a time bounds the oscillation of $f$ by $\sum_q\delta_q(f)$; the bound on the variance is Popoviciu's inequality [77]. Part (ii) is Föllmer's covariance estimate [41] under Dobrushin's condition [34, 35], with the coefficients $C_{qq'}$ read off from the cross-ratios of $k$. For (iii), by induction $(C^n)_{qq'} \le \theta^n e^{-\lambda\ell(q,q')}$: the case $n = 0$ is $\ell(q,q) = 0$, and $(C^{n+1})_{qq'} = \sum_k C_{qk}(C^n)_{kq'} \le \theta^n e^{-\lambda\ell(q,q')}\sum_k C_{qk}e^{\lambda\ell(q,k)}$ by the triangle inequality; summing over $n$ gives the bound, and the hypothesis of (iii) contains that of (ii). □

**Theorem 215** (many islands)**.** *Let $s \geq 1$, let the data be those above, assume hypotheses (a)–(c) of Theorem 209 with $W'$ as in Lemma 213, and suppose $Z_{ij} > 0$ for $i,j \in \{0,1\}$. Put $S_u = \sum_{q \in \mathscr{Q}} \Delta_{uq}$ for $u \in \{a,b\}$, let $C$ be the $\mathscr{Q} \times \mathscr{Q}$ matrix with $C_{qq} = 0$ and $C_{qq'} = \tanh(\Delta_{qq'}/4)$, and let*

$$\Delta_* = \Delta_{ab} + \frac{1}{4} S_a S_b;$$

*when $\sum_{q'} C_{qq'} < 1$ for every $q$, let instead, with $D = \sum_{n \geq 0} C^n$,*

$$\Delta_* = \Delta_{ab} + \frac{1}{4} \min\Big\{ S_a S_b,\ \sum_{q,q' \in \mathscr{Q}} \Delta_{aq} D_{qq'} \Delta_{q'b} \Big\}.$$

*Then*

$$\big| \log Z_{11} + \log Z_{00} - \log Z_{10} - \log Z_{01} \big| \leq \Delta_*,$$

*the two island labels, the pattern of $I_a$ among $\omega_a^0, \omega_a^1$ and that of $I_b$ among $\omega_b^0, \omega_b^1$, have conditional covariance at most $\frac{1}{4}\tanh(\Delta_*/4)$ given the labels of all other blocks of $\Lambda$, the coarse field and $U^{\mathrm{ext}}$, and for every bounded measurable $F \geq 0$ of the links of $L_a \cup p_a \cup x_a$,*

$$e^{-\Delta_*} \leq \frac{\mathbb{E}_{i1}[F]}{\mathbb{E}_{i0}[F]} \leq e^{\Delta_*} \qquad (i = 0,1)$$

*whenever the four expectations are positive. For $\mathscr{Q} = \emptyset$, $\Delta_* = \Delta_{ab}$, which for $w_{ab} \geq 2s+8$ is the bound of Corollary 210.*

*Proof.* By Lemma 212, $Z_{ij} = \iint K \, da_i \, db_j$ with $K(x_a, x_b) = \int \tilde{K}(x) \prod_q \gamma_q(dx_q)$; $Z_{ij} > 0$ forces every $\gamma_q \neq 0$, and $K$ is smooth and positive on $D_a \times D_b$. Fix $x_a^0, x_a^1 \in D_a$, $x_b^0, x_b^1 \in D_b$, and let $x_a(\vartheta) = x_a^0 + \vartheta(x_a^1 - x_a^0)$, $x_b(\varphi)$ likewise, which stay in the convex sets $D_a, D_b$. The logarithm of the cross-ratio of $K$ at these points is $\int_0^1 \int_0^1 \partial_\vartheta \partial_\varphi \log K \, d\vartheta \, d\varphi$. With $P$ the law on $\prod_q D_q$ proportional to $\tilde{K}(x_a(\vartheta), x_b(\varphi), \cdot) \prod_q \gamma_q$, differentiation under the integral sign gives

$$\partial_\vartheta \partial_\varphi \log K = \mathbb{E}_P\big[\partial_\vartheta \partial_\varphi \log \tilde{K}\big] + \mathrm{Cov}_P\big(\partial_\vartheta \log \tilde{K}, \partial_\varphi \log \tilde{K}\big).$$

By Lemma 213 with $h_a = x_a^1 - x_a^0$, $h_b = x_b^1 - x_b^0$, the first term has modulus at most $\Delta_{ab}$. Apply Lemma 214 with $X_q = D_q$, $k = \tilde{K}(x_a(\vartheta), x_b(\varphi), \cdot)$, $f = \partial_\vartheta \log \tilde{K}$ and $g = \partial_\varphi \log \tilde{K}$. Changing $x_q$ along the segment between two points of $D_q$ and integrating the mixed derivative of Lemma 213 for the pair $(a,q)$ gives $\delta_q(f) \leq \Delta_{aq}$, likewise $\delta_q(g) \leq \Delta_{bq}$, and the last statement of Lemma 213 for the pair $(q,q')$ shows that $\kappa_{qq'} = \Delta_{qq'}$ is admissible, so the matrix of Lemma 214 is $C$. Parts (i) and (ii) of that lemma bound the covariance by $\Delta_* - \Delta_{ab}$, uniformly in $(\vartheta, \varphi)$. So the cross-ratio of $K$ is at most $e^{\Delta_*}$ and at least $e^{-\Delta_*}$ on $D_a \times D_b$, and Lemma 186 gives the first bound. The joint law of the two labels given the others is $Z_{ij}/\sum Z$, so Corollary 187 gives the second. For the third, $F\Phi_a^i$ is a product of $a$-factors, so the identity holds with $Fa_i$ in place of $a_i$ and the same $K$ and $b_j$, and Lemma 186 applied to $Fa_i$, $a_i$ and $b_1$, $b_0$ gives it, since $Z(Fa_i, b_j)/Z(a_i, b_j) = \mathbb{E}_{ij}[F]$. □

The first form of $\Delta_*$ needs nothing of the other islands: their hard constraints, branches and masses are integrated in the $\gamma_q$ and never compared. Its mediated term $\frac{1}{4} S_a S_b$ does not decay with $w_{ab}$ when other islands lie near both $I_a$ and $I_b$, since it treats every collar coordinate as able to reach every other. The second form keeps the interaction between the islands, through the matrix $C$, and under a weighted form of the Dobrushin condition $\sum_{q'} \tanh(\Delta_{qq'}/4) < 1$ it decays along chains of islands.

**Corollary 216** (decay along chains of islands)**.** *Under the hypotheses of Theorem 215, let $\ell$ be the pseudometric on $\mathscr{I}$ given by the least value of $\sum_i (w_{u_i u_{i+1}} - 2s - 8)_+$ over the chains $u = u_0, u_1, \ldots, u_k = v$ in $\mathscr{I}$, let $0 \leq \vartheta \leq 1$, and suppose*

$$\theta = \max_{q \in \mathscr{Q}} \sum_{q' \in \mathscr{Q} \setminus \{q\}} \tanh \frac{\Delta_{qq'}}{4} \, e^{2\vartheta \alpha_G \ell(q,q')} < 1.$$

*Then*

$$\Delta_* \leq \Delta_{ab} + \frac{e^{-2\vartheta\alpha_G \ell(a,b)}}{4(1-\theta)} T_a T_b, \qquad T_u = \sum_{q \in \mathscr{Q}} \Delta_{uq} e^{2\vartheta \alpha_G \ell(u,q)},$$

$$T_u \leq \frac{1920}{\beta_W} \Lambda_2^2 r^2 \sqrt{n_u} \sum_{q \in \mathscr{Q}} \sqrt{n_q} \, e^{-2(1-\vartheta)\alpha_G (w_{uq} - 2s - 8)_+}.$$

*Proof.* $\ell$ is a pseudometric since it is a least sum over chains, and $\ell(u,v) \leq (w_{uv} - 2s - 8)_+$. Lemma 214(iii) on $\mathscr{Q}$ with $\lambda = 2\vartheta\alpha_G$ gives $D_{qq'} \leq e^{-\lambda \ell(q,q')}/(1-\theta)$ and the hypothesis of the second form of $\Delta_*$, and $-\ell(q,q') \leq -\ell(a,b) + \ell(a,q) + \ell(q',b)$ gives the first bound. The second is $\ell(u,q) \leq (w_{uq} - 2s - 8)_+$ in each term. □

A chain crosses each island at no cost in $\ell$: an island is a set on which nothing is known, and the corollary charges only the good layers between consecutive islands of the chain. With $\vartheta = 0$ the hypothesis is the Dobrushin condition $\sum_{q'} C_{qq'} \leq \theta < 1$.

The sums $S_u$ and $T_u$ are sums over the islands near $I_u$, and the rarity of Theorem 199 controls them in the mean. For a bad block $y$ let $|I(y)|$ be the number of blocks of its island, and $|I(y)| = 0$ for a good block.

**Proposition 217** (the island sums in the mean)**.** *Form the islands with the joining distance $2s+8$, so that $w_{uv} \geq 2s+8$ for distinct islands, let $N_s = (4s+17)^4 - 1$ be the number of neighbours of a block, $\Gamma \geq 1$, and $\beta_W \geq \beta^*(\rho, s, \Gamma)$ of Theorem 199(iii) with $N_s$ neighbours. Then:*

(i) *for every block $y$, $\mathbb{E}_{\mu_L} |I(y)| \leq 2\Gamma(2\Gamma - 1)^{-2} N_s^{-2}$;*

(ii) *for every set $J$ of blocks of the torus and every $\kappa > 0$,*

$$\mathbb{E}_{\mu_L} \sum_{y \notin J} |I(y)| e^{-\kappa \, \mathrm{dist}(J,y)} \leq \frac{2\Gamma}{(2\Gamma - 1)^2 N_s^2} \, |J| \, \sigma(\kappa),$$

*where, with $x = e^{-\kappa}$,*

$$\sigma(\kappa) = \sum_{y \in \mathbb{Z}^4} e^{-\kappa \|y\|_\infty} = 1 + \frac{64x(1 + 4x + x^2)}{(1-x)^4} + \frac{16x}{(1-x)^2};$$

(iii) *in the setting of Theorem 215, when every $I_q$, $q \in \mathcal{Q}$, is an island, with $B' = \bigcup_q I_q$, $0 \le \vartheta \le 1$ and $\kappa = 2(1-\vartheta)\alpha_G$,*

$$T_u \le \frac{1920}{\beta_W} \Lambda_2^2 r^2 \sqrt{45\, n_u}\, (2s+7)^2 e^{\kappa(2s+9)} \times \sum_{y \in B'} |I(y)| e^{-\kappa \operatorname{dist}(I_u, y)}.$$

*At $\kappa = 2\alpha_G = 5 \cdot 10^{-4}$, $\sigma(\kappa) = 6.144 \cdot 10^{15}$, and at $\Gamma = 1$ the bound of (ii) is $3.249 \cdot 10^5 |J|$ for $s = 1$ and $3.498 \cdot 10^3 |J|$ for $s = 5$.*

*Proof.* (i) The island paragraph of the proof of Theorem 199 gives $\mu_L\{|I(y)| = n\} \le N_s^{2(n-1)} A^{n/H}$, and its proof of (iii) gives $\Gamma N_s^2 A^{1/H} \le \frac{1}{2}$ at $\beta_W \ge \beta^*(\rho, s, \Gamma)$, so $\mu_L\{|I(y)| = n\} \le N_s^{-2} (2\Gamma)^{-n}$ and $\mathbb{E}|I(y)| \le N_s^{-2} \sum_n n (2\Gamma)^{-n} = 2\Gamma (2\Gamma - 1)^{-2} N_s^{-2}$. (ii) By linearity it suffices that $\sum_y e^{-\kappa \operatorname{dist}(J,y)} \le |J| \sigma(\kappa)$; since $e^{-\kappa \operatorname{dist}(J,y)} \le \sum_{z \in J} e^{-\kappa \operatorname{dist}(z,y)}$, it suffices that $\sum_y e^{-\kappa \operatorname{dist}(z,y)} \le \sigma(\kappa)$ on the torus, which holds because each block of the torus has a preimage in $\mathbb{Z}^4$ at $\ell^\infty$ distance $\operatorname{dist}(z,y)$ from a preimage of $z$, distinct blocks having distinct preimages. The closed form counts the $(2k+1)^4 - (2k-1)^4 = 64k^3 + 16k$ points of $\mathbb{Z}^4$ at distance $k \ge 1$ and sums $\sum k x^k$ and $\sum k^3 x^k$. (iii) In the second bound of Corollary 216, $n_q \le 45(2s+7)^4 |I_q|$, since every $x_q$-link is based in a block at distance at most $s+3$ from $I_q$, so $\sqrt{n_q} \le \sqrt{45}(2s+7)^2 |I_q|$; and $(w_{uq} - 2s - 8)_+ \ge \operatorname{dist}(I_u, I_q) - 2s - 9$, while $|I_q| e^{-\kappa \operatorname{dist}(I_u, I_q)}$ is the term of $\sum_{y \in I_q} |I(y)| e^{-\kappa \operatorname{dist}(I_u, y)}$ at a block of $I_q$ nearest to $I_u$. The numbers are computed in the archive. □

The proposition is annealed: it bounds the mean of the island sums under $\mu_L$ for a fixed set $J$, while Theorem 215 is a statement given all the labels, and it says nothing about the law of the islands near $I_u$ given that $I_u$ is an island. At $\Gamma = 1$ its bound exceeds one, because the weighted volume $\sigma(2\alpha_G) = 6.1 \cdot 10^{15}$ of the decay length $1/(2\alpha_G) = 2000$ blocks of Theorem 209 exceeds $N_s^2$; for $\Gamma \ge 2$ the bound of (ii) is at most $|J| \sigma(\kappa) / (\Gamma N_s^2)$, and Theorem 199(iii) reaches every $\Gamma$ at the coupling $\beta^*(\rho, s, \Gamma)$.

With Theorem 215 the comparison of item (ii) of Sec. XIII P holds with any number of islands, given the labels of all of them: the cross-ratio of two islands is at most $\Delta_*$, explicit in the two-island quantities $\Delta_{uv}$ of Corollary 210, and the observables of one island change by a factor in $[e^{-\Delta_*}, e^{\Delta_*}]$ under a flip of the other. The prefactor $\Lambda_2$ is bounded by Lemma 211, whose proof uses only the plaquettes at one collar and applies to each island $I_u$ when $r \le 8 \cdot 10^{-5}$. Two things remain. The Dobrushin condition of the second form and the smallness of $S_u$ or $T_u$ are conditions on the configuration of the islands, which Proposition 217 controls only in the mean. And the covariance of the two labels with the other labels averaged out differs from the conditional covariance by the covariance of the conditional probabilities of the two labels given the others, which is the term of Proposition 189 and is not bounded here.

The first of them, the smallness of the island sums together with the Dobrushin condition, holds in every window of bounded size outside an event whose probability is at most the size of the window times the product of two rarities: that of one bad block and that of a second bad block or island. Two facts give this. The bound of Proposition 217(i) charges an island of $n$ blocks the factor $A^{n/H}$, while Lemma 184 gives $A^{\lceil n/H \rceil}$, which is $A$ for $n \le H$; with the ceiling, the mean mass of the island of a block is at most $G_0 < 2H 2^{-H} N_s^{-2}$ below, instead of $2N_s^{-2}$. And two bad blocks at distance at least six have disjoint witness boxes, so that two islands near each other cost the product of two rarities. For two islands $I_u \ne I_v$ of the torus, formed with the joining distance $2s+8$, put

$$\bar{n}_u = 45 \big| \{ y : \operatorname{dist}(y, I_u) \le s+3 \} \big|,$$
$$\bar{\Lambda}_2 = 99.0001\, \beta_W + 4.65 \cdot 10^6,$$
$$\bar{\Delta}_{uv} = \frac{1920}{\beta_W} \bar{\Lambda}_2^2 r^2 \sqrt{\bar{n}_u \bar{n}_v}\, e^{-2\alpha_G (w_{uv} - 2s - 8)},$$

and, for $0 \le \vartheta < 1$, $\bar{T}_u = \sum_v \bar{\Delta}_{uv} e^{2\vartheta \alpha_G (w_{uv} - 2s - 8)}$, the sum over all islands $I_v \ne I_u$ of the torus. These are functions of the labels alone. In the setting of Theorem 215, when the sets $I_u$ are islands, $n_u \le \bar{n}_u$, Lemma 211, applied to each island as above, gives $\Lambda_2 \le \bar{\Lambda}_2$, and $\ell(u,v) \le w_{uv} - 2s - 8$; hence $\Delta_{uv} \le \bar{\Delta}_{uv}$, and $S_u \le \bar{T}_u$ and $T_u \le \bar{T}_u$ for the $T_u$ of Corollary 216 with the same $\vartheta$. Write $W^{+t}$ for the set of blocks at distance at most $t$ from a set $W$ of blocks.

**Theorem 218** (the island sums in a window)**.** *Let $G = \mathrm{SU}(2)$, $s \ge 1$, $2758\rho \le 6.0 \cdot 10^{-13}$, $r = 43\rho$, $\eta = 92\rho$, form the islands of the torus with the joining distance $2s+8$, and let $N_s = (4s+17)^4 - 1$. Let $\beta^*$ be the largest solution of the equation of Theorem 199 with $N_s$ in place of $\Delta$, $\log q^* = -H(\log 2 + 2\log N_s) - \log(eMH)$ its right side, $C_0$ that of Lemma 146, $\beta_W \ge \beta^*$, and*

$$G_0 = \frac{H 2^{-H}}{(N_s^2 - 1)(1 - 2^{-H})^2}.$$

(i) *For every block $y$, $\mathbb{E}_{\mu_L} |I(y)| \le G_0$, and Proposition 217(ii) holds at $\Gamma = 1$ with $2\Gamma(2\Gamma-1)^{-2} N_s^{-2}$ replaced by $G_0$.*

(ii) *Let $0 \le \vartheta < 1$, $\kappa = 2(1-\vartheta)\alpha_G$, let $R \ge 2s + 13$ be an integer, $\sigma > 0$, $W$ a set of blocks,*

$$c_2 = \frac{1920}{\beta^*} \bar{\Lambda}_2(\beta^*)^2 r^2 \cdot 45 (2s+12)^2 (2s+7)^2 e^{\kappa(2s+14)},$$
$$\tau = \min\big\{ \sigma(\kappa),\, e^{-\kappa R/2} \sigma(\kappa/2) \big\},$$

*with $\bar{\Lambda}_2(\beta^*)$ the value of $\bar{\Lambda}_2$ at $\beta_W = \beta^*$ and $\sigma(\cdot)$ as in Proposition 217. Outside an event of $\mu_L$-probability at most*

$$\varepsilon = |W| M q^* \Big( 11^4 (2R+1)^4 M q^* + \frac{c_2 G_0 \tau}{\sigma} \Big),$$

*every island $I_u$ meeting $W$ has $\ell^\infty$ diameter at most 5, hence at most $6^4$ blocks, lies at distance more than $R$ from every other island, and satisfies $\bar{T}_u \le \sigma$.*

(iii) *Outside the same event, in the setting of Theorem 215 with $I_a$, $I_b$ and every $I_q$ islands meeting $W$: $S_a, S_b \le \sigma$*

*and $\Delta_* \le \Delta_{ab} + \frac14\sigma^2 \le \sigma + \frac14\sigma^2$; if $\sigma < 4$, the weighted Dobrushin condition of Corollary 216 holds with $\theta \le \sigma/4$, and*

$$\Delta_* \le \Delta_{ab} + \frac{e^{-2\vartheta\alpha_G \ell(a,b)}\sigma^2}{4-\sigma} \le \sigma + \frac{\sigma^2}{4-\sigma}.$$

*At $\rho = 2.17\cdot 10^{-16}$ and $s = 1$: $\beta^* = 1.12206\cdot 10^{37}$, $|\log q^*| = 32481.67$, $Mq^* \le 10^{-14102.4}$, the bound A of Theorem 199(i) is at most $10^{-14098.8}$, $G_0 \le 10^{-397.6}$, and the bound of (i) for Proposition 217(ii) at $\kappa = 2\alpha_G$ is $G_0\sigma(2\alpha_G)|J| \le 10^{-381.8}|J|$, in place of $3.249\cdot 10^5|J|$. For $\vartheta = 0$, $c_2 \le 10^{22.13}$; with $R = 15$, $\varepsilon \le |W|(10^{-28194} + 10^{-14462}/\sigma)$, and with $R = 2\cdot 10^8$, $\varepsilon \le 10^{-28166}|W|$ for every $\sigma \in [10^{-1000}, 4)$; for $\vartheta = \frac12$ and $R = 4\cdot 10^8$, $\varepsilon \le 10^{-28165}|W|$ on the same range. At $s = 5$: $\beta^* = 1.32163\cdot 10^{37}$, $Mq^* \le 10^{-16652.7}$, $G_0 \le 10^{-399.5}$, and with $\vartheta = 0$, $R = 2\cdot 10^8$, $\varepsilon \le 10^{-33267}|W|$ for $\sigma \in [10^{-1000}, 4)$.*

*Proof.* Write $p = q_\rho(\beta_W)$ and $B$ for the set of bad blocks, and let $Q_y$ be the set of unit cubes with corner at $\ell^\infty$ distance at most five from $2y$. By the witness paragraph of the proof of Theorem 199, which applies since $\rho \le 10^{-8}$, a bad block $y$ has a cube of $Q_y$ on which the event of Theorem 147 holds; $|Q_y| \le M$, and a cube lies in at most $H$ of the sets $Q_y$. For a set $S$ of blocks put $Q_S = \bigcup_{y\in S} Q_y$.

*Step 1: separated sets.* Let $S_1, \dots, S_m$ be nonempty sets of blocks with $\operatorname{dist}(S_i, S_j) \ge 6$ for $i \ne j$, and $k_i = \lceil |S_i|/H \rceil$. Then

$$\mu_L\Big(\bigcup_i S_i \subset B\Big) \le \prod_i \binom{M|S_i|}{k_i} p^{k_i} \le \prod_i A^{k_i},$$

and the factor of a set of one block is $Mp$. Two blocks at distance at least six have a coordinate in which the corners $2y$ differ by at least twelve modulo the side of the torus, while a cube of $Q_y \cap Q_{y'}$ would put $2y$ and $2y'$ within ten of each other in every coordinate; so the sets $Q_{S_i}$ are pairwise disjoint. On the event, $Q_{S_i}$ contains at least $k_i$ distinct bad cubes, since one cube witnesses at most $H$ blocks; choosing $k_i$ of them in each $Q_{S_i}$ gives $\sum_i k_i$ distinct bad cubes, and the union bound over the choices with Theorem 147 gives the first inequality. The second is that of Lemma 184.

*Step 2: the mass of an island.* Let $x = N_s^2 A^{1/H}$, which is at most $\frac12$ by Theorem 199(i). There are at most $N_s^{2(n-1)}$ connected sets of $n$ blocks containing a block $y$ in the graph of the joining distance, and for $(k-1)H < n \le kH$, $N_s^{2(n-1)}A^k = x^{kH}N_s^{-2(kH+1-n)}$. Hence

$$\begin{aligned}\sum_{C\ni y} |C| A^{\lceil |C|/H\rceil} &\le \sum_{k\ge1} kH\, x^{kH} \sum_{j\ge1} N_s^{-2j} \\ &= \frac{Hx^H}{(N_s^2-1)(1-x^H)^2} \le G_0,\end{aligned} \tag{29}$$

the sum over connected sets $C$ containing $y$, the last inequality since $t/(1-t)^2$ increases on $[0,1)$. Since $\mu_L\{I(y) = C\} \le \mu_L\{C \subset B\} \le A^{\lceil |C|/H\rceil}$ by Lemma 184, $\mathbb{E}|I(y)| \le G_0$. The proof of Proposition 217(ii) uses only this mean, which gives (i).

*Step 3: no near pair.* Let $\mathscr{G}$ be the event that no two bad blocks $y \in W^{+5}$ and $y'$ satisfy $6 \le \operatorname{dist}(y,y') \le R$. On $\mathscr{G}$ let $I_u$ be an island meeting $W$ at $y_0$. Every block of $I_u$ is within distance 5 of $y_0$: otherwise a path $y_0 = z_0, z_1, \dots$ in $I_u$ with steps at most $2s+8$ has a first block $z_k$ at distance more than 5 from $y_0$, and $6 \le \operatorname{dist}(y_0, z_k) \le 5 + 2s + 8 \le R$. So $I_u \subset W^{+5}$, two blocks of $I_u$ are at distance at most $10 < R$, hence at most 5, and $I_u$ lies in a box of side 6. If another island $I_q$ had $\operatorname{dist}(I_u, I_q) \le R$, a pair realising this distance would consist of a bad block of $W^{+5}$ and a bad block at distance between $2s+9 \ge 6$ and $R$. This proves the first two assertions of (ii) on $\mathscr{G}$.

*Step 4: the island sums on $\mathscr{G}$.* For $y \in B$ put

$$F(y) = \sum_{y'} |I(y')| e^{-\kappa \operatorname{dist}(y,y')},$$

the sum over $y' \in B \setminus I(y)$ with $\operatorname{dist}(y,y') > R$, and let $c_2(\beta_W)$ be $c_2$ with $\beta_W$ in place of $\beta^*$. On $\mathscr{G}$, let $I_u$ meet $W$ at $y$. For an island $I_q \ne I_u$ let $y_q \in I_q$ realise $\operatorname{dist}(I_u, I_q)$; then $\operatorname{dist}(y, y_q) \le \operatorname{dist}(I_u, I_q) + 5$, $\operatorname{dist}(y, y_q) > R$ by Step 3, and $w_{uq} - 2s - 8 = \operatorname{dist}(I_u, I_q) - 2s - 9 \ge \operatorname{dist}(y, y_q) - 2s - 14$. The box of side $2s+12$ around $I_u$ contains every block at distance at most $s+3$ from it, so $\bar n_u \le 45(2s+12)^4$, and $\bar n_q \le 45(2s+7)^4|I_q|$ as in the proof of Proposition 217(iii). With $\sqrt{|I_q|} \le |I_q|$ and distinct islands giving distinct $y_q$,

$$\bar T_u = \frac{1920}{\beta_W}\bar\Lambda_2^2 r^2 \sqrt{\bar n_u} \sum_q \sqrt{\bar n_q}\, e^{-\kappa(w_{uq} - 2s - 8)} \le c_2(\beta_W) F(y).$$

*Step 5: probabilities.* By Step 1 with two single blocks, at most $11^4|W|$ blocks of $W^{+5}$ and at most $(2R+1)^4$ blocks within distance $R$ of each,

$$\mu_L(\mathscr{G}^c) \le 11^4 (2R+1)^4 |W| (Mp)^2.$$

For $y \in W$, $\mathbb{E}[\mathbf{1}_{y\in B} F(y)]$ is a sum over $y'$ with $\operatorname{dist}(y,y') > R$ of $e^{-\kappa\operatorname{dist}(y,y')}$ times a sum over the connected sets $C' \ni y'$ with $y \notin C'$ of $|C'|\mu_L\{y \in B, I(y') = C'\}$. On that event $y$ and $C'$ lie in different islands, so $\operatorname{dist}(y, C') \ge 2s+9 \ge 6$, and Step 1 with $S_1 = \{y\}$, $S_2 = C'$ bounds the probability by $MpA^{\lceil |C'|/H\rceil}$; by (29) the sum over $C'$ is at most $MpG_0$. The sum over $y'$ is at most $\sum_z e^{-\kappa\|z\|_\infty}$ over $z \in \mathbb{Z}^4$ with $\|z\|_\infty > R$, by the preimages of the proof of Proposition 217(ii), which is at most $\sigma(\kappa)$ and, by $e^{-\kappa\|z\|_\infty} \le e^{-\kappa R/2} e^{-\kappa\|z\|_\infty/2}$ there, at most $e^{-\kappa R/2}\sigma(\kappa/2)$. So $\mathbb{E}[\mathbf{1}_{y\in B} F(y)] \le MpG_0\tau$, and by Step 4, Markov's inequality and a union bound over $W$,

$$\begin{aligned}&\mu_L\big(\mathscr{G} \cap \{\bar T_u > \sigma \text{ for some } I_u \text{ meeting } W\}\big) \\ &\qquad\qquad \le \frac{c_2(\beta_W)}{\sigma}|W| MpG_0\tau.\end{aligned}$$

*Step 6: the coupling.* By Theorem 147, $p \le e^{f(\beta_W)}$ with $f$ of Theorem 199, which decreases on $[96/\rho^2, \infty)$, so $p \le e^{f(\beta^*)} = q^*$. Since $\bar\Lambda_2 \ge 99.0001\,\beta_W$, the logarithmic derivative of $c_2(\beta)e^{f(\beta)}$ is at most $2/\beta - 1/\beta + 6/\beta - \rho^2/16$, which is negative for $\beta \ge \beta^*$ because $\beta^*\rho^2/16 = |\log q^*| + 6(\log\beta^* + C_0) > 7$. Hence $c_2(\beta_W)p \le c_2 q^*$, and Step 5 gives the bound $\varepsilon$ on the complement of the event of (ii).

*Step 7: the consequences.* Outside that event, $\Delta_{ab} \le \bar\Delta_{ab} \le \bar T_a \le \sigma$, since $\bar\Delta_{ab}$ is a term of $\bar T_a$ with a factor at least one, and $S_a, S_b \le \sigma$, which gives the first form of $\Delta_*$ in Theorem 215. By $\tanh t \le t$ and $\ell(q,q') \le w_{qq'} - 2s - 8$, the sum of Corollary 216 at $q$ is at most $\frac14 \bar T_q \le \frac14 \sigma$, so $\theta \le \sigma/4 < 1$, and the corollary with $T_a, T_b \le \sigma$ gives the last display.

*The numbers* are enclosed in interval arithmetic in the archive: the roots by $f(\beta_{\mathrm{lo}}) - \log q^* > 0 > f(\beta_{\mathrm{hi}}) - \log q^*$ to relative width $10^{-20}$, and every power of ten through its decimal logarithm, with $c_2$ evaluated at $\beta_{\mathrm{hi}}$. □

The bound is a union bound over the window and is linear in $|W|$; for $|W| \le 10^{28000}$ it is at most $10^{-166}$. Its content lies in the windows where Theorem 199 does not exclude islands: the probability that $W$ contains a bad block at all is at most $|W|A \le 10^{-14098.8}|W|$, so for $10^{14099} \le |W| \le 10^{28000}$ that theorem does not exclude bad blocks in $W$, and Theorem 218 shows that, outside probability $10^{-166}$, every island they form is a cluster of diameter at most 5, at distance more than $2 \cdot 10^8$ from every other island, and meets the conditions of Theorem 215 and Corollary 216 with $\sigma = 10^{-1000}$. No coupling beyond the threshold $\beta^*$ of the joining distance $2s+8$ and no factor $\Gamma > 1$ is needed: the weighted volume $\sigma(2\alpha_G) = 6.1 \cdot 10^{15}$ of the decay length is beaten by $G_0 \le 10^{-397.6}$. The bound is void for windows of more than $10^{28167}$ blocks, and it is a statement about the labels under $\mu_L$, not about the law of the labels near $I_u$ given that $I_u$ is an island; the covariance of the conditional probabilities of two labels with the others averaged, the second open point above, is not touched.

### M. The one-step error in gauge-invariant form

Proposition 195 requires the one-step error to be a relative bound by a gauge-invariant form. This subsection constructs such a bound at the trivial background on a box of blocks, from an absolute bound on the gauge complement taken as a hypothesis. The error of the covariant Hessian of a gauge-invariant induced action vanishes on the coarse gauge directions and is at most a multiple $\varepsilon'$ of the coarse Wilson form. On a box of side $n$, $\varepsilon'$ is of order $n^2$ times the absolute error, and this order cannot be improved.

*Setting.* On the coarse lattice of side $L$ let $W = \{y : a_\nu \le y_\nu \le a_\nu + n - 1\}$ be a box of $n^4$ blocks, $n+2 \le L$, $K = \{(y,\mu) : y \in W\}$ and $W^\circ = \{y \in W : y - e_\mu \in W \text{ for every } \mu\}$. As in Theorem 131, the links off $A_W$ are fixed and the averages of the pairs that enter $W$ from outside are integrated. For $w \in \mathfrak{g}^K$ put $|w|^2 = \sum_k \|w_k\|^2$, let $Q_1(w)$ be the coarse plaquette form of $w$ extended by zero off $K$, and put

$$S_W(w) = \inf\big\{ Q(X) : X \in \mathfrak{g}^{A_W},\ V_k(X) = w_k \ (k \in K) \big\}.$$

At $U = I$, $2\lambda Q_1$ is the Hessian at $I$ of the coarse Wilson action $\lambda Q_1$ with the links off $K$ fixed. $2\lambda S_W$ is the Hessian of the Gaussian block-spin step, the minimum of the Hessian $2\lambda Q$ of the fine action at given averages. The *coarse gauge group* $\Gamma_W = G^{W^\circ}$ acts on $G^K$ by $(h \cdot V)_{y,\mu} = h_y V_{y,\mu} h^{-1}_{y+e_\mu}$, with $h_y = I$ for $y \notin W^\circ$. This is the action on the cut-off averages of the fine gauge transformations that are constant on blocks and equal to $I$ off the blocks of $W^\circ$. These transformations preserve the tree gauge by Lemma 93(ii), and they fix the links off $A_W$: a link with base in a block $y' \notin W$ ends in $y'$ or $y' + e_\mu$, and neither lies in $W^\circ$. They also preserve the Wilson action and Haar measure, and the cut-off average is covariant (Theorem 117). For $V_0 \in G^K$ and $\eta \in \mathfrak{g}^{W^\circ}$, extended by zero, put

$$(d_{V_0}\eta)_{y,\mu} = \mathrm{Ad}(V_{0,y,\mu}^{-1})\eta_y - \eta_{y+e_\mu},$$

let $\mathscr{O}_{V_0} = d_{V_0}(\mathfrak{g}^{W^\circ})$ be the gauge directions at $V_0$ and $\Pi_{V_0}$ the orthogonal projection of $\mathfrak{g}^K$ onto $\mathscr{O}_{V_0}^\perp$. At $V_0 = I$ write $d$, $\mathscr{O}$ and $\Pi$.

**Lemma 219** (the window off the gauge directions)**.** *Let $W$ be the box above.*

(i) *The map $d$ is injective on $\mathfrak{g}^{W^\circ}$, and $\mathscr{O} = \ker Q_1 = \ker S_W$.*

(ii) *For every $w \in \mathfrak{g}^K$, $|\Pi w|^2 \le C_n Q_1(w)$ with*

$$C_n = 1 + \frac{1}{4\sin^2(\pi/2n)} \le 1 + \frac{n^2}{4}.$$

(iii) *$c_* Q_1 \le S_W \le 4Q_1$ on $\mathfrak{g}^K$, with $c_* = (1+\sqrt{21/8})^{-2} > 0.1456$.*

(iv) *There is $w \in \mathscr{O}^\perp \setminus \{0\}$ with $|w|^2 = C_n^- Q_1(w)$, where $C_n^- = (12\sin^2(\pi/(2n+2)))^{-1} \ge (n+1)^2/(3\pi^2)$.*

(v) *If $W$ is the whole coarse torus, the fields that are constant in each direction lie in $\mathscr{O}^\perp \cap \ker Q_1 \cap \ker S_W$, so no bound of the form* (ii) *holds.*

*Proof.* Let $\Lambda = \prod_\nu [a_\nu, a_\nu + n]$, with boundary $\partial\Lambda$. The interior sites of $\Lambda$ form $W^\circ$. Call $(y,\mu) \in K$ *inner* if $y_\nu > a_\nu$ for every $\nu \ne \mu$, and a *face link* otherwise. The inner links are the links of $\Lambda$ not contained in $\partial\Lambda$; call a plaquette of $\Lambda$ not contained in $\partial\Lambda$ inner. The links of $\partial\Lambda$ that are not face links have base outside $W$, so they are fixed, and $w$ vanishes on them.

(ii) The cochain complex of $\Lambda$ relative to $\partial\Lambda$, with values in $\mathfrak{g}$, has the interior sites, inner links and inner plaquettes as cells of degrees $0,1,2$. On these, $d_0 = d$, and $d_1$ is the curvature on the inner plaquettes of a field on the inner links extended by zero. This complex is the fourfold tensor product of the complex of $[0,n]$ relative to its endpoints. In that complex $C^0 = \mathbb{R}^{n-1}$, $C^1 = \mathbb{R}^n$ and $d$ is the difference operator. With the product inner products, the Laplacian $\Delta = dd^* + d^*d$ of the product is the sum of the Laplacians of the factors. On $C^0$, $d^*d$ is the Dirichlet Laplacian, with eigenvalues $4\sin^2(\pi j/2n)$ for $1 \le j \le n-1$. On $C^1$, $dd^*$ has the same nonzero eigenvalues and one zero eigenvalue. A 1-cochain of $\Lambda$ is a sum of products with exactly one factor of degree one, so $\Delta_1 \ge \gamma_n = 12\sin^2(\pi/2n)$. On $\ker d_0^* = \mathscr{O}^\perp$ this gives $|d_1 u|^2 \ge \gamma_n |u|^2$.

Write $w = u + f$, with $u$ on the inner links and $f$ on the face links. Since $\mathscr{O}$ lies on the inner links, $\Pi w = \Pi u + f$ and $|\Pi w|^2 = |\Pi u|^2 + |f|^2$. Let $f_k \ne 0$ at a face link $k = (y,\mu)$, and let $\nu \ne \mu$ be the least direction with $y_\nu = a_\nu$. The plaquette at $y - e_\nu$ in the plane $\{\mu,\nu\}$ has $k$ as its only link in $K$, since its other three links have base outside $W$. So its curvature

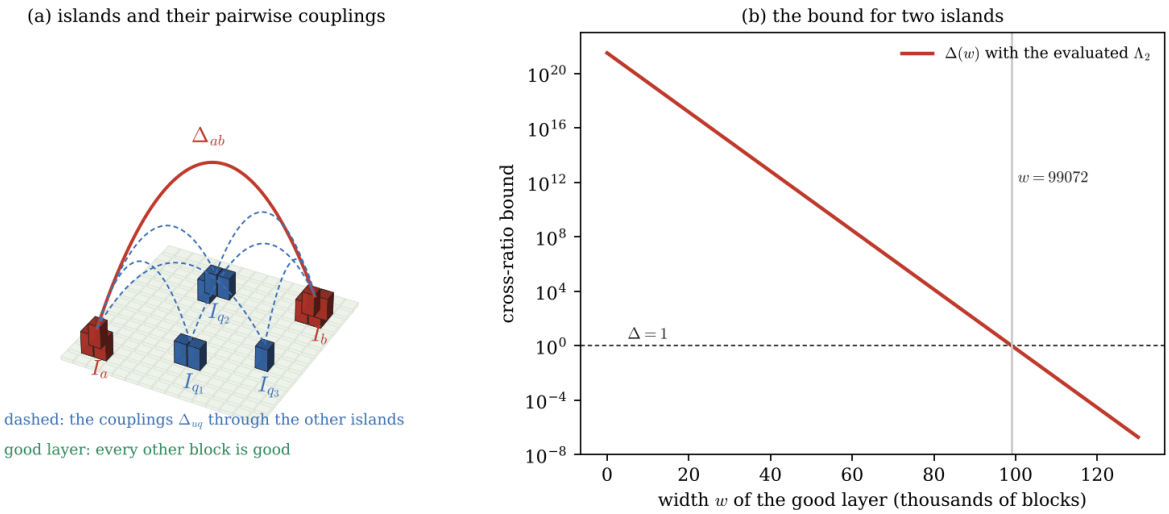



FIG. 24. (a) Islands of bad blocks (red: the two islands $I_a$, $I_b$ compared; blue: other islands $I_q$) in a window whose remaining blocks are good. The solid arc is the two-island bound $\Delta_{ab}$ of Corollary 210; the dashed arcs are the bounds $\Delta_{uq}$ through which the other islands enter $\Delta_*$ in Theorem 215. (b) The bound $\Delta$ for two islands against the width $w$ of the good layer between them, with $\Lambda_2$ from Lemma 211 at $\beta_W = 1.13\cdot 10^{37}$; it falls below one at $w = 99072$.

is $\pm f_k$, and distinct face links give distinct such plaquettes, none of them inner. Consider next an inner plaquette $P$ in the plane $\{\mu,\nu\}$ at $z$. It has $z_\rho > a_\rho$ for $\rho \notin \{\mu,\nu\}$, so its face links are among $(z,\mu)$, if $z_\nu = a_\nu$, and $(z,\nu)$, if $z_\mu = a_\mu$. Thus $P$ contains at most two face links, and a face link lies in at most one inner plaquette. Hence $(d_1 u)_P^2 \le 3\big((dw)_P^2 + \sum_{k\in P} f_k^2\big)$, where the sum runs over the face links of $P$. Since $d_1\Pi u = d_1 u$,

$$|\Pi w|^2 \le \gamma_n^{-1}|d_1 u|^2 + |f|^2 \le \frac{3}{\gamma_n}\Big(\sum_{P\ \text{inner}} (dw)_P^2 + |f|^2\Big) + |f|^2 \le C_n Q_1(w),$$

because the inner plaquettes and the plaquettes of the face links are distinct terms of $Q_1(w)$. Finally $\sin x \ge 2x/\pi$ on $[0,\pi/2]$ gives $C_n \le 1 + n^2/4$.

(iii) Let $X \in \mathfrak{g}^{A_W}$ with $V(X) = w$ on $K$. Extended by zero, $X$ vanishes on the tree links, and its averages are $w$ on $K$, zero on the coarse links with both blocks outside $W$, and $v'_{y-e_\mu,\mu} = S_y^\mu/16$ on a pair entering $y \in W$ from $y - e_\mu \notin W$, by Lemma 93(iii). By Proposition 99(i), $Q_1(w+v') \le Q(X)$. The sum $S_y^\mu$ has at most seven active links, and every link inside a block enters one such sum. So $\sum |v'|^2 \le \frac{7}{256} I \le \frac{7}{32} Q(X)$, by the bound $I \le 8Q_{\text{in}}$ in the proof of Theorem 94, which uses only the tree links. A coarse plaquette contains at most two entering pairs, since $(z,\mu)$ and $(z+e_\mu,\nu)$ cannot both enter, nor can $(z,\nu)$ and $(z+e_\nu,\mu)$. A coarse link lies in six plaquettes, so $Q_1(v') \le 12\sum |v'|^2 \le \frac{21}{8} Q(X)$. By the triangle inequality for $Q_1^{1/2}$, $Q_1(w)^{1/2} \le (1+\sqrt{21/8})Q(X)^{1/2}$, and the infimum over $X$ gives $S_W \ge c_* Q_1$. For the upper bound, the field $\varphi^*(w)$ of Proposition 99(ii), with $w$ extended by zero, lies in $\mathfrak{g}^{A_W}$, has averages $w$ on $K$ and $Q(\varphi^*(w)) = 4Q_1(w)$.

(i) If $d\eta = 0$, follow the links of direction 0 from $y \in W^\circ$ to the first site outside $W^\circ$, where $\eta$ vanishes; so $\eta_y = 0$. Extended by zero, $d\eta$ is exact on the whole lattice, so $Q_1(d\eta) = 0$, and $\ker Q_1 \subset \mathscr{O}$ by (ii). The fine field $\varphi = d(\eta \circ b)$, with $b(x)$ the block of $x$, vanishes on the tree links and has $Q(\varphi) = 0$. It has $V(\varphi) = d\eta$ by Lemma 93(ii). It lies in $\mathfrak{g}^{A_W}$, since a link on which it is nonzero meets a block of $W^\circ$ and so has base in $W$. Hence $S_W(d\eta) = 0$, and $\ker S_W \subset \ker Q_1$ by (iii).

(iv) Let $f$ be the product of sines on $\{a_\nu,\dots,a_\nu+n-1\}^3$, $\nu = 1,2,3$, whose eigenvalue for the Dirichlet Laplacian of the cube is $12\sin^2(\pi/(2n+2))$. Let $w_{y,0} = f(y_1,y_2,y_3)Z$ for a unit $Z \in \mathfrak{g}$, and $w = 0$ in the other directions. The sum over $y_0$ telescopes, so $\langle w, d\eta\rangle = 0$ for every $\eta$. The curvature of a plaquette in the plane $\{0,\nu\}$ is a difference of $f$ along $e_\nu$, the plaquettes in the other planes vanish, and for each of the $n$ values of $y_0$ the sum is the Dirichlet form of $f$. Hence $Q_1(w)/|w|^2 = 12\sin^2(\pi/(2n+2)) \le 3\pi^2/(n+1)^2$.

(v) Then $W^\circ = W$ and every coarse link is in $K$. A constant field has $dw = 0$, and $\langle w, d\eta\rangle = \sum_y \sum_\mu \langle w_\mu, \eta_y - \eta_{y+e_\mu}\rangle = 0$. The field $\varphi^*(w)$ of Proposition 99(ii) has $Q = 4Q_1(w) = 0$, so $S_W(w) = 0$. □

The gauge directions at a background $V_0$ near $I$ are close to those at $I$, and the flat form keeps its gap off them.

**Lemma 220** (the window at a curved background)**.** *Let $V_0 \in G^K$ with $\|\mathrm{Ad}(V_{0,k}^{-1}) - 1\| \le \theta$ for every $k \in K$, and put $s_n = 2\sin(\pi/2n)$. If $\theta < s_n$, then every $w \in \mathscr{O}_{V_0}^\perp$ satisfies*

$$|w - \Pi w| \le \frac{\theta}{s_n}|w|, \qquad |w|^2 \le \Big(1 - \frac{\theta}{s_n}\Big)^{-2} C_n Q_1(w).$$

*Proof.* Write $d_{V_0} = d + R$ with $(R\eta)_{y,\mu} = (\mathrm{Ad}(V_{0,y,\mu}^{-1}) - 1)\eta_y$. Every site of $W^\circ$ is the base of the four links $(y,\mu)$, all in $K$, so $|R\eta|^2 \le 4\theta^2|\eta|^2$ and $\|R^*\| \le 2\theta$. The form $|d\eta|^2$ is the Dirichlet form of the box $\Lambda$ of the proof of Lemma 219 relative to

its boundary, the sum of four one-dimensional Dirichlet Laplacians, so $d^*d \geq 4s_n^2$ and $\|d(d^*d)^{-1}\| = \|(d^*d)^{-1/2}\| \leq (2s_n)^{-1}$. For $w \perp \mathscr{O}_{V_0}$ we have $d_{V_0}^* w = 0$, that is $d^*w = -R^*w$, and $w - \Pi w = d(d^*d)^{-1}d^*w = -d(d^*d)^{-1}R^*w$. Hence $|w - \Pi w| \leq \theta|w|/s_n$, so $|w| \leq (1-\theta/s_n)^{-1}|\Pi w|$, and Lemma 219(ii) gives the second bound. □

For a smooth function $f$ near $V_0 \in G^K$ write $\operatorname{Hess} f(0)$ and $\nabla_k f(0)$ for the Hessian and the gradient in the coordinates $\omega \mapsto f(V_0 e^\omega)$, $e^\omega = (e^{\omega_k})_k$.

**Lemma 221** (invariant functions and the slice)**.** *Let $f$ be smooth and $\Gamma_W$-invariant near $V_0 \in G^K$.*

(i) *For $w \in \mathfrak{g}^K$ and $\eta \in \mathfrak{g}^{W^\circ}$, with $a_k = \operatorname{Ad}(V_{0,k}^{-1})\eta_y$ and $b_k = \eta_{y+e_\mu}$ at $k = (y,\mu)$,*

$$\operatorname{Hess} f(0)[w, d_{V_0}\eta] = \frac{1}{2}\sum_k \langle \nabla_k f(0), [w_k, a_k + b_k]\rangle.$$

(ii) *$d_{V_0}$ is injective, and $\Upsilon(s,\eta) = e^\eta \cdot (V_0 e^s)$, on $\mathscr{O}_{V_0}^\perp \times \mathfrak{g}^{W^\circ}$, is a diffeomorphism from a neighbourhood of $0$ onto a neighbourhood of $V_0$, with $f(\Upsilon(s,\eta)) = f(V_0 e^s)$. The* covariant Hessian

$$\mathscr{H} f(w,w') = \operatorname{Hess}(f\circ\Upsilon)(0)\big[D\Upsilon(0)^{-1}w, D\Upsilon(0)^{-1}w'\big]$$

*equals* $\operatorname{Hess} f(0)[\Pi_{V_0} w, \Pi_{V_0} w']$. *It vanishes if $w$ or $w'$ lies in $\mathscr{O}_{V_0}$, and it equals* $\operatorname{Hess} f(0)$ *if $\nabla f(0) = 0$.*

*Proof.* (i) For the action of $e^{t\eta}$,

$$(e^{t\eta}\cdot V_0 e^\omega)_k = V_{0,k} e^{\omega_k} e^{t\operatorname{Ad}(e^{-\omega_k})a_k} e^{-tb_k},$$

and $\frac{d}{dt}\log(e^\omega e^{t\xi})|_{t=0} = \Phi(\operatorname{ad}_\omega)\xi$ with $\Phi(z) = z/(1-e^{-z})$. So the orbit through $V_0 e^\omega$ has the tangent

$$\zeta(\omega)_k = \Phi(\operatorname{ad}_{\omega_k})\big(\operatorname{Ad}(e^{-\omega_k})a_k - b_k\big),$$

with $\zeta(0) = d_{V_0}\eta$ and $D\zeta(0)w = -\frac{1}{2}([w_k, a_k+b_k])_k$. Invariance gives $Df(\omega)[\zeta(\omega)] = 0$ near 0. Differentiating along $w$ at 0 gives $\operatorname{Hess} f(0)[w,\zeta(0)] = -Df(0)[D\zeta(0)w]$.

(ii) Injectivity follows as in Lemma 219(i), with $\eta_y = \operatorname{Ad}(V_{0,k})\eta_{y+e_\mu}$ along the links $k$ of direction 0. $D\Upsilon(0)(\sigma,\eta) = \sigma + d_{V_0}\eta$ is an isomorphism, and the inverse function theorem applies. Invariance gives $f\circ\Upsilon(s,\eta) = f(V_0 e^s)$, whose Hessian at 0 is $\operatorname{Hess} f(0)$ on the $s$-variables and zero in $\eta$. Since $D\Upsilon(0)^{-1}w = (\Pi_{V_0}w, \eta)$ with $d_{V_0}\eta = w - \Pi_{V_0}w$, the formula follows. If $\nabla f(0) = 0$, (i) shows that $\operatorname{Hess} f(0)$ vanishes on $\mathfrak{g}^K \times \mathscr{O}_{V_0}$, so $\operatorname{Hess} f(0)[w,w'] = \operatorname{Hess} f(0)[\Pi_{V_0}w, \Pi_{V_0}w']$. □

At a point that is not critical, $\operatorname{Hess} f(0)$ does not vanish on the gauge directions. The Hessian in exponential coordinates of an invariant function, and with it any absolute bound on it, is therefore not gauge-invariant; $\mathscr{H} f$ is the gauge-invariant object. Let $j(\omega) = \det\big((1-e^{-\operatorname{ad}_\omega})/\operatorname{ad}_\omega\big)$ be the density of Haar measure in exponential coordinates. Since $\operatorname{ad}_\omega$ is skew, $\log j(\omega) = \frac{1}{24}\operatorname{Tr}(\operatorname{ad}_\omega^2) + O(\|\omega\|^4)$. Put

$$\delta_G = \frac{1}{12}\sup_{\|x\|=1}|\operatorname{Tr}(\operatorname{ad}_x^2)| \leq \frac{\dim G}{3},$$

which is $\frac{1}{3}$ for $\mathrm{SU}(2)$, where $\|\operatorname{ad}_x\|_{HS} = 2\|x\|$.

**Proposition 222** (relative one-step bound)**.** *Let $U = I$, so that $V_0 = I$. Let $A$ be smooth near $0 \in \mathfrak{g}^K$ and suppose that, for some $\varepsilon \geq 0$,*

$$\big|\operatorname{Hess} A(0)[\sigma,\sigma] - 2\lambda S_W(\sigma)\big| \leq \lambda\varepsilon|\sigma|^2 \quad \text{for } \sigma \in \mathscr{O}^\perp. \tag{H}$$

*Let $A^{\mathrm{gi}}$ be the $\Gamma_W$-invariant function with $A^{\mathrm{gi}}(\Upsilon(s,\eta)) = A(s) + \sum_k \log j(s_k)$, and let $E = \mathscr{H}A^{\mathrm{gi}} - 2\lambda S_W$ as bilinear forms. Then:*

(i) *$E(w,g) = 0$ for every $w \in \mathfrak{g}^K$ and $g \in \mathscr{O}$;*

(ii) *for every $w \in \mathfrak{g}^K$,*

$$|E(w,w)| \leq (\lambda\varepsilon + \delta_G)|\Pi w|^2 \leq (\lambda\varepsilon+\delta_G)C_n Q_1(w)$$
$$\leq \varepsilon' \cdot 2\lambda S_W(w), \qquad \varepsilon' = \Big(\varepsilon + \frac{\delta_G}{\lambda}\Big)\frac{C_n}{2c_*},$$

*so that $(1-\varepsilon')2\lambda S_W \leq \mathscr{H}A^{\mathrm{gi}} \leq (1+\varepsilon')2\lambda S_W$;*

(iii) *the factor $C_n$ is necessary up to a constant. The $\Gamma_W$-invariant $F$ with $F(\Upsilon(s,\eta)) = \lambda S_W(s) + \frac{1}{2}\lambda\varepsilon|s|^2$ has $\operatorname{Hess} F(0) = 2\lambda S_W + \lambda\varepsilon|\cdot|^2$ on $\mathscr{O}^\perp$ and $\mathscr{H}F - 2\lambda S_W = \lambda\varepsilon|\Pi\cdot|^2$. At the field $w$ of Lemma 219*(iv) *this is at least $\frac{1}{8}\varepsilon C_n^- \cdot 2\lambda S_W(w)$.*

*Proof.* (i) $\mathscr{H}A^{\mathrm{gi}}(w,g) = 0$ by Lemma 221(ii). $S_W$ is positive semidefinite with $g \in \ker S_W$ by Lemma 219(i), so $S_W(w,g) = 0$ by Cauchy–Schwarz. (ii) By Lemma 221(ii), $\mathscr{H}A^{\mathrm{gi}}(w,w)$ is the Hessian at 0 of $s \mapsto A(s) + \sum_k \log j(s_k)$ at $\Pi w$, that is, $\operatorname{Hess} A(0)[\Pi w, \Pi w] + \frac{1}{12}\sum_k \operatorname{Tr}(\operatorname{ad}^2_{(\Pi w)_k})$. Also $S_W(w) = S_W(\Pi w)$, by (i). Hypothesis (H) at $\sigma = \Pi w$ gives the first inequality. Lemma 219(ii) gives the second, and Lemma 219(iii), $Q_1 \leq S_W/c_*$, gives the third. (iii) The Hessian of $s \mapsto \lambda S_W(s) + \frac{1}{2}\lambda\varepsilon|s|^2$ is $2\lambda S_W + \lambda\varepsilon|\cdot|^2$. That $w$ lies in $\mathscr{O}^\perp$ and has $|w|^2 = C_n^- Q_1(w) \geq \frac{1}{4}C_n^- S_W(w)$, by Lemma 219(iii). □

*The induced action.* Let $e^{-A(\omega)}$ be the density in $\omega$ of the cut-off averages of a window under the Wilson measure restricted by a cut-off. The small-field restriction is an example: it is a soft box in the coordinates of the chart of Theorem 131, which is not $\Gamma_W$-invariant. Then $e^{-A^{\mathrm{gi}}}$ is the density with respect to Haar measure on $G^K$ of the averages under the measure whose cut-off on the fibre over $\Upsilon(s,\eta)$ is the transport by the block-constant gauge transformation $e^\eta$ of the cut-off on the fibre over $e^s$. The transport maps fibre to fibre (Theorem 117) and preserves the Wilson weight and Haar measure, so the fibre integrals agree. On the slice $\mathscr{O}^\perp$ the two cut-offs coincide. A cut-off defined through gauge-invariant quantities gives $A^{\mathrm{gi}} = A + \sum_k \log j$ everywhere.

*Remark* 223 (inputs and what is missing). Proposition 222 rests on one input and leaves the following open.

(a) *Input.* The bound (H) on the Hessian of the induced action at one point is an absolute bound, of the kind by which Theorems 125 and 127 control the non-Gaussian terms, and only its restriction to $\mathscr{O}^\perp$ is used. It is not proved in this paper. For $\mathrm{SU}(2)$, $\lambda = 10^{100}$ and $\varepsilon = 1.63\cdot 10^{-9}$,

interval arithmetic gives $\varepsilon' \le 8.40\cdot 10^{-9}$ at $n=2$, $1.52\cdot 10^{-8}$ at $n=4$, $1.52\cdot 10^{-7}$ at $n=16$ and $5.95\cdot 10^{-4}$ at $n=1024$, and $\varepsilon' \le 0.1$ for every $n \le 8455$.

(b) *The cut-off.* The construction replaces a non-invariant cut-off off the slice by its gauge transport. Bałaban defines the small-field regions by gauge-invariant conditions on plaquettes [7, 8, 32]. For such a cut-off the transport is not needed.

(c) *Points that are not critical.* $\mathscr{H}$ is taken at one point. An iteration needs the bound as a function on a neighbourhood. There the gradient terms of Lemma 221(i) and the third derivatives enter, and Proposition 222 says nothing about them. Bałaban expands around the minimiser of the action with given averages, the background field, at which the first-order terms vanish [6, 32].

(d) *Curved backgrounds.* Lemma 221 holds at every $V_0$. The Gaussian reference $S_W^U(w) = \inf\{Q_U(X) : BX = w\}$ need not vanish on $\mathscr{O}_{V_0}$, where $B$ is the derivative of the cut-off averages at $U$. The fine gauge variation $X_\eta$ of $e^\eta$ has $BX_\eta = d_{V_0}\eta$ and $D_p(X_\eta) = (1-\mathrm{Ad}\,U_p)\eta_{y(p)}$, with $y(p)$ the block of the base of $p$. Hence $S_W^U(d_{V_0}\eta) \le 384\kappa^2|\eta|^2$ when $\delta(U_p,I) \le \kappa$, where 96 plaquettes have their base in a block. The reference for an invariant action is $S_W^U \circ \Pi_{V_0}$. By Lemma 220 its gap on $\mathscr{O}_{V_0}^\perp$ follows from a comparison $S_W^U \ge c_1 Q_1 - c_2\kappa|\cdot|^2$ there, with the gap $c_1(1-\theta/s_n)^2 C_n^{-1} - c_2\kappa$; that comparison is not proved here.

(e) *The action.* The induced action is not a plaquette action. Its Hessian is compared with $2\lambda S_W$, not with $2\lambda Q_1$, and Theorem 131 does not apply to it as it stands at the next scale. The large fields, which Bałaban treats separately [8], are removed by the cut-off and not estimated.

*Remark* 224 (relative errors over scales). The proof of Proposition 194 uses only the order of forms, so it holds for positive semidefinite forms. A bound $(1\pm\varepsilon')\,2\lambda S_W$ on the covariant Hessian therefore passes to the block infimum of the next step with factor one. Proposition 195 does not apply to it. That proposition concerns an error bounded by the Euclidean form, which does not vanish on $\mathscr{O}$. The block infimum turns that error into a term at least $4\lambda\varepsilon(1+2724\kappa)^{-1}|w|^2$, also where $S_W$ vanishes.

The two statements agree. A window of side $n$ in blocks of one scale has side $n/2$ in the blocks of the next, and $C_n/C_{n/2} \to 4$. Converting an absolute error at one scale costs the factor $C_n$. Carrying it one step as an absolute error multiplies it by at least four, after which converting costs $C_{n/2}$. Both routes have the same order, and the factor four of Proposition 195 is the loss of the gap of the window.

With windows of side $n_k = n_1 2^{1-k}$ at the scales $k=1,\dots,K$, $\sum_k C_{n_k} \le K + n_1^2/3$, so

$$\sum_k \varepsilon_k' \le \sup_k\Big(\varepsilon_k + \frac{\delta_G}{\lambda_k}\Big)\frac{K+n_1^2/3}{2c_*}.$$

The part $n_1^2/3$ is the conversion cost. It is the absolute error carried to the last scale, $4^{K-1} \approx n_1^2$, so it is finite for a fixed window but not uniform in the volume. Removing it needs a gauge-invariant localisation of the one-step bound, which is part of Bałaban's analysis [8]. The part $K$ is marginal. The Haar term $\delta_G/\lambda_k$ is of order $g_k^2$, and $\sum_k g_k^2$ diverges logarithmically along the one-loop coupling by Remark 109(c). The order $g^2$ of the one-step error must therefore be absorbed into the running coupling and the normalisation of the field. Only a remainder of order $g^4$ can be carried as $\varepsilon_k'$. For $n_1 = 2^{12}$, $K=13$, $\varepsilon_k = 1.63\cdot 10^{-9}$ and $\lambda_k = 10^{100}$, the values of $C_{n_k}$ give $\sum_k \varepsilon_k' \le 0.0127$, against $4^{12}\varepsilon_1 \le 0.028$ for the absolute error of the first scale carried to the last. No effective action is constructed here beyond the first scale, and item (iv) of Sec. XIII P remains open.

### N. Every compact simple group

The constructions of this section were written for SU(2). This subsection records which of them hold for a general group, and with which constants. Let $G$ be a compact connected group whose Lie algebra $\mathfrak{g}$ is simple, with a faithful unitary representation in which $G \subset \mathrm{U}(N)$, the metric $\langle X,Y\rangle = -\mathrm{Tr}(XY)$ on $\mathfrak{g}$, the least Ricci curvature $\kappa_G$ of Theorem 5, $d = \dim G$ and $K = \kappa_G/(d-1)$. The distance $\delta(g,h) = \|g-h\|$ is the Hilbert–Schmidt distance, $d(g,h)$ the geodesic distance of $G$, the Wilson weight is $\exp(-\beta_W \sum_p s_p)$ with

$$s_p = 1 - \tfrac{1}{N}\mathrm{Re}\,\mathrm{Tr}\,U_p = \frac{\delta(U_p,I)^2}{2N}, \qquad \beta_W = N^2\beta,$$

which is the normalisation of this section for $N=2$, and $\rho_G$ is the radius of Sec. XII. Put $I_d(x) = \int_0^x \sin^{d-1} u\,du$.

**Lemma 225** (spectral constants)**.** *Let $X,Y \in \mathfrak{g}$ and $U \in G$.*

(i) *The operator $\mathrm{ad}X$ on $\mathfrak{g}$ is skew, with spectrum in $i[-\sqrt{2}\|X\|, \sqrt{2}\|X\|]$.*

(ii) $-\mathrm{Tr}\big((\mathrm{ad}X)^2\big) = 4\kappa_G\|X\|^2$*; in particular $\delta_G = \kappa_G/3$ in Proposition 222.*

(iii) *With $s(V) = 1 - \frac{1}{N}\mathrm{Re}\,\mathrm{Tr}\,V$, the derivative of $s(Ue^{tX})$ at $t=0$ has modulus at most $\|X\|/\sqrt{N}$; on $\mathrm{U}(N)$, $\delta(g,h) \le 2\sqrt{N}$.*

(iv) *Let $j$ be the density of Haar measure in exponential coordinates,*

$$j(X) = \det\big((1-e^{-\mathrm{ad}X})/\mathrm{ad}X\big).$$

*If $\|X\| \le 10^{-3}$, then*

$$|\partial_Y \log j(X)| \le \tfrac{\kappa_G}{3}(1+10^{-6})\|X\|\|Y\|.$$

*Proof.* (i) Invariance of the trace makes $\mathrm{ad}X$ skew. On $\mathfrak{u}(N)$, in a basis in which $X = \mathrm{diag}(i\lambda_1,\dots,i\lambda_N)$, $\mathrm{ad}X$ has eigenvalues $i(\lambda_j-\lambda_k)$, and $\mathfrak{g}$ is an invariant subspace; $(\lambda_j-\lambda_k)^2 \le 2(\lambda_j^2+\lambda_k^2) \le 2\|X\|^2$. (ii) $\mathrm{Ric}(X,X) = -\frac{1}{4}\mathrm{Tr}((\mathrm{ad}X)^2)$ is an

Ad-invariant symmetric form. Since $\mathfrak{g}\otimes\mathbb{C}$ is simple, the adjoint representation is irreducible over $\mathbb{C}$ and every invariant symmetric form is a multiple of $\langle\cdot,\cdot\rangle$, so $\mathrm{Ric}=\kappa_G\langle\cdot,\cdot\rangle$. Then $\delta_G=\frac{1}{12}\cdot 4\kappa_G$. (iii) The derivative is $-\frac{1}{N}\operatorname{Re}\operatorname{Tr}(UX)$. Since $\overline{\operatorname{Tr}(UX)}=-\operatorname{Tr}(U^*X)$, $\operatorname{Re}\operatorname{Tr}(UX)=\frac12\operatorname{Tr}((U-U^*)X)$, and $\|U-U^*\|^2=4\sum_j\sin^2\theta_j\le 4N$ for the eigenangles $\theta_j$ of $U$. The diameter bound is $\|g-h\|^2=2N-2\operatorname{Re}\operatorname{Tr}(g^*h)\le 4N$. (iv) $\operatorname{Tr}\operatorname{ad}X=0$, so $\log j(X)=\operatorname{Tr}\Phi(\operatorname{ad}X)$ with $\Phi(z)=\log(\sinh(z/2)/(z/2))$, and $\partial_Y\log j(X)=\operatorname{Tr}(\Phi'(\operatorname{ad}X)\operatorname{ad}Y)$. For real $y$, $\Phi'(iy)=i(\frac1y-\frac12\cot\frac y2)$ and $0\le\frac1y-\frac12\cot\frac y2\le\frac{y}{12}(1+\frac{y^2}{50})$ for $0\le y\le 1$; by (i) the spectral radius of $\operatorname{ad}X$ is at most $\sqrt2\cdot10^{-3}$, so $\|\Phi'(\operatorname{ad}X)\|_{HS}\le\frac{1}{12}(1+10^{-6})\|\operatorname{ad}X\|_{HS}$, and (ii) together with the Cauchy–Schwarz inequality gives the bound. □

For SU(2) the spectrum in (i) is $\{0,\pm i\sqrt2\|X\|\}$, and the estimates of Lemma 174, Proposition 179, Lemma 182 and Step 4 of Lemma 211 that use the spectrum of an operator $\operatorname{ad}W$ use only that it lies in $i[-\sqrt2\|W\|,\sqrt2\|W\|]$: the bounds $\|L_X\|\le\theta/\sin\theta$, $\|E_Y\|\le1$, $\|\varpi(\operatorname{ad}W)^{-1}\|\le m(\theta)$, the divided-difference bounds and $\kappa_0<0.8935$ hold for $G$ unchanged. The Hilbert–Schmidt norm $\|\operatorname{ad}W\|_{HS}=2\|W\|$ becomes $2\sqrt{\kappa_G}\|W\|$, a trace over $\mathfrak{g}$ is bounded by $d$ times the operator norm instead of 3 times it, and the pair $(\|X\|/\sqrt2,2\sqrt2)$ of (iii) becomes $(\|X\|/\sqrt N,2\sqrt N)$.

**Lemma 226** (the logarithm radius of a stabiliser)**.** *Let $t$ be a tensor in $(\mathbb{C}^N)^{\otimes a}\otimes(\mathbb{C}^{N*})^{\otimes b}$ with $m=a+b\ge1$, and let $G$ be the identity component of the stabiliser of $t$ in $\mathrm{U}(N)$. If $g\in G$ and $\delta(g,I)<2\sin\frac{\pi}{2m}$, the principal logarithm of $g$ lies in $\mathfrak{g}$.*

*Proof.* Let $e^{i\theta_j}$, $|\theta_j|\le\pi$, be the eigenvalues of $g$ and $X=\log g$. Since $4\sin^2(\theta_j/2)\le\delta(g,I)^2<4\sin^2\frac{\pi}{2m}$, every $|\theta_j|<\pi/m$. The induced action of $e^{sX}$ on the tensor space is $e^{sD}$, where the derivation $D$ induced by $X$ is skew-Hermitian with eigenvalues $i$ times sums of $m$ terms $\pm\theta_j$, so its spectrum lies in $i(-\pi,\pi)$. Decompose $t=\sum_\mu t_\mu$ over the eigenspaces of $D$. From $e^Dt=t$, $e^{i\mu}t_\mu=t_\mu$ with $|\mu|<\pi$, so $t_\mu=0$ for $\mu\ne0$ and $Dt=0$. The one-parameter group $e^{sX}$ then fixes $t$ and lies in its identity component $G$, so $X\in\mathfrak{g}$. □

$\mathrm{SO}(N)$ and $\mathrm{Sp}(n)$ are the identity components of the stabilisers of a symmetric and of a skew form, $m=2$ and $2\sin\frac{\pi}{4}>1$, so $\rho_G=1$. $G_2\subset\mathrm{U}(7)$, $F_4\subset\mathrm{U}(26)$, $E_6\subset\mathrm{U}(27)$ and $E_8\subset\mathrm{U}(248)$ are the identity components of the stabilisers of tensors of degree at most three: the three-form of the octonions, the quadratic and cubic invariants of the traceless exceptional Jordan algebra, its cubic determinant, and the Lie bracket [13]. Each $\rho_G<1$ is admissible for them, for instance $\rho_G=0.99$. $E_7\subset\mathrm{U}(56)$ is the identity component of the stabiliser of the symplectic and the quartic form of the Freudenthal triple system [13], $m=4$, so $\rho_G=0.76<2\sin\frac{\pi}{8}$ is admissible. For $\mathrm{SU}(N)$, $\rho_G=\min\{1,5/\sqrt N\}$.

**Lemma 227** (small balls)**.** *For $r>0$, the Haar measure of $\{g\in G:d(g,I)\le r\}$ is at least $I_d(\sqrt{K}r)/I_d(\pi)$, with equality for* SU(2)*. If $\sqrt{K}r\le\frac14$, it is at least $(\frac{95}{96})^{d-1}(\sqrt{K}r)^d/(d\,I_d(\pi))$.*

*Proof.* The Ricci curvature is at least $(d-1)K$, so the diameter is at most $\pi/\sqrt{K}$ by Myers' theorem, and the relative volume comparison of Bishop and Gromov [76] bounds the ratio of the volume of the ball of radius $r$ to that of the ball of radius $\pi/\sqrt{K}$, which is $G$, from below by the same ratio on the sphere of curvature $K$ and dimension $d$, $I_d(\sqrt{K}r)/I_d(\pi)$. For SU(2) the group is that sphere, of radius $\sqrt2$. The second bound follows from $\sin u\ge u(1-u^2/6)\ge\frac{95}{96}u$ on $[0,\frac14]$. □

Lemma 227 takes the place of the angle density $\frac{2}{\pi}\sin^2\theta$ used in Lemma 146. With it, put

$$C_0(G)=\frac12+\frac23\Big(\log\big(d\,I_d(\pi)\big)+\frac{d}{2}\log\frac{16}{KN}-(d-1)\log\frac{95}{96}\Big).$$

**Lemma 228.** *For $\beta_W\ge KN$, the normalising constant of the Wilson measure of $G$ on $\Lambda_L$ satisfies*

$$-\log Z_L\le 6L^4\Big(\frac{d}{3}\log\beta_W+C_0(G)\Big),$$

*and $\mathbb{E}S\le 6L^4(\frac{d}{3}\log\beta_W+C_0(G))/\beta_W$. If $KN\le16$, then $C_0(G)>0$.*

*Proof.* Put $r=\sqrt N/(4\sqrt{\beta_W})$, so that $\sqrt{K}r\le\frac14$. If every link is within geodesic distance $r$ of $I$, every plaquette has $\delta(U_p,I)\le d(U_p,I)\le 4r$ and $s_p\le 8r^2/N=1/(2\beta_W)$, so $\beta_W S\le 3L^4$. By Lemma 227 the $4L^4$ links do so with probability at least

$$\Big(\big(\tfrac{95}{96}\big)^{d-1}\big(\tfrac{KN}{16\beta_W}\big)^{d/2}\big/\big(d\,I_d(\pi)\big)\Big)^{4L^4},$$

which gives the first bound; the second follows from the concavity of $-\log Z_L$ in the coupling, as in Lemma 146. The integrals satisfy $I_dI_{d+1}=2\pi/d$, by induction from $I_{d+2}=\frac{d}{d+1}I_d$, and $I_d\ge I_{d+1}$, so $d\,I_d(\pi)\ge\sqrt{2\pi d}>1$; the other terms of $C_0(G)$ are nonnegative when $KN\le16$. □

For SU(2), $C_0(G)=4.32001$, below the constant 4.90817 of Lemma 146. For $\mathrm{SU}(N)$, $KN=N^2/(2(N^2-2))\le1$; for every group of Table I, $KN\le1$, so the lemma holds at every $\beta_W\ge1$.

**Theorem 229.** *For $\beta_W\ge KN$, $0<\rho\le1$ and every set $A$ of unit cubes, Theorem 147 and Corollary 148 hold for the Wilson measure of $G$ with*

$$q_\rho^G(\beta_W)=\min\Big\{1,\exp\Big(2d\log\beta_W+6C_0(G)-\frac{\beta_W\rho^2}{8N}\Big)\Big\}.$$

*Proof.* Reflection in a hyperplane through sites preserves Haar measure on $G$ and maps $\operatorname{Re}\operatorname{Tr}U_p$ to $\operatorname{Re}\operatorname{Tr}U_p$ or $\operatorname{Re}\operatorname{Tr}U_p^{-1}=\operatorname{Re}\operatorname{Tr}U_p$, so the chessboard estimate holds as in Theorem 147. A bad plaquette has $s_p>\rho^2/(2N)$, and if every cube is bad there are at least $L^4/4$ bad plaquettes, so $S\ge\rho^2L^4/(8N)$. Lemma 228 bounds $Z_L$. The corollary uses only the product bound. □

**Theorem 230** (Section XIII for a compact simple group)**.** *Let $G \subset \mathrm{U}(N)$ be as above, with $\kappa_G \le 2500$ in (iii). Replace* SU(2) *by G, the hypothesis $\rho \le 10^{-8}$ by $\rho \le 10^{-8}\rho_G$, and $C_0$, $q_\rho$ by $C_0(G)$, $q_\rho^G$. Then:*

(i) *Lemmas 173, 174, 184, 200, Theorems 199 and 201* (i), (ii) *hold with*

$$f_G(\beta) = 2d\log\beta + 6C_0(G) - \frac{\beta\rho^2}{8N}$$

*in place of f (and $\rho_0$ in place of $\rho$ at scale j), and with $\beta^* > 16dN/\rho^2$ in place of $\beta^* > 96/\rho^2$. The numbers M, H, Δ, $M_j$, $H_j$ and $\log q^*$ do not depend on G, and*

$$\beta_G^* = \frac{8N|\log q^*|}{\rho^2}(1+\varepsilon_G), \quad \varepsilon_G = \frac{2d\log\beta_G^* + 6C_0(G)}{|\log q^*|}.$$

(ii) *Theorems 127, 131, Propositions 128 and 133 hold as stated, since they are stated for every compact $G \subset \mathrm{U}(r)$: $c_3 = \max\{1.5\cdot 10^4\sqrt{N}, d\}$, $c_6 = 6.81\cdot 10^5 d\, c_3^2$, $\alpha_G = 2.5\cdot 10^{-4}$.*

(iii) *Theorem 209 holds with $\lambda = \beta_W/(2N)$, hypothesis (b) read as $\beta_W \ge 480Nc_6$, and $\mathscr{A} = 120/\lambda = 240N/\beta_W$. Corollary 210 holds with*

$$\Delta = \frac{960N}{\beta_W}\Lambda_2^2 r^2 \sqrt{n_x n_y}\, e^{-2\alpha_G(w-2s-8)},$$

*and its hypothesis (a) holds for $\rho \le 1/(2758\cdot 7.8\cdot 10^7 c_3)$.*

(iv) *Lemma 211 holds with $\frac{2}{N}c_W$ and $\frac{d}{3}c_\psi$ in place of $c_W$ and $c_\psi$; under hypotheses (a) and (c) of Theorem 209,*

$$\Lambda_2 \le \frac{198.0002}{N}\beta_W + \frac{4.65\cdot 10^6}{3}d.$$

(v) *Proposition 179 holds with the density bounds $(1+\kappa_0)^{-d}$ and $(1-\kappa_0)^{-d}$, and with radius $\rho_G/32$ when $\rho_G < 1$; Lemma 181 holds with the Jacobian bound $\sigma_0^d$; Lemma 182 holds with $64.94\cdot\frac{d}{3}\sqrt{\kappa_G}$ and $32.47\cdot\frac{d}{3}\sqrt{\kappa_G}$; Theorem 183 holds with $24n_p\beta_W$, $197\beta_W + a_G$ and $87\beta_W + b_G$, where*

$$a_G = 1558.56\sqrt{N}\,\tfrac{d}{3}\sqrt{\kappa_G} + 1.1986,$$
$$b_G = 779.28\sqrt{N}\,\tfrac{d}{3}\sqrt{\kappa_G} + 0.5993.$$

(vi) *Theorem 215, Corollary 216, Proposition 217 and Theorem 218 hold with the threshold $\beta_G^*$ of (i), the condition $2758\rho \le 1/(7.8\cdot 10^7 c_3)$, $\frac{960N}{\beta_W}$ in place of $\frac{1920}{\beta_W}$, and $\bar\Lambda_2 = \frac{198.0002}{N}\beta_W + \frac{4.65\cdot 10^6}{3}d$.*

(vii) *The statements of Subsections XIII C to XIII G, other than Proposition 192, of Subsection XIII L and of Subsection XIII M hold for G as stated, with $\delta_G = \kappa_G/3$.*

*For* SU(3), *$d = 8$, $N = 3$, $\kappa_G = \frac{3}{2}$, $\rho_G = 1$, $C_0(G) = 10.44721$, $f_G(\beta) = 16\log\beta + 6C_0(G) - \beta\rho^2/24$, and $\beta^*(10^{-8},1) = 5.269\cdot 10^{21}$, $\beta^*(10^{-8},5) = 8.231\cdot 10^{21}$, $\beta^*(10^{-9},1) = 5.286\cdot 10^{23}$, $\beta^*(10^{-9},5) = 8.248\cdot 10^{23}$, $\beta^*(10^{-8},5,10^{30}) = 2.972\cdot 10^{22}$, with $0.026 < \varepsilon_G < 0.045$ in the first four. At scale j, for $s = 5$ and $\rho = 10^{-8}$, $\beta_1^* = 4.368\cdot 10^{23}$, $\beta_2^* = 6.991\cdot 10^{24}$, $\beta_3^* = 1.786\cdot 10^{26}$ and $16^{-j}\beta_j^* \le 4.38\cdot 10^{22}$ for $1 \le j \le 12$. In (iii) for* SU(3), *$c_3 \le 25980.8$, $c_6 \le 3.678\cdot 10^{15}$, (b) holds for $\beta_W \ge 5.30\cdot 10^{18}$, (a) for $\rho \le 1.789\cdot 10^{-16}$, and at $\rho = 1.78\cdot 10^{-16}$ the threshold with the joining distance $2s+8$ is $\beta^* = 2.570\cdot 10^{37}$ for $s = 1$ and $3.015\cdot 10^{37}$ for $s = 5$, where (b) holds; $\Lambda_2 \le 66.0001\beta_W + 1.24\cdot 10^7$, and the constant $c_2$ of Theorem 218 at $s = 1$, $\vartheta = 0$ is at most $10^{22.14}$.*

*Proof.* (i) The witness paragraph of the proof of Theorem 199 uses Lemma 124, which is stated for every $G$ with $\kappa \le 10^{-8}\rho_G$; Lemma 116, a statement on $\mathrm{U}(N)$; and $d(g,h) \le (1+10^{-12})\delta(g,h)$ for $\delta(g,h) \le 10^{-6}$, which holds on $G$ because there $\delta \le \rho_G$, the principal logarithm of $g^{-1}h$ lies in $\mathfrak{g}$ and $d(g,h) = \|\log(g^{-1}h)\|$. In Lemma 173 the constants are those of Lemma 116 and the triangle inequality. In Lemma 174 the cut-off is nonzero only where $\max_x \delta(v_x, I) < \rho_G$, so the logarithms lie in $\mathfrak{g}$, and the contraction bound uses the spectrum of $\mathrm{ad}X$ only through Lemma 225(i). Lemmas 184 and 200 count cubes; Theorem 152, used in the latter, is stated for every $G$. The threshold paragraph applies to $f_G$, which is concave with maximum $2d\log(16dN/\rho^2) + 6C_0(G) - 2d > 0 > \log q^*$ at $16dN/\rho^2$, by Lemma 228 and $\rho \le 10^{-8}$. The identity for $\varepsilon_G$ is $f_G(\beta_G^*) = \log q^*$. (ii) There is nothing to prove. (iii) In the proof of Theorem 209 the Wilson action enters as $\lambda\|\cdot\|^2$ with $\lambda = \beta_W/(2N)$, the dimension enters through $\|A_v\|_1 \le 2d\, c_3\|x_v\|_1$ and $\|A_a\|_2^2 \le 2d\, c_3^2\|\bar P\|^2|a|^2$, which are absorbed in $c_6$, and the Haar part is the logarithmic gradient of $j$ at $\|X_e\| \le 10^{-3}$, at most $\frac{\kappa_G}{3}(1+10^{-6})10^{-3} \le 0.84$ per unit of $\|x_v\|_1$ by Lemma 225(iv) and $\kappa_G \le 2500$. The rest of the proof is a statement about the chart constants of (ii). The corollary is Proposition 208 with $4\mathscr{A} = 960N/\beta_W$, and $\eta + 62r = 2758\rho$. (iv) The Wilson term of a plaquette is $\beta_W(1 - \frac{1}{N}\mathrm{Re}\,\mathrm{Tr}\,U_p)$, so every bound of (27) carries $\frac{1}{N}$ in place of $\frac{1}{2}$; Steps 2 to 6 use the spectrum of $\mathrm{ad}W$ through Lemma 225(i) only. In Step 7, $|\mathrm{Tr}A| \le d\|A\|$ on $\mathfrak{g}$ replaces $|\mathrm{Tr}A| \le 3\|A\|$. (v) In the proof of Proposition 179 the density of a linear map $T$ with $\|T - I\| \le \kappa_0$ on $\mathfrak{g}$ lies between $(1-\kappa_0)^d$ and $(1+\kappa_0)^d$; the cut-off is inactive below $\rho_G/2$, which gives the radius $\rho_G/32$. Lemma 135 is stated with $\rho_G$, and $JJ^* \ge \sigma_0^2$ on the $d$-dimensional $\mathfrak{g}$ gives $\sqrt{\det JJ^*} \ge \sigma_0^d$. In the proof of Lemma 182, $\mathrm{tr}(T^{-1}\dot T)$ is at most $d\|T^{-1}\|\|\dot T\|$ and $\|\mathrm{ad}W\|_{HS} = 2\sqrt{\kappa_G}\|W\|$ by Lemma 225(ii), while the spectra lie in the segments used there by Lemma 225(i). In the proof of Theorem 183 the oscillation of a site function is at most $24\cdot 2\sqrt{N}$ and $|\dot s_p| \le v/\sqrt{N}$ by Lemma 225(iii), so the first term is at most $24\sqrt{N}\sup|\partial_t \log h_t|$; the factors $\sqrt{N}$ cancel in the Wilson part, which gives $24n_p\beta_W$, $24\cdot 8.1916\beta_W < 197\beta_W$ and $24\cdot 6\cdot 0.5993\,\beta_W < 87\beta_W$, and remain in the density part, which gives $a_G$ and $b_G$. For SU(2), $a_G < 2206$ and $b_G < 1103$, the constants of the theorem. (vi) These statements use the preceding ones through the constants listed, and the threshold through $q^G$. (vii) These statements concern the label vari-

ables, the coupling of labels, the island combinatorics and the quadratic forms of a box. The group enters them only through the results of Sec. XII, stated for every $G$, through the constants of (iii) and (iv), and through $\delta_G$ in Subsection XIII M, which is $\kappa_G/3$ by Lemma 225(ii). The numbers are the roots of $f_G = \log q^*$ and the values of the constants above, evaluated with outward-rounded interval arithmetic; each root is bracketed by two points of relative distance $10^{-25}$ at which $f_G - \log q^*$ has opposite signs, to the right of the maximum of $f_G$. □

*Remark* 231. Four parts of this section do not transfer. Subsection XIII J (Lemma 203, Propositions 204 and 202, Theorem 205) measures frames by the angle of a unit quaternion; on $\mathrm{SU}(N)$, $N \geq 3$, the operator-norm angle is subadditive and conjugation invariant, and the constants of the subsection have not been computed for it. Proposition 180 uses every preimage of the average on $\mathrm{SU}(2)$, and Proposition 192 is a computation with unit quaternions. The examples of Propositions 172, 176 and 177 are stated on $\mathrm{SU}(2)$ and are not claimed for other groups.

*Remark* 232. For $\mathrm{SU}(3)$ the plaquette blocks reach $\beta_W \leq 0.77$ (Theorem 7), in place of 0.351 for $\mathrm{SU}(2)$. For other groups the block constant (15) needs the modulus $\Lambda(r)$ of (8), a gap of the one-link measure and bounds on the Hessian of $\log Z$, and none of them has been certified. The modulus that holds for every $G$ does not suffice: the one-link measure $\nu_b$ satisfies the Bakry–Émery condition with constant $\kappa_G - \|b\|_{\mathrm{op}} \geq \kappa_G - r$, so $\Lambda(r) \leq 1/(\kappa_G - r)$; with this bound the right side of (8) is below one only for $N\beta < \kappa_G/18$, and, since $K_3 \geq s \geq \Lambda(r)$, every value of $\alpha_3$ computed from it is at least $10N\beta/(\kappa_G - 6N\beta)$, below one only for $N\beta < \kappa_G/16$. Both ranges lie inside $N\beta < 2\kappa_G/29$ of Theorem 5.

### O. One step of the block map: the large fields

This subsection carries out, for $G = \mathrm{SU}(2)$, the large-field part of one step of the block map $U \mapsto c = V^\sharp(U)$ on the whole torus. It proves three things. A set of rough blocks is suppressed by $e^{-\lambda\kappa^2}$ per block, either absolutely or in the mean. The weight of a coarse field is an exact sum over large-field regions of kernels with this suppression. The coarse action lies between $\lambda/(1.65\cdot 10^6)$ and $8\lambda$ times the coarse curvature, up to a volume term. It also states which bound the next step would need, and how far Theorem 205 supplies it. The small-field part and its locality are the subject of item (iv) of Sec. XIII P and are not treated here. The constants are enclosed in interval arithmetic and the lattice counts are exact enumerations.

Throughout, $\Lambda_L$ is the torus of side $L = 2^m \geq 32$ in the tree gauge, $T$ is the coarse torus of the blocks $B(y)$, $|T| = L^4/16$, $\lambda = \beta_W/4$, so that the Wilson weight is $e^{-\lambda Q}$ with $Q = \sum_p \delta(U_p, I)^2$, and $\nu_c$ is the fibre measure of Sec. XII E with $W = T$. For a coarse field $c$ write $c_P$ for the holonomy of a coarse plaquette and $Q_1(c) = \sum_P \delta(c_P, I)^2$.

**Definition 233.** The plaquette at $x$ in the plane $\{\mu,\nu\}$ is *owned* by the block containing $x$; $\Pi(y)$ is the set of plaquettes owned by $y$, and $\Pi(X) = \bigcup_{y\in X}\Pi(y)$. For $\kappa > 0$ a block $y$ is *calm* if $\delta(U_p, I) \leq \kappa$ for every $p \in \Pi(y)$, and *rough* otherwise; $R(U)$ is the set of rough blocks. For a block $y$ let $I_y$ be the 17 active links inside $B(y)$, $P_{y,\mu}$ the eight private links of the pair $(y,\mu)$, and $\Xi_y = (I_y, P_{y,0}, \dots, P_{y,3})$.

**Lemma 234.** (i) *Every plaquette has exactly one owner, and* $|\Pi(y)| = 96$.

(ii) *Every* $p \in \Pi(y)$ *in the plane* $\{\mu,\nu\}$ *contains an active link with base in* $B(y)$, *and each of its links has its base in* $B(y)$, $B(y+e_\mu)$ *or* $B(y+e_\nu)$.

(iii) *The variables* $\Xi_y$, $y \in T$, *are the active links, each once. The factor* $f_y = \exp(-\lambda \sum_{p\in\Pi(y)} \delta(U_p, I)^2)$ *and the indicator* $r_y$ *of* $\{y \text{ rough}\}$ *are functions of* $\Xi_y$ *and* $\Xi_{y+e_\mu}$, $0 \leq \mu \leq 3$, *and* $e^{-\lambda Q} = \prod_y f_y$.

*Proof.* (i) A plaquette has one corner, and each of the 16 sites of a block is the corner of six plaquettes. (ii) Let $o$ be the offset of the corner $x$. If $o_\mu = 1$ the link $(x,\mu)$ leaves $B(y)$ and is private, and likewise if $o_\nu = 1$. If $o_\mu = o_\nu = 0$ the four vertices lie in $B(y)$ and the plaquette is a cycle in the block, which the fifteen tree links, a tree by Lemma 93(i), cannot contain; so one of its links, all based in $B(y)$, is active. The links of $p$ are based at $x$, $x+e_\mu$ and $x+e_\nu$. (iii) A link based in $B(y)$ is a tree link, an active link inside $B(y)$, or a private link of a pair $(y,\mu)$, by Lemma 93(i); the rest follows from (ii). □

An enumeration of the 96 plaquettes of a block confirms (i) and (ii).

*Suppression.* For a set $X$ of blocks let $A_X$, $Q_X$, $\nu_c^X$, $\pi_E$ and $Q_*(E)$ be as in Sec. XII E with $W = X$, so that $\nu_c^X$ lives on the fibre of $X$ with the links off $A_X$ fixed.

**Theorem 235.** *Let* $G = \mathrm{SU}(2)$, $\kappa > 0$, $\lambda > 0$ *and let* $X$ *be a set of blocks.*

(i) *For every* $c$ *and every choice of the links off* $A_X$,

$$\int \mathbf{1}\{X \subset R(U)\}\, e^{-\lambda Q_X}\, d\nu_c^X \leq M^{4|X|} e^{-\lambda\kappa^2|X|},$$

*with* $M < 2.3621\cdot 10^4$ *the constant of Lemma 137.*

(ii) *Let* $\kappa \leq 1$ *and* $0 < \tau \leq 1$. *If the exterior* $E$ *of* $X$ *has* $Q_*(E) \leq (1-\tau)\kappa^2|X|$, *then*

$$\pi_E\{X \subset R\} \leq \Big(\frac{5.9\cdot 10^{14}}{(\tau\kappa^2)^3}\Big)^{49|X|} e^{-3\lambda\tau\kappa^2|X|/4}.$$

(iii) *If* $\beta_W \geq 1$ *and* $\kappa \leq 1$, *then* $\mu_L\{X \subset R\} \leq (16\, q_\kappa(\beta_W))^{|X|}$, *with* $q_\kappa$ *as in Theorem 147.*

*Proof.* On $\{X \subset R\}$ each block of $X$ owns a plaquette with $\delta(U_p, I) > \kappa$; these plaquettes are distinct and, by Lemma 234(ii), contain links of $A_X$. Hence $Q_X > \kappa^2|X|$ there.

(i) The integrand is at most $e^{-\lambda\kappa^2|X|}$, and $\nu_c^X$ has mass at most $M^{4|X|}$ by (24).

(ii) With $t = \tau\kappa^2|X|$, $\{X \subset R\} \subset \{Q_X > Q_*(E) + t\}$. The proof of Proposition 143 uses $t \le 1$ only to have $\rho = t/(90.6 n_P) < 1.5 \cdot 10^{-4}$. Each of the $49|X|$ links of $A_X$ lies in six plaquettes and a plaquette contains at most four of them, so $n_P \ge 73.5|X|$ and $\rho \le \tau\kappa^2/6659 < 1.51 \cdot 10^{-4}$, which still gives $16\sqrt{2}\rho + 16\rho^2 < 22.63\rho$. The proof therefore gives the bound $(14 \cdot 90.6^3 n_P^3/t^3)^{49|X|} e^{-3\lambda t/4}$, and $n_P \le 384|X|$ with $14 \cdot 90.6^3 \cdot 384^3 < 5.9 \cdot 10^{14}$.

(iii) The plaquette at $x$ lies in the unit cube $\mathscr{C}_x$, so a rough block $y$ lies in $\bigcup_{x \in B(y)} B_x$. Expanding the intersection over $y \in X$ of these unions gives at most $16^{|X|}$ intersections $\bigcap_{y \in X} B_{x(y)}$ with $x(y) \in B(y)$, each over $|X|$ distinct cubes, and Theorem 147 bounds each by $q_\kappa^{|X|}$. □

Part (i) is uniform in the exterior and in $c$, but it is an absolute bound: it does not compare rough with calm completions. The comparison (ii) needs its hypothesis. For the frustrated exterior described after Corollary 153, a plaquette through the link $e$ is bad with conditional probability tending to one. No conditional bound uniform in the exterior holds.

*Coarse curvature.* For a coarse plaquette $P$ let $N(P)$ be the fine plaquettes with a vertex in one of the 144 blocks at $\ell^\infty$ distance at most one from the four blocks of $P$, the witness region of Lemma 173, and let $N_{\mathrm{ov}}$ be the largest number of witness regions that contain one fine plaquette. An enumeration gives $N_{\mathrm{ov}} = 1441$, and 2310 witness regions meet $\Pi(y)$.

**Lemma 236.** *Let $G = \mathrm{SU}(2)$, $C_E = 8/\kappa_c^2 < 1140.26$ and $C_* = 1441\, C_E < 1.6432 \cdot 10^6$. For every field $U$ in the tree gauge, $c = V^\sharp(U)$ and every coarse plaquette $P$,*

$$\delta(c_P, I)^2 \le C_E \max_{p \in N(P)} \delta(U_p, I)^2,$$

*and if $\kappa < \kappa_c$ and $\delta(c_P, I) > 16.003\,\kappa$, some $p \in N(P)$ has $\delta(U_p, I) > \kappa$. Consequently $Q_1(c) \le C_*\, Q(U)$, and*

$$Q_1^{>}(c) := \sum_{\delta(c_P, I) > 16.003\kappa} \delta(c_P, I)^2 \le C_* \sum_{\delta(U_p, I) > \kappa} \delta(U_p, I)^2.$$

*In particular $Q(U) \ge r^2/1140.26$ whenever some coarse plaquette has $\delta(c_P, I) \ge r$.*

*Proof.* Let $m$ be the maximum over $N(P)$. If $m < \kappa_c$, then Proposition 204(iv), applied with every $\kappa \in (m, \kappa_c)$, gives $\delta(c_P, I) \le 16.003\, m$, and $16.003^2 < 256.1 < C_E$. If $m \ge \kappa_c$, then $\delta(c_P, I) \le 2\sqrt{2}$, the diameter of SU(2), so $\delta(c_P, I)^2 \le 8 \le C_E m^2$. The second statement is the same proposition. Summing $\delta(c_P, I)^2 \le C_E \sum_{p \in N(P)} \delta(U_p, I)^2$ over $P$ counts each fine plaquette at most $N_{\mathrm{ov}}$ times; for $Q_1^{>}$ keep in each sum only the plaquettes above $\kappa$, among which the maximum lies. □

The last statement is the first step of Proposition 166 for SU(2), with 1140.26 in place of 1154; it is a bound for one step only.

*The coarse action.* Since $h = \int \nu_c\, dh(c)$, the law of $c = V^\sharp(U)$ under the Wilson measure is $Z_L^{-1} e^{-A_1(c)}\, dh(c)$ with

$$e^{-A_1(c)} = Z(c) = \nu_c\big(e^{-\lambda Q}\big).$$

Let $S_1(c) = \inf_{F_c} Q$. In the reconstruction $R(c)$ of Sec. XII B, with $c_{y,\mu}$ on the private links of $(y, \mu)$ and $I$ on the links inside the blocks, every path of $V^\sharp_{y,\mu}$ has holonomy $c_{y,\mu}$. So $v_x = I$ at every site, $R(c)$ is regular, and $V^\sharp(R(c)) = c$. A plaquette at $x$ in the plane $\{\mu, \nu\}$ of $R(c)$ is $c_P$ when $o_\mu = o_\nu = 1$ and $I$ otherwise, so $Q(R(c)) = 4Q_1(c)$.

**Theorem 237.** *Let $G = \mathrm{SU}(2)$. For every coarse field $c$:*

(i) *$A_1(c) \ge \lambda S_1(c) - 4|T| \log M$, and $S_1(c) \ge Q_1(c)/C_*$;*

(ii) *for $\lambda \ge 100$, $\eta > 0$ and every regular $U \in F_c$, $A_1(c) \le (1+\eta)\lambda Q(U) + c_1(\lambda, \eta)|T|$, where*

$$c_1(\lambda, \eta) = 12 \log 1.957 + 45 \log 14 + 67.5\big(1 + \log(11520(1+\eta^{-1})\lambda)\big);$$

*in particular $A_1(c) \le 8\lambda Q_1(c) + c_1(\lambda, 1)|T|$.*

*Proof.* (i) $Z(c) \le \nu_c(F_c) e^{-\lambda S_1(c)}$ and $\nu_c(F_c) \le M^{4|T|}$ by (24); then Lemma 236.

(ii) Let $\rho^2 = 135/(2 \cdot 96 \cdot 8100(1+\eta^{-1})\lambda)$, so $\rho \le 9.32 \cdot 10^{-4}$ and $46\rho < 0.043$. The proof of Proposition 141 with $W = T$ uses $t \le 1$ only through this bound on $46\rho$. It gives a set $\Gamma(D) \subset F_c$ with $\nu_c(\Gamma(D)) \ge 1.957^{-12|T|} (\rho^3/14)^{45|T|}$ on which each of the $n_P = 96|T|$ plaquettes moves by at most $90\rho$. There $\delta(U'_p, I) \le \delta(U_p, I) + 90\rho$, so

$$Q(U') \le (1+\eta) Q(U) + (1+\eta^{-1})\, 96|T|\, 8100\rho^2.$$

Hence $A_1(c) \le (1+\eta)\lambda Q(U) + |T| g(\rho)$ with $g(\rho) = (1+\eta^{-1}) 777600\lambda\rho^2 + 135 \log \rho^{-1} + 12 \log 1.957 + 45 \log 14$, and the chosen $\rho$ minimises $g$, with value $c_1(\lambda, \eta)$. For the last statement take $U = R(c)$ and $\eta = 1$. □

So the coarse action is $\lambda$ times a quantity between $Q_1/C_*$ and $8Q_1$, up to $O(\log \lambda)$ per block; the volume term $67.5 \log \lambda$ is the Gaussian count of the 135 real fluctuation coordinates of a block. Neither constant is near one. For example $c_1(10^{12}, 1) < 2737.5$ and $4 \log M < 40.28$.

*The large-field representation.* Write $s_y = 1 - r_y$. For $X \subset T$ let $X_1, \dots, X_n$ be its components for $\ell^\infty$ adjacency of blocks, so that distinct components are at $\ell^\infty$ distance at least two, and let $\partial^- X_i$ be the blocks $z \notin X$ with $z + e_\mu \in X_i$ for some $\mu$. By the definition of $\nu_c$,

$$\nu_c(d\Xi) = \prod_y h(dI_y) \prod_{y,\mu} \nu_c^{y,\mu}(dP_{y,\mu} \mid I_y, I_{y+e_\mu}),$$

where $\nu_c^{y,\mu}$ is the measure of (23) on the private links of the pair, of mass at most $M$.

**Theorem 238.** *Let $G = \mathrm{SU}(2)$ and $\kappa > 0$.*

(i) *For every $c$, $Z(c) = \sum_{X \subset T} Z_X(c)$ with*

$$Z_X(c) = \int \nu_{c,X}^{\mathrm{out}}(d\Xi) \prod_{y \notin X \cup \partial^- X} f_y s_y \prod_i K_{X_i}(c; \Xi).$$

*Here $\nu_{c,X}^{\mathrm{out}}$ is the product of the factors of $\nu_c$ for the links inside the blocks off $X$ and for the pairs with*

*both blocks off $X$. The kernel $K_{X_i}$ is the integral of $\prod_{y\in X_i} f_y r_y \prod_{z\in\partial^- X_i} f_z s_z$ against the factors of $\nu_c$ for the links inside $X_i$, the pairs from $X_i$ and the pairs $(z,\mu)$ with $z\in\partial^- X_i$ and $z+e_\mu\in X_i$. It depends on the integration variable of $\nu_{c,X}^{\mathrm{out}}$ only through the blocks at $\ell^\infty$ distance one from $X_i$, and no two kernels share a factor or a variable.*

(ii) *For $\kappa<\kappa_c$, uniformly in the outer variables,*

$$0\le K_{X_i}\le M^{8|X_i|}\exp\Big(-\lambda\max\Big\{\kappa^2|X_i|,\frac{Q^{(i)}(c)}{C_*}\Big\}\Big),$$

*where $Q^{(i)}(c)$ is the sum of $\delta(c_P,I)^2>(16.003\kappa)^2$ over the coarse plaquettes $P$ with $\emptyset\neq N(P)\cap\Pi(X)\subset\Pi(X_i)$.*

(iii) *With $N_\lambda=e^{-c_1(\lambda,1)}$, $K_{X_i}\le N_\lambda^{|X_i|}e^{-a(\lambda,\kappa)|X_i|}$, where*

$$a(\lambda,\kappa)=\lambda\kappa^2-8\log M-c_1(\lambda,1),$$

*and $Z(c)\ge N_\lambda^{|T|}e^{-8\lambda Q_1(c)}$ for $\lambda\ge 100$.*

*Proof.* (i) Expand $\prod_y(f_y r_y+f_y s_y)$ and group the terms by $X=\{y: r_y\}$; then apply Fubini to the nonnegative integrands. By Lemma 234(iii), $f_y$ and $r_y$ involve $\Xi_y$ and $\Xi_{y+e_\mu}$, and $\nu_c^{y,\mu}$ involves $P_{y,\mu}$, $I_y$ and $I_{y+e_\mu}$. A factor or measure involving a variable of $X_i$ therefore belongs to a block of $X_i\cup\partial^- X_i$ or to a pair from it. Such a factor cannot involve a variable of $X_j$ for $j\neq i$, since $z+e_\mu\in X_i$ and $z+e_\nu\in X_j$ would put $X_i$ and $X_j$ at distance one. (ii) The measures in $K_{X_i}$ have mass at most $M$ for each of the $4|X_i|$ pairs from $X_i$ and the at most $4|X_i|$ pairs into it, and $h$ is a probability. The integrand is at most $e^{-\lambda Q_{\Pi(X_i)}}$ and vanishes unless every block of $X_i$ is rough. Then $Q_{\Pi(X_i)}>\kappa^2|X_i|$, and every fine plaquette above $\kappa$ is owned by a rough block. So for the $P$ counted in $Q^{(i)}$ the plaquettes above $\kappa$ in $N(P)$ lie in $\Pi(X_i)$, and Lemma 236 gives $Q^{(i)}(c)\le C_* Q_{\Pi(X_i)}$. (iii) The first bound is (ii) with the definition of $a$; the second is Theorem 237(ii) with $U=R(c)$. □

The representation is exact, and every kernel carries $e^{-\lambda\kappa^2}$ per block, uniformly in its boundary values and in $c$. When the coarse field is curved its kernel also carries $e^{-\lambda Q^{(i)}/C_*}$. The small-field factor $\prod f_y s_y$, however, is not a product of local factors once the outer variables are integrated. The comparison of a kernel with the calm weight $S_{X_i}$, the same integral with $s_y$ in place of $r_y$ on $X_i$, would give the relative activities $K_{X_i}/S_{X_i}$. It fails where $S_{X_i}=0<K_{X_i}$, that is, where the boundary values admit no calm completion; Theorem 235(ii) bounds the ratio only under its hypothesis.

**Theorem 239.** *Call two nonempty $\ell^\infty$-connected sets of blocks compatible when their $\ell^\infty$ distance is at least two, and let $w$ be activities with $|w(X)|\le e^{-9|X|}$.*

(i) *The criterion of Kotecký and Preiss [60] holds with $a(X)=|X|/10$. So $\log\sum\prod_i w(X_i)$, the sum over families of pairwise compatible sets, is the absolutely convergent sum of the connected clusters. The clusters incompatible with a given block contribute at most $1/10$ in absolute value, and the logarithm is at most $|T|/10$ in modulus.*

(ii) *The rate of Theorem 238(iii) satisfies $a(\lambda,\kappa)\ge 9$ for $\lambda\ge\max\{\lambda^*(\kappa),100\}$, with $\beta^*=4\lambda^*$ given below.*

| $\kappa$ | $\beta^*$ | $\lambda^*\kappa^2$ |
|---|---|---|
| 0.08 | $1.13074\cdot10^6$ | 1809.2 |
| $10^{-2}$ | $8.39984\cdot10^7$ | 2100.0 |
| $10^{-4}$ | $1.09584\cdot10^{12}$ | 2739.6 |
| $10^{-8}$ | $1.60348\cdot10^{20}$ | 4008.7 |

*Proof.* (i) A connected set of $n$ blocks through a given block is recovered from a walk of length $2(n-1)$ around a spanning tree, with 80 neighbours at each step, as in Corollary 148; so there are at most $6400^{n-1}$. A set incompatible with $X$ contains one of the at most $81|X|$ blocks within distance one of $X$. Hence

$$\sum_{X'\not\sim X}|w(X')|e^{|X'|/10}\le 81|X|\frac{u}{1-6400u},$$

with $u=e^{-8.9}$, and the right side is below $0.0870|X|<|X|/10$. This is the hypothesis of the theorem of Kotecký and Preiss, whose conclusion is the statement. (ii) $a$ is increasing for $\lambda>67.5/\kappa^2$, and the thresholds are enclosed by bisection in interval arithmetic to relative width $10^{-8}$. □

Theorem 239 applies to the large fields once the small-field part is in local form. Then each region costs its kernel divided by the calm weight of the same region, and $a\ge 9$ beats the entropy of connected regions if that calm weight is at least $N_\lambda$ per block. This lower bound on the calm weight with boundary values is not proved. It is the small-field cluster expansion of item (iv) of Sec. XIII P, and it must have the relative and local form required by Propositions 194 and 195 and Lemma 193.

*The next step.* To repeat the step at the coarse scale with constants $(\lambda_1,\kappa_1)$, the proofs above need three properties of the weight $e^{-A_1(c)}$ in place of $e^{-\lambda Q}$: (H1) it is a product of nonnegative factors attached to the coarse blocks, each depending on $c$ within distance one, up to a polymer expansion of the kind just described; (H2) the factor of a block that owns a coarse plaquette with $\delta(c_P,I)>\kappa_1$ is at most $e^{-\lambda_1\kappa_1^2}$ times the calm weight, with $\lambda_1\kappa_1^2-8\log M-c_1(\lambda_1,1)\ge 9$; (H3) the averages of calm coarse fields are calm. Item (H3) holds at every step for $\kappa_1<\kappa_c$, since Proposition 204(iv) is a statement about one field on any lattice; the threshold then grows by 16.003 per step, or by $4\cdot1.023$ along the composed average of Theorem 152. The suppression actually available for (H2) is inherited from the fine scale, as follows.

**Proposition 240.** *Let $G=\mathrm{SU}(2)$, $\kappa<\kappa_c$ and $j\ge 1$.*

(i) *If $\delta(c_P,I)>16.003\kappa$, then $N(P)$ meets $\Pi(R(U))$. Each rough block meets at most 2310 witness regions, so at most $2310|R(U)|$ coarse plaquettes exceed $16.003\kappa$. On them the kernels carry $e^{-\lambda Q^{(i)}/C_*}$, that is, the coupling $\lambda/C_*$.*

(ii) *Let $\rho\le 1.023\cdot10^{-4}$, $\rho_0=\rho/(1.01\cdot1.023\cdot4^j)$, and call a fine block rough at level $\rho_0$. By Lemma 200 and Theorem 205(i), every bad block of scale $j$ has a rough fine*

*block among the $m_j = (17\cdot 2^{j-1}-2)^4$ blocks that meet its box. A fine block meets the boxes of at most $H_j$ blocks of scale $j$, with $H_j$ as in that lemma. Hence for a set $Y$ of blocks of scale $j$ and $a_0>0$, $\sum_X e^{-a_0|X|}\le e^{-A_j|Y|}$, where the sum runs over the sets $X$ of fine blocks inside the union of the boxes of $Y$ that meet every box, and*

$$A_j=\max_{s>0}\Big(\frac{a_0-s}{H_j}-m_j\log(1+e^{-s})\Big).$$

(iii) *With $a_0=a(\beta_W/4,\rho_0)$ and $\rho=10^{-4}$, $A_j\ge 9$ holds for $\beta_W\ge\beta_j^{\mathrm{pw}}$, where $\beta_1^{\mathrm{pw}}=2.7321\cdot10^{14}$, $\beta_2^{\mathrm{pw}}=4.39218\cdot10^{15}$, $\beta_3^{\mathrm{pw}}=1.09464\cdot10^{17}$, and $1.70\cdot10^{13}\le 16^{-j}\beta_j^{\mathrm{pw}}\le 2.90\cdot10^{13}$ for $1\le j\le 30$. At each of the 2001 couplings of Theorem 205(iii), the largest such $j$ satisfies $k_*-j\in\{1,2\}$.*

*Proof.* (i) is Lemma 236 and Theorem 238(ii), with the count from the enumeration. (ii) The box of Lemma 200 has $17\cdot2^j-4$ points per coordinate, from an even to an odd coordinate, so it meets $\frac12(17\cdot2^j-4)$ fine blocks per coordinate. A witness plaquette with corner in the box is owned by one of these. The number of boxes met by a fine block is the count of that lemma for the two cubes of the block per coordinate, which the enumeration confirms to be $H_j$. A set $X$ meeting all the boxes of $Y$ has $|X|\ge|Y|/H_j$ and lies in a set of at most $m_j|Y|$ blocks. So the sum is at most $\sum_{k\ge|Y|/H_j}\binom{m_j|Y|}{k}e^{-a_0k}\le e^{-(a_0-s)|Y|/H_j}(1+e^{-s})^{m_j|Y|}$. (iii) The maximum is at $e^s=m_jH_j-1$, and the thresholds and $k_*$ of Corollary 153 are enclosed in interval arithmetic. □

So the next step stands as follows. Theorem 205 gives (H3) up to $a_{k_*}/8$ as a statement about probabilities under the Wilson measure: the labels of the composed average are rare. It gives neither (H1) nor (H2), which concern the weight $e^{-A_1}$ for every coarse field. Proposition 240(iii) reaches $a_{k_*}/4$, but it bounds sums of fine kernels with entropy of $\ell^\infty$ adjacency rather than of the joining distance of the islands, and only for activities of the form $e^{-a_0|X|}$ relative to the calm weight, the hypothesis that is not proved. In both cases the suppression at scale $j$ is the fine one at the radius $\rho_0\propto4^{-j}$, that is, the coupling $\lambda/16^j$ in units of the scale-$j$ threshold. By Proposition 202, fine witnesses carry it no further than $a_{k_*}$. Iterating the step instead loses the factor $C_*$ of Theorem 237(i), or $H_j\ge4096$ in Proposition 240(ii), at every scale. What the iteration needs is (H2) with $\lambda_1\ge(1-\varepsilon_1)\lambda$ and summable $\varepsilon_k$, a lower bound $A_1\ge(1-\varepsilon)\lambda S_1$ on large fields, local and relative to the calm weight. That requires a coarse energy for SU(2) at every scale, of the kind Proposition 166 gives for torus-valued fields. It is not proved here, and this subsection gives no bound on the mass gap.

### P. What remains

The results of this section leave the following statements, which together with Theorem 131 and Sec. XII G are what the cube condition (D) needs for SU(2) past the Dobrushin ceiling.

(i) *The bad side.* For a pair whose non-path links are bad, a bound on the Kantorovich response of the law of its eight private links, given the inside links of its two blocks, uniform in the exterior. Proposition 180 shows that the response of the path-link law alone to its non-path links is infinite, so the non-path links must be integrated over the bad set; Lemma 178 bounds the response of that law to every link that enters only the action, and Proposition 179 settles the good side to radius $1/32$. The inside links move the fibre $\{V_k^\sharp=c_k\}$ while the bad set does not; Theorem 183 puts them among the variables and bounds the response of the law of the 24 links of the site of the pair to every other link by at most $197\beta_W+2206$, uniformly in the exterior, at the cost of one block of collar on each side of an island. Whether the law of the eight private links alone is Lipschitz in the inside links is not decided.

(ii) *The island comparison.* For two islands in a good region, Theorem 207 factorises their weight through one kernel on the good layer between them, Theorem 209 proves the covariance bound (26) for the law of that layer in the coordinates of the labels, and Corollary 210 bounds the cross-ratio by $\Delta$, exponentially small in the width. Lemma 211 evaluates the prefactor, $\Lambda_2\le99.0001\beta_W+4.65\cdot10^6$, so that $\Delta\le1$ at the threshold needs a good layer of width at least 99072. Theorem 215 performs the induction over all the islands of a window given their labels. Theorem 218 gives the smallness of its island sums and its Dobrushin condition in every window of at most $10^{28000}$ blocks, outside probability $10^{-166}$. What is not done is the same statement in larger windows, which a union bound cannot give, and the covariance of the conditional probabilities of two labels when the others are averaged. Proposition 189 shows that no rarity bound replaces this comparison.

(iii) *The island threshold.* Theorem 199 gives the explicit coupling $\beta^*(\rho,s)$ past which the islands of bad blocks at one scale are rare enough to beat their entropy, at the exponent $1/H$ of Proposition 185 and the collar of Proposition 175. What is not given is the same statement at the scale of the cube: $\beta^*$ is of order $10^{13}$ at the witness radius $10^{-4}$ of Theorem 205, and of order $10^{37}$ under the calmness of Corollary 210, far beyond the range in which the chessboard estimate reaches that scale, so that (iv) is needed before the threshold bears on the gap.

(iv) *The scales.* An effective action at every scale whose one-step errors are summable in the sense of Proposition 194 and whose locality norms obey Lemma 193, down to the last $O(1)$ scales where the running coupling is of order one; this is the large-field expansion of Bałaban, and it is not carried out here. Theorem 201 and Theorem 205 settle the entropy count of the islands at every scale up to $a_{k_*}/8$, and Proposition 202 shows that fine witnesses carry it no further; the law of the iterates given their labels, which the effective action is for, is untouched. Proposition 195 shows that the one-

step errors must be relative bounds by a gauge-invariant form: an error bounded by a multiple of the Euclidean form grows by a factor at least $4/(1+2724\kappa)$ at each step. Proposition 222 gives such a relative bound for one step at the trivial background on a box of side $n$, from an absolute bound on the Hessian of the induced action off the gauge directions; its factor $C_n$, of order $n^2$, is necessary (Remark 224) and is removed only by a gauge-invariant localisation, and the bound at curved backgrounds and away from critical points is not proved. Sec. XIII O carries out the large-field part of one step for SU(2): each rough block costs $e^{-\lambda\kappa^2}$, the weight of a coarse field is an exact sum over large-field regions, and Theorem 239 sums the regions once the small-field part is local with calm weight at least $N_\lambda$ per block, which is not proved; the coarse action is comparable to $\lambda$ times the coarse curvature only up to the factor $C_*$, so the step does not iterate.

The averaged covariance of item (ii), with the island sums in windows of more than $10^{28000}$ blocks, is the single-scale part of (D) that remains open, item (i) being settled for the site of a pair by Theorem 183 and item (ii) given the labels of all islands by Theorem 215; item (iii) is settled by Theorem 199 at one scale and by Theorem 205 at every scale up to $a_{k_*}/8$; item (iv) is the multiscale part. Sec. XIII N carries the single-scale results of this section to every compact simple group, with explicit constants, except the parts listed in Remark 231. On the continuum side, Lemmas 196 and 197 reduce the nontriviality of the limit to a lower bound on one ratio of lattice two-point functions at a fixed physical time, uniform along the tuning, which is the open input of Theorem 48; Sec. XI B states it exactly as the condition $(\mathrm{N}_u)$, shows that neither the strong-coupling range nor the cube estimate in the form of Theorem 45 supplies it, and reduces it to the trial states of Corollary 71.

## ACKNOWLEDGEMENTS

This paper rests on the work of many earlier scientists. Lattice gauge theory is due to Wilson [93] and its Hamiltonian form to Kogut and Susskind [61]; reflection positivity and the reconstruction of the transfer matrix to Osterwalder and Schrader [73, 74], and their lattice form, with the first proof of a gap at strong coupling, to Osterwalder and Seiler [75, 82]. The uniqueness method is Dobrushin's [34–36], the curvature method is that of Bakry and Émery [4, 5] in the form of Shen, Zhu and Zhu [84], the covariance estimate under Dobrushin's condition is Föllmer's [41], the chessboard estimate is that of Fröhlich, Israel, Lieb and Simon [42] and Fröhlich and Spencer [43], disagreement percolation is due to van den Berg and Maes [89], the Brascamp–Lieb inequality to Brascamp and Lieb [21], the correlation representation to Helffer and Sjöstrand [53] and Naddaf and Spencer [72], the perturbation lemma to Holley and Stroock [52], the derivative of a matrix function to Daleckii and Krein [33], and the strict positivity of the transfer matrix to Lüscher [64], the gauge fixing on a forest is Creutz's [30], and the curvature of Lie groups is Milnor's [70]; the localisation estimate of Sec. XII A is that of Combes and Thomas [31]. The analysis of the single-link law uses results of Barta [10], Biernacki and Krzyż [23], McShane [69], Stein and Weiss [87] and Watson [91], and the continuum problem is the one formulated by Jaffe and Witten [56] and studied by Bałaban [6–8] through the renormalisation group, in the account of Dimock [32]; gauge fields with cut-offs were constructed by Brydges, Fröhlich and Seiler [25] and by Magnen, Rivasseau and Sénéor [67]; recent probabilistic work on lattice Yang–Mills theory is due to Chatterjee [28, 29], Basu and Ganguly [11], Adhikari and Cao [1], and Cao, Nissim and Sheffield [27], and the two-dimensional theory was constructed by Driver [37], Gross, King and Sengupta [47] and Lévy [62].

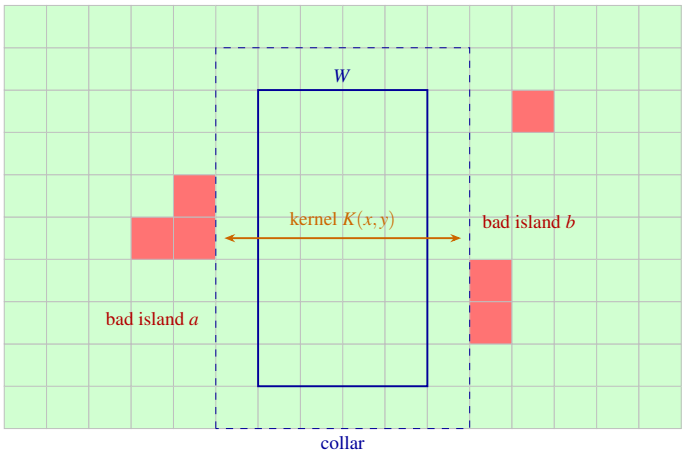
## DATA AND CODE AVAILABILITY

The proofs are analytic except for the evaluations in Propositions 34 and 35 and Lemma 38 and the numerical bounds of Lemma 16, which are computations in interval arithmetic with rigorous error bounds, and for Proposition 95 and Theorem 267, whose certificates are checked in exact arithmetic. The programs of these steps, in Python with python-flint (FLINT and Arb) and a C kernel for products of Taylor polynomials, their outputs and the list of certified boxes are in the repository `https://github.com/creelie/4Y-M`, release v4.32.0 (doi:10.5281/zenodo.23933856), together with a Lean 4 certificate of the line field, Lean 4 files for the comparison arithmetic, the coupling lemmas of Sec. XIII E and the finite arithmetic of Theorem 209, Corollary 210, Proposition 195, Proposition 204 and Theorem 205, independent exact or numerical checks of the other finite statements, our checks of the certificate of [79], and the certificate of Proposition 95 with its exact check. The code was written by Deep Bhattacharjee, James Montgomery Lockwood and Tristan Romanov.

## DECLARATIONS

*Author contributions.* Deep Bhattacharjee conceived the project, proved the main results, wrote the programs of the certified computations and wrote the paper. Ushashi Bhattacharya checked the proofs and the references and edited the text. James Montgomery Lockwood contributed the analysis of constrained cubes [63] and its code. Tristan Romanov contributed the certified bound on the forest-gauge Hessian [79], the first form of Sec. XIII and its code. All four authors approved the final text.

*Use of artificial intelligence.* The computations, the checks and parts of the drafting were done with AI assistance (Claude Opus 5.5, Claude Fable 5.1, GPT-6.1 Sol Ultra and GPT-6 Astra Ultra) under the direction of Deep Bhattacharjee, who checked every statement and is responsible for the text. The combination of the forest gauge with the sharp quarter behind Theorem 1 was first assembled by Anaxia, the reasoning engine of the BLLM system [14, 15]; the line field, the free-boundary limit, the ceiling, the curvature bound, the SU(3) part, the




| label | witness and result |
|---|---|
| non-path link off $R(c)$ | fine defect, Theorem 117 |
| coarse plaquette $> \eta$ | fine defect, Lemma 173 |
| path link | none: determined, Lemma 174 |
| bad non-path links | fold, Proposition 180 |
| bad pair | site, Theorem 183 |
| all links | nonconvex, Proposition 176 |
| good window | graph law, Proposition 175 |
| $n$ bad labels | $A^{n/H}$, Lemma 184 |
| two islands | one kernel, Theorem 207 |
| layer law | chart bound, Theorem 209 |
| two islands | $e^{-2\alpha_G(w-2s-8)}$, Corollary 210 |
| two islands | $\frac{1}{4}\tanh(\Delta/4)$, Corollary 187 |
| many islands | $\Delta_*$, Theorem 215 |
| rarity alone | floor $q^2/16$, Proposition 189 |
| island of $n$ blocks | $2^{-n}$ past $\beta^*$, Theorem 199 |
| islands at scale $j$ | $2^{-n}$ past $\beta_j^*$, Theorem 201 |

FIG. 25. Left: a window $W$ of coarse blocks (solid) with its collar (dashed), good blocks in green and bad islands in red. On the good blocks the conditional law is the graph law of Proposition 175; the weight of two islands is compared through a kernel on the good layer between them, whose cross-ratio bounds their covariance (Lemma 186, Corollary 187). Right: the labels considered in Sec. XIII, their fine witnesses, and the estimate or the obstruction that each produces.

plaquette blocks and the certified constants are the authors' own.

*Funding.* None. *Competing interests.* None. *Ethics approval.* Not applicable: the work involves no human participants, animals or personal data.

### Appendix A: The line field

Table III lists, for each residue class of $x$ modulo two, the direction $\sigma(x)$ and the permutation $\pi_x(\mu) = \sigma(x+e_\mu)$. In every row $\pi_x$ fixes $\sigma(x)$ and is a three-cycle on the other directions, which is the condition of Lemma 8.

### Appendix B: The comparison lemma

Let $(G,\delta)$ be a compact group with a bi-invariant metric of diameter $D$, $\Omega = G^A$, and $C_{ee'}$ the coefficients of Sec. III, with row and column sums at most $\alpha < 1$ and $C_{ee'} = 0$ unless $e, e'$ share a plaquette, so that $|e - e'| \le 1$ for the distance of link midpoints. Fix $\lambda > 0$ with $\theta = \alpha e^{\lambda} < 1$.

**Lemma 241.** *Let $S \subset A$, possibly infinite, and let $\mu_1, \mu_2$ be probability measures on $\Omega$ invariant under $\gamma_e^F$ for every $e \notin S$. Let $T \subset A \setminus S$ be finite, $w_b = e^{-\lambda\,\mathrm{dist}(b,T)}$, and suppose $\sum_{z\in S} w_z < \infty$. There is a coupling of $\mu_1, \mu_2$ whose link distances $\rho(b) = \mathbb{E}\,\delta(\sigma_b, \sigma'_b)$ satisfy*

$$\sum_{b\notin S} w_b \rho(b) \le \frac{\theta D}{1-\theta} \sum_{z\in S} w_z .$$

*If $S = \emptyset$ then $\mu_1 = \mu_2$.*

TABLE III. The line field: $x$ modulo two (digits $x_0x_1x_2x_3$), the direction $\sigma(x)$ and the values $\sigma(x+e_\mu)$ for $\mu = 0,1,2,3$.

| $x$ | $\sigma(x)$ | $\mu=0$ | $\mu=1$ | $\mu=2$ | $\mu=3$ |
|---|---|---|---|---|---|
| 0000 | 0 | 0 | 2 | 3 | 1 |
| 1000 | 0 | 0 | 3 | 1 | 2 |
| 0100 | 2 | 3 | 0 | 2 | 1 |
| 1100 | 3 | 2 | 0 | 1 | 3 |
| 0010 | 3 | 1 | 2 | 0 | 3 |
| 1010 | 1 | 3 | 1 | 0 | 2 |
| 0110 | 2 | 1 | 3 | 2 | 0 |
| 1110 | 1 | 2 | 1 | 3 | 0 |
| 0001 | 1 | 2 | 1 | 3 | 0 |
| 1001 | 2 | 1 | 3 | 2 | 0 |
| 0101 | 1 | 3 | 1 | 0 | 2 |
| 1101 | 3 | 1 | 2 | 0 | 3 |
| 0011 | 3 | 2 | 0 | 1 | 3 |
| 1011 | 2 | 3 | 0 | 2 | 1 |
| 0111 | 0 | 0 | 3 | 1 | 2 |
| 1111 | 0 | 0 | 2 | 3 | 1 |

*Proof.* The couplings form a weakly compact set and $\Psi(\pi) = \sum_b w_b \mathbb{E}_\pi \delta(\sigma_b, \sigma'_b)$ is weakly continuous because $\sum_b w_b < \infty$; let $\pi^*$ be a minimiser. For $e \notin S$ replace the pair $(\sigma_e, \sigma'_e)$ by a sample of an optimal coupling of $\gamma_e^F(\cdot \mid \sigma)$ and $\gamma_e^F(\cdot \mid \sigma')$, chosen measurably [90, Cor. 5.22]. Both $\mu_i$ are invariant, so the result is again a coupling, and changing the finitely many links that enter $\gamma_e^F$ one at a time gives $W_1 \le \sum_{e'} C_{ee'} \delta(\sigma_{e'}, \sigma'_{e'})$. Minimality gives $\rho(e) \le \sum_{e'} C_{ee'} \rho(e')$. Multiply by $w_e$ and

sum over $e \notin S$; where $C_{ee'} \neq 0$, $w_e \le e^{\lambda} w_{e'}$, so

$$\sum_{e \notin S} w_e \rho(e) \le \theta \Big[ \sum_{e' \notin S} w_{e'} \rho(e') + \sum_{z \in S} w_z \rho(z) \Big],$$

all sums are finite, and $\rho \le D$ on $S$. If $S = \emptyset$, $\rho \equiv 0$. □

**Corollary 242.** *Let $\mu$ be the Gibbs measure of $\gamma^F$, $F$ a local function with support $S_F$, $L_F$-Lipschitz in each link, and $H$ bounded with finite support $S_H$ disjoint from $S_F$. Then*

$$|\operatorname{Cov}_\mu(F,H)| \le \|H\|_\infty L_F \frac{\theta D}{1-\theta} |S_H| \, e^{-\lambda \operatorname{dist}(S_F, S_H)}.$$

*Proof.* Condition on the links of $S_H$ [59, Ch. 3]; the conditional measures $\mu^\sigma = \mu(\cdot \mid \sigma_{S_H})$ are invariant under $\gamma_e^F$ for $e \notin S_H$. Lemma 241 with $S = S_H$ and $T = S_F$, for which $w_b = 1$ on $S_F$ and $w_z \le e^{-\lambda \operatorname{dist}(S_F, S_H)}$ on $S_H$, gives $|\mu^\sigma(F) - \mu^{\sigma'}(F)| \le L_F \sum_{b \in S_F} \rho(b) \le L_F \frac{\theta D}{1-\theta} |S_H| e^{-\lambda \operatorname{dist}(S_F, S_H)}$ for any two boundary values. Since $H$ depends only on $\sigma_{S_H}$, $\operatorname{Cov}_\mu(F,H) = \iint H(\sigma) \big(\mu^\sigma(F) - \mu^{\sigma'}(F)\big) \, d\mu(\sigma) \, d\mu(\sigma')$, and the bound follows. □

**Lemma 243.** *Let the blocks partition $A$, let $C_{Bj}$ be the block coefficients of Sec. V, with $\sum_B C_{Bj} \le \alpha < 1$ for every $j$ and $C_{Bj} = 0$ unless every link of $B$ lies within distance 2 of $j$, and fix $\lambda > 0$ with $\theta = \alpha e^{2\lambda} < 1$. Let $S \subset A$ be a union of blocks and let $\mu_1, \mu_2$ be invariant under $\gamma_B^{\mathscr{B}}$ for every block $B \not\subset S$. With $T$, $w_b$ as in Lemma 241 and $\sum_{z \in S} w_z < \infty$, there is a coupling with $\sum_{b \notin S} w_b \rho(b) \le \frac{\theta D}{1-\theta} \sum_{z \in S} w_z$, and $\mu_1 = \mu_2$ if $S = \emptyset$.*

*Proof.* Take a minimiser $\pi^*$ of $\Psi$ as before. For a block $B \not\subset S$ replace $(\sigma_B, \sigma'_B)$ by an optimal coupling, for the metric $d$, of the two block laws; it is again a coupling, and changing the boundary links one at a time bounds the expected new $d$-distance by $\sum_j C_{Bj} \rho(j)$. Put $\bar w_B = \max_{i \in B} w_i$. Minimality gives

$$\sum_{i \in B} w_i \rho(i) \le \bar w_B \sum_j C_{Bj} \rho(j),$$

and $\bar w_B \le e^{2\lambda} w_j$ whenever $C_{Bj} \neq 0$. Summing over the blocks outside $S$ and using the column sums,

$$\sum_{b \notin S} w_b \rho(b) \le \theta \Big[ \sum_{j \notin S} w_j \rho(j) + \sum_{z \in S} w_z \rho(z) \Big],$$

which gives the claim as in Lemma 241. □

**Lemma 244.** *Let $n \ge 2$ be even, let $V_x$, $x \in 2\mathbb{Z}^4$, be the translates of Sec. X, and suppose that $D(n,\alpha)$ holds.*

(a) *If a probability measure $\mu$ on $G^A$ is invariant under $\gamma_e^F$ for every $e$ in a finite set $V$, it is invariant under $\gamma_V$.*

(b) *Fix $\lambda > 0$ with $\theta = \alpha e^{\lambda \rho_n} < 1$. Let $S \subset A$, and let $\mu_1, \mu_2$ be invariant under $\gamma_{V_x}$ for every $x$ with $V_x \cap S = \emptyset$. With $T$, $w_b$ as in Lemma 241 and $\sum_{z \in S} w_z < \infty$, there is a coupling with*

$$\sum_{b \notin S} w_b \rho(b) \le \frac{D(1 + c_n e^{\lambda \rho_n})}{1-\theta} \sum_{z \in S} w_z,$$

*and $\mu_1 = \mu_2$ if $S = \emptyset$.*

*Proof.* (a) Let $\nu_\eta$ be the conditional law of $Q_V$ given the links $\eta$ outside $V$. For bounded $f$ on $G^V$ and bounded $g$ of $\eta$, $\gamma_e^F(fg) = g\,\gamma_e^F f$ for $e \in V$, so invariance gives $\int g(\eta) \big(\nu_\eta(\gamma_e^F f) - \nu_\eta(f)\big) \, d\mu = 0$ for all $g$; taking $f$ in a countable dense set of continuous functions, $\nu_\eta$ is invariant under every $\gamma_e^F(\cdot \mid \cdot, \eta)$, $e \in V$, for $\mu$-almost every $\eta$. So is $\gamma_V(\cdot \mid \eta)$, whose conditional laws of single links are the $\gamma_e^F$. The composition $P_\eta$ of the kernels $\gamma_e^F$, $e \in V$, in a fixed order has a density with respect to Haar measure on $G^V$ bounded below by a positive constant, because each single-link density is at least $e^{-12N^2\beta}$ times its maximum over the group. A Markov kernel with such a lower bound has at most one invariant probability measure (Doeblin's condition), so $\nu_\eta = \gamma_V(\cdot \mid \eta)$.

(b) Let $\pi^*$ minimise $\Psi$ as in Lemma 241 and call $x$ admissible if $V_x \cap S = \emptyset$. For admissible $x$ replace $(\sigma_{V_x}, \sigma'_{V_x})$ by an optimal coupling, for $d_{V_x}$, of the two block laws; both $\mu_i$ are invariant, so the result is a coupling, and changing the boundary links one at a time bounds the new expected $d_{V_x}$-distance by $\sum_j C_{V_x,j} \rho(j)$. With $\bar w_x = \max_{i \in V_x} w_i$, minimality gives

$$\sum_{i \in V_x} w_i \rho(i) \le \bar w_x \sum_{j \notin V_x} C_{V_x,j} \rho(j),$$

and $\bar w_x \le e^{\lambda \rho_n} w_j$ whenever $C_{V_x,j} \neq 0$. Let $S'$ be the set of links outside $S$ that lie in a translate meeting $S$; each is within distance $\rho_n$ of a link of $S$, so $\sum_{i \in S'} w_i \le c_n e^{\lambda \rho_n} \sum_{z \in S} w_z$. A link outside $S \cup S'$ lies in $N_n$ translates, all admissible. Summing over admissible $x$ and using $\sum_{x:\, j \notin V_x} C_{V_x,j} = K_{c(j)} \le \alpha N_n$,

$$N_n \sum_{i \notin S \cup S'} w_i \rho(i) \le \theta N_n \Big[ \sum_{i \notin S \cup S'} w_i \rho(i) + \sum_{i \in S \cup S'} w_i \rho(i) \Big].$$

All sums are finite and $\rho \le D$, so $\sum_{i \notin S \cup S'} w_i \rho(i) \le \frac{\theta D}{1-\theta} (1 + c_n e^{\lambda \rho_n}) \sum_{z \in S} w_z$; adding $\sum_{i \in S'} w_i \rho(i) \le D c_n e^{\lambda \rho_n} \sum_{z \in S} w_z$ gives the bound. If $S = \emptyset$ then $S' = \emptyset$ and $\rho \equiv 0$. □

The covariance bound of Corollary 242 follows from part (b) in the same way, with $\theta D/(1-\theta)$ replaced by $D(1 + c_n e^{\lambda \rho_n})/(1-\theta)$, since the conditional measures $\mu^\sigma$ there are invariant under $\gamma_{V_x}$ for $V_x \cap S_H = \emptyset$ by part (a).

## Appendix C: The plaquette blocks

Table IV lists the block plaquettes of Sec. V modulo $2\mathbb{Z}^4$: the plaquette at $x$ in the plane $\{\mu, \nu\}$ is a block plaquette exactly when $x$ modulo two and $\{\mu, \nu\}$ appear in the table. Eight lie in the plane $\{0,1\}$ and eight in $\{2,3\}$; together they contain the 48 active links of a cell of side two, each once.

In closed form, the plaquette at $x$ in $\{0,1\}$ is a block plaquette when $x_0 \equiv x_1$ and the plaquette at $x$ in $\{2,3\}$ when $x_2 \equiv x_3$; the blocks do not depend on the line field beyond the fact that every plaquette contains one fixed link. The bounds of Lemma 16 are evaluated in ball arithmetic at 128 bits: the series of $Z_0$ and of its first three derivatives in $s = r^2$ is summed to forty-eight terms, and the tail is bounded for $s \le 5$ by twice the first omitted term, because consecutive terms decrease by a factor below one half. The program also writes the constants,

TABLE IV. The sixteen block plaquettes modulo two: the base point $x$ (digits $x_0x_1x_2x_3$ modulo two) and the plane.

| $x$ | plane | $x$ | plane |
|---|---|---|---|
| 0000 | $\{0,1\}$ | 0000 | $\{2,3\}$ |
| 0001 | $\{0,1\}$ | 0011 | $\{2,3\}$ |
| 0010 | $\{0,1\}$ | 0100 | $\{2,3\}$ |
| 0011 | $\{0,1\}$ | 0111 | $\{2,3\}$ |
| 1100 | $\{0,1\}$ | 1000 | $\{2,3\}$ |
| 1101 | $\{0,1\}$ | 1011 | $\{2,3\}$ |
| 1110 | $\{0,1\}$ | 1100 | $\{2,3\}$ |
| 1111 | $\{0,1\}$ | 1111 | $\{2,3\}$ |

rounded outward to $10^{-12}$, for a check of the chain (3) at $\beta_W = \frac{1}{3}$, 0.345, 0.35, 0.351 in exact rational arithmetic by the Lean kernel (`gauge-fixed/blocks/sharp`). The constants of (3) and $\alpha$ are then evaluated at a given $\beta_W$ from exact input.

For SU(3) the bounds of Lemma 38 are evaluated on the 841 boxes of side $\frac{3}{16}$ at $r = \frac{3}{2}$ and the 784 boxes of side $\frac{19}{100}$ at $r = \frac{38}{25}$ and the 826 boxes of side $\frac{77}{400}$ at $r = \frac{77}{50}$ with the moments of Appendix E, and the constants of (15) and $\alpha_3$ are evaluated from these bounds and the cover of Proposition 34 or Proposition 35 in ball arithmetic at 120 bits. The cover of Proposition 35 is computed as in Appendix E with $r = \frac{38}{25}$ and the threshold 1.03, and with $r = \frac{77}{50}$ and the threshold 1.0275.

### Appendix D: The blocks of Lemma 20

*Diagonal blocks.* Let $w_a = e^{i\phi}$. On the diagonal part, $f_p = -\cos\phi\,(y_1+y_2-y_3-y_4)^2$ and $c_p = 2(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$, so

$$\mathscr{F} = \sum_i \alpha_i y_i^2 + \cos\phi\,(y_1+y_2-y_3-y_4)^2 - \tfrac{9}{5}(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$$

with $y_k = 0$ and weights $\alpha_{k\pm1} = \frac{21}{10}$, $\alpha_{k+2} = \frac{5}{4}$. It is affine in $\cos\phi$, so positivity at $\cos\phi = \pm1$ suffices. For $k = 1$ and $\cos\phi = 1$ the matrix in $(y_2, y_3, y_4)$ is

$$\begin{pmatrix} \frac{31}{10} & -\frac{19}{10} & -1 \\ -\frac{19}{10} & \frac{9}{4} & \frac{19}{10} \\ -1 & \frac{19}{10} & \frac{31}{10} \end{pmatrix},$$

and Table V lists the leading principal minors for every position; all are positive.

*Pair blocks.* Let $w_a = e^{i(\chi+\vartheta)}$ and $w_b = e^{i(\chi-\vartheta)}$. The terms of $f_p$ carry $w_a + \bar{w}_b = 2\cos\chi\, e^{i\vartheta}$ or its conjugate, and $c_p$ carries $w_a\bar{w}_b = e^{2i\vartheta}$, so the Hermitian matrix $M(m,\vartheta)$ of $\mathscr{F}$, $m = \cos\chi$, is affine in $m$ with $M(-1,\vartheta) = M(1,\vartheta+\pi)$, and $m = 1$ suffices. For the fixed link $e_2$, in the variables $(z_1, z_3, z_4)$ and

TABLE V. Leading principal minors of the diagonal blocks of Lemma 20.

| Fixed link | $\cos\phi = 1$ | $\cos\phi = -1$ |
|---|---|---|
| $e_1, e_4$ | $\frac{31}{10}, \frac{673}{200}, \frac{8421}{2000}$ | $\frac{11}{10}, \frac{53}{200}, \frac{21}{2000}$ |
| $e_2$ | $\frac{31}{10}, \frac{861}{100}, \frac{8421}{2000}$ | $\frac{11}{10}, \frac{21}{100}, \frac{21}{2000}$ |
| $e_3$ | $\frac{9}{4}, \frac{673}{200}, \frac{8421}{2000}$ | $\frac{1}{4}, \frac{53}{200}, \frac{21}{2000}$ |

with $c = \cos\vartheta$,

$$M = \begin{pmatrix} \frac{21}{5} + 2c & -2e^{i\vartheta} & -\frac{9}{5} - 2e^{i\vartheta} \\ -2e^{-i\vartheta} & \frac{21}{5} + 2c & \frac{9}{5} + 2e^{-i\vartheta} \\ -\frac{9}{5} - 2e^{-i\vartheta} & \frac{9}{5} + 2e^{i\vartheta} & \frac{5}{2} + 2c \end{pmatrix}.$$

With $u = \cos^2(\vartheta/2) \in [0,1]$ its leading principal minors are

$$\frac{20u+11}{5}, \quad \frac{(20u+1)(20u+21)}{25}, \quad \frac{400u(40u^2-23u+4)+21}{250},$$

and $40u^2 - 23u + 4$ has negative discriminant $529 - 640$. The other positions give the same determinant and the leading minors $\frac{20u+11}{5}$ or $\frac{8u+1}{2}$ and $\frac{800u^2-180u+53}{50}$, whose discriminant $32400 - 169600$ is negative. Every block is positive definite.

The inequality is not tight. In floating point the least eigenvalue of the whole form over $W \in \mathrm{U}(N)$, $N \le 4$, is 0.0149, which is also the least eigenvalue of the exact blocks; this slack is what the choice of the weights $\frac{21}{10}$, $\frac{5}{4}$, $\frac{9}{10}$ trades against the constant $\frac{29}{2}$.

### Appendix E: The certified computation

*Truncation.* For a corner $b$ of a box the moments of Lemma 32 are derivatives of $Z$ at $(b, \bar{b})$. The truncated series is $\mathbb{E}_H[A_t(UD)B_t(\bar{U}D')]$ with $A_t$ the part of $\sum_\lambda c_\lambda s_\lambda(UD)$ over partitions with $\lambda_1 - \lambda_3 \le 34$ and $\lambda_3 \le 11$, and $B_t$ likewise. Every omitted partition has $|\lambda| > 34$, because $3 \cdot 12 > 34$, and the omitted part is at most $T(\|D\|) = \sum_{n>34}(\frac{3}{2}\|D\|)^n/n!$ in modulus. On the polydisc $|D_{ij} - b_{ij}| \le \rho$, $\|D\| \le \|b\| + 3\rho$, and Cauchy's estimate bounds the $\alpha$-th derivative of the omitted part by $\alpha!\rho^{-|\alpha|}T(\|b\|+3\rho)$. The derivatives of the full exponential are at most $2^{-|\alpha|}e^{v/2}$ with $v = \sum_k |b_k|$, since $|\mathrm{Tr}(Ub)| \le \sum_k |b_k|$ for diagonal $b$. With $\varepsilon_\alpha = 2^{|\alpha|}\alpha!\rho^{-|\alpha|}T(\|b\|+3\rho)$ the truncated moment differs from the true one by at most $\varepsilon_\alpha e^{v/2} + (e^{v/2} + \varepsilon_\alpha)\varepsilon_\beta$, and $\rho$ is chosen to minimise $\rho^{-6}T$. The largest error bound of an unnormalised moment at any corner is $1.7 \times 10^{-11}$.

*Weight zero.* Conjugation by the diagonal torus $T$ preserves Haar measure and $\mathrm{Re}\,\mathrm{Tr}(Ub)$ for diagonal $b$, and multiplies $u_{ij}$ by $t_i t_j^{-1}$; a moment of a monomial on which $T$ acts by a nontrivial character is therefore 0, and only 8278 of the 134596 monomials of degree at most six have the trivial character. The functions $\mathrm{Tr}(X)$, $e_2(X)$, $\det X$ of $X = DD'^{\top}$ and $\det D$, $\det D'$ are invariant under the same action on the variables, so after these are formed in the full ring of Taylor polynomials of

degree six in the 18 variables, the series is computed in the subring of weight-zero monomials.

*Rounding.* The truncated series is a Taylor polynomial of total degree six in the 36 real directions of $(D-b, D'-\bar b)$, computed in double precision. Every operation is repeated on absolute values and carries an operation count $n$, so that the computed coefficient differs from the exact one by at most $\gamma_n = nu/(1-nu)$, $u = 2^{-53}$, times the absolute-value coefficient [51, Sec. 3.1]. With $C$ in $2^{-30}\mathbb{Z}$, the polynomials $2^{31}q_h$, $X_a(2^{31}q_h)$, $12\cdot 2^{31}(\varphi_h + \Delta q_h)$ and $12\cdot 2^{31}\langle \nabla\varphi_{F_k}, \nabla q_h\rangle$ for the coordinate vectors $h$ have coefficients in $\mathbb{Z}[i]$, and every partial sum in their computation is bounded, through the same operations on absolute values, by less than $2^{50}$; they are therefore computed exactly. Normalisation by $Z$, the products with the corner coordinates, the quadratic forms and their error matrices are formed with the same bookkeeping, and the largest eigenvalues are bounded in ball arithmetic [57].

*Separation.* The functional $c$ of a discarded box is the dual solution of a linear program for the gauge of the centre of the box with respect to a $72\times 72$ grid of $T$. Only the final inequality is used: $\max_T \ell$ is bounded by its maximum over a $512\times 512$ grid plus $\tfrac12 H\delta^2/2$, with $\delta = 2\pi/512$ and $H = |c_1| + |c_2| + 2|c_3|$ a bound for its Hessian on the torus, and the inequality is checked with relative margin $10^{-12}$.

## Appendix F: The transfer matrix

This appendix recalls the Osterwalder–Schrader reconstruction for the lattice [73–75] in the form used for the gaps, following our earlier paper [16].

Let $\omega$ be a state on the gauge-invariant local functions of $\mathbb{Z}^4$ that is invariant under translations and the hypercubic group and reflection positive for reflections in planes through sites and in planes bisecting links; by Proposition 11(iii)–(iv) and its analogues in the other theorems, the states of this paper are of this kind. The reconstruction gives a Hilbert space with a vector $\Omega$, a map $F \mapsto \Psi_F$ with dense range, and a self-adjoint $\mathbb{T}$ with $0 \le \mathbb{T} \le 1$, $\mathbb{T}\Omega = \Omega$ and $\langle \Psi_F, \mathbb{T}^t \Psi_H\rangle = \omega(\vartheta(\bar F)\tau_t H)$, where $\vartheta$ is the time reflection and $\tau_t$ the time translation.[26]

**Proposition 245.** *For $m > 0$ the following are equivalent:*

(a) *the spectrum of $\mathbb{T}$ on $\Omega^\perp$ lies in $[0, e^{-m}]$;*

(b) *for every $F$ in a class of gauge-invariant local functions supported at times in a bounded interval and dense in $L^2(\omega)$ on each such support, there is $C_F < \infty$ with $0 \le \omega(\vartheta(\bar F)\tau_t F) - |\omega(F)|^2 \le C_F e^{-mt}$ for all $t \ge 0$.*

*Under* (a) *one may take $C_F = \|F\|_\infty^2$ for every bounded $F$.*

[26] $\Psi_F$ is the class of $F$ in the completion of the positive-time local functions for the form $\langle F, H\rangle = \omega(\vartheta(\bar F)H)$, which is positive semidefinite by reflection positivity. Since $|\omega(\vartheta(\bar F)F)| \le \omega(|F|^2)$ by the Cauchy–Schwarz inequality and the invariance of $\omega$ under $\vartheta$, functions that are close in $L^2(\omega)$ give close vectors.

*Proof.* If (a) holds, then

$$\omega(\vartheta(\bar F)\tau_t F) - |\omega(F)|^2 = \langle \Psi'_F, \mathbb{T}^t \Psi'_F\rangle$$

with $\Psi'_F = \Psi_F - \langle \Omega, \Psi_F\rangle\Omega \in \Omega^\perp$; this number lies in $[0, e^{-mt}\|\Psi_F\|^2]$, and $\|\Psi_F\|^2 \le \|F\|_\infty^2$. Conversely, let $\sigma$ be the spectral measure of $\mathbb{T}$ at $\Psi'_F$ and let $\delta > 0$. Then

$$\sigma\big((e^{-m+\delta}, 1]\big)\, e^{(-m+\delta)t} \le C_F e^{-mt}$$

for all $t$, so $\sigma((e^{-m+\delta}, 1]) = 0$. The spectral projection of $(e^{-m}, 1]$ restricted to $\Omega^\perp$ therefore annihilates the vectors $\Psi'_F$ of the class, which are dense by the footnote above, and it vanishes. □

The class in (b) cannot be replaced by one observable. On $\mathbb{C}^3$ let $\mathbb{T} = \mathrm{diag}(1, e^{-m}, e^{-m/2})$, $\Omega = e_1$ and $\Psi'_F = e_2$: the function in (b) is $e^{-mt}$, while the spectrum on $\Omega^\perp$ reaches $e^{-m/2}$. The decay of one two-point function places only the spectral measure of $\Psi'_F$ in $[0, e^{-m}]$, and the gap follows when such vectors are dense in $\Omega^\perp$, as in (b) and in Theorem 46. Proposition 8.8 of [39] draws the gap on $\Omega^\perp$ from the two-point function of one fixed observable.

## Appendix G: Spectral measures and Laplace transforms

This appendix collects the facts on spectral measures that the continuum step uses, with proofs.

**Lemma 246.** *Let $T$ be a self-adjoint operator on a Hilbert space with $0 \le T \le 1$ and let $\psi$ be a unit vector. There is a unique probability measure $\nu$ on $[0,1]$ with $\langle \psi, h(T)\psi\rangle = \int h\, d\nu$ for every bounded Borel function $h$ on $[0,1]$. If $T\psi' = \psi'$ implies $\langle \psi', \psi\rangle = 0$, then $\nu(\{1\}) = 0$. If $\mu$ is the image of $\nu$ under $\lambda \mapsto -c^{-1}\log\lambda \in [0,\infty]$ for some $c > 0$, then $\langle \psi, T^{\sigma/c}\psi\rangle = \int_{[0,\infty]} e^{-\sigma E} d\mu(E)$ for $\sigma > 0$, with the integrand $0$ at $E = \infty$.*

*Proof.* The spectral theorem [78, Sec. VII.2] gives the measure $\nu$, carried by the spectrum of $T$, which lies in $[0,1]$; uniqueness holds because the moments $\int \lambda^n d\nu = \langle \psi, T^n\psi\rangle$ determine a measure on a compact interval, polynomials being dense in $C[0,1]$. The spectral projection $P = \mathbf{1}_{\{1\}}(T)$ projects onto the eigenvectors of $T$ with eigenvalue 1, so $\nu(\{1\}) = \|P\psi\|^2 = \langle P\psi, \psi\rangle = 0$ under the hypothesis. The last identity is the change of variables $\lambda^{\sigma/c} = e^{-\sigma E}$ for $\lambda = e^{-cE}$, with $\lambda = 0$ corresponding to $E = \infty$. □

**Lemma 247.** *Every sequence of probability measures on a compact metric space $K$ has a subsequence that converges weakly, that is, against every continuous function, to a probability measure on $K$.*

*Proof.* $C(K)$ is separable; let $(h_j)$ be dense. The numbers $\int h_j\, d\mu_k$ are bounded by $\|h_j\|_\infty$, so a diagonal subsequence makes all of them converge. By density the limit $\Lambda(h) = \lim \int h\, d\mu_k$ exists for every $h \in C(K)$; $\Lambda$ is linear, positive and $\Lambda(1) = 1$, so by the Riesz representation theorem it is integration against a probability measure. □

**Lemma 248.** *Let $\mu$ be a probability measure on $[0,\infty]$ and $\rho(\sigma)=\int e^{-\sigma E}d\mu(E)$ for $\sigma>0$.*

(a) *$\rho$ is nonincreasing and log-convex: $\rho(\sigma)^2\le\rho(\sigma-s)\rho(\sigma+s)$ for $0<s<\sigma$.*

(b) *$\rho(\sigma)>0$ for one $\sigma>0$ if and only if $\mu([0,\infty))>0$, and then $\rho(\sigma)>0$ for every $\sigma>0$.*

(c) *$\rho(\sigma)\to\mu([0,\infty))$ as $\sigma\downarrow 0$.*

(d) *If $\mu$ is carried by $[m,\infty]$, then $\rho(\sigma)\le e^{-m\sigma}$.*

(e) *For $M>0$, $\mu([0,M])\ge\rho(\sigma)-e^{-\sigma M}$.*

(f) *If $\mu_k\to\mu$ weakly on $[0,\infty]$, then $\int e^{-\sigma E}d\mu_k\to\rho(\sigma)$ for every $\sigma>0$.*

*Proof.* (a) The integrand decreases in $\sigma$. By the Cauchy–Schwarz inequality,

$$\Big(\int e^{-\sigma E}d\mu\Big)^2\le\int e^{-(\sigma-s)E}d\mu\int e^{-(\sigma+s)E}d\mu.$$

(b) The integrand is positive exactly on $[0,\infty)$. (c) The integrand increases to $\mathbf{1}_{[0,\infty)}$ as $\sigma\downarrow 0$, and monotone convergence applies. (d) The integrand is at most $e^{-m\sigma}$ on $[m,\infty]$. (e) Split the integral at $M$: it is at most $\mu([0,M])+e^{-\sigma M}$. (f) For $\sigma>0$ the function $E\mapsto e^{-\sigma E}$, extended by 0 at $\infty$, is continuous on $[0,\infty]$. □

By (b) and (e), a lower bound $\rho(\sigma)\ge c>0$ at one $\sigma$ places mass at least $c/2$ on the bounded interval $[0,\sigma^{-1}\log(2/c)]$; for the measures $\mu_k$ of Theorem 48 a bound uniform in $k$ therefore makes them tight on $[0,\infty)$ with a fixed fraction of their mass.

Part (b) also excludes a positive Laplace representation for a spectral cutoff of compact support. If $\chi(\lambda)=\int_{[0,\infty)}e^{-s\lambda}d\nu(s)$ with $\nu$ a positive measure and $\chi(0)=\nu([0,\infty))>0$, then (b), applied to $\nu/\chi(0)$, gives $\chi(\lambda)>0$ for every $\lambda>0$, so $\chi$ vanishes nowhere. The cutoff $\chi_\sigma$ of [39, Sec. 9, Eqs. (9.11), (9.12)] equals 1 on $[0,\sigma]$ and 0 on $[2\sigma,\infty)$ and is asserted to have such a representation.

### Appendix H: The character expansion

Let $G$ be a compact connected Lie group with normalised Haar measure $dU$. For an irreducible unitary representation $\rho_\lambda$ of dimension $d_\lambda$ write $\chi_\lambda=\operatorname{Tr}\rho_\lambda$. The characters form an orthonormal basis of the square-integrable class functions (Peter–Weyl and Schur orthogonality [26, Chs. 2–4]).

**Lemma 249.** *Let $w\in L^1(G)$ be a class function with $w(U^{-1})=w(U)$, and put*

$$\hat c_\lambda=d_\lambda^{-1}\int w\,\chi_\lambda\,dU.$$

*Then $\int w(U)\rho_\lambda(U)\,dU=\hat c_\lambda I$, and $\int w(U)\chi_\lambda(U^{-1}V)\,dU=\hat c_\lambda\chi_\lambda(V)$.*

*Proof.* $M=\int w(U)\rho_\lambda(U)dU$ commutes with every $\rho_\lambda(V)$, since $w(VUV^{-1})=w(U)$ and Haar measure is invariant; by Schur's lemma $M=cI$, and taking traces gives $c=\hat c_\lambda$. Then $\int w(U)\chi_\lambda(U^{-1}V)dU=\operatorname{Tr}(\int w(U)\rho_\lambda(U^{-1})dU\,\rho_\lambda(V))$, and the substitution $U\mapsto U^{-1}$ with $w(U^{-1})=w(U)$ turns the integral into $M$. □

**Lemma 250.** *Let $G=\mathrm{SU}(N)$ and $\beta>0$, and let*

$$w_\beta(U)=e^{\frac{\beta}{N}\operatorname{Re}\operatorname{Tr}U}\Big/\int e^{\frac{\beta}{N}\operatorname{Re}\operatorname{Tr}V}dV.$$

*Then $\hat c_\lambda(\beta)=d_\lambda^{-1}\int w_\beta\chi_\lambda\,dU$ is real, $0\le\hat c_\lambda(\beta)\le 1$, and $\hat c_\lambda(\beta)<1$ for $\lambda\neq 0$. Moreover*

$$\hat c_\lambda(\beta)=1-\frac{NC_2(\lambda)}{\beta}+O(\beta^{-2})\qquad(\beta\to\infty),$$

*where $-\sum_a\rho_\lambda(T_a)^2=2C_2(\lambda)I$ for a basis $(T_a)$ of $\mathfrak{su}(N)$ orthonormal for $|X|^2=-\operatorname{Tr}X^2$, so that $C_2=(N^2-1)/(2N)$ on the fundamental representation.*

*Proof.* Since $w_\beta(\bar U)=w_\beta(U)$ and $\chi_\lambda(\bar U)=\overline{\chi_\lambda(U)}$, and complex conjugation preserves Haar measure, $\int w_\beta\chi_\lambda=\int w_\beta\bar\chi_\lambda$, which is the complex conjugate of the left side; so $\hat c_\lambda$ is real. Next, $w_\beta$ is a multiple of $\exp(\frac{\beta}{2N}(\chi_{\mathrm F}+\bar\chi_{\mathrm F}))=\sum_m\frac{1}{m!}(\frac{\beta}{2N})^m(\chi_{\mathrm F}+\bar\chi_{\mathrm F})^m$, and each power is the character of a representation, hence a sum of irreducible characters with nonnegative integer coefficients. By orthonormality $\int w_\beta\bar\chi_\lambda\ge 0$, so $\hat c_\lambda\ge 0$. As $|\chi_\lambda(U)|\le d_\lambda$, with $\operatorname{Re}\chi_\lambda(U)=d_\lambda$ only where $\rho_\lambda(U)=I$, a proper closed subgroup when $\lambda\neq 0$ and so a Haar null set, and $w_\beta>0$ everywhere, $\hat c_\lambda=d_\lambda^{-1}\int w_\beta\operatorname{Re}\chi_\lambda<1$ for $\lambda\neq 0$.

For the expansion, $\operatorname{Re}\operatorname{Tr}U\le N$ with equality only at $U=I$, so outside a neighbourhood $\{U=e^X:|X|<\delta\}$ both integrals $\int e^{\frac{\beta}{N}\operatorname{Re}\operatorname{Tr}U}\chi_\lambda$ and $\int e^{\frac{\beta}{N}\operatorname{Re}\operatorname{Tr}U}$ are smaller than their values near $I$ by a factor $e^{-c\beta}$. Near $I$ write $U=e^X$; the Haar density in these coordinates is $j(X)=1+O(|X|^2)$, even in $X$. For antihermitian $X$, $\operatorname{Tr}X^3$ is imaginary, so

$$\tfrac{1}{N}\operatorname{Re}\operatorname{Tr}e^X=1-\tfrac{1}{2N}|X|^2+r(X),\qquad 0\le r(X)\le C|X|^4,$$

and $d_\lambda^{-1}\chi_\lambda(e^X)=1+d_\lambda^{-1}\operatorname{Tr}\rho_\lambda(X)+\frac{1}{2d_\lambda}\operatorname{Tr}\rho_\lambda(X)^2+c_3(X)+O(|X|^4)$, with $c_3$ cubic and $\operatorname{Tr}\rho_\lambda(X)=0$, since $\mathfrak{su}(N)$ is perfect. Let $\langle\cdot\rangle$ denote the average against the Gaussian weight $e^{-\beta|X|^2/(2N)}$, under which the coordinates $x_a$ of $X=\sum x_aT_a$ are independent with variance $N/\beta$. With $h=je^{\beta r}$,

$$\begin{aligned}\hat c_\lambda(\beta)&=\frac{\langle d_\lambda^{-1}\chi_\lambda(e^X)\,h\rangle}{\langle h\rangle}+O(e^{-c\beta})\\&=1+\frac{\langle\frac{1}{2d_\lambda}\operatorname{Tr}\rho_\lambda(X)^2h\rangle}{\langle h\rangle}+\frac{\langle(c_3(X)+O(|X|^4))h\rangle}{\langle h\rangle},\end{aligned}$$

the averages being taken over $|X|<\delta$. Under $\langle\cdot\rangle$, $|X|^{2m}$ has mean of order $\beta^{-m}$ and $\beta|r(X)|\le C\beta|X|^4$ has mean of order $\beta^{-1}$, so $h=1+O(\beta^{-1})$ in every moment. The weight $h$ is even in $X$, since $\operatorname{Tr}X^k$ is imaginary for odd $k$ and $j$ is even, so the

odd term $c_3$ contributes nothing; the quartic remainder and the correlation of the quadratic term with $h-1$ contribute $O(\beta^{-2})$. Finally $\langle \operatorname{Tr}\rho_\lambda(X)^2\rangle = \frac{N}{\beta}\sum_a \operatorname{Tr}\rho_\lambda(T_a)^2 = -\frac{2NC_2(\lambda)d_\lambda}{\beta}$, which gives the expansion. For $\mathrm{SU}(2)$ and the representation of spin $j$ the integrals are explicit, $\hat{c}_j(\beta) = I_{2j+1}(\beta)/I_1(\beta)$ with the modified Bessel functions $I_n$, in agreement with $C_2 = j(j+1)$. □

## Appendix I: The single-link law of SU(2)

This appendix proves the sharp quarter, $W_1(\nu_b,\nu_{b'}) \le \frac{1}{4}|b-b'|$ for the one-link law of $\mathrm{SU}(2)$, which Theorems 1 and 3 use, and the exact value at $b=0$, which Theorem 2 uses. The argument is that of our earlier paper [16]; it is given in full so that the present paper is self-contained.

### 1. The von Mises–Fisher law

Identify $\mathrm{SU}(2)$ with the unit quaternions, that is with the unit sphere $S^3 \subset \mathbb{R}^4$, so that $\operatorname{Re}\operatorname{Tr} U = 2x_0(U)$ and $\operatorname{Re}\operatorname{Tr}(UV) = 2\langle u,\bar{v}\rangle$ for the Euclidean inner product of $\mathbb{R}^4$ and quaternion conjugation $v \mapsto \bar{v}$. Fix a link $e$ and the field off $e$. Every plaquette through $e$ can be rotated cyclically so that its trace reads $\operatorname{Re}\operatorname{Tr}(Q_e A_p)$ with $A_p \in \mathrm{SU}(2)$, the staple of $e$ in $p$. The action therefore depends on $Q_e$ only through

$$N\beta \sum_{p\ni e} \operatorname{Re}\operatorname{Tr}(Q_e A_p) = \langle x, b\rangle, \qquad b = 4\beta \sum_{p\ni e} \bar{A}_p \in \mathbb{R}^4,$$

with $x = Q_e \in S^3$. Since there are six staples, each of norm one,

$$\kappa := |b| \le y := 24|\beta|. \tag{I1}$$

The conditional law of $Q_e$ is the von Mises–Fisher law[27]

$$\nu_b(dx) = Z(\kappa)^{-1} e^{\langle x,b\rangle}\, \sigma(dx) \tag{I2}$$

on the unit sphere, with $\sigma$ the normalised surface measure. We use coordinates in which $b = \kappa e_1$. The density of $t = x_1$ is proportional to $(1-t^2)^{1/2}e^{\kappa t}$ on $[-1,1]$, so $Z(\kappa) = 2I_1(\kappa)/\kappa$ with the modified Bessel function $I_1$, normalised so that $Z(0) = 1$.

**Lemma 251.** *For every $\kappa \ge 0$ the law $\nu_b$ on the unit sphere $S^3$ satisfies the Poincaré inequality $\operatorname{Var}_{\nu_b}(g) \le \frac{1}{3}\int |\nabla g|^2 d\nu_b$, and the constant $\frac{1}{3}$ is attained at $\kappa = 0$.*

*Proof.* In the polar angle $\theta \in (0,\pi)$ from $e_1$ the generator of the reversible diffusion of $\nu_b$ is $L = \partial_\theta^2 + b(\theta)\partial_\theta + \sin^{-2}\theta\,\Delta_{S^2}$ with $b(\theta) = 2\cot\theta - \kappa\sin\theta$. On spherical harmonics of degree $\ell$ on $S^2$ it acts as a Sturm–Liouville operator $L_\ell$ with potential $\ell(\ell+1)\sin^{-2}\theta$. A ground state comparison of Barta's type [10] with the trial function $\psi = \sin\theta\, e^{a\kappa\cos\theta}$ gives $-L_1\psi \ge 3\psi$ for $a = -\frac{1}{5}$, and the same trial function with $a = -\frac{2}{5}$ gives the bound 3 for the derivatives of zonal eigenfunctions; Appendix J carries out both computations. Hence every nonzero eigenvalue of $-L$ is at least 3. At $\kappa = 0$ the first eigenvalue of the Laplacian of the unit $S^3$ is 3. □

### 2. Moments

Let $A(\kappa) = \mathbb{E}_{\nu_b} x_1 = Z'(\kappa)/Z(\kappa) = I_2(\kappa)/I_1(\kappa)$.

**Lemma 252.** *For every $\kappa > 0$, in coordinates with $b = \kappa e_1$,*

(i) $\mathbb{E}x_2^2 = \mathbb{E}x_3^2 = \mathbb{E}x_4^2 = A/\kappa$, $\mathbb{E}(1-x_1^2) = 3A/\kappa$, *the covariance matrix of $x$ is* $\operatorname{diag}(A', A/\kappa, A/\kappa, A/\kappa)$, *and* $A' = 1 - A^2 - 3A/\kappa$;

(ii) $$\frac{\kappa}{2+\sqrt{\kappa^2+4}} \le A(\kappa) \le \frac{\kappa}{1+\sqrt{\kappa^2+9}};$$

(iii) $$A' \le A/\kappa \le \left(1+\sqrt{\kappa^2+9}\right)^{-1} \le \tfrac{1}{4};$$

(iv) $$A(\kappa) \ge \kappa\big/\left(\tfrac{3}{2} + \sqrt{\kappa^2 + 25/4}\right).$$

*Proof.* (i) Integrating by parts against $\frac{d}{dt}(1-t^2)^{3/2} = -3t(1-t^2)^{1/2}$ gives $\kappa\mathbb{E}(1-x_1^2) = 3\mathbb{E}x_1$, and the three coordinates $x_2, x_3, x_4$ are exchangeable under $\nu_b$, with $\sum_i x_i^2 = 1$. The off-diagonal covariances vanish by the reflections $x_i \mapsto -x_i$, $i \ge 2$, and $A' = \operatorname{Var} x_1$ because $A = (\log Z)'$. (ii) By (i), $A$ solves the Riccati equation $A' = 1 - A^2 - 3A/\kappa$ with $A(\kappa) = \kappa/4 + O(\kappa^3)$. Appendix J shows that the left side of (ii) is a subsolution and the right side a supersolution of this equation, with the same behaviour at 0, and concludes by comparison. (iii) The middle inequality is (ii). For the first, put $s = \sqrt{\kappa^2+4}$. By (ii), $4A/\kappa \ge 4/(2+s)$ and $1 - A^2 \le 1 - \kappa^2/(2+s)^2$, and

$$1 - \frac{\kappa^2}{(2+s)^2} \le \frac{4}{2+s} \iff (2+s)^2 - \kappa^2 \le 4(2+s)$$
$$\iff s^2 - \kappa^2 \le 4,$$

which holds with equality. So $A' = 1 - A^2 - 3A/\kappa \le A/\kappa$. (iv) is the comparison of (ii) with a third barrier, which improves the lower bound of (ii); it is carried out in the same appendix. □

### 3. The Lipschitz rate of the family

Let $\delta$ be the chordal metric $\delta(x,x') = |x-x'|$ on $S^3$ and $W_1$ the Kantorovich distance for $\delta$. For $c \in \mathbb{R}^4$ and a unit vector $h \in \mathbb{R}^4$ put

$$L(c,h) = \sup_g \operatorname{Cov}_{\nu_c}\big(g, \langle h, x\rangle\big), \tag{I3}$$

[27] On the sphere $S^{p-1} \subset \mathbb{R}^p$ the von Mises–Fisher law with mean direction $\mu$ and concentration $\kappa$ has density proportional to $e^{\kappa\langle x,\mu\rangle}$ with respect to the surface measure, and its normalising constant is $(2\pi)^{p/2} I_{p/2-1}(\kappa)/\kappa^{p/2-1}$; it is the maximum entropy law with prescribed mean resultant, and it plays on the sphere the role of the Gaussian in $\mathbb{R}^p$. For $p = 4$ the Bessel index is 1. The mean resultant length $\mathbb{E}\langle x,\mu\rangle = I_{p/2}(\kappa)/I_{p/2-1}(\kappa)$ is the function $A$ of Lemma 252. See Mardia and Jupp [68], Ch. 9, for these facts and for the tangent-normal decomposition used in Appendix J.

the supremum over 1-Lipschitz functions $g$ for $\delta$. Since $\frac{d}{dt}\nu_{c+th}(g)=\mathrm{Cov}_{\nu_{c+th}}(g,\langle h,x\rangle)$, the Kantorovich–Rubinstein duality gives, for all $b,b'$,

$$W_1(\nu_b,\nu_{b'})\le |b-b'|\sup_{c\in[b,b']} L\Big(c,\frac{b'-b}{|b'-b|}\Big). \tag{I4}$$

By rotation invariance $L(c,h)$ depends only on $\kappa=|c|$ and on the angle $\varphi$ between $h$ and $c^{\perp}$, and we write $L(\kappa,\varphi)$.

**Lemma 253.** *For all $c$ and $h$, $L(c,h)\le 1/(2\sqrt{3})$. Moreover:*

(i) *at $c=0$, $L(0,h)=\frac14$ for every $h$;*

(ii) *if $h\perp c$, then $L(c,h)=\mathbb{E}_{\nu_c}x_2^2=A(\kappa)/\kappa\le\frac14$;*

(iii) *if $\kappa\ge\sqrt{88}/3\approx 3.127$, then $L(c,h)\le\frac14$ for every $h$.*

*Proof.* A 1-Lipschitz function for $\delta$ is 1-Lipschitz for the geodesic distance, which is larger, so $|\nabla g|\le 1$ almost everywhere, and Lemma 251 gives $\mathrm{Var}\,g\le\frac13$. By Lemma 252(i), (iii) the largest eigenvalue of the covariance of $x$ is $A/\kappa\le\frac14$, so $\mathrm{Var}\langle h,x\rangle\le A/\kappa$. The Cauchy–Schwarz inequality gives

$$L(c,h)\le\big(\tfrac13 A/\kappa\big)^{1/2}\le\Big(3\big(1+\sqrt{\kappa^2+9}\big)\Big)^{-1/2}\le\frac{1}{2\sqrt{3}}. \tag{I5}$$

(i) and (ii) are reflection arguments. Let $R$ be the reflection in the hyperplane $h^{\perp}$, and suppose $\nu_c$ is $R$-invariant, which is the case when $c=0$ or $h\perp c$. Then $\mathbb{E}\langle h,x\rangle=0$ and

$$\begin{aligned}\mathrm{Cov}(g,\langle h,x\rangle)&=\mathbb{E}\big[(g(x)-g(Rx))\langle h,x\rangle;\ \langle h,x\rangle>0\big]\\&\le\mathbb{E}\big[2\langle h,x\rangle^2;\ \langle h,x\rangle>0\big]=\mathbb{E}\langle h,x\rangle^2,\end{aligned}$$

since $\delta(x,Rx)=2|\langle h,x\rangle|$. Equality holds for $g=\langle h,x\rangle$, which is 1-Lipschitz. At $c=0$, $\mathbb{E}\langle h,x\rangle^2=\frac14$; for $h\perp c$, $\mathbb{E}\langle h,x\rangle^2=A/\kappa$. (iii) By (I5), $L\le\frac14$ as soon as $1+\sqrt{\kappa^2+9}\ge\frac{16}{3}$, that is $\kappa^2\ge\frac{88}{9}$. □

The reflection settles the directions $h\perp c$ and the point $c=0$; for an oblique $h$ the reflection no longer preserves $\nu_c$, and the Cauchy–Schwarz bound (I5) is too weak for $\kappa<\sqrt{88}/3$. The next theorem closes this case.

**Theorem 254** (The sharp quarter)**.** *For every $c\in\mathbb{R}^4$ and every unit vector $h\in\mathbb{R}^4$,*

$$L(c,h)\le\tfrac14,$$

*with equality at $c=0$. Equivalently, $W_1(\nu_b,\nu_{b'})\le\frac14|b-b'|$ for all $b,b'\in\mathbb{R}^4$, and the constant $\frac14$ is the best possible.*

The equivalence is (I4) in one direction and the derivative formula above it in the other; the sharpness is Lemma 253(i). The metric matters.[28] The proof has four steps. A Lipschitz function on $S^3$ extends to the unit ball $B\subset\mathbb{R}^4$, and Green's formula in $B$ bounds the covariance by a negative Sobolev norm of the density (Lemma 255). This norm splits into a zonal and a transverse part, both explicit series in modified Bessel functions (Lemma 256). The transverse part is at most $\frac14$ because a certain power series has nonnegative coefficients (Proposition 258), and the zonal part is at most $\frac14$ by a Riccati comparison (Proposition 259).

### 4. Harmonic extension

Let $\sigma$ be the normalised surface measure of $S^3$ and $H_k$ the space of spherical harmonics of degree $k$, the restrictions to $S^3$ of harmonic homogeneous polynomials of degree $k$ on $\mathbb{R}^4$. The spaces $H_k$ are orthogonal in $L^2(\sigma)$ and span it; we write $u_k$ for the component of $u$ in $H_k$, and for $u$ with $\int u\,d\sigma=0$ we put

$$\mathscr{Q}(u)=\sum_{k\ge1}\frac1k\,\|u_k\|^2_{L^2(\sigma)}. \tag{I6}$$

This is $\langle u,\Lambda^{-1}u\rangle$ for the Dirichlet-to-Neumann operator $\Lambda$ of $B$, which acts on $H_k$ as multiplication by $k$.

**Lemma 255.** *Let $u$ be a real-analytic function on $S^3$ with $\int u\,d\sigma=0$ and let $g$ be 1-Lipschitz for the chordal metric. Then*

$$\int_{S^3} g\,u\,d\sigma\le\tfrac12\,\mathscr{Q}(u)^{1/2}.$$

*Proof.* The chordal metric is the Euclidean metric of $\mathbb{R}^4$ restricted to $S^3$, so McShane's extension $G(y)=\inf_{x\in S^3}\big(g(x)+|x-y|\big)$ is 1-Lipschitz on $\mathbb{R}^4$ and equals $g$ on $S^3$ [69]; by Rademacher's theorem $|\nabla G|\le1$ almost everywhere. Let $\psi=\sum_{k\ge1}k^{-1}r^k u_k(\omega)$ in polar coordinates $y=r\omega$. Each term is a harmonic polynomial, the series converges with all derivatives on the closed ball because $u$ is real-analytic,[29] and $\partial_r\psi=u$ on $S^3$. Green's formula for the Lipschitz function $G$ and the smooth harmonic function $\psi$ reads $\int_{S^3}G\,\partial_r\psi\,dS=\int_B\nabla G\cdot\nabla\psi\,dy$. With $|S^3|=2\pi^2$ and $|B|=\pi^2/2$ the Cauchy–Schwarz inequality gives

$$\int g\,u\,d\sigma=\frac{1}{2\pi^2}\int_B\nabla G\cdot\nabla\psi\,dy\le\frac{|B|^{1/2}}{2\pi^2}\Big(\int_B|\nabla\psi|^2dy\Big)^{1/2}.$$

Green's formula once more gives $\int_B|\nabla\psi|^2=\int_{S^3}\psi\,\partial_r\psi\,dS=2\pi^2\sum_k k^{-1}\|u_k\|^2=2\pi^2\mathscr{Q}(u)$, and the right side becomes $\frac12\mathscr{Q}(u)^{1/2}$. □

The lemma is a form of Thomson's principle. By Kantorovich–Rubinstein duality, $\sup_g\int g\,u\,d\sigma$ is the least cost

[28] In the geodesic metric of $S^3$ the statement is false already at $c=0$: the function $\arcsin x_2$ is geodesically 1-Lipschitz and its covariance with $x_2$ under the uniform law is $\frac{8}{9\pi}\approx0.283>\frac14$. A proof therefore has to use that the chordal distance between two points is shorter than their geodesic distance, that is, it has to see the sphere from the ball it bounds. The proof below does exactly this: it extends a chordally Lipschitz function to the ball and integrates by parts there.

[29] For real-analytic $u$ the norms $\|u_k\|$ decay geometrically in $k$, and the supremum of a function in $H_k$ is at most $(\dim H_k)^{1/2}=k+1$ times its $L^2(\sigma)$ norm, so the series for $\psi$ and for each of its derivatives converge uniformly on $\bar B$.

of a flow in $\mathbb{R}^4$ that carries the positive part of $u\,d\sigma$ to its negative part, the cost being the total length travelled by the mass.[30] The harmonic field $\nabla\psi$ is one such flow, through the interior of the ball, and the lemma computes its cost. At $c=0$ the flow is optimal: for $u=\langle h,x\rangle$ one has $\psi=\langle h,y\rangle$, the field $\nabla\psi=h$ is constant, the Cauchy–Schwarz inequality is an equality, and $\mathscr{Q}(u)=\|u\|^2=\frac14$.

### 5. Spectral decomposition

Fix $c=\kappa e_1$ with $\kappa>0$ and write $h=\cos\varphi\,e_1+\sin\varphi\,e_2$, which is no loss of generality by rotation invariance. Let $\rho=d\nu_c/d\sigma=e^{\kappa x_1}/Z(\kappa)$, $Z(\kappa)=\int e^{\kappa x_1}d\sigma=2I_1(\kappa)/\kappa$, and

$$u=\big(\langle h,x\rangle-\mathbb{E}_{\nu_c}\langle h,x\rangle\big)\rho,$$

so that $\operatorname{Cov}_{\nu_c}(g,\langle h,x\rangle)=\int g u\,d\sigma$ and $\int u\,d\sigma=0$. Let $U_n$ be the Chebyshev polynomials of the second kind and put

$$d_n(\kappa)=\frac{(n+1)\,I_{n+1}(\kappa)}{I_1(\kappa)},\qquad n\ge 0. \tag{I7}$$

**Lemma 256.** *With this notation*

$$\mathscr{Q}(u)=\cos^2\varphi\,\mathscr{Q}_z(\kappa)+\sin^2\varphi\,\mathscr{Q}_x(\kappa),$$

*where*

$$\mathscr{Q}_z=\sum_{n\ge1}\frac{d_n'(\kappa)^2}{n},\qquad \mathscr{Q}_x=\frac{1}{3\kappa^2}\sum_{n\ge1}(n+2)\,d_n(\kappa)^2.$$

*Proof.* The functions $U_n(x_1)$, $n\ge0$, are the zonal harmonics: $U_n(x_1)\in H_n$, and they are orthonormal in $L^2(\sigma)$ (Appendix K). The expansion $e^{\kappa t}=\frac{2}{\kappa}\sum_n(n+1)I_{n+1}(\kappa)U_n(t)$ of the same appendix gives $\rho=\sum_n d_n(\kappa)U_n(x_1)$, with $d_0=1$.

Write $u=\cos\varphi\,u_z+\sin\varphi\,u_x$ with $u_z=(x_1-A)\rho$ and $u_x=x_2\rho$; here $\mathbb{E}_{\nu_c}x_1=A$ and $\mathbb{E}_{\nu_c}x_2=0$. The reflection $x_2\mapsto-x_2$ maps each $H_k$ to itself, fixes $u_z$ and reverses $u_x$, so the components of $u_z$ and $u_x$ in each $H_k$ are orthogonal and the cross term in $\mathscr{Q}(u)$ vanishes.

Since $A=Z'/Z$, we have $u_z=\partial_\kappa\rho=\sum_n d_n'(\kappa)U_n(x_1)$, and the component of $u_z$ in $H_n$ is $d_n'U_n(x_1)$, of norm $|d_n'|$. This gives $\mathscr{Q}_z$.

For $u_x$, let $y(t)=\cos t\,e_1+\sin t\,e_2$. The law $\nu_{\kappa y(t)}$ has density $\rho_t=e^{\kappa\langle y(t),x\rangle}/Z(\kappa)=\sum_n d_n(\kappa)U_n(\langle y(t),x\rangle)$, and $\partial_t\rho_t|_{t=0}=\kappa x_2\rho$. The function $U_n(\langle y(t),x\rangle)$ lies in $H_n$ for every $t$, hence so does its derivative $x_2U_n'(x_1)$ at $t=0$, and the component of $u_x$ in $H_n$ is $\kappa^{-1}d_nx_2U_n'(x_1)$. Its norm follows from the eigenvalue $n(n+2)$ of the Laplacian of $S^3$ on $H_n$. For $f=U_n(x_1)$,

$$\begin{aligned}n(n+2)=n(n+2)\|f\|^2&=\int|\nabla_{S^3}f|^2d\sigma\\&=\int U_n'(x_1)^2(1-x_1^2)\,d\sigma,\end{aligned}$$

since the tangential gradient of $x_1$ has squared length $1-x_1^2$. Given $x_1$, the point $(x_2,x_3,x_4)$ is uniform on the sphere of radius $(1-x_1^2)^{1/2}$, so $\mathbb{E}[x_2^2\mid x_1]=\frac13(1-x_1^2)$ and $\|x_2U_n'(x_1)\|^2=\frac13n(n+2)$. Summing $n^{-1}\kappa^{-2}d_n^2\cdot\frac13n(n+2)$ over $n\ge1$ gives $\mathscr{Q}_x$. □

By Lemmas 255 and 256,

$$L(\kappa,\varphi)\le\tfrac12\max\big(\mathscr{Q}_z(\kappa),\mathscr{Q}_x(\kappa)\big)^{1/2}, \tag{I8}$$

and Theorem 254 follows once both parts are at most $\frac14$. Both are $\frac14$ at $\kappa=0$, by continuity from the values $d_1\sim\kappa/2$, $d_1'\to\frac12$ and $d_n=O(\kappa^n)$.

### 6. Three Bessel sums

The series of Lemma 256 reduce to closed forms. Write $I_j$ for $I_j(\kappa)$.

**Lemma 257.** *For every $\kappa>0$,*

$$\sum_{j\ge1}j^2I_j^2=\frac{\kappa}{4}I_1(2\kappa),\qquad \sum_{j\ge1}j^3I_j^2=\frac{\kappa^2}{4}\big(I_0^2+I_1^2\big),$$
$$I_1(2\kappa)=2\sum_{j\ge1}I_{j-1}I_j.$$

The first and third identities follow from Parseval's formula for $e^{\kappa\cos\theta}=\sum_{j\in\mathbb{Z}}I_je^{ij\theta}$, and the second from the recurrences of $I_j$ by telescoping; Appendix K gives the proofs.

### 7. The transverse part

**Proposition 258.** *$\mathscr{Q}_x(\kappa)\le\frac14$ for every $\kappa\ge0$.*

*Proof.* By (I7), $\mathscr{Q}_x=(3\kappa^2I_1^2)^{-1}\sum_{j\ge2}j^2(j+1)I_j^2$, and the sum equals $\sum_{j\ge1}(j^3+j^2)I_j^2-2I_1^2$. By Lemma 257, $\mathscr{Q}_x\le\frac14$ is equivalent to

$$\mathscr{D}(\kappa):=(8+2\kappa^2)I_1^2-\kappa^2I_0^2-\kappa I_1(2\kappa)\ge0. \tag{I9}$$

Each term is an entire function of $\kappa$ that is even after the factor $\kappa^2$ is removed, so $\mathscr{D}(\kappa)=\sum_{n\ge0}\mathscr{D}_n\kappa^{2n}$. Lemma 261 of Appendix K computes the coefficients: $\mathscr{D}_0=\mathscr{D}_1=0$, and for $n=m+1\ge2$

$$\mathscr{D}_n=\frac{N_m}{4^m\,m!\,(m+1)!\,(m+1)(m+2)},$$

[30] This is the Beckmann formulation of the transport problem with cost $|x-y|$: the optimal cost equals the least $\int|V|$ over vector measures $V$ on $\mathbb{R}^4$ with divergence $u\,d\sigma$. Restricting the flow to the closed ball does not change the infimum, since projecting onto $\bar B$ shortens paths. The gradient $\nabla\psi$ on $B$ is one admissible flow. It minimises the energy $\int|V|^2$ rather than the cost $\int|V|$, and the Cauchy–Schwarz inequality converts one into the other at the price of the factor $|B|^{1/2}$.

with

$$N_m = \binom{2m}{m}\big(4(2m+1)+(m-1)(m+1)(m+2)\big) - 4^m(m+1)(m+2).$$

The numerator $N_m$ vanishes for $m=0,1,2$ and is positive for $m\ge 3$ (Lemma 261). So every coefficient of $\mathscr{D}$ is nonnegative and $\mathscr{D}\ge 0$. □

The first nonzero coefficient is $\mathscr{D}_4 = \frac{1}{2304}$, so $\frac14 - \mathscr{Q}_x = \mathscr{D}/(12\kappa^2 I_1^2)$ is of order $\kappa^4$ at 0: the transverse part leaves the value $\frac14$ very slowly, which is why no estimate that loses a constant can prove the proposition.

### 8. The zonal part

Let $A = I_2/I_1$ as before, so that $d_1 = 2A$ and, by Lemma 252(i), $A' = 1 - A^2 - 3A/\kappa$.

**Proposition 259.** *For $0<\kappa\le\sqrt{88}/3$,*

$$\mathscr{Q}_z(\kappa) \le \frac14 + 4A'^2 + \frac{A^2}{4} - \frac{A}{2\kappa} - \frac{2A^2}{\kappa^2} < \frac14.$$

*Proof. The higher modes.* With $r_\nu = I_{\nu+1}/I_\nu$ the recurrence $I'_\nu = I_{\nu+1} + \frac{\nu}{\kappa} I_\nu$ gives

$$\frac{d'_n}{d_n} = \frac{I'_{n+1}}{I_{n+1}} - \frac{I'_1}{I_1} = \frac{n}{\kappa} + r_{n+1} - r_1.$$

By Lemma 262 of Appendix K, $r_{n+1}\le r_1$, and $r_1 = A \le \kappa/4$ by Lemma 252(iii). Hence $-n/\kappa \le n/\kappa - \kappa/4 \le d'_n/d_n \le n/\kappa$ for $n\ge 2$ and $\kappa^2\le 16$, so $d'^2_n \le n^2 d_n^2/\kappa^2$. With $d'_1 = 2A'$,

$$\mathscr{Q}_z \le 4A'^2 + \frac{1}{\kappa^2}\sum_{n\ge 2} n\, d_n^2 = 4A'^2 + \frac{S_3 - S_2 - 4I_2^2}{\kappa^2 I_1^2},$$

where $S_2 = \sum_{j\ge1} j^2 I_j^2$ and $S_3 = \sum_{j\ge1} j^3 I_j^2$; the second form uses $n\, d_n^2 I_1^2 = (j-1)j^2 I_j^2$ with $j = n+1$.

*The closed form.* The recurrences $I_0 - I_2 = \frac{2}{\kappa} I_1$ and $I_1 - I_3 = \frac{4}{\kappa} I_2$ give $I_0/I_1 = 2/\kappa + A$ and $I_2 I_3/I_1^2 = A - 4A^2/\kappa$. By Lemma 257, $S_3/(\kappa^2 I_1^2) = \frac14(2/\kappa + A)^2 + \frac14$, and, keeping three terms of the third identity,

$$\frac{S_2}{\kappa^2 I_1^2} \ge \frac{I_0 I_1 + I_1 I_2 + I_2 I_3}{2\kappa I_1^2} = \frac{1}{2\kappa}\Big(\frac{2}{\kappa} + 3A - \frac{4A^2}{\kappa}\Big).$$

Collecting terms, $\mathscr{Q}_z \le \frac14 + 4A'^2 + T(A)$ with $T(a) = \frac{a^2}{4} - \frac{a}{2\kappa} - \frac{2a^2}{\kappa^2}$, which is the first inequality.

*The barrier.* Let $s = \sqrt{\kappa^2 + 25/4}$ and $A_\flat = \kappa/(\frac32 + s)$. By Lemma 252(iv), $A_\flat \le A \le \kappa/4$. The function $a \mapsto 1 - a^2 - 3a/\kappa$ decreases on $a\ge 0$, and $A'\ge 0$ because $A' = \operatorname{Var} x_1$, so $0 \le A' \le \alpha := 1 - A_\flat^2 - 3A_\flat/\kappa$. For $\kappa^2 \le 12$ the quadratic $T$ decreases on $[0,\kappa/4]$, since

$$T(a) - T(a') = (a'-a)\,\frac{2\kappa - (a+a')(\kappa^2-8)}{4\kappa^2}$$

and $(a+a')(\kappa^2-8) \le \frac{\kappa}{2}(\kappa^2-8) \le 2\kappa$ there. So $\mathscr{Q}_z \le \frac14 + 4\alpha^2 + T(A_\flat)$. With $\kappa^2 = s^2 - \frac{25}{4}$ and $A_\flat/\kappa = 2/(2s+3)$ both terms are rational in $s$: $\alpha = 16/(2s+3)^2$, and

$$4\alpha^2 + T(A_\flat) = \frac{(2s-5)\,p(s)}{4(2s+3)^4}, \qquad p(s) = 8s^3 + 28s^2 - 98s - 695. \tag{I10}$$

For $0<\kappa\le\sqrt{88}/3$ we have $\frac52 < s \le \sqrt{577}/6 < 4.004$. On $s\ge\frac52$ the cubic $p$ increases, since $p'(s) = 24s^2 + 56s - 98 > 0$, and $p(4.004) < -120$. So the right side of (I10) is negative. □

### 9. Proof of Theorem 254

At $c=0$ the theorem is Lemma 253(i), and for $|c|\ge\sqrt{88}/3$ it is Lemma 253(iii). For $0<|c|<\sqrt{88}/3$, Propositions 258 and 259 give $\mathscr{Q}_x, \mathscr{Q}_z \le \frac14$, and (I8) gives $L\le\frac14$. □

## Appendix J: Computations for the single-link law

### 1. Barta's inequality

Let $\rho(\theta) = \sin^2\theta\, e^{\kappa\cos\theta}$ on $(0,\pi)$, let $V$ be a continuous potential, and let

$$T = -\rho^{-1}\partial_\theta\,\rho\,\partial_\theta + V.$$

If $\psi>0$ is smooth on $(0,\pi)$ with $\rho\psi^2 = O(\sin^4\theta)$ at the end points and $T\psi \ge \lambda_0\psi$, then

$$\int (|h'|^2 + Vh^2)\rho\, d\theta \ge \lambda_0 \int h^2\rho\, d\theta$$

for every smooth $h$ vanishing to first order at 0 and $\pi$. Indeed, writing $h = \psi u$ and integrating by parts, the boundary terms $\rho\psi\psi' u^2$ vanish and

$$\int (|h'|^2 + Vh^2)\rho = \int \rho\psi^2|u'|^2 + \int \frac{T\psi}{\psi} h^2\rho \;\ge\; \lambda_0 \int h^2\rho.$$

This is the inequality of Barta [10] in the form of a ground state transformation.

### 2. The computations of Lemma 251

The generator of the reversible diffusion of $\nu_b$ on $S^3$ is $L = \Delta_{S^3} + \kappa\nabla x_1\cdot\nabla$. In the polar angle $\theta$ from $e_1$ the round metric is $d\theta^2 + \sin^2\theta\, g_{S^2}$, so $\Delta_{S^3} = \partial_\theta^2 + 2\cot\theta\,\partial_\theta + \sin^{-2}\theta\,\Delta_{S^2}$, and $\nabla x_1\cdot\nabla = -\sin\theta\,\partial_\theta$. On $h(\theta)Y_\ell$ with a spherical harmonic $Y_\ell$ of degree $\ell$ on $S^2$ the operator acts as $L_\ell = \partial_\theta^2 + b\,\partial_\theta - \ell(\ell+1)\sin^{-2}\theta$, $b = 2\cot\theta - \kappa\sin\theta = \rho'/\rho$, so $-L_\ell = T$ with $V = \ell(\ell+1)\sin^{-2}\theta$.

*Degree $\ell\ge 1$.* Take $\psi = \sin\theta\, e^{a\kappa\cos\theta}$. Then $\psi'/\psi = \cot\theta - a\kappa\sin\theta$ and $\psi''/\psi = (\psi'/\psi)^2 + (\psi'/\psi)'$, and a short computation gives

$$\frac{-L_1\psi}{\psi} = 3 + (1+5a)\kappa\cos\theta - a(1+a)\kappa^2\sin^2\theta.$$

For $a=-\frac{1}{5}$ this is $3+\frac{4}{25}\kappa^2\sin^2\theta\ge 3$. Since $V=\ell(\ell+1)\sin^{-2}\theta\ge 2\sin^{-2}\theta$ for $\ell\ge 1$, Barta's inequality gives $-L_\ell\ge 3$ on every sector with $\ell\ge 1$.

*Degree $\ell=0$.* Let $g(\theta)$ be a zonal eigenfunction, $-L_0 g=\lambda g$, $g$ not constant, so that $g'(0)=g'(\pi)=0$. Differentiating $g''+bg'=-\lambda g$ gives, for $h=g'$,

$$-\big(h''+bh'\big)-b'h=\lambda h,\qquad -b'=\frac{2}{\sin^2\theta}+\kappa\cos\theta,$$

so $h$ is an eigenfunction of $T$ with $V=2\sin^{-2}\theta+\kappa\cos\theta$, with eigenvalue $\lambda$, and $h\not\equiv 0$ vanishes at both ends. With the same $\psi$,

$$\frac{T\psi}{\psi}=3+(2+5a)\kappa\cos\theta-a(1+a)\kappa^2\sin^2\theta,$$

which for $a=-\frac{2}{5}$ is $3+\frac{6}{25}\kappa^2\sin^2\theta\ge 3$. Barta's inequality applied to $h$ gives $\lambda\ge 3$. Together with the case $\ell\ge 1$ every nonzero eigenvalue of $-L$ is at least 3, which is Lemma 251; the spectrum is discrete because $S^3$ is compact and the drift is smooth.

### 3. Moments

The density of $t=x_1$ under $\nu_b$ is proportional to $(1-t^2)^{1/2}e^{\kappa t}$. For every smooth $\phi$ on $[-1,1]$,

$$\frac{d}{dt}\Big[(1-t^2)^{3/2}e^{\kappa t}\Big]=(1-t^2)^{1/2}e^{\kappa t}\big(\kappa(1-t^2)-3t\big),$$

and integration by parts gives $\mathbb{E}[\phi'(t)(1-t^2)]=\mathbb{E}\big[\phi(t)\big(3t-\kappa(1-t^2)\big)\big]$. With $\phi=1$ this is $\kappa\mathbb{E}(1-x_1^2)=3\mathbb{E}x_1$, the identity of Lemma 252(i). With $\phi=t^k$ it gives the recursion

$$m_{k+2}=m_k-\frac{k+3}{\kappa}m_{k+1}+\frac{k}{\kappa}m_{k-1},\qquad m_k=\mathbb{E}x_1^k,$$

from $m_0=1$ and $m_1=A$; all polynomial moments of $\nu_b$ are rational in $\kappa$ and $A(\kappa)$.[31]

### 4. The Riccati barriers

By Lemma 252(i), $A$ solves $A'=\mathscr{F}(\kappa,A):=1-A^2-3A/\kappa$ on $(0,\infty)$, and $A(\kappa)=\frac{\kappa}{4}-\frac{\kappa^3}{96}+O(\kappa^5)$ from the series of $I_1$ and $I_2$. Let $s=\sqrt{\kappa^2+4}$, $u=\sqrt{\kappa^2+9}$, $\phi_-=\kappa/(2+s)$ and $\phi_+=\kappa/(1+u)$. Direct differentiation gives

$$\phi_-'-\mathscr{F}(\kappa,\phi_-)=-\frac{\kappa^2}{s(s+2)^2}\le 0,$$

$$\phi_+'-\mathscr{F}(\kappa,\phi_+)=\frac{(u-3)^2}{u(1+u)^2}\ge 0,$$

and $\phi_\pm=\frac{\kappa}{4}+O(\kappa^3)$.[32] The comparison is singular at 0 because of the term $3A/\kappa$, and we carry it out with an integrating factor. Let $D=A-\phi_-$. Then

$$\begin{aligned}D'&=\mathscr{F}(\kappa,A)-\mathscr{F}(\kappa,\phi_-)-\big(\phi_-'-\mathscr{F}(\kappa,\phi_-)\big)\\&=-\Big(A+\phi_-+\frac{3}{\kappa}\Big)D+R,\end{aligned}$$

with $R=\kappa^2/(s(s+2)^2)\ge 0$. Put $\mu(\kappa)=\kappa^3\exp\int_0^\kappa(A+\phi_-)$. Then $(\mu D)'=\mu R\ge 0$, and $\mu D=O(\kappa^6)\to 0$ as $\kappa\to 0$, so $\mu D\ge 0$ and $A\ge\phi_-$. The same argument with $D=\phi_+-A$ and $R=(u-3)^2/(u(1+u)^2)$ gives $A\le\phi_+$. This is Lemma 252(ii).

For (iv) let $s=\sqrt{\kappa^2+25/4}$ and $\phi_\flat=\kappa/(\frac{3}{2}+s)$. Since $s^2-\frac{9}{4}=\kappa^2+4$, the same differentiation gives

$$\phi_\flat'-\mathscr{F}(\kappa,\phi_\flat)=-\frac{5\,(s-\frac{5}{2})}{2s\,(\frac{3}{2}+s)^2}\le 0,$$

so $\phi_\flat$ is a subsolution, and $\phi_\flat=\frac{\kappa}{4}-\frac{\kappa^3}{80}+O(\kappa^5)$.[33] Hence $A-\phi_\flat=\frac{\kappa^3}{480}+O(\kappa^5)$, and the integrating-factor argument above, with $D=A-\phi_\flat$, $R=\frac{5(s-5/2)}{2s(3/2+s)^2}\ge 0$ and $\mu=\kappa^3\exp\int_0^\kappa(A+\phi_\flat)$, gives $\mu D=O(\kappa^6)$ at 0, $(\mu D)'=\mu R\ge 0$, and so $A\ge\phi_\flat$. Since $\frac{3}{2}+\sqrt{\kappa^2+25/4}<2+\sqrt{\kappa^2+4}$ for $\kappa>0$, this improves the lower bound of (ii).

### Appendix K: Bessel functions

This appendix proves the auxiliary statements used in Appendix I for Theorem 254. Throughout, $I_j=I_j(\kappa)$ with $\kappa>0$, and $I_{-j}=I_j$ for integers $j$. We use the series

$$I_\nu(\kappa)=\sum_{i\ge 0}\frac{(\kappa/2)^{2i+\nu}}{i!\,(i+\nu)!},$$

the integral $I_j(\kappa)=\pi^{-1}\int_0^\pi e^{\kappa\cos\theta}\cos j\theta\,d\theta$, the Fourier series $e^{\kappa\cos\theta}=\sum_{j\in\mathbb{Z}}I_j e^{ij\theta}$ and the recurrences

$$I_{j-1}-I_{j+1}=\frac{2j}{\kappa}I_j,\quad I_j'=I_{j+1}+\frac{j}{\kappa}I_j,\tag{K1}$$

all of which are in Watson's treatise [91], §§3.7, 3.71, 2.22. All series below converge absolutely, because $0<I_j\le(\kappa/2)^j e^\kappa/j!$.

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[31] For instance $m_2=1-3A/\kappa$, $m_3=A-3/\kappa+12A/\kappa^2$ and $m_4=m_2-5m_3/\kappa+2A/\kappa$. The joint moments with $x_2,x_3,x_4$ follow from exchangeability: $\mathbb{E}[\phi(x_1)x_2^2]=\frac{1}{3}\mathbb{E}[\phi(x_1)(1-x_1^2)]$.

[32] For the first identity write $\phi_-=(s-2)/\kappa$, which follows from $s^2-4=\kappa^2$; then $\phi_-'=(\kappa^2/s-s+2)/\kappa^2$ and $1-\phi_-^2-3\phi_-/\kappa=(\kappa^2-(s-2)^2-3(s-2))/\kappa^2$, and the difference simplifies with $s^2=\kappa^2+4$. The second is the same computation with $\phi_+=\kappa(u-1)/(u^2-1)=\kappa(u-1)/(\kappa^2+8)$ in place of $\phi_-$ and $u^2=\kappa^2+9$ in place of $s^2=\kappa^2+4$. After clearing the denominator $\kappa^2(1+u)^2$ the numerator becomes $\kappa^2(u-3)^2/u$, which is the stated residual.

[33] With $\phi_\flat=\kappa(s-\frac{3}{2})/(\kappa^2+4)$ one has $\phi_\flat'=(\frac{3}{2}s+\frac{25}{4})/(s(\frac{3}{2}+s)^2)$, using $ds/d\kappa=\kappa/s$; and $\mathscr{F}(\kappa,\phi_\flat)=1-\phi_\flat^2-3/(\frac{3}{2}+s)$. Clearing the denominator $s(\frac{3}{2}+s)^2$ and replacing $\kappa^2$ by $s^2-\frac{25}{4}$ leaves $-\frac{5}{2}s+\frac{25}{4}$ in the numerator, which is the stated residual. The expansion follows from $s=\frac{5}{2}+\frac{\kappa^2}{5}+O(\kappa^4)$.

### 1. Zonal harmonics

Under $\sigma$ the coordinate $x_1$ has the density $\frac{2}{\pi}(1-t^2)^{1/2}$ on $[-1,1]$, the area of the two-sphere $\{x_1=t\}$ being proportional to $1-t^2$ and its normal speed to $(1-t^2)^{-1/2}$. The Chebyshev polynomials $U_n(\cos\theta)=\sin((n+1)\theta)/\sin\theta$ satisfy

$$\int U_n U_m \, d\sigma = \frac{2}{\pi}\int_0^\pi \sin((n+1)\theta)\sin((m+1)\theta)\,d\theta = \delta_{nm},$$

and $U_n = C_n^{(1)}$ is the Gegenbauer polynomial of index $\frac{4-2}{2}=1$, so $U_n(\langle y,x\rangle)$ is the zonal harmonic of degree $n$ on $S^3$ with pole $y$: $|x|^n U_n(\langle y,x\rangle/|x|)$ is a harmonic polynomial on $\mathbb{R}^4$ [87], Ch. IV, §2.

**Lemma 260.** *$e^{\kappa t} = \frac{2}{\kappa}\sum_{n\ge 0}(n+1)I_{n+1}(\kappa)\,U_n(t)$ in $L^2\big(\frac{2}{\pi}(1-t^2)^{1/2}dt\big)$, and uniformly on $[-1,1]$.*

*Proof.* With $t=\cos\theta$ and $2\sin\theta\sin((n+1)\theta)=\cos n\theta-\cos(n+2)\theta$, the $n$-th coefficient is

$$\frac{2}{\pi}\int_0^\pi e^{\kappa\cos\theta}\sin((n+1)\theta)\sin\theta\,d\theta = I_n - I_{n+2} = \frac{2(n+1)}{\kappa}I_{n+1},$$

by (K1). Since $|U_n|\le n+1$ and $I_{n+1}$ decays factorially, the series converges uniformly. □

### 2. Proof of Lemma 257

Let $F(\theta)=e^{\kappa\cos\theta}=\sum_j I_j e^{ij\theta}$. Then

$$F'(\theta) = -\kappa\sin\theta\,F(\theta) = \sum_j ijI_j e^{ij\theta},$$

and Parseval's formula gives

$$\sum_{j\in\mathbb{Z}} j^2 I_j^2 = \frac{\kappa^2}{2\pi}\int_0^{2\pi}\sin^2\theta\, e^{2\kappa\cos\theta}d\theta = \frac{\kappa^2}{2}\big(I_0(2\kappa)-I_2(2\kappa)\big) = \frac{\kappa}{2}I_1(2\kappa),$$

the last step by (K1) at the argument $2\kappa$. Halving gives the first identity. The coefficient of $e^{i\theta}$ in $F^2=e^{2\kappa\cos\theta}$ is $I_1(2\kappa)=\sum_{j\in\mathbb{Z}}I_jI_{1-j}$, and the terms with $j\le 0$ repeat those with $j\ge 1$; this is the third identity.

For the second, $jI_j=\frac{\kappa}{2}(I_{j-1}-I_{j+1})$ by (K1), so

$$\sum_{j\ge1} j^3 I_j^2 = \frac{\kappa^2}{4}\sum_{j\ge1} j\big(I_{j-1}^2+I_{j+1}^2-2I_{j-1}I_{j+1}\big).$$

Shifting indices, $\sum_{j\ge1} j(I_{j-1}^2+I_{j+1}^2) = I_0^2+\sum_{m\ge1}2mI_m^2$, and $\sum_{m\ge1}2mI_m^2=\kappa\sum_{m\ge1}(I_{m-1}I_m-I_mI_{m+1})=\kappa I_0I_1$ by telescoping. For the mixed term write $j=\frac12((j-1)+(j+1))$ and apply (K1) to $(j-1)I_{j-1}$ and to $(j+1)I_{j+1}$, using $I_{-1}=I_1$ at $j=1$:

$$jI_{j-1}I_{j+1} = \frac{\kappa}{4}\big[(I_{j-1}I_j - I_jI_{j+1}) + (I_{j-2}I_{j+1}-I_{j-1}I_{j+2})\big].$$

Both brackets telescope, to $I_0I_1$ and to $I_{-1}I_2=I_1I_2$, so $\sum_{j\ge1}jI_{j-1}I_{j+1}=\frac{\kappa}{4}(I_0I_1+I_1I_2)$. Altogether

$$\sum_{j\ge1} j^3I_j^2 = \frac{\kappa^2}{4}\Big(I_0^2+\frac{\kappa}{2}I_1(I_0-I_2)\Big) = \frac{\kappa^2}{4}\big(I_0^2+I_1^2\big). \qquad \square$$

### 3. The coefficients of $\mathscr{D}$

**Lemma 261.** *Write $\mathscr{D}(\kappa)=\sum_{n\ge0}\mathscr{D}_n\kappa^{2n}$ for the function of (I9). Then $\mathscr{D}_0=\mathscr{D}_1=0$, and for $m\ge1$*

$$\mathscr{D}_{m+1} = \frac{N_m}{4^m\,m!\,(m+1)!\,(m+1)(m+2)},$$

*with $N_m=\binom{2m}{m}\big(4(2m+1)+(m-1)(m+1)(m+2)\big)-4^m(m+1)(m+2)$. Moreover $N_m=0$ for $m\le2$ and $N_m>0$ for $m\ge3$.*

*Proof. The coefficients.* Multiplying the series of $I_0$ and $I_1$ and collecting powers,

$$I_0^2=\sum_{m\ge0}\alpha_m\kappa^{2m},\qquad I_1^2=\sum_{m\ge0}\beta_m\kappa^{2m+2},$$
$$\kappa I_1(2\kappa)=\sum_{m\ge0}\frac{\kappa^{2m+2}}{m!\,(m+1)!},$$

where

$$\alpha_m=\frac{1}{4^m}\sum_{i=0}^{m}\frac{1}{i!^2(m-i)!^2},$$
$$\beta_m=\frac{1}{4^{m+1}}\sum_{i=0}^{m}\frac{1}{i!\,(i+1)!\,(m-i)!\,(m-i+1)!}.$$

By Vandermonde's identity, $\sum_i\binom{m}{i}^2=\binom{2m}{m}$ and $\sum_i\binom{m}{i}\binom{m+2}{i+1}=\binom{2m+2}{m+1}$, so

$$\alpha_m=\frac{\binom{2m}{m}}{4^m\,m!^2},\qquad \beta_m=\frac{\binom{2m+2}{m+1}}{4^{m+1}m!\,(m+2)!}.$$

The coefficient of $\kappa^{2m+2}$ in $\mathscr{D}$ is $\mathscr{D}_{m+1}=8\beta_m+2\beta_{m-1}-\alpha_m-1/(m!(m+1)!)$, with $\beta_{-1}=0$; for $m=0$ it is $\frac84-1-1=0$, and $\mathscr{D}_0=0$ because every term of $\mathscr{D}$ is $O(\kappa^2)$. For $m\ge1$ put $\Delta=4^m\,m!\,(m+1)!\,(m+1)(m+2)$ and $b=\binom{2m}{m}$. With $\binom{2m+2}{m+1}=\frac{2(2m+1)}{m+1}b$,

$$8\beta_m=\frac{4(2m+1)\,b}{\Delta},\qquad \alpha_m=\frac{(m+1)^2(m+2)\,b}{\Delta},$$
$$2\beta_{m-1}=\frac{2m(m+1)(m+2)\,b}{\Delta},$$
$$\frac{1}{m!(m+1)!}=\frac{4^m(m+1)(m+2)}{\Delta},$$

and $2m-(m+1)=m-1$ gives $\mathscr{D}_{m+1}=N_m/\Delta$.

*The sign.* Directly, $N_0=N_1=N_2=0$ ($2-2$, $24-24$, $192-192$), $N_3=80$, $N_4=1140$ and $N_5=10416$. For $m\ge 6$ we first show $16^m\le 4m\binom{2m}{m}^2$ for $m\ge 1$ by induction: it is an equality at $m=1$, and $(m+1)\binom{2m+2}{m+1}=2(2m+1)\binom{2m}{m}$ gives

$$4(m+1)\binom{2m+2}{m+1}^2=\frac{16(2m+1)^2}{m+1}\binom{2m}{m}^2\ge\frac{16(2m+1)^2}{4m(m+1)}16^m\ge 16^{m+1}.$$

Since $(m-1)^2\ge 4m$ for $m\ge 6$, it follows that $\binom{2m}{m}^2(m-1)^2\ge 16^m$, that is $\binom{2m}{m}(m-1)\ge 4^m$, and

$$N_m=\Big(\binom{2m}{m}(m-1)-4^m\Big)(m+1)(m+2)+4(2m+1)\binom{2m}{m}>0.$$

□

### 4. Monotonicity of Bessel ratios

**Lemma 262.** *For $n\ge 1$ and $\kappa>0$, $I_{n+2}(\kappa)/I_{n+1}(\kappa)\le I_2(\kappa)/I_1(\kappa)$.*

*Proof.* We use a lemma of Biernacki and Krzyż [23]: if $f=\sum a_jx^j$ and $g=\sum b_jx^j$ have positive coefficients and $a_j/b_j$ is nonincreasing in $j$, then $f/g$ is nonincreasing on $x>0$. Indeed

$$f'g-fg'=\sum_{i,j}(i-j)a_ib_jx^{i+j-1}=\frac12\sum_{i,j}(i-j)b_ib_j\Big(\frac{a_i}{b_i}-\frac{a_j}{b_j}\Big)x^{i+j-1}\le 0.$$

Take $x=\kappa^2$, $f=I_{n+1}(\kappa)/\kappa^{n+1}$ and $g=I_1(\kappa)/\kappa$. Then $a_j/b_j=2^{-n}(j+1)!/(j+n+1)!$ decreases in $j$, so $I_{n+1}/(\kappa^nI_1)$ is nonincreasing in $\kappa$. Its logarithmic derivative is, by the second recurrence of (K1),

$$\frac{n+1}{\kappa}+\frac{I_{n+2}}{I_{n+1}}-\frac{n}{\kappa}-\frac{1}{\kappa}-\frac{I_2}{I_1}=\frac{I_{n+2}}{I_{n+1}}-\frac{I_2}{I_1}\le 0.$$

□

## Appendix L: The certificate bound

This appendix derives the bound $h_F\le 14.420001$ of Remark 26 from the data of [79]. The argument is the proof of the bound in Theorem 4 with two changes. The divergence identity of Lemma 19, taken there with the factor $\frac{9}{10}$, is replaced by a positive semidefinite form at each site, and the weights $\frac{21}{10}$ and $\frac54$ of Lemma 20 are replaced by weights chosen plaquette by plaquette. Everything is proved by hand except the positivity of ninety-six matrices of trigonometric polynomials and the arithmetic of finitely many rational numbers.[34]

### 1. Ends and corners

At a site $y$ the eight *ends* are

$$o_\mu(y)=X_{y,\mu},\qquad i_\mu(y)=Q^*_{y-e_\mu,\mu}X_{y-e_\mu,\mu}Q_{y-e_\mu,\mu},$$

for $\mu=0,1,2,3$, as in the proof of Lemma 19; the end of a link in $F$ or outside the box is 0. Each link $e=(y,\mu)$ has the end $o_\mu$ at its tail $y$ and the end $i_\mu$ at its head $y+e_\mu$, both of norm $|X_e|$.

**Lemma 263.** *Let $p$ be the plaquette at $x$ in the plane $\mu<\nu$. Then*

$$\begin{aligned}\langle Y_1,Y_4\rangle&=\langle o_\mu,o_\nu\rangle(x),\\ \langle Y_1,Y_2\rangle&=\langle i_\mu,o_\nu\rangle(x+e_\mu),\\ \langle Y_3,Y_4\rangle&=\langle o_\mu,i_\nu\rangle(x+e_\nu),\\ \langle Y_2,WY_3W^*\rangle&=\langle i_\nu,i_\mu\rangle(x+e_\mu+e_\nu).\end{aligned}$$

*Every pair of ends at one site in two different directions occurs in exactly one of these identities, for exactly one plaquette.*

*Proof.* The first identity is the definition of the ends. The inner product is invariant under conjugation by a unitary matrix, so $\langle X_{e_1},Q_{e_1}X_{e_2}Q^*_{e_1}\rangle=\langle Q^*_{e_1}X_{e_1}Q_{e_1},X_{e_2}\rangle$, which is the second identity, and the third follows in the same way from $\langle Q_{e_4}X_{e_3}Q^*_{e_4},X_{e_4}\rangle=\langle X_{e_3},Q^*_{e_4}X_{e_4}Q_{e_4}\rangle$. For the fourth, $WQ_{e_4}=Q_{e_1}Q_{e_2}Q^*_{e_3}$, so that $WY_3W^*=RX_{e_3}R^*$ with $R=Q_{e_1}Q_{e_2}Q^*_{e_3}$, and conjugating both vectors by $(Q_{e_1}Q_{e_2})^*$ gives $\langle Q^*_{e_2}X_{e_2}Q_{e_2},Q^*_{e_3}X_{e_3}Q_{e_3}\rangle$. The links $e_2$ and $e_3$ both have their head at $x+e_\mu+e_\nu$. For the last statement, the pairs $(o_\mu,o_\nu)$, $(i_\mu,o_\nu)$, $(o_\mu,i_\nu)$ and $(i_\mu,i_\nu)$ at $y$ come from the plaquettes in the plane $\mu\nu$ at $y$, $y-e_\mu$, $y-e_\nu$ and $y-e_\mu-e_\nu$ respectively. There are $4\binom42=24$ such pairs at each site and $6\cdot4=24$ corners of plaquettes per site. □

For each class of sites modulo 2 let $G_y$ be a real symmetric $8\times8$ matrix indexed by the ends, with $G_y[o_\mu,i_\mu]=0$ for every $\mu$. For the plaquette $p$ at $x$ in the plane $\mu<\nu$ put

$$\begin{aligned}c_p^G=\;&2G_x[o_\mu,o_\nu]\langle Y_1,Y_4\rangle+2G_{x+e_\mu}[i_\mu,o_\nu]\langle Y_1,Y_2\rangle\\&+2G_{x+e_\nu}[o_\mu,i_\nu]\langle Y_3,Y_4\rangle\\&+2G_{x+e_\mu+e_\nu}[i_\mu,i_\nu]\langle Y_2,WY_3W^*\rangle.\end{aligned}$$

The matrix $G^0$ with $G^0[a,a]=2$, $G^0[o_\mu,o_\nu]=G^0[i_\mu,i_\nu]=1$ and $G^0[o_\mu,i_\nu]=-1$ for $\mu\ne\nu$ is the matrix of the form $|\sum_\mu(o_\mu-i_\mu)|^2+\sum_\mu|o_\mu+i_\mu|^2$ of Lemma 19, and $c_p^{G^0}$ is the corner form $c_p$ of (7).

---

[34] The constant 14.420001 is not the best possible for this kind of certificate: by the search reported in Remark 26 it cannot go below 14.384, while the true value of $h_F$ is at least 13.123.

**Lemma 264.** *On $\mathbb{Z}^4$, or on a box with free boundary, for every field $Q$ and every square-summable $X$,*

$$\sum_y \sum_{a,b} G_y[a,b]\,\langle \mathrm{end}_a, \mathrm{end}_b\rangle(y) = \sum_p c_p^G + \sum_e \big(G_{t(e)}[o,o] + G_{h(e)}[i,i]\big)\,|X_e|^2,$$

*where $t(e)$ and $h(e)$ are the tail and the head of $e$, and $o$, $i$ are its two ends. If every $G_y$ is positive semidefinite, the left side is nonnegative.*

*Proof.* Split the sum over $a,b$ at each site into the diagonal terms, the pairs $(o_\mu, i_\mu)$, whose coefficient is 0, and the pairs of ends in different directions, each counted twice. The diagonal terms give the last sum, since each link has one end at its tail and one at its head, and the pairs in different directions give $\sum_p c_p^G$ by Lemma 263. In a box the argument at the end of the proof of Lemma 19 applies. At each site the left side is the trace of $G_y$ times the Gram matrix of the eight ends, and the trace of a product of two positive semidefinite matrices is nonnegative. □

### 2. From plaquettes to the Hessian

Number the positions of a plaquette as in (5) and let $k_p$ be the position of its link in $F$; by Lemma 9 there is exactly one.

**Lemma 265.** *Let every $G_y$ be positive semidefinite with $G_y[o_\mu, i_\mu] = 0$, and let $a_{p,r} \geq 0$ for the three positions $r \neq k_p$ of each plaquette. Suppose that for every plaquette $p$, every $N$, every $W \in \mathrm{U}(N)$ and all $Y_1, \dots, Y_4 \in \mathfrak{u}(N)$ with $Y_{k_p} = 0$,*

$$f_p + c_p^G \leq \sum_{r \neq k_p} a_{p,r} |Y_r|^2. \tag{L1}$$

*Then $|\operatorname{Hess} S_0(v,v)| \leq h|v|^2$ for every field $Q$ and every $v = (X_e Q_e)$ with $X_e = 0$ for $e \in F$, on $\mathbb{Z}^4$ and on every box, where*

$$h = \max_{e \notin F} \Big( \sum_{p \ni e} a_{p,e} + G_{t(e)}[o,o] + G_{h(e)}[i,i] \Big)$$

*and $a_{p,e}$ is the weight of the position of $e$ in $p$.*

*Proof.* Summing (L1) over the plaquettes and adding the identity of Lemma 264, whose left side is nonnegative,

$$\begin{aligned} \operatorname{Hess} S_0(v,v) = \sum_p f_p &\leq \sum_p \Big( \sum_r a_{p,r} |Y_r|^2 - c_p^G \Big) \\ &\leq \sum_{e \notin F} \Big( \sum_{p \ni e} a_{p,e} + G_{t(e)}[o,o] + G_{h(e)}[i,i] \Big) |X_e|^2 \\ &\leq h|v|^2, \end{aligned}$$

since $|Y_r| = |X_e|$ for the link $e$ at position $r$. In a box the links near the boundary lie in fewer plaquettes, and the weights are nonnegative. For the lower bound, apply (L1) at $-W$. By (6) this changes the sign of $f_p$, while $c_p^G$ does not change, and the same chain gives $-\operatorname{Hess} S_0(v,v) \leq h|v|^2$. □

The hypothesis is needed on $\mathrm{U}(N)$ and not only on $\mathrm{SU}(N)$, since $-W \notin \mathrm{SU}(N)$ when $N$ is odd.

**Lemma 266.** *If (L1) holds for $N = 2$ and every $W = \operatorname{diag}(e^{i\varphi_1}, e^{i\varphi_2})$, it holds for every $N$ and every $W \in \mathrm{U}(N)$.*

*Proof.* Both sides are invariant under conjugation of $W$ and all $Y_i$ by one unitary matrix, so we may take $W = \operatorname{diag}(w_1, \dots, w_N)$. Every term of $f_p$, of $c_p^G$ and of $|Y_r|^2$ is a multiple of $\operatorname{Re}\operatorname{Tr}(Y_i D_1 Y_j D_2)$ with $D_1, D_2 \in \{I, W, W^*\}$, and

$$\operatorname{Re}\operatorname{Tr}(Y_i D_1 Y_j D_2) = \operatorname{Re} \sum_{a,b} (Y_i)_{ab}\,(D_1)_{bb}\,(Y_j)_{ba}\,(D_2)_{aa}.$$

The difference of the two sides of (L1) is therefore a sum of one term for each $a$, in the entries $(Y_i)_{aa}$, and one term for each pair $a < b$, in the entries $(Y_i)_{ab}$ and $(Y_i)_{ba} = -\overline{(Y_i)_{ab}}$, as in the proof of Lemma 20. The term of the pair $a < b$ is the same difference for $N = 2$ at $W' = \operatorname{diag}(w_a, w_b)$, evaluated on the matrices $Y_i' \in \mathfrak{u}(2)$ with off-diagonal entries $(Y_i)_{ab}$, $(Y_i)_{ba}$ and zero diagonal. The term of $a$ is the difference at $W'$, for any $b \neq a$, evaluated on $Y_i' = \operatorname{diag}((Y_i)_{aa}, 0)$. In both cases $Y_{k_p}' = 0$, so each term is nonnegative. □

In the orthonormal basis $iI/\sqrt{2}$, $i\sigma_1/\sqrt{2}$, $i\sigma_2/\sqrt{2}$, $i\sigma_3/\sqrt{2}$ of $\mathfrak{u}(2)$, with the Pauli matrices $\sigma_j$, the difference of the two sides of (L1) at $N = 2$ is a quadratic form on the three free positions, with a real symmetric $12 \times 12$ matrix $M_p(\varphi)$. Its entries are affine combinations of 1, $\cos\varphi_a$, $\sin\varphi_a$, $\cos(\varphi_1 - \varphi_2)$ and $\sin(\varphi_1 - \varphi_2)$: the form $f_p$ is linear in $W$, and $W$ enters $c_p^G$ only through $\langle Y_2, W Y_3 W^* \rangle$. Since $W$ commutes with $I$ and $\sigma_3$, $M_p(\varphi)$ splits into two blocks of size six, one on the span of $iI, i\sigma_3$ and one on the span of $i\sigma_1, i\sigma_2$, at every position. By Lemmas 265 and 266, a certificate is a choice of $G_y$ and $a_{p,r}$ such that every $M_p(\varphi)$ is positive semidefinite at every $\varphi$, and it gives $h_F \leq h$.

### 3. The certificate

The fixed set $F$ and the field of ends have period two, so it suffices to give $G_y$ for the sixteen classes of sites and $a_{p,r}$ for the ninety-six classes of plaquettes, six planes at each of sixteen sites. The certificate of [79] consists of sixteen positive definite matrices $G_y$ of size eight with $G_y[o_\mu, i_\mu] = 0$, and of nonnegative weights $a_{p,r}$, all rational with denominator $10^6$. We checked that the set of fixed links used there is our $F$ up to a symmetry of the lattice. Of the sixty-four links of a period cell, sixteen lie in $F$, one ending at each site. For each of the other forty-eight the bracket in the definition of $h$ is computed exactly, and the largest is $14420001/10^6$.

**Theorem 267** (Computer-assisted)**.** *For every $N$, every field $Q$ and every $v = (X_e Q_e)$ with $X_e = 0$ for $e \in F$, on $\mathbb{Z}^4$ and on every box,*

$$|\operatorname{Hess} S_0(v,v)| \leq 14.420001\,|v|^2.$$

*Proof.* We show that every $M_p(\varphi)$ is positive semidefinite, which proves the theorem by Lemmas 265 and 266. Write

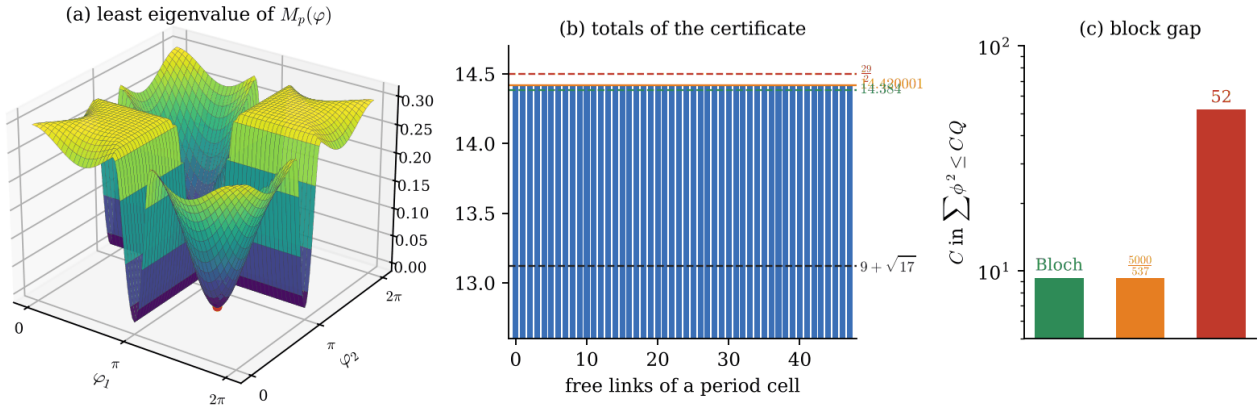



FIG. 26. The certificate of Appendix L. (a) The least eigenvalue of $M_p(\varphi)$ over the torus of phases for the plaquette class with the smallest minimum, about 0.0057 on a grid of $96^2$ phases (red dot); Theorem 267 shows it stays positive. (b) The forty-eight totals in the definition of $h$, in increasing order, against $\frac{29}{2}$, the value 14.420001 of the largest, the floor 14.384 for certificates on the period-two cell, and the lower bound $9+\sqrt{17}$ for $h_F$. (c) The constant $C$ in $\sum \phi^2 \le CQ$ for the block gauge: 52 from Theorem 94, $5000/537$ from Proposition 95, and the bulk Bloch value $1/0.10749 = 9.3034$, the best constant when $W$ is the set of all blocks. The eigenvalues in (a) and the Bloch value are in floating point.

$z = (1, \cos\varphi_1, \sin\varphi_1, \cos\varphi_2, \sin\varphi_2)$. For each of the 192 blocks of size six there is a positive definite integer symmetric matrix $P$ of size thirty such that

$$2^{24} \cdot 4 \cdot 10^6 M(\varphi)_{jk} = \sum_{a,b=0}^{4} z_a z_b P_{6a+j,\,6b+k}$$

for all $j,k$, as polynomials modulo $\cos^2\varphi_a + \sin^2\varphi_a = 1$. For $x \in \mathbb{R}^6$ put $w = z \otimes x$. Then $2^{24} \cdot 4 \cdot 10^6 x^{\mathsf{T}} M(\varphi) x = w^{\mathsf{T}} P w$, which is positive for $x \neq 0$, since $z_0 = 1$ gives $w \neq 0$. So every block is positive definite at every $\varphi$. □

The proof of Theorem 267 rests on computations of three kinds. The positivity of the $G_y$, the signs of the weights and the forty-eight totals are finite statements about rational numbers. The identities between $M(\varphi)$ and the matrices $P$, and the positive definiteness of the $P$, are finite statements about integers. All of these are checked by the Lean 4 kernel in [79], on no axioms beyond `propext` and `Quot.sound`; the program there also checks the positivity at every phase a second time, by interval arithmetic on a subdivision of the torus of phases. Lemmas 263 to 266 and the formula (6) for the Hessian are the written proofs above and are not formalised.[35] We checked the finite data in exact arithmetic and the positivity at every phase by eigenvalues on an adaptive grid with a Lipschitz bound, in floating point, and J. M. Lockwood checked it again in exact rational arithmetic with outward-rounded interval elimination, finding least eigenvalue above $559/716800$ for every plaquette.

With 14.420001 in place of $\frac{29}{2}$ the proofs of Theorems 4 and 5 go through unchanged, which gives the ranges stated in Remark 26. Those theorems are stated with $\frac{29}{2}$ because its proof uses no computer.

[35] Formalising them needs matrices over $\mathbb{C}$, the group $\mathrm{U}(N)$ and second derivatives along geodesics, that is, a library such as Mathlib; the Lean files of [79] use only the core library.


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